

PROBABILITY AND STATISTICS

**For B. Tech
FIRST YEAR (SECOND SEMESTER)
Students of JNTU ANANTAPUR
as per New Syllabus (R19) 2019 - 20**

(Common to CSE, IT and Food Technology)

[Including Chapter Wise Previous GATE Questions]

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PREFACE

We are happy to bring out the revised edition of the book “*Probability and Statistics*”. The earlier editions have been received well by the student and teacher community. This edition has been thoroughly reviewed in light of the latest syllabus **2019-20 (R19)** of **B. Tech. I Year II Semester** students of Jawaharlal Nehru Technological University, **ANANTAPUR**.

Questions from Previous Examination papers of J.N.T.U, Anantapur have been inserted at proper places. The **Quiz** questions have been given at the end of each unit. An important highlight of this edition is the inclusion of previous GATE Questions at the end of each chapter for the benefit of the students. We hope this edition will meet the requirements of students and teachers.

We invite constructive criticism from the teachers for further improvement of the book.

We are thankful to the Management Team and the Editorial Department of S Chand And Company Limited, New Delhi for all the help and support in the publication of this book.

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PROBABILITY AND STATISTICS

(Common to C. S. E., I. T., & Food Technology)
(19A54202)

UNIT - I : Descriptive Statistics and Methods for Data Science

- ◆ Data science, Statistics Introduction, Population vs Sample, Collection of data, primary and secondary data, Type of variable: dependent and independent Categorical and Continuous variables, Data visualization, Measures of Central tendency, Measures of Variability (spread or variance) Skewness Kurtosis, correlation, correlation coefficient, rank correlation, regression coefficients, principle of least squares, method of least squares, regression lines.

UNIT - II : Probability

- ◆ Probability, probability axioms, addition law and multiplicative law of probability, conditional probability, Baye's theorem, random variables (discrete and continuous), probability density functions, properties, mathematical expectation.

UNIT - III : Probability Distributions

- ◆ Probability distribution - Binomial, Poisson approximation to the binomial distribution and normal distribution-their properties.

UNIT - IV : Estimation and Testing of Hypothesis (Large Sample Tests)

- ◆ Estimation-parameters, statistics, sampling distribution, point estimation, Formulation of null hypothesis, alternative hypothesis, the critical and acceptance regions, level of significance, two types of errors and power of the test. Large Sample Tests: Test for single proportion, difference of proportions, test for single mean and difference of means. Confidence interval for parameters in one sample and two sample problems.

UNIT - V : Small Sample Tests

- ◆ Student t - distribution (test for single mean, two means and paired t -test), testing of equality of variances (F-test), χ^2 - test for goodness of fit, χ^2 - test for independence of attributes.



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UNIT - I

DESCRIPTIVE STATISTICS AND METHODS FOR DATA SCIENCE

1. *Descriptive Statistics*

- ♦ *Data Science, Statistics*
- ♦ *Collection of Data*
- ♦ *Population, Sample, Types of Variables*
- ♦ *Data Visualization*

2. *Measures of Central Tendency and Measures of Variability*

- ♦ *Mean, Median, Mode*
- ♦ *Geometric Mean, Harmonic Mean*
- ♦ *Range, Mean Deviation*
- ♦ *Variance and Standard Deviation*
- ♦ *Skewness and Kurtosis*

3. *Correlation*

- ♦ *Types of Correlation*
- ♦ *Covariance, Coefficient of Correlation*
- ♦ *Rank Correlation Coefficient*

4. *Regression*

- ♦ *Principle of Least Squares*
- ♦ *Regression Coefficients and Regression Equation*
- ♦ *Angle Between Two Regression Lines*



CHAPTER - 1

DESCRIPTIVE STATISTICS

1.1 DATA SCIENCE

Introduction :

Data science is the practice of using statistical techniques, regression models, machine learning and deep learning algorithms to produce advanced insights and build predictive applications. All software applications are intended to increase productivity and efficiency by automating human activity. Traditionally, these tasks needed to be repetitive in nature and based on a deterministic set of rules. An example would be an accounting system that can take sales and expenses and automatically create a balance sheet. The intent of data science applications is to automate tasks that require human judgement and are not driven by deterministic rules. An example would be a predictive application that automatically determines if a customer feedback statement is positive or negative, or if an email is a spam email or not.

Sometimes, data science is also to provide insights that may not be otherwise available. An example would be an application that Toyota may use to predict whether an existing customer is ready to purchase a new car. This is something that an experienced sales person may also not be able to determine with high accuracy.

All applications work by creating and operating a digital model of the real-life scenario that they are trying to automate. Data science applications use historical data to “learn” the chief characteristics of the real-life scenario and then create the digital model on that basis. Typically, this model is mathematical in nature, but it can take other forms.

Since data is being used to build non-deterministic models, such techniques are grouped under the umbrella term “Data science”. Other terms that are applied are “Machine learning” and “Artificial Intelligence”. However, Artificial Intelligence can have broader connotations and is sometimes treated as a distinct category from Data science. But the popular AI techniques have many things in common with Machine Learning techniques, making these disciplines a continuum than having sharp boundaries.

Data Science Techniques : There are a variety of techniques that come in the scope of data science.

- ❖ Regression techniques are used to build a mathematical equation that uses input variables to predict the value of an output variable. For example, factors such location, number of rooms, age of the house can be used to predict the price of a house by using regression techniques.
- ❖ Machine learning methods create computational models with input data and use the models to determine the “class” of a variable. For example, an ML model can be used to determine if an email is “spam” or “regular”. Here, “spam” and “regular” are called classes and such algorithms are called classification algorithms. Machine learning methods rely on iterative

algorithms that refine the computational model and can increase in sophistication based on the computational power available. Some well-known algorithms are Decision Trees, Random Forest, XG Boost

- ❖ There are many other techniques that can be used, depending on the available data and the use case. Clustering algorithms, Naïve Bayes, Support Vector Machines are some examples
- ❖ Neural networks are considered AI algorithms and require large amounts of data and computational power. But they are very powerful
- ❖ Sometimes, fitting a statistical distribution may result in a good model for the scenario

Data Science Methodology :

Data science applications require a particular methodology and skills.

- ❖ **Exploratory Data Analysis (EDA):** These applications work by identifying underlying patterns in the data and codifying that into a computational model that represent the real-life scenario (or the domain). So it is critical for a data science practitioner to understand the domain and how the data represents the domain. He needs to understand the basic properties of the data (mean, variance, range, distribution etc.) correlation between the variable that he is trying to predict and the input variables, as well as the relationship between the input variables themselves. The practitioner needs to know how to handle extreme values in the data (called outliers), how to assess the quality of the data and what to do with missing values. Often, pattern detection may require slicing or aggregation of data by various dimensions. So the data science practitioner needs to be aware of techniques to programmatically explore the data and draw insights that are meaningful for modeling
- ❖ **Data Visualization:** Usually, the practitioner needs to deal with large volumes of data that is analyzable across several dimensions. Data visualization techniques are vital to be able to abstract data and detect patterns. The practitioner needs to be familiar with various kinds of plots and when they are to be used, and the ability to generate such plots programmatically from the data
- ❖ **Data Manipulation:** Data science often needs to merge multiple datasets to create a common dataset. The student needs to be able to summarize data at various levels and in general be very well versed with data merging, splitting and computing relevant measures
- ❖ **Feature Selection and Extraction:** A data science application aims to create a digital model by choosing the key characteristics of the real-life scenario. Since real-life objects have many characteristics, the data science practitioner needs to select those characteristics that are relevant to the prediction. For example, the color of the house door is a characteristic of the house but may not be relevant to the price of the house, whereas the square footage of the garden may be a factor that influences price. Such characteristics are called features of the machine learning model. The data science practitioner should use the insights gained from the EDA and visualization exercise to determine the correct features that have predictive power
- ❖ **Model Development:** The data science student needs to be able to determine the correct ML algorithm that can be used for a scenario. Sometimes, an empirical approach is needed to determine the best model for prediction. The student needs to be familiar with common algorithms and the merits and demerits of those algorithms
- ❖ **Model Performance Management:** once a model is built, its predictions need to be checked for accuracy. The practitioner needs to be aware of the various metrics of

prediction and use the appropriate metric for the situation. The student also needs to be able to boost performance by using additional features or other techniques.

Many of these techniques are available as software libraries in languages such as R programming and Python. The practitioner should be conversant with these languages and the libraries.

Conclusion : Data science is a powerful discipline that can deliver great value to enterprises. It can be applied to a variety of domains and there are specialized domain specific techniques available. But data science problems are open-ended and require experimentation and an active spirit of enquiry. Thus, a practitioner can benefit from a knowledge of these techniques but also should exhibit thorough analytical skills, a comfort with data manipulation and should also be creative in crafting the correct model for the situation.

1.2 STATISTICS

Def. : Statistics is a tool in the hands of mankind to translate complex facts into simple and understandable statement of facts.

The word statistics is derived from the Italian word stato and it means a political state. In the singular sense statistics is as defined as a science which deals with scientific methods of collection, organization, summarization, presentation, analysis and interpretation of numerical data. Statistical methods are applied for investigation in every important science.

Statistical Methods :

1. **Collection of Data :** The first step of an investigation is the collection of data. Careful collection is needed because further analysis is based on this.
2. **Organisation of Data :** The large mass of figures that are collected from a survey needs organisation.
3. **Presentation of Data :** The collected data must be edited very carefully so that irrelevant answers and wrong computations must be corrected or adjusted.

The collected data must be classified and tabulated before they can be analysed.

4. **Analysis of Data :** After presentation of the data the next step is to analyse the presented data. Analysis includes condensation, summarisation, conclusion etc., through means of Measures of Central Tendencies, Dispersion, Skewness, Kurtosis, correlation and Regression etc.
5. **Interpretation of Data :** Valid conclusions must be drawn on the basis of analysis. Correct interpretation leads to valid conclusion.

The statistical generalisation provides the estimates of the characteristic behaviour of population, but not of individual person.

The real purpose of statistical methods is to make sense out of facts and figures, prove the unknown and to cast light upon the situation. The statistical methods are employed as a tool for comparison between past and present results with a view to find out the reasons for changes. Statistics has become so much indispensable in all phases of human endeavour.

1.3 COLLECTION OF DATA

The basic problem of statistical enquiry is to collect facts and figures relating to a particular phenomenon under study. The investigator is a person who conducts the statistical enquiry. He is a trained and efficient statistician. The statistician counts and measures the characteristic under study for further statistical analysis. The respondents or informants are the persons from whom the information is collected. The statistical units are the items on which the measurement

is taken. Collection of data is the process of enumeration together with the proper recording of results. The success of an enquiry depends on the proper collection of data.

Primary and Secondary Data :

Statistical data may be classified as primary and secondary.

Primary Data : If an individual or an officer collects the data to study a particular problem, the data are the raw materials of the enquiry. They are the primary data collected by the investigator himself to study any particular problem.

Secondary Data are those which are already collected by someone for some purpose and are available for the present study. For example, the data collected during Census operations are primary data to the department of Census and the same data, if used by a research worker for some study are secondary data.

Sources of Secondary Data :

1. **Published sources :** Such as international publications, official publications of central and state governments, semi-official publications of semi-government institutions like municipal corporations, panchayats, etc., publications of research institutions, publications of commercial and financial institutions, reports of various committees, Journals and news papers.
2. **Unpublished Sources :** They are records maintained by various government and private offices, the research carried out by individual research scholars in the universities or research institutes.

Precautions in the use of Secondary Data :

Before using the secondary data, we must take into consideration

- a. **The Suitability of Data :** The investigator must satisfy himself that the data available is suitable for the purpose of enquiry.
- b. **Adequacy of Data :** If the data are suitable for the purpose of investigation, then we must consider whether the data is useful or adequate for the present analysis.
- c. **Reliability of Data :** The reliability of data can be tested by finding whether the collecting agents used proper methods or not. If the methods are proper, the data can be relied on.

Without knowing the meanings and limitations, we cannot accept the secondary data.

1.4 POPULATION VS. SAMPLE

In Statistical enquiry, all the items, which fall within the purview of enquiry are known as **Universe or Population**. That is population is a complete set of all possible observations of the type which is to be investigated. This is a statistical usage and the term population does not necessarily refer to people.

1. Finite and Infinite Population :

Population can be either finite population or infinite population. When the number of observations can be counted and definite, it is known as “finite population”. For example, when we are studying the economic background of students of a college, all the students of the college will constitute population and this number will be finite. When the number of observations cannot be counted and is infinite, it is known as infinite population. For example, the number of stars in the sky is infinite population.

2. Hypothetical and Existant Population :

Universes can be classified as extent and hypothetical. An Universe containing persons or objects is known as existant or real population. The examples are the students of a college, population of a city, the employees of a factory.

Hypothetical Universe which is also known as therotical population, is one which does not consist of concrete objects. This population exists only in imagination. For example, if we toss a coin infinite times, the result is a hypothetical population. Information on population can be collected in two ways. Census method and sample method.

a) Census Method : The object of census or complete enumeration is to collect information for each and every unit of the population.

In Census method every element of the population is included in the investigation. When we make a complete enumeration of all items in the population, it is known as census method of collection of data. For example, if we study the average expenditure of the students of University, which has 20,000 students, we must study the expenditure of all the 20,000 students. In the Census method complete enumeration is done.

This method requires a large number of enumeration and is a costly method. Then only government alone can use this method for conducting population Census, production Census etc.

b) Sample Method : In the case of sample enquiry, only a part of the whole group of population will be studied. We can study the charecteristics of a population from sampling. A study of the sample will give a correct idea of the Universe or population.

Merits :

1. It saves time, because fewer items are collected and processed.
2. When the results are urgently required, this method is very helpful.
3. It reduces the cost of the enumeration.
4. More reliable results can be obtained, since there are fewer chances of sampling errors.
5. Expert and trained people can be employed for scientific processing and analysis.

METHODS OF SAMPLING :

1. Random Sampling Method (Probability Sampling)

A random sample is one where each item in the universe has an equal chance of known opportunity.

2. Non-Random Sampling :

This can be done in three methods.

a) Judgement or Purposive sampling : The choice of the sample items depends on the judgement of the investigation.

b) Quota sampling : To collect data, the universe is divided into quota according to some characteristics.

a) Convenience sampling or check sampling : The sampling is obtained by selecting convenient population units.

1. It is suitable when the population is not clearly defined.
2. Sample is not clear.
3. Complete source list is not available.

A sample obtained from automobile registration, telephone directories etc. is a convenient sample. They are unsatisfactory. They are biased. But they are used for pilot studies.

1.5 TYPES OF VARIABLES

All experiments examine some kind of variable(s). A variable is not only something that we measure, but also something that we can manipulate and something we can control for. First, we illustrate the role of dependent and independent variables. Finally, we explain how variables can be characterised as either categorical or continuous.

Dependent and Independent Variables

An independent variable, sometimes called an experimental or predictor variable, is a variable that is being manipulated in an experiment in order to observe the effect on a dependent variable, sometimes called an outcome variable.

Imagine that a tutor asks 100 students to complete a maths test. The tutor wants to know why some students perform better than others. Whilst the tutor does not know the answer to this, she thinks that it might be because of two reasons: (1) some students spend more time revising for their test; and (2) some students are naturally more intelligent than others. As such, the tutor decides to investigate the effect of revision time and intelligence on the test performance of the 100 students. The dependent and independent variables for the study are:

Dependent Variable: Test Mark (measured from 0 to 100)

Independent Variables: Revision time (measured in hours) Intelligence (measured using IQ score)

The dependent variable is simply that, a variable that is dependent on an independent variable(s). For example, in our case the test mark that a student achieves is dependent on revision time and intelligence. Whilst revision time and intelligence (the independent variables) may (or may not) cause a change in the test mark (the dependent variable), the reverse is implausible; in other words, whilst the number of hours a student spends revising and the higher a student's IQ score may (or may not) change the test mark that a student achieves, a change in a student's test mark has no bearing on whether a student revises more or is more intelligent.

Categorical and Continuous Variables. Categorical variables are also known as discrete or qualitative

- ❖ Nominal variables are variables that have two or more categories, but which do not have an intrinsic order. For example, a real estate agent could classify their types of property into distinct categories such as houses, condos, co-ops or bungalows. So "type of property" is a nominal variable with 4 categories called houses, condos, co-ops and bungalows. Of note, the different categories of a nominal variable can also be referred to as groups or levels of the nominal variable. Another example of a nominal variable would be classifying where people live in the USA by state. In this case there will be many more levels of the nominal variable (50 in fact).
- ❖ Dichotomous variables are nominal variables which have only two categories or levels. For example, if we were looking at gender, we would most probably categorize somebody as either "male" or "female". This is an example of a dichotomous variable (and also a nominal variable). Another example might be if we asked a person if they owned a mobile phone. Here, we may categorise mobile phone ownership as either "Yes" or "No". In the real estate agent example, if type of property had been classified as either residential or commercial then "type of property" would be a dichotomous variable.

- ❖ Ordinal variables are variables that have two or more categories just like nominal variables only the categories can also be ordered or ranked. So if you asked someone if they liked the policies of the Democratic Party and they could answer either "Not very much", "They are OK" or "Yes, a lot" then you have an ordinal variable. Why? Because you have 3 categories, namely "Not very much", "They are OK" and "Yes, a lot" and you can rank them from the most positive (Yes, a lot), to the middle response (They are OK), to the least positive (Not very much). However, whilst we can rank the levels, we cannot place a "value" to them; we cannot say that "They are OK" is twice as positive as "Not very much"
- Continuous variables are also known as quantitative variables. Continuous variables can be further categorized as either interval or ratio variables.
- ❖ Interval variables are variables for which their central characteristic is that they can be measured along a continuum and they have a numerical value (for example, temperature measured in degrees Celsius or Fahrenheit). So the difference between 20°C and 30°C is the same as 30°C to 40°C. However, temperature measured in degrees Celsius or Fahrenheit is NOT a ratio variable.
 - ❖ Ratio variables are interval variables, but with the added condition that 0 (zero) of the measurement indicates that there is none of that variable. So, temperature measured in degrees Celsius or Fahrenheit is not a ratio variable because 0°C does not mean there is no temperature. However, temperature measured in Kelvin is a ratio variable as 0 Kelvin (often called absolute zero) indicates that there is no temperature whatsoever. Other examples of ratio variables include height, mass, distance and many more. The name "ratio" reflects the fact that you can use the ratio of measurements. So, for example, a distance of ten metres is twice the distance of 5 metres.

1.6 DATA VISUALISATION

Data visualization is the graphical representation of information and data. By using visual elements like charts, graphs and maps, data visualization tools provide an accessible way to see and understand trends, outliers, and patterns in data.

Data visualization is the act of taking information (data) and placing it into a visual context, such as a map or graph. Data visualizations make big and small data easier for the human brain to understand, and visualization also makes it easier to detect patterns, trends and outliers in groups of data.

5 types of Big data visualization categories.

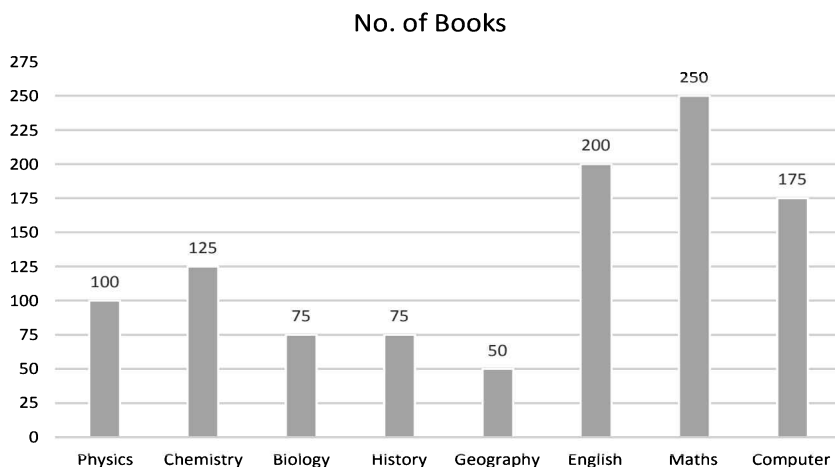
1. Bar Chart
2. Line chart
3. Scatter plot
4. Sparkline
5. Pie Chart

1. Bar Graph/Chart : A bar graph is a pictorial representation of the numerical data by a number of bars of uniform width with different heights, erected horizontally or vertically with equal spacing between them.

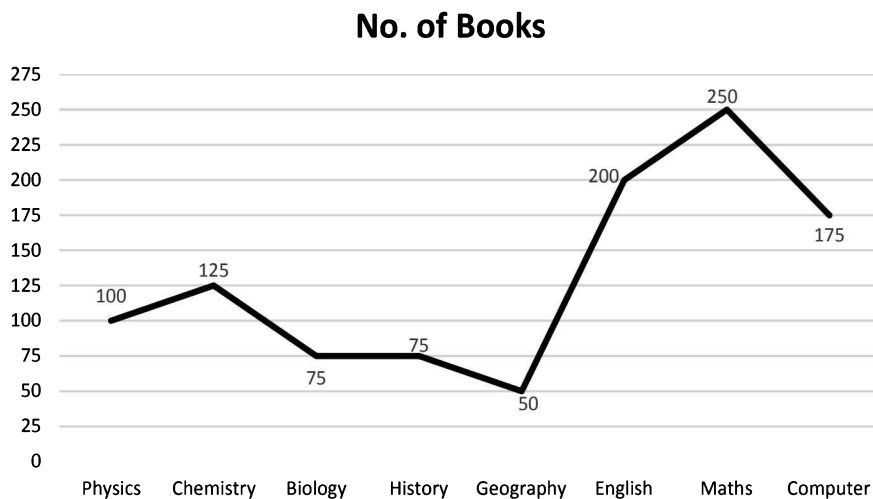
Ex. The following table shows the number of books of different subjects in a library.

Subject	Phy.	Chem.	Bio.	Hist.	Geography	Eng.	Maths	Comp.
No. of Books	100	125	75	75	50	200	250	175

Solution. Take the subjects along the X-axis and number of books along the Y- axis.
Construct the bars of same width, with same distance between them.
Take the scale as 25 books = 5 small divisions or $\frac{1}{2}$ cm.

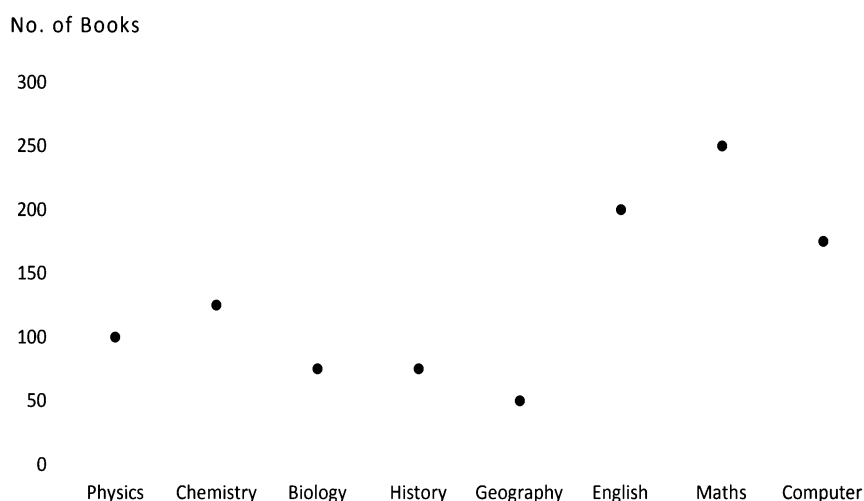


2. Line Graph/Chart : A line chart is a type of chart which displays information as a series of data points called markers connected by straight line segment.



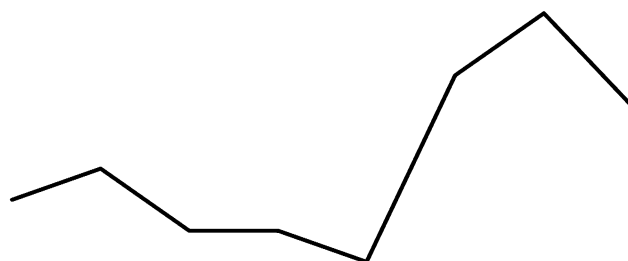
3. Scatter Graph/Chart : A scatter plot is a type of plot or mathematical diagram using cartesian coordinates to display values for two variables for a set of data. With scatter plots we often talk about how the variables relate to each other. This is called correlation.

A scatter plot is also called scatter graph, scatter chart or scatter diagram.



4. Sparkline : A sparkline is a very small line chart drawn without axes or coordinates. It present the general shape of the variation in some measurement, such as temperature or stock market price in a simple and highly condensed way.

Spark Line chart



5. Pie Chart/Graph : A pie chart is a way of showing how something is shared or divided. This pie chart shows how 36 pupils usually come to school :

The number of pupils is 36 and the whole group of 36 pupils is represented by the complete angle 360° . The angles at the centre are in proportion to each category. Thus, the angle of 120° at the centre corresponds to the walk group.

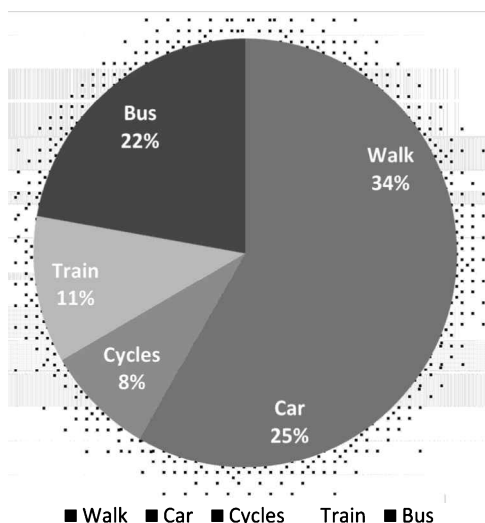
Since angle of 360° at the centre corresponds to the whole group of 36 pupils.

$\therefore 1^\circ$ corresponds to $\frac{36}{360}$ of the whole group.

$\therefore 120^\circ$ corresponds to $\frac{36}{360} \times 120$ i.e. 12 pupils.

So, 12 pupil walk to school.

Similarly, number of pupils coming by car $= \frac{36}{360} \times 90 = 9$



REVIEW QUESTIONS

1. Distinguish between primary data and secondary data
2. Discuss various methods of collecting primary data.
3. Explain the precautions to be taken in using secondary data.
4. Explain the importance and significance of sample.
5. Discuss about dependent variables.
6. Explain about independent variables.
7. Discuss the catagorical variables.
8. Explain about continuous variables.
9. Discuss about Pie diagram.
10. Explain bar chart with an example.

CHAPTER - 2

MEASURES OF CENTRAL TENDENCY AND VARIABILITY

2.1. INTRODUCTION.

The word statistics is derived from the Italian word *stato* and it means a political state. In the singular sense statistics is as defined as a science which deals with scientific methods of collection, organization, summarization, presentation, analysis and interpretation of numerical data. Statistical methods are applied for investigation in every important science.

2.2. MEASURES OF CENTRAL TENDENCY

The importance of statistical analysis is to find a number which represents in some definite way the entire data. Such a representative number is called the central value or an average. The value of an average lies somewhere in between the two extreme items possibly in the centre where most of the items concentrate. Hence an average constitutes a measure of the central tendency of the series. Measures of central tendency enable us to compare different groups of data.

The following are some important measures of central tendency in common use.

1. *Arithmetic Mean* (A.M.)
2. *Median*
3. *Mode*
4. *Geometric Mean* (G.M)
5. *Harmonic Mean* (H.M.)

An outline of formulae for the calculation of these measures of central tendency is given below.

2.3. ARITHMETIC MEAN

The arithmetic mean of a series of values of a variate is defined as the quantity obtained by dividing the sum of the value of variates by their number.

Arithmetic Mean of an Ungrouped Data

Let $x_1, x_2, \dots, x_n$ be the n values of a variate x . Then the arithmetic mean is given by

$$\text{A.M.} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$

Arithmetic Mean of a Grouped Data

Let $x_1, x_2, \dots, x_n$ be the midvalues of n classes of a frequency distribution with frequencies $f_1, f_2, \dots, f_n$ respectively.

Since the value x_1 occurs f_1 times, the x_2 occurs f_2 times ... the value x_n occurs f_n times, the arithmetic mean by definition is given by

$$\text{A.M.} = \frac{f_1x_1 + f_2x_2 + \dots + f_nx_n}{f_1 + f_2 + \dots + f_n} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i}$$

$$\text{A.M.} = \frac{1}{N} = \sum_{i=1}^n f_i x_i \quad \text{where } N = \sum_{i=1}^n f_i = \text{Total frequency.}$$

Arithmetic Mean of the Combined Data

The arithmetic means of two sets of data containing m, n items are $\bar{x}$ and $\bar{y}$ respectively. If the two sets of data are combined to get one set of $(m+n)$ items, then the arithmetic mean $\bar{z}$ of this set is given by $\bar{z} = \frac{m\bar{x} + n\bar{y}}{m+n}$.

2.4. MEDIAN

Median of a distribution is defined as the value of the variate which divides it into two equal parts when the variates are arranged in ascending or descending order of magnitude.

Median of an Ungrouped Data

Arrange the variates in ascending or descending order of magnitude. Determine the total number n of variates. If n is odd, then the median is the value of $\left(\frac{n+1}{2}\right)$ th variate. If n is even, the median is the average value of $\frac{n}{2}$ th and $\left(\frac{n}{2}+1\right)$ th variates.

$$\text{i.e., If } n \text{ is odd, Median} = \frac{\left(\frac{n}{2}\right)^{\text{th}} \text{ term} + \left(\frac{n}{2}+1\right)^{\text{th}} \text{ term}}{2}$$

Median of a Grouped Data

In a continuous frequency distribution find the value of $\frac{N}{2}$ where N is the total frequency. Calculate the less than cumulative frequencies of the classes.

The class corresponding to the cumulative frequency just greater than $\frac{N}{2}$ is called the median class. The value of the median is calculated by the following formula :

$$\text{Median} = l + \frac{\frac{N}{2} - m}{f} \times c$$

where l = Lower limit of the median class

m = cumulative frequency of the class preceding the median class

N = Total frequency

f = Frequency of the median class

c = Width of the median class.

2.5. MODE

Mode or modal value of the distribution is defined as that value of the variate for which frequency is maximum.

Mode of an Ungrouped Data

When individual items are given, mode is the value of the item which occurs most frequently. However it is better to convert these individual items into a frequency table to calculate mode.

Mode of a Grouped Data

In a grouped data the class interval having the maximum frequency is called the modal class. Even if the frequency distribution has class intervals of unequal magnitude the mode can be calculated provided the modal class and the classes preceding and succeeding it are of the same magnitude.

Some frequency distributions have two or more classes having the highest frequency. They are said to be *multimodal* distribution. The mode is not a good measure in such cases. Sometimes the midpoint of the class having the maximum frequency is taken as mode. In such cases it is called *crude* mode.

The mode of a grouped data is computed by the following formula:

$$\text{Mode} = l + \frac{f - f_1}{2f - f_1 - f_2} \times c$$

where l = Lower limit of the modal class

f = Frequency of the modal class

f_1 = Frequency of the class preceding the modal class

f_2 = Frequency of the class succeeding the modal class

c = Width of modal class.

2.6. GEOMETRIC MEAN

Geometric mean is defined as the n th root of the product of the values of a distribution none of them being zero.

Geometric Mean of an Ungrouped Data

Let $x_1, x_2, \dots, x_n$ be the n values of the variate x none of them being zero. Then the geometric mean G is defined by $G = \sqrt[n]{x_1 x_2 \dots x_n} = (x_1 x_2 \dots x_n)^{1/n}$

The calculations can be easily made by taking logarithms both sides.

$$\log G = \log(x_1 x_2 \dots x_n)^{1/n}$$

$$= \frac{1}{n} (\log x_1 + \log x_2 + \dots + \log x_n) = \frac{1}{n} \sum_{i=1}^n \log x_i$$

$$\therefore G = \text{Antilog} \left[\frac{1}{n} \sum_{i=1}^n \log x_i \right]$$

Geometric Mean of a Grouped Data

Let $x_1, x_2, \dots, x_n$ be the mid-values of n classes of a frequency distribution with frequencies $f_1, f_2, \dots, f_n$. Let $\sum_{i=1}^n f_i = N$. Then the geometric mean is defined as the N th root of the product of the mid-values of the classes raised to the powers of their respective frequencies.

$$\therefore \text{G.M.} = \sqrt[N]{x_1^{f_1} x_2^{f_2} \dots x_n^{f_n}} = (x_1^{f_1} x_2^{f_2} \dots x_n^{f_n})^{(1/N)}$$

Geometric Mean of the Combined Data

If m items have a geometric mean x and n items have a geometric mean y , then the geometric mean G of the combined data is given by

$$\log G = \frac{m \log x + n \log y}{m + n}$$

2.7. HARMONIC MEAN

The harmonic mean (H.M.) of a series of values is defined as the reciprocal of the arithmetic mean of the reciprocal of individual items of which no item is equal to zero.

Harmonic mean of an ungrouped data

Let $x_1, x_2, \dots, x_n$ be n terms of the data.

Their reciprocals are $\frac{1}{x_1}, \frac{1}{x_2}, \dots, \frac{1}{x_n}$.

$$\therefore \text{A.M. of these reciprocals} = \frac{1/x_1 + 1/x_2 + \dots + 1/x_n}{n} = \frac{1}{n} \sum_{i=1}^n \frac{1}{x_i}$$

$\therefore$ Harmonic Mean = Reciprocal of the A.M. of the reciprocals

$$\text{i.e., } H.M. = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}}$$

2.8 HARMONIC MEAN OF A GROUPED DATA

Let $x_1, x_2, \dots, x_n$ be the non-zero mid-values of n classes with frequencies $f_1, f_2, \dots, f_n$.

Let $\sum_{i=1}^n f_i = N$.

Then $H.M.$ is calculated by the following formula.

$$H.M. = \frac{\frac{f_1}{x_1} + \frac{f_2}{x_2} + \dots + \frac{f_n}{x_n}}{\sum_{i=1}^n \frac{f_i}{x_i}} = \frac{N}{\sum_{i=1}^n \frac{f_i}{x_i}}$$

SOLVED EXAMPLES

Example 1 : Calculate the A.M. of the following data

Roll No.s	1	2	3	4	5	6	7	8	9	10
Marks (X)	40	50	55	78	58	60	73	35	43	48

Solution :

Roll No.s	Marks (X)
1	40
2	50
3	55
4	78
5	58
6	60
7	73
8	35
9	43
10	48
$n = 10$	$\sum X = 540$

$$\text{Arithmetic Mean} = \bar{X} = \frac{\sum X}{n} = \frac{540}{10} = 54$$

Example 2 : From the following, find the mean profits

Profits per shop (Rs)	120 – 200	200 – 300	300 – 400	400 – 500	500 – 600	600 – 700	700 – 800
Number of Shops	10	18	20	26	30	28	18

Solution :

Calculation of Mean

Profit per shop	Mid value	$d = m - 450$	f	fd
100-200	150	– 300	10	–3000
200-300	250	–200	18	–3600
300-400	350	–100	20	–2000
400-500	450	0	26	0
500-600	550	100	30	3000
600-700	650	200	28	5600
700-800	750	300	18	5400
			$\sum f = 150$	$\sum fd = 5400$

$$\text{Mean} = \bar{X} = A + \frac{\sum fd}{N}$$

$$= 450 + \frac{5400}{150} = 450 + 36 = 486$$

$\therefore$ Average profit is Rs. 486

Example 2 : Find the median from the following : 57, 58, 61, 42, 38, 65, 72, 66

Solution : Arranging in ascending order, we get

38, 42, 57, 58, 61, 65, 66, 72. Here $n = 8$ (even)

$$\therefore \text{Median} = \frac{\left(\frac{n}{2}\right)^{\text{th}} \text{ term} + \left(\frac{n}{2} + 1\right)^{\text{th}} \text{ term}}{2} = \frac{4^{\text{th}} \text{ term} + 5^{\text{th}} \text{ term}}{2} = \frac{58 + 61}{2} = \frac{119}{2} = 59.5$$

Example 2 : Calculate the median from the following data

Marks	10–25	25–40	40–55	55–70	70–85	85–100
Frequency	6	20	44	26	3	1

Solution : We will tabulate the values as follows :

Computation of the Median

Marks	Frequency	Cumulative Frequency
10-25	6	6
25-40	20	26
40-55	44	70
55-70	26	96
70-85	3	99
85-100	1	100

Here $\frac{N}{2} = \frac{100}{2} = 50$. Here the class just greater than $\frac{N}{2}$ is 40–55.

$\therefore$ Median class is 40–55.

$$\text{Median} = l + \frac{\frac{N}{2} - m}{f} \times c$$

where l = Lower end of the Median class = 40

$$\frac{N}{2} = \frac{100}{2} = 50$$

m = cumulative frequency of the class preceding the median class = 26

c = length of the class interval = 15

$$\therefore \text{Median} = 40 + \frac{50 - 26}{44} \times 15 = 40 + 8.18 = 48.18$$

Example 2 : Find the mode of the following salaries :

(i) 850, 750, 600, 825, 850, 725, 600, 850, 640, 530

(ii) 40, 45, 48, 57, 78

(iii) 45, 55, 50, 45, 40, 55, 45, 55

Solution : (i) 85 is repeated 3 times. Therefore, modal salary = Rs. 850

(ii) No value is repeated. Thus there is no mode.

(iii) 45 is repeated 3 times. 55 is also repeated 3 times. Therefore there are two modes.

Mode (i) = 45, mode (ii) = 55

Note :

- (i) When we calculate mode from the data, if there is only one mode, then the series is called unimodal.
- (ii) If there are two modes it is called bimodal
- (iii) If there are 3 modes it is called trimodal.
- (iv) If there are more than 3 modes it is called multimodal.

Example 2 : Find the mode of the following distribution.

Class interval	0–10	10–20	20–30	30–40	40–50	50–60	60–70	7–80
Frequency	5	8	7	12	28	20	10	10

Solution : Here the maximum frequency is 28.

Thus 40-50 is the modal class

We use the formula, $\text{Mode} = l + \frac{(f - f_1)}{2f - f_1 - f_2} \times c$

where, l = lower end of the modal class = 40

c = length of the class interval = 10

f = frequency of the modal class = 28

f_1 = frequency of the class preceding modal class = 12

f_2 = frequency of the class succeeding the modal class = 20

$$\therefore \text{Mode} = 40 + \frac{10(28-12)}{2 \times 28 - 12 - 20} = 40 + 6.666 = 46.666$$

Example 2 : Find the geometric mean of the following data :

Yield of wheat (kg)	7.5–10.5	10.5–13.5	13.5–16.5	16.5–19.5	19.5–22.5	22.5–25.5	25.5–28.5
No. of farms	5	9	19	23	7	4	1

Solution : The G.M. is calculated using the formula

G. M. = $\text{Antilog} \left(\frac{\sum f \log m}{N} \right)$ where m is the mid-value of the class interval.

We tabulate the values as follows :

Mid value (m)	$\log m$	f	$f \log m$
9	0.9542	5	4.7710
12	1.0792	9	9.7128
15	1.1761	19	22.3459
18	1.2553	23	28.8719
21	1.3222	7	9.2554
24	1.3802	4	8.5208
27	1.4314	1	1.4314
		$N = 68$	$\sum f \log m = 81.9092$

$$\therefore \text{G. M.} = \text{Antilog} \left(\frac{\sum f \log m}{N} \right) = \text{Antilog} \left(\frac{81.9092}{68} \right)$$

$$= \text{Antilog} (1.2045) = 16.02 \text{ kgs.}$$

Example 8 : Calculate the H. M. of the following data

Size of the items	6	7	8	9	10	11
Frequency	4	6	9	5	2	8

Solution : We tabulate the values as follows :

Size of item (X)	Frequency	$\frac{1}{X}$	$f \left(\frac{1}{X} \right)$
6	4	0.1667	0.6668
7	6	0.1429	0.8574
8	9	0.1250	1.1250
9	5	0.1111	0.5555
10	2	0.1000	0.2000
11	8	0.0909	0.7292
	$N = \sum f_i = 34$		$\sum \left(\frac{f}{X} \right) = 4.13$

$$\therefore \text{H. M.} = \frac{N}{\sum \frac{f_i}{x_i}} = \frac{34}{4.1319} = 8.23$$

Example 9 : Calculate the H.M. of the following data

Marks	30-40	40-50	50-60	60-70	70-80	80-90	90-100
Freq.	15	13	8	6	15	7	6

Solution : Calculation of H. M.

Marks	Mid value (m)	Freq.	$\frac{1}{m}$	$f \left(\frac{1}{m} \right)$
30-40	35	15	0.02857	0.42855
40-50	45	13	0.02222	0.28886
50-60	55	8	0.01818	0.14544
60-70	65	6	0.01534	0.09204
70-80	75	15	0.01333	0.19995
80-90	85	7	0.01176	0.08232
90-100	95	6	0.01053	0.06318
		$N = \sum f_i = 70$		$\sum \left(\frac{f}{m} \right) = 1.30034$

$$\text{H. M.} = \frac{N}{\sum \left(\frac{f_i}{m_i} \right)} = \frac{70}{1.30034} = 53.83$$

2.9. MEASURES OF DISPERSION

Introduction. If the items within a distribution differ from one another in magnitude the term *dispersion* or *scatteredness* is used to indicate the difference. The distributions differ from one another in respect of two main characteristics.

1. They may differ in measures of central tendency.
2. They may have the same measure of central tendency but have wide disparities in the formation of distributions.

Consider the two series 3, 4, 5, 6, 7 and 12, 13, 14, 15, 16. The arithmetic means of the two series are 5 and 14. Although the means are different the items in the two series are scattered in the same way around the means. Next consider two other series 5, 8, 10, 4, 3 and 6, 15, 0, 7, 2. These two series have the same arithmetic mean 6 but the scatteredness of the various items in the two series about their mean is different. From the above two examples we infer that the average fails to give us an idea how the various items are scattered and how the distributions are constituted. It is therefore necessary to have measures of scatteredness or dispersion in order to study the distribution fully. Measures of dispersion enable us to compare the variability of two or more frequency distributions.

The measures of dispersion in common use are:

1. Range
2. Mean deviation
3. Standard deviation.

2.10 RANGE

Range for an ungrouped data is defined as the difference between the greatest and the least values of the variate.

For a grouped data range is defined as the difference between the upper limit of the largest class and lower limit of the smallest class.

Ex. 1. Find the range of marks of students in a class given as

60, 72, 96, 28, 35, 10, 40, 9, 85, 25.

Range = Largest value – Smallest value = $96 - 9 = 87$.

Ex. 2 : The following table gives the daily sales (Rs.) of Two firms A and B for five days.

Firm A	Firm B
5050	4900
5025	3100
4950	2200
4835	1800
5140	13000
$\bar{X}_A = 5000$	$\bar{X}_B = 5000$

The sales of both the firms in average is same but distribution pattern is not similar. There is a great amount of variation in the daily sales of the firm B than that of the firm A.

Range of sales of firm A = Greatest value – Smallest value = $5140 - 4835 = 305$

Range of sales of firm B = Greatest value – Smallest value = $13000 - 1800 = 11200$

2.11 MEAN DEVIATION

Mean deviation is defined as arithmetic average of absolute values of the deviations of the variates measured from an average (median, mode or mean).

The absolute value of the deviation denoted by |deviation| is the numerical value of the deviation with positive sign.

Note . Mean deviation can be similarly calculated by taking deviations from the median or mode.

2.12. MEAN DEVIATION FROM MEAN OF AN UNGROUPED DATA

Let $x_1, x_2, x_3, \dots, x_n$ be the values of n variates and $\bar{x}$ be their arithmetic mean. Let $|x_i - \bar{x}|$ be the absolute value of the deviation of the variate x_i from $\bar{x}$.

$$\therefore \text{Mean deviation} = \frac{\sum_{i=1}^n |x_i - \bar{x}|}{n}$$

SOLVED EXAMPLES

Example 1 : Calculate the mean deviation of the variates 40, 62, 54, 68, 76 from A.M.

Solution : A.M. $= \bar{x} = \frac{40 + 62 + 54 + 68 + 76}{5} = \frac{300}{5} = 60$

$$\begin{aligned} \therefore \text{Mean deviation} &= \frac{\sum_{i=1}^n |x_i - \bar{x}|}{n} \\ &= \frac{|40 - 60| + |62 - 60| + |54 - 60| + |68 - 60| + |76 - 60|}{5} = \frac{52}{5} = 10.4. \end{aligned}$$

Example 1 : Find the mean deviation from the mean for the following data : 38, 70, 48, 40, 42, 55, 63, 46, 54, 44.

Solution : Mean $= \bar{x} = \frac{38 + 70 + 48 + 40 + 42 + 55 + 63 + 46 + 54 + 44}{10} = \frac{500}{10} = 50.$

The deviations of the given observations from $\bar{x}$:

x_i	38	70	48	40	42	55	63	46	54	44
$x_i - \bar{x}$	38 - 50	70 - 50	48 - 50	40 - 50	42 - 50	55 - 50	63 - 50	46 - 50	54 - 50	44 - 50

$$\therefore \text{Mean deviation from the mean} = \frac{\sum_{i=1}^{10} |x_i - \bar{x}|}{10} = \frac{84}{10} = 8.4.$$

Example 2 : Find the main deviation about the (a) mean (b) median for the following distribute data 6, 7, 10, 12, 13, 4, 12, 16.

Solution : The arithmetic mean of the given data is

$$(a) \quad \bar{x} = \frac{6+7+10+12+13+4+12+16}{8} = \frac{80}{8} = 10$$

The absolute values of deviation from A. M. are

$$(6-10), (7-10), (10-10), (12-10), (13-10), (14-10), (12-10), (16-10)$$

i.e., 4, 3, 0, 2, 3, 6, 2, 6

$$\begin{aligned} \therefore \text{Mean deviation from mean} &= \frac{\sum |x_i - \bar{x}|}{n} \\ &= \frac{4+3+0+2+3+6+2+6}{8} = \frac{26}{8} = 3.25 \end{aligned}$$

- (b) Writing the data in ascending order magnitude, we get
4, 6, 7, 10, 12, 12, 13, 16.

$$\text{The median } b \text{ of these observations is } = \frac{10+12}{2} = \frac{22}{2} = 11$$

The absolute values of the deviations from the median are

$$|4-11|, |6-11|, |7-11|, |10-11|, |12-11|, |12-11|, |13-11|, |16-11|$$

i.e., 7, 5, 4, 1, 11, 2, 5

$$\therefore \text{Mean deviation from median} = \frac{\sum |x_i - b|}{n} = \frac{26}{8} = 3.25$$

Example 4 : Find the mean deviation from the median for the data 34, 66, 30, 38, 44, 50, 40, 60, 42, 51.

Solution : Arranging the data in ascending order, we have :

0, 34, 38, 40, 42, 44, 50, 51, 60, 66. ($n = 10$ terms)

$$\text{Now Median} = \frac{\left(\frac{n}{2}\right)^{\text{th}} \text{ term} + \left(\frac{n}{2} + 1\right)^{\text{th}} \text{ term}}{2} = \frac{42 + 44}{2} = 43.$$

x_i	30	34	38	40	42	44	50	51	60	66
$x_i - \text{Med}$	30-43	34-43	38-43	40-43	42-43	55-43	63-43	46-43	54-43	44-43

$$\therefore \text{Mean deviation from the median} = \frac{\sum |x_i - \text{Median}|}{n} = \frac{\sum |x_i - 43|}{n} = \frac{87}{10} = 8.7.$$

2.13 MEAN DEVIATION FOR A GROUPED DATA

We know that data can be arranged as a frequency distribution in two ways

- Discrete Frequency Distribution and
- Continuous Frequency Distribution.

Mean Deviation about Mean of a Grouped Data with Discrete Frequency Distribution.

Let $x_1, x_2, \dots, x_n$ be the midvalues of n class intervals with frequencies $f_1, f_2, \dots, f_n$ of a frequency distribution. Let $\bar{x}$ be the arithmetic mean of the distribution. Let $|x_i - \bar{x}|$ be the absolute value of the deviation of the midvalue x_i from the arithmetic mean $\bar{x}$.

Then the mean deviation about the arithmetic mean

$$= \frac{|x_1 - \bar{x}|f_1 + |x_2 - \bar{x}|f_2 + \dots + |x_n - \bar{x}|f_n}{f_1 + f_2 + \dots + f_n}$$

$$= \frac{\sum_{i=1}^n f_i |x_i - \bar{x}|}{\sum_{i=1}^n f_i} = \frac{1}{n} \sum_{i=1}^n f_i |x_i - \bar{x}| \text{ where } \sum_{i=1}^n f_i = N$$

SOLVED EXAMPLES

Example 1 : Find the mean deviation about the mean for the following data

x_i	2	5	7	8	10	35
f_i	6	8	10	6	8	2

Solution : We will tabulate the values as follows :

x_i	f_i	$f_i x_i$	$ x_i - \bar{x} $	$f_i x_i - \bar{x} $
2	6	12	6	36
5	8	40	3	24
7	10	70	1	10
8	6	48	0	0
10	8	80	2	16
35	2	10	27	54
	$\Sigma f_i = N$ = 40	$\Sigma f_i x_i =$ 320		140

$$\text{Thus A. M.} = \bar{x} = \frac{\Sigma f_i x_i}{\Sigma f} = \frac{320}{40} = 8$$

$$\therefore \text{Mean deviation} = \frac{\Sigma f_i |x_i - \bar{x}|}{\Sigma f_i} = \frac{140}{40} = 3.5$$

Example 1 : Find the mean deviation from the median for the following data

x_i	6	9	3	12	15	13	21	22
f_i	4	5	3	2	5	4	4	3

Solution : We write observations in ascending order to get the table as follows :

x_i	3	6	9	12	13	15	21	22
f_i	3	4	5	2	4	5	4	3

$$\text{Here } N = \Sigma f_i = 30$$

$\therefore$ Median is the mean of 15th and 16th observations which is equal to 13.

Now we tabulate the absolute values of the deviations.

$ x_i - med $ $= x_i - 13 $	10	7	4	1	0	2	8	9
f_i	3	4	5	2	4	5	4	3
$f_i x_i - med $	30	28	20	2	0	10	32	27

Thus $\sum f_i |x_i - med| = 149$

$$\therefore \text{Mean deviation from median} = \frac{\sum f_i |x_i - med|}{\sum f_i} = \frac{149}{30} = 4.97$$

Example 3 : Calculate the mean deviation from median from the following data

Size of item	6	7	8	9	10	11	12
freq.	3	6	9	13	8	5	4

Solution : We tabulate the values as follows

size	freq.	cumulative freq.	deviations from Median $ x_i - med $	$f_i x_i - med $
6	3	3	3	9
7	6	9	2	12
8	9	18	1	9
9	13	31	0	0
10	8	39	1	8
11	5	44	2	10
12	4	48	3	12

Here no. of values, $n = 7$ (odd). $\therefore \text{Median} = \frac{\left(\frac{n}{2}\right)^{\text{th}} \text{ term} + 1}{2} = \frac{7+1}{2} = 4^{\text{th}} \text{ term} = 9$.

$$\text{Mean deviation from median} = \frac{\sum f_i |x_i - med|}{\sum f_i} = \frac{60}{48} = 1.25$$

Example 4 : Find the mean deviation from the mean for the following data

x_i	5	10	15	20	25
f_i	7	4	6	3	5

Solution : Calculations of Mean Deviation about Mean

x_i	f_i	$f_i x_i$	$ x_i - \bar{x} $	$f_i x_i - \bar{x} $
5	7	35	$ 5 - 14 = 9$	63
10	4	40	$ 10 - 14 = 4$	16
15	6	90	$ 15 - 14 = 1$	6
20	3	60	$ 20 - 14 = 6$	18
25	5	125	$ 25 - 14 = 11$	55
	$\sum f_i = 25$	$\sum f_i x_i = 350$		158

$$\text{Mean, } \bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{350}{25} = 14$$

$$\text{Mean deviation from mean} = \frac{\sum f_i |x_i - \bar{x}|}{\sum f_i} = \frac{158}{25} = 6.32$$

Mean Deviation From the Mean for a Continuous Frequency Distribution

A continuous frequency distribution is a series in which the data is classified into different class intervals along with respective frequency. We calculate the A.M. of a continuous frequency distribute, we take x_i as the mid value of the class interval.

SOLVED EXAMPLES

Example 1 : The following table gives the sales of 100 companies. Find the mean deviation from the mean.

Sales in thousands	40 – 50	50 – 60	60 – 70	70 – 80	80 – 90	90 – 100
Number of companies	5	15	25	30	20	5

Solution : We shall construct the following table for the given data.

Sales	Number of companies f_i	Midpoint of the class x_i	$f_i x_i$	$ x_i - \bar{x} $	$f_i x_i - \bar{x} $
40 – 50	5	45	225	26	130
50 – 60	15	55	825	16	240
60 – 70	25	65	1625	6	150
70 – 80	30	75	2250	4	120
80 – 90	20	85	1700	14	280
90 – 100	5	95	475	24	120
	$\sum f_i = N$ = 100		$\sum f_i x_i$ = 7100		$\sum f_i x_i - \bar{x} $ = 1040

$$\text{Now } \bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{7100}{100} = 71$$

$$\therefore \text{Mean Deviation from mean} = \frac{\sum f_i |x_i - \bar{x}|}{\sum f_i} = \frac{1040}{100} = 10.4$$

Example 3 : Find the mean deviation of the following frequency distribution.

Class interval	0 – 4	4 – 8	8 – 12	12 – 16	16 – 20	20 – 24
Freq.	8	12	35	25	13	7

Class Interval	Mid value x_i	Freq. f_i	$f_i x_i$	$ x_i - \bar{x} $ $= x_i - 11.76 $	$f_i x_i - \bar{x} $
0–4	2	8	16	9.76	78.08
4–8	6	12	72	5.76	69.12
8–12	10	35	350	1.76	61.60
12–16	14	25	350	2.24	56.00
16–20	18	13	234	6.24	81.12
20–24	22	7	154	10.24	71.68
		$\sum f_i = N$ $= 100$	$\sum f_i x_i$ $= 1176$		$\sum f_i x_i - \bar{x} $ $= 417.60$

$$\text{Here } \bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{1176}{100} = 11.76$$

$$\therefore \text{Mean deviation} = \frac{1}{N} \cdot \sum f_i |x_i - \bar{x}| = \frac{417.60}{100} = 4.176$$

Step Deviation Method (Short Cut method)

Suppose in the given data the midpoints of the class intervals x_i and their associated frequencies are numerically large. Then the computations become tedious. To avoid large calculations, we take an assumed mean a which lies in the middle or close to it in the data and take the deviations of the mid points x_i from this assumed mean. This is equal to shifting the origin from 0 to assumed mean on the number line.

Again, if there is a common factor of all the deviations, we divide them by their common factor (h) to further simplify the deviations. These are known as **Step Deviations**. This amounts to change of scale on the number line.

With the assumed mean a and a common factor h we define a new variable, $d_i = \frac{x_i - a}{h}$.

$$\text{Then A. M.} = \bar{x} = \left(\frac{\sum f_i d_i}{N} \right) h$$

We illustrate the simplified procedure with some examples.

SOLVED EXAMPLES

Example 1 : Find the mean deviation from the mean for the following data.

Classes	0–100	100–200	200–300	300–400	400–500	500–600	600–700	700–800
Freq.	4	8	9	10	7	5	4	3

Solution : We tabulate the data as follows :

classes	Mid values (x_i)	d_i	frequencies (f_i)	$f_i d_i$	$ x_i - \bar{x} $	$f_i x_i - \bar{x} $
0–100	50	–3	4	–12	308	1232
100–200	150	–2	8	–16	208	1664
200–300	250	–1	9	–9	108	972
300–400	350	0	10	0	8	80
400–500	450	1	7	7	92	644
500–600	550	2	5	10	192	960
600–700	650	3	4	12	290	1168
700–800	750	4	3	12	392	1176
			50	4		7896

$$d_i = \frac{x_i - \text{assumed mean}}{\text{class size}} = \frac{x_i - 350}{100}$$

$$\text{Now } \bar{x} = a + \frac{\sum f_i d_i}{\sum f_i} \times \text{class size} = 350 + \frac{4}{50} \times 100 = 358$$

$$\therefore \text{Mean deviation from mean} = \frac{\sum f_i |x_i - \bar{x}|}{\sum f_i} = \frac{7896}{50} = 157.92$$

Example 2 : Find the mean deviation about the mean finite following data

Marks obtained	0–10	10–20	20–30	30–40	40–50
No. of students	5	8	15	16	6

Solution : We form the following table

classes	Mid values (x_i)	freq. (f_i)	d_i	$f_i d_i$	$ x_i - \bar{x} $	$f_i x_i - \bar{x} $
0–10	5	5	–2	–10	22	110
10–20	15	8	–1	–8	12	96
20–30	25	15	0	0	2	30
30–40	35	16	1	16	8	128
40–50	40	5	2	12	18	108
		$\sum f_i = N$ = 50		$\sum f_i d_i$ = 10		$\sum f_i x_i - \bar{x} $ = 472

$$\text{Here } \bar{x} = a + \frac{\sum f_i d_i}{N} \times c = 25 + \frac{10}{50} \times 10 = 27 \text{ and } d_i = \frac{x_i - \bar{x}}{h} = \frac{x_i - 27}{10}$$

$$\therefore \text{Mean deviation from mean} = \frac{\sum f_i |x_i - \bar{x}|}{N} = \frac{472}{50} = 9.44$$

Example 3 : Find Mean deviation from mean of the following data, using the step deviation method.

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70
No. of students	6	5	8	15	7	6	3

Solution : We will construct the following table to calculate the required values

classes	Mid point x_i	No. of students (f_i)	d_i $= \frac{x_i - 35}{10}$	$f_i d_i$	$ x_i - \bar{x} $	$f_i x_i - \bar{x} $
0-10	5	6	-3	-18	28.4	170.4
10-20	15	5	-2	-10	18.4	92
20-30	25	8	-1	-8	8.4	67.2
30-40	35	15	0	0	1.6	24.0
40-50	45	7	1	7	11.6	81.2
50-60	55	6	2	12	21.6	129.6
60-70	65	3	3	9	31.6	94.8

$$\text{Mean } (\bar{x}) = A + h \frac{\sum f_i d_i}{N} = 35 + \frac{10(-8)}{50} = 33.4$$

$$\text{Mean deviation from mean} = \frac{1}{N} \sum f_i |x_i - \bar{x}| = \frac{659.2}{50} = 13.18$$

Example 4 : Find Mean deviation from median for the following data

Age of workers	20-25	25-30	30-35	35-40	40-45	45-50	50-55	55-60
No. of workers	120	125	175	160	150	140	100	30

Solution : We form the following table for the given data

class interval	Midpoint x_i	Freq. f_i	Cumulative freq. (c.f.)	$ x_i - \bar{x} $	$f_i x_i - \bar{x} $
20-25	22.5	120	120	$ 22.5 - 37.5 = 15$	1800
25-30	27.5	125	245	$ 27.5 - 37.5 = 10$	1250
30-35	32.5	175	420	$ 32.5 - 37.5 = 5$	875
35-40	37.5	160	580	$ 37.5 - 37.5 = 0$	0
40-45	42.5	150	730	$ 42.5 - 37.5 = 5$	750
45-50	47.5	140	870	$ 47.5 - 37.5 = 10$	1400
50-55	52.5	100	970	$ 52.5 - 37.5 = 15$	1500
55-60	57.5	30	1000	$ 57.5 - 37.5 = 20$	600
		$N = 1000$			8175

Here $\frac{N}{2} = \frac{1000}{2} = 500$. The c.f. just greater than $\frac{N}{2}$ is 580.

The corresponding class interval is 35–40. This is the median class.

$$\therefore \text{Median} = l + \left(\frac{\frac{N}{2} - m}{f} \right) \times c = 35 + \left(\frac{500 - 420}{160} \right) \times 5 = 35 + \frac{400}{160} = 35 + 2.5 = 37.5$$

$$\text{and Mean deviation about the Median} = \frac{\sum f_i |x_i - \text{med.}|}{\sum f_i} = \frac{8175}{1000} = 8.175$$

Example 5 : Calculate the mean deviation from the median of the following data

Wages / week (Rs.)	10–20	20–30	30–40	40–50	50–60	60–70	70–80
No. of workers	4	6	10	20	10	6	4

Solution :

wages / week (in Rs.)	Mid values(x_i)	Freq. f_i	Cumulative freq. $c.f$	$ x_i - 45 $	$f_i x_i - 45 $
10–20	15	4	4	30	120
20–30	25	6	10	20	120
30–40	35	10	20	10	100
40–50	45	20	40	0	0
50–60	55	10	50	10	100
60–70	65	6	56	20	120
70–80	75	4	60	30	120
		$N = \sum f_i = 60$			$\sum f_i x_i - 45 = 680$

Here $N = 60$; $\frac{N}{2} = 30$

The cumulative frequencies just greater than $\frac{N}{2} = 30$ is 40 and the corresponding class is 40–50. So 40–50 is the median class.

Here $l = 40$, $f = 20$, $h = 10$, P.C.F = 20, $N = 60$

$$\therefore \text{Median} = l + \frac{\frac{N}{2} - m}{f} \times c = 40 + \frac{30 - 20}{20} \times 10 = 45$$

$$\text{and Mean deviation from Median} = \frac{680}{60} = 11.33$$

2.14. VARIANCE AND STANDARD DEVIATION OF UNGROUPED /GROUPED DATA

Earlier in calculating the mean deviation about mean or median, we were taking the absolute values of the deviations, so that they may not cancel among themselves. Again we adopt another method to avoid the difficulty that arise due to the signs of the deviations. We consider the squares of the deviations to make them non-negative. Thus if $x_1, x_2, \dots, x_n$ are n observations and $\bar{x}$ is their mean, we consider $\sum(x_i - \bar{x})^2$.

The following cases may arise :

Case (i) : If $\sum(x_i - \bar{x})^2 = 0$, then each $(x_i - \bar{x}) = 0$ which implies that all observations are equal to the mean $\bar{x}$ and there is no dispersion.

Case (ii) : If $\sum(x_i - \bar{x})^2$ is small, then it shows that each observation x_i is very close to the mean $\bar{x}$ and hence the degree of dispersion is very low.

Case (iii) : If $\sum(x_i - \bar{x})^2$ is large, then it indicates that a higher degree of dispersion of observations from the mean $\bar{x}$.

If we take the mean of the squared deviations from the mean i.e., $\frac{1}{n} \sum(x_i - \bar{x})^2$, then it is found that this number leads to a proper measure of dispersion. The number is called variance and is denoted by σ^2 . Then σ the standard deviation is given by the positive square root of variance.

$$\text{Variance} = \sigma^2 = \frac{1}{n} \sum(x_i - \bar{x})^2$$

$$\text{Standard deviation} = \sigma = \sqrt{\frac{1}{n} \sum(x_i - \bar{x})^2}$$

SOLVED EXAMPLES

Example 1 : Find the variance and standard deviation of the following data :
5, 12, 3, 18, 6, 8, 2, 10

Solution : We construct the following table to calculate variance and standard deviation.

x_i	5	12	3	18	6	8	2	10
$x_i - \bar{x}$	-3	4	-5	10	-2	0	-6	2
$(x_i - \bar{x})^2$	9	16	25	100	4	0	36	4

Here $\sum(x_i - \bar{x})^2 = 194$

$$\text{Variance } (\sigma^2) = \frac{\sum(x_i - \bar{x})^2}{n} = \frac{194}{8} = 24.25$$

$$\text{Standard deviation} = \sigma = \sqrt{24.25} = 4.95 \text{ (approx.)}$$

Example 4 : Find the variance and standard deviation for the following data :
45, 60, 62, 60, 50, 65, 58, 68, 44, 48

Solution : Mean $\bar{x} = \frac{45+60+62+60+50+65+58+68+44+48}{10} = \frac{560}{10} = 56$

x_i	45	60	62	60	50	65	58	68	44	48
$x_i - \bar{x}$	-11	4	6	4	-6	9	2	12	-12	-8
$(x_i - \bar{x})^2$	121	16	36	16	36	81	4	144	144	64

$$\therefore \text{Variance, } \sigma^2 = \frac{\sum(x_i - \bar{x})^2}{n} = \frac{662}{10} = 66.2$$

$$\text{Standard deviation } (\sigma) = \sqrt{66.2} = 8.136$$

Variance and Standard Deviation for a Discrete Frequency Distribution.

Example 3 : Calculate the standard deviation for the following distribution

x_i	4	8	11	17	20	24	32
f_i	3	5	9	5	4	3	1

Solution : We construct following table for computing the required values.

x_i	f_i	$f_i x_i$	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$f_i x_i - \bar{x} ^2$
4	3	12	-10	100	300
8	5	40	-6	36	180
11	9	99	-3	9	81
17	5	85	3	9	45
20	4	80	6	36	144
24	3	72	10	100	300
32	1	32	18	324	324
	$\sum f_i = N = 30$	$\sum f_i x_i = 420$			$\sum f_i (x_i - \bar{x})^2 = 1374$

$$\text{Here } N = 30, \bar{x} = \text{AM} = \frac{\sum f_i x_i}{\sum f_i} = \frac{420}{30} = 14$$

$$\text{Variance } (\sigma^2) = \frac{\sum f_i (x_i - \bar{x})^2}{\sum f_i} = \frac{1374}{30} = 45.8$$

$$\text{Standard deviation } (\sigma) = \sqrt{45.8} = 6.77$$

Example 4 : Find the mean standard deviation of the following frequency distribution.

x_i	6	10	14	18	24	28	30
f_i	2	4	7	12	8	4	3

Solution : We have

x_i	f_i	$f_i x_i$	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$f_i x_i - \bar{x} ^2$
6	2	12	-13	169	338
10	4	40	-9	81	324
14	7	95	-5	25	175
18	12	216	-1	1	12
24	8	192	5	25	200
28	4	112	9	81	324
30	3	90	11	121	363
	$\Sigma f_i = N$ = 40	$\Sigma f_i x_i$ = 760			$\Sigma f_i (x_i - \bar{x})^2$ = 1736

Here $N = \Sigma f_i = 40$, $\bar{x} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{760}{40} = 19$ and $\Sigma f_i (x_i - \bar{x})^2 = 1736$

$$\therefore \text{Variance} = \sigma^2 = \frac{\Sigma f_i (x_i - \bar{x})^2}{N} = \frac{1736}{40} = 43.4$$

$$\text{Standard deviation} = \sigma = \sqrt{43.4} = 6.59$$

Variance and Standard deviation of a Continuous Frequency Distribution

If there are n classes in given distribution, each class represented by its mid point x_i and corresponding frequency f_i , then we calculate standard deviation using the formula

$$\sigma = \sqrt{\frac{\Sigma f_i (x_i - \bar{x})^2}{\Sigma f_i}} \text{ where } N = \Sigma f_i \text{ and } \bar{x} \text{ is the mean of the distribution.}$$

ANOTHER METHOD :

To avoid the tediousness of calculation and to simplify the calculation, we adopt the following alternative method.

$$\begin{aligned} \text{We know that variance } \sigma^2 &= \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^2 \\ &= \frac{1}{N} \sum f_i (x_i^2 + \bar{x}^2 - 2x_i \bar{x}) \\ &= \frac{1}{N} \left[\sum_{i=1}^n f_i x_i^2 + \sum_{i=1}^n f_i \bar{x}^2 - \sum_{i=1}^n 2\bar{x} f_i x_i \right] \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{N} \left[\sum_{i=1}^n f_i x_i^2 + \bar{x}^2 \cdot N - 2\bar{x} \sum_{i=1}^n f_i x_i \right] \\
&= \frac{1}{N} \sum_{i=1}^n f_i x_i^2 + \bar{x}^2 - 2\bar{x}^2 \\
&= \frac{1}{N} \sum_{i=1}^n f_i x_i^2 - (\bar{x}^2) = \frac{1}{N} \sum_{i=1}^n f_i x_i^2 - \left(\frac{\sum_{i=1}^n f_i x_i}{N} \right)^2
\end{aligned}$$

$$\therefore \text{Standard Deviation} = \sigma = \sqrt{\frac{1}{N} \sum_{i=1}^n f_i x_i^2 - \left(\frac{\sum_{i=1}^n f_i x_i}{N} \right)^2}$$

Step Deviation Method (Short Cut Method) :

If the midvalues x_i in the continuous distribution are large, the calculation of mean and variance becomes difficult. In such cases we apply the step deviation method, as described below

Let h be the length of the class interval and A is the assumed mean.

We define $y_i = \frac{x_i - A}{h}$, $i = 1, 2, \dots, n$. Then $x_i = A + hy_i$

$$\begin{aligned}
\bar{x} &= \frac{\sum_{i=1}^n f_i x_i}{N} = \frac{\sum_{i=1}^n f_i (A + hy_i)}{N} \\
&= \frac{1}{N} \left[\sum_{i=1}^n A f_i + \sum_{i=1}^n h f_i y_i \right] = \frac{1}{N} \left[A \sum_{i=1}^n f_i + h \sum_{i=1}^n f_i y_i \right] \\
&= A + h \sum_{i=1}^n f_i y_i = A + h\bar{y}
\end{aligned}$$

$$\begin{aligned}
\sigma_x^2 &= \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^2 = \frac{1}{N} \sum_{i=1}^n f_i (A + hy_i - A - h\bar{y})^2 \\
&= \frac{1}{N} \sum_{i=1}^n f_i h^2 (y_i - \bar{y})^2 = h^2 \left[\frac{1}{N} \sum_{i=1}^n f_i (y_i - \bar{y})^2 \right] \\
&= h^2 \sigma_y^2 \text{ (or) } \sigma_x = h \sigma_y
\end{aligned}$$

$$\begin{aligned}
\text{But we know that, S. D. } (\sigma_x) &= \frac{1}{N} \sqrt{N \sum_{i=1}^n f_i x_i^2 - (\sum_{i=1}^n f_i x_i)^2} \\
&= \frac{1}{N} \sqrt{N \sum_{i=1}^n f_i y_i^2 - (\sum_{i=1}^n f_i y_i)^2}
\end{aligned}$$

$$\therefore \sigma_x = \frac{h}{N} \sqrt{N \sum_{i=1}^n f_i y_i^2 - (\sum_{i=1}^n f_i y_i)^2}$$

Example 4 : Calculate the variance and standard deviation of the following continuous frequency distribution.

class interval	30–40	40–50	50–60	60–70	70–80	80–90	90–100
Freq.	3	7	12	15	8	3	2

Solution : We tabulate the following way.

class interval (C.I.)	Freq. (f_i)	Mid point (x_i)	$y_i = \frac{x_i - A}{h}$ $A = 65, h = 10$	y_i^2	$f_i y_i$	$f_i y_i^2$
30–40	3	35	–3	9	–9	27
40–50	7	45	–2	4	–14	25
50–60	12	55	–1	1	–12	12
60–70	15	65	0	0	0	0
70–80	8	75	1	1	8	8
80–90	3	85	2	4	6	12
90–100	2	95	3	9	6	18
	$N =$ $\Sigma f_i = 50$				$\Sigma f_i y_i$ $= -15$	$\Sigma f_i y_i^2$ $= 105$

Assumed mean = $A = 65$,

length of the class interval = $h = 10$

$$\text{Mean} = \bar{x} = A + \frac{\Sigma f_i y_i}{N} \times h = 65 - \left(\frac{15}{50} \times 10 \right) = 62$$

$$\begin{aligned} \text{Variance } \sigma^2 &= \frac{h^2}{N^2} [N \Sigma f_i y_i^2 - (\Sigma f_i y_i)^2] \\ &= \frac{100}{2500} [50(105) - (-15)^2] = 201 \end{aligned}$$

$\therefore$ Standard deviation = 14.18

Example 4 : Find the standard deviation of the following data (use the step deviation method)

wages (Rs.)	125–175	175–225	225–275	275–325	325–375	375–425	425–475	475–525	525–575
no. of workers	2	22	19	14	3	4	6	1	1

Solution : Length of the class interval $h = 50$, assumed mean = $a = 300$, $y_i = \frac{x_i - a}{h}$

Mid points of C. I. (x_i)	Frequencies (f_i)	y_i	y_i^2	$f_i y_i$	$f_i y_i^2$
150	2	-3	9	-6	18
200	22	-2	4	-44	88
250	19	-1	1	-19	19
300	14	0	0	0	0
350	3	1	1	3	3
400	4	2	4	8	16
450	6	3	9	18	54
500	1	4	16	4	16
550	1	5	25	5	25
	$\Sigma f_i = N = 72$			$\Sigma f_i y_i = -31$	$\Sigma f_i y_i^2 = 239$

$$\text{Mean} = \bar{x} = A + \left(\frac{\Sigma f_i y_i}{N} \right) \times h = 300 + \left(\frac{-31}{72} \right) 50 = 300 - \frac{1550}{72} = 278.47$$

$$\begin{aligned} \text{Variance } \sigma^2 &= \frac{h^2}{N^2} [N \Sigma f_i y_i^2 - (\Sigma f_i y_i)^2] \\ &= \frac{2500}{72^2} [72 (239) - (31)^2] = 2500 \left[\frac{239}{72} - \left(\frac{31}{72} \right)^2 \right] \end{aligned}$$

$$\therefore \sigma_x = 88.52$$

Example 3 : Find the mean deviation from the mean and standard deviation of the series $a, a + d, a + 2d, \dots, a + 2nd$.

Solution : Number of terms in the series = $2n + 1$

$$\bar{x} = A.M. = \frac{a + (a + d) + (a + 2d) + \dots + (a + 2nd)}{2n + 1}$$

$$= \frac{2n + 1}{2} [2a + (2n + 1 - 1)d] = a + nd$$

Series is $a, a + d, a + 2d, \dots, a + (n - 1)d, a + nd, a + (n + 1)d, \dots, a + (2n - 1)d, a + 2nd$.

$$\begin{aligned} \text{Mean deviation} &= \frac{\Sigma |x_i - \bar{x}|}{2n + 1} \\ &= \frac{nd + (n - 1)d + (n - 2)d + \dots + d + 0 + d + \dots + (n - 1)d + nd}{2n + 1} \\ &= \frac{2[d + \dots + (n - 2)d + (n - 1)d + nd + 0 + d + \dots + (n - 1)d + nd]}{2n + 1} \\ &= \frac{2d(1 + 2 + \dots + n)}{2n + 1} = \frac{2d}{2} \cdot \frac{n(n + 1)}{2n + 1} = \frac{n(n + 1)}{2} d. \end{aligned}$$

Let the assumed mean be A. Then

$$\begin{aligned}
 \sigma^2 &= \frac{1}{n} \sum (x_i - A)^2 - \left[\frac{\sum (x_i - A)}{n} \right]^2 \\
 &= \frac{d^2 + 2^2 d^2 + \dots + (2n)^2 d^2}{2n+1} - \left[\frac{d + 2d + \dots + 2nd}{2n+1} \right]^2 \\
 &= \frac{d^2}{2n+1} \frac{2n(2n+1)(4n+1)}{6} - d^2 \left[\frac{2n(2n+1)}{2(2n+1)} \right]^2 = \frac{d^2 n(4n+1)}{3} - d^2 n^2 \\
 &= nd^2 \left(\frac{4n+1}{3} - n \right) = \frac{nd^2(4n+1-3n)}{3} = \frac{n(n+1)}{3} d^2 \\
 \therefore \sigma &= d \sqrt{\frac{n(n+1)}{3}}.
 \end{aligned}$$

Example 8 : Given that $\bar{x}$ is the mean and σ^2 is the variance of n observations $x_1, x_2, \dots, x_n$. Prove that the mean and variance of the observations $ax_1, ax_2, \dots, ax_n$ are $a\bar{x}$ and $a^2\sigma^2$ respectively. ($a \neq 0$)

Solution : We have $\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$

$$\text{Mean of } ax_1, ax_2, \dots, ax_n = \frac{ax_1 + ax_2 + \dots + ax_n}{n} = \frac{a(x_1 + x_2 + \dots + x_n)}{n} = a\bar{x}$$

Also we have $\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n}$ and

$$\therefore \text{Variance of } ax_1, ax_2, \dots, ax_n = \frac{\sum (ax_i - a\bar{x})^2}{n} = a^2 \frac{\sum (x_i - \bar{x})^2}{n} = a^2 \sigma^2$$

Hence the result.

Example 9 : The variance of 20 observations is 5. If each of the observations is multiplied by 2, find the variance of the resulting observation.

Solution : Here $a = 2$ and $\sigma^2 = 5$

$$\therefore \text{Variance of the resulting observations} = a^2 \sigma^2 = 2^2 \times 5 = 20, \text{ using Example 8}$$

Example 10 : If each of the observations $x_1, x_2, \dots, x_n$ is increased by k , k is a positive or negative number, then the variance remains unchanged.

Solution : Let $\bar{x}$ be the mean of $x_1, x_2, \dots, x_n$.

$$\text{Variance} = \sigma_1^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$$

Let the new observations be $y_i = x_i + k$ where k is +ve or -ve number.

$$\begin{aligned}
 \text{Then, mean of the new observations} &= \bar{y} = \frac{1}{n} \sum y_i = \frac{1}{n} \sum (x_i + k) = \frac{1}{n} [\sum x_i + \sum k] \\
 &= \frac{1}{n} \sum x_i + \frac{1}{n} \sum k = \bar{x} + \frac{1}{n} (kn) = \bar{x} + k
 \end{aligned}$$

The variance of the new observations $= \sigma_2^2$

$$= \frac{1}{n} \sum (y_i - \bar{y})^2 = \frac{1}{n} \sum [(x_i + k) - (\bar{x} + k)]^2 = \frac{1}{n} \sum (x_i - \bar{x})^2 = \sigma_1^2$$

Note : Adding (or subtracting) a positive number to (or from) each of given set of observations do not affect the variance.

2.15 COEFFICIENT OF VARIATION

A measure of dispersion is expressed in the same units as the variate in question. For example standard deviation of weights expressed in grams is also expressed in kilograms. So it becomes difficult to compare the variability of two distributions whose variates are expressed in different units. Hence it becomes necessary to find out a relative measure of dispersion which is purely a number and independent of units of measurement. Coefficient of variation is one such relative measure.

Coefficient of variation (C.V.) is defined as the ratio of the standard deviation σ to the arithmetic mean $\bar{x}$ and it is often expressed as a percentage.

$$\therefore \text{Coefficient of variation} = \frac{\sigma}{\bar{x}} \times 100 \text{ where } \bar{x} \neq 0.$$

Note : The distribution having greater coefficient of variation is said to be more variable than the other. The distribution having lesser coefficient of variation is said to be more consistent than the other.

Analysis / Comparison of Two Frequency Distributions with Equal Means :

Suppose two distributions are having same mean $\bar{x}_1 = \bar{x}_2 = \bar{x}$ but different standard deviations σ_1 and σ_2 respectively. Then coefficient of variations are given by $\left(\frac{\sigma_1}{\bar{x}} \times 100\right)$ and $\left(\frac{\sigma_2}{\bar{x}} \times 100\right)$. Thus the C.V.'s can be compared using σ_1 and σ_2 only. Here, the series with lower value of standard deviation is said to be more consistent than the other series with greater standard deviation. The series with greater standard deviation is called more dispersed than other.

SOLVED EXAMPLES

Example 1 : Students of two sections A and B of a class show the following performance in a test (for 100 marks). Which section of students has greater variability in performance ?

	Section A	Section B
No. of students	50	60
Average marks in test	45	45
Variance of distributions of marks	64	81

Solution : Given variances are 64 and 81.

$\therefore$ Standard deviations are 8 and 9 i.e., $\sigma_1 = 8$ and $\sigma_2 = 9$.

They are having same mean. Thus $\bar{x}_1 = \bar{x}_2 = \bar{x} = 45$.

Hence section B has greater variability in the performance.

Example 2 : Goals scored by two teams A and B in foot ball season are as follows :

Number of goals scored in match	Number of matches	
	Team A	Team B
0	24	25
1	9	9
2	8	6
3	5	5
4	4	5

By calculating the standard deviation in each case find which team be consider more consistent.

Solution :

Team A					Team B				
x_i	f_i	$d_i = x_i - a$ $= x_i - 2$	$f_i d_i$	$f_i d_i^2$	x_i	f_i	$d_i = x_i - a$ $= x_i - 2$	$f_i d_i$	$f_i d_i^2$
0	24	-2	-48	96	0	25	-2	-50	100
1	9	-1	-9	9	1	9	-1	-9	9
2	8	0	0	0	2	6	0	0	6
3	5	1	5	5	3	5	1	5	5
4	4	2	8	16	4	5	2	10	20
	50		$\Sigma f_i d_i$ $= -44$	$\Sigma f_i d_i^2$ $= 126$		50		-44	140

$$\text{Mean} = \bar{x} = a + \frac{f_i d_i}{f_i} = 2 - \frac{44}{50} = 1.12$$

$$\text{and } \sigma = \sqrt{\frac{1}{N} \Sigma f_i d_i^2 - \left(\frac{1}{N} \Sigma f_i d_i \right)^2}$$

$$= \sqrt{\frac{126}{50} - \left(\frac{-44}{50} \right)^2} = \sqrt{2.52 - 0.7774} = 1.32$$

$$\text{Mean} = \bar{x} = a + \frac{f_i d_i}{f_i} = 2 - \frac{44}{50} = 1.12$$

$$\text{and } \sigma = \sqrt{\frac{1}{N} \Sigma f_i d_i^2 - \left(\frac{1}{N} \Sigma f_i d_i \right)^2}$$

$$= \sqrt{2.8 - 0.7744} = 1.42$$

Here, we find means are equal but S. D of team A < S. D. of team B.

Hence team A is more consistent.

Example 3 : Lives of two models of refrigerators A and B, are given below. Which refrigeration model would you suggest to purchase?

Life in years	Model A	Model B
0-2	5	2
2-4	16	7
4-6	13	12
6-8	7	19
8-10	5	9

Solution :

Class interval	Mid point x_i	x_i^2	Model A			Model B		
			f_i	$f_i x_i$	$f_i x_i^2$	f_i	$f_i x_i$	$f_i x_i^2$
0-2	1	1	5	5	5	2	2	2
2-4	3	9	16	48	144	7	21	63
4-6	5	25	13	65	325	12	60	300
6-8	7	49	7	49	343	19	133	931
8-10	9	81	5	45	405	9	81	729
			$\Sigma f_i = N$ $= 46$	212	1221	$N = 49$	297	2025

For model A, $\bar{x}_A = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{212}{46} = 4.6$

and $\sigma_A = \sqrt{\frac{f_i x_i^2}{N} - \left(\frac{f_i x_i}{N}\right)^2} = \sqrt{5.38} = 2.319$

For model B, $\bar{x}_B = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{297}{49} = 6.06$

and $\sigma_B = \sqrt{\frac{2025}{49} - \left(\frac{297}{49}\right)^2} = \sqrt{4.61} = 2.147$

Coefficient of variation for model A $= \frac{\sigma_A \times 100}{\bar{x}_A} = \frac{2.319}{4.6} \times 100 = 50.41$

Coefficient of variation for model B $= \frac{\sigma_B \times 100}{\bar{x}_B} = \frac{2.147}{6.06} \times 100 = 35.43$

Hence the model A is suggested to purchase.

Example 4 : In example 6 find the coefficient of variance.

Solution : We have mean $\bar{x} = 278.47$ and standard deviation $= \sigma_x = 88.52$

$\therefore$ Coefficient of variation $= \frac{88.52}{278.47} \times 100 = 31.79$

Example 4 : The following data gives the analysis of month wise wages paid to the workers of two firms A and B belonging to the same industry

(i) which firm has greater variability in individual works ?

(ii) which firm has larger wage bill ?

	Firm A	Firm B
Number of workers	500	600
Average daily wage	186	175
Variance of distribution of wages	81	100

Solution : (i) For the firm A variance of distribution of wages, $\sigma_1^2 = 81 \Rightarrow \sigma_1 = 9$

Similarly, for B, $\sigma_2^2 = 100 \Rightarrow \sigma_2 = 10$

$\therefore$ C. V. of the distribution of wages of firm A $= \frac{\sigma_1}{\bar{x}_1} \times 100 = \frac{9}{180} \times 100 = 4.84$

and C. V. of the distribution of wages of firm B $= \frac{\sigma_2}{\bar{x}_2} \times 100 = \frac{10}{175} \times 100 = 5.71$

Since C. V. of firm B > C. V of firm A, we conclude that the firm B has greater variability in individual wages.

(ii) Number of workers for firm A (x_1) = 500

Daily average wage $= \bar{x}_1 = \text{Rs.}186$

Total daily wages paid to workers of A $= n_1 x_1 = 500 \times 186 = \text{Rs.} 93,000$

For the firm B, number of wage earners $= n_2 = 60$

Average daily wage $= \bar{x}_2 = \text{Rs.}175$

Total daily wages paid to workers of firm B $= n_2 \bar{x}_2 = 600 \times 175 = \text{Rs.}1,05,000$

$\therefore$ Firm B has larger wage bill.

Example 2 : The scores of two cricketers A and B in 10 innings are given here. Find who is a better run getter and who is more a consistent player.

Scores of A x_i	40	25	19	80	38	8	67	121	66	76
Scores of B y_i	28	70	31	0	14	111	66	31	25	4

Solution : For cricketer A, average $\bar{x} = \frac{540}{10} = 54$

For cricketer B, average $\bar{y} = \frac{380}{10} = 38$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	y_i	$(y_i - \bar{y})$	$(y_i - \bar{y})^2$
40	-14	196	28	-10	100
25	29	841	70	32	1024
19	-35	1225	31	-7	49
80	26	676	0	-38	1444
38	-16	256	14	-24	576
8	-46	2116	111	73	5329
67	13	169	66	28	784
121	67	4489	31	-7	49
66	12	144	25	-13	169
76	22	484	4	-34	1156
Σx_i $= 540$		10596	$\Sigma y_i = 380$		10,674

$$\text{Standard deviation of A} = \sigma_x = \sqrt{\frac{1}{n}(x_i - \bar{x})^2} = \sqrt{\frac{10596}{10}} = 32.55$$

$$\text{Standard deviation of B} = \sigma_y = \sqrt{\frac{1}{n}(y_i - \bar{y})^2} = \sqrt{\frac{10674}{10}} = \sqrt{1067.4} = 32.67$$

$$\text{C. V. of A} = \frac{\sigma_x}{\bar{x}} \times 100 = \frac{32.55}{54} \times 100 = 60.28$$

$$\text{C. V. of B} = \frac{\sigma_y}{\bar{y}} \times 100 = \frac{32.67}{38} \times 100 = 86$$

Since $\bar{x} > \bar{y}$, cricketer A is a better rungetter

But C. V of A < C. V. of B, $\therefore$ cricketer A is more consistent player.

EXERCISE

- Find the mean deviation from the mean for the following data :
(i) 6, 7, 10, 12, 13, 4, 8, 12 (ii) 3, 6, 10, 4, 9, 10
- Find the mean deviation about the median for the following data :
(i) 22, 24, 30, 27, 29, 31, 25, 28, 41, 42 (ii) 4, 6, 9, 3, 10, 13, 2
- Find the mean deviation from the mean for the following data :

(i)

x_i	10	30	50	70	90
f_i	4	24	28	16	8

(ii)

x_i	10	11	12	13
f_i	3	12	18	12

- Calculate the mean deviation from the median.

(i)

classes	0-10	10-20	20-30	30-40	40-50
Frequencies	5	10	20	5	10

(ii)

Marks obtained	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
No. of boys	5	8	7	12	28	20	10	10

- Find the mean deviation from mean for the following data :

(i)

Classes	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Freq.	2	3	8	14	8	3	2

(ii)

Height (in cms)	95–105	105–115	115–125	125–135	135–145	145–155
No. of boys	9	13	26	30	12	10

6. Find the variance for the discrete data given below

(i) 4, 5, 2, 8, 7 (ii) 6, 7, 10, 12, 13, 4, 8, 12

7. Find the variance and standard deviation for the following frequency distribution.

(i)

x_i	6	10	14	18	24	28	30
f_i	2	4	7	12	8	4	3

(ii)

x_i	60	61	62	63	64	65	66	67	68
f_i	2	1	12	29	25	12	10	4	5

8. (i) Find the mean and variance using step deviation method for the following data.

Age in years	20–30	30–40	40–50	50–60	60–70	70–80	80–90
No. of numbers	3	61	132	153	140	51	2

(ii)

Classes	0–30	30–60	60–90	90–120	120–150	150–180	180–210
Freq.	2	3	5	10	3	5	2

9. From the prices of shares X and Y given below, for 10 days of trading, which share is more stable?

X	35	54	52	53	56	58	52	50	51	49
Y	108	107	105	105	106	107	104	103	104	101

10. The coefficients of variations of two distributions are 60 and 70 and their standard deviations are 21 and 16 respectively. Find their arithmetic means.
11. The mean of 5 observations is 4.4. Their variance is 8.24. If three of the observations are 1, 2, 6. Find the other two observations.
12. The arithmetic mean and standard deviation of a set of 9 items are 43 and 5 respectively. If an item of value 63 is added to that set find the mean and standard deviation of 10 items set given.

ANSWERS

- | | | |
|-----------------------|-----------------------------------|----------------------|
| 1. (i) 2.79 (ii) 2.67 | 2. (i) 4.7 (ii) 3.28 | 3. (i) 16 (ii) 0.71 |
| 4. (i) 9 (ii) 170.58 | 5. (i) 14.284 (ii) 11.288 | 6. (i) 4.72, 2.172 |
| (ii) 10.125, 3.181 | 7. (i) 43.4, 6.59 (ii) 2.86, 1.69 | 8. (i) 140.89, 11.86 |
| (ii) 227.61, 47.708 | 11. 4, 9 | 12. 58.5, 7.648 |

2.16 SKEWNESS

The measures of Central Tendency and Dispersion do not indicate whether the distribution is symmetric or not. Measures of skewness gives the direction and the extent of skewness. In symmetrical distribution the mean, median and mode are identical. The more the mean moves away from the mode, the larger the asymmetry or skewness. Thus skewness is the lack of symmetry. The measures of central tendency and dispersion are inadequate to characterise a distribution completely. They may be supported by two more measures **Skewness and Kurtosis**.

A distribution which is not symmetrical is called a skewed distribution. In such distributions the mean, the mode and median will not coincide. The values are pulled apart.

Test of Skewness :

The absence of asymmetry or skewness can be stated under the following conditions.

If the distribution is Symmetric, the following conditions are observed :

- 1) The values of mean, mode, median coincide (the values are equal)
- 2) $Q_3 - \text{Median} = \text{Median} - Q_1$
- 3) The sum of positive deviations = The sum of negative deviations.
- 4) The frequencies on either side of the mode are equal.

Similarly, a skewed distribution will have following characteristics :

1. Mean $\neq$ Median $\neq$ Mode
2. $Q_3 - \text{Median} \neq \text{Median} - Q_1$
3. The sum of positive deviations $\neq$ The sum of negative deviations.

Measures of Skewness

Absolute skewness = Mean – Mode = ('+' positive skewness) = ('-' negative skewness)

If the value of Mean is greater than Mode, then the skewness is positive.

If the value of Mode is greater than Mean, the skewness is negative.

The absolute measure of skewness will not be proper measure for comparison. Hence in each series a relative measure or coefficient of skewness will have to be computed.

There are three important measures of relative skewness.

1. Karl Pearson's coefficient of skewness.
2. Bowley's coefficient of skewness.
3. Kelly's coefficient of skewness.

Generally Karl Pearson method is widely used.

$$\text{Karl Pearson coefficient of skewness } (S_{k_p}) = \frac{\bar{X} - \text{mode}}{\sigma}$$

In case mode is illdefined.

$$\text{The coefficient of skewness } (S_{k_p}) = \frac{3(\text{Mean} - \text{Median})}{\sigma} = \frac{3(\bar{X} - M)}{\sigma}$$

SOLVED EXAMPLES

Example 4 : Calculate Karl Pearson's coefficient of skewness for the following data :
25, 15, 23, 40, 27, 25, 23, 25, 20.

Solution :**Computation table of Mean and Standard Deviation**

Size	Deviation from A = 25 (d) = size - A	Square of deviation (d ²)
25	0	0
15	-10	100
23	-2	4
40	15	225
27	2	4
25	0	0
23	-2	4
25	0	0
20	-5	25
	$\Sigma d = -2$	$\Sigma d^2 = 362$

Here $n = 9$, Mean = $A \pm \frac{\Sigma d}{n} = 25 - \frac{2}{9} = 24.78$, Mode = 25

$$\text{and S. D.} = \sqrt{\frac{\Sigma d^2}{N} - \left(\frac{\Sigma d}{N}\right)^2} = \sqrt{\frac{362}{9} - \left(\frac{-2}{9}\right)^2} = 6.3$$

$$\text{Karl Pearson coefficient of skewness} = \frac{\text{Mean} - \text{Mode}}{\text{S.D.}} = \frac{24.78 - 25}{6.3} = -0.03$$

Example 2 : Calculate Karl Pearson's coefficient of skewness for the following data :

Variable	0-5	5-10	10-15	15-20	20-25	25-30	30-35	35-40
Frequency	2	5	7	13	21	16	8	3

Solution : Computation of mean and standard deviation. Take $A = 22.5$.

Variable X	Mid value (m)	Freq. (f)	Deviation $d' = \frac{(m - 22.5)}{5}$	fd'	d'^2	fd'^2
0-5	2.5	2	-4	-8	16	32
5-10	7.5	5	-3	-15	9	45
10-15	12.5	7	-2	-14	4	28
15-20	17.5	13 f_1	-1	-13	1	13
20-25	22.5	21 f	0	0	0	0
25-30	27.5	16 f_2	1	16	1	16
30-35	32.5	8	2	16	4	32
35-40	37.5	3	3	9	9	27
		$\Sigma f = N = 75$		$\Sigma fd' = -9$		$\Sigma fd'^2 = 193$

Here c = Class interval = 5

$$\text{Mean } \bar{X} = A + \frac{\sum fd'}{N} \times c = 22.5 + \frac{-9}{75} \times 5 = 21.9$$

$$\begin{aligned} \text{S. D. } = \sigma &= \left(\sqrt{\frac{\sum fd'^2}{N} - \left(\frac{\sum fd'}{N} \right)^2} \right) \times c = \sqrt{\frac{193}{75} - \left(\frac{-9}{75} \right)^2} \times 5 = \sqrt{2.573 - 0.014} \times 5 \\ &= \sqrt{2.558} \times 5 = 1.599 \times 5 = 7.9950 = 8 \end{aligned}$$

$$\text{Mode} = l + \frac{f - f_1}{2f - f_1 - f_2} \times c = 20 + \frac{21 - 13}{2 \times 21 - 13 - 16} \times 5 = 20 + \frac{40}{13} = 23.08$$

$$\therefore \text{Pearson's coefficient of skewness} = \frac{\text{Mean} - \text{Mode}}{\text{S.D.}} = \frac{21.9 - 23.08}{8} = \frac{-1.18}{8} = -0.148$$

2.17 KURTOSIS

The expression **Kurtosis** is used to describe the peakedness of curve. As far as the measurement of a shape is concerned, we have two characteristics. Skewness which refers to asymmetry of a series and Kurtosis which measures the peakedness of a normal curve. All the frequency curves expose different degrees of flatness or peakedness.

This characteristic of frequency curve is termed as **Kurtosis**. Measures of Kurtosis denote the shape of the top of a frequency curve.

Measures of Kurtosis

The measures of Kurtosis of a frequency distribution are based upon the fourth moment about the mean of the distribution.

$$\text{Symmetrically, } \beta_2 = \frac{\mu_4}{\mu_2^2}$$

where μ_4 = fourth moment, μ_2 = second moment

If $\beta_2 = 3$, the distribution is said to be normal (neither flat nor peaked) and the curve is **normal curve (mesokurtic)**

If $\beta_2 > 3$, the distribution is said to be more peaked and the curve is **lepekurtic**.

If $\beta_2 < 3$, the distribution is said to be flatter than normal curve and the curve is **platykurtic**.

SOLVED EXAMPLES

Example 2 : From the following distribution, calculate

- (i) First 4 moments about the mean
- (ii) Skewness based on moments
- (iii) Kurtosis

Income (Rs)	0-10	10-20	20-30	30-40
Freq.	1	3	4	2

Solution :**Computation of Moments, Skewness and Kurtosis**

Income	Mid value (m)	Frequency (f)	$d = \frac{m-15}{10}$	fd	fd^2	fd^3	fd^4
0-10	5	1	-1	-1	1	-1	1
10-20	15	3	0	0	0	0	0
20-30	25	4	1	4	4	4	4
30-40	35	2	2	8	8	16	32
	N = 10			$\sum fd = 7$	$\sum fd^2 = 13$	$\sum fd^3 = 19$	$\sum fd^4 = 37$

(i) Moments about mean

$$\mu'_1 = \frac{\sum fd}{N} \times c = \frac{7}{10} \times 10 = 7$$

$$\mu'_2 = \frac{\sum fd^2}{N} \times c^2 = \frac{13}{10} \times 10^2 = 130$$

$$\mu'_3 = \frac{\sum fd^3}{N} \times c^3 = \frac{19}{10} \times 10^3 = 1900$$

$$\mu'_4 = \frac{\sum fd^4}{N} \times c^4 = \frac{37}{10} \times 10^4 = 37000$$

$$\text{First moment about mean, } \mu_1 = \mu'_1 - \mu'_1 = 7 - 7 = 0$$

$$\text{Second moment about mean, } \mu_2 = \mu'_2 - (\mu'_1)^2 = 130 - 7^2 = 81$$

$$\text{Third moment about mean, } \mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2\mu_1^3 = 1900 - 3(130)(7) + 2(7)^3 = -144$$

$$\text{Fourth moment, } \mu_4 = \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2\mu_2^2 - 3\mu_1^4 = 37000 - 53200 + 38200 - 7203 = 14817$$

(ii) Skewness based on moments is studied by β_1

$$\therefore \beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{(-144)^2}{(81)^3} = 0.039$$

(iii) Kurtosis is studied by β_2

$$\therefore \beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{14817}{(81)^2} = 2.26$$

Since $\beta_2 < 3$, the curve is more peaked and is platykurtic.

REVIEW QUESTIONS

1. (i) Explain Skewness. (ii) Define Skewness
2. Distinguish between positive and negative Skewness
3. What are the tests of Skewness.
4. (i) Define Kurtosis (ii) What is Kurtosis. How does it differ from skewness.
5. Explain the terms : (i) Lepto Kurtic (ii) Meso Kurtic (iii) Platy Kurtic

CHAPTER - 3

CORRELATION

3.1 INTRODUCTION

Previously we are used to study the characteristics of only one variable like marks, weights, heights, prices, ages, sales etc. This type of analysis is called **Univariate Analysis**. If there exists some relationship between two variables then the statistical analysis of such data is called **Bivariate Analysis**. We may be interested to find any relation between the two variables under study. For example, we can easily conclude that there is some relation between the price and the sale, price and the production of a commodity.

Correlation refers to the relationship of two or more variables. We know that there exists relationship between the heights of a Father and a son, wage and price index. The study of the relation is called **correlation**. It measures the closeness of the relationship between the variables.

3.2 DEFINITION

Correlation is a statistical analysis which measures and analyses the degree or extent to which two variables fluctuate with reference to each other.

The correlation expresses the relationship or interdependence of two setsof variables upon each other. One variable may be called the **subject** (independent) and the other **relative** (dependent).

3.3 TYPES OF CORRELATION

Correlation is classified into many types.

1. Positive and negative
2. Simple and multiple
3. Partial and total
4. Linear and non-linear

1. Positive and Negative Correlation :

Positive and Negative correlation depend upon the direction of change of the variables. If two variables tend to move together in the same direction i.e. an increase in the value of one variable is accompanied by an increase in the value of the other variable; or a decrease in the value of one variable is accompanied by a decrease in the value of the other variable, then the correlation is called **positive or direct correlation**. Height and weight, rainfall and yield of crops, price and supply are examples of positive correlation.

If two variables, tend to move together in opposite directions so that an increaseor decrease in the values of one variable is accompanied by a decrease or increase in the value of the other variable, then the correlation is called **negative or inverse correlation**.

2. Simple and Multiple correlation :

When we study only two variables, the relationship is described as simple correlation; example are quantity of money and price level, demand and price, etc. But in a multiple correlation we study more than two variables simultaneously; example is the relationship of price, demand and supply of a commodity.

3. Partial and Total correlation :

The study of two variables excluding some other variables is called **partial correlation**. For example, we study price and demand, eliminating the supply side. In **total correlation**, all the facts are taken into account.

4. Linear and non-linear correlation :

If the ratio of change between two variables is uniform, then there will be **linear correlation** between them. Consider the following

A	2	7	12	17
B	3	9	15	21

We can see that the ratio of change between the variables is the same. If we plot these on the graph, we get a straight line.

In a **curvilinear or non - linear correlation**, the amount of change in one variable does not bear a constant ratio of the amount of change in the other variables. The graph of non-linear or curvilinear relationship will be a curve.

3.4 METHODS OF STUDYING CORRELATION

There are 2 different methods for finding out the relationship between variables. They are (1) Graphic methods (2) Mathematical methods.

1. Graphic Methods are

- (a) Scatter diagram or scattergram
- (b) Simple graph

2. Mathematical Methods are

- (a) Karl Pearson's coefficient of correlation
- (b) Spearman's Rank coefficient of correlation.
- (c) Coefficient of concurrent deviation.
- (d) Method of least squares

We will now discuss these methods.

3.5 COVARIANCE

Definition 1 : Let X and Y be the two random variables with means $\bar{X}$ and $\bar{Y}$ respectively. Then the covariance between two variables X and Y is defined by the relation

$$\text{Cov}(X, Y) = E[(X - \bar{X})(Y - \bar{Y})]$$

Note 1. The above relation can be put in the form

$$\text{Cov}(X, Y) = E(XY) - E(X) \cdot E(Y)$$

2. The covariance of two independent random variables is zero.

i.e. if X and Y are two independent random variables, then

$$\text{Cov}(X, Y) = 0$$

$$\begin{aligned} \text{Proof : } \text{Cov}(X, Y) &= E[(X - \bar{X})(Y - \bar{Y})] \\ &= E(X - \bar{X}) \cdot E(Y - \bar{Y}) \\ &= [E(X) - E(\bar{X})][E(Y) - E(\bar{Y})] \\ &= (\bar{X} - \bar{X})(\bar{Y} - \bar{Y}) \\ &= 0 \end{aligned}$$

3. If a and b are constants, then

$$(i) \text{ Cov}(X + a, Y + b) = \text{Cov}(X, Y) \quad (ii) \text{ Cov}(aX, bY) = ab \text{ Cov}(X, Y)$$

Definition 2 : If $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ be n pairs of observations on two variables X and Y, then the covariance of X and Y, written as $\text{cov}(X, Y)$ is defined by

$$\text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \quad \dots (1)$$

where $\bar{x}$ and $\bar{y}$ are means of X and Y series respectively. Covariance is an absolute measure.

Covariance of X and Y *i.e.* $\text{Cov}(X, Y)$ is a measure of joint variation in X and Y. Just like

$\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ is the variance of X and $\frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2$ is the variance of Y, similarly

$\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$ is the covariance of X and Y. Further, as variance gives an idea of the

variation in a single variable X or Y, covariance gives an idea of joint variation in both the variables taken together.

Another convenient form of the formula (1) for computational work is as follows :

$$\begin{aligned} \text{Cov}(X, Y) &= \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \\ &= \frac{1}{n} \sum_{i=1}^n (x_i y_i - x_i \bar{y} - \bar{x} y_i + \bar{x} \bar{y}) \\ &= \frac{1}{n} \sum_{i=1}^n x_i y_i - \bar{y} \cdot \frac{1}{n} \sum_{i=1}^n x_i - \bar{x} \cdot \frac{1}{n} \sum_{i=1}^n y_i + \bar{x} \bar{y} \\ &= \frac{1}{n} \sum_{i=1}^n x_i y_i - \bar{y} \cdot \bar{x} - \bar{x} \cdot \bar{y} + \bar{x} \bar{y} = \frac{1}{n} \sum_{i=1}^n x_i y_i - \bar{x} \cdot \bar{y} \end{aligned}$$

$$\text{Hence } \text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \frac{1}{n} \sum_{i=1}^n x_i y_i - \bar{x} \cdot \bar{y}$$

3.6 COEFFICIENT OF CORRELATION

Correlation is a statistical technique used for analysing the behavior of two or more variables. Its analysis deals with the association, between two or more variables. Statistical measures of correlation relates to covariation between series but not of function or casual relationship.

3.7 KARL PEARSON'S COEFFICIENT OF CORRELATION

Karl Pearson (1867 - 1936) a British Biometrician and Statistician suggested a mathematical method for measuring the magnitude of linear relationship between two variables. This is known as Pearsonian Coefficient of correlation. It is denoted by ' r '. This method is most widely used. It is also called **Product - Moment correlation coefficient**.

Karl Pearson's coefficient of correlation or correlation coefficient r between the variables X and Y is defined by the formula

$$r = \frac{\text{Covariance of X and Y}}{(\text{Standard deviation of X})(\text{Standard deviation of Y})}$$

$$\text{i.e., } r = \frac{\text{Cov}(X, Y)}{\sigma_x \sigma_y} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n \sigma_X \sigma_Y} \text{ where } \sigma_X = \text{S. D. of X and } \sigma_Y = \text{S. D. of Y}$$

$$\text{We know } \sigma_X = \sqrt{\frac{1}{n} \sum (x_i - \bar{x})^2} \text{ and } \sigma_Y = \sqrt{\frac{1}{n} \sum (y_i - \bar{y})^2}$$

Different Forms of the Formula for the correlation coefficient r .

$$1. \text{ We know that } \text{Cov}(X, Y) = \frac{1}{n} \sum (x_i - \bar{x})(y_i - \bar{y}) = \frac{1}{n} \sum x_i y_i - \bar{x} \bar{y}$$

$$\text{and } \sigma_X^2 = \frac{1}{n} \sum (x_i - \bar{x})^2 = \frac{1}{n} \sum x_i^2 - \bar{x}^2$$

$$\sigma_Y^2 = \frac{1}{n} \sum (y_i - \bar{y})^2 = \frac{1}{n} \sum y_i^2 - \bar{y}^2$$

$$\therefore r = \frac{\text{Cov}(X, Y)}{\sigma_x \sigma_y} = \frac{\frac{1}{n} \sum x_i y_i - \bar{x} \bar{y}}{\sqrt{\frac{1}{n} \sum x_i^2 - (\bar{x})^2} \times \sqrt{\frac{1}{n} \sum y_i^2 - (\bar{y})^2}}$$

Alternate form of $r(X, Y)$:

$$r(X, Y) = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{\sqrt{n \sum x_i^2 - (\sum x_i)^2} \cdot \sqrt{n \sum y_i^2 - (\sum y_i)^2}} \quad \left(\because \bar{x} = \frac{1}{n} \sum x_i, \bar{y} = \frac{1}{n} \sum y_i \right)$$

This formula saves a lot of computational labour.

2. Direct Method :

If X and Y are two variates having their means $\bar{x}$ and $\bar{y}$ respectively, then

$$r(X, Y) = \frac{\sum (dx \times dy)}{\sqrt{\sum dx^2 \times \sum dy^2}}$$

where $dx = x_i - \bar{x}$ = deviation of X from its mean

$dy = y_i - \bar{y}$ = deviation of Y from its mean

$$dx^2 = (x_i - \bar{x})^2 \text{ and } dy^2 = (y_i - \bar{y})^2$$

It can also be written as

$$r(X, Y) = \frac{\sum (dx \times dy)}{n\sigma_X \cdot \sigma_Y}$$

where n is the number of observations in X or Y and σ_X, σ_Y are standard deviations of X and Y respectively.

$$3. \quad r(X, Y) = \frac{\sum dx \cdot dy - \frac{(\sum dx)(\sum dy)}{n}}{\sqrt{\sum dx^2 - \frac{(\sum dx)^2}{n}} \times \sqrt{\sum dy^2 - \frac{(\sum dy)^2}{n}}}$$

where $dx = x - a$, a is the assumed mean of X - series, $dy = y - b$, b is the assumed mean of Y-series and n is the number of observations of X or Y.

We apply this formula when

(i) the terms of the series X and Y are big.

(ii) the means $\bar{x}$ or $\bar{y}$ are not integers.

4. Step Deviation Method :

The direct method becomes very lengthy and tedious if the means $\bar{x}$ and $\bar{y}$ of the two series X and Y are fractions. In such cases use the step deviation method. If u and v are step deviations from the assumed means, then

$$r(u, v) = \frac{n \sum uv - \sum u \cdot \sum v}{\sqrt{n \sum u^2 - (\sum u)^2} \cdot \sqrt{n \sum v^2 - (\sum v)^2}}$$

where $u = \frac{x - a}{h}$, $v = \frac{y - b}{k}$.

Here a, b, h, k are constants (a is the assumed mean of X series and b is the assumed mean of Y series).

3.8 PROPERTIES OF CORRELATION COEFFICIENT

Property I : Show that the maximum value of rank correlation coefficient is 1.

(OR)

The coefficient of correlation lies between -1 and $+1$. Symbolically, $-1 \leq r \leq +1$ or $|r| \leq 1$

Note : 1. $-1 \leq r \leq 1$ i.e. correlation coefficient can not exceed one numerically.

2. If $r = 1$ correlation is perfect and positive.

3. If $r = -1$ correlation is perfect and negative.

4. If $r = 0$, then there is no relationship between the variables.

Property II :

The coefficient of correlation is independent of the change of origin and scale of measurements.

Property III : If X, Y are random variables and a, b, c, d are any numbers such that

$a \neq 0, c \neq 0$ then

$$r(aX + b, cY + d) = \frac{ac}{|ac|} r(X, Y)$$

Property IV : Two independent variables are uncorrelated. That is, if X and Y are independent variables then $r(X, Y) = 0$.

Note. The converse of the above is not true, i.e., two uncorrelated variables may not be independent as the following examples.

e.g.1 :

X	-3	-2	-1	1	2	3	$\Sigma X = 0$
Y	9	4	1	1	4	9	$\Sigma Y = 28$
XY	-27	-8	-1	1	8	27	$\Sigma XY = 0$

$$\bar{X} = \frac{1}{n} \Sigma X = 0,$$

$$\text{cov}(X, Y) = \frac{1}{n} \Sigma XY - \bar{X}\bar{Y} = 0$$

$$r(X, Y) = \frac{\text{cov}(X, Y)}{\sigma_x \sigma_y} = 0$$

Thus X and Y are uncorrelated. But on examination, we can see that X and Y are connected by the relation $Y = X^2$ and hence are not independent.

e.g.2 : Consider the following data :

$x:$	1	2	3	4	5	6	7
$y:$	9	4	1	0	1	4	9

Here $\Sigma x = 28, \Sigma y = 28, \Sigma xy = 112$

$$\therefore \text{cov}(x, y) = \frac{1}{n} \left[\sum xy - \frac{(\sum x)(\sum y)}{n} \right] = \frac{1}{7} \left[112 - \frac{28 \times 28}{7} \right] = 0$$

Thus, $r_{xy} = 0$.

But the variables x and y are related by the relation $y = (x - 4)^2$

Hence the converse is not true.

SOLVED EXAMPLES

Example 1 : Calculate coefficient of correlation from the following data

X	12	9	8	10	11	13	7
Y	14	8	6	9	11	12	3

Solution : In both series items are in small number.

So there is no need to take deviations.

We use the formula $r = \frac{\text{Covariance of } X, Y}{\sigma_X \cdot \sigma_Y}$

Computation of Coefficient of Correlation

X	Y	X^2	Y^2	XY
12	14	144	196	168
9	8	81	64	72
8	6	64	36	48
10	9	100	81	90
11	11	121	121	121
13	12	169	144	156
7	3	49	9	21
$\Sigma X = 70$	$\Sigma Y = 63$	$\Sigma X^2 = 728$	$\Sigma Y^2 = 651$	$\Sigma XY = 676$

$$\text{i.e., } r = \frac{\Sigma XY}{\Sigma X^2 \cdot \Sigma Y^2} \quad (\text{or}) \quad r = \frac{(\Sigma XY \times N) - (\Sigma X \times \Sigma Y)}{\sqrt{(\Sigma X^2 \times N - (\Sigma X)^2)(\Sigma Y^2 \times N - (\Sigma Y)^2)}}$$

Here $N = \text{no. of observations} = 7$

$$\begin{aligned} \therefore r &= \frac{(676 \times 7) - (70 \times 63)}{\sqrt{728 \times 7 - (70)^2} \sqrt{651 \times 7 - (63)^2}} = \frac{4732 - 4410}{\sqrt{5096 - 4900} \sqrt{4557 - 3969}} \\ &= \frac{322}{\sqrt{196 \times 588}} = \frac{322}{339.48} = 0.95 \end{aligned}$$

Example 2 : Find the coefficient of correlation between the two variables.

x	50	50	55	60	65	65	65	60	60	60
y	11	13	14	16	16	15	15	14	13	13

Solution :**Caluculation of coefficient of correlation**

x	y	$X = x - \bar{x}$	$Y = y - \bar{y}$	X^2	Y^2	XY
50	11	-9	-3	81	9	27
50	13	-9	-1	81	1	9
55	14	-4	0	16	0	0
60	16	1	2	1	4	2
65	16	6	2	36	4	12
65	15	6	1	36	1	6
65	15	6	1	36	1	6
60	14	1	0	1	0	0
60	13	1	-1	1	1	-1
60	13	1	-1	1	1	-1
590	140	0	0	290	22	60

$$\bar{x} = \frac{590}{10} = 59, \bar{y} = \frac{140}{10} = 14$$

$$\begin{aligned} \text{Correlation coefficient} = r &= \frac{\sum XY - \frac{\sum X \sum Y}{N}}{\sqrt{\sum X^2 - \frac{(\sum X)^2}{N}} \sqrt{\sum Y^2 - \frac{(\sum Y)^2}{N}}} = \frac{60 - 0}{\sqrt{290 - 0} \sqrt{22 - 0}} \\ &= \frac{60}{79.87} = 0.7512 \end{aligned}$$

Example 2 : Find if there is any significant correlation between the heights and weights given below.

Height in inches	57	59	62	63	64	65	55	58	57
Weight in lbs	113	117	126	126	130	129	111	116	112

Solution :**Computation of Coefficient of Correlation**

Height in Inches x	Deviation from Mean (60) $X = x - \bar{x}$	Square of deviations X^2	Weight in lbs y	Deviation from Mean $Y = y - \bar{y}$	Square of deviations Y^2	Product of deviations of X and Y series (XY)
57	-3	9	113	-7	49	21
59	-1	1	117	-3	9	3
62	2	4	126	6	36	12
63	3	9	126	6	36	18
64	4	16	130	10	100	40
65	5	25	129	9	81	45
55	-5	25	111	-9	81	45
58	-2	4	116	-4	16	8
57	-3	9	112	-8	64	24
540	0	102	1080	0	472	216

$$\therefore \text{Coefficient of correlation, } r = \frac{\Sigma XY}{\sqrt{\Sigma X^2 \times \Sigma Y^2}}$$

$$= \frac{216}{\sqrt{102 \times 471}} = 0.98$$

Example 2 : Psychological tests of intelligence and of engineering ability were applied to 10 students. Here is a record of ungrouped data showing intelligence ratio (I. R) and Engineering ratio (E. R) calculate the co-efficient of correlation.

Student	A	B	C	D	E	F	G	H	I	J
I.R	105	104	102	101	100	99	98	96	93	92
E.R	101	103	100	98	95	96	104	92	97	94

Solution : We construct the following table :

Student	Intelligence ratio		Engineering ratio				
	x	$x - \bar{x} = X$	y	$y - \bar{y} = Y$	X^2	Y^2	XY
A	105	6	101	3	36	9	18
B	104	5	103	5	25	25	25
C	102	3	100	2	9	4	6
D	101	2	98	0	4	0	0
E	100	1	95	-3	1	9	-3
F	99	0	96	-2	0	4	0
G	98	-1	104	6	1	36	-6
H	96	-3	92	-6	9	36	18
I	93	-6	97	-1	36	1	6
J	92	-7	94	-4	49	16	28
Totals	990	0	980	0	170	140	92

From this table, we have mean of x , i.e. $\bar{x} = 990/10 = 99$;

mean of y , i.e., $\bar{y} = 980/10 = 98$, $\Sigma X^2 = 170$, $\Sigma Y^2 = 140$ and $\Sigma XY = 92$

Substituting these values in $r = \frac{\Sigma XY}{\sqrt{(\Sigma X^2 \times \Sigma Y^2)}} = \frac{92}{\sqrt{(170 \times 140)}} = \frac{92}{154.3} = 0.59$

Example 1 : Find Karl Pearson's coefficient of correlation from the following data

Wages	100	101	102	102	100	99	97	98	96	95
Cost of living	98	99	99	97	95	92	95	94	90	91

Solution : Computation of Karl Pearson's Coefficient of Correlation

Wages (x)	$X = x - \bar{X}$ $\bar{X} = 99$	X^2	y Cost of living	$Y = y - \bar{Y}$ $\bar{Y} = 95$	Y^2	XY
100	+1	1	98	+3	9	+3
101	+2	4	99	+4	16	+8
102	+3	9	99	+4	16	+12
102	+3	9	97	+2	4	+6
100	+1	1	95	0	0	0
99	0	0	92	-3	9	0
97	-2	4	95	0	0	0
98	-1	1	94	-1	1	+1
96	-3	9	90	-5	25	+15
95	-4	16	91	-4	16	+16
$\Sigma x = 990$	$\Sigma X = 0$	$\Sigma X^2 = 54$	$\Sigma y = 950$	$\Sigma Y = 0$	$\Sigma Y^2 = 96$	$\Sigma XY = 61$

$$r = \frac{\Sigma XY}{\sqrt{\Sigma X^2 \times \Sigma Y^2}} = \frac{61}{\sqrt{54 \times 96}} = \frac{61}{\sqrt{5184}} = \frac{61}{72} = 0.847$$

$$\therefore r = +0.847$$

Example 1 : Calculate the coefficient of correlation between age of cars and annual maintenance cost and comment :

Age of cars (years)	2	4	6	7	8	10	12
Annual maintenance cost (Rupees)	1600	1500	1800	1900	1700	2100	2000

Solution : Let age of cars = x , Annual maintenance = y

Computation of coefficient of correlation

x	y	$X = x - \bar{X}$ $\bar{X} = 7$	$Y = (y - \bar{Y}) / 100$ $\bar{Y} = 1800$	X^2	Y^2	XY
2	1600	-5	-2	25	4	10
4	1500	-3	3	9	9	9
6	1800	-1	0	1	0	0
7	1900	0	1	0	1	0
8	1700	1	-1	1	1	-1
10	2100	3	3	9	9	9
12	2000	5	2	25	4	10
$\Sigma x = 49$	$\Sigma y = 12600$	$\Sigma X = 0$	$\Sigma Y = 0$	$\Sigma X^2 = 70$	$\Sigma Y^2 = 28$	$\Sigma XY = 37$

$$\therefore r = \frac{\Sigma XY}{\sqrt{\Sigma X^2 \Sigma Y^2}} = \frac{37}{\sqrt{70 \times 28}} = 0.836$$

Comment : We can observe that there is a high degree of positive correlation between age of cars and annual maintenance cost.

Example 2 : With the following data in 6 cities, calculate the coefficient of correlation by Pearson's method between the density of population and the death rate :

Cities	Area in Sq. K.m.	Population '(000)	No. of deaths
A	150	30	300
B	180	90	1440
C	100	40	560
D	60	42	840
E	120	72	1224
F	80	24	312

Solution : To solve, we have to find the density of population and death rate

$$\text{So, Density} = \frac{\text{Population}}{\text{Area}}, \text{Death rate} = \frac{\text{No.of deaths}}{\text{Population}}$$

Density (x)	200	500	400	700	600	300
Death rate (y)	10	16	14	20	17	13

x	y	$X = x - \bar{X}$ $= x - 450$	$Y = y - \bar{Y}$ $= y - 15$	X^2	Y^2	XY
200	10	-250	-5	62500	25	1250
500	16	50	1	2500	1	50
400	14	-50	-1	2500	1	50
700	20	250	5	62500	25	1250
600	17	150	2	22500	4	300
300	13	-150	-2	22500	4	300
$\Sigma x = 2700$	$\Sigma y = 90$	$\Sigma X = 0$	$\Sigma Y = 0$	$\Sigma X^2 = 175000$	$\Sigma Y^2 = 60$	$\Sigma XY = 3200$

$$\text{Coefficient of correlation } r = \frac{\Sigma XY}{\sqrt{\Sigma X^2 \times \Sigma Y^2}} = \frac{3200}{\sqrt{175000 \times 60}} = \frac{3200}{3240.37} = 0.988$$

3.12 WHEN DEVIATIONS ARE TAKEN FROM AN ASSUMED MEAN

When actual mean is not a whole number, but a fraction or when the series is large, the calculation by direct method will involve a lot of time. To avoid such tedious calculation, we can use the assumed mean method.

$$\therefore r = \frac{\Sigma XY - \frac{\Sigma X \Sigma Y}{N}}{\sqrt{\left[\Sigma X^2 - \frac{(\Sigma X)^2}{N}\right]} \sqrt{\left[\Sigma Y^2 - \frac{(\Sigma Y)^2}{N}\right]}}$$

where

X = Deviation of the items of x - series from an assumed mean A i.e. $X = (x - A)$

Y = Deviation of the items of y - series from an assumed mean A i.e. $Y = (y - A)$

N = Number of items

ΣXY = The total of the product of the deviations of x and y -series from their assumed mean.

ΣX^2 = The total of the squares of the deviations of x - series from an assumed mean.

ΣY^2 = The total of the squares of the deviations of y - series from an assumed mean.

ΣX = Total of the deviations of x - series from assumed mean.

ΣY = Total of the deviations of y - series from assumed mean.

SOLVED EXAMPLES

Example 1 : Calculate Karl Pearson's correlation coefficient for the following paired data.

X	28	41	40	38	35	33	40	32	36	33
Y	23	34	33	34	30	26	28	31	36	38

what inference would you draw from the estimate.

Solution : Here $N=10$. Take $\bar{x} = 35$ and $\bar{y} = 31$

Computation of correlation coefficient

x	Deviation from assumed mean	Square of deviation		Deviation from assumed mean	Square of deviations	Product of deviation
	$X = x - 35$ $(X = x - \bar{x})$	X^2	y	$Y = y - 31$ $(Y = y - \bar{y})$	Y^2	$X Y$
28	-7	49	23	-8	64	56
41	+6	36	34	+3	9	18
40	+5	25	33	+2	4	10
38	+3	9	34	+3	9	9
35	0	0	30	-1	1	0
33	-2	4	26	-5	25	10
40	+5	25	28	-3	9	-15
32	-3	9	31	0	0	0
35	+1	1	36	+5	25	5
33	-2	4	38	+7	49	-4
$\Sigma x = 355$ $\Sigma X = 6$		$\Sigma X^2 = 162$	$\Sigma y = 313$	$\Sigma Y = 3$	$\Sigma Y^2 = 195$	$\Sigma XY = 79$

$$\therefore r = \frac{\Sigma XY - \frac{\Sigma X \cdot \Sigma Y}{N}}{\sqrt{\left[\Sigma X^2 - \frac{(\Sigma X)^2}{N} \right] \times \left[\Sigma Y^2 - \frac{(\Sigma Y)^2}{N} \right]}}$$

$$= \frac{79 - \frac{6 \times 3}{10}}{\sqrt{162 - \frac{6^2}{10}} \sqrt{195 - \frac{3^2}{10}}} = \frac{77.2}{\sqrt{1584} \sqrt{194.9}} = 0.45$$

Example 2 : Find a suitable coefficient of correlation for the following data :

Fertiliser used (tonnes)	15	18	20	24	30	35	40	50
Productivity (tonnes) :	85	93	95	105	120	130	150	160

Solution : Here $N = 8$. Take $\bar{x} = 29$ and $\bar{y} = 119$

Computation of coefficient of correlation

Fertiliser used x			Productivity y			
Fertiliser used	Deviation from assumed mean	Square of deviation	Productivity	Deviation from assumed mean	Square of deviation	Productivity of deviation
x	$X = x - \bar{X}$ $= x - 29$	X^2	y	$Y = y - \bar{Y}$ $= y - 119$	Y^2	XY
15	-14	196	85	-34	1156	476
18	-11	121	93	-26	676	286
20	-9	81	95	-24	576	216
24	-5	25	105	-14	196	70
30	1	1	120	1	1	1
35	6	36	130	11	121	66
40	11	121	150	31	961	341
50	21	441	160	41	1681	861
	$\Sigma X = 0$	$\Sigma X^2 = 1022$		$\Sigma Y = -14$	$\Sigma Y^2 = 5368$	$\Sigma XY = 2317$

$$\therefore r = \frac{\Sigma XY \times N - (\Sigma X)(\Sigma Y)}{\sqrt{(\Sigma X)^2 \times N - (\Sigma X)^2 \times (\Sigma Y)^2 \times N - (\Sigma Y)^2}} = \frac{2317}{\sqrt{1022 \times 5343.5}} = \frac{2317}{2336.89} = 0.99$$

Example 1 : The following table gives the distribution of the total population and those who are totally and partially blind among them. Find out if there is any relation between age and blindness.

Age	No. of Persons '(000)	Blind
0–10	100	55
10–20	60	40
20–30	40	40
30–40	36	40
40–50	24	36
50–60	11	22
60–70	6	18
70–80	3	15

Solution : In order to make the data comparable it is necessary to find out the number of blinds out of a fixed number. We have to find out the number of blind persons corresponding to one lakh, in each age group :

The first figure : $\frac{55}{100000} \times 100000 = 55$

The second figure : $\frac{40}{60000} \times 100000 = 67$

The third figure : $\frac{40}{40000} \times 100000 = 100 \dots$ so on.

We can get the other values continuing in the same manner.

Age x	Mid Value m	$X = \frac{m-45}{10}$	X^2	Blind per lakh y	Deviation from assumed mean (180) Y	Y^2	XY
0–10	5	–4	16	55	–125	15625	500
10–20	15	–3	9	67	–113	12769	339
20–30	25	–2	4	100	–80	6400	160
30–40	35	–1	1	111	–69	4761	69
40–50	45	0	0	150	–30	900	0
50–60	55	1	1	200	20	400	20
60–70	65	2	4	300	120	14400	240
70–80	75	3	9	500	320	102400	960
		$\Sigma X = -4$	$\Sigma X^2 = 44$		$\Sigma Y = 43$	$\Sigma Y^2 = 157655$	$\Sigma XY = 2288$

$$\begin{aligned}
 \therefore \text{Coefficient of correlation, } r &= \frac{\Sigma XY - \frac{(\Sigma X) \times (\Sigma Y)}{N}}{\sqrt{\left\{ \Sigma X^2 - \frac{(\Sigma X)^2}{N} \right\} \left\{ \Sigma Y^2 - \frac{(\Sigma Y)^2}{N} \right\}}} \\
 &= \frac{2288 - \left(\frac{-4 \times 43}{8} \right)}{\sqrt{\left\{ 44 - \frac{(-4)^2}{8} \right\} \left\{ 157655 - \frac{(43)^2}{8} \right\}}} \\
 &= \frac{2309.5}{2571.34} = 0.898
 \end{aligned}$$

Example 4 : Find out the coefficient of correlation in the following case.

Height of father (in inches)	65	66	67	67	68	69	71	73
Height of son (in inches)	67	68	64	68	72	70	69	70

Solution : Here N = No. of observations = 8. Take $\bar{X} = 67$ and $\bar{Y} = 68$

Computation of coefficient of correlation

Height of father in inches x	Deviations from assumed mean (67) $X = x - \bar{X}$	Square of deviations X^2	Height of son in inches y	Deviations from assumed mean (68) $Y = y - \bar{Y}$	Square of deviations Y^2	Product of deviations x and y XY
65	-2	4	67	-1	1	2
66	-1	1	68	0	0	0
67	0	0	64	-4	16	0
67	0	0	68	0	0	0
68	1	1	72	4	16	4
69	2	4	70	2	4	4
71	4	16	69	1	1	4
73	6	36	70	2	4	12
$\Sigma x = 546$	$\Sigma X = 10$	$\Sigma X^2 = 62$	$\Sigma y = 548$	$\Sigma Y = 4$	$\Sigma Y^2 = 42$	$\Sigma XY = 26$

$$\begin{aligned}
 \therefore \text{Coefficient of correlation, } r &= \frac{\Sigma XY - \frac{(\Sigma X) \times (\Sigma Y)}{N}}{\sqrt{\left\{ \Sigma X^2 - \frac{(\Sigma X)^2}{N} \right\} \left\{ \Sigma Y^2 - \frac{(\Sigma Y)^2}{N} \right\}}}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{(\Sigma XY) \times N - (\Sigma X)(\Sigma Y)}{\sqrt{((\Sigma X)^2 \times N - (\Sigma X)^2) \times ((\Sigma Y)^2 \times N - (\Sigma Y)^2)}} \\
 &= \frac{(26 \times 8) - (10 \times 4)}{\sqrt{(62 \times 8 - 10^2) \times (42 \times 8 - 4^2)}} = \frac{168}{355.98} = 0.472
 \end{aligned}$$

Example 2 : Calculate Karl Pearson's correlation coefficient for the following paired data.

X	38	45	46	38	35	38	46	32	36	38
Y	28	34	38	34	36	26	28	29	25	36

What inference would you draw from estimate ?

Solution : **Computation of coefficient of correlation**

X	$x = X - 38$	x^2	Y	$y = Y - 34$	y^2	xy
38	0	0	28	-6	36	0
45	7	49	34	0	0	0
46	8	64	38	4	16	32
38	0	0	34	0	0	0
35	-3	9	36	2	4	-6
38	0	0	26	-8	64	0
46	8	64	28	-6	36	-48
32	-6	36	29	-5	25	30
36	-2	4	25	-9	81	18
38	0	0	36	2	4	0
	$\Sigma x = 12$	$\Sigma x^2 = 226$		$\Sigma y = -26$	$\Sigma y^2 = 266$	$\Sigma xy = 26$

$$\begin{aligned}
 \therefore r &= \frac{\Sigma xy - \frac{\Sigma x \Sigma y}{N}}{\sqrt{\Sigma x^2 - \frac{(\Sigma x)^2}{N}} \sqrt{\Sigma y^2 - \frac{(\Sigma y)^2}{N}}} \\
 &= \frac{26 - \frac{12 \times (-26)}{10}}{\sqrt{226 - \frac{144}{10}} \sqrt{266 - \frac{676}{10}}}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{26 + 31.2}{\sqrt{226 - 14.4} \sqrt{266 - 676}} \\
 &= \frac{57.2}{\sqrt{211.6} \sqrt{198.4}} \\
 &= \frac{57.2}{(14.54)(14.08)} = \frac{57.2}{204.72} \\
 &= 0.2794
 \end{aligned}$$

Example 6 : Calculate the correlation coefficient for the following heights (in inches) of fathers (X) and their sons (Y) :

X	65	66	67	67	68	69	70	72
Y	67	68	65	68	72	72	69	71

Solution : **Computation of Correlation Coefficient**

X	Y	X ²	Y ²	XY
65	67	4225	4489	4355
66	68	4356	4624	4488
67	65	4489	4225	4355
67	68	4489	4624	4556
68	72	4624	5184	4896
69	72	4761	5184	4968
70	69	4900	4761	4830
72	71	5184	5041	5112
ΣX = 544	ΣY = 552	ΣX ² = 37028	ΣY ² = 38132	ΣXY = 37560

$$\text{Now } \bar{X} = \frac{1}{N} \Sigma X = \frac{544}{8} = 68, \quad \bar{Y} = \frac{1}{N} \Sigma Y = \frac{552}{8} = 69$$

$$\begin{aligned}
 \therefore r &= \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\frac{1}{N} \Sigma XY - (\bar{X})(\bar{Y})}{\sqrt{\left(\frac{1}{N} \Sigma X^2 - (\bar{X})^2\right) \left(\frac{1}{N} \Sigma Y^2 - (\bar{Y})^2\right)}} \\
 &= \frac{\frac{1}{8} \times 37560 - (68 \times 69)}{\sqrt{\left(\frac{37028}{8} - (68)^2\right) \left(\frac{38132}{8} - (69)^2\right)}} = 0.603
 \end{aligned}$$

Example 6 : Find the coefficient of correlation between X and Y

X	1	2	3	4	5	6	7	8	9
Y	12	11	13	15	14	17	16	19	18

Solution :

X	Y	X ²	Y ²	XY
1	12	1	144	12
2	11	4	121	22
3	13	9	169	39
4	15	16	225	60
5	14	25	196	70
6	17	36	289	102
7	16	49	256	112
8	19	64	361	132
9	18	81	324	162
45	135	285	2085	731

$$\text{Now } \bar{X} = \frac{45}{9} = 5; \quad \bar{Y} = \frac{135}{9} = 15$$

$$\begin{aligned} \therefore r &= \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} \\ &= \frac{\Sigma XY - N(\bar{X})(\bar{Y})}{\sqrt{(\Sigma X^2 - N(\bar{X})^2)(\Sigma Y^2 - N(\bar{Y})^2)}} \\ &= \frac{731 - 9(5)(15)}{\sqrt{(285 - 9 \times 25)(2085 - 9 \times 225)}} = 0.933 \end{aligned}$$

Example 8 : Find coefficient of correlation between X and Y for the following data

X	10	12	18	24	23	27
Y	13	18	12	25	30	10

Solution : We write the values in the following table

X	Y	XY	X ²	Y ²
10	13	130	100	269
12	18	216	144	324
18	12	216	324	144
24	25	600	476	625
23	30	690	529	900
27	10	270	729	100
$\Sigma X = 144$	$\Sigma Y = 108$	$\Sigma XY = 2122$	$\Sigma X^2 = 2402$	$\Sigma Y^2 = 2362$

We have, Coefficient of correlation, $r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\frac{1}{n} \sum XY - \bar{X}\bar{Y}}{\sigma_X \sigma_Y}$

where, $\bar{X} = \frac{1}{n} \sum X = \frac{1}{6} \times 114 = 19$, $\bar{Y} = \frac{1}{n} \sum Y = \frac{1}{6} \times 108 = 18$

$$\sigma_X^2 = \frac{1}{n} \sum X^2 - \bar{X}^2 = \frac{1}{6} \times 2402 - 361 = 400.33 - 361 = 39.33$$

$$\text{and } \sigma_Y^2 = \frac{1}{n} \sum Y^2 - \bar{Y}^2 = \left(\frac{1}{6} \times 2362 \right) - 324 = 393.67 - 324 = 69.67$$

$$\text{Thus } \text{Cov}(X, Y) = \frac{1}{6} (2122) - (19 \times 18) = 353.67 - 342 = 11.67$$

$$\therefore r = \frac{11.67}{\sqrt{39.33} \sqrt{69.67}} = 0.223$$

Example 9 : Calculate coefficient of correlation for the following data :

X	9	8	7	6	5	4	3	2	1
Y	15	16	14	13	11	12	10	8	9

Solution : We write the values in the table as follows

X	Y	XY	X ²	Y ²
9	15	135	81	225
8	16	128	64	256
7	14	98	49	196
6	13	78	36	169
5	11	55	25	121
4	12	48	16	144
3	10	30	9	100
2	8	16	4	64
1	9	9	1	81
$\sum X = 45$	$\sum Y = 108$	$\sum XY = 597$	$\sum X^2 = 285$	$\sum Y^2 = 1356$

We know that,

$$\bar{X} = \frac{1}{n} \sum X = \frac{45}{9} = 5; \quad \bar{Y} = \frac{1}{n} \sum Y = \frac{1}{9} \times 108 = 12$$

$$\sigma_X^2 = \frac{1}{n} \sum X^2 - \bar{X}^2 = \left(\frac{1}{9} \times 285 \right) - 25 = \frac{95}{3} - 25 = \frac{20}{3}$$

$$\sigma_Y^2 = \frac{1}{n} \sum Y^2 - \bar{Y}^2 = \left(\frac{1}{9} \times 1356 \right) - 644 = \frac{20}{3}$$

$$\text{We have, } \text{Cov}(X, Y) = \frac{1}{n} \sum XY - \bar{X}\bar{Y} = \frac{1}{9} \times 597 - 60 = \frac{199}{3} - 60 = \frac{19}{3}$$

$$\therefore r_{XY} = \frac{\frac{19}{3}}{\sqrt{\frac{20}{3}} \times \sqrt{\frac{20}{3}}} = \frac{19}{20} = 0.95$$

Example 10 : Find the coefficient of correlation between X and Y for the following data

X	1	2	3	4	5	6	7	8	9
Y	10	11	12	14	13	15	16	17	18

Solution :

Calculation of Coefficient of Correlation

X	Y	X ²	Y ²	XY
1	10	1	100	10
2	11	4	121	22
3	12	9	144	36
4	14	16	196	56
5	13	25	169	65
6	15	36	225	90
7	16	49	256	112
8	17	64	289	136
9	18	81	324	162
45	126	285	1824	689

$$\text{Here } \bar{X} = \frac{45}{9} = 5, \bar{Y} = \frac{126}{9} = 14$$

$\therefore r = \text{Coefficient of correlation}$

$$\begin{aligned}
 &= \frac{\sum XY - N(\bar{X})(\bar{Y})}{\sqrt{\sum X^2 - N(\bar{X})^2} \sqrt{\sum Y^2 - N(\bar{Y})^2}} = \frac{689 - 9(5)(14)}{\sqrt{(285 - 9(25))} \sqrt{(1824 - 9(196))}} \\
 &= \frac{689 - 9(5)(14)}{\sqrt{(285 - 9(25))} \sqrt{1824 - 9(196)}} = \frac{689 - 636}{\sqrt{285 - 225} \cdot \sqrt{1824 - 1764}} \\
 &= \frac{53}{\sqrt{60} \sqrt{60}} = \frac{53}{60} = 0.8833
 \end{aligned}$$

Example 11 : Given $n = 10, \sigma_x = 5.4, \sigma_y = 6.2$ and sum of the product of deviations from the mean of X and Y is 66 find the correlation co-efficient. [JNTU (H) Dec. 2011 (Set No. 4)]

Solution : Given $n = 10 ; \sigma_x = 5.4 ; \sigma_y = 6.2$

$$\text{We have } \sigma_x^2 = \frac{1}{n} \sum (x - \bar{x})^2$$

$$\Rightarrow (5.4)^2 10 = \sum (x - \bar{x})^2 \quad \dots (1)$$

$$\text{Also we have } \sigma_y^2 = \frac{1}{n} \sum (y - \bar{y})^2$$

$$\Rightarrow (6.2)^2 10 = \sum (y - \bar{y})^2 \quad \dots (2)$$

$$\text{Also given } \sum (x - \bar{x})(y - \bar{y}) = 66 \quad \dots (3)$$

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

$$= \frac{66}{(5.4)(6.2)}$$

$$= \frac{66}{334.8} = 0.1971$$

3.9 RANK CORRELATION COEFFICIENT

A British Psychologist Charles Adward Spearman found out the method of finding the coefficient of correlation by ranks. This method is based on rank and is useful in dealing with qualitative characteristics such as morality, character, intelligence and beauty. It can not be measured quantitatively as in the case of Pearson's coefficient of correlation. It is based on the ranks given to the observations. Rank correlation is applicable only to the individual observations. The formula for Spearman's rank correlation is given by

$$\rho = 1 - \frac{6 \sum D^2}{N(N^2 - 1)}$$

where $\rho \rightarrow$ Rank coefficient of correlation

$D^2 \rightarrow$ Sum of the squares of the differences of two ranks.

$N \rightarrow$ Number of paired observations

3.10 PROPERTIES OF RANK CORRELATION COEFFICIENT

1. The value of ρ lies between +1 and -1
2. If $\rho = 1$, there is complete agreement in the order of the ranks and the direction of the rank is same.

3. If $\rho = -1$ then there is complete disagreement in the order of the ranks and they are in opposite directions.

Procedure to solve problems :

1. When the ranks are given.

Step. 1. Compute the difference of two ranks and denote it by D .

Step. 2. Square D and get ΣD^2

Step. 3. Obtain ρ by substituting the figures in the formula.

2. When the ranks are not given, but actual data are given, then we must give ranks. We can give ranks by taking the highest as 1 or the lowest value as 1, next to the highest (lowest) as 2 and follow the same procedure for both the variables.

SOLVED EXAMPLES

Example 1 : Following are the ranks obtained by 10 students in two subjects, Statistics and Mathematics. To what extent the knowledge of the students in two subjects is related ?

Statistics	1	2	3	4	5	6	7	8	9	10
Mathematics	2	4	1	5	3	9	7	10	6	8

Solution :

Rank in Statistics (x)	Rank in Mathematics (y)	$D = (x - y)$	D^2
1	2	-1	1
2	4	-2	4
3	1	+2	4
4	5	-1	1
5	3	+2	4
6	9	-3	9
7	7	0	0
8	10	-2	4
9	6	+3	9
10	8	+2	4
			$\Sigma D^2 = 40$

$$\therefore \rho = 1 - \frac{6 \Sigma D^2}{N(N^2 - 1)} = 1 - \frac{6 \times 40}{10(10^2 - 1)} = 1 - \frac{240}{10(100 - 1)} = 1 - \frac{240}{990} = 1 - 0.24 = 0.76$$

Example 2 : The ranks of 16 students in Mathematics and Statistics are as follows :

(1,1) (2,10) (3,3) (4,4) (5,5) (6,7) (7,2) (8,6) (9,8) (10,11)
(11,15) (12,9) (13,14) (14,12) (15,16) (16,13)

Calculate the rank correlation coefficient for proficiencies of this group in Mathematics and Statistics.

Solution :

Ranks in Maths (X)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
Ranks in Statistics (Y)	1	10	3	4	5	7	2	6	8	11	15	9	14	12	16	13	
D = X – Y	0	–8	0	0	0	–1	5	2	1	–1	–4	3	–1	2	–1	3	ΣD = 0
D ²	0	64	0	0	0	1	25	4	1	1	16	9	1	4	1	9	ΣD ² = 136

$$\text{Rank correlation coefficient, } \rho = 1 - \frac{6 \Sigma D^2}{N(N^2 - 1)} = 1 - \frac{6 \times 136}{16 \times 255} = 1 - \frac{1}{5} = \frac{4}{5} = 0.8$$

Example 1 : A random sample of 5 college students is selected and their grades in Mathematics and statistics are found to be

	1	2	3	4	5
Mathematics	85	60	73	40	90
Statistics	93	75	65	50	80

Calculate Pearman's rank correlation coefficient.

Solution :

Marks in Mathematics X	Ranks x	Marks in Statistics Y	Rank y	Rank difference D = x - y	D ²
85	2	93	1	1	1
60	4	75	3	1	1
73	3	65	4	-1	1
40	5	50	5	0	0
90	1	80	2	-1	1
					$\Sigma D^2 = 4$

Here $N = 5, \Sigma D^2 = 4$

$$\begin{aligned} \therefore \text{Rank correlation, } \rho &= 1 - \frac{6 \Sigma D^2}{N(N^2 - 1)} = 1 - \frac{6 \times 4}{5(5^2 - 1)} \\ &= 1 - \frac{24}{5 \times 24} = 1 - \frac{1}{5} = 1 - 0.2 \\ &= 0.8 \end{aligned}$$

Example 4 : Ten competitors in a musical test were ranked by the three judges A, B and C in the following order.

Ranks by A	1	6	5	10	3	2	4	9	7	8
Ranks by B	3	5	8	4	7	10	2	1	6	9
Ranks by C	6	4	9	8	1	2	3	10	5	7

Using rank correlation method, discuss which pair of judges has the nearest approach to common likings in music.

Solution : Here $N = 10$.

Ranks by A (X)	Ranks by B (Y)	Ranks by C (Z)	$D_1 = X - Y$	$D_2 = X - Z$	$D_3 = Y - Z$	D_1^2	D_2^2	D_3^2
1	3	6	-2	-5	-3	4	25	9
6	5	4	1	2	1	1	4	1
5	8	9	-3	-4	-1	9	16	1
10	4	8	6	2	-4	36	4	16
3	7	1	-4	2	6	16	4	36
2	10	2	-8	0	8	64	0	64
4	2	3	2	1	-1	4	1	1
9	1	10	8	-1	-9	64	1	81
7	6	5	1	2	1	1	4	1
8	9	7	-1	1	2	1	1	4
Total			$\Sigma D_1 = 0$	$\Sigma D_2 = 0$	$\Sigma D_3 = 0$	$\Sigma D_1^2 = 200$	$\Sigma D_2^2 = 60$	$\Sigma D_3^2 = 214$

$$\therefore \rho_1(X, Y) = 1 - \frac{6 \Sigma D_1^2}{N(N^2 - 1)} = 1 - \frac{6 \times 200}{10 \times 99} = 1 - \frac{40}{33} = \frac{-7}{33}$$

$$\rho_2(X, Z) = 1 - \frac{6 \Sigma D_2^2}{N(N^2 - 1)} = 1 - \frac{6 \times 60}{10 \times 99} = 1 - \frac{4}{11} = \frac{7}{11}$$

$$\text{and } \rho_3(Y, Z) = 1 - \frac{6 \Sigma D_3^2}{N(N^2 - 1)} = 1 - \frac{6 \times 214}{10 \times 99} = 1 - \frac{214}{165} = -\frac{49}{165}$$

Since $\rho_2(X, Z)$ is maximum, we conclude that the pair of judges A and C has the nearest approach to common likings in music.

3.15 EQUAL OR REPEATED RANKS

If any two or more persons are bracketed equal in any classification or if there is more than one item with the same value in the series then the Spearman's Formula for calculating the rank correlation coefficient breaks down. In this case common ranks are given to repeated items. The common rank is the average of the ranks which these items would have assumed, if they were different from each other and the next item will get the rank next to ranks all ready assumed.

For example : If two individuals are placed in the 7th place, each of them are given the rank 7.5 and the next rank will be 9. Similarly if 3 are ranked equal at the 7th place then they are given the rank $\frac{7+8+9}{3} = 8$ which is the common rank assigned to each, and the next rank will be 10. We use a slightly different formula.

$$\rho = 1 - 6 \left\{ \frac{\Sigma D^2 + \frac{1}{12}(m^3 - m) + \frac{1}{12}(m^3 - m) \dots}{N^3 - N} \right\}$$

where m = the number of items whose ranks are common.

SOLVED EXAMPLES

Example 2 : From the following data calculate the rank correlation coefficient after making adjustment for tied ranks.

X	48	33	40	9	16	16	65	24	16	57
Y	13	13	24	6	15	4	20	9	6	19

Solution : First we have to assign ranks to the variables.

X	Rank (x)	Y	Rank (y)	D = x - y	D ²
48	8	13	5.5	2.5	6.25
33	6	13	5.5	0.5	0.25
40	7	24	10	-3	9.00
9	1	6	2.5	-1.5	2.25
16	3	15	7	4	16.00
16	3	4	1	2	4.00
65	10	20	9	1	1.00
24	5	9	4	1	1.00
16	3	6	2.5	5	0.25
57	9	19	8	1	1.00
					$\Sigma D^2 = 41$

16 is repeated 3 times in X items hence $m = 3$. Since 13 and 6 are repeated twice in Y items; hence $m = 2$.

$$\begin{aligned} \therefore \rho &= 1 - \frac{6 \left[\Sigma D^2 + \frac{1}{12}(m^3 - m) + \frac{1}{12}(m^3 - m) + \frac{1}{12}(m^3 - m) \right]}{N^3 - N} \\ &= 1 - \frac{6 \left[41 + \frac{1}{12}(3^3 - 3) + \frac{1}{12}(2^3 - 2) + \frac{1}{12}(2^3 - 2) \right]}{990} \\ &= +0.733 \end{aligned}$$

Example 2 : Obtain the rank correlation coefficient for the following data

X :	68	64	75	50	64	80	75	40	55	64
Y :	62	58	68	45	81	60	68	48	50	70

Solution :

Computation of Rank Correlation

X	Y	Rank X (x)	Rank Y (y)	D = x - y	D ²
68	62	4	5	-1	1
64	58	6	7	-1	1
75	68	2.5	3.5	-1	1
50	45	9	10	-1	1
64	81	6	1	5	25
80	60	1	6	-5	25
75	68	2.5	3.5	-1	1
40	48	10	9	1	1
55	50	8	8	0	0
64	70	6	2	4	16
				$\Sigma D = 0$	$\Sigma D^2 = 72$

In X - series, 75 occurs two times, the ranks should be allotted are 2 and 3.

Therefore the average of 2 and 3 i.e. 2.5 is given to both.

Next value 68, so rank = 4

64 occurs 3 times. The ranks should be 5, 6 and 7. So rank = $\frac{5+6+7}{3} = 6$

To ΣD^2 we add $\frac{m(m^2-1)}{12}$ for each value repeated, so for 75, $m = 2$, for 64, $m = 3$.

So for X series, correlation is $\frac{2(4-1)}{12} + \frac{3(9-1)}{12} = \frac{5}{2}$

In Y - series, 68 occurs twice, so rank = $\frac{3+4}{2} = 3.5$

68 occurs twice so $m = 2$

So for Y series, correlation is $\frac{2(4-1)}{12} = \frac{1}{2}$

$$\therefore \rho = \frac{1 - 6 \left(\Sigma D^2 + \frac{5}{2} + \frac{1}{2} \right)}{N(N^2 - 1)} = \frac{1 - 6(72 + 3)}{10 \times 99} = 0.545$$

Example 2 : A sample of 12 fathers and their elder sons gave the following data about their elder sons. Calculate the coefficient of rank correlation.

Fathers	65	63	67	64	68	62	70	66	68	67	69	71
Sons	68	66	68	65	69	66	68	65	71	67	68	70

Solution : We write the given values in the following table.

Fathers	Sons	Ranks	Ranks	$d_i = x_i - y_i$	d_i^2
65	68	9	5.5	3.5	12.25
63	66	11	9.5	1.5	2.25
67	68	6.5	5.5	1.0	1
64	65	10	11.5	-1.5	2.25
68	69	4.5	3	1.5	2.25
62	66	12	9.5	2.5	6.25
70	68	2	5.5	-3.5	12.25
66	65	8	11.5	3.5	12.25
68	71	4.5	1	-3.5	12.25
67	67	6.5	8	-1.5	2.25
69	68	3	5.5	-2.5	6.25
71	70	1	2	-1	1
					$\Sigma d_i^2 = 72.5$

Repeated values are given common rank, which is the mean of the ranks then would have got. In X series 68 and 67 appear twice.

$$\text{Correction factor in X series} = \frac{2(2^2 - 1)}{12} + \frac{2(2^2 - 1)}{12} = 1$$

In Y series 68 appears 4 times, 66 appears twice and 65 appears twice.

$$\text{Correction factor in Y series} = \frac{4(4^2 - 1)}{12} + \frac{2(2^2 - 1)}{12} + \frac{2(2^2 - 1)}{12} = 6$$

$$\therefore \text{Rank correlation coefficient} = r = \frac{1 - 6(72.5 + 7)}{12(12^2 - 1)} = 0.722$$

Example 1 : The difference between the ranks are 0.5, -6, -4.5, -3, -5, -1, 3, 0, 5,

5.5, 0, -0.5. For refracted ranks x and y . $\frac{\Sigma m(m^2 - 1)}{12} = 3.5$. $r = 0.44$. Find the number of terms.

Solution : Given Differences (d_i) : 0.5, -6, -4.5, -3, -5, -1, 3, 0, 5, 5.5, 0, -0.5

$$\therefore d_i^2 = 0.25, 36, 20.25, 9, 25, 1, 9, 0, 25, 30.25, 0, 0.25$$

$$\text{and } \Sigma d_i^2 = 156$$

$$\therefore r = 1 - 6 \left\{ \frac{\sum d_i^2 + \frac{\sum m(m^2 - 1)}{12}}{(N^2 - N)} \right\} = \frac{1 - (159.5) 6}{N^2 - N} = 1 - \frac{957}{N^2 - N}$$

$$\Rightarrow 0.44 = 1 - \frac{957}{N^2 - N}$$

$$\Rightarrow \frac{957}{N^2 - N} = 1 - 0.44 = 0.56 \Rightarrow \frac{957}{0.56} = N^2 - N$$

$$\Rightarrow 1708.92 = N^2 - N \Rightarrow 41.84 \times 40.84 = N(N - 1)$$

$$\therefore N = 42.$$

EXERCISE

1. Calculate the coefficient of correlation between X and Y from the following data :

X:	1	2	3	4	5	6	7
Y:	2	4	5	3	8	6	7

2. The following data gives the marks obtained by 10 students in accountancy and statistics.

R. No.	Accountancy	Statistics
1	45	35
2	70	90
3	65	70
4	30	40
5	90	95
6	40	40
7	50	80
8	75	80
9	85	80
10	60	50

Find the coefficient of correlation and interpret it.

3. Calculate coefficient of correlation between age and playing habit from the following data.

Age in years	Population	No. of players
15 - 20	1500	1200
20 - 25	2000	1560
25 - 30	4000	2280
30 - 35	3000	1500
35 - 40	2500	1000
40 - 45	1000	300
45 - 50	800	200
50 - 55	500	50
55 - 60	200	6

4. The following table gives the distribution of the total population and those who are totally and partially blind among them. Find the coefficient of correlation.

Age	No. of Persons	Blind
0 - 10	100	55
10 - 20	60	40
20 - 30	40	40
30 - 40	36	40
40 - 50	24	36
50 - 60	11	22
60 - 70	6	18
70 - 80	3	15

5. The following table gives the scores obtained by 11 students in English and Telugu Translation. Find the rank correlation coefficient.

Scores in English	40	46	54	60	70	80	82	85	85	90	95
Scores in Telugu	45	45	50	43	40	75	55	72	65	42	70

6. The ranks of the 15 students in two subjects A and B are given below, the two numbers within the brackets denoting the ranks of the same student in A and B respectively.

(1,10), (2,7), (3,2), (4,6), (5,4), (6,8), (7,3), (8,1), (9,11), (10,15), (11,9), (12,5), (13,14), (14,12), (15,13)

Use Spearman's formula to find the rank correlation coefficient.

ANSWERS

1. $r = 0.79$ 2. $r = 0.903$ 3. $r = -0.993$ 4. $r = 0.898$
 5. $r_s = 0.359$ 6. $r_s = 0.514$

CHAPTER - 4

REGRESSION

4.1 INTRODUCTION

The study of correlation measures the direction and strength of the relationship between two variables. In correlation we can estimate the value of one variable, when the value of the other variable is given. Since price and supply are correlated, we can find out the expected amount of supply for a given price or we can find out the required price to get a given amount of supply.

In regression, we can estimate the value of one variable with the value of the other variable which is known. The statistical method which helps us to estimate the unknown value of one variable from the known value of the related variable is called **regression**. The line described in the average relationship between two variables is known as **line of regression**. Now a days we are using the term **estimating line** instead of regression line.

4.2 USES

1. It is used to estimate the relation between two economic variables like Income and Expenditure.
2. It is a highly valuable tool in Economics and Business.
3. It is widely used for prediction purpose.
4. We can calculate coefficient of correlation and coefficient of determination with the help of the regression coefficient.
5. It is useful in statistical estimation of demand curves, supply curves, production function, cost function and consumption function etc.

4.3 COMPARISON BETWEEN CORRELATION AND REGRESSION

The correlation coefficient is a measure of degree of covariability between two variables, while the regression establishes a functional relation between dependent and independent variables, so that the former can be predicted for a given value of the later. In correlation, both the variables x and y are random variables, where as in Regression, x is a random variable and y is fixed variable. The coefficient of correlation is a relative measure where as Regression coefficient is an absolute figure.

4.4 METHODS OF STUDYING REGRESSION

We have two methods for studying regression

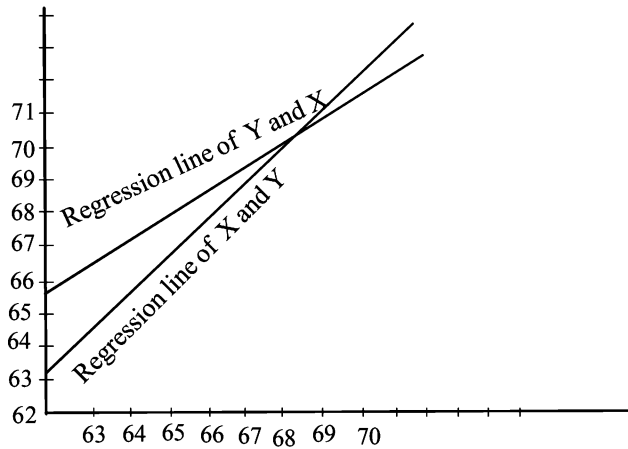
1. Graphic method
2. Algebraic method.

1. Graphic Method :

In this method, the points representing the pairs of values of the variables are plotted on a graph. The independent variable is taken on X - axis and the dependent variables on Y - axis. These points form a scatter diagram. A regression line is drawn between these points by free hand.

e.g. Fit a regression line on the scatter diagram for the following data.

X	Y
65	68
67	68
62	66
70	68
67	67
69	68
71	70



2. Algebraic Method :

Regression Line : A regression line is a straight line fitted to the data by the method of least squares. It indicates the best possible mean value of one variable corresponding to the mean value of the other. There are always two regression lines constructed for the relationship between two variables X and Y. Thus one regression line shows the regression of X upon Y and the other shows the regression of Y on X.

4.5 LEAST - SQUARES CURVE FITTING PROCEDURES

With an experimental data, the data is plotted on a graph paper and a straight line is drawn through the plotted points. This is the usual method to fit a mathematical equation to experimental data. The method of least squares is the most systematic procedure to fit a unique curve through the given data points. Its application is wide in practical computations.

Let the set of data points be $(x_i, y_i), i = 1, 2, \dots, m$. Suppose the curve $y = f(x)$ is fitted to this data. Let the observed value at $x = x_i$ is y_i and the corresponding value on the curve is $f(x_i)$. Let e_i be the error of approximation at $x = x_i$. Then we have

$$e_i = y_i - f(x_i) \quad \dots(1)$$

$$\begin{aligned} \text{Consider } S &= [y_1 - f(x_1)]^2 + [y_2 - f(x_2)]^2 + \dots + [y_m - f(x_m)]^2 \\ &= e_1^2 + e_2^2 + \dots + e_m^2 \end{aligned} \quad \dots(2)$$

The method of least squares consists of minimising S.

4.5.1 FITTING A STRAIGHT LINE

Let $y = a_0 + a_1x$ be the straight line to be fitted for the given data. Then

$$S = [y_1 - (a_0 + a_1x_1)]^2 + [y_2 - (a_0 + a_1x_2)]^2 + \dots + [y_m - (a_0 + a_1x_m)]^2 \quad \dots(3)$$

If S is to be minimum, we must have

$$\begin{aligned} \frac{\partial S}{\partial a_0} = 0 &\Rightarrow -2[y_1 - (a_0 + a_1x_1)] - 2[y_2 - (a_0 + a_1x_2)] - \dots \\ &\quad - 2[y_m - (a_0 + a_1x_m)] = 0 \end{aligned} \quad \dots(4)$$

$$\begin{aligned} \frac{\partial S}{\partial a_1} = 0 &\Rightarrow -2x_1[y_1 - (a_0 + a_1x_1)] - 2x_2[y_2 - (a_0 + a_1x_2)] - \dots \\ &\quad - 2x_m[y_m - (a_0 + a_1x_m)] = 0 \end{aligned} \quad \dots(5)$$

Simplifying these equations, we get

$$ma_0 + a_1(x_1 + x_2 + \dots + x_m) = y_1 + y_2 + \dots + y_m$$

$$\text{and } a_0(x_1 + x_2 + \dots + x_m) + a_1(x_1^2 + x_2^2 + \dots + x_m^2) = x_1y_1 + x_2y_2 + \dots + x_my_m \quad \dots(6)$$

Writing in simplified form, we have

$$ma_0 + a_1 \sum_{i=1}^m x_i = \sum_{i=1}^m y_i \quad \dots(7)$$

$$\text{and } a_0 \sum_{i=1}^m x_i + a_1 \sum_{i=1}^m x_i^2 = \sum_{i=1}^m x_i y_i \quad \dots(8)$$

Since we know x_i and y_i , equations (4) and (5) can be solved for two unknowns a_0 and a_1 . Equations (7) and (8) are called *normal equations*.

These equations can be written as

$$\Sigma y = ma_0 + a_1 \Sigma x$$

$$\Sigma xy = a_0 \Sigma x + a_1 \Sigma x^2$$

Differentiating (4) and (5) once again w.r.t. a_0 and a_1 we get, $\frac{\partial^2 S}{\partial a_0^2}$ and $\frac{\partial^2 S}{\partial a_1^2}$.

We find that these two will be positive at a_0 and a_1 . Thus S is minimum at a_0 and a_1 .

Dividing the equation (7) with m we obtain, $a_0 + a_1 \bar{x} = \bar{y}$ where $(\bar{x}, \bar{y})$ is a centroid of the given data points. Thus we can conclude that the fitted straight line passes through centroid of the data points.

Result : Derive the normal equations to fit the straight line $y = a + bx$.

Sol. : Taking $a_0 = a$ and $a_1 = b$ in the above, we get the result.

4.6 REGRESSION EQUATION

Regression equation is an algebraic expression of the regression line. It can be classified into Regression equation, Regression coefficient, individual observation and group discussion.

The standard Form of the Regression equation is $Y = a + bX$ where a, b are called constants. ' a ' indicates the value of Y when $X = 0$. It is called **Y-intercept**. ' b ' indicates the value of slope of the regression line and gives a measure of change of Y for a unit change in X . It is also called the **Regression Coefficient of Y on X**. Similarly, if the equation of the line is $X = a + bY$ then the constants ' a ' indicates the X - intercept and ' b ' indicates the slope of the regression line and is known as **Regression Coefficient of X on Y**.

Thus if we know the value of a and b we can easily compute the value of Y for any given value of X . The values of a and b are found with the help of the following Normal equations.

Regression Equation of Y on X :

Normal equations are

$$\Sigma Y = Na + b \Sigma X$$

$$\Sigma XY = a \Sigma X + b \Sigma X^2$$

Regression Equation of X on Y :

Normal equations are

$$\Sigma X = Na + b \Sigma Y$$

$$\Sigma XY = a \Sigma Y + b \Sigma Y^2$$

Illustration : Determine the equation of a straight line which best fits the data.

X :	10	12	13	16	17	20	25
Y :	10	22	24	27	29	33	37

Solution : Straight line is $Y = a + bx$

The two normal equations are

$$\Sigma Y = b \Sigma X + N a$$

$$\Sigma XY = b \Sigma X^2 + a \Sigma X$$

X	X^2	Y	XY
10	100	10	100
12	144	22	264
13	169	24	312
16	256	27	432
17	289	29	493
20	400	33	660
25	625	37	925
$\Sigma X = 113$	$\Sigma X^2 = 1938$	$\Sigma Y = 182$	$\Sigma XY = 3186$

Substituting the values in the above normal equations, we get

$$113b + 7a = 182 \quad \dots (1)$$

$$1983b + 113a = 3186 \quad \dots (2)$$

Solving (1) and (2), we get $a = 0.82, b = 1.56$

Thus the equation of the straight line is $Y = a + bX$

$$\therefore Y = 0.82 + 1.56 X$$

This is called the regression equation of Y on X.

4.7 DEVIATION TAKEN FROM ARITHMETIC MEAN OF X ON Y

This method is easier and simpler than the previous method to find the values of a and b. We can find out the deviations of X and Y series from their respective means.

Regression equation of X on Y is given by $X - \bar{X} = r \frac{\sigma_x}{\sigma_y} (Y - \bar{Y})$, where

$\bar{X}$ = Mean of X series, $\bar{Y}$ = Mean of Y series

The regression coefficient of X on Y $= b_{XY} = r \frac{\sigma_x}{\sigma_y} = \frac{\Sigma xy}{\Sigma y^2}$, where $x = X - \bar{X}$, $y = Y - \bar{Y}$

The regression coefficient of Y on X $= b_{YX} = r \frac{\sigma_y}{\sigma_x} = \frac{\Sigma xy}{\Sigma x^2}$, where $x = X - \bar{X}$, $y = Y - \bar{Y}$

Thus $r^2 = b_{XY} \times b_{YX}$

4.8 RESULT

Prove that regression line always pass through the point $(\bar{X}, \bar{Y})$.

Sol. Regression line of X on Y is given by the equation

$$X - \bar{X} = r \frac{\sigma_x}{\sigma_y} (Y - \bar{Y})$$

If we take the value of Y to be equal to $\bar{Y}$, then this equation becomes

$$X - \bar{X} = 0 \text{ or } X = \bar{X}$$

When $Y = \bar{Y}$, we have $X = \bar{X}$.

It follows that there is a point $(\bar{X}, \bar{Y})$ on the regression line of X on Y.

In the same manner, by taking the regression equation of Y on X, we can show that there is a point $(\bar{X}, \bar{Y})$ on the regression line of Y on X.

This means that there is a point $(\bar{X}, \bar{Y})$ which appears on both the regression lines. There can be only one such point on the scatter diagram and it is this common point $(\bar{X}, \bar{Y})$ where two lines intersect.

If we find out the values of X and Y by selecting, the two regression equations as simultaneous equations, we shall see that the values of X and Y are the values of $\bar{X}$ and $\bar{Y}$ respectively.

4.9 RESULT

If the two regression lines of y on x and x on y are respectively

$$a_1x + b_1y + c_1 = 0 \text{ and } a_2x + b_2y + c_2 = 0 \text{ prove that } a_1b_2 < a_2b_1.$$

Sol. Regression line of y on x is

$$\begin{aligned} b_1y &= -a_1x + c_1 \\ \Rightarrow y &= \frac{-a_1}{b_1}x - \frac{c_1}{b_1} \quad \dots (1) \end{aligned}$$

Here b_{yx} = coefficient of $y = -\frac{a_1}{b_1}$

Regression line of x on y is

$$\begin{aligned} a_2x + b_2y + c_2 &= 0 \\ \Rightarrow a_2x &= -b_2y - c_2 \Rightarrow x = \frac{-b_2y}{a_2} - \frac{c_2}{a_2} \end{aligned}$$

Here b_{xy} = coefficient of $x = -\frac{b_2}{a_2}$

We have $r = \sqrt{b_{yx} \times b_{xy}}$

$$\therefore r = \sqrt{\frac{-a_1}{b_1} \times \frac{-b_2}{a_2}} = \sqrt{\frac{a_1 b_2}{a_2 b_1}} < 1 \Rightarrow a_1 b_2 < a_2 b_1$$

SOLVED EXAMPLES

Example 1 : For the following data, find equations of the two regression lines.

X	1	2	3	4	5
Y	15	25	35	45	55

Solution : Calculation of regression coefficients

X	Y	$x = X - \bar{X}$	x^2	$y = Y - \bar{Y}$	y^2	xy
1	15	-2	4	-20	400	40
2	25	-1	1	-10	100	10
3	35	0	0	0	0	0
4	45	1	1	10	100	10
5	55	2	4	20	400	40
15	175		10		1000	100

We have $\bar{X} = \frac{\sum X}{5} = \frac{15}{5} = 3$

and $\bar{Y} = \frac{\sum Y}{5} = \frac{175}{5} = 35$.

Now $b_{yx} = \frac{\sum xy}{\sum x^2} = \frac{100}{10} = 10$

$\therefore$ Regression equation of y on x is given by ,

$$y - \bar{y} = b_{yx}(x - \bar{x}) \Rightarrow y - 35 = 10(x - 3)$$

Now $b_{xy} = \frac{\sum xy}{\sum y^2} = \frac{100}{1000} = \frac{1}{10}$

$\therefore$ Regression equation of x on y is given by ,

$$x - \bar{x} = b_{xy}(y - \bar{y}) \Rightarrow x - 3 = \frac{1}{10}(y - 35)$$

Example 1 : In the following table S is weight of Potassium bromide which will dissolve in 100 grms. of water at $V^\circ \text{C}$. Fit an equation of the form $S = mT + b$ by the method of least squares. Use this relation to estimate S when $T = 50^\circ$.

T	0	20	40	60	80
S	54	65	75	85	96

Solution :

T	$d_T = \frac{T-40}{10}$	d_T^2	S	$d_S = \frac{S-75}{10}$	d_S^2	$d_T d_S$
0	-4	16	54	-2.1	4.41	8.4
20	-2	4	65	-1.0	1.00	2.0
40	0	0	75	0	0.00	3.0
60	+2	4	85	+1.0	1.00	2.0
80	+4	16	96	+2.1	4.41	8.4
200	0	40	375	0	10.82	20.8

$$\text{Now, } m = \frac{\sum d_T d_S}{\sum d_T^2} = \frac{20.8}{40} = 0.52$$

and b is given by the equation $\sum S = m \sum T + Nb$

$$\therefore 375 = (0.52)(200) + 5b \Rightarrow b = 54.2$$

When $T = 50^\circ \text{C}$, $S = 0.52 \times 50 + 54.2 = 80.2$

Example 1 : A panel of two judges P and Q graded seven dramatic performances by independently awarding marks as follows :

Performance	1	2	3	4	5	6	7
Marks by P	46	42	44	40	43	41	45
Marks by Q	40	38	36	35	39	37	41

The eight performance, which judge Q would not attend, was awarded 37 marks by judge P. If judge Q had also been present, how many marks would be expected to have been awarded by him to the eight performance.

Solution : Calculation of Regression Coefficients

Performance	X (Marks by P)	$x = X - \bar{X}$	x^2	Y (Marks by Q)	$y = Y - \bar{Y}$	xy
1	46	+3	9	40	+2	+6
2	42	-1	1	38	0	0
3	44	+1	1	36	-2	2
4	40	-3	1	35	-3	+9
5	43	0	0	39	+1	0
6	41	-2	4	37	-1	+2
7	45	+2	4	41	+3	+6
	301		28	266		21

$$\bar{X} = \frac{301}{7} = 43; \bar{Y} = \frac{266}{7} = 38$$

Regression equation of Y on X is given by

$$Y - \bar{Y} = b_{yx}(X - \bar{X}) \quad \dots (1)$$

$$\text{Now } b_{yx} = \frac{\sum xy}{\sum x^2} = \frac{21}{28} = 0.75$$

Substituting the values in equation (1), we get

$$Y - 38 = 0.75(X - 43) \Rightarrow Y = 0.75X + 5.75$$

$$\text{If } X = 37, \text{ then } Y = 0.75(37) + 5.75 = 33.5$$

Hence if the judge Q would have been present, he would have awarded 33.5 marks to the eight performance.

Example 4 : Find the most likely production corresponding to a rainfall 40 from the following data:

	Rain fall (X)	Production (Y)
Average	30	500 Kgs
Standard deviation	5	100 Kgs
Coefficient of correlation	0.8	

Solution : We have to calculate the value of Y when $X = 40$.

So we have to find the regression equation of Y on X.

Mean of X series, $\bar{X} = 30$; Mean of Y series, $\bar{Y} = 500$

σ of X series, $\sigma_x = 5$; σ of Y series, $\sigma_y = 100$

Regression of Y on X is

$$Y - \bar{Y} = r \cdot \frac{\sigma_x}{\sigma_y}(X - \bar{X})$$

$$\Rightarrow Y - 500 = (0.8) \frac{5}{100}(X - 30)$$

$$\text{When } X = 40, Y - 500 = \frac{4}{100}(40 - 30) = \frac{40}{100}$$

$$\Rightarrow Y = 500 + \frac{4}{10} = 500.4$$

Hence the expected value of Y is 500.4 kg.

Example 4 : From a sample of 200 pairs of observation the following quantities were calculated.

$$\Sigma X = 11.34, \Sigma Y = 20.78, \Sigma X^2 = 12.16, \Sigma Y^2 = 84.96, \Sigma XY = 22.13$$

From the above data show how to compute the coefficients of the equation $Y = a + bX$.

Solution : We can compute the coefficients of the equation $Y = a + bX$ by solving the normal equations :

$$\Sigma Y = na + b\Sigma X \text{ and } \Sigma XY = a\Sigma X + a\Sigma X^2$$

Substituting the values

$20.72 = 200a + 11.34b$ $\Rightarrow a = \frac{20.72 - 11.34b}{200}$ $= 0.1036 - 0.0567b \dots (1)$ $= 0.1036 - 0.0567(1.82)$ $\therefore a = 0.0005$	and	$22.13 = 11.13a + 12.16b$ $\Rightarrow 22.13 = 11.34(0.1036 - 0.567b) + 12.16b, \text{ using (1)}$ $\Rightarrow 22.13 = 1.175 - 0.643b + 12.16b$ $\Rightarrow 20.955 = 11.517b$ $\therefore b = \frac{20.955}{11.517} = 1.82$
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Example 6 : Criticise the following :

Regression coefficient of Y on X is 0.7 and that of X on Y is 3.2

Solution : $r = \sqrt{b_{yx} \cdot b_{xy}} = \sqrt{3.3 \times 0.7} = \sqrt{2.24} = 1.5$ (approximately)

But correlation coefficient can not exceed 1.

Hence there is some inconsistency in the information given.

Example 6 : Write the relation between correlation and regression coefficients. Is it possible to have two variables x and y with regression coefficient as 2.8 and -0.5 ? Explain.

Solution : We have correlation coefficient, $r = \sqrt{\sigma_{xy} \times \sigma_{yx}}$ where, σ_{xy} and σ_{yx} are regression coefficients.

If we take $\sigma_{xy} = 2.8$ and $\sigma_{yx} = -0.5$, then

$$r = \sqrt{\sigma_{xy} \times \sigma_{yx}} = \sqrt{2.8 \times (-0.5)} = \sqrt{-1.4} \text{ which is not a real number.}$$

Thus it is not possible.

4.10 DEVIATIONS TAKEN FROM THE ASSUMED MEAN

If the actual mean is fraction this method is used.

In this method we take deviations from the assumed mean instead of Arithmetic Mean.

$$X - \bar{X} = r \frac{\sigma_x}{\sigma_y} (Y - \bar{Y})$$

We can find out the value of $r \frac{\sigma_x}{\sigma_y}$ by applying the following formula

$$r \frac{\sigma_x}{\sigma_y} = \frac{\Sigma dx dy - \frac{\Sigma dx \times \Sigma dy}{N}}{\Sigma dy^2 - \frac{(\Sigma dy)^2}{N}}, \text{ where } dx = X - A; dy = Y - A$$

The regression equations of Y on X is $Y - \bar{Y} = r \frac{\sigma_y}{\sigma_x} (X - \bar{X})$

We can find out the value of $r \frac{\sigma_y}{\sigma_x}$ by applying the following formula :

$$b_{yx} = r \frac{\sigma_y}{\sigma_x} = \frac{\Sigma dx dy - \frac{\Sigma dx \times \Sigma dy}{N}}{\Sigma dx^2 - \frac{(\Sigma dx)^2}{N}}$$

SOLVED EXAMPLES

Example 1 : Price indices of cotton and wool are given below for the 12 months of a year. Obtain the equations of lines of regression between the indices.

Price index of cotton (X)	78	77	85	88	87	82	81	77	76	83	97	93
Price index of wool (Y)	84	82	82	85	89	90	88	92	83	89	98	99

Solution :

Calculation of Regression Equation

Price index of cotton (X)	(X - 84) = dx	dx ²	Price index of wool (Y)	(Y - 88) = dy	dy ²	dx dy
78	-6	36	84	-4	16	24
77	-7	49	82	-6	36	42
85	1	1	82	-6	36	-6
88	4	16	85	-3	9	-12
87	3	9	89	1	1	3
82	-2	4	90	2	4	-4
81	-3	9	88	0	0	0
77	-7	49	92	4	16	-28
76	-8	64	83	-5	25	40
83	-1	1	89	1	1	-1
97	13	169	98	10	100	130
93	9	81	99	11	121	99
Total = 1004	-4	488	1061	5	365	287

$$\text{Now } b_{xy} = \frac{\Sigma dx dy - \left(\frac{\Sigma dx \times \Sigma dy}{N} \right)}{\Sigma dy^2 - \frac{(\Sigma dy)^2}{N}} = \frac{287 - \left(\frac{-4 \times 5}{12} \right)}{365 - \frac{(5)^2}{12}} = 0.795$$

Now Regression equation of X on Y : $X - \bar{X} = b_{xy} (Y - \bar{Y})$

$$\Rightarrow X - 83.67 = 0.795 (Y - 88.42) \Rightarrow X = 0.795Y + 13.38$$

$$\text{Now } b_{yx} = \frac{\sum dx \, dy - \left(\frac{\sum dx \times \sum dy}{N} \right)}{\sum dx^2 - \frac{(\sum dx)^2}{N}} = \frac{287 - \left(\frac{-4 \times 5}{12} \right)}{488 - \frac{(-4)^2}{12}} = 0.59$$

Regression equation of Y on X : $Y - \bar{Y} = b_{yx} (X - \bar{X})$

$$\Rightarrow Y - 88.42 = 0.59 (X - 83.67)$$

$$\Rightarrow Y = 0.59X + 39.05$$

Example 4 : From the following data, calculate

(i) Correlation coefficient (ii) Standard deviation of Y (σ_y)

$$b_{xy} = 0.85 ; b_{yx} = 0.89 ; \sigma_x = 3 .$$

Solution : (i) Coefficient of correlation :

$$r = \sqrt{b_{xy} \times b_{yx}} = \sqrt{0.85 \times 0.89} = 0.87$$

(ii) Standard deviation of Y :

$$r \times \frac{\sigma_x}{\sigma_y} = 0.85 = 0.87 \times \frac{3}{\sigma_y} = 0.85 \Rightarrow \sigma_y = 3.07$$

Example 3 : Given the following data, calculate the expected value of Y when $X = 12$

	x	y
Average	7.6	14.8
Standard deviation	3.6	2.5
$r = 0.99$		

Solution : We have to calculate the expected value of Y when X is 12.

So we have to find out regression equation of Y on X.

Mean of X series ($\bar{X}$) = 7.6

Mean of Y series ($\bar{Y}$) = 14.8

σ of X series = 3.6

σ of Y series = 2.5

Coefficient of correlation (r) = 0.99

Regression of Y on X is

$$Y - \bar{Y} = r \frac{\sigma_y}{\sigma_x} (X - \bar{X})$$

$$\Rightarrow Y - 14.8 = 0.99 \times \frac{2.5}{3.6} (X - 7.6) = 0.688X + 9.57$$

When $X = 12$, $Y = 0.688(12) + 9.57 = 17.826$

Hence the expected value of Y is 17.83.

Example 4 : Determine the equation of a straight line which best fits the data.

X :	10	12	13	16	17	20	25
Y :	10	22	24	27	29	33	37

Solution : Let the required straight line be $Y = a + bX$

The two normal equations are

$$\Sigma Y = b \Sigma X + Na$$

$$\Sigma XY = b \Sigma X^2 + a \Sigma X$$

X	X^2	Y	XY
10	100	10	100
12	144	22	264
13	169	24	312
16	256	27	432
17	289	29	493
20	400	33	660
25	625	37	925
$\Sigma X = 113$	$\Sigma X^2 = 1938$	$\Sigma Y = 182$	$\Sigma XY = 3186$

Substituting the values in the above normal equations, we get

$$113b + 7a = 182 \quad \dots (1)$$

$$1983b + 113a = 3186 \quad \dots (2)$$

Multiplying (1) by 113;

$$12769b + 791a = 20566 \quad \dots (3)$$

Multiplying (2) by 7;

$$13881b + 791a = 22302 \quad \dots (4)$$

$$(4) - (3) \text{ gives } b = \frac{1736}{1112} = 1.56 \Rightarrow a = 0.82$$

Hence the equation of the required straight line is

$$Y = a + bX \text{ i.e., } Y = 0.82 + 1.56X$$

This is called regression equation of Y on X.

Example 5 : Calculate the regression equations of Y on X from the data given below, taking deviations from actual means of X and Y.

Price (RS.)	10	12	13	12	16	15
Amount Demanded	40	38	43	45	37	43

Estimate the likely demand when the price is Rs. 20.

Solution : Calculation of Regression Equation

x	$(x-13) = X$	X^2	y	$(y-41) = Y$	Y^2	XY
10	-3	9	40	-1	1	3
12	-1	1	38	-3	9	3
13	0	0	43	2	4	0
12	-1	1	45	4	16	-4
16	3	9	37	-4	16	-12
15	2	4	43	2	4	4

Regression equation of Y on X is $Y - \bar{Y} = r \frac{\sigma_y}{\sigma_x} (X - \bar{X})$ (1)

$$\text{Now } r \frac{\sigma_y}{\sigma_x} = \frac{\sum XY}{\sum X^2} = -0.25$$

$$(1) \Rightarrow Y - 41 = -0.25(X - 13) \Rightarrow Y = -0.25X + 44.25$$

When X is 20; Y = 39.25

When the price is Rs. 20, the likely demand is 39.25.

Example 6 : The heights of mothers and daughters are given in the following table. From the two tables of regression estimate the expected average height of daughter when the height of the mother is 64.5 inches.

Height of the mother(inches)	62	63	64	64	65	66	68	70
Height of the daughter (inches)	64	65	61	69	67	68	71	65

Solution : Let X = Height of the mother
and Y = Height of the daughter

Let $dx = X - 65$ and $dy = Y - 67$. Then

$$\sum X = 522, \sum dx = 2, \sum dx^2 = 50, \sum Y = 530$$

$$\sum dy = -6, \sum dy^2 = 74, \sum dx dy = 20$$

$$\text{Now } \bar{X} = \frac{\sum X}{N} = \frac{522}{8} = 66.25, \bar{Y} = \frac{\sum Y}{N} = \frac{530}{8} = 65.25$$

$$\therefore b_{yx} = \frac{\frac{\sum dx dy - \sum dx \sum dy}{N}}{\frac{\sum dx^2 - \frac{(\sum dx)^2}{N}}} = \frac{20 - \frac{2 \times -6}{8}}{50 - \frac{2^2}{8}} = 0.434$$

Hence Regression equation of Y on X is $Y - \bar{Y} = b_{yx}(X - \bar{X})$

$$\Rightarrow Y = 37.93 + 0.434X \text{ when } X = 64.5 \text{ then } Y = 65.923$$

Example 6 : The following data, based on 450 students, are given for marks in statistics are Economics at a certain examination.

Mean marks in statistics = 40

Mean marks in Economics = 8

S. D. of marks in statistics = 12

Variance of marks (Economics) = 256

Sum of the products of deviations of marks from this respective mean 42075. Give the equations of the two lines of regression and estimate the average marks in Economics of candidates who obtained 50 marks in statistics.

Solution : Given $\bar{X}$ = Mean of marks in Statistics = 40

$\bar{Y}$ = Mean of marks in Economics = 8

σ_X = S. D. of marks in Statistics = 12

σ_Y = S. D. of marks in Economics = 16

Coefficient of correlation, $r = \frac{\sum XY}{N\sigma_X\sigma_Y} = \frac{42075}{450 \times 12 \times 16} = 0.49$

Regression equation of X on Y is

$$X - \bar{X} = \frac{r\sigma_X}{\sigma_Y}(Y - \bar{Y})$$

$$\Rightarrow X = 0.37Y + 22.24$$

Regression equation of Y on X is

$$Y - \bar{Y} = \frac{r\sigma_Y}{\sigma_X}(X - \bar{X})$$

$$\Rightarrow Y = 0.65X + 22$$

When $X = 50$, $Y = 54.5$

Example 8 : The following calculations have been made for prices of 12 stocks (X) in stock exchange, on a certain day along with the volume of the sales in thousands of shares (Y). From these calculations find the regression equation of prices of stocks, on the volume of the sales of shares.

$$\sum X = 580, \sum Y = 370, \sum XY = 11499, \sum X^2 = 41658, \sum Y^2 = 17.206$$

Solution : We have

$$\text{Mean} = \bar{X} = \frac{\sum X}{N} = \frac{580}{12} = 48.33$$

$$\text{and } \bar{Y} = \frac{\sum Y}{N} = \frac{370}{12} = 30.83$$

$$\therefore b_{xy} = \frac{\sum XY - N\bar{X}\bar{Y}}{\sum Y^2 - N(\bar{Y})^2} = \frac{11494 - 12 \times 48.33 \times 30.83}{17206 - 12(30.83)^2} = -1.101$$

Thus regression equation of X on Y is

$$X - 48.33 = (-1.101)(Y - 30.83)$$

$$\Rightarrow X = -1.101Y + 82.27$$

Example 9 : Given the following information regarding a distribution $N = 5, \bar{X} = 10, \bar{Y} = 20, \sum(X - Y)^2 = 100, (Y - 10)^2 = 160$. Find the regression coefficients and hence the coefficient of correlation.

Solution : Here $dx = X - 4, dy = Y - 10$

$$\bar{X} = A + \frac{\sum dx}{N} \Rightarrow 10 = 4 + \frac{\sum dx}{5} \Rightarrow \sum dx = 30 \quad (\text{here } A = 4)$$

$$\text{Also } \bar{Y} = B + \frac{\sum dy}{N} \Rightarrow 20 = 10 + \frac{\sum dy}{5} \Rightarrow \sum dy = 50, (B = 10)$$

We know that

$$b_{YX} = \frac{\sum dxdy - \frac{\sum dx \sum dy}{N}}{\sum dx^2 - \frac{(\sum dx)^2}{N}} = \frac{80 - \frac{30 \times 50}{5}}{100 - \frac{(30)^2}{5}} = \frac{80 - 300}{100 - 180} = \frac{-220}{-80} = 2.75$$

$$b_{XY} = \frac{\sum dxdy - \frac{\sum dx \sum dy}{N}}{\sum dy^2 - \frac{(\sum dy)^2}{N}} = \frac{80 - \frac{30 \times 50}{5}}{160 - \frac{(50)^2}{5}} = \frac{-220}{-340} = 0.65$$

$$\text{Coefficient of correlation} = \pm \sqrt{b_{XY} + b_{YX}} = \sqrt{0.65 \times 2.75} = \sqrt{1.7875} = 1.337$$

We have b_{XY} and b_{YX} is positive. So r is also positive.

Here we get the coefficient of correlation as more than 1. The given data is inconstant.

Example 10 : Given $n = 10, \sigma_x = 5.4, \sigma_y = 6.2$ and sum of the product of deviations from the mean of X and Y is 66 find the correlation co-efficient.

Solution : Given $n = 10; \sigma_x = 5.4; \sigma_y = 6.2$

$$\text{We have } \sigma_x^2 = \frac{1}{n} \sum (x - \bar{x})^2$$

$$\Rightarrow (5.4)^2 10 = \sum (x - \bar{x})^2 \quad \dots (1)$$

$$\text{and } \sigma_y^2 = \frac{1}{n} \sum (y - \bar{y})^2$$

$$\Rightarrow (6.2)^2 10 = \sum (y - \bar{y})^2 \quad \dots (2)$$

$$\text{Given } \sum (x - \bar{x})(y - \bar{y}) = 66 \quad \dots (3)$$

$$\therefore r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}} = \frac{66}{(5.4)(6.2)} = \frac{66}{334.8} = 0.1971$$

4.11 ANGLE BETWEEN TWO REGRESSION LINES

Let the lines of regression of X on Y and Y on X are respectively given by

$$x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y}) \quad \dots (1) \quad \text{and} \quad y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x}) \quad \dots (2)$$

$$\text{Slope of the line (1)} = m_1 = \frac{1}{r} \frac{\sigma_y}{\sigma_x}; \quad \text{Slope of the line (2)} = m_2 = r \frac{\sigma_y}{\sigma_x}$$

Let θ be the angle between two regression lines X on Y and Y on X. Then

$$\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2} = \frac{\frac{1}{r} \frac{\sigma_y}{\sigma_x} - r \frac{\sigma_y}{\sigma_x}}{1 + \left(\frac{1}{r} \frac{\sigma_y}{\sigma_x} \right) \left(r \frac{\sigma_y}{\sigma_x} \right)}$$

$$= \frac{\frac{\sigma_y}{\sigma_x} \left(\frac{1}{r} - r \right)}{1 + \frac{\sigma_y^2}{\sigma_x^2}} = \frac{\sigma_y}{\sigma_x} \left(\frac{1 - r^2}{r} \right) \cdot \frac{\sigma_x^2}{\sigma_x^2 + \sigma_y^2} = \left(\frac{1 - r^2}{r} \right) \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}$$

Note :

$$1. \text{ If } \theta \text{ is acute, } \tan \theta = \left(\frac{1 - r^2}{r} \right) \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \quad (\because -1 \leq r \leq 1)$$

$$2. \text{ If } \theta \text{ is obtuse, } \tan \theta = \left(\frac{r^2 - 1}{r} \right) \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}$$

$$3. \text{ If } r = 0 \text{ then } \tan \theta = \infty \Rightarrow \theta = \pi / 2$$

Thus if there is no relationship between the two variables (*i.e.* they are independent) then $\tan \theta = \pi / 2$.

$$4. \text{ If } r = \pm 1 \text{ then } \tan \theta = 0 \Rightarrow \theta = 0 \text{ or } \pi$$

Hence the two regression lines are parallel or coincident. The correlation between two variables is perfect.

SOLVED EXAMPLES

Example 4 : If θ is the angle between two regression lines and S.D. of Y is twice the S. D. of X and $r = 0.25$, find $\tan \theta$.

Solution : Given $\sigma_y = 2\sigma_x$ and $r = 0.25$

If θ is the angle between two regression lines, then

$$\begin{aligned}\tan \theta &= \left(\frac{1-r^2}{r} \right) \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} = \left(\frac{1-(0.25)^2}{0.25} \right) \frac{\sigma_x (2\sigma_x)}{\sigma_x^2 + 4\sigma_x^2} \\ &= \frac{1-0.0625}{0.25} \cdot \frac{2}{5} = 3.75 \times \frac{2}{5} = 1.5\end{aligned}$$

Example 1 : If $\sigma_x = \sigma_y = \sigma$ and the angle between the regression lines is $\tan^{-1}\left(\frac{4}{3}\right)$. Find r .

Solution : $\tan \theta = \left(\frac{1-r^2}{r} \right) \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}$

$$\Rightarrow \theta = \tan^{-1} \left[\frac{1-r^2}{r} \cdot \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \right]$$

Here $\sigma_x = \sigma_y = \sigma$

$$\theta = \tan^{-1} \left[\frac{1-r^2}{r} \cdot \frac{\sigma^2}{2\sigma^2} \right] = \tan^{-1} \left(\frac{1-r^2}{2r} \right) \quad \dots (1)$$

$$\text{By data, } \theta = \tan^{-1} \left(\frac{4}{3} \right) \quad \dots (2)$$

From (1) and (2), we have

$$\therefore \frac{1-r^2}{2r} = \frac{4}{3} \Rightarrow 3-3r^2-8r=0$$

$$\Rightarrow 3r^2+8r-3=0 \Rightarrow (3r-1)(r+3)=0 \Rightarrow r=\frac{1}{3} \text{ or } r=-3$$

Since $-1 \leq r \leq 1$, we cannot have $r = -3$.

$$\therefore r = 1/3$$

Example 3 : The tangent of the angle between two regression lines is 0.6 and if $\sigma_x = \frac{1}{2}\sigma_y$, find the correlation coefficient between x and y .

Solution : We know that $\tan \theta = \frac{1-r^2}{r} \cdot \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}$

$$\Rightarrow 0.6 = \frac{1-r^2}{r} \cdot \frac{0.5 \sigma_y^2}{\frac{\sigma_y^2}{4} + \sigma_y^2} \Rightarrow 0.6 = \frac{1-r^2}{r} \cdot \frac{\sigma_y^2(0.5)}{\sigma_y^2(1.25)}$$

$$\Rightarrow \frac{1-r^2}{r} = 1.5 = \frac{3}{2} \Rightarrow 2r^2 + 3r - 2 = 0$$

$$\Rightarrow r = \frac{3 \pm \sqrt{9+16}}{4} = \frac{-3}{4} \pm \frac{5}{4} = -2 \text{ or } \frac{1}{2}$$

$$\therefore r = \frac{1}{2}. (\because r = -2 \text{ is not possible.})$$

Example 1 : Find the mean values of the variables X and Y and correlation coefficient from the following regression equations.

$$2Y - X - 50 = 0$$

$$3Y - 2X - 10 = 0$$

Solution : Given equations are

$$2Y - X = 50 \quad \dots (1)$$

$$3Y - 2X = 10 \quad \dots (2)$$

$$(1) \times 2 \Rightarrow 4Y - 2X = 100 \quad \dots (3)$$

Solving (2) and (3), we get $Y = 90$.

Substituting the value of Y in (1), we get $X = 130$.

The regression lines pass through $(\bar{x}, \bar{y})$.

$\therefore$ We get means $\bar{x} = 130$ and $\bar{y} = 90$

Rewriting (1) and (2), we have

$$Y = \frac{1}{2}X + 25 \text{ and } X = \frac{3}{2}Y - 5$$

$$\therefore r = \frac{\sigma_X}{\sigma_Y} = \frac{3}{2} \quad \dots (4) \text{ and } r = \frac{\sigma_Y}{\sigma_X} = \frac{1}{2} \quad \dots (5)$$

$$(4) \times (5) \Rightarrow r^2 = \frac{3}{4} \Rightarrow r = 0.866$$

Example 1 : Test whether the equations $2x + 3y = 4$ and $x - y = 5$ represent valid regression lines.

Solution : Let $2x + 3y = 4$ be the regression line of x on y and $x - y = 5$ be the regression line of y on x.

$$\text{From } x - y = 5, \text{ we have } y = x - 5. \text{ So } r \frac{\sigma_y}{\sigma_x} = 1 \quad \dots (1)$$

Similarly, we can write $2x + 3y = 4$ as $x = -\frac{3}{2}y + 2$

$$\therefore r \frac{\sigma_x}{\sigma_y} = -\frac{3}{2} \quad \dots(2)$$

$$(1) \times (2) \Rightarrow r^2 = \frac{-3}{2} \times 1 = -\frac{3}{2}$$

In this case, r^2 is negative.

Hence, the given equations does not represent valid regression lines.

Example 6 : If $X = 2Y + 3$ and $Y = kX + 6$ are the regression lines of X on Y and Y on X respectively.

(a) Show that $0 \leq k \leq \frac{1}{2}$ (b) If $k = \frac{1}{8}$ find r and $(\bar{x}, \bar{y})$

Solution : (a) Given $X = 2Y + 3$ is the regression line of X on Y .

$$\therefore \frac{r\sigma_X}{\sigma_Y} = 2$$

The regression line of Y on X is $Y = kX + 6$

$$\therefore \frac{r\sigma_Y}{\sigma_X} = k$$

Multiplying these two equations, we get $r^2 = 2k$

$$\text{We have } 0 \leq r^2 \leq 1 \Rightarrow 0 \leq 2k \leq 1 \Rightarrow 0 \leq k \leq \frac{1}{2}$$

(b) If $k = \frac{1}{8}$ then $y = \frac{1}{8}x + 6$

$$r^2 = 2k = \frac{2}{8} = \frac{1}{4} \Rightarrow r = \frac{1}{2}$$

Since both regression coefficients are positive, we take positive values for r .

$$\therefore r = \frac{1}{2}$$

Since the regression lines pass through $(\bar{x}, \bar{y})$ we have

$$\bar{x} = 2\bar{y} + 3 \quad \dots (1) \text{ and } \bar{y} = \frac{1}{8}\bar{x} + 6 \quad \dots (2)$$

From (2), $\bar{y} = \frac{1}{8}(2\bar{y} + 3) + 6$, using (1)

$$= \frac{2\bar{y} + 3 + 48}{8}$$

$$\Rightarrow 8\bar{y} = 2\bar{y} + 51 \Rightarrow 6\bar{y} = 51$$

$$\Rightarrow \bar{y} = \frac{51}{6} \text{ and } \bar{x} = 2\bar{y} + 3 = \frac{102}{6} + 3 = \frac{120}{6} = 20$$

Example 6 : For 20 army personal the regression of weight of kidneys (Y) on weight of heart (X) is $Y = 399X + 6.394$ and the regression of weight of heart on weight of kidneys is $X = 1.212Y + 2.461$. Find the correlation coefficient between the two variable and also their means.

Solution : Given $Y = .399X + 6.394$

$$X = 1.212Y + 2.461$$

Let $\bar{X}$ and $\bar{Y}$ be the means. Then we have

$$\bar{Y} = .399\bar{X} + 6.394$$

$$\bar{X} = 1.212\bar{Y} + 2.461$$

They can be written as

$$\bar{Y} - .399\bar{X} = 6.394 \quad \dots (1)$$

$$-1.212\bar{Y} + \bar{X} = 2.461 \quad \dots (2)$$

(1) $\times 1.212$ gives

$$1.212\bar{Y} - .483588\bar{X} = 7.7495$$

$$-1.212\bar{Y} + \bar{X} = 2.461$$

$$\hline 0.516412\bar{X} = 10.2105$$

$$\Rightarrow \bar{X} = 197720.03$$

$$\therefore \bar{Y} = 78890.28$$

Regression coefficient of Y on X is (0.399) and that of X on Y is 1.212.

$\therefore$ Coefficient of correlation is geometric mean between the regression coefficients is

$$r = \sqrt{0.399 \times 1.212} = \sqrt{0.483588} = 0.6953$$

Example 4 : The equations of two regression lines are $7x - 16y + 9 = 0$ and $5y - 4x - 3 = 0$. Find the coefficient of correlation and the means of x and y.

Solution : Given equations are,

$$7x - 16y + 9 = 0 \quad \dots (1)$$

$$5y - 4x - 3 = 0 \quad \dots (2)$$

$$(1) \times 4 \text{ gives } 28x - 64y + 36 = 0$$

$$(2) \times 7 \text{ gives } -28x + 35y - 21 = 0$$

$$\text{Adding, } -29y + 15 = 0 \Rightarrow y = \frac{15}{29} = 0.5172$$

$$(1) \Rightarrow 7x = 16y - 9 = \frac{240}{29} - 9 = \frac{240 - 261}{29} = \frac{-21}{29}$$

$$\therefore x = \frac{-3}{29} = 0.1034$$

Since the regression line passes through $(\bar{x}, \bar{y})$, we have

$$\therefore \bar{x} = \text{mean} = 0.1034$$

$$\bar{y} = \text{mean} = 0.5172$$

$$\text{From (1), } x = \frac{16}{7}y - \frac{9}{7}$$

$$\text{From (2), } y = \frac{4}{5}x + \frac{3}{5}$$

$$\therefore r \frac{\sigma_x}{\sigma_y} = \frac{16}{7} \text{ and } r \frac{\sigma_y}{\sigma_x} = \frac{4}{5}$$

Multiplying these two equations, we get

$$r^2 = \frac{16}{7} \cdot \frac{4}{5} = \frac{64}{35} \Rightarrow r = \frac{8}{\sqrt{35}}$$

Example 9 : Given that $x = 4y + 5$ and $y = Kx + 4$ are the regression lines of x on y and y on x respectively. Show that $0 \leq k \leq 0.25$.

Solution : Given $x = 4y + 5$ is regression line of x on y .

$$\therefore r \frac{\sigma_x}{\sigma_y} = 4 \quad \dots (1)$$

The regression line of y on x is $y = kx + 4$

$$\therefore r \frac{\sigma_y}{\sigma_x} = k \quad \dots (2)$$

Multiplying (1) and (2), we get, $r^2 = 4k$

Since $0 \leq r \leq 1$, we have

$$0 \leq r^2 \leq 1 \Rightarrow 0 \leq 4k \leq 1 \Rightarrow 0 \leq k \leq \frac{1}{4}$$

Example 10 : Given that $X = 4Y + 5$ and $Y = kX + 4$ are the lines of regression of X on Y and Y on X respectively. Show that $0 < 4k < 1$. If $k = \frac{1}{16}$, find the means of the two variables and coefficient of correlation between them.

Solution : Given lines are $X = 4Y + 5 \quad \dots (1)$

and $Y = kX + 4 \quad \dots (2)$

$$\text{From (1) and (2), } r \frac{\sigma_X}{\sigma_Y} = 4 \text{ and } r \frac{\sigma_Y}{\sigma_X} = k$$

Multiplying these two equations, we get $r^2 = 4k$

Since $0 \leq r^2 \leq 1$, we have $0 \leq 4K \leq 1 \Rightarrow 0 \leq K \leq \frac{1}{4}$

If $K = \frac{1}{16}$ then we have

$$X = 4Y + 5 \quad \dots (1) \quad \text{and} \quad Y = \frac{1}{16}X + 4 \quad \dots (2)$$

$$(1) \Rightarrow X - 4Y - 5 = 0$$

$$(2) \times 4 \Rightarrow -\frac{1}{4}X + 4Y - 16 = 0$$

$$\text{Adding we get, } \frac{3}{4}X - 21 = 0 \Rightarrow X = 28$$

$$\text{From (2), we get } Y = \frac{23}{4}$$

The regression lines pass through $(\bar{x}, \bar{y})$

$$\text{We get means } \bar{x} = 28 \text{ and } \bar{y} = \frac{23}{4}.$$

$$\text{We have } r^2 = 4k = 4 \cdot \frac{1}{16} = \frac{1}{4} \Rightarrow r = \pm \frac{1}{2}$$

We consider positive value and take $r = \frac{1}{2}$

EXERCISE

1. Given the bi-variate data :

X:	1	5	3	2	1	1	7	3
Y:	6	1	0	0	1	2	1	5

(a) Find the Regression line of Y on X and hence predict Y if $X = 10$.

(b) Fit a Regression line of X on Y and hence predict X if $Y = 2.5$.

(c) Calculate Karl Pearson's correlation coefficient.

2. Calculate the correlation coefficient and regression coefficient for the following data.

X:	2	4	6	8	10	12	14
Y:	4	2	5	10	4	11	12

Find the estimate of y when $x = 13$

3. Find the most likely production corresponding to a rainfall 40" from the following data :

	Rainfall	Production
Average	30"	500 kgs
Standard deviation	5"	100 kgs
Coefficient of Correlation	0.8	

4. Heights of fathers and sons are given in inches :

Height of father	65	66	67	67	68	69	71	73
Height of son	67	68	64	68	72	70	69	70

Form the two lines of regression and calculate the expected average height of the son when the height of the father is 67.5 inches.

ANSWERS

- (a) $Y = 2.874 - 0.304X$, when $X = 10, Y = -0.166$
 (b) $X = 3.431 - 0.278Y$, when $Y = 2.5, X = 2.736$
 (c) $r = -0.29$
- Regression coefficient of X on $Y = 0.847$,
 Regression coefficient of Y on $X = 0.732$
 Coefficient of correlation $= 0.787$
 Correlation equation is $Y = 0.732X + 1.001$
 when $X = 13, Y = 10.571$
- The most likely production corresponding to a rainfall of 40" is 660 kgs.
- $Y = 0.421X + 39.77$; $X = 0.524Y + 32.9$
 when $X = 67.5", Y = 68.19"$

OBJECTIVE TYPE QUESTIONS

- The coefficient of correlation

(a) cannot be positive (b) cannot be negative
 (c) can be either positive or negative (d) none
- Which of the following is the highest range of r ?

(a) 0 and 1 (b) -1 and 0 (c) -1 and 1 (d) 1 and 1
- The coefficient of correlation is independent of

(a) change of scale only (b) change of origin only
 (c) both change of scalar and origin (d) no change
- The value of r^2 for a particular situation is 0.81. What is coefficient of correlation

(a) 0.81 (b) 0.9 (c) 0.09 (d) 0.085
- The coefficient of correlation =

(a) has no limits (b) can be less than 1
 (c) can be more than 1 (d) varies between ± 1

6. Coefficient of correlation =
(a) $b_{xy} \times b_{yx}$ (b) $\sqrt{b_{xy} \times b_{yx}}$ (c) $\sqrt{b_{xy}}$ (d) $\sqrt{b_{yx}}$
7. One regression coefficient is positive then the other regression coefficient is
(a) positive (b) negative (c) equal to zero (d) cannot say
8. Regression coefficient is independent of
(a) origin (b) scale (c) origin and scale (d) none
9. When two regression lines coincide then r is
(a) 0 (b) -1 (c) 1 (d) 0.5
10. The regression lines cut each other at the point of
(a) average of X and Y (b) average of X only
(c) average of Y only (d) none

ANSWERS

- | | | | |
|--------|---------|--------|--------|
| 1. c | 2. c | 3. c | 4. b |
| 5. d | 6. b | 7. a | 8. a |
| 9. c | 10. a | | |



UNIT - II

PROBABILITY

5. *Probability*

- ♦ *Sample Space and Events*
- ♦ *Axioms of Probability*
- ♦ *Conditional Probability*
- ♦ *Baye's Theorem*

6. *Random Variables and Distribution Functions*

- ♦ *Random Variables (Discrete and Continuous)*
- ♦ *Distribution Function*
- ♦ *Discrete and Continuous Probability Distributions*
- ♦ *Mathematical Expectation and Properties*



CHAPTER - 5

PROBABILITY

5.1 REVIEW OF PROBABILITY

5.1.1 INTRODUCTION

The theory of probability is one of the most useful and interesting branches of modern mathematics. It is becoming prominent by its application in many fields of learning, such as Insurance, Statistics, Biological Sciences, Physical Sciences, Engineering, etc.

An Italian mathematician, Galileo (1564 - 1642), attempted a quantitative measure of probability while dealing with some problems related to gambling. In the middle of 17th Century, two French mathematicians, Pascal and Fermat, laid down the first foundation of the mathematical theory of probability while solving the famous ‘Problem of Points’ posed by **Chevalier-De-Mere**. Other mathematicians from several countries also contributed in no small measure to the theory of probability. Outstanding of them were two Russian mathematicians, A. Kintchine and A. Kolmogoroff, who axiomised the calculus of probability.

If an experiment is repeated under similar and homogeneous conditions, we generally come across two types of situations.

- (i) The net result, what is generally known as ‘outcome’ is unique or certain.
- (ii) The net result is not unique but may be one of the several possible outcomes.

The situations covered by (i) are known as ‘deterministic’ or ‘predictable’ and situations covered by (ii) are known as ‘probabilistic’ or ‘unpredictable’.

‘Deterministic’ means the result can be predicted with certainty. For example, if r is the radius of the sphere then its volume is given by $V = \frac{4}{3}\pi r^3$ which gives uniquely the volume of the sphere.

There are some situations which do not lend themselves to the deterministic approach and they are known as ‘Probabilistic’.

For example, by looking at the sky, one is not sure whether the rain comes or not.

In such cases we talk of chances or probability which can be taken as a quantitative measure of certainty. We will now introduce the concept of probability with definitions.

5.1.2 RANDOM EXPERIMENT

[JNTU 2007 Sup]

Definition. If an ‘experiment’ is conducted, any number of times, under essentially identical conditions, there is a set of all possible outcomes associated with it. If the result is not certain and is anyone of the several possible outcomes, the experiment is called **a random trial or a random experiment**. The outcomes are known as **elementary events** and a set of outcomes is an **event**. Thus an elementary event is also an event.

6.1.3 EQUALLY LIKELY EVENTS

Events are said to be equally likely when there is no reason to expect anyone of them rather than anyone of the others.

- e. g. When a card is drawn from a pack, any card may be obtained. In this trial, all the 52 elementary events are equally likely.

6.1.4 EXHAUSTIVE EVENTS

All possible events in any trial are known as exhaustive events.

- e. g. 1. In tossing a coin, there are two exhaustive elementary events, viz. head and tail.
 2. In throwing a die, there are six exhaustive elementary events (i.e.), getting 1 or 2 or 3 or 4 or 5 or 6.
 3. In drawing 3 balls out of 9 balls in a box, there are 9C_3 exhaustive elementary events.

6.1.5 MUTUALLY EXCLUSIVE EVENTS

[JNTU Aug 2007 Sup, (A) Dec. 2017]

Events are said to be mutually exclusive, if the happening of any one of the events in a trial excludes the happening of any one of the others i.e., if no two or more of the events can happen simultaneously in the same trial.

6.1.6 CLASSICAL DEFINITION OF PROBABILITY

In a random experiment, let there be n mutually exclusive and equally likely elementary events. Let E be an event of the experiment. If m elementary events form event E (are favourable to E), then the probability of E (Probability of happening of E or chance of E), is defined as

$$P(E) = \frac{m}{n} = \frac{\text{Number of elementary events in } E}{\text{Total number of elementary events in the random experiment}}$$

If $\bar{E}$ denotes the event of non-occurrence of E , then the number of elementary events in $\bar{E}$ is $n - m$ and hence the probability of $\bar{E}$ (non-occurrence of E) is

$$P(\bar{E}) = \frac{n - m}{n} = 1 - \frac{m}{n} = 1 - P(E) \Rightarrow P(E) + P(\bar{E}) = 1$$

Since m is a non-negative integer, n is a positive integer and $m \leq n$, we have

$$0 \leq \frac{m}{n} \leq 1.$$

Hence $0 \leq P(E) \leq 1$ and $0 \leq P(\bar{E}) \leq 1$.

- Note :** 1. Suppose we say that occurrence of an event E as success, and non-occurrence $\bar{E}$ as failure. Probability $P(E)$ of the happening of the event E is known as the **probability of success** and the probability $P(\bar{E})$ of non-happening of the event is known as the **probability of failure**.
 2. If $P(E) = 1$, the event E is called **certain event** and if $P(E) = 0$, the event E is called an **impossible event**.

3. Suppose A is an event with $P(\bar{A}) < P(A)$

$$\Rightarrow 1 - P(A) < P(A) \Rightarrow 1 < 2P(A)$$

$$\Rightarrow \frac{1}{2} < P(A) \Rightarrow P(A) > \frac{1}{2}$$

SOLVED EXAMPLES

Example 1 : What is the probability for a leap year to have 52 Mondays and 53 Sundays?

Solution : A leap year has 366 days *i.e.*, 52 weeks and 2 days.

These two days can be any one of the following 7 ways

(i) Mon & Tue (ii) Tues & Wed (iii) Wed & Thurs (iv) Thurs & Fri
(v) Fri & Sat (vi) Sat & Sun (vii) Sun & Mon

Let E be the event of having 52 Mondays and 53 Sundays in the year.

Total number of possible cases is $n = 7$

Number of favourable cases to E is $m = 1$

(Sat & Sun is the only favourable case)

$$\therefore P(E) = \frac{m}{n} = \frac{1}{7}$$

Example 1 : Five digit numbers are formed with 0, 1, 2, 3, 4 (not allowing a digit being repeated in any number). Find the probability of getting 2 in the ten's place and 0 in the units place always.

Solution : Total number of 5 digit numbers using the digits 0, 1, 2, 3, 4 is
 $= n = 4 \times 4 \times 3 \times 2 \times 1 = 96$ (or) $5! - 4! = 96$.

Let E be the event of getting a number having 2 in 10's place and 0 in the units place.

So the number of numbers favourable to $E = m = 3.2.1.1.1 = 6$

$$\therefore P(E) = \frac{m}{n} = \frac{6}{96} = \frac{1}{16}$$

Example 1 : In a class there are 10 boys and 5 girls. A committee of 4 students is to be selected from the class. Find the probability for the committee to contain at least 3 girls.

Solution : A committee of 4 students out of 15 can be formed in ${}^{15}C_4$ ways

$$\text{i.e., } n = {}^{15}C_4$$

Let E be the event of forming a committee with at least 3 girls.

Now the committee can have 1 boy, 3 girls or no boy, 4 girls. So the number of ways of forming the committee = The number of favourable ways to E

$$= {}^{10}C_1 \times {}^5C_3 + {}^{10}C_0 \times {}^5C_4 = 100 + 5 = 105$$

$$\therefore P(E) = \frac{m}{n} = \frac{105}{{}^{15}C_4} = 0.0769$$

Example 1 : A class consists of 6 girls and 10 boys. If a committee of 3 is chosen at random from the class, find the probability that (i) 3 boys are selected (ii) exactly 2 girls are selected. [JNTU 2008S (Set No.2)]

Solution : Total number of students = 16

$$n(S) = \text{no. of ways of choosing 3 from 16} = {}^{16}C_3$$

(i) Suppose 3 boys are selected. This can be done in ${}^{10}C_3$ ways.

$$\text{Here } n(E) = {}^{10}C_3$$

$$\therefore P(E) = \text{The probability that 3 boys are selected} = \frac{n(E)}{n(S)}$$

$$= \frac{{}^{10}C_3}{{}^{16}C_3} = \frac{10 \times 9 \times 8}{16 \times 15 \times 14} = \frac{3}{14} = 0.2143$$

(ii) Suppose exactly 2 girls are selected. Then

$$n(E) = {}^6C_2 \times {}^{10}C_1$$

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{{}^6C_2 \times {}^{10}C_1}{{}^{16}C_3} = \frac{15}{56} = 0.2678$$

Example 5 : A and B throw alternately with a pair of ordinary dice. A wins if he throws 6 before B throws 7 and B wins if he throws 7 before A throws 6. If A begins, show that his chance of winning is 30/61. [JNTU April 2004 (Set No. 3)]

Solution : When two dice are thrown, we have $n(s) = 36$

$$\text{The probability of } A \text{ throwing '6'} = \frac{5}{36} \text{ i.e., } P(A) = \frac{5}{36}$$

The probability of A not throwing '6' is and is given by

$$P(\bar{A}) = 1 - P(A) = 1 - \frac{5}{36} = \frac{31}{36}$$

$$\text{The probability of } B \text{ throwing '7'} = \frac{6}{36} \text{ i.e., } P(B) = \frac{6}{36} = \frac{1}{6}$$

$$\text{The probability of } B \text{ not throwing 7 is } P(\bar{B}) = 1 - P(B) = 1 - \frac{1}{6} = \frac{5}{6}$$

$\therefore$ Chances of winning of 'A' is

$$\begin{aligned} &= P(A) + P(\bar{A})P(\bar{B})P(A) + P(\bar{A})P(\bar{B})P(\bar{A})P(\bar{B})P(A) + \dots \\ &= \left(\frac{5}{36}\right) + \left(\frac{31}{36} \times \frac{5}{6} \times \frac{5}{36}\right) + \left(\frac{31}{36} \times \frac{5}{6} \times \frac{31}{36} \times \frac{5}{6} \times \frac{5}{36}\right) + \dots \\ &= \frac{5}{36} \left[1 + \left(\frac{31}{36} \times \frac{5}{6}\right) + \left(\frac{31}{36} \times \frac{5}{6}\right)^2 + \dots \right] = \frac{5}{36} \left[\frac{1}{1 - \left(\frac{31}{36} \times \frac{5}{6}\right)} \right] = \frac{30}{61} \end{aligned}$$

Example 5 : A, B and C in order toss a coin. The first one to toss head wins the game. What are the probabilities of winning, assuming that the game may continue indefinitely.

[JNTU August 2007S, (H) Apr. 2012 (Set No. 1), Dec. 2012]

Solution : A may win in the first round with probability $1/2$.

He may win in the second round, after all of them failed in the first round, with the probability $\left(\frac{1}{2}\right)^3 \cdot \frac{1}{2}$, in the third round with probability $= \left(\frac{1}{2}\right)^6 \cdot \frac{1}{2}$ and so on.

$$\begin{aligned}\therefore \text{The chance of A's success} &= \frac{1}{2} + \left(\frac{1}{2}\right)^3 \cdot \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)^6 \cdot \frac{1}{2} + \dots \\ &= \frac{1}{2} \left[1 + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^6 + \dots \right] \\ &= \frac{1}{2} \cdot \frac{1}{1 - (1/2)^3} \quad [\because \text{The series is in G. P.}] \\ &= \frac{4}{7}\end{aligned}$$

B may win the first round after A's failure with probability $= \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{4}$,

Second round probability $= \left(\frac{1}{2}\right)^3 \cdot \frac{1}{4}, \dots$

$$\therefore B's \text{ chance of success} = \frac{1}{4} + \frac{1}{4} \left(\frac{1}{2}\right)^3 + \dots = \frac{\frac{1}{4}}{1 - \left(\frac{1}{2}\right)^3} = \frac{2}{7}$$

Similarly, the chance of winning the game for C $= \frac{\frac{1}{8}}{1 - \left(\frac{1}{2}\right)^3} = \frac{1}{7}$

Example 7 : Two cards are selected at random from 10 cards numbered 1 to 10. Find the probability that the sum is even if (i) The two cards are drawn together. (ii) The two cards drawn one after other with replacement. [JNTU Nov. 2006 (Set No. 3)]

Solution : (i) Suppose two cards are drawn at a time.

$\therefore$ Number of ways of drawing two cards at a time from 10 cards $= {}^{10}C_2 = 45$.

For the sum on both the cards to be even, both cards should be even numbered or both the cards should be odd numbered.

Two even numbered cards can be chosen from five even numbered cards in ${}^5C_2 = 10$ ways

$\therefore$ Total no. of favourable outcomes = $10 + 10 = 20$.

Hence required probability = $\frac{20}{45} = \frac{4}{9}$.

(ii) Suppose the two cards are chosen one after another with replacement.

This can be done in $10 \times 10 = 100$ ways.

For the sum to be even both the cards must be even or both the cards must be odd.

Number of ways of selecting two even cards = $5 \times 5 = 25$.

Similarly, number of ways of selecting two odd cards = $5 \times 5 = 25$.

$\therefore$ Required Probability = $\frac{25+25}{100} = \frac{1}{2}$.

Example 8 : A box contains n tickets marked 1 through n . Two tickets are drawn in succession without replacement. Determine the probability that the number on the tickets are consecutive integers. [JNTU 2004 Sup (Set No. 4), Feb 2008 Sup (Set No. 3)]

Solution : The box contains n tickets marked 1 through n .

The number of ways in which two tickets can be drawn without replacement
= $n(n-1)$

The total number of favourable cases in which the numbers on the tickets are consecutive integers = $n-1$.

$\therefore$ Required probability = $\frac{n-1}{n(n-1)} = \frac{1}{n}$.

Example 9 : Two cards are selected at random from 10 each numbered 1 to 10.

Find the probability that the sum is odd if

(i) Two cards are drawn.

(ii) Two cards are drawn one after another with replacement.

(iii) Two cards are drawn one after another without replacement [JNTU 2003 S]

Solution : (i) Two cards can be drawn at a time from 10 cards in ${}^{10}C_2 = 45$ ways

Let E_1 denote the event of the two cards are such that the sum is odd.

We must have one card even and another odd.

Number of ways of doing it = ${}^5C_1 \times {}^5C_1 = 25$

$\therefore$ Required probability = $\frac{25}{45} = \frac{5}{9}$

(ii) Let E_2 = The sum is even when two cards are drawn one after another with replacement.

The no. of favourable cases = 50.

$\therefore$ The no. of ways in which two cards can be drawn one after another

with replacement = ${}^{10}C_1 \times {}^{10}C_1 = 100$

$$\therefore \text{Required probability} = \frac{50}{100} = \frac{1}{2}$$

(iii) The no. of favourable cases = 50

The number of cases that the two cards can be drawn one after another without replacement = ${}^{10}C_1 \times {}^9C_1 = 90$

$$\therefore \text{Required probability} = \frac{50}{90} = \frac{5}{9}$$

Example 10 : Determine the probability for each of the following events : A non- defective bolt will be found if out of 600 bolts already examined 12 were defective.

[JNTU Dec. 2005, April 2006 (Set No. 1)]

Solution : The probability of defective bolt, $P(D) = \frac{12}{600} = \frac{1}{50}$.

$$\begin{aligned} \therefore \text{The probability of finding a non-defective bolt} &= P(\bar{D}) = 1 - P(D) \\ &= 1 - \frac{1}{50} = \frac{49}{50} = 0.98 \end{aligned}$$

Example 11 : What is the probability of picking an ace and a king from a 52 cards deck?

[JNTU (A) Nov. 2011 (Set No. 3)]

Solution : Number of ways of picking 2 cards from 52 cards = ${}^{52}C_2$

Number of ways of picking one ace and a king = ${}^4C_1 \times {}^4C_1 = 16$

$$\text{Probability of picking one ace and a king} = \frac{16}{{}^{52}C_2} = \frac{16}{26 \times 51} = \frac{8}{663}$$

5.1.7 AXIOMATIC DEFINITION OF PROBABILITY

The classical definition of probability breaks down when we do not have a complete priori analysis i.e., when the outcomes of the trial are not equally likely or when the total number of trials is infinite or when the enumeration of all equally likely events is not possible. So the necessity of the following definition viz., statistical or empirical definition of probability.

5.1.8 VON MISES'S STATISTICAL DEFINITION OF PROBABILITY

Let a trial be repeated any number of times under essentially identical conditions and let E be an event of it. The ratio of the number of times (m) the event E happens to the number of trials (n) i.e., m/n is called the **relative frequency of the event E** and is denoted by $R(E)$. Then the probability $P(E)$ of the event E is the limit approached by $R(E)$ as the number of trials (n) increases indefinitely, it being assumed a unique limit value exists.

Symbolically, we write = $\lim_{n \rightarrow \infty} R(E) = P(E)$.

The statistical definition of probability, although is of great use from practical point of view, is not conducive for mathematical approach since an actual limiting number may not really exist. Hence another definition is thought of based on axiomatic approach. This definition leads to the development of calculus of probability.

5.1.9 SIMPLE EVENT

[JNTU 2007 Sup]

An event in a trial that cannot be further split is called a **simple event** or an **elementary event**.

5.1.10 SAMPLE SPACE

[JNTU 2007 Sup, (A) Dec. 2017]

The set of all possible simple events in a trial is called a **sample space** for the trial. Each element of a sample space is called a **sample point**. Any subset of a sample space is an **event**. It is generally denoted by E . Thus a simple event is a sample point.

Sample space is denoted by S .

e.g : Two coins are tossed.

Then the possible simple events of the trial are HH, HT, TH, TT .

$\therefore$ The sample space of the trial = $S = \{HH, HT, TH, TT\}$

If an event (E) is that either two heads or two tails appear then $E = \{HH, TT\}$.

Clearly the elements of E , are sample points and $E \subseteq S$.

5.1.11 MUTUALLY EXCLUSIVE EVENTS

[JNTU (A) Dec. 2017]

Two events E_1, E_2 of a sample space S are said to be mutually exclusive if they have no sample points in common i.e., $E_1 \cap E_2 = \phi$.

Mutually exclusive events are sometimes called as **Disjoint Events**.

5.1.12 COMPLEMENTARY EVENTS

Two events of a sample space whose intersection is ϕ and whose union is the entire sample space are called complementary events. Thus if E is an event of a sample space S , its complement is denoted by E' or $\bar{E}$ and $E \cap \bar{E} = \phi$, $E \cup \bar{E} = S$, $\bar{E} = S - E$.

5.1.13 PROBABILITY-AXIOMATIC APPROACH

Definition : Let S be a finite sample space. A real valued function P from the power set of S into R is called a probability function on S if the following three axioms are satisfied.

Axioms of Probability : [JNTU (A) 2009 (Set No. 3), (H) 2009 (Set No. 1), (A) 2010 S]

(i) **Axiom of positivity :** $P(E) \geq 0$ for every subset E of S .

(ii) **Axiom of certainty :** $P(S) = 1$

(iii) **Axiom of union :** If E_1, E_2 are disjoint subsets of S , then

$$P(E_1 \cup E_2) = P(E_1) + P(E_2)$$

The image $P(E)$ of E is called the **Probability of the Event E** .

Note : If $E_1, E_2, \dots, E_n$ are disjoint subsets of S , then

$$P(E_1 \cup E_2 \cup \dots \cup E_n) = P(E_1) + P(E_2) + \dots + P(E_n)$$

This is the generalisation of axiom (iii).

5.1.14 PROBABILITY OF AN EVENT IN A SAMPLE SPACE CONTAINING EQUALLY LIKELY SIMPLE EVENTS

Let a sample space S consists of n equally likely simple events $E_1, E_2, \dots, E_n$.

$$\therefore P(E_1) = P(E_2) = \dots = P(E_n)$$

Since $S = E_1 \cup E_2 \cup \dots \cup E_n$, we have $P(S) = P(E_1) + P(E_2) + \dots + P(E_n)$

$$\therefore 1 = P(E_1) + P(E_2) + \dots + P(E_n) \quad \text{i.e., } 1 = n P(E_1) \Rightarrow P(E_1) = \frac{1}{n}$$

$$\text{Hence } P(E_1) = P(E_2) = \dots = P(E_n) = \frac{1}{n}$$

Let an event $E \subseteq S$ be such that $E = E_1 \cup E_2 \cup \dots \cup E_m$, $m < n$.

$$\therefore P(E) = P(E_1) + P(E_2) + \dots + P(E_m) = \frac{1}{n} + \frac{1}{n} + \dots \text{ } m \text{ times} = \frac{m}{n}$$

Since $E = E_1 \cup E_2 \cup \dots \cup E_m$, the number of sample points favourable to $E = n(E) = m$.

Total no. of sample points in $S = n(S) = n$

$$\begin{aligned} \therefore P(E) &= \frac{m}{n} = \frac{n(E)}{n(S)} \\ &= \frac{\text{Number of simple events (sample points) favourable to } E}{\text{Total Number of Simple Events (sample Points) in } S} \end{aligned}$$

Note : While calculating $P(E)$ with formula $P(E) = \frac{n(E)}{n(S)}$, we must be sure that S consists of only simple events which are equally likely.

SOLVED EXAMPLES

Example 1 : What is the probability that a card drawn at random from the pack of playing cards may be either a queen or a king.

Solution : Let S be the sample space associated with the drawing of a card.

$$\therefore n(S) = {}^{52}C_1 = 52$$

Let E_1 be the event of the card drawn being a queen.

$$\therefore n(E_1) = {}^4C_1 = 4$$

Let E_2 be the event of the card drawn being a king

$$\therefore n(E_2) = {}^4C_1 = 4$$

But E_1, E_2 are mutually exclusive events.

Since $E_1 \cup E_2$ is the event of drawing either a queen or a king, we have

$$\begin{aligned} P(E_1 \cup E_2) &= P(E_1) + P(E_2) \\ &= \frac{n(E_1)}{n(S)} + \frac{n(E_2)}{n(S)} = \frac{4}{52} + \frac{4}{52} = \frac{2}{13} \end{aligned}$$

Example 1 : In a group there are 3 men and 2 women. Three persons are selected at random from this group. Find the probability that one man and two women or two men and one woman are selected.

Solution : Let S be the sample space associated with the selection of 3 persons out of 5.

$$\therefore n(S) = {}^5C_3 = 10$$

Let E_1 be the event of selecting 1 man and 2 women.

$$\therefore n(E_1) = {}^3C_1 \times {}^2C_2 = 3 \times 1 = 3$$

Let E_2 be the event of selecting 2 men and a women.

$$\therefore n(E_2) = {}^3C_2 \times {}^2C_1 = 6$$

But $E_1 \cap E_2 = \phi$. i.e., E_1, E_2 are mutually exclusive.

Now $E_1 \cup E_2$ is the event of selecting 1 man and 2 women or 2 men or 1 woman.

$$\therefore P(E_1 \cup E_2) = P(E_1) + P(E_2) = \frac{n(E_1)}{n(S)} + \frac{n(E_2)}{n(S)} = \frac{3}{10} + \frac{6}{10} = \frac{9}{10}$$

Example 3 : In a sample of 446 cars, stopped at a road block, only 67 of the drivers had their seat belts fastened. Estimate the probability that a driver stopped on that road, will have his or her seat belt fastened. [JNTU 2003S]

Solution : Number of favourable outcomes = 67

Exhaustive events = 446

$$\therefore \text{Required probability} = \frac{67}{446}$$

Example 4 : Twelve balls are distributed at random among three boxes. What is the probability that the first box will contain 3 balls ?

Solution : Since each ball can go to any one of the three boxes there are three ways in which a ball can go to any one of the three boxes. Hence there are 3^{12} ways in which 12 balls can be placed in the three boxes.

Number of ways in which 3 balls out of 12 can go to the first box is ${}^{12}C_3$.

Now the remaining 9 balls are to be placed in remaining 2 boxes and their can be done in 2^9 ways.

$$\therefore \text{Total number of favourable cases} = {}^{12}C_3 \times 2^9$$

$$\text{Hence required probability} = \frac{{}^{12}C_3 \times 2^9}{3^{12}} = 0.212$$

Example 4 : What is the probability that at least two out of n people have the same birthday ? Assume 365 days in a year and that all days are equally likely.

Solution : Since the birthday of a person can fall on any of 365 days, the number of exhaustive cases for n persons = $365 \times 365 \times 365 \times \dots \times 365$ (n times)

If the birthdays of all the n persons fall on different days of the year, then the number of favourable cases would be $365(365-1)(365-2)\dots\dots(365-\overline{n-1})$

Hence, required probability

$$= 1 - \text{probability that no two persons have birthdays of the same day}$$

$$= 1 - \frac{365 \times (365-1) \times (365-2) \times \dots \times (365-\overline{n-1})}{365 \times 365 \times \dots \times 365}$$

$$= 1 - \left(1 - \frac{1}{365}\right) \left(1 - \frac{2}{365}\right) \dots \left(1 - \frac{n-1}{365}\right)$$

Example 6 : Out of 15 items 4 are not in good condition 4 are selected at random. Find the probability that (i) All are not good (ii) Two are not good .

[JNTU 2008, (H) Nov. 2010, May 2012 (Set No. 1)]

Solution : Total number of items = 15

Number of ways of picking 4 items = ${}^{15}C_4$

(i) Suppose 4 items are chosen which are not good.

Number of ways of selecting 4 items which are not good = 4C_4

$$\therefore \text{The probability all are not good} = \frac{{}^4C_4}{{}^{15}C_4} = \frac{1}{1365}$$

(ii) Suppose two items are not good

Number of ways of selecting of 2 bad items = 4C_2

Number of ways of selecting 2 good items = ${}^{11}C_2$

$\therefore$ Required probability of getting two items which are not good

$$= P(2 \text{ good and } 2 \text{ bad items}) = \frac{{}^4C_2 \times {}^{11}C_2}{{}^{15}C_4} = \frac{99}{455}$$

Example 7 : Five persons in a group 20 are engineers. If three persons are selected at random, determine the probability that all are engineers and the probability that at least one being an engineer.

[JNTU (A) Nov. 2011 (Set No. 2)]

Solution : Number of Engineers in a group of 20 people = 5

Number of ways selecting 3 persons from 20 people = ${}^{20}C_3$

(i) Number of ways in which all are engineers = 5C_3

$$\therefore \text{The probability that all the three are engineers} = \frac{{}^5C_3}{{}^{20}C_3} = \frac{5 \times 4 \times 3}{20 \times 19 \times 18} = \frac{1}{114}$$

(ii) Number of ways in which atleast one engineer is selected

$$= {}^5C_1 + {}^5C_2 + {}^5C_3 = 5 + 10 + 10 = 25$$

$$\therefore \text{The probability that atleast 1 engineer is selected} = \frac{25}{{}^{20}C_3} = \frac{5}{228}$$

Example 8 : From a pack of 52 cards, three cards are drawn at random. Find the probability that two are aces and another is a king. [JNTU (A) Nov. 2011 (Set No. 2)]

Solution : The number of ways of drawing 3 cards from a pack of 52 cards = ${}^{52}C_3$

The number of ways in which two cards are aces and another is a king = ${}^4C_2 \times {}^4C_1 = 24$

$\therefore$ The probability two cards are aces and another is king

$$= \frac{24}{{}^{52}C_3} = \frac{24}{\frac{52 \times 51 \times 50}{1 \times 2 \times 3}} = \frac{24}{26 \times 17 \times 50} = \frac{6}{13 \times 17 \times 25}$$

Example 9 : Prove the Identity : If $B \subset A$, then $P(A \cap B') = P(A) - P(B)$

[JNTU (A) Apr. 2012 (Set No. 1)]

Solution : $B \subseteq A \Rightarrow A \cap B = B$.

Now $P(A \cap B') = P(A - (A \cap B))$

$$= P(A) - P(A \cap B) = P(A) - P(B)$$

5.1.15 ADDITION THEOREM ON PROBABILITY

Theorem : If S is a sample space, and E_1, E_2 are any events in S then

$$P(E_1 \text{ or } E_2) = P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

[JNTU 2007 (Set No. 1), 2009 Sup (Set No. 2); (A) 2009, June 2012, (H) Dec. 2013]

Proof : Case (i) : $E_1 \cap E_2 \neq \phi$. Let E_1, E_2 contain the sample points

$a_1, a_2, \dots, a_k, a_{k+1}, \dots, a_{k+l}$ and $a_{k+1}, \dots, a_{k+l}, a_{k+l+1}, \dots, a_{k+l+m}$ respectively.

$$\therefore E_1 = \{a_1, a_2, \dots, a_k, a_{k+1}, \dots, a_{k+l}\} \text{ and}$$

$$E_2 = \{a_{k+1}, a_{k+2}, \dots, a_{k+l}, a_{k+l+1}, \dots, a_{k+l+m}\}$$

$$\therefore E_1 \cup E_2 = \{a_1, a_2, \dots, a_k, a_{k+1}, \dots, a_{k+l}, a_{k+l+1}, \dots, a_{k+l+m}\}$$

$$\text{and } E_1 \cap E_2 = \{a_{k+1}, a_{k+2}, \dots, a_{k+l}\}$$

$$P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

$$\begin{aligned} &= P(a_1) + P(a_2) + \dots + P(a_k) + P(a_{k+1}) + \dots + P(a_{k+l}) \\ &\quad + P(a_{k+1}) + \dots + P(a_{k+l}) + P(a_{k+l+1}) + \dots + P(a_{k+l+m}) \\ &\quad - [P(a_{k+1}) + P(a_{k+2}) + \dots + P(a_{k+l})] \end{aligned}$$

$$= P(a_1) + \dots + P(a_k) + P(a_{k+1}) + \dots + P(a_{k+l}) + P(a_{k+l+1}) + \dots + P(a_{k+l+m})$$

$$= P(E_1 \cup E_2)$$

Case (ii) : $E_1 \cap E_2 = \phi$

$$\begin{aligned}\therefore P(E_1 \cup E_2) &= P(E_1) + P(E_2) - P(E_1 \cap E_2) \\ &= P(E_1) + P(E_2) - 0 \\ &= P(E_1) + P(E_2) - P(\phi) \\ &= P(E_1) + P(E_2) - P(E_1 \cap E_2)\end{aligned}$$

Cor. If E_1, E_2 are two mutually exclusive events, then

$$(i) P(E_1 \cup E_2) = P(E_1) + P(E_2) \text{ since } P(E_1 \cap E_2) = P(\phi) = 0 \text{ [Axiom (iii)]}$$

$$\begin{aligned}(ii) P(\overline{E_1 \cup E_2}) &= P(S - (E_1 \cup E_2)) = P(S) - P(E_1 \cup E_2) \\ &= 1 - \{P(E_1) + P(E_2)\} = P(\overline{E_1}) - P(E_2)\end{aligned}$$

eg. 1: If $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{4}$ and $P(A \cap B) = \frac{1}{8}$, then

$$P(A \cup B) = \frac{1}{2} + \frac{1}{4} - \frac{1}{8} = \frac{5}{8}$$

eg. 2: If $P(A) = 0.25$, $P(B) = 0.50$ and $P(A \cup B) = 0.59$, then

$$P(A \cap B) = 0.25 + 0.50 - 0.59 = 0.16$$

SOLVED EXAMPLES

Example 1 : A card is drawn from a well shuffled pack of cards. What is the probability that it is either a spade or an ace ? [JNTU Nov. 2003S (Set No. 2)]

Solution : Let S is the sample space of all the simple events.

$$\therefore n(s) = 52$$

Let A denote the event of getting a spade and B denotes the event of getting an ace

Then $A \cup B$ = The event of getting a spade or an ace

$A \cap B$ = The event of getting a spade and an ace

$$P(A) = \frac{13}{52}, P(B) = \frac{4}{52}, P(A \cap B) = \frac{1}{52}$$

By Addition Theorem,

$$\begin{aligned}P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= \frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}\end{aligned}$$

Example 1 : Three students A, B, C are in running race. A and B have the same probability of winning and each is twice as likely to win as C. Find the probability that B or C wins. [JNTU April 2006, (A) Nov. 2010 (Set No. 4)]

Solution : $A \cup B \cup C = S =$ Sample space of race

By data, $P(A) = P(B)$ and $P(A) = 2 P(C)$... (1)

We have $P(A) + P(B) + P(C) = 1 \Rightarrow 2P(C) + 2 P(C) + P(C) = 1$ [by (1)]

$$\Rightarrow P(C) = \frac{1}{5}, P(A) = \frac{2}{5} \text{ and } P(B) = \frac{2}{5}$$

The probability that B or C wins $= P(B \cup C)$

$$= P(B) + P(C) - P(B \cap C)$$

$$= \frac{2}{5} + \frac{1}{5} - 0 = \frac{3}{5}$$

Example 1 : From a city 3 news papers A, B, C are being published. A is read by 20%, B is read by 16%, C is ready by 14% both A and B are read by 8%, both A and C are read by 5% both B and C are read by 4% and all three A, B, C are read by 2%. What is the percentage of the population that read at least one paper.

Solution : Given $P(A) = \frac{20}{100}$, $P(B) = \frac{16}{100}$, $P(C) = \frac{14}{100}$

and $P(A \cap B) = \frac{8}{100}$, $P(A \cap C) = \frac{5}{100}$, $P(B \cap C) = \frac{4}{100}$ and $P(A \cap B \cap C) = \frac{2}{100}$

$\therefore P(A \cup B \cup C)$

$$= P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

$$= \frac{20}{100} + \frac{16}{100} + \frac{14}{100} - \frac{8}{100} - \frac{4}{100} - \frac{5}{100} + \frac{2}{100} = \frac{35}{100}$$

$\therefore$ Percentage of the population that read at least one paper $= \frac{35}{100} \times 100 = 35$

5.1.16 CONDITIONAL EVENT

Def : If E_1, E_2 are events of a sample space S and if E_2 occurs after the occurrence of E_1 , then the event of occurrence of E_2 after the event E_1 is called conditional event of E_2 given E_1 . It is denoted by $\frac{E_2}{E_1}$. Similarly we define $\frac{E_1}{E_2}$.

- eg. 1.** Two coins are tossed. The event of getting two tails given that there is at least one tail is a conditional event.
- eg. 2.** Two unbiased dice are thrown. If the sum of the numbers thrown on them is 7, the event of getting 1 on anyone of them is a conditional event.
- eg. 3.** A die is thrown 3 times. The event of getting the sum of the numbers thrown is 15 when it is known that the first throw was a 5 is a conditional event.

5.2 CONDITIONAL PROBABILITY

Def : If E_1 and E_2 are two events in a sample space S and $P(E_1) \neq 0$, then the probability of E_2 , after the event E_1 has occurred, is called the conditional probability of the event of E_2 given

E_1 and is denoted by $P\left(\frac{E_2}{E_1}\right)$ or $P(E_2 / E_1)$ and we define $P\left(\frac{E_2}{E_1}\right) = \frac{P(E_1 \cap E_2)}{P(E_1)}$.

Similarly, we define $P\left(\frac{E_1}{E_2}\right) = \frac{P(E_1 \cap E_2)}{P(E_2)}$

$$\begin{aligned} \text{We have } P\left(\frac{E_2}{E_1}\right) &= \frac{P(E_1 \cap E_2)}{P(E_1)} = \frac{n(E_1 \cap E_2) / n(S)}{n(E_1) / n(S)} = \frac{n(E_1 \cap E_2)}{n(E_1)} \\ &= \frac{\text{Number of elements in } E_1 \cap E_2}{\text{Number of elements in } E_1} \end{aligned}$$

Here $E_1 \subset S$ and $P(E_1) > 0$.

We agreed to consider only those elements (sample points) of the event E_1 as the sample space and E_2 to be another subset of S . Since E_1 is the new sample space, the only elements of E_2 that concern us are also the elements of E_1 i.e., the elements of $E_2 \cap E_1$.

Then we define the probability of E_2 relative to the new sample space E_1 i.e., the conditional probability of E_2 given E_1 .

Thus some authors define the conditional probability of E_2 given E_1

$$\text{i.e., } P\left(\frac{E_2}{E_1}\right) \text{ as } \frac{n(E_2 \cap E_1)}{n(E_1)}.$$

5.2.1 IF A, B, C ARE THREE EVENTS OF A SAMPLE SPACE WE HAVE SOME USEFUL RESULTS

1. $P\left(\frac{A}{A}\right) = \frac{P(A \cap A)}{P(A)} = \frac{P(A)}{P(A)} = 1, (P(A) > 0)$
2. $P\left(\frac{\phi}{A}\right) = \frac{P(\phi \cap A)}{P(A)} = \frac{P(\phi)}{P(A)} = \frac{0}{P(A)} = 0, (P(A) > 0)$
3. $A - B = A - (A \cap B) \Rightarrow P(A - B) = P(A - (A \cap B))$
 $\Rightarrow P(A - B) = P(A) - P(A \cap B) [\because A \cap B \subseteq A]$
4. $A \subseteq B \Rightarrow A \cap C \subseteq B \cap C \Rightarrow P(A \cap C) \leq P(B \cap C)$
 $\Rightarrow \frac{P(A \cap C)}{P(C)} \leq \frac{P(B \cap C)}{P(C)} \Rightarrow P\left(\frac{A}{C}\right) \leq P\left(\frac{B}{C}\right) (\because P(C) > 0)$

5. A, B are mutually exclusive events and $P(B) > 0$

$$\Rightarrow P(A \cap B) = 0 \text{ and } P(B) > 0 \text{ results } P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{0}{P(B)} = 0$$

6. $P(A \cap B) = 0$ and $P(B) \neq 1$

$$\Rightarrow P\left(\frac{A}{\bar{B}}\right) = \frac{P(A \cap \bar{B})}{P(\bar{B})} = \frac{P(A - (A \cap B))}{P(\bar{B})} = \frac{P(A) - P(A \cap B)}{1 - P(B)} = \frac{P(A) - 0}{1 - P(B)}$$

$$\Rightarrow P\left(\frac{A}{\bar{B}}\right) = \frac{P(A)}{1 - P(B)}$$

7. $P(A) + P(B) - P(A \cap B) = P(A \cup B) \Rightarrow P(A) + P(B) - P(A \cap B) \leq 1$

$$\Rightarrow P(A) + P(B) - 1 \leq P(A \cap B) \Rightarrow \frac{P(A \cap B)}{P(B)} \geq \frac{P(A) + P(B) - 1}{P(B)}$$

$$\Rightarrow P\left(\frac{A}{B}\right) \geq \frac{P(A) + P(B) - 1}{P(B)}$$

8. $P\left(\frac{A}{B}\right) \cdot P(B) + P\left(\frac{A}{\bar{B}}\right) \cdot P(\bar{B}) = P(A \cap B) + P(A \cap \bar{B})$

$$= P(A \cap B) + P(A - (A \cap B))$$

$$= P(A \cap B) + P(A) - P(A \cap B) = P(A)$$

$$\therefore P\left(\frac{A}{B}\right) \cdot P(B) + P\left(\frac{A}{\bar{B}}\right) \cdot P(\bar{B}) = P(A)$$

9. $P\left(\frac{\bar{A}}{\bar{B}}\right) = \frac{P(\bar{A} \cap \bar{B})}{P(\bar{B})} = \frac{P(S - (A \cup B))}{1 - P(B)} = \frac{P(S) - P(A \cup B)}{1 - P(B)} = \frac{1 - P(A \cup B)}{1 - P(B)}$

10. A and B are mutually exclusive and $P(A \cup B) \neq 0 \Rightarrow P(A \cap B) = 0$ and $P(A \cup B) \neq 0$.

$$\text{Now } P\left(\frac{A}{A \cup B}\right) = \frac{P(A \cap (A \cup B))}{P(A \cup B)} = \frac{P(A)}{P(A) + P(B) - P(A \cap B)} = \frac{P(A)}{P(A) + P(B)}$$

11. B, C are mutually exclusive events $\Rightarrow P(B \cap C) = 0$

$$\text{Now } P\left(\frac{B \cup C}{A}\right) = \frac{P((B \cup C) \cap A)}{P(A)} = \frac{P((B \cap A) \cup (C \cap A))}{P(A)}$$

$$= \frac{P(B \cap A) + P(C \cap A)}{P(A)}$$

($\because B \cap A, C \cap A$ are mutually exclusive)

$$= \frac{P(B \cap A)}{P(A)} + \frac{P(C \cap A)}{P(A)}$$

$$\Rightarrow P\left(\frac{B \cup C}{A}\right) = P\left(\frac{B}{A}\right) + P\left(\frac{C}{A}\right)$$

12. For any three events A, B and C

$$(i) \quad P\left(\frac{A \cup B}{C}\right) = P\left(\frac{A}{C}\right) + P\left(\frac{B}{C}\right) - P\left(\frac{A \cap B}{C}\right)$$

$$(ii) \quad P\left(\frac{A \cap \bar{B}}{C}\right) + P\left(\frac{A \cap B}{C}\right) = P\left(\frac{A}{C}\right)$$

Proof : (i) We have $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Replace A and B by the compound events $A \cap C$ and $B \cap C$ respectively.

$$\therefore P[(A \cap C) \cup (B \cap C)] = P(A \cap C) + P(B \cap C) - P(A \cap B \cap C)$$

$$\text{Hence } \frac{P[(A \cap C) \cup (B \cap C)]}{P(C)} = \frac{P(A \cap C)}{P(C)} + \frac{P(B \cap C)}{P(C)} - \frac{P(A \cap B \cap C)}{P(C)}, P(C) \neq 0$$

$$\therefore \frac{P[(A \cup B) \cap C]}{P(C)} = P\left(\frac{A}{C}\right) + P\left(\frac{B}{C}\right) - P\left(\frac{A \cap B}{C}\right)$$

$$\Rightarrow P\left(\frac{A \cup B}{C}\right) = P\left(\frac{A}{C}\right) + P\left(\frac{B}{C}\right) - P\left(\frac{A \cap B}{C}\right)$$

$$\begin{aligned} (ii) \quad P\left(\frac{A \cap \bar{B}}{C}\right) + P\left(\frac{A \cap B}{C}\right) &= \frac{P(\bar{A} \cap \bar{B} \cap C)}{P(C)} + \frac{P(A \cap B \cap C)}{P(C)} \\ &= \frac{P(A \cap \bar{B} \cap C) + P(A \cap B \cap C)}{P(C)} = \frac{P(A \cap C)}{P(C)} = P\left(\frac{A}{C}\right) \end{aligned}$$

eg. 1 : Let a die be rolled and $A = \{1, 3, 5\}$, $B = \{2, 3\}$ and $C = \{2, 3, 4, 5\}$ be the events associated with the random experiment.

Sol. Now $P(1) = \dots = P(6) = \frac{1}{6}$

$$P\left(\frac{B}{A}\right) = \frac{P(B \cap A)}{P(A)} = \frac{P(\{3\})}{P(\{1, 3, 5\})} = \frac{\frac{1}{6}}{\frac{1}{6} + \frac{1}{6} + \frac{1}{6}} = \frac{1}{3}$$

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{P(\{3\})}{P(\{2, 3\})} = \frac{\frac{1}{6}}{\frac{1}{6} + \frac{1}{6}} = \frac{1}{2}$$

$$P\left(\frac{C}{A}\right) = \frac{P(C \cap A)}{P(A)} = \frac{P(\{3,5\})}{P(\{1,3,5\})} = \frac{\frac{1}{6} + \frac{1}{6}}{\frac{1}{6} + \frac{1}{6} + \frac{1}{6}} = \frac{2}{3}$$

$$P\left(\frac{A}{C}\right) = \frac{P(A \cap C)}{P(C)} = \frac{\frac{1}{6} + \frac{1}{6}}{\frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6}} = \frac{1}{2}$$

$$P\left(\frac{C}{B}\right) = \frac{P(C \cap B)}{P(B)} = \frac{P(\{2,3\})}{P(\{2,3\})} = \frac{\frac{1}{6} + \frac{1}{6}}{\frac{1}{6} + \frac{1}{6}} = 1$$

$$P\left(\frac{B}{C}\right) = \frac{P(B \cap C)}{P(C)} = \frac{P(\{2,3\})}{P(\{2,3,4,5\})} = \frac{\frac{1}{6} + \frac{1}{6}}{\frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6}} = \frac{1}{2}$$

eg. 2 : If $P(A^c) = \frac{3}{8}$, $P(B^c) = \frac{1}{2}$ and $P(A \cap B) = \frac{1}{4}$, then find

$$(i) P\left(\frac{A}{B}\right) \text{ and } P\left(\frac{B}{A}\right) \quad (ii) P\left(\frac{A^c}{B^c}\right) \text{ and } P\left(\frac{B^c}{A^c}\right).$$

Sol. $P(A) = 1 - P(A^c) = 1 - \frac{3}{8} = \frac{5}{8}$, $P(B) = 1 - P(B^c) = 1 - \frac{1}{2} = \frac{1}{2}$

and $P(A \cap B) = \frac{1}{4}$

$$(i) P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{1/4}{1/2} = \frac{1}{2} \text{ and } P\left(\frac{B}{A}\right) = \frac{P(B \cap A)}{P(A)} = \frac{1/4}{5/8} = \frac{2}{5}$$

$$(ii) P\left(\frac{A^c}{B^c}\right) = \frac{P(A^c \cap B^c)}{P(B^c)} = \frac{1 - P(A \cup B)}{P(B^c)} = \frac{1 - [P(A) + P(B) - P(A \cap B)]}{P(B^c)}$$

$$= \frac{1 - \left[\frac{5}{8} + \frac{1}{2} - \frac{1}{4}\right]}{\frac{1}{2}} = \frac{1}{4}$$

$$\text{Now } P\left(\frac{B^c}{A^c}\right) = \frac{P(B^c \cap A^c)}{P(A^c)} = \frac{1 - P(B \cup A)}{P(A^c)} = \frac{1 - \left[\frac{1}{2} + \frac{5}{8} - \frac{1}{4}\right]}{\frac{3}{8}} = \frac{1}{3}$$

5.2.2 MULTIPLICATION THEOREM OF PROBABILITY

Statement : In a random experiment if E_1, E_2 are two events such that $P(E_1) \neq 0$ and $P(E_2) \neq 0$, then

$$P(E_1 \cap E_2) = P(E_1) \cdot P(E_2/E_1),$$

$$P(E_2 \cap E_1) = P(E_2) \cdot P(E_1/E_2)$$

Proof : Let S be the sample space associated with the random experiment. Let E_1, E_2 be two events of S such that $P(E_1) \neq 0, P(E_2) \neq 0$. Since $P(E_1) \neq 0$, by the definition of conditional probability of E_2 given E_1 ,

$$P\left(\frac{E_2}{E_1}\right) = \frac{P(E_1 \cap E_2)}{P(E_1)}$$

$$\therefore P(E_1 \cap E_2) = P(E_1) \cdot P\left(\frac{E_2}{E_1}\right)$$

Since $P(E_2) \neq 0$, we have $P\left(\frac{E_1}{E_2}\right) = \frac{P(E_1 \cap E_2)}{P(E_2)}$ and hence $P(E_2 \cap E_1) = P(E_2) \cdot P\left(\frac{E_1}{E_2}\right)$

eg. 1. If $P(A \cap B) = \frac{1}{6}$, $P(A) = \frac{1}{2}$, then $P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{1/6}{1/2} = \frac{1}{3}$

eg. 2. If A, B are two events ($\neq \phi$) then $P(\overline{A} \cup \overline{B}) = 1 - P(A \cap B) = 1 - P(A) \cdot P(B/A)$

Note : Multiplication theorem can be extended three events E_1, E_2, E_3 as :

$$P(E_1 \cap E_2 \cap E_3) = P(E_1) \cdot P\left(\frac{E_2}{E_1}\right) \cdot P\left(\frac{E_3}{E_1 \cap E_2}\right)$$

This result can be extended to 4 or more events.

5.2.3 COMPOUND EVENT, INDEPENDENT AND DEPENDENT EVENTS

Compound Event : When two or more events occur in conjunction with each other, their joint occurrence is called compound event.

eg. 1. If 2 balls are drawn from a bag containing 4 green, 6 black, 7 white balls, the event of drawing 2 green balls or 2 white balls is a compound event.

eg. 2. When a die and a coin are tossed the event of getting utmost 4 on the die and head on the coin is a compound event and separately they are independent events.

eg. 3. From a pack of cards if a spade card is drawn and not replaced and if again a club card is drawn, the two drawings form a compound event and separately they are dependent events.

Note : Multiplication theorem of probability is also called theorem of compound probability.

5.2.4 INDEPENDENT AND DEPENDENT EVENTS

If the occurrence of the event E_2 is not affected by the occurrence or non-occurrence of the event E_1 , then the event E_2 is said to be independent of E_1 and $P(E_2 / E_1) = P(E_2)$.

If $P(E_1) \neq 0, P(E_2) \neq 0$ and E_2 is independent of E_1 , then E_1 is independent of E_2 . In this case we say that E_1, E_2 are **mutually independent** or **simply independent**.

If the occurrence of the event E_2 is effected by the occurrence of E_1 , then the events E_1, E_2 are dependent and $P(E_2 / E_1) \neq P(E_2)$.

THEOREM :

The event E is independent of ϕ and S

Proof : (i) $E \cap \phi = \phi \Rightarrow P(E \cap \phi) = P(\phi) = 0 = P(E) \cdot 0 = P(E) \cdot P(\phi)$

$\Rightarrow E, \phi$ are independent events.

(ii) $E \cap S = E \Rightarrow P(E \cap S) = P(E) = P(E) \cdot 1 = P(E) \cdot P(S)$

$\therefore E, S$ are independent events.

THEOREM :

If E_1 and E_2 are independent events of a sample space S , then (i) $\bar{E}_1$ and $\bar{E}_2$ are independent. (ii) $\bar{E}_1$ and E_2 are independent (iii) E_1 and $\bar{E}_2$ are independent.

(OR) If A and B are independent events show that $(A^c, B^c), (A, B^c)$ and (A^c, B) are also independent.

[JNTU Nov 2004, JNTU (H) 2009 (Set 4), (K) 2009 (Set 2)]

Proof :

$$\begin{aligned} (i) \quad P(\bar{E}_1 \cap \bar{E}_2) &= P(\overline{E_1 \cup E_2}) = P(S - (E_1 \cup E_2)) \\ &= P(S) - P(E_1 \cup E_2) \\ &= 1 - \{P(E_1) + P(E_2) - P(E_1 \cap E_2)\} \\ &= 1 - P(E_1) - P(E_2) + P(E_1) \cdot P(E_2) \\ &= (1 - P(E_1)) (1 - P(E_2)) = P(\bar{E}_1) \cdot P(\bar{E}_2) \end{aligned}$$

$\Rightarrow \bar{E}_1$ and $\bar{E}_2$ are independent

$$\begin{aligned} (ii) \quad P(\bar{E}_1 \cap E_2) &= P(E_2 - (E_1 \cap E_2)) \\ &= P(E_2) - P(E_1 \cap E_2) = P(E_2) - P(E_1) \cdot P(E_2) \\ &= P(E_2) (1 - P(E_1)) = P(E_2) \cdot P(\bar{E}_1) \\ &= P(\bar{E}_1) \cdot P(E_2) \end{aligned}$$

$\Rightarrow \bar{E}_1$ and E_2 are independent

(iii) Proceed as in (ii).

5.2.5 PAIRWISE INDEPENDENT EVENTS

E_1, E_2, E_3 are events of a sample space S . They are said to be pairwise independent if

$$\begin{aligned} P(E_1 \cap E_2) &= P(E_1) \cdot P(E_2), \quad P(E_2 \cap E_3) = P(E_2) \cdot P(E_3), \\ P(E_1 \cap E_3) &= P(E_1) \cdot P(E_3) \text{ when } P(E_1) \neq 0, P(E_2) \neq 0, P(E_3) \neq 0. \end{aligned}$$

Let E_1, E_2, E_3 be three events of a sample space. It is possible to have E_1 and E_2 independent, E_1 and E_3 independent, and E_2 and E_3 independent and still to have $E_1 \cap E_2$ and E_3 , say, dependent or to have E_1, E_2 and E_3 dependent. That is, it is possible

for three events E_1, E_2 and E_3 dependent. That is, it is possible for three events E_1, E_2, E_3 to be pairwise independent and not be mutually independent. Hence we say :

Three events E_1, E_2 and E_3 are mutually independent or simply independent if

$$(i) P(E_1 \cap E_2) = P(E_1) \cdot P(E_2), P(E_2 \cap E_3) = P(E_2) \cdot P(E_3), P(E_1 \cap E_3) = P(E_1) \cdot P(E_3)$$

$$(ii) P(E_1 \cap E_2 \cap E_3) = P(E_1) \cdot P(E_2) \cdot P(E_3)$$

Note : 1. In applications it appears that practically all events which are pairwise independent are also independent themselves.

2. If E_1, E_2 and E_3 are independent events then $\bar{E}_1, \bar{E}_2, \bar{E}_3$ are independent events and hence $P(\bar{E}_1 \cap \bar{E}_2 \cap \bar{E}_3) = P(\bar{E}_1) \cdot P(\bar{E}_2) \cdot P(\bar{E}_3)$

THEOREM :

If E_1, E_2, E_3 are mutually independent events of a sample space S , then $E_1 \cup E_2$ and E_3 are also independent events.

Proof : $P[(E_1 \cup E_2) \cap E_3]$

$$\begin{aligned} &= P[(E_1 \cap E_3) \cup (E_2 \cap E_3)] \\ &= P(E_1 \cap E_3) + P(E_2 \cap E_3) - P((E_1 \cap E_3) \cap (E_2 \cap E_3)) \\ &= P(E_1) \cdot P(E_3) + P(E_2) \cdot P(E_3) - P(E_1 \cap E_2 \cap E_3) \\ &= P(E_1) \cdot P(E_3) + P(E_2) \cdot P(E_3) - P(E_1) \cdot P(E_2) \cdot P(E_3) \\ &= [P(E_1) + P(E_2) - P(E_1) \cdot P(E_2)] P(E_3) \\ &= [P(E_1) + P(E_2) - P(E_1 \cap E_2)] P(E_3) = P(E_1 \cup E_2) \cdot P(E_3) \end{aligned}$$

$\Rightarrow E_1 \cup E_2, E_3$ are independent events.

Note : 1. Probability of the occurrence of at least one of the independent events E_1, E_2, E_3

$$= P(E_1 \cup E_2 \cup E_3)$$

$$\text{For : } P(E_1 \cup E_2 \cup E_3) = 1 - P(\overline{E_1 \cup E_2 \cup E_3}) = 1 - P(\bar{E}_1 \cap \bar{E}_2 \cap \bar{E}_3)$$

$$= 1 - P(\bar{E}_1) \cdot P(\bar{E}_2) \cdot P(\bar{E}_3)$$

[since $\bar{E}_1, \bar{E}_2, \bar{E}_3$ are independent events.]

2. $E_1, E_2, \dots, E_n$ are n independent events with probabilities $p_1, p_2, \dots, p_n$.

Probability of the occurrence of at least one of independent events $E_1, E_2, \dots, E_n$

$$\begin{aligned} &= P(E_1 \cup E_2 \cup \dots \cup E_n) = 1 - P(\overline{E_1 \cup E_2 \cup \dots \cup E_n}) \\ &= 1 - P(\bar{E}_1 \cap \bar{E}_2 \cap \bar{E}_3 \cap \dots \cap \bar{E}_n) = 1 - P(\bar{E}_1) \cdot P(\bar{E}_2) \cdot \dots \cdot P(\bar{E}_n) \\ &= 1 - (1 - p_1)(1 - p_2) \dots (1 - p_n) \end{aligned}$$

THEOREM :

Prove that if E_1, E_2 and E_3 are random events in a sample space and if E_1, E_2, E_3 are pairwise independent and E_1 is independent of $(E_2 \cup E_3)$, then E_1, E_2 and E_3 are mutually independent.

Proof : Since E_1, E_2, E_3 are pairwise independent, we have

$$P(E_1 \cap E_2) = P(E_1)P(E_2); P(E_2 \cap E_3) = P(E_2)P(E_3); P(E_1 \cap E_3) = P(E_1)P(E_3)$$

$$\text{and } P[E_1 \cap (E_2 \cup E_3)] = P(E_1)P(E_2 \cup E_3) \quad \dots (1)$$

$$\begin{aligned} \text{Now } P[E_1 \cap (E_2 \cup E_3)] &= P[(E_1 \cap E_2) \cup (E_1 \cap E_3)] \\ &= P(E_1 \cap E_2) + P(E_1 \cap E_3) - P[(E_1 \cap E_2) \cap (E_1 \cap E_3)] \\ &= P(E_1)P(E_2) + P(E_1)P(E_3) - P(E_1 \cap E_2 \cap E_3) \end{aligned}$$

$$\begin{aligned} \text{Also } P(E_1)P(E_2 \cup E_3) &= P(E_1)[P(E_2) + P(E_3) - P(E_2 \cap E_3)] \\ &= P(E_1)P(E_2) + P(E_1)P(E_3) - P(E_1)P(E_2 \cap E_3) \end{aligned}$$

Hence using (1), we have

$$\begin{aligned} P(E_1)P(E_2) + P(E_1)P(E_3) - P(E_1 \cap E_2 \cap E_3) \\ = P(E_1)P(E_2) + P(E_1)P(E_3) - P(E_1)P(E_2 \cap E_3) \\ \Rightarrow P(E_1 \cap E_2 \cap E_3) = P(E_1)P(E_2)P(E_3) \end{aligned}$$

SOLVED EXAMPLES

Example 1 : A bag contains 8 red and 6 blue balls. Two drawing of each 2 balls are made. Find the probability that the first drawing gives two red balls and second drawing gives 2 blue balls, if the balls drawn first are replaced before the second draw.

Solution : Let E_1 be the event of drawing 2 red balls in the first draw from the bag containing 8 red and 6 blue balls.

Then $P(E_1) = \frac{{}^8C_2}{{}^{14}C_2}$. The two drawn balls are replaced.

Let E_2 be the event of drawing 2 blue balls in the second draw from the bag.

Then $P(E_2) = \frac{{}^6C_2}{{}^{14}C_2}$

Now $E_1 \cap E_2$ = Event of drawing 2 red balls in the first draw and another drawing of 2 blue balls in the second draw after the balls are replaced.

Also E_1, E_2 are independent.

$$\begin{aligned} \therefore P(E_1 \cap E_2) &= P(E_1) \cdot P(E_2) = \frac{{}^8C_2}{{}^{14}C_2} \times \frac{{}^6C_2}{{}^{14}C_2} \\ &= \frac{8 \times 7}{14 \times 13} \times \frac{6 \times 5}{14 \times 13} = \frac{60}{1183} \end{aligned}$$

Example 4 : Find the probability of drawing 2 red balls in succession from a bag containing 4 red and 5 black balls when the ball that is drawn first is (i) not replaced (ii) replaced.

Solution : Let E_1 be the event of drawing a red ball in the first draw and E_2 be the event of drawing a red ball in second draw also.

(i) After the first draw the ball is not replaced. The first ball can be drawn in 9 ways and the second in 8 ways since the first ball is not replaced. Then both the balls can be drawn in 9×8 ways.

There are 4 ways in which E_1 can occur and 3 ways in which E_2 can occur, so that E_1 and E_2 can occur in 4×3 ways.

$$\begin{aligned} P\left(\frac{E_2}{E_1}\right) &= P(E_2, \text{ given the probability of } E_1) \\ &= P(\text{2nd ball is red, given that first ball is red}) = \frac{3}{8} \\ \therefore P(E_1 \cap E_2) &= P(E_1) \times P\left(\frac{E_2}{E_1}\right) = \frac{4}{9} \times \frac{3}{8} = \frac{1}{6} \quad \left[\because P(E_1) = \frac{4}{9} \right] \end{aligned}$$

(ii) Suppose the ball is replaced after the first draw.

$$\text{Then } P(E_1 \cap E_2) = \frac{4}{9} \cdot \frac{4}{9} = \frac{16}{81}$$

Example 1 : Three students are chosen at random from a class consisting of 12 boys and 4 girls. Find the probability for three students chosen one after another in succession to be boys.

Solution : Let A , B , C be the events of choosing a boy each time in succession.

$\therefore$ Probability for 3 students chosen one after another in succession to be

$$\text{boys} = P(A \cap B \cap C) = P(A) \cdot P\left(\frac{B}{A}\right) \cdot P\left(\frac{C}{A \cap B}\right) = \frac{12}{16} \times \frac{11}{15} \times \frac{10}{14} = \frac{11}{28}$$

Example 4 : A class has 10 boys and 5 girls. Three students are selected at random one after another. Find the probability that (i) first two are boys and third is girl (ii) First and third are of same sex and the second is of opposite sex. [JNTU 2004S, Feb. 2008S (Set No. 3)]

Solution : Total no. of students = $10 + 5 = 15$

(i) The probability that first two are boys and the third is girl is

$$P(E_1 \cap E_2 \cap E_3) = \frac{10}{15} \cdot \frac{9}{14} \cdot \frac{5}{13} = \frac{15}{91}$$

- (ii) Suppose the first and third are boys and second is a girl

$$\text{Probability of the event} = P(E_1) = \frac{10}{15} \cdot \frac{5}{14} \cdot \frac{9}{13} = \frac{15}{91} = 0.1648$$

Suppose first and third are girls and second is boy.

$$\text{Then the probability of the event} = P(E_2) = \frac{5}{15} \cdot \frac{10}{14} \cdot \frac{4}{13} = \frac{20}{273}$$

$$\begin{aligned} \therefore \text{Required probability} &= P(E_1) + P(E_2) \\ &= \frac{15}{91} + \frac{20}{273} = \frac{45+20}{273} = \frac{65}{273} = 0.238 \end{aligned}$$

Example 5 : Two aeroplanes bomb a target in succession. The probability of each correctly scoring a hit is 0.3 and 0.2 respectively. The second will bomb only if the first misses the target. Find the probability that (i) target is hit (ii) both fails to score hits.

[JNTU 2004S (Set No. 3)]

Solution : Let A be the event of first plane hitting the target and B be the event of 2nd plane hitting the target

The probability of 1st plane hitting the target $= P(A) = 0.3$

The probability of 2nd plane hitting the target $= P(B) = 0.2$

$$P(\bar{A}) = 1 - P(A) = 0.7 \text{ and } P(\bar{B}) = 1 - P(B) = 0.8$$

- (i) $P(\text{target is hit}) = P[(A \text{ hits}) \text{ (or) } (A \text{ fails and } B \text{ hits})]$

$$= P[A \cup (\bar{A} \cap B)]$$

$$= P(A) + P(\bar{A} \cap B) \text{ (By Addition Theorem)}$$

$$= P(A) + P(\bar{A}) \cdot P(B)$$

(By multiplication theorem, since A and B are independent)

$$= 0.3 + (0.7)(0.2) = 0.44$$

- (ii) $P(\text{both fails}) = P(A \text{ fails and } B \text{ fails})$

$$= P(\bar{A} \cap \bar{B}) = P(\bar{A}) \cdot P(\bar{B}) = (0.7)(0.8) = 0.56$$

Example 6 : Determine (i) $P\left(\frac{B}{A}\right)$ (ii) $P\left(\frac{A}{B^c}\right)$ if A and B are events with $P(A) = \frac{1}{3}$,

$$P(B) = \frac{1}{4}, P(A \cup B) = \frac{1}{2}.$$

[JNTU Dec. 2004S, (A) Dec. 2013 (Set No. 1)]

Solution : Given $P(A) = \frac{1}{3}, P(B) = \frac{1}{4}, P(A \cup B) = \frac{1}{2}$

$$\text{Now } P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

$$= \frac{1}{3} + \frac{1}{4} - \frac{1}{2} = \frac{1}{12}$$

$$(i) \quad P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{1/12}{1/3} = \frac{1}{4}$$

$$(ii) \quad P(B^c) = 1 - P(B) = 1 - \frac{1}{4} = \frac{3}{4}$$

$$P(A \cap B^c) = P(A) - P(A \cap B) = \frac{1}{3} - \frac{1}{12} = \frac{1}{4}$$

$$\therefore P\left(\frac{A}{B^c}\right) = \frac{P(A \cap B^c)}{P(B^c)} = \frac{1/4}{3/4} = \frac{1}{3}$$

Example 7 : In a certain town 40% have brown hair, 25% have brown eyes and 15% have both brown hair and brown eyes. A person is selected at random from the town.

(i) If he has brown hair, what is the probability that he has brown eyes also ?

(ii) If he has brown eyes, determine the probability that he does not have brown hair ?

Solution : Given $n(S) = 100$

Let A = The set of people who have brown hair

Then $n(A) = 40$

$$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{40}{100}$$

Let B = The set of people who have brown eyes. Then, $n(B) = 25$

$$\therefore P(B) = \frac{25}{100}$$

$A \cap B$ = The set of people who have both brown hair and brown eyes.

$$n(A \cap B) = 15; P(A \cap B) = \frac{15}{100}$$

$$(i) \quad P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{15/100}{40/100} = \frac{3}{8}$$

$$(ii) \quad P\left(\frac{\bar{A}}{B}\right) = \frac{P(\bar{A} \cap B)}{P(B)} = \frac{P(B) - P(A \cap B)}{P(B)} = 1 - \frac{P(A \cap B)}{P(B)}$$

$$= 1 - \frac{15/100}{25/100} = 1 - \frac{3}{5} = \frac{2}{5}$$

$$\therefore P\left(\frac{\bar{A}}{B}\right) = \frac{2}{5}$$

Example 8 : Box A contains 5 red and 3 white marbles and box B contains 2 red and 6 white marbles. If a marble is drawn from each box, what is the probability that they are both of same colour. [JNTU May 2006 (Set No. 3), Aug 2007 S, (A) Nov. 2010 (Set No. 2)]

Solution :

Suppose E_1 = The event that the marble is drawn from box A and is red.

$$\therefore P(E_1) = \frac{1}{2} \cdot \frac{5}{8} = \frac{5}{16}$$

and E_2 = The event that the marble is from box B and is red

$$\therefore P(E_2) = \frac{1}{2} \cdot \frac{2}{8} = \frac{1}{8}$$

The probability that both the marbles are red is

$$P(E_1 \cap E_2) = P(E_1) \cdot P(E_2) = \frac{5}{16} \cdot \frac{1}{8} = \frac{5}{128}$$

Let E_3 = The event that the marble drawn from box A and is white

$$\therefore P(E_3) = \frac{1}{2} \cdot \frac{3}{8} = \frac{3}{16}$$

Let E_4 = The event of marble drawn from box B is white

$$\therefore P(E_4) = \frac{1}{2} \cdot \frac{6}{8} = \frac{3}{8}$$

$$\text{and } P(E_3 \cap E_4) = \frac{3}{16} \cdot \frac{3}{8} = \frac{9}{128}$$

The probability that the marbles are of same colour

$$= P(E_1 \cap E_2) + P(E_3 \cap E_4) = \frac{5}{128} + \frac{9}{128} = \frac{14}{128} = \frac{7}{64} = 0.109$$

Example 9 : Two marbles are drawn in succession from a box containing 10 red, 30 white, 20 blue and 15 orange marbles, with replacement being made after each draw. Find the probability that (i) both are white (ii) first is red and second is white.

[JNTU 2005S, 2006, 2007, Nov. 2008, 2008S, (A) Nov. 2010 (Set No.1)]

Solution : Total no. of marbles in the box = 75

(i) Let E_1 = Event of the first drawn marble is white

$$\therefore P(E_1) = \frac{30}{75}$$

Let E_2 = The event of second drawn marble is also white

$$\therefore P(E_2) = \frac{30}{75}$$

The probability that both marbles are white (with replacement)

$$= P(E_1 \cap E_2) = P(E_1) \cdot P\left(\frac{E_2}{E_1}\right) = \frac{30}{75} \cdot \frac{30}{75} = \frac{4}{25}$$

(ii) Let E_1 = The event that the first drawn marble is red

$$\therefore P(E_1) = \frac{10}{75} = \frac{2}{15}$$

Let E_2 = The event that the drawn marble is white

$$\therefore P\left(\frac{E_2}{E_1}\right) = \frac{30}{75} = \frac{2}{5}$$

The probability that the first marble is red and second marble is white

$$\begin{aligned} &= P(E_1 \cap E_2) = P(E_1) \cdot P\left(\frac{E_2}{E_1}\right) \\ &= \frac{2}{15} \cdot \frac{2}{5} = \frac{4}{75} \end{aligned}$$

Example 10 : Three boxes, practically indistinguishable in appearance have two drawers each. Box 1 contains a gold coin in first and silver coin in the other drawer, Box 2 contains a gold coin in each drawer and Box 3 contains a silver coin in each drawer. One box is chosen at random and one of its drawers is opened at random and a gold coin is found. What is the probability that the other drawer contains a coin of silver. [JNTU 2004 (Set No.2)]

Solution : Let E_i denote the event that the box is chosen, $i = 1, 2, 3$.

$$P(E_i) = \frac{1}{3} \text{ for } i = 1, 2, 3$$

Let A be the event that the gold coin is chosen. Then

$$P\left(\frac{A}{E_i}\right) = \text{Probability that a gold coin is chosen from the box } i = 1, 2, 3.$$

$$\therefore P\left(\frac{A}{E_1}\right) = \frac{1}{2} \quad (\because \text{the total no. of coins in box 1 is 2})$$

$$P\left(\frac{A}{E_2}\right) = \frac{2}{2} = 1 \quad (\text{there are two gold coins in box 2})$$

$$\text{and } P\left(\frac{A}{E_3}\right) = \frac{0}{2} = 0 \quad (\text{There is no gold coin in box 3})$$

(i) The probability that the drawn coin is gold

$$P(A) = P(E_1) P\left(\frac{A}{E_1}\right) + P(E_2) P\left(\frac{A}{E_2}\right) + P(E_3) P\left(\frac{A}{E_3}\right)$$

$$= \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot 1 + 0 = \frac{1}{6} + \frac{1}{3} = \frac{1}{2}$$

(ii) The probability that the drawn coin is silver

$$P(B) = 1 - P(A) = 1 - \frac{1}{2} = \frac{1}{2}$$

Example 11 : The probabilities that students A, B, C, D solve a problem are $\frac{1}{3}$, $\frac{2}{5}$, $\frac{1}{5}$

and $\frac{1}{4}$ respectively. If all of them try to solve the problem, what is the probability that the problem is solved. [JNTU 2007 (Set No. 4)]

Solution : Given the probability of A, B, C, D solving the problem is

$$P(A) = \frac{1}{3}, P(B) = \frac{2}{5}, P(C) = \frac{1}{5}, P(D) = \frac{1}{4}$$

The probability that the problem is not solved by A, B, C, D are

$$P(\bar{A}) = \frac{2}{3}, P(\bar{B}) = \frac{3}{5}, P(\bar{C}) = \frac{4}{5} \text{ and } P(\bar{D}) = \frac{3}{4}$$

The probability that the problem is not solved when A, B, C, D try together (independently)

$$\begin{aligned} &= P(\bar{A} \cap \bar{B} \cap \bar{C} \cap \bar{D}) = P(\bar{A}) \cdot P(\bar{B}) \cdot P(\bar{C}) \cdot P(\bar{D}) \\ &= \frac{2}{3} \cdot \frac{3}{5} \cdot \frac{4}{5} \cdot \frac{3}{4} = \frac{6}{25} \end{aligned}$$

$$\therefore \text{The probability that the problem is solved} = 1 - \frac{6}{25} = \frac{19}{25}$$

Example 11 : A can hit a target 3 times in 5 shots, B hits target 2 times in 5 shots, C hits target 3 times in 4 shots. Find the probability of the target being hit when all of them try. [JNTU 2003S (Set No. 3)]

Solution : Let $P(A)$ be the probability of A hitting the target,

$P(B)$ be the probability of B hitting the target

and $P(C)$ be the probability of C hitting the target.

$$\text{Given } P(A) = \frac{3}{5}, P(B) = \frac{2}{5}, P(C) = \frac{3}{4}$$

$$\Rightarrow P(\bar{A}) = \frac{2}{5}, P(\bar{B}) = \frac{3}{5} \text{ and } P(\bar{C}) = \frac{1}{4}$$

The probability that none of A, B, C hits the target

$$= P(\bar{A} \cap \bar{B} \cap \bar{C}) = P(\bar{A}) \cdot P(\bar{B}) \cdot P(\bar{C})$$

$$= \frac{2}{5} \cdot \frac{3}{5} \cdot \frac{1}{4} = \frac{3}{50}$$

$\therefore$ Required Probability = $P(A \cup B \cup C)$

= The probability of atleast one of A, B, C hitting the target.

$$= 1 - P(\bar{A} \cap \bar{B} \cap \bar{C}) = 1 - \frac{3}{50} = \frac{47}{50}$$

Example 13 : The students in a class are selected at random one after the other for an examination. Find the probability that the boys and girls sit alternately if there are (i) 5 boys and 4 girls (ii) 4 boys and 4 girls. [JNTU 2006 (Set No. 2)]

Solution : Total number of students = 9.

They can be arranged themselves in $9!$ ways.

(i) Since there are 5 boys and 4 girls, they can sit alternately as shown below.

BGBGBGBGB

Let E be the event that 5 boys and 4 girls sit alternately.

There are 5 places for 5 boys and 4 places for 4 girls. So they can be arranged in $5! \times 4!$ ways.

$$\therefore P(E) = \frac{5! \times 4!}{9!} = \frac{1}{126}$$

(ii) Since there are 4 boys and 4 girls, they can sit alternately in the following two mutually exclusive ways.

(a) $E_1 = BGBGBGBG$

(b) $E_2 = GBGBGBGB$

Let $E = E_1 \cup E_2$. Then

$$\begin{aligned} P(E) &= P(E_1 \cup E_2) = P(E_1) + P(E_2) = \frac{4! \times 4!}{8!} + \frac{4! \times 4!}{8!} \\ &= \frac{1}{70} + \frac{1}{70} = \frac{1}{35} \end{aligned}$$

Example 13 : A problem in Statistics is given to the 3 students A, B, C whose chances of solving it are $\frac{1}{2}$, $\frac{3}{4}$ and $\frac{1}{4}$ respectively. What is the probability that the problem is solved?

[JNTU 2003S (Set No.4), 2007, (A) Dec. 2013 (Set No. 1)]

Solution : Let $P(A)$ denote the probability that the problem is solved by A.

Similarly let $P(B)$ and $P(C)$ denote the probabilities of B and C solving the problem respectively.

$$\text{By data } P(A) = \frac{1}{2}, P(B) = \frac{3}{4} \text{ and } P(C) = \frac{1}{4}$$

$$\therefore P(\bar{A}) = 1 - \frac{1}{2} = \frac{1}{2}; P(\bar{B}) = 1 - \frac{3}{4} = \frac{1}{4}$$

$$P(\bar{C}) = 1 - \frac{1}{4} = \frac{3}{4}$$

The probability that the problem is not solved is given by

$$\begin{aligned} P(\bar{A} \cap \bar{B} \cap \bar{C}) &= P(\bar{A}) \cdot P(\bar{B}) \cdot P(\bar{C}) \\ &= \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{32} \end{aligned}$$

The probability that the problem is solved

$$\begin{aligned} &= P(A \cup B \cup C) \\ &= 1 - P(\bar{A} \cap \bar{B} \cap \bar{C}) \\ &= 1 - \frac{3}{32} = \frac{29}{32} = 0.90625 \end{aligned}$$

Example 13 : Two digits are selected at random from the digits 1 through 9.

- (i) If the sum is odd, what is the probability that 2 is one of the numbers selected ?
- (ii) If 2 is one of the digits selected, what is the probability that the sum is odd ?

[JNTU 2005, 2006S (Set No. 1)]

Solution : The given set consists of five odd digits (1, 3, 5, 7, 9) and four even digits (2, 4, 6, 8)

We know that

even + even = even

even + odd = odd

odd + even = odd

odd + odd = even

- (i) Favourable events for selecting '2' = 1

$\therefore$ Total events = 4

$$\text{Hence } P(2) = \frac{1}{4}$$

- (ii) If '2' is selected, then the probability that the sum is odd = $P\left(\frac{\text{sum is odd}}{2 \text{ is selected}}\right)$

If 2 is selected, then the remaining number of digits = 8

$\therefore$ Total events = 8

$$P\left(\frac{\text{sum is odd}}{2 \text{ is selected}}\right) = \frac{\text{Favourable cases for odd}}{\text{Total events}} = \frac{5}{8}$$

Example 13 : The probabilities of 3 students to solve a problem in Mathematics are $1/2$, $1/3$, $1/4$ respectively, find the probability that the problem to be solved.

[JNTU Nov. 2008 (Set No. 4), (K) 2009 (Set No. 4)]

Solution : Given $P(E_1) = \frac{1}{2}$; $P(E_2) = \frac{1}{3}$; $P(E_3) = \frac{1}{4}$

$\therefore P(\bar{E}_1)$ = The probability that the problem is not solved by the 1st student.

$$= 1 - P(E_1) = \frac{1}{2}$$

$$\text{Similarly, } P(\bar{E}_2) = 1 - \frac{1}{3} = \frac{2}{3}$$

$$\text{and } P(\bar{E}_3) = 1 - \frac{1}{4} = \frac{3}{4}$$

The probability that the problem is not solved = $P(\bar{E}_1 \cap \bar{E}_2 \cap \bar{E}_3)$

$$= P(\bar{E}_1) \cdot P(\bar{E}_2) \cdot P(\bar{E}_3) = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} = \frac{1}{4}$$

$\therefore$ The probability that the problem is solved = $P(E_1 \cup E_2 \cup E_3)$

$$= 1 - P(\bar{E}_1 \cap \bar{E}_2 \cap \bar{E}_3) = 1 - \frac{1}{4} = \frac{3}{4}$$

Example 13 : A, B, C, are aiming to shoot a balloon. A will succeed 4 times out of 5 attempts. The chance of B to shoot the balloon is 3 out of 4 and that of C is 2 out of 3. If the three aim the balloon simultaneously, then find the probability that at least two of them hit the balloon.

[JNTU Nov. 2008 (Set No. 3)]

Solution : The probability of A hitting the target = $P(A) = \frac{4}{5}$

The probability of B hitting the target = $P(B) = \frac{3}{4}$

The probability of C hitting the target = $P(C) = \frac{2}{3}$

$\therefore$ The probabilities of A, B, C not hitting the target respectively are

$$P(\bar{A}) = \frac{1}{5}, P(\bar{B}) = \frac{1}{4}, P(\bar{C}) = \frac{1}{3}$$

Now the probability that exactly two will hit the balloon

$$= P(A \cap B \cap \bar{C}) + P(A \cap \bar{B} \cap C) + P(\bar{A} \cap B \cap C)$$

$$\begin{aligned}
&= P(A) \cdot P(B) \cdot P(\bar{C}) + P(A) \cdot P(\bar{B}) \cdot P(C) + P(\bar{A}) \cdot P(B) \cdot P(C) \\
&= \frac{4}{5} \cdot \frac{3}{4} \cdot \frac{1}{3} + \frac{4}{5} \cdot \frac{1}{4} \cdot \frac{2}{3} + \frac{1}{5} \cdot \frac{3}{4} \cdot \frac{2}{3} \\
&= \frac{12+8+6}{60} = \frac{26}{60} = \frac{13}{30}
\end{aligned}$$

The probability that all will hit the balloon

$$= P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C) = \frac{4}{5} \cdot \frac{3}{4} \cdot \frac{2}{3} = \frac{2}{5}$$

$\therefore$ The probability that at least two of them will hit the target

$$= \frac{13}{30} + \frac{2}{5} = \frac{13+12}{30} = \frac{25}{30} = \frac{5}{6}$$

Example 10 : There are 12 cards numbered 1 to 12 in a box, if two cards are selected what is the probability that sum is odd

(i) With replacement

(ii) Without replacement

[JNTU Apr. 2009 (Set No. 2)]

Solution : The cards having even number = {2, 4, 6, 8, 10, 12}

The cards with odd numbers = {1, 3, 5, 7, 9, 11}

With replacement : Suppose we select a card place it back and select one card.

This can happen in $12 \cdot 12 = 144$ ways.

To have an odd sum, (i) the first card has to be odd and the second has to be even or

(ii) the first card has to be even and the second has to be odd.

Number of these favourable ways = $6 \cdot 6 + 6 \cdot 6 = 72$

$$\therefore P(\text{sum is odd}) = \frac{72}{144} = \frac{1}{2}$$

Without replacement : The total number of outcomes = $(12)(11) = 132$

The sum can be odd as below :

(i) The card in first draw is even and on the second is odd.

$$\text{Probability (for (i))} = \frac{6}{12} \cdot \frac{6}{11} = \frac{36}{132}$$

(ii) The card in first draw is odd and on the second is even.

$$P(\text{for (ii)}) = \frac{6}{12} \cdot \frac{6}{11} = \frac{36}{132}$$

$$\begin{aligned}
\therefore P(\text{the sum is odd}) &= \frac{36}{132} + \frac{36}{132} = \frac{72}{132} \\
&= \frac{6}{11}
\end{aligned}$$

Example 13 : Three machines I, II, III produce 40%, 30%, 30% of the total number of items of factory. The percentages of defective items of these machines are 4%, 2%, 3%. If an item is selected at random, find the probability that the item is defective.

[JNTU Apr. 2009 (Set No. 2)]

Solution : Let A, B and C be the events that the machines I, II and III be chosen respectively and let D be the event which denotes the defective item.

$$\text{Given that } P(A) = \frac{40}{100} = 0.40, P(B) = \frac{30}{100} = 0.30, P(C) = \frac{30}{100} = 0.30.$$

From the given data, we have

$$P\left(\frac{D}{A}\right) = \frac{4}{100} = 0.04, P\left(\frac{D}{B}\right) = \frac{2}{100} = 0.02, P\left(\frac{D}{C}\right) = \frac{3}{100} = 0.03$$

$\therefore$ The probability that the selected item at random is defective is

$$\begin{aligned} P(D) &= P(A).P\left(\frac{D}{A}\right) + P(B).P\left(\frac{D}{B}\right) + P(C).P\left(\frac{D}{C}\right) \\ &= \frac{40}{100} \times \frac{4}{100} + \frac{30}{100} \times \frac{2}{100} + \frac{30}{100} \times \frac{3}{100} = \frac{16+16+9}{1000} = \frac{41}{1000} \end{aligned}$$

Example 10 : Two dice are thrown. Let A be the event that the sum of the points on the faces is 9. Let B be the event that at least one number is 6. Find

(i) $P(A \cap B)$ (ii) $P(A \cup B)$ (iii) $P(A^C \cup B^C)$. [JNTU Apr. 2009 (Set No. 3)]

Solution : There are 36 simple outcomes when two dice are thrown.

The event A (= that a sum 9) occurs in the following way :

$$A = \{(3, 6), (4, 5), (5, 4), (6, 3)\}$$

$$\therefore P(A) = \frac{4}{36}$$

The event B that at least one number is 6 occurs in the following way :

$$B = \{(6,1), (6,2), (6,3), (6,4), (6,5), (6,6), (1,6), (2,6), (3,6), (4,6), (5,6)\}$$

$$\therefore P(B) = \frac{11}{36}$$

$$\text{Now } A \cap B = \{(3,6), (6,3)\}$$

$$(i) \quad P(A \cap B) = \frac{2}{36} = \frac{1}{18}$$

$$\begin{aligned} (ii) \quad P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= \frac{4}{36} + \frac{11}{36} - \frac{2}{36} = \frac{13}{36} \end{aligned}$$

$$(iii) \quad P(A^C \cup B^C) = P((A \cap B)^C) = 1 - P(A \cap B) = \frac{17}{18}$$

Example 13 : In a state election in year 2002 there were three major parties X, Y, Z fighting for the claim of Chief Minister ship. The chances of winning the election of the 3 parties are in the ratio 1 : 2 : 3 respectively. The probability that the party X if selected, will introduce total arrack prohibition in that state is $1/2$. The probability that the party Y if selected, will introduce total arrack prohibition in the state is $1/4$ and the prohibition that the party Z if selected, will introduce total arrack prohibition in the state is $3/4$. What is the probability that there will be a total prohibition in the state after the election in year 2002.

[JNTU Apr. 2009 (Set No. 3)]

Solution : Let $P(X \text{ wins}) = P(X)$

$P(Y \text{ wins}) = P(Y)$ and $P(Z \text{ wins}) = P(Z)$

Since $P(X) : P(Y) : P(Z) :: 1 : 2 : 3$

$$\therefore P(X) = \frac{1}{6} ; P(Y) = \frac{2}{6} ; P(Z) = \frac{3}{6}$$

Let A be the event of introducing total arrack prohibition. It is given that

$$P(A/X) = \frac{1}{2} ; P(A/Y) = \frac{1}{4} ; P(A/Z) = \frac{3}{4}$$

$$P(A) = P(A/X) \cdot P(X) + P(A/Y) \cdot P(Y) + P(A/Z) \cdot P(Z)$$

$$= \frac{1}{2} \cdot \frac{1}{6} + \frac{1}{4} \cdot \frac{2}{6} + \frac{3}{4} \cdot \frac{3}{6} = \frac{1}{12} + \frac{1}{12} + \frac{9}{24} = \frac{13}{24}$$

Example 11 : In a group consisting of equal number men and women 10% of the men and 45% of the women are unemployed. If a person is selected randomly from the group then find the probability that the person is an employee.

[JNTU (A) Dec. 2009 (Set No. 3)]

Solution : We have $P(\text{Men}) = P(M) = \frac{1}{2}$, $P(\text{Women}) = P(W) = \frac{1}{2}$

Let A be the event of employed and E be the unemployed event.

$$\text{Then } P\left(\frac{A}{M}\right) = \frac{10}{100} = \frac{1}{10}, P\left(\frac{E}{M}\right) = \frac{9}{10}$$

$$P\left(\frac{A}{W}\right) = \frac{45}{100} = \frac{4.5}{10}, P\left(\frac{E}{W}\right) = \frac{5.5}{10}$$

$$\text{Probability that the person is employed} = P(M) \cdot P\left(\frac{A}{M}\right) + P(W) \cdot P\left(\frac{A}{W}\right)$$

$$= \frac{1}{2} \left(\frac{9}{10} + \frac{4.5}{10} \right) = \frac{13.5}{20} = 0.675$$

Example 13 : If the probability that a communication system will have high fidelity is 0.81 and the probability that it will have high fidelity and selectivity is 0.18. What is the probability that a system with high fidelity will also have high selectivity ?

[JNTU (A) Dec. 2009 (Set No. 4)]

Solution : Let F be the event of communication system will have high fidelity and S be the event of communication system will have high fidelity and selectivity

Then $P(F) = 0.81$ and $P(F \cap S) = 0.18$

$\therefore$ The probability that a system with high fidelity will have high selectivity also

$$= P\left(\frac{S}{F}\right) = \frac{P(S \cap F)}{P(F)} = \frac{0.18}{0.81} = \frac{2}{9}$$

Example 24 : A can hit a target once in five shots B can hit two targets in 3 shots. C can hit one target in 4 shots. What is the probability that 2 shots hit the target ?

[JNTU (A) Dec. 2009 (Set No. 4)]

Solution : $P(A) = \frac{1}{5}, P(B) = \frac{2}{3}, P(C) = \frac{1}{4}$

The probability that two shots hit the target

$$\begin{aligned} &= P(A \cap B \cap \bar{C}) + P(\bar{A} \cap B \cap C) + P(A \cap \bar{B} \cap C) \\ &= \frac{1}{5} \cdot \frac{2}{3} \cdot \frac{3}{4} + \frac{4}{5} \cdot \frac{2}{5} \cdot \frac{1}{4} + \frac{1}{5} \cdot \frac{1}{5} \cdot \frac{1}{4} = \frac{6+8+1}{60} = \frac{15}{60} = \frac{1}{4} \end{aligned}$$

Example 25 : If A and B are two events such that $P(A) = 1/3$, $P(B) = 3/4$ and $P(A \cup B) = 11/12$ find $P(A|B)$ and $P(B|A)$. [JNTU (K) 2009 (Set No.1), (A) June 2016]

Solution : We have $P(A \cap B) = P(A) + P(B) - P(A \cup B)$

$$= \frac{1}{3} + \frac{3}{4} - \frac{11}{12} = \frac{4+9-11}{12} = \frac{2}{12} = \frac{1}{6}$$

$$\text{Now } P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{1/6}{1/3} = \frac{1}{2}$$

$$\text{and } P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{1/6}{3/4} = \frac{4}{18} = \frac{2}{9}$$

Example 24 : A fair die is tossed twice. Find the probability of getting a 4,5 or 6 on the first toss and 1,2,3 or 4 on the second toss. [JNTU (K) 2009 (Set No.1)]

Solution : Let E be the required event.

Total number of sample points = $6 \times 6 = 36$

The favourable sample points are

(4,1), (4,2), (4,3), (4,4)

(5,1), (5,2), (5,3), (5,4)

(6,1), (6,2), (6,3), (6,4)

$$\text{Here } n(E) = 12, n(S) = 36 \quad \therefore P(E) = \frac{12}{36} = \frac{1}{3}$$

Example 13 : Two cards are drawn from a well shuffled pack of 52 cards. Find the probability that they are both aces if the first card is (i) replaced (ii) not replaced.

[JNTU (K) 2009 (Set No. 2)]

Solution : We understand that 2 cards are drawn one after another.

Let E be the event that the two cards are aces.

- (i) When the card drawn first is replaced, we have $n(S) = 52 \times 52$

$$P(E) = \frac{4}{52} \times \frac{4}{52} = \frac{1}{13 \times 13} = \frac{1}{169}$$

- (ii) When the card drawn first is not replaced, we have

$$P(E) = \frac{4}{52} \times \frac{3}{51} = \frac{1}{13 \times 17} = \frac{1}{221}$$

Example 18 : A class has 12 boys and 8 girls. Three students are selected at random one after the another. Find the probability that (i) first two are boys and third is girl (ii) first and third are of the same sex and the second is of opposite sex. [JNTU (K) 2009 (Set No. 3)]

Solution : Total number of students in the class = 20

Number of ways of selecting 3 students one after another = $n(S) = 20 \times 19 \times 18$

- (i) Let E = First two are boys and the third is a girl. Then $n(E) = 12 \times 11 \times 8$

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{12 \times 11 \times 8}{20 \times 19 \times 18} = \frac{44}{285}$$

- (ii) E = Event of first and third are of same sex and second is of opposite sex. This can be that first and third are boys and second is a girl or first and third are girls and the second is a boy.

$$\therefore n(E) = 20 \times 8 \times 19 + 8 \times 20 \times 7$$

$$\text{Hence } P(E) = \frac{n(E)}{n(S)} = \frac{20 \times 19 \times 8 + 20 \times 8 \times 7}{20 \times 19 \times 18} = \frac{64}{342} = \frac{32}{171}$$

Example 25 : The probability that India wins a cricket test match against West Indies is known to be $\frac{2}{5}$. If India and West Indies play three test matches, what is the probability that (i) India will lose all the three matches (ii) India will win at least one test match (iii) India will win at most one match ? [JNTU (K) 2009 (Set No. 4)]

Solution : Given that probability of India winning against West - Indies = $P(W) = \frac{2}{5}$

Probability of losing against West - Indies = $P(\bar{W}) = \frac{3}{5}$

- (i) The Probability that India will loose all the three matches

$$= P(\bar{W}) \times P(\bar{W}) \times P(\bar{W}) = \frac{3}{5} \times \frac{3}{5} \times \frac{3}{5} = \frac{27}{125}$$

(ii) The Probability that India will win at least one test match

$$= 1 - P(\text{India will win no test matches}) = 1 - \frac{27}{125} = \frac{98}{125}$$

(iii) The Probability that India will win at most one match

= The Probability that India will win one match or no matches

$$= P(W) \cdot P(\bar{W}) \cdot P(W) + P(\bar{W}) \cdot P(\bar{W}) \cdot P(\bar{W})$$

$$= \frac{2}{5} \cdot \frac{3}{5} \cdot \frac{3}{5} + \frac{3}{5} \cdot \frac{3}{5} \cdot \frac{3}{5} = \frac{45}{125} = \frac{9}{25}$$

Example 30 : A, B, C are events in a sample space. If A is independent of events B, $B \cup C$ and $B \cap C$ then prove that A and C are independent. [JNTU 2009 (Set No. 3)]

Solution : Let A be the given event.

Given that A is independent of events B, $B \cup C$ and $B \cap C$.

$$\therefore P(A \cap B) = P(A) \cdot P(B) \quad \dots (1)$$

$$P(A \cap (B \cup C)) = P(A) \cdot P(B \cup C) \quad \dots (2)$$

$$P(A \cap (B \cap C)) = P(A) \cdot P(B \cap C) \quad \dots (3)$$

$$\text{Consider } P(A \cap (B \cup C)) = P(A) \cdot P(B \cup C)$$

$$\Rightarrow P[(A \cap B) \cup (A \cap C)] = P(A)[P(B) + P(C) - P(B \cap C)]$$

$$\Rightarrow P(A \cap B) + P(A \cap C) - P((A \cap B) \cap (A \cap C))$$

$$= P(A) \cdot P(B) + P(A) \cdot P(C) - P(A) \cdot P(B \cap C)$$

$$\Rightarrow P(A \cap C) - P(A \cap B \cap C) = P(A) \cdot P(C) - P(A \cap B \cap C)$$

$$\Rightarrow P(A \cap C) = P(A) \cdot P(C) \quad \text{Hence A and C are independent.}$$

Example 11 : A student appears for tests I, II and III. The student is successful if he passes either in test I and II or tests I and III. The probability of the student passing in test I, II, III are p, q and 1/2 respectively. If the probability that the student is successful is 1/2 then find the relation between p and q. [JNTU (A) Nov. 2011 (Set No.1)]

Solution : Given, $P(I) = p$, $P(II) = q$ and $P(III) = \frac{1}{2}$

$$\text{By data, } P(I) \cdot P(II) + P(I) \cdot P(III) = \frac{1}{2}$$

$$\Rightarrow pq + \frac{1}{2}p = \frac{1}{2} \Rightarrow p + 2pq = 1.$$

This gives the required relation between p and q.

Example 13 : One card is drawn from a regular deck of 52 cards. What is the probability of card being either red or a king ? [JNTU (A) April 2012 (Set No. 4)]

Solution : Number of ways of drawing one card from a pack of 52 cards = 52.

Number of ways in which it is red = 26.

$$\text{The probability of drawing red card} = P(E_1) = \frac{26}{52}$$

Number of ways in which it is a king = 4

$$\text{The probability of drawing a king} = P(E_2) = \frac{4}{52}$$

Number of ways in which it is a king and red = 2.

$$\therefore P(E_1 \cap E_2) = \frac{2}{52}$$

By addition theorem, required probability = $P(E_1) + P(E_2) - P(E_1 \cap E_2)$

$$= \frac{26}{52} + \frac{4}{52} - \frac{2}{52} = \frac{15}{26}$$

Example 13 : A bag contains 12 balls numbered from 1 to 12. If a ball is taken at random what is the probability of having a ball with a number which is a multiple of either 2 or 3. [JNTU (A) April 2012 (Set No. 4)]

Solution : Number of ways of drawing a ball from the given bag = 12

Number of ways in which it will be a multiple of 2 = 6 (i.e., 2, 4, 6, 8, 10, 12)

$$\therefore P(E_1) = \frac{6}{12}$$

Number of ways in which the number on the ball is a multiple of 3 = 4 (i.e., 3, 6, 9, 12)

$$\therefore P(E_2) = \frac{4}{12}$$

Number of ways in which the number on the ball is a multiple of 2 and 3 = 2 (i.e., 6, 12)

$$\therefore P(E_1 \cap E_2) = \frac{2}{12}$$

Using Addition theorem of probability,

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2) = \frac{6+4-2}{12} = \frac{2}{3}$$

REVIEW QUESTIONS

- | | |
|---|--|
| <ol style="list-style-type: none"> 1. Define (i) Sample space (ii) Mutually exclusive events. 2. State classical (Mathematical) definition of Probability. 3. State Addition Theorem of Probability. 4. Define conditional Probability. 5. State Multiplication Theorem of Probability. 6. Define (i) Compound event (ii) Dependent and Independent events. | <p>[JNTU (A) Dec. 2017]</p>

<p>[JNTU (A) Dec. 2015]</p> |
|---|--|

EXERCISE 5(A)

1. A bag contains 2 white, 3 black and 4 green balls. The first ball is taken at random from the bag. The second ball is taken at random from the remaining balls. What is the probability that the first ball is white and the second is black ?
2. Mr. X is selected for interview for 3 posts. For the first post there are 5 candidates, for the second there are 4 and for the third there are 6. If the selection of each candidate is equally likely, find the chance that Mr. X will be selected for atleast one post.
3. An urn A contains 8 black balls and 5 white balls. A second urn B contains 6 black and 7 white balls. Find the probability that blind folded person in one draw shall obtain a white ball from an urn.
4. A husband and wife appear in an interview for the same post. The probability of husband's selection is $\frac{1}{7}$ and that of wife's selection is $\frac{1}{5}$. What is the probability that
(a) both of them will be selected ? (b) none of them will be selected ?
(c) only one of them will be selected ?
5. An anti-aircraft gun can take a maximum of 4 shots at an enemy plane moving away from it. The probabilities of hitting the plane at the first, second, third and fourth shots are 0.4, 0.3, 0.2 and 0.1 respectively. What is the probability that the gun hits the plane?
6. Two persons A and B toss a dice. The person who first throws 4 or 5 wins. A starts the game. Show that the probabilities of A's and B's winning are in the ration 3 : 2.
7. Box A contains 5 red and 3 white marbles and Box B contains 2 red and 6 white marbles.
(a) If a marble is drawn from each box, what is the probability that they are both of the same color ?
(b) If 2 marbles are drawn from each box, what is the probability that all four marbles are of same color ?
8. A die is tossed. If the number is odd, what is the probability that it is prime ?
9. What is the probability of getting an even number in the throw of a die ?
[JNTU (A) June 2018]
10. If the two outcomes A and B are independent where $P(A) = 0.4$ and $P(A \cup B) = 0.7$ then find $P(B)$.
[JNTU (A) June 2019]

ANSWERS

1. $\frac{2}{9} \times \frac{3}{8}$
2. $1 - \frac{4}{5} \times \frac{3}{4} \times \frac{5}{6}$
3. $\frac{1}{2} \times \frac{5}{13} + \frac{1}{2} \times \frac{7}{13}$
4. (a) $\frac{1}{7} \times \frac{1}{5}$ (b) $\frac{6}{7} \times \frac{4}{5}$ (c) $\frac{1}{7} \times \frac{4}{5} + \frac{6}{7} \times \frac{1}{5}$
5. $1 - (0.6)(0.7)(0.8)(0.9)$
7. (a) $\frac{7}{16}$ (b) $\frac{55}{784}$
8. $\frac{2}{3}$

5.3 BAYE'S THEOREM [JNTU 2007S, 2008, (A), (H) Dec. 2009, (A) Nov. 2010 (Set No. 4)]

$E_1, E_2, \dots, E_n$ are n mutually exclusive and exhaustive events such that $P(E_i) > 0$ ($i = 1, 2, \dots, n$) in a sample space S and A is any other event in S intersecting with every E_i (i.e., A can only occur in combination with any one of the events $E_1, E_2, \dots, E_n$) such that $P(A) > 0$.

If E_i is any of the events of $E_1, E_2, \dots, E_n$ where $P(E_1), P(E_2), \dots, P(E_n)$ and $P(A/E_1), P(A/E_2), \dots, P(A/E_n)$ are known, then

$$P(E_K / A) = \frac{P(E_K) \cdot P(A/E_K)}{P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2) + \dots + P(E_n) \cdot P(A/E_n)}$$

Proof : $E_1, E_2, \dots, E_n$ are n events of S such that $P(E_i) > 0$ and $E_i \cap E_j = \phi$ for $i \neq j$ where $i, j = 1, 2, \dots, n$. Also $E_1, E_2, \dots, E_n$ are exhaustive events of S and A is any other event of S where $P(A) > 0$.

$$S = E_1 \cup E_2 \cup \dots \cup E_n \text{ and}$$

$$A = A \cap S = A \cap (E_1 \cup E_2 \cup \dots \cup E_n)$$

$$= (A \cap E_1) \cup (A \cap E_2) \cup \dots \cup (A \cap E_n)$$

Here $A \cap E_1, A \cap E_2, \dots, A \cap E_n$ are mutually exclusive events.

$$\begin{aligned} \text{Then } P(E_K / A) &= \frac{P(E_K \cap A)}{P(A)} = \frac{P(E_K \cap A)}{P[(A \cap E_1) \cup (A \cap E_2) \cup \dots \cup (A \cap E_n)]} \\ &= \frac{P(E_K \cap A)}{P(A \cap E_1) + P(A \cap E_2) + \dots + P(A \cap E_n)} \\ &= \frac{P(E_K) \cdot P(A/E_K)}{P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2) + \dots + P(E_n) \cdot P(A/E_n)} \end{aligned}$$

Note : Baye's theorem is also known as formula for the **Probability of "Causes"**, i.e., probability of a particular (cause) E_i given that event A has happened (already).

$P(E_i)$ is 'a priori probability' known even before the experiment, $P(A/E_i)$ "**Likelihoods**" and $P(E_i/A)$ "**Posteriori Probabilities**" determined after the result of the experiment.

SOLVED EXAMPLES

Example 1 : In a certain college, 25% of boys and 10% of girls are studying mathematics. The girls constitute 60% of the student body. (a) What is the probability that mathematics is being studied? (b) If a student is selected at random and is found to be studying mathematics, find the probability that the student is a girl? (c) a boy? [JNTU 2004 (Set No. 2)]

Solution : Given $P(\text{Boy}) = P(B) = \frac{40}{100} = \frac{2}{5}$

and $P(\text{Girl}) = P(G) = \frac{60}{100} = \frac{3}{5}$

Probability that mathematics is studied given that the student is a boy

$$= P(M / B) = \frac{25}{100} = \frac{1}{4}$$

Probability that mathematics is studied given that the student is a girl

$$= P(M / G) = \frac{10}{100} = \frac{1}{10}$$

(a) Probability that the student studied Mathematics

$$= P(M) = P(G) P(M/G) + P(B) P(M/B)$$

$\therefore$ By total probability theorem,

$$P(M) = \frac{3}{5} \cdot \frac{1}{10} + \frac{2}{5} \cdot \frac{1}{4} = \frac{4}{25}$$

(b) By Baye's theorem, probability of mathematics student is a girl

$$= P(G / M) = \frac{P(G) P(M / G)}{P(M)} = \frac{\frac{3}{5} \cdot \frac{1}{10}}{\frac{4}{25}} = \frac{3}{8}$$

(c) Probability of maths student is a boy

$$= P(B / M) = \frac{P(B) P(M / B)}{P(M)} = \frac{\frac{2}{5} \cdot \frac{1}{4}}{\frac{4}{25}} = \frac{5}{8}$$

Example 1 : The chance that doctor A will diagnose a disease x correctly is 60%. The chance that a patient will die by his treatment after correct diagnosis is 40% and the chance of death by wrong diagnosis is 70%. A patient of doctor A, who had disease x, died. What is the chance that his disease was diagnosed correctly.

Solution : Let E_1 be the event that "disease x is diagnosed correctly by doctor A" and E_2 be the event that "a patient of doctor A who has disease x died".

$$\text{Then } P(E_1) = \frac{60}{100} = 0.6, P(E_2 / E_1) = \frac{40}{100} = 0.4$$

$$\text{Now } P(\bar{E}_1) = 1 - 0.6 = 0.4 \text{ and } P(E_2 / E_1) = \frac{70}{100} = 0.7$$

$\therefore$ By Baye's theorem,

$$\begin{aligned} P(E_1 / E_2) &= \frac{P(E_1) \cdot P(E_2 / E_1)}{P(E_1) \cdot P(E_2 / E_1) + P(\bar{E}_1) \cdot P(E_2 / \bar{E}_1)} \\ &= \frac{0.6 \times 0.4}{0.6 \times 0.4 + 0.4 \times 0.7} = \frac{6}{13} \end{aligned}$$

Example 3 : A bag A contains 2 white and 3 red balls and a bag B contains 4 white and 5 red balls. One ball is drawn at random from one of the bags and it is found to be red. Find the probability that the red ball drawn is from bag B. [JNTU (A) Nov. 2011 (Set No. 4)]

Solution : Let A and B denote the events of selecting bag A and bag B respectively.

$$\text{Then } P(A) = \frac{1}{2}; P(B) = \frac{1}{2}.$$

Let R denote the event of drawing a red ball.

$$\text{Having selected bag } A, \text{ the probability to draw a red ball from } A = P(R / A) = \frac{3}{5}$$

$$\text{Similarly, } P(R / B) = \frac{5}{9}$$

One of the bags is selected at random and from it a ball is drawn at random.

It is found to be red. Then the probability that the selected bag is B

$$\begin{aligned} &= P(B / R) = \frac{P(B) \cdot P(R / B)}{P(A) \cdot P(R / A) + P(B) \cdot P(R / B)} \\ &= \frac{\frac{1}{2} \cdot \frac{5}{9}}{\frac{1}{2} \cdot \frac{3}{5} + \frac{1}{2} \cdot \frac{5}{9}} = \frac{25}{52} \end{aligned}$$

Example 4 : First box contains 2 black, 3 red, 1 white balls; second box contains 1 black, 1 red, 2 white balls and third box contains 5 black, 3 red, 4 white balls. Of these a box is selected at random. From it a red ball is randomly drawn. If the ball is red, find the probability that it is from second box.

Solution : Let x, y, z be the first, second and third boxes

$$\therefore P(x) = \frac{1}{3}, P(y) = \frac{1}{3}, P(z) = \frac{1}{3}$$

Let R be the event of drawing a red ball from a box.

$$\text{So } P(R / x) = \frac{3}{6}, P(R / y) = \frac{1}{4}, P(R / z) = \frac{3}{12}$$

$\therefore$ By Baye's theorem, the required probability

$$\begin{aligned} &= P(y/R) = \frac{P(y) \cdot P(R / y)}{P(x) \cdot P(R / x) + P(y) \cdot P(R / y) + P(z) \cdot P(R / z)} \\ &= \frac{\frac{1}{3} \times \frac{1}{4}}{\frac{1}{3} \times \frac{3}{6} + \frac{1}{3} \times \frac{1}{4} + \frac{1}{3} \times \frac{3}{12}} = \frac{1}{4} \end{aligned}$$

Example 3 : Suppose 5 men out of 100 and 25 women out of 10,000 are colour blind. A colour blind person is chosen at random. What is the probability of the person being a male (Assume male and female to be in equal numbers) ?

[JNTU April 2004, (A) Nov. 2010 (Set No. 2)]

Solution : Given that 5 men out of 100 and 25 women out of 10000 are color blind. A colour blind person is chosen at random.

The probability that the chosen person is male $= P(M) = \frac{1}{2}$

Similarly the probability that the chosen person is female $= P(W) = \frac{1}{2}$

Let B represent a blind person. Then

$$P(B / M) = \frac{5}{100} = 0.05$$

$$P(B / W) = \frac{25}{10000} = \frac{1}{400} = .0025$$

The probability that the chosen person is male is given by

$$\begin{aligned} P(M / B) &= \frac{P(B / M) \cdot P(M)}{P(M) \cdot P(B / M) + P(W) \cdot P(B / W)} \\ &= \frac{0.05 \times 0.5}{(.05 \times .5) + (.5 \times .0025)} = 0.95 \end{aligned}$$

Example 3 : In a bolt factory machines A, B, C manufacture 20%, 30% and 50% of the total of their output and 6%, 3% and 2% are defective. A bolt is drawn at random and found to be defective. Find the probabilities that it is manufactured from (i) Machine A. (ii) Machine B (iii) Machine C.

[JNTU 2007 (Set No. 4), JNTU (A) Apr. 2012, (Set No. 2)]

Solution : Let $P(A)$, $P(B)$, $P(C)$ be the probabilities of the events that the bolts are manufactured by the machines A, B, C respectively. Then

$$P(A) = \frac{20}{100} = \frac{1}{5}, P(B) = \frac{30}{100} = \frac{3}{10}, P(C) = \frac{50}{100} = \frac{1}{2}$$

Let D denote that the bolt is defective. Then

$$P(D / A) = \frac{6}{100}, P(D / B) = \frac{3}{100}, P(D / C) = \frac{2}{100}$$

(i) If bolt is defective, then the probability that it is from machine A

$$= P(A / D) = \frac{P(D / A) \cdot P(A)}{P(D / A) \cdot P(A) + P(D / B) \cdot P(B) + P(D / C) \cdot P(C)} = \frac{12}{31}$$

$$\text{Similarly, (ii) } P(B / D) = \frac{9}{31} \quad \text{(iii) } P(C / D) = \frac{10}{31}$$

Example 3 : Of the three men, the chances that a politician, a business man or an academician will be appointed as a vice-chancellor (V.C) of a University are 0.5, 0.3, 0.2 respectively. Probability that research is promoted by these persons if they are appointed as V.C are 0.3, 0.7, 0.8 respectively.

- (i) Determine the probability that research is promoted
- (ii) If research is promoted, what is the probability that V.C is an academician?

[JNTU 2007, 2007S, 2008S (Set No. 1)]

Solution : Let A , B , C be the events that a politician, businessmen or an academician will be appointed as V.C of the three men. Then

$$P(A) = 0.5, P(B) = 0.3, P(C) = 0.2$$

The probabilities that research is promoted if they are appointed as V.Cs are

$$P(R/A) = 0.3, P(R/B) = 0.7, P(R/C) = 0.8$$

- (i) The probability that the research is promoted

$$\begin{aligned} &= P(A).P(R/A) + P(B).P(R/B) + P(C).P(R/C) \\ &= (0.5)(0.3) + (0.3)(0.7) + (0.2)(0.8) = 0.52 \end{aligned}$$

- (ii) The probability that research is promoted when the V.C is an academician

$$\begin{aligned} &= P(C / R) = \frac{P(C).P(R / C)}{P(C).P(R / C) + P(B).P(R / B) + P(A).P(R / A)} \\ &= \frac{0.16}{0.15 + 0.21 + 0.16} = \frac{4}{13} = 0.30769 \end{aligned}$$

Example 8 : A businessman goes to hotels X, Y, Z, 20%, 50%, 30% of the time respectively. It is known that 5%, 4%, 8% of the rooms in X, Y, Z hotels have faulty plumbings. What is the probability that business man's room having faulty plumbing is assigned to hotel Z?

[JNTU 2007, 2008, (K) 2009, (A) Nov. 2010 (Set No. 3)]

Solution : Let the probabilities of business man going to hotels X , Y , Z be respectively $P(X)$, $P(Y)$, $P(Z)$. Then

$$\begin{aligned} P(X) &= \frac{20}{100} = \frac{2}{10}; & P(Y) &= \frac{50}{100} = \frac{5}{10} \\ P(Z) &= \frac{30}{100} = \frac{3}{10} \end{aligned}$$

Let E be the event that the hotel room has faulty plumbing. Then the probabilities that hotels X , Y , Z have faulty plumbing are

$$\begin{aligned} P(E / X) &= \frac{5}{100} = \frac{1}{20}; & P(E / Y) &= \frac{4}{100} = \frac{1}{25} \\ P(E / Z) &= \frac{8}{100} = \frac{2}{25} \end{aligned}$$

The probability that the business man's room having faulty plumbing is assigned to hotel Z

$$\begin{aligned}
 &= P(Z / E) = \frac{P(Z) \cdot P(E / Z)}{P(Z) \cdot P(E / Z) + P(Y) \cdot P(E / Y) + P(X) \cdot P(E / X)} \\
 &= \frac{\frac{3}{10} \times \frac{2}{25}}{\frac{3}{10} \cdot \frac{2}{25} + \frac{5}{10} \cdot \frac{1}{25} + \frac{2}{10} \cdot \frac{1}{20}} = \frac{4}{9}
 \end{aligned}$$

Example 9 : There are two boxes. in box I, 11 cards are there numbered 1 to 11 and in box II, 5 cards are there numbered 1 to 5. A box is chosen and a card is drawn. If the card shows an even number then another card is drawn from the same box. If card shows an odd number another card is drawn from the other box. Find the probability that (i) both are even (ii) both are odd (iii) if both are even what is the probability that they are from box I ?

[JNTU 2007, 2007S, 2008S]

Solution : Number of cards in box I	= 11
Number of cards with even number	= 5
Number of cards with odd number	= 6
Number of cards in box II	= 5
Number of cards with even number	= 2
Number of cards with odd number	= 3

$$\text{The probability of choosing any one box} = \frac{1}{2}$$

(i) Let E be the event that both the cards are even.

For this a box is chosen and a card is picked, if the first card is even then the second card is also picked from the same box and that card is also even.

Let E_1 be the event that both the cards are from box I

$$\therefore P(E_1) = \frac{1}{2} \left(\frac{5}{11} \right) \left(\frac{4}{10} \right) = \frac{1}{11}$$

Let E_2 be the event that both the cards are from box II. Then

$$P(E_2) = \frac{1}{2} \left(\frac{2}{5} \right) \left(\frac{1}{4} \right) = \frac{1}{20}$$

$$\therefore P(E) = P(E_1) + P(E_2) = \frac{1}{11} + \frac{1}{20} = \frac{31}{220}$$

(ii) Let E be the event that both cards are odd.

Then a box is chosen.

First card is odd and second card is picked from another box and that is also odd.

Let E_1 be the event that the first card is odd from box I and second card is odd from box II.

$$\therefore P(E_1) = \frac{1}{2} \left(\frac{6}{11} \right) \left(\frac{3}{5} \right) = \frac{9}{55}$$

Let E_2 be the event that first card is odd from box II and second card is odd from box I.

$$\text{Then } P(E_2) = \frac{1}{2} \left(\frac{3}{5} \right) \left(\frac{6}{11} \right) = \frac{9}{55}$$

$$\therefore P(E) = P(E_1) + P(E_2) = \frac{9}{55} + \frac{9}{55} = \frac{18}{55}$$

$$(iii) \quad \text{The probability that both cards from box I are even} = \frac{1}{2} \cdot \frac{5}{11} \cdot \frac{4}{10} = \frac{1}{11}$$

$$\text{The probability that both cards from box II are even} = \left(\frac{1}{2} \right) \left(\frac{2}{5} \right) \left(\frac{1}{4} \right) = \frac{1}{20}$$

Probability that if both cards are even then they are from box I

$$= \frac{\frac{1}{2} \cdot \frac{5}{11} \cdot \frac{4}{10}}{\frac{1}{2} \cdot \frac{5}{11} \cdot \frac{4}{10} + \frac{1}{2} \cdot \frac{2}{5} \cdot \frac{1}{4}} = \frac{\frac{1}{11}}{\frac{1}{11} + \frac{1}{20}} = \frac{20}{31}$$

Example 10 : Two bolts are drawn from a box containing 4 good and 6 bad bolts. Find the probability that the second bolt is good if the first one is found to be bad.

[JNTU 2004 (Set No. 1)]

Solution : Since in the problem, it is given that “probability that the second bolt is good, if the first one is found to be bad”, we have to understand that the two bolts are drawn in succession (without replacement).

$$\text{Probability that the first bolt is found to be bad} = \frac{6}{10}$$

$$\text{Probability that the bolt drawn now, is found to be good} = \frac{4}{9}$$

$$P(\text{first is bad and second is good}) = \frac{6}{10} \cdot \frac{4}{9} = \frac{4}{15}$$

Example 11 : In a factory, machine A produces 40% of the output and machine B produces 60%. On the average, 9 items in 1000 produced by A are defective and 1 item in 250 produced by B is defective. An item drawn at random from a day's output is defective. What is the probability that it was produced by A or B ? [JNTU 2004S, (A) Nov. 2010 (Set No. 1)]

Solution : Output produced by A = 40%

$$\therefore P(A) = 0.4$$

Output produced by B = 60%

$$\therefore P(B) = 0.6$$

$P(D / A)$ = Probability that items produced by A are defective

$$= \frac{9}{1000} = .009$$

Similarly, $P(D / B) = \frac{1}{250} = 0.004$

$P(A / D)$ = Probability that the bolt is produced by A given that it is defective

$$\begin{aligned} &= \frac{P(A) \times P(D / A)}{P(A) \times P(D / A) + P(B) \times P(D / B)} \\ &= \frac{0.4 \times 0.009}{0.4 \times 0.009 + 0.6 \times 0.004} \\ &= \frac{0.0036}{0.0036 + 0.0024} = \frac{0.0036}{0.006} = 0.6 \end{aligned}$$

$P(B / D)$ = Probability of manufacturing the defective bolt by machine B

$$= \frac{0.6 \times 0.004}{0.006} = \frac{0.0024}{0.006} = 0.4$$

(or) since $P(A / D) = 0.6$, therefore, $P(B / D) = 0.4$

Note : Strictly speaking, the probability that it is produced by A or B = 1.

Example 11 : A manufacturer firm produces steel pipes in three plants, with daily production volume of 500; 1000 and 2000 units respectively. According to past experience; it is known that the fraction of defective outputs produced by these plants are respectively 0.005; 0.008 and 0.010. If a pipe is selected from day's total production and found to be defective, find out (i) from which plant the pipe came (ii) what is the probability that it came from the first plant?

Solution : Let A_1, A_2, A_3 be the events that a pipe is manufactured in plant 1, 2 or 3 respectively

$$\therefore P(A_1) = \frac{500}{3500} = \frac{1}{7}, P(A_2) = \frac{1000}{3500} = \frac{2}{7}, P(A_3) = \frac{2000}{3500} = \frac{4}{7}.$$

Let B be the event that a defective pipe is drawn.

Then $P(B / A_1) = 0.005, P(B / A_2) = 0.008, P(B / A_3) = 0.010$

Baye's theorem gives

$$\begin{aligned} P\left(\frac{A_1}{B}\right) &= \frac{P(B / A_1) \cdot P(A_1)}{P(B / A_1) \cdot P(A_1) + P(B / A_2) \cdot P(A_2) + P(B / A_3) \cdot P(A_3)} \\ &= \frac{\frac{0.005}{7}}{\left(0.005 \times \frac{1}{7}\right) + \left(0.008 \times \frac{2}{7}\right) + \left(0.010 \times \frac{4}{7}\right)} = \frac{5}{61} \end{aligned}$$

$$\text{Similarly, } P(A_2 / B) = \frac{0.016 / 7}{0.061 / 7} = \frac{16}{61}; \quad P(A_3 / B) = \frac{0.040 / 7}{0.061 / 7} = \frac{40}{61}$$

Thus it is most probable that the defective pipe has been drawn from the output of the third plant.

Example 13 : A man is known to speak the truth 3 out of 4 times. He throws a die and reports that it is a six. Find the probability that it is actually a six.

[JNTU (A) Nov. 2011 (Set No. 4)]

Solution : Let E_1 be the event that throws 6 when rolling a die. Then

$$P(E_1) = \frac{1}{6} \text{ let } E_2 = \bar{E}_1 \text{ and } P(E_2) = 1 - P(E_1) = \frac{5}{6}$$

Let A be the event that the man is telling the truth. Then $P(A) = 3/4$

Clearly, (A / E_1) is event that the man is speaking the truth, it is being known that 6 has fallen and (A / E_2) is the event that the man does not speak the truth.

$$P(A / E_1) = 3/4 \text{ and } P(A / E_2) = 1 - \frac{3}{4} = \frac{1}{4}$$

$\therefore (E_1 / A)$ is the event that it is actually 6, when it is known that the man reports 6.

Using Bayes' theorem,

$$P(E_1 / A) = \frac{P(A / E_1) \cdot P(E_1)}{P(A / E_1) \cdot P(E_1) + P(A / E_2) \cdot P(E_2)} = \frac{\frac{1}{6} \times \frac{3}{4}}{\frac{1}{6} \times \frac{3}{4} + \frac{5}{6} \times \frac{1}{4}} = \frac{3}{8}$$

REVIEW QUESTIONS

1. State Baye's Theorem of probability. [JNTU (A) June 2016]
2. State and prove Baye's Theorem. [JNTU (A) Dec. 2015]

EXERCISE 5(B)

1. A test of multiple choice questions with four choices is held. A candidate appearing for the test either guesses or recalls or computes the answer for any question. The probability that he makes a guess is $1/6$ and the probability that he recalls the answer is $1/3$. The probability that his answer is correct, given that he recalls it is $1/8$. Find the probability that he computes the answer to the question, given that he answered it correctly.
2. A box contains 4 balls. Two balls are drawn from it and are found to be white. Find the probability that all the balls in the bag are white.
3. In a certain college, 40% of men and 10% of women are taller than 1.8 metres. Further more in the college 60% of students are women. If a student is selected at random and is taller than 1.8 metres find the probability that the selected student is a woman.

4. Companies B_1, B_2, B_3 produce 30%, 45% and 25% of the cars respectively. It is known that 2%, 3% and 2% of the cars produced from B_1, B_2 and B_3 are defective.
 (a) What is the probability that a car purchased is defective ?
 (b) If a car purchased is found to be defective what is the probability that this car is produced by company B_3 ? **[JNTU Nov 2005 (Set 2)]**
5. Suppose three companies X, Y, Z produce T.V's X produce twice as many as Y while Y and z produce the same number. It is known that 2% of X , 2% of Y and 4% of y are defective. All the T.V's produced and put into one shop and then one T.V. is chosen at random.
 (a) What is the probability that the TV is defective ?
 (b) Suppose a T.V chosen is defective, what is the probability that this T.V is produced by company X ?
6. Box I contains 1 white, 2 red, 3 green balls, Box II contains 2 white, 3 red, 1 green balls, Box III contains 3 white, 1 red, 2 green balls. Two balls are drawn from a box chosen at random. These are found to be one white and one red. Determine the probability that the balls so drawn come from box II.
7. In a bolt factory machine A, B and C manufacture respectively 25%, 35% and 40% of the total of their output 5, 4, 2 percents are defective bolts. A bolt is drawn at random from the product and is found to be defective. What are the probabilities that it was manufactured by machines A, B and C ? **[JNTU (A) June 2016]**

ANSWERS

- | | |
|--|--|
| <p>1. $\frac{6}{7}$ [Hint : $P(E) = \frac{\frac{1}{2} \times 1}{\frac{1}{6} \cdot \frac{1}{4} + \frac{1}{3} \cdot \frac{1}{8} + \frac{1}{8} \cdot 1}$]</p> | <p>2. $\frac{3}{5}$ [Hint : $\frac{\frac{1}{3} \times 1}{\frac{1}{3} \times \frac{1}{6} + \frac{1}{3} \times \frac{3}{6} + \frac{1}{3} \times 1}$]</p> |
| <p>3. $\frac{0.6 \times 0.1}{0.4 \times 0.4 + 0.6 \times 0.1}$</p> | <p>4. (a) 0.0245 (b) $\frac{10}{49}$</p> |
| <p>5. (a) 0.025 (b) 0.40</p> | <p>6. $\frac{6}{11}$</p> |

QUIZ

1. If a coin is tossed twice, the probability of getting at least one head is
 (a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{3}{4}$ (d) none
2. The probability of getting a number greater than 2 or an even number in a single throw of a fair die is
 (a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $\frac{5}{6}$ (d) none

3. A bag contains 3 red balls, 4 white balls and 7 black balls. The probability of drawing a red or a black ball is
- (a) $\frac{2}{7}$ (b) $\frac{5}{7}$ (c) $\frac{3}{7}$ (d) $\frac{4}{7}$
4. The odds in favour of drawing a king or a diamond from a well shuffled pack are
- (a) 9 : 4 (b) 4 : 9 (c) 5 : 9 (d) 9 : 5
5. The probability that a leap year will have 53 Tuesdays is
- (a) $\frac{1}{7}$ (b) $\frac{3}{7}$ (c) $\frac{2}{7}$ (d) $\frac{5}{7}$
6. The probability of at least one of the events A and B occurs is 0.6. If A and B occurs simultaneously with probability .2 then $P(A') + P(B') =$
- (a) 0.4 (b) 1.2 (c) 0.8 (d) 1.4
7. A and B' are two independent events such that $P(A' \cap B) = \frac{8}{25}$ and $P(A \cap B') = \frac{3}{25}$, then $P(A)$ is
- (a) $\frac{2}{5}$ (b) $\frac{4}{5}$ (c) $\frac{1}{5}$ (d) $\frac{3}{5}$
8. For two events A and B if $P(A) = P\left(\frac{A}{B}\right) = \frac{1}{4}$ and $P\left(\frac{B}{A}\right) = \frac{1}{2}$, then
- (a) A and B are mutually exclusive (b) A and B are independent
- (c) $P\left(\frac{A'}{B}\right) = \frac{3}{4}$ (d) none
9. If a coin is tossed 6 times in succession, the probability of getting at least one head is
- (a) $\frac{1}{64}$ (b) $\frac{3}{32}$ (c) $\frac{63}{64}$ (d) none
10. If p_1 and p_2 are the probabilities of passing an examination respectively, then the probability of only one failing in the examination is
- (a) $p_1 + p_2 - 2p_1 p_2$ (b) $p_1 + p_2$
- (c) $p_1 + p_2 - p_1 p_2$ (d) none
11. A coin is tossed n times. The probability that the head will present itself an odd number of times is
- (a) $\frac{1}{2^n}$ (b) $\frac{1}{2n}$ (c) $\frac{1}{2}$ (d) $\frac{4}{13}$

12. Two cards are drawn at random from a pack of 52 cards. The probability of these two being aces is
(a) $\frac{2}{52C_2}$ (b) $\frac{1}{221}$ (c) $\frac{1}{663}$ (d) none
13. If a card is drawn from a well shuffled pack of 52 cards, then the probability that it is a spade or a queen is
(a) $\frac{17}{52}$ (b) $\frac{4}{13}$ (c) $\frac{13}{52} + \frac{4}{51}$ (d) none
14. If two balls are drawn from a bag containing 3 white 4 black and 5 red balls, then the probability that the balls drawn are of different colours is
(a) $\frac{47}{66}$ (b) $\frac{10}{33}$ (c) $\frac{5}{22}$ (d) $\frac{2}{11}$
15. Six boys and six girls sit round a table randomly. The probability that all the six girls sit together is
(a) $\frac{2}{77}$ (b) $\frac{1}{77}$ (c) $\frac{2! \times 6! \times 6!}{11!}$ (d) $\frac{2! \times 6! \times 6!}{12!}$
16. From 20 tickets marked 1, 2, 20 one is drawn at random. The probability that the number on it is a multiple of 2 or 3 is
(a) $\frac{16}{20}$ (b) $\frac{15}{20}$ (c) $\frac{14}{20}$ (d) $\frac{13}{20}$
17. 3 letters are written to different persons and addresses on the 3 envelopes are also written. Without looking at the addresses, the probability that the letters go into the right envelopes is
(a) 1 (b) $\frac{1}{6}$ (c) $\frac{1}{3}$ (d) none
18. The probability of solving a problem by three students A, B, C respectively are $\frac{1}{3}, \frac{1}{4}, \frac{1}{5}$. Then the probability that the problem will be solved is
(a) $\frac{59}{60}$ (b) $\frac{3}{5}$ (c) $\frac{2}{5}$ (d) none
19. If $P(A \cup B) = 0.75$ and $P(A \cap B) = 0.15$ then $P(\bar{A}) + P(\bar{B}) =$
(a) 0.9 (b) 0.6 (c) 1.25 (d) 1.1
20. If $P(A) = 0.4$, $P(A \cup B) = 0.7$, and A, B are independent events, then $P(B) =$
(a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{1}{4}$ (d) $\frac{1}{5}$

21. An experiment yields three mutually exclusive events A, B, C if $P(A) = 2 P(B) = 3 P(C)$, then $P(A)$
- (a) $\frac{2}{11}$ (b) $\frac{3}{11}$ (c) $\frac{6}{11}$ (d) $\frac{5}{11}$
22. The chance that a non-leap year contains 53 Mondays is
- (a) $\frac{1}{7}$ (b) $\frac{2}{7}$ (c) $\frac{1}{365}$ (d) $\frac{2}{365}$

ANSWERS

- | | | | |
|---------|-------------|-------------|-------------|
| 1. c | 2. c | 3. b | 4. b |
| 5. c | 6. $c \& d$ | 7. $b \& c$ | 8. $b \& c$ |
| 9. c | 10. a | 11. c | 12. b |
| 13. d | 14. a | 15. b | 16. d |
| 17. b | 18. b | 19. d | 20. b |
| 21. c | 22. a | | |

PREVIOUS GATE QUESTIONS

1. In a housing society, half of the families have a single child per family, while the remaining half have two children per family. The probability that a child picked at random, has a sibling is _____ [GATE 2014 (EC)]
2. A box contains 25 parts of which 10 are defective. Two parts are being drawn simultaneously in a random manner from the box. The probability of both the parts being good is [GATE 2014 (ME)]
- (A) $\frac{7}{20}$ (B) $\frac{42}{125}$ (C) $\frac{25}{20}$ (D) $\frac{5}{9}$
3. A group consists of equal number of men and women. Of this group 20% of men and 50% of the women are unemployed. If a person is selected at random from this group, the probability of the selected person being employed is _____ [GATE 2014 (ME)]
4. A fair (unbiased) coin was tossed four times in succession and resulted in the following outcomes :
- (i) Head (ii) Head (iii) Head (iv) Head . The probability of obtaining a 'Tail' when the coin is tossed again is [GATE 2014 (CE)]
- (A) 0 (B) $\frac{1}{2}$ (C) $\frac{4}{5}$ (D) $\frac{1}{5}$

5. The probability that a given positive integer lying between 1 and 100 (both inclusive) is NOT divisible by 2, 3, 5 is _____ [GATE 2014 (CS)]
6. Let S be a sample space and two mutually exclusive events A and B such that $A \cup B = S$. If $P(\cdot)$ denotes the probability of the events, the maximum value of $P(A)P(B)$ is _____ [GATE 2014 (CS)]
7. Suppose A and B are two independent events with probabilities $P(A) \neq 0$ and $P(B) \neq 0$. Let $\bar{A}$ and $\bar{B}$ be their complements. Which one of the following statements is FALSE? [GATE 2015 (EC)]
- (A) $P(A \cap B) = P(A)P(B)$ (B) $P(A | B) = P(A)$
(C) $P(A \cup B) = P(A) + P(B)$ (D) $P(\bar{A} \cap \bar{B}) = P(\bar{A})P(\bar{B})$
8. Two coins R and S are tossed. The 4 joint events $H_R H_S, T_R T_S, H_R T_S, T_R H_S$ have probabilities 0.28, 0.18, 0.30, 0.24 respectively, where H represent head and T represents tail. Which one of the following is TRUE? [GATE 2015 (EE)]
- (A) The coin tosses are independent (B) R is fair, S is not
(C) S is fair, R is not (D) The coin tosses are dependent
9. The chance of a student passing an exam is 20%. The chance of a student passing the exam and getting above 90% marks in it is 5%. Given that a student passes the examination. The probability that the student gets above 90% marks is [GATE 2015 (ME)]
- (A) $\frac{1}{18}$ (B) $\frac{1}{4}$ (C) $\frac{2}{9}$ (D) $\frac{5}{18}$
10. If $P(X) = \frac{1}{4}$, $P(Y) = \frac{1}{3}$ and $P(X \cap Y) = \frac{1}{12}$, the value of $P(Y | X)$ is [GATE 2015 (ME)]
- (A) $\frac{1}{4}$ (B) $\frac{4}{25}$ (C) $\frac{1}{3}$ (D) $\frac{29}{50}$
11. Four cards are randomly selected from a pack of 52 cards. If the first two cards are kings, what is the probability that the third card is a king? [GATE 2015 (CE)]
- (A) $\frac{4}{52}$ (B) $\frac{2}{50}$ (C) $\frac{1}{52} \times \frac{1}{52}$ (D) $\frac{1}{52} \times \frac{1}{52} \times \frac{1}{50}$
12. Candidates were asked to come to an interview with 3 pens each. Black, blue, green and red were the permitted pen colours that the candidate could bring. The probability that a candidate comes with all 3 pens having the same colour is ____ [GATE 2016 (EE)]

13. X and Y are two random independent events. It is known that $P(X) = 0.40$ and $P(X \cup Y^c) = 0.7$. Which one of the following is the value of $P(X \cup Y)$?
(A) 0.7 (B) 0.5 (C) 0.4 (D) 0.2 [GATE 2016(CE)]
14. Consider the following experiment : [GATE 2016(CS)]
Step 1 : Flip a fair coin twice
Step 2 : If the outcomes are (TAILS, HEADS) then output Y and stop.
Step 3 : If the outcomes are either (HEADS, HEADS) or (HEADS, TAILS) then output N and S .
Step 4 : If the outcomes are (TAILS, TAILS) then go to Step 1
The probability that the output of the experiment is Y is (upto two decimal places)

15. Suppose that a shop has an equal number of LED bulbs of two different types. The probability of an LED bulb lasting more than 100 hours given that it is of Type 1 is 0.7, and given that it is of Type 2 is 0.4. The probability that an LED bulb chosen uniformly at random lasts more than 100 hours is _____ [GATE 2016(CS)]
16. The probability of getting a "head" in a single toss of a biased coin is 0.3. The coin is tossed repeatedly till a "head" is obtained. If the tosses are independent, then the probability of getting "head" for the first time in the fifth toss is _____
[GATE 2016(EC)]
17. Three cards were drawn from a pack of 52 cards. The probability that they are a king, a queen and a jack is _____ [GATE 2016(ME)]
(A) $\frac{16}{5525}$ (B) $\frac{64}{2197}$ (C) $\frac{3}{13}$ (D) $\frac{8}{16575}$
18. An urn contains 5 red and 7 green balls. A ball is drawn at random and its colour is noted. The ball is placed back into the urn along with another ball of the same colour. The probability of getting a red ball in the next draw is _____ [GATE 2016(IN)]
(A) $\frac{65}{156}$ (B) $\frac{67}{156}$ (C) $\frac{79}{156}$ (D) $\frac{89}{156}$
19. Three fair cubical dice are thrown simultaneously. The probability that all three dice have the same number of dots on the faces showing up is (upto third decimal place)
_____ [GATE 2017 (EC)]

20. Two coins are tossed simultaneously. The probability (upto two decimal points accuracy) of getting at least one head is _____ [GATE 2017 (ME)]
21. A two-faced fair coin has its faces designated as head (H) and tail (T). This coin is tossed three times in succession to record the following outcomes : H, H, H. If the coin is tossed one more time, the probability (up to one decimal place) of obtaining H again, given the previous realizations of H, H and H, would be _____ [GATE 2017 (CE)]
22. P and Q are considering to apply for a job. The probability that P applies for the job is $\frac{1}{4}$. The probability that P applies for the job given that Q applies for the job is $\frac{1}{2}$, and the probability that Q applies for the job given that P applies for the job is $\frac{1}{3}$. Then the probability that P does not apply for the job given that Q does not apply for the job is _____ [GATE 2017 (CS)]
- (A) $\frac{4}{5}$ (B) $\frac{5}{6}$ (C) $\frac{7}{8}$ (D) $\frac{11}{12}$
23. The probability that a communication system will have high fidelity is 0.81. The probability that the system will have both high fidelity and high selectivity is 0.18. The probability that a given system with high fidelity will have high selectivity is _____ [GATE 2017 (IN)]
- (A) 0.181 (B) 0.191 (C) 0.222 (D) 0.826
24. A cab was involved in a hit and run accident at night. Your are given the following data about the cabs in the city and the accident.
- (i) 85% of cabs in the city are green and the remaining cabs are blue.
- (ii) A witness identified the cab involved in the accident as blue.
- (iii) It is known that a witness can correctly identified the cab colour only 80% of the time.
- Which of the following is closest to the probability that the accident was caused by a blue cab ? [GATE 2018 (EC)]
- (A) 12% (B) 15% (C) 41% (D) 80%
25. A class of twelve children has two more boys than girls. A group of three children are randomly picked from this class to accompany the teacher on a field trip. What is the probability that the group accompanying the teacher contains more girls than boys ? [GATE 2018 (EE)]
- (A) 0 (B) $\frac{325}{864}$ (C) $\frac{525}{864}$ (D) $\frac{5}{12}$

26. Four red balls, four green balls and four blue balls are put in a box. Three balls are pulled out of the box at random one after another without replacement. The probability that all the three balls are red is [GATE 2018 (ME)]
- (A) $\frac{1}{72}$ (B) $\frac{1}{55}$ (C) $\frac{1}{36}$ (D) $\frac{1}{27}$
27. Two people P and Q decide to independently roll two identical dice, each with 6 faces numbered 1 to 6. The person with the lower number wins. In case of a tie, they roll the dice repeatedly until there is no tie. Define a trial as a throw of the dice by P and Q. Assume that all 6 numbers on each dice are equi-probable and that all trials are independent. The probability (rounded to 3 decimal places) that one of them wins on the third trial is _____ [GATE 2018 (CS)]
28. Two bags A and B have equal number of balls. Bag A has 20% red balls and 80% green balls. Bag B has 30% red balls, 60% green balls and 10% yellow balls. Contents of Bags A and B are mixed thoroughly and a ball is randomly picked from the mixture. What is the chance that the ball picked is red ? [GATE 2018 (IN)]
- (A) 20% (B) 25% (C) 30% (D) 40%
29. The probabilities of occurrence of events F and G are $P(F) = 0.3$ and $P(G) = 0.4$, respectively. The probability that both events occur simultaneously is $P(F \cap G) = 0.2$. The probability of occurrence of atleast one event $P(F \cup G)$ is _____ [GATE 2018 (PI)]
30. The probability that the annual maximum flood discharge will exceed $2500 \text{ m}^3/\text{s}$, at least once in next 5 years is found to be 0.25. The return period of this flood to be 0.25. The return period of this flood event (in years, rounded off to 1 decimal place) is _____ [GATE 2019 (CE-FN)]
31. An array of 25 distinct elements is to be sorted using quick sort. Assume that the pivot element is chosen uniformly at random. The probability that the pivot element gets placed in the worst possible location in the first round of partitioning (rounded off to 2 decimal places) is _____ [GATE 2019 (CS)]
32. Two numbers are chosen independently and uniformly at random from the set $\{1, 2, 3, \dots, 13\}$. The probability (rounded off to 3 decimal places) that their 4 bit (unsigned) binary representations have the same most significant bit is _____ [GATE 2019 (CS)]

33. A box has 8 red balls and 8 green balls. Two balls are drawn randomly in succession from the box without replacement. The probability that the first ball drawn is red and the second ball drawn is green is [GATE 2019 (IN)]

(A) $\frac{4}{15}$

(B) $\frac{7}{16}$

(C) $\frac{1}{2}$

(D) $\frac{8}{15}$

ANSWERS

1. 0.667

2. A

3. 0.65

4. B

5. 0.26

6. 0.25

7. C

8. D

9. B

10. C

11. B

12. 0.01818

13. A

14. 0.33

15. 0.55

16. 0.07

17. A

18. A

19. 0.028

20. 0.75

21. 0.5

22. A

23. C

24. C

25. B

26. B

27. 0.023

28. B

29. 0.5

30. 17.9

31. 0.08

32. 0.461

33. A

CHAPTER - 6

RANDOM VARIABLES AND DISTRIBUTION FUNCTIONS

6.1 INTRODUCTION

Previously we considered the concept of random experiment, events, sample space and sample points. The events described on a random experiment may be numerical or non-numerical (descriptive). For example, the outcomes that we obtain, when we throw a die are numerical. (We get the outcomes as 1, 2, 3, 4, 5 or 6). Hence the sample space is numerical. But, the outcomes we obtain, when we toss a coin are non-numerical. We get the outcomes as head or tail. Here the sample - space is non-numerical or descriptive.

It is inconvenient to deal with these descriptive outcomes mathematically. Hence, for ease of manipulation, we may assign a real number to each of the outcomes using a fixed rule or mapping. For example, when we toss a coin we get two outcomes, namely head or tail; we can assign numerical values, say '1' to head and '0' to tail. This interpretation is easy and attractive from mathematical point of view and also practically meaningful.

This rule or mapping from the original sample space (numerical or non-numerical) to a numerical (real) sample space, subjected to certain constraints is called a random variable. Thus Random variable is a real valued function which maps the numerical or non-numerical sample space (domain) of the random experiment to real values (codomain or range).

6.2 RANDOM VARIABLE

We know that a variable is a quality which changes or varies - the change may occur due to time factor or any factor. For example, the height of a person vary with age, but the age of a person changes with time. Every variable has a range in which it can take any value. Examples are: marks scored by a student in an examination, number of children in a family, height and weight of a person, etc. The variable could be **continuous or discrete**. A continuous variable takes all possible values in its range. For example, the height of a person can take any value in a certain range, but for the sake of convenience, it is measured only up to the accuracy of inches or cms. Similarly, the weight of a person can be any value, but it is usually expressed in kgs. A discrete variable takes only certain values in a range. For example, number of children in a family, number on a dice, etc can take only integral values.

A variable is called a random variable when there is a chance factor associated with the various values, which it can take in its range. For example, the life of a person is a random variable and there are probabilities associated with various age values.

Def. : A real variable X whose value is determined by the outcome of a random experiment is called a random variable. A random variable X can also be regarded as a real - value function defined on the sample space S of a random experiment such that for each point x of the sample space, $f(x)$ is the probability of occurrence of the event represented by x .

Illustration: The sample space corresponding to tossing of two coins.

When two coins are tossed, its outcomes or sample points can be (Head, Head), (Head, Tail), (Tail, Head), (Tail, Tail) i.e., $S = \{HH, HT, TH, TT\}$. After the performance of the experiment, we count the number of tails and denote it by X . The first outcome HH has 0 Tail, so $X = 0$. Similarly $X = 1$, denote the outcomes HT or TH and $X = 2$, represents the outcome TT .

Thus X takes the values 0,1,2 i.e., $X = 0,1,2$.

A random variable is also called a stochastic variable or simply a variable or a chance variable. The name random variable is given to the variable X because it is defined on a sample space, the outcomes of which are uncertain and hence depend on chance.

The random variable assigns a real value $X(w)$ such that

1. The set $\{w \mid X(w) \leq x\}$ is an event for every $x \in R$, for which a probability is defined. This condition is called as measurability.
2. The probabilities of the events $\{w \mid X(w) = \infty\}$ and $\{w \mid X(w) = -\infty\}$ are equal to zero.
i.e. $P(X = \infty) = P(X = -\infty) = 0$
3. For every set $A \subset S$ there corresponds a set $T \subset R$ called the image of A . Also for every $T \subset R$ there exists in S the inverse image $X^{-1}(T)$.
i.e. $X^{-1}(T) = \{w \in S \mid X(w) \in T\}$

This set is an event in S which has a probability $P[X^{-1}(T)]$.

Notation : Random variables are usually denoted by capital letters of English alphabets and particular values which the random variable takes are denoted by the corresponding small letters.

Ex. 1. Consider a random experiment consisting of tossing a coin twice. The sample space $S = \{s_1, s_2, s_3, s_4\}$, where $s_1 = HH$, $s_2 = HT$, $s_3 = TH$ and $s_4 = TT$ consists of four elements (sample points). Define a function $X: S \rightarrow R$ by $X(s) = \text{"number of heads"}$.

Then $X(s_1) = 2$, $X(s_2) = 1$, $X(s_3) = 1$ and $X(s_4) = 0$

$$\begin{aligned}\text{Range of } X &= \{X(s) : s \in S\} \\ &= \{X(s_1), X(s_2), X(s_3), X(s_4)\} \\ &= \{0, 1, 2\}\end{aligned}$$

Ex. 2. Consider the random experiment of throwing a pair of dice and noting the sum. The sample space consists of points $S = \{(1, 1), (1, 2), \dots, (1, 6), (2, 1), (2, 2), \dots, (2, 6), \dots, (6, 1), (6, 2), \dots, (6, 6)\}$.

If X is the “sum of numbers appearing on the faces of dice”, X is a random variable.

Here $X : S \rightarrow R$ is defined through $X((i, j)) = i + j$ for $(i, j) \in S$.

If in a random experiment, the event corresponding to a number a occurs, then the corresponding random variable X is said to assume the value a and the probability of the event is denoted by $P(X = a)$. Similarly, the probability of the event X assuming any value between a and b is denoted by $P(a < X < b)$. The probability of the event $X \leq c$ (X assumes any value smaller than or equal to c) is denoted by $P(X \leq c)$ and the probability of the event $X > c$ is denoted by $P(X > c)$. That is, $P(X > c)$ is simply the probability of the set of outcomes s for which $X(s) > c$ or $P(X > c) = P\{s | X(s) > c\}$.

For the example (1) given above, we have

$$P(X \leq 1) = P\{HH, HT, TH\} = \frac{3}{4}$$

If X and Y are random variables on the sample space S , then $X + Y$, $X + K$, KX and KY (K is a constant) are also random variables on S and are defined by $(X + Y)(s) = X(s) + Y(s)$, $(X + K)(s) = X(s) + K$, $KX(s) = K \cdot X(s)$, etc.

- Note :**
1. $P(X \leq c) + P(X > c) = P(-\infty < X < \infty)$
 2. $P(X > c) = 1 - P(X \leq c)$
 3. $P(X = a \text{ or } X = b) = P\{(X = a) \cup (X = b)\}$
 4. $P(X = a \text{ and } X = b) = P\{(X = a) \cap (X = b)\}$

6.3 TYPES OF RANDOM VARIABLES

Random variable is of two types :

- (i) Discrete Random Variable
- (ii) Continuous Random Variable

1. Discrete Random Variable : A random variable X which can take only a finite number of discrete values in an interval of domain is called a *discrete random variable*. In other words, if the random variable takes the values only on the set $\{0, 1, 2, \dots, n\}$ is called a Discrete Random variable.

Tossing of a coin, throwing a dice, the number of defectives in a sample of electric bulbs, the number of printing mistakes in each page of a book, the number of telephone calls received by the telephone operator are examples of Discrete Random variables.

Thus to each outcome ‘ s ’ of a random experiment there corresponds a real number $X(s)$ which is defined for each point of the sample S .

A few examples are :

- (i) In the example (1) , $X(s) = \{s : s = 0, 1, 2\}$ or Range of $X = \{0, 1, 2\}$.

The random variable X is a discrete random variable.

- (ii) The random variable denoting the number of students in a class is

$$X(x) = \{x : x \text{ is a positive integer}\}$$

2. Continuous Random Variable : A random variable X which can take values continuously i.e., which takes all possible values in a given interval is called a *continuous random variable*.

For example, the height, age and weight of individuals are examples of continuous random variable. Also temperature and time are continuous random variables.

6.4 PROBABILITY FUNCTION

Illustration 1 : Distribution of Probability over various numbers on Die

When a single dice is thrown, the probability associated with each number on the top face of the die can take the values with probability being same is $1/6$ for all the numbers 1 to 6. This can be depicted with the help of a diagram as follows:

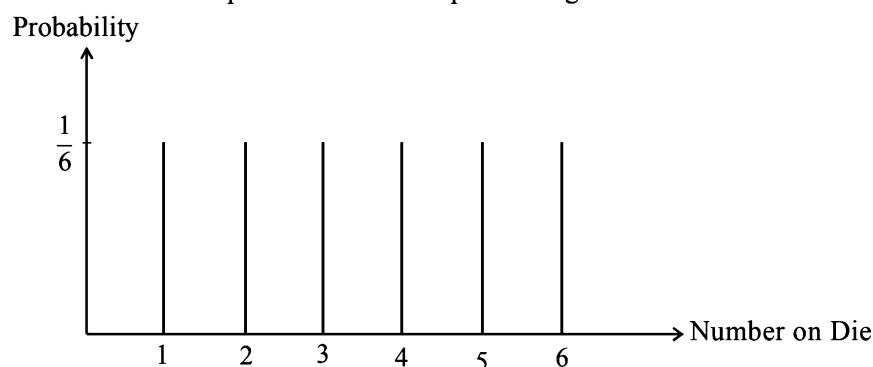
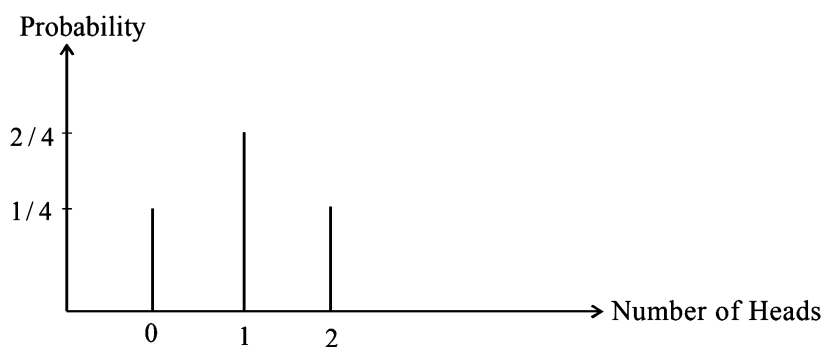


Illustration 2: Distribution of Probability over number of Heads

When two coins are tossed, the number of Heads is a random variable which can take the values 0, 1 or 2. The associated probabilities are $\frac{1}{4}$, $\frac{1}{2}$ and $\frac{1}{4}$ and is shown below:



It may be noted that total of all the three probabilities $\frac{1}{4}$, $\frac{1}{2}$ and $\frac{1}{4}$ is 1. This is so because the random variable is bound to take one of the three values 0, 1 or 2 in its range.

The above diagrams shows how the total probability ($= 1$) is distributed over the range of the variable. So, these diagrams are referred to graphs of probability distributions. In many situations, these graphs are described by a mathematical function which is referred to as a Probability Function or Statistical distribution.

6.4.1 PROBABILITY FUNCTION OF A DISCRETE RANDOM VARIABLE

If for a discrete random variable X , the real valued function $p(x)$ is such that

$$P(X = x) = p(x)$$

then $p(x)$ is called probability function or probability density function of a discrete random variable X . Probability function $p(x)$ gives the measure of probability for different values of X .

Properties of a Probability Function :

If $p(x)$ is a probability function of a random variable X , then it possesses the following properties:

1. $p(x) \geq 0$ for all x .
2. $\sum p(x) = 1$, summation is taken over for all values of x .
3. $p(x)$ cannot be negative for any value of x .

6.4.2 PROBABILITY DISTRIBUTION FUNCTION

Let X be a random variable. Then the probability distribution function associated with X is defined as the probability that the outcome of an experiment will be one of the outcomes for which $X(s) \leq x, x \in \mathbb{R}$. That is, the function $F(x)$ [i.e., $F_X(x)$] defined by

$$F_X(x) = P(X \leq x) = P\{s : X(s) \leq x\}, -\infty < x < \infty \text{ is called the distribution function of } X.$$

Properties of Distribution Function :

1. If F is the distribution function of a random variable X and if $a < b$, then
 - (i) $P(a < X \leq b) = F(b) - F(a)$
 - (ii) $P(a \leq X \leq b) = P(X = a) + [F(b) - F(a)]$
 - (iii) $P(a < X < b) = [F(b) - F(a)] - P(X = b)$
 - (iv) $P(a \leq X < b) = [F(b) - F(a)] - P(X = b) + P(X = a)$

Note : If $P(X = a) = P(X = b) = 0$ then

$$P(a < X \leq b) = P(a \leq X \leq b) = P(a < X < b) = P(a \leq X < b) = F(b) - F(a)$$

2. All distribution functions are monotonically increasing and lie between 0 and 1. That is, if F is the distribution function of the random variable X , then

$$(i) 0 \leq F(x) \leq 1 \quad (ii) F(x) < F(y) \text{ when } x < y.$$

$$3. (i) F(-\infty) = \lim_{x \rightarrow -\infty} F(x) = 0 \quad (ii) F(\infty) = \lim_{x \rightarrow \infty} F(x) = 1$$

Probability Distributions :

- (i) Discrete Probability Distribution
- (ii) Continuous Probability Distribution

6.5 DISCRETE PROBABILITY DISTRIBUTION (PROBABILITY MASS FUNCTION)

Probability distribution of a random variable is the set of its possible values together with their respective probabilities. Suppose X is a discrete random variable with possible outcomes (values) $x_1, x_2, x_3, \dots$. The probability of each possible outcome x_i is $p_i = P(X = x_i) = p(x_i)$ for $i = 1, 2, 3, \dots$

If the numbers $p(x_i)$, $i = 1, 2, 3, \dots$ satisfy the two conditions

$$(i) p(x_i) > 0 \text{ for all values of } i; 0 < p_i \leq 1$$

$$(ii) \sum p(x_i) = 1, i = 1, 2, 3, \dots,$$

then the function $p(x)$ is called the *probability mass function* of the random variable X and the set $\{p(x_i)\}$, $i = 1, 2, \dots$ is called the discrete probability distribution of the discrete random variable X .

The probability distribution of the random variable X is given by means of the following table :

X	x_1	x_2	x_3		x_i		x_n
$P(X)$	p_1	p_2	p_3		p_i		p_n

$$\text{Further } P(X < x_i) = p(x_1) + p(x_2) + \dots + p(x_{i-1})$$

$$P(X \leq x_i) = p(x_1) + p(x_2) + \dots + p(x_{i-1}) + p(x_i)$$

$$\text{and } P(X > x_i) = 1 - P(X \leq x_i)$$

Ex. 1. In tossing a coin two times, $S = \{TT, HT, TH, HH\}$

$P(X = 0)$ = Probability of getting two tails (no heads)

$$= P(\{T, T\}) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$P(X = 1)$ = Probability of getting one head

$$= P(\{H, T\}, \{T, H\}) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$P(X = 2)$ = Probability of getting two heads

$$= P(\{H, H\}) = \frac{1}{4}$$

Thus the total probability 1 is distributed into three parts $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$ according to whether $X = 0$ or 1 or 2. This probability distribution is given in the following table :

$X = x_i$	0	1	2
$P(X = x_i)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

Ex. 2. Let X be the random variable which represents the sum of numbers of points on throwing two unbiased dice. If the point shown in each die is equal to one, the minimum sum of numbers is equal to two. If both dice show six, then the sum of numbers at the maximum is 12.

Thus if two unbiased dice are thrown, then the sum X of the two numbers which turn up must be an integer between 2 and 12. For $X = 2$, there is only one favourable point (1, 1) and hence $P(X = 2) = 1/36$, since there are 36 sample points in all. For $X = 3$, there are two favourable sample points (1, 2) and (2, 1) and hence $P(X = 3) = 2/36$. Similarly for $X = 4$, there are three favourable sample points (1, 3), (2, 2), (3, 1) and hence $P(X = 4) = 3/36$ and so on. Now the probability distribution in this case is given by the following table :

$X = x_i$	2	3	4	5	6	7	8	9	10	11	12
$P(X = x_i)$	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

Clearly $\sum p(x_i) = p_1 + p_2 + \dots + p_{11} = 1$

From the above table, we have

$$P(X \geq 10) = \frac{3}{36} + \frac{2}{36} + \frac{1}{36} = \frac{6}{36} = \frac{1}{6}$$

$$P(X \leq 6) = \frac{1}{36} + \frac{2}{36} + \frac{3}{36} + \frac{4}{36} + \frac{5}{36} = \frac{15}{36} = \frac{5}{12}$$

$$\text{and } P(7 < X < 12) = \frac{5}{36} + \frac{4}{36} + \frac{3}{36} + \frac{2}{36} = \frac{14}{36} = \frac{7}{18}$$

In the above example, X takes only a finite number of values as such it is a discrete random variable.

Note : The concept of **probability distribution function** is analogous to that of **frequency distribution**. A frequency distribution tells us how the total frequency is distributed among different values of the variable, whereas a probability distribution tells us how total probability 1 is distributed among the values which the random variable can take.

Cumulative Distribution Function of a Discrete Random Variable

There are many occasions in which it is of interest to know the probability that the value of a random variable is less than or equal to some real number x .

Suppose that X is a discrete random variable. Then the Discrete Distribution function or Cumulative Distribution function $F(x)$ is defined by

$$F(x) = P(X \leq x) = \sum_{(i: x_i \leq x)} p(x_i) = \sum_{i=1}^x p(x_i) \text{ where } x \text{ is any integer.}$$

(OR) $F(x) = P(X \leq x) = \sum_{t \leq x} f(t)$, where $-\infty < x < \infty$ and $f(t)$ is the value of the probability distribution of X at t .

Aliter : If $p(x)$ i.e., $f(x)$ is the probability function or probability distribution, then the value of $\sum_{x=0}^x p(x)$ i.e., $\sum_{x=0}^x f(x)$ denoted by $F(x)$ is called the **Cumulative Distribution Function** or simply **Distribution Function**. It is similar to cumulative frequency i.e., the cumulative frequency upto a given point or value. It gives the probability that the random variable takes any value from 0 (lowest value that the variable can take) to a given value.

For example,

$$\begin{aligned} F(2) &= \sum_{x=0}^2 f(x) = f(0) + f(1) + f(2) = p(0) + p(1) + p(2) \\ &= \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1 \text{ (in Ex. 1 above)} \end{aligned}$$

Suppose if X takes only a finite number of values $x_1, x_2, \dots, x_n$ then the distribution function is given by

$$F(x) = \begin{cases} 0, & -\infty < x < x_1 \\ p(x_1), & x_1 \leq x < x_2 \\ p(x_1) + p(x_2), & x_2 \leq x < x_3 \\ \dots & \dots \\ \dots & \dots \\ p(x_1) + p(x_2) + \dots + p(x_n), & x_n \leq x < \infty \end{cases}$$

Probability Density Function :

The probability density function $f_X(x)$ is defined as the derivative of the probability distribution function, $F_X(x)$ of the random variable X .

$$\text{Thus } f_X(x) = \frac{d}{dx}[F_X(x)]$$

6.6 EXPECTATION (MEAN), VARIANCE AND STANDARD DEVIATION OF A PROBABILITY DISTRIBUTION

The behaviour of a random variable is completely characterized by the distribution function $F(x)$ or density function $f(x)$ or $P(x_i)$. Instead of a function, a more compact description can be made by a single numbers such as mean, median and mode known as measures of central tendency of the random variable X .

EXPECTATION

In this section, We discuss the application of the concepts of probability theory to real life situations when the decisions are based on expectations about the value of a variable like life of an item, say electric bulb, steel, cement, scooter battery, etc.

When a dice is thrown, we know that the variable representing the top number on the dice can be any value from 1 to 6 with probability $1/6$. Now, suppose a person is not interested in listening to this statement giving the range of the variable, but all we want to know is that single number which is expected to come up when the dice is thrown. The answer to these types of situations when one wants the reply as a single value, is provided by the theory of expectation. The theory is discussed separately for discrete and continuous variables.

Expectation theory plays a very important role in decision - making because most of the time we take decisions based on what is expected to happen.

(1) Expectation or Mean of a Discrete Variable :

As defined earlier, a discrete variable takes only some finite values like number on a dice, number of children in a family, etc.

Suppose a random variable X assumes the values $x_1, x_2, \dots, x_n$ with respective probabilities $p_1, p_2, \dots, p_n$. Then the Mathematical Expectation or Mean or Expected value of X , denoted by $E(X)$ or μ , is defined as the sum of products of different values of x and the corresponding probabilities.

$$\therefore E(X) = x_1 p_1 + x_2 p_2 + \dots + x_n p_n$$

$$\text{i.e., } E(X) = \sum_{i=1}^n p_i x_i$$

$$\text{Similarly, } E(x^r) = \sum_{i=1}^n p_i \cdot x_i^r$$

In general, the expected value of any function $g(x)$ of a random variable X is defined as

$$E[g(x)] = \sum_{i=1}^n p_i g(x_i)$$

Note : Expected value of X is a population mean. If population mean is μ then $E(X) = \mu$.

Some Important Results on Expectation :

(i) If X is a random variable and K is a constant, then

$$E(X + K) = E(X) + K \quad [\text{JNTU 2005S, 2005, 2007S, (H) Nov. 2009 (Set No.1)}]$$

Proof : By definition, we have

$$\begin{aligned} E(X + K) &= \sum_{i=1}^n (x_i + K) p_i = \sum_{i=1}^n x_i p_i + K \sum_{i=1}^n p_i \\ &= E(X) + K(1) = E(X) + K \left[\because \sum_{i=1}^n p_i = 1 \right] \end{aligned}$$

Note 1 : Expected value of constant term is constant, that is, if K is constant, then

$$E(K) = K \quad [\because E(K) = \sum p_i K = K \sum p_i = K(1) = K].$$

Note 2 : If K is constant, then $E(KX) = KE(X)$

$$E(KX) = \sum_{i=1}^n (K x_i) p_i = K \sum_{i=1}^n x_i p_i = K E(X), \text{ where } K \text{ is a constant.}$$

(ii) If X is a random variable and a and b are constants, then

$$E(aX \pm b) = a E(X) \pm b$$

Proof : Proceed as in (i)

(iii) If X and Y are any two random variables, then

$$E(X + Y) = E(X) + E(Y) \text{ provided } E(X) \text{ and } E(Y) \text{ exist.}$$

[JNTU 2005 S, 2005, 2007S, (H) Nov. 2009 (Set No.1)]

Proof : Let X assume the values $x_1, x_2, \dots, x_n$ and Y assume the values $y_1, y_2, \dots, y_m$.

Then by definition,

$$E(X) = \sum_{i=1}^n p_i x_i \text{ and } E(Y) = \sum_{j=1}^m p_j y_j$$

$$\text{Let } p_{ij} = P(X = x_i \cap Y = y_j) = p(x_i, y_j)$$

[This is called the joint probability function of X and Y]

The sum $(X + Y)$ is also a random variable which can take $m \times n$ values $(x_i + y_j)$, $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, m$.

By definition,

$$\begin{aligned} E(X + Y) &= \sum_{i=1}^n \sum_{j=1}^m p_{ij} (x_i + y_j) = \sum_{i=1}^n \sum_{j=1}^m p_{ij} x_i + \sum_{i=1}^n \sum_{j=1}^m p_{ij} y_j \\ &= \sum_{i=1}^n \left[x_i \sum_{j=1}^m p_{ij} \right] + \sum_{j=1}^m \left[y_j \sum_{i=1}^n p_{ij} \right] \\ &= \sum_{i=1}^n x_i p_i + \sum_{j=1}^m y_j p_j = E(X) + E(Y) \end{aligned}$$

Note : 1. $E(X + Y + Z) = E(X + (Y + Z)) = E(X) + E(Y + Z) = E(X) + E(Y) + E(Z)$

Similarly proceeding, we have in general by the method of induction, if

$X_1, X_2, \dots, X_n$ are ' n ' random variables, then

$$E(X_1 + X_2 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n)$$

i.e., The mathematical expectation of the sum of random variables is equal to the sum of their expectations, provided all the expectations exist.

2. $E(aX + bY) = a E(X) + b E(Y)$, where a and b are constants

3. $E(X - \bar{X}) = 0$

(iv) If X and Y are two independent random variables, then

$$E(XY) = E(X) E(Y)$$

Proof : Let X assume the values $x_1, x_2, \dots, x_n$ with respective probabilities $p_1, p_2, \dots, p_n$ and Y assume the values $y_1, y_2, \dots, y_m$ with probabilities $p'_1, p'_2, \dots, p'_m$.

Then by definition,

$$E(X) = \sum_{i=1}^n p_i x_i \text{ and } E(Y) = \sum_{j=1}^m p'_j y_j$$

$$\begin{aligned}
 \text{Let } p_{ij} &= P(X = x_i \cap Y = y_j) \\
 &= P(X = x_i) \cdot P(Y = y_j) \quad (\text{since } X \text{ and } Y \text{ are independent}) \\
 &= p_i p'_j
 \end{aligned}$$

The product XY is a random variable which can assume $m \times n$ values $x_i y_j$, $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, m$.

$\therefore$ By definition,

$$\begin{aligned}
 E(XY) &= \sum_{i=1}^n \sum_{j=1}^m x_i y_j p_{ij} = \sum_{i=1}^n \sum_{j=1}^m x_i y_j p_i p'_j \\
 &= \left(\sum_{i=1}^n p_i x_i \right) \left(\sum_{j=1}^m p'_j y_j \right) = E(X) E(Y)
 \end{aligned}$$

Note : $E(XYZ) = E(X.YZ) = E(X) \cdot E(YZ) = E(X) E(Y) E(Z)$

Most of the concepts discussed with the frequency distributions apply equally well to distribution functions.

Important Observation : $E\left(\frac{1}{X}\right)$ and $\frac{1}{E(X)}$ are not same.

(2) Mean :

The mean value μ of the discrete distribution function is given by

$$\mu = \frac{\sum p_i x_i}{\sum p_i} = \sum p_i x_i = E(X)$$

Note : If $E(X) = \mu$, then $E(X - \mu) = 0$.

(3) Variance :

Variance characterizes the variability in the distributions since two distributions with same mean can still have different dispersion of data about their means.

Variance of the probability distribution of a random variable X is the mathematical expectation of $[X - E(X)]^2$. Then

$$\text{Var}(X) = E[X - E(X)]^2 \text{ i.e., } \text{Var}(X) = \sum_{i=1}^n \left\{ [x_i - E(X)]^2 \times p(x_i) \right\}$$

Another Form of Variance :

If X is a random variable, then the mathematical expectation of $(X - \mu)^2$ is defined to be the variance of the random variable X . Then

$$\begin{aligned}
 \text{Var}(X) &= E(x - \mu)^2 = \sum_{i=1}^n p_i (x_i - \mu)^2 \\
 &= \sum (x_i^2 - 2\mu x_i + \mu^2) p_i \\
 &= \sum x_i^2 p_i - 2\mu \sum x_i p_i + \mu^2 \sum p_i \\
 &= E(X^2) - 2\mu \cdot \mu + \mu^2 \cdot 1 \quad [\because \mu = \sum x_i p_i, \sum p_i = 1]
 \end{aligned}$$

$$\therefore \sigma^2 = E(X^2) - \mu^2$$

$$\text{i.e., } \text{Var}(X) = E(X^2) - [E(X)]^2$$

Note : The variance of a random variable X is also denoted by $V(X)$.

(4) Standard Deviation : It is the positive square root of the variance.

$$\therefore \text{S.D} = \sigma = \sqrt{\sum_{i=1}^n p_i x_i^2 - \mu^2} = \sqrt{E(X^2) - \mu^2} = \sqrt{E(X^2) - [E(X)]^2}$$

Some Important Results on Variance :

1. Variance of constant is zero i.e., $V(K) = 0$
2. If K is a constant, then $V(KX) = K^2 V(X)$
3. If X is a random variable and K is a constant, then $V(X + K) = V(X)$
4. If X is a discrete random variable, then $V(aX + b) = a^2 V(X)$, where $V(X)$ is variance of X and a, b are constants. **[JNTU (A) June 2016]**

Proof. Let $Y = aX + b$... (1)

Then $E(Y) = E(aX + b) = aE(X) + b$... (2)

(1) - (2) gives $Y - E(Y) = a[X - E(X)]$

Squaring and taking expectation of both sides, we get

$$E[\{Y - E(Y)\}^2] = a^2 E[\{X - E(X)\}^2] \quad \text{i.e., } V(Y) = a^2 V(X)$$

or $V(aX + b) = a^2 V(X)$

Cor. (i) If $b = 0$, then $V(aX) = a^2 V(X)$

(ii) If $a = 0$, then $V(b) = 0$

(iii) If $a = 1$, then $V(X + b) = V(X)$

5. If X and Y are two independent random variables, then $V(X \pm Y) = V(X) \pm V(Y)$.

SOLVED EXAMPLES

Example 1 : Let X denote the number of heads in a single toss of 4 fair coins. Determine

- (i) $P(X < 2)$ (ii) $P(1 < X \leq 3)$.

[JNTU 2006 (Set No. 4)]

Solution : The required probability distribution is

X	0	1	2	3	4
$P(X)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

$$(i) P(X < 2) = P(X = 0) + P(X = 1) = \frac{1}{16} + \frac{4}{16} = \frac{5}{16}$$

$$(ii) P(1 < X \leq 3) = P(X = 2) + P(X = 3) = \frac{6}{16} + \frac{4}{16} = \frac{10}{16} = \frac{5}{8}$$

Example 2 : Two dice are thrown. Let X assign to each point (a, b) in S the maximum of its numbers i.e., $X(a, b) = \max(a, b)$. Find the probability distribution. X is a random variable with $X(s) = \{1, 2, 3, 4, 5, 6\}$. Also find the mean and variance of the distribution.

[JNTU 2004, 2007, 2008S, (A) Dec. 2009, Nov. 2010 (Set No. 4)]

(OR) A random variable X has the following distribution ?

x	1	2	3	4	5	6
$P(x)$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

Find (a) the mean (b) variance (c) $P(1 < x < 6)$ [JNTU (H) Apr. 2012 (Set No. 1)]

Solution : The total number of cases are $6 \times 6 = 36$.

The maximum number could be 1, 2, 3, 4, 5, 6 i.e., $X(s) = X(a, b) = \max(a, b)$.

The number 1 will appear only in one case (1, 1). So $p(1) = P(X = 1) = P(1, 1) = \frac{1}{36}$

For maximum 2, favourable cases are (2, 1), (2, 2), (1, 2)

So $p(2) = P(X = 2) = 3/36$

For maximum 3, favourable cases are (1, 3), (3, 1), (2, 3), (3, 2), (3, 3).

So $p(3) = P(X = 3) = 5/36$

For maximum 4, favourable cases are (1, 4), (4, 1), (2, 4), (4, 2), (3, 4), (4, 3), (4, 4)

So $p(4) = P(X = 4) = 7/36$

Similarly, $p(5) = P(X = 5)$

$$= P((1, 5), (5, 1), (2, 5), (5, 2), (3, 5), (5, 3), (4, 5), (5, 4), (5, 5))$$

$$= \frac{9}{36}$$

and $p(6) = P(X = 6)$

$$= P((1, 6), (6, 1), (2, 6), (6, 2), (3, 6), (6, 3), (4, 6), (6, 4), (5, 6), (6, 5), (6, 6))$$

$$= \frac{11}{36}$$

$\therefore$ The required discrete probability distribution is

$X = x_i$	1	2	3	4	5	6
$P(X = x_i) = p(x_i)$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

$$(i) \quad \text{Mean, } \mu = \sum_{i=1}^6 p_i x_i = 1 \cdot \frac{1}{36} + 2 \cdot \frac{3}{36} + 3 \cdot \frac{5}{36} + 4 \cdot \frac{7}{36} + 5 \cdot \frac{9}{36} + 6 \cdot \frac{11}{36}$$

$$= \frac{1}{36} (1 + 6 + 15 + 28 + 45 + 66) = \frac{161}{36} = 4.47$$

$$(ii) \quad \text{Variance, } \sigma^2 = \sum_{i=1}^6 p_i x_i^2 - \mu^2$$

$$\begin{aligned}
&= \frac{1}{36} (1)^2 + \frac{3}{36} (2)^2 + \frac{5}{36} (3)^2 + \frac{7}{36} (4)^2 + \frac{9}{36} (5)^2 + \frac{11}{36} (6)^2 - (4.47)^2 \\
&= \frac{1}{36} (1 + 12 + 45 + 112 + 225 + 396) - (4.47)^2 \\
&= \frac{791}{36} - 19.9808 = 21.97 - 19.981 = 1.9912
\end{aligned}$$

Example 1 : A random variable X has the following probability function :

x	0	1	2	3	4	5	6	7
$p(x)$	0	K	$2K$	$2K$	$3K$	K^2	$2K^2$	$7K^2 + K$

- (i) Determine K (ii) Evaluate $P(X < 6)$, $P(X \geq 6)$, $P(0 < X < 5)$ and $P(0 \leq X \leq 4)$
 (iii) if $P(X \leq K) > \frac{1}{2}$, find the minimum value of K and, (iv) Determine the distribution function of X (v) Mean (vi) Variance.

[JNTU 04S, 05S, 08S, (A) Nov. 11, (H) Dec. 11, (K) May 10S, Nov. 2012, Mar. 2014 (Set No. 2)]

(or) The probability function of a variate X is

X	0	1	2	3	4	5	6	7
$P(X)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$k + 7k^2$

Find k , and $P(X < 6)$

[JNTU (A) Dec. 2018]

Solution :

- (i) Since $\sum_{x=0}^7 p(x) = 1$, we have

$$K + 2K + 2K + 3K + K^2 + 2K^2 + 7K^2 + K = 1$$

$$i.e., 10K^2 + 9K - 1 = 0 \quad i.e., (10K - 1)(K + 1) = 0$$

$$\therefore K = \frac{1}{10} = 0.1 \quad (\text{since } p(x) \geq 0, \text{ so } K \neq -1)$$

- (ii) $P(X < 6) = P(X = 0) + P(X = 1) + \dots + P(X = 5)$
 $= 0 + K + 2K + 2K + 3K + K^2 = 8K + K^2 = 0.8 + 0.01 = 0.81 \quad [\because K = 0.1]$

$$P(X \geq 6) = 1 - P(X < 6) = 1 - 0.81 = 0.19$$

$$P(0 < X < 5) = P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)$$

$$= K + 2K + 2K + 3K = 8K = \frac{8}{10} = 0.8$$

$$P(0 \leq X \leq 4) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)$$

$$= 0 + K + 2K + 2K + 3K = 8K = 8(0.1) = 0.8$$

Note : 1. $P(X \leq 5) = P(X = 0) + P(X = 1) + \dots + P(X = 5)$

$$= 0.81 \quad (\text{Refer (ii)})$$

$$2. P(X > 5) = 1 - P(X \leq 5) = 1 - 0.81 = 0.19$$

$$3. P(0 < X < 6) = P(X = 1) + P(X = 2) + \dots + P(X = 5) = 0.81$$

(iii) The required minimum value of K is obtained as below.

$$P(X \leq 1) = P(X = 0) + P(X = 1) = 0 + K = \frac{1}{10} = 0.1$$

$$P(X \leq 2) = [P(X = 0) + P(X = 1)] + P(X = 2) = \frac{1}{10} + \frac{2}{10} = \frac{3}{10} = 0.3$$

$$P(X \leq 3) = [P(X = 0) + P(X = 1) + P(X = 2)] + P(X = 3) = 0.3 + 0.2 = 0.5$$

$$P(X \leq 4) = P(X \leq 3) + P(X = 4) = 0.5 + \frac{3}{10} = 0.8 > 0.5 = \frac{1}{2}$$

$\therefore$ The minimum value of K for which $P(X \leq K) > \frac{1}{2}$ is $K = 4$

(iv) The distribution function of X is given by the following table :

X	$F(x) = P(X \leq x)$
0	0
1	$K = 1/10$
2	$3K = 3/10$
3	$5K = 5/10$
4	$8K = 8/10$
5	$8K + K^2 = 81/100$
6	$8K + 3K^2 = 83/100$
7	$9K + 10K^2 = 1$

$$\begin{aligned}
 \text{(v) Mean, } \mu &= \sum_{i=0}^7 p_i x_i \\
 &= 0(0) + 1(K) + 2(2K) + 3(2K) + 4(3K) + 5(K^2) + 6(2K^2) + 7(7K^2 + K) \\
 &= 66K^2 + 30K = \frac{66}{100} + \frac{30}{10} = 0.66 + 3 = 3.66 \left(\because K = \frac{1}{10} \right)
 \end{aligned}$$

$$\begin{aligned}
 \text{(vi) Variance} &= \sum_{i=0}^7 p_i x_i^2 - \mu^2 \\
 &= K + 8K + 18K + 48K + 25K^2 + 72K^2 + 343K^2 + 49K - (3.66)^2 \\
 &= 440K^2 + 124K - (3.66)^2 = \frac{440}{100} + \frac{124}{10} - (3.66)^2 \\
 &= 4.4 + 12.4 - 13.3956 = 3.4044
 \end{aligned}$$

Example 4 : A random variable X has the following probability distribution.

$X :$	1	2	3	4	5	6	7	8
$P(X) :$	K	$2K$	$3K$	$4K$	$5K$	$6K$	$7K$	$8K$

Find the value of

(i) K (ii) $P(X \leq 2)$ (iii) $P(2 \leq X \leq 5)$ [JNTU 2008 (Set No. 1)]

Solution: (i) Since $\sum_{i=1}^{\infty} p(x_i) = 1$, we have

$$K + 2K + 3K + 4K + 5K + 6K + 7K + 8K = 1 \Rightarrow 36K = 1.$$

$$\therefore K = \frac{1}{36}$$

$$(ii) P(X \leq 2) = P(X = 1) + P(X = 2)$$

$$= K + 2K = 3K = \frac{3}{36} = \frac{1}{12}$$

$$(iii) P(2 \leq X \leq 5) = P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5)$$

$$= 2K + 3K + 4K + 5K = 14K = \frac{14}{36} = \frac{7}{18}$$

Example 4 : A random variable X has the following probability distribution.

Values of x	0	1	2	3	4	5	6	7	8
$P(x)$	a	$3a$	$5a$	$7a$	$9a$	$11a$	$13a$	$15a$	$17a$

(i) Determine the value of a .

(ii) Find $p(x < 3)$, $p(x \geq 3)$ and $p(0 < x < 5)$

(iii) Find the distribution function $F(x)$.

[JNTU(K) Dec. 2013 (Set No.1)]

Solution : (i) Since $\sum_{i=0}^8 p(x_i) = 1$, we have

$$a + 3a + 5a + 7a + 9a + 11a + 13a + 15a + 17a = 1$$

$$i.e., 81a = 1 \text{ or } a = \frac{1}{81}$$

$$(ii) p(x < 3) = p(x = 0) + p(x = 1) + p(x = 2)$$

$$= a + 3a + 5a = 9a = \frac{9}{81} = \frac{1}{9}$$

$$p(x \geq 3) = 1 - p(x < 3) = 1 - \frac{1}{9} = \frac{8}{9}$$

$$p(0 < x < 5) = p(x = 1) + p(x = 2) + p(x = 3) + p(x = 4)$$

$$= 3a + 5a + 7a + 9a = 24a = \frac{24}{81} = \frac{8}{27}$$

(iii) The distribution function $F(x)$ of the discrete variate X is defined by

$$F(x) = P(X \leq x) = \sum_{i=1}^x p(x_i) \text{ where } x \text{ is any integer}$$

X	F(X) = P(X ≤ x)
0	$a = \frac{1}{81}$
1	$4a = \frac{4}{81}$
2	$9a = \frac{9}{81}$
3	$16a = \frac{16}{81}$
4	$25a = \frac{25}{81}$
5	$36a = \frac{36}{81}$
6	$49a = \frac{49}{81}$
7	$64a = \frac{64}{81}$
8	$81a = \frac{81}{81} = 1$

Example 6 : The probability density function of a variate X is

X	0	1	2	3	4	5	6
P(X)	k	3k	5k	7k	9k	11k	13k

- (i) Find k (ii) Find P(X < 4), P(X ≥ 5), P(3 < X ≤ 6) (iii) What will be the minimum value of k so that P(X ≤ 2) > 0.3 ? [JNTU (K) Nov. 2009, Dec. 2013 (Set No. 3, 4)]

Solution : (i) Since $\sum_{i=0}^6 p(x_i) = 1$ so $k + 3k + 5k + 7k + 9k + 11k + 13k = 1$

i.e., $49k = 1$ or $k = \frac{1}{49}$

(ii) $P(X < 4) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$

$$= k + 3k + 5k + 7k = 16k = \frac{16}{49} \left[\because k = \frac{1}{49} \right]$$

$$P(X \geq 5) = P(X = 5) + P(X = 6) = 11k + 13k = 24k = \frac{24}{49}$$

$$P(3 < X \leq 6) = P(X = 4) + P(X = 5) + P(X = 6) = 9k + 11k + 13k = 33k = \frac{33}{49}$$

(iii) $P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2) = k + 3k + 5k = 9k$

$$P(X \leq 2) > 0.3 \Rightarrow 9k > 0.3 \text{ or } k > \frac{1}{30}. \quad \therefore \text{The minimum value of } k \text{ is } \frac{1}{30}.$$

Example 2 : For the discrete probability distribution

X	0	1	2	3	4	5	6
P(X)	0	2K	2K	3K	K ²	2K ²	7K ² +K

Find (i) k (ii) mean (iii) Variance [JNTU (H) 2009 (Set No. 1)]

Solution : (i) Since the total probability is unity, we have $\sum_{x=0}^6 p(x) = 1$

$$\text{i.e., } 0 + 2K + 2K + 3K + K^2 + 2K^2 + (7K^2 + K) = 1$$

$$\text{i.e., } 10K^2 + 8K - 1 = 0$$

$$\therefore K = \frac{-8 \pm \sqrt{64 + 40}}{20} = \frac{-8 \pm \sqrt{104}}{20} = \frac{-8 \pm 2\sqrt{26}}{20} = \frac{-4 \pm \sqrt{26}}{10}$$

$$\text{Since, } p(x) \geq 0, \therefore K = \frac{-4 + \sqrt{26}}{10} = 0.1099$$

$$(ii) \text{ Mean, } \mu = \sum_{i=0}^6 p_i x_i$$

$$\begin{aligned} &= (0)(0) + (1)(2K) + (2)(2K) + (3)(3K) + (4)(K^2) + (5)(2K^2) + (6)(7K^2 + K) \\ &= 2K + 4K + 9K + 4K^2 + 10K^2 + 42K^2 + 6K \\ &= 56K^2 + 21K = K(56K + 21) \\ &= (0.1099)[56(0.1099) + 21] = 2.9842 \end{aligned}$$

$$(iii) \text{ Variance} = \sum_{i=0}^6 p_i x_i^2 - \mu^2$$

$$\begin{aligned} &= [0 + 2K(1)^2 + 2K(2)^2 + 3K(3)^2 + K^2(4)^2 + 2K^2(5)^2 + (7K^2 + K)(6)^2] - (2.9842)^2 \\ &= 2K + 8K + 27K + 16K^2 + 50K^2 + 252K^2 + 36K - 8.9054 \\ &= 318K^2 + 73K - 8.9054 = K(318K + 73) - 8.9054 \\ &= (0.1099)[318(0.1099) + 73] - 8.9054 = 2.9581 \end{aligned}$$

Example 8 : A random variables X has the following probability function

X_i	-3	-2	-1	0	1	2	3
$P(X_i)$	K	0.1	K	0.2	2k	0.4	2k

Find (i) K (ii) Mean (iii) Variance [JNTU (H) 2009 (Set No. 2)]

Solution : (i) We know that, if X is a random variable, then $\sum p(x_i) = 1$

$$\text{i.e., } K + 0.1 + K + 0.2 + 2K + 0.4 + 2K = 1$$

$$\text{i.e., } 6K + 0.7 = 1 \quad \text{i.e., } 6K = 0.3 \quad \text{or } K = \frac{0.3}{6} = \frac{0.1}{2}$$

$$\begin{aligned} \text{(ii) Mean, } \mu &= \sum_{i=0}^6 p_i x_i \\ &= (-3)(K) + (-2)(0.1) + (-1)(K) + 0(0.2) + 1(2K) + 2(0.4) + 3(2K) \\ &= -3K - 0.2 - K + 0 + 2K + 0.8 + 6K \\ &= 4K + 0.6 = 4\left(\frac{0.1}{2}\right) + 0.6 = 0.8 \end{aligned}$$

$$\begin{aligned} \text{(iii) Variance} &= \sum_{i=0}^6 p_i x_i^2 - \mu^2 \\ &= K(-3)^2 + (0.1)(-2)^2 + K(-1)^2 + 0.2(0)^2 + 2K(1)^2 + (0.4)(2)^2 + 2K(3)^2 - (0.8)^2 \\ &= 9K + 0.4 + K + 0 + 2K + 1.6 + 18K - 0.64 \\ &= 30K + 2 - 0.64 = 30\left(\frac{0.1}{2}\right) + 1.36 = 1.5 + 1.36 = 2.86 \end{aligned}$$

Example 9 : From a lot of 10 items containing 3 defectives, a sample of 4 items is drawn at random. Let the random variable X denote the number of defective items in the sample. Find the probability distribution of X when the sample is drawn without replacement.

[JNTU (K) May 2013 (Set No. 4)]

Solution : Obviously X can take the values 0, 1, 2 or 3.

Given total number of items = 10

No. of good items = 7

No. of defective items = 3

$$P(X = 0) = P(\text{no defective}) = \frac{{}^7C_4}{{}^{10}C_4} = \frac{7!}{4!3!} \times \frac{4!6!}{10!} = \frac{1}{6}$$

$$P(X = 1) = P(\text{one defective and 3 good items})$$

$$= \frac{{}^3C_1 \times {}^7C_3}{{}^{10}C_4} = \frac{3 \times 7!}{3!4!} \times \frac{4!6!}{10!} = \frac{1}{2}$$

$$P(X = 2) = P(2 \text{ defective and 2 good items})$$

$$= \frac{{}^3C_2 \times {}^7C_2}{{}^{10}C_4} = \frac{3}{10}$$

$$P(X = 3) = P(3 \text{ defective and 1 good item})$$

$$= \frac{{}^3C_3 \times {}^7C_1}{{}^{10}C_4} = \frac{7}{8 \times 9 \times 10} = \frac{1}{30}$$

∴ The probability distribution of random variable X is as follows :

X	0	1	2	3
$P(X)$	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{3}{10}$	$\frac{1}{30}$

Example 10 : Let X denote the minimum of the two numbers that appear when a pair of fair dice is thrown once. Determine the

- (i) Discrete probability distribution (ii) Expectation
(iii) Variance [JNTU 2006, (K) Nov. 2009, 2010S (Set No. 1)]

Solution : When two dice are thrown, total number of outcomes is $6 \times 6 = 36$

$$\text{In this case, sample space } S = \left\{ \begin{array}{l} (1,1), (1,2), (1,3), (1,4), (1,5), (1,6) \\ (2,1), (2,2), (2,3), (2,4), (2,5), (2,6) \\ (3,1), (3,2), (3,3), (3,4), (3,5), (3,6) \\ (4,1), (4,2), (4,3), (4,4), (4,5), (4,6) \\ (5,1), (5,2), (5,3), (5,4), (5,5), (5,6) \\ (6,1), (6,2), (6,3), (6,4), (6,5), (6,6) \end{array} \right\}$$

If the random variable X assigns the minimum of its number in S , then the sample

$$\text{space } S = \left\{ \begin{array}{cccccc} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 & 3 & 3 \\ 1 & 2 & 3 & 4 & 4 & 4 \\ 1 & 2 & 3 & 4 & 5 & 5 \\ 1 & 2 & 3 & 4 & 5 & 6 \end{array} \right\}$$

The minimum number could be 1, 2, 3, 4, 5, 6.

For minimum 1, favourable cases are (1, 1), (1, 2), (2, 1), (1, 3), (3, 1), (1, 4), (4, 1), (1, 5), (5, 1), (1, 6), (6, 1).

$$\text{So } P(X=1) = \frac{11}{36}$$

For minimum 2, favourable cases are (2, 2), (2, 3), (3, 2), (2, 4), (4, 2), (2, 5), (5, 2), (2, 6), (6, 2).

$$\text{So } P(X=2) = \frac{9}{36}$$

$$\text{Similarly, } P(X=3) = P((3, 3), (3, 4), (4, 3), (3, 5), (5, 3), (3, 6), (6, 3)) = \frac{7}{36}$$

$$P(X=4) = P((4, 4), (4, 5), (5, 4), (4, 6), (6, 4)) = \frac{5}{36}$$

$$P(X = 5) = P((5, 5), (5, 6), (6, 5)) = \frac{3}{36}$$

$$P(X = 6) = P((6, 6)) = \frac{1}{36}$$

∴ The probability distribution is

X	1	2	3	4	5	6
$P(X)$	$\frac{11}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{5}{36}$	$\frac{3}{36}$	$\frac{1}{36}$

(ii) Expectation = Mean = $\sum p_i x_i$

$$i.e., E(X) = 1 \cdot \frac{11}{36} + 2 \cdot \frac{9}{36} + 3 \cdot \frac{7}{36} + 4 \cdot \frac{5}{36} + 5 \cdot \frac{3}{36} + 6 \cdot \frac{1}{36}$$

$$\text{or } \mu = \frac{1}{36} (11 + 18 + 21 + 20 + 15 + 6) = \frac{91}{36} = 2.5278$$

(iii) Variance = $\sum p_i x_i^2 - \mu^2$

$$= \frac{11}{36} \cdot 1 + \frac{9}{36} \cdot 4 + \frac{7}{36} \cdot 9 + \frac{5}{36} \cdot 16 + \frac{3}{36} \cdot 25 + \frac{1}{36} \cdot 36 - \left(\frac{91}{36}\right)^2$$

$$= \frac{1}{36} (11 + 36 + 63 + 80 + 75 + 36) - \left(\frac{91}{36}\right)^2$$

$$i.e., \sigma^2 = 8.3611 - 6.3898 = 1.9713$$

Note : Standard deviation, $\sigma = \sqrt{1.9713} = 1.404$

Example 11 : Calculate expectation and variance of X, if the probability distribution of the random variable X is given by

X	-1	0	1	2	3
f	0.3	0.1	0.1	0.3	0.2

[JNTU 2006 (Set No.2)]

Solution :

(i) Expectation = Mean = $\sum f_i x_i$

$$i.e., E(X) = (-1)(0.3) + 0(0.1) + 1(0.1) + 2(0.3) + 3(0.2)$$

$$\text{or } \mu = -0.3 + 0.1 + 0.6 + 0.6 = 1$$

(ii) Variance = $\sum f_i x_i^2 - \mu^2 = \sum x_i^2 f_i - [E(X)]^2$

$$= (-1)^2 (0.3) + 0(0.1) + 1(0.1) + 2^2(0.3) + 3^2(0.2) - 1$$

$$= 0.3 + 0.1 + 1.2 + 1.8 - 1 = 2.4$$

Example 12 : Calculate the mean for the following distribution.

$X = x :$	0.3	0.2	0.1	0	1	2	3
$P(X = x) :$	0.05	0.10	0.30	0	0.30	0.15	0.1

[JNTU 2008 (Set No. 4)]

Solution : The mean value of the probability distribution of a variate X is commonly known as its expectation and is denoted by $E(X)$. It is given by

$$\begin{aligned}
 E(X) &= \sum_i x_i f(x_i) \\
 &= 0.3(0.05) + 0.2(0.10) + 0.1(0.30) + 0(0) + 1(0.30) + 2(0.15) + 3(0.1) \\
 &= 0.015 + 0.02 + 0.03 + 0 + 0.30 + 0.30 + 0.3 = 0.965
 \end{aligned}$$

Example 13 : A random variable X is defined as the sum of the numbers on the faces when two dice are thrown. Find the mean of X . [JNTU (A) 2009 (Set No. 4)]

Solution : Let x be the sum of the numbers on the faces when two dice are thrown. x is a discrete random variable whose probability distribution is given by

x_i	2	3	4	5	6	7		11	12
$p(x_i)$	1/36	2/36	3/36	4/36	5/36	6/36		2/36	1/36

$$\begin{aligned}
 \therefore \text{Mean of } X &= E(X) = \sum_i p_i x_i \\
 &= 2\left(\frac{1}{36}\right) + 3\left(\frac{2}{36}\right) + 4\left(\frac{3}{36}\right) + 5\left(\frac{4}{36}\right) + 6\left(\frac{5}{36}\right) + 7\left(\frac{6}{36}\right) \\
 &\quad + 8\left(\frac{5}{36}\right) + 9\left(\frac{4}{36}\right) + 10\left(\frac{3}{36}\right) + 11\left(\frac{2}{36}\right) + 12\left(\frac{1}{36}\right) \\
 &= \frac{1}{36}(2 + 6 + 12 + 20 + 30 + 42 + 40 + 36 + 30 + 22 + 12) \\
 &= \frac{252}{36} = 7
 \end{aligned}$$

Example 14 : Find the mean and variance of the uniform probability distribution given by $f(x) = \frac{1}{n}$ for $x = 1, 2, 3, \dots, n$.

[JNTU 2001, (H) Nov. 2009, Nov. 2010, Dec. 2011 (Set No. 3)]

Solution : The probability distribution is

x	1	2	3	...	n
$f(x)$	$\frac{1}{n}$	$\frac{1}{n}$	$\frac{1}{n}$	...	$\frac{1}{n}$

$$(i) \quad \text{Mean} = \sum_{i=1}^n x_i f(x_i) = 1 \cdot \frac{1}{n} + 2 \cdot \frac{1}{n} + \dots + n \cdot \frac{1}{n}$$

$$\text{or } E(X) = \mu = \frac{1}{n} (1 + 2 + 3 + \dots + n) = \frac{1}{n} \cdot \frac{n(n+1)}{2} = \frac{n+1}{2}$$

$$(ii) \quad \text{Variance} = \sum_{i=1}^n x_i^2 f(x_i) - \mu^2$$

$$= 1^2 \cdot \frac{1}{n} + 2^2 \cdot \frac{1}{n} + 3^2 \cdot \frac{1}{n} + \dots + n^2 \cdot \frac{1}{n} - \left(\frac{n+1}{2} \right)^2$$

$$= \frac{1}{n} (1^2 + 2^2 + 3^2 + \dots + n^2) - \frac{1}{4} (n+1)^2$$

$$= \frac{1}{n} \cdot \frac{n(n+1)(2n+1)}{6} - \frac{1}{4} (n+1)^2$$

$$= \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} = \frac{(n+1)}{2} \left(\frac{2n+1}{3} - \frac{n+1}{2} \right)$$

$$= \frac{n+1}{12} (4n+2-3n-3) = \frac{(n+1)(n-1)}{12} = \frac{n^2-1}{12}$$

Example 13 : A sample of 4 items is selected at random from a box containing 12 items of which 5 are defective. Find the expected number E of defective items.

[JNTU (H) Nov. 2009 (Set No. 4), (A) Nov. 2010 (Set No. 1)]

Solution : Let X denote the number of defective items among 4 items drawn from 12 items.

Obviously X can take the values 0, 1, 2, 3, or 4.

No. of good items = 7

No. of defective items = 5

$$P(X=0) = P(\text{no defective}) = \frac{{}^7C_4}{{}^{12}C_4} = \frac{{}^7C_4}{495} = \frac{35}{495} = \frac{7}{99}$$

$P(X=1) = P(\text{one defective and 3 good items})$

$$= \frac{{}^7C_3 \times {}^5C_1}{{}^{12}C_4} = \frac{175}{495} = \frac{35}{99}$$

$P(X=2) = P(2 \text{ defective and 2 good items})$

$$= \frac{{}^7C_2 \times {}^5C_2}{{}^{12}C_4} = \frac{210}{495} = \frac{42}{99}$$

$P(X=3) = P(3 \text{ defective and 1 good item})$

$$= \frac{{}^7C_1 \times {}^5C_3}{{}^{12}C_4} = \frac{70}{495} = \frac{14}{99}$$

$$P(X = 4) = P(\text{all are defective})$$

$$= \frac{{}^5C_4}{{}^{12}C_4} = \frac{5}{495} = \frac{1}{99}$$

Discrete probability distribution is

$X = x_i$	0	1	2	3	4
$P(X = x_i) = f(x_i)$	$\frac{7}{99}$	$\frac{35}{99}$	$\frac{42}{99}$	$\frac{14}{99}$	$\frac{1}{99}$

Expected number of defective items = $E(X) = \sum x_i f(x_i)$

$$= 0 \cdot \frac{7}{99} + 1 \cdot \frac{35}{99} + 2 \cdot \frac{42}{99} + 3 \cdot \frac{14}{99} + 4 \cdot \frac{1}{99} = \frac{165}{99}$$

Example 13 : Show that the variance of a random variable X is given by

$$\sigma^2 = E(X^2) - [E(X)]^2$$

Solution : We know that

$$\begin{aligned}\sigma^2 &= E[(X - \mu)^2] = E(X^2 - 2X\mu + \mu^2) = E(X^2) - 2\mu E(X) + E(\mu^2) \\ &= E(X^2) - 2\mu \cdot \mu + \mu^2 = E(X^2) - \mu^2 = E(X^2) - [E(X)]^2\end{aligned}$$

Example 13 : Given that $f(x) = K / 2x$, is a probability distribution for a random variable X that can take on the values $x = 0, 1, 2, 3$ and 4.

(i) Find K (ii) Mean and variance of X. [JNTU (A) Sup., 2010]

Solution : Given $f(x) = K / 2x$, where $x = 0, 1, 2, 3, 4$. If $x = 0$, $f(x)$ is not defined.

Hence the problem is not correct.

We shall delete $x = 0$ and workout the problem.

(i) If $f(x)$ is a probability distributive function, then

$$1. \quad f(x) \geq 0 \Rightarrow f(x) = \frac{K}{2x} \geq 0 \Rightarrow K \geq 0$$

$$\begin{aligned}2. \quad \sum_{\forall x} f(x) &= 1 \Rightarrow \sum_{x=1}^4 \frac{K}{2x} = 1 \Rightarrow \frac{K}{2} \cdot \sum_{x=1}^4 \frac{1}{x} = 1 \\ &\Rightarrow \frac{K}{2} \left[1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} \right] = 1 \Rightarrow \frac{K}{2} \left(\frac{25}{12} \right) = 1 \\ \therefore K &= \frac{24}{25}\end{aligned}$$

The probability distribution is given in the following table :

x	1	2	3	4
$f(x)$	$\frac{K}{2}$	$\frac{K}{4}$	$\frac{K}{6}$	$\frac{K}{8}$

$$\begin{aligned}
 (ii) \text{ Mean} &= E(X) = \sum_{x=1}^4 x \cdot f(x) \\
 &= 1 \times \frac{K}{2} + 2 \times \frac{K}{4} + 3 \times \frac{K}{6} + 4 \times \frac{K}{8} \\
 &= \frac{K}{2} + \frac{K}{2} + \frac{K}{2} + \frac{K}{2} = 4 \left(\frac{K}{2} \right) = 2K \\
 &= 2 \left(\frac{24}{25} \right) = \frac{48}{25} \left[\because K = \frac{24}{25} \right]
 \end{aligned}$$

$$\begin{aligned}
 \text{Now } E(X^2) &= \sum_{x=1}^4 x^2 \cdot f(x) \\
 &= 1 \times \frac{K}{2} + 4 \times \frac{K}{4} + 9 \times \frac{K}{6} + 16 \times \frac{K}{8} \\
 &= \frac{K}{2} + K + \frac{3K}{2} + 2K = 5K
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Variance} &= E(X^2) - [E(X)]^2 \\
 &= 5K - (2K)^2 = 5K - 4K^2 = K(5 - 4K) \\
 &= \frac{24}{25} \left(5 - \frac{96}{25} \right) \left[\because K = \frac{24}{25} \right] \\
 &= \frac{24}{25} \left(\frac{29}{25} \right) = \frac{696}{625} = 1.1136
 \end{aligned}$$

Example 18 : A variate X has the following probability distribution

x	-3	6	9
$p(x)$	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{1}{3}$

Find (a) (i) $E(X)$ (ii) $E(X^2)$ (iii) $E[(2X + 1)^2]$

(b) Find the mean and variance of the above probability distribution.

[JNTU (A) June 2016]

Solution :

$$(a) \quad (i) \quad E(X) = \sum_{i=1}^3 p_i x_i = (-3) \left(\frac{1}{6} \right) + 6 \cdot \frac{1}{2} + 9 \cdot \frac{1}{3} = -\frac{1}{2} + 3 + 3 = \frac{11}{2}$$

$$\begin{aligned}
 (ii) \quad E(X^2) &= \sum_{i=1}^3 p_i x_i^2 = (-3)^2 \left(\frac{1}{6}\right) + (6)^2 \cdot \frac{1}{2} + (9)^2 \cdot \frac{1}{3} \\
 &= \frac{9}{6} + \frac{36}{2} + \frac{81}{3} = \frac{93}{2}
 \end{aligned}$$

$$\begin{aligned}
 (iii) \quad E[(2X + 1)^2] &= E(4X^2 + 4X + 1) = 4E(X^2) + 4E(X) + E(1) \\
 &= 4 \cdot \frac{93}{2} + 4 \cdot \frac{11}{2} + 1 = \frac{418}{2} = 209
 \end{aligned}$$

(b) From a (i), (ii), we have

$$\text{Mean} = E(X) = \frac{11}{2} \text{ and } E(X^2) = \frac{93}{2}$$

$$\therefore \text{Variance} = E(X^2) - [E(X)]^2 = \frac{93}{2} - \frac{121}{4} = \frac{372 - 121}{4} = \frac{251}{4}$$

Example 13 : A player wins if he gets 5 on a single throw of a die, he loses if he gets 2 or 4. If he wins, he gets Rs. 50, if he loses he gets Rs. 10, otherwise he has to pay Rs. 15. Find the value of the game to the player. Is it favourable ?

Solution : Range of $X = \{-15, 10, 50\}$

Since there are six numbers in a die, out of these 5 is only one number, the probability of getting Rs. 50 is

$$P(X = 50) = \frac{1}{6}$$

Similarly, the probability of getting two numbers (2 or 4) to win Rs. 10 is

$$P(X = 10) = \frac{2}{6} = \frac{1}{3}$$

The probability of getting three numbers (1 or 3 or 6) to lose Rs. 15 is

$$P(X = -15) = \frac{3}{6} = \frac{1}{2}$$

Discrete probability distribution is

$X = x$	- 15	10	50
$P(X) = p(x)$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{6}$

$$\text{Expected value of the game} = E(X) = \sum x_i p(x_i) = (-15) \cdot \frac{1}{2} + 10 \cdot \frac{1}{3} + 50 \cdot \frac{1}{6} = \text{Rs. } \frac{25}{6}$$

Game is favourable to the player since $E > 0$.

Example 20 : A player tosses 3 fair coins. He wins Rs. 500 if 3 heads appear, Rs. 300 if 2 heads appear, Rs. 100 if 1 head occurs. On the other hand, he loses Rs. 1500 if 3 tails occur. Find the expected gain of the player.

Solution : Let X denote the gain. Then the range of X is $\{-1500, 100, 300, 500\}$

The sample space $S = n(s) = 2^3 = 8$

$$= \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

The probability of all 3 heads (getting Rs. 500) = $P(X = 3) = \frac{1}{8}$

The probability of getting 2 heads (getting Rs. 300) = $P(X = 2) = \frac{3}{8}$

The probability of getting one head (getting Rs. 100) = $P(X = 1) = \frac{3}{8}$

The probability of getting 3 tails (lossing Rs. 1500) = $P(X = 0) = \frac{1}{8}$

Discrete probability distribution is

$X = x_i$	- 1500	100	300	500
$P(X = x_i) = p(x_i)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

$\therefore$ Expected value of $X = E(X) = \sum x_i p(x_i)$

$$= (-1500) \cdot \frac{1}{8} + 100 \cdot \frac{3}{8} + 300 \cdot \frac{3}{8} + 500 \cdot \frac{1}{8}$$

$$= \frac{1}{8}(-1500 + 300 + 900 + 500) = \text{Rs. } 25.$$

Example 21 : A fair coin is tossed until a head or five tails occurs. Find the expected number E of tosses of the coin. [JNTU (H) Nov. 2009 (Set No. 3)]

Solution : If head occurs first time there will be only one toss. On the other hand, if first one is tail, second occurs. If head occurs there will be only two tosses. Suppose second one is also tail third occurs. If head occurs there will be three tosses and so on.

$$\therefore p(1) = p(H) = \frac{1}{2}, \quad p(2) = p(TH) = \frac{1}{4},$$

$$p(3) = p(TTH) = \frac{1}{8}, \quad p(4) = p(TTTH) = \frac{1}{16},$$

$$p(5) = p(TTTTH) + p(TTTTT) = \frac{1}{32} + \frac{1}{32} = \frac{1}{16}$$

The probability distribution function of X is

x	1	2	3	4	5
$p(x)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{16}$

Hence $E(X) = \sum_i p_i x_i$

$$= 1 \times \frac{1}{2} + 2 \times \frac{1}{4} + 3 \times \frac{1}{8} + 4 \times \frac{1}{16} + 5 \times \frac{1}{16} = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \frac{5}{16}$$

$$= \frac{8+8+6+4+5}{16} = \frac{31}{16} = 1.9375$$

Example 22 : A player tosses two fair coins. He wins Rs.100/- if head appears, Rs. 200/- if two heads appear. On the other hand he loses Rs.500/- if no head appears. Determine the expected value E of the game and is the game favourable to the player?

[JNTU (K) Nov. 2009 (Set No. 3)]

Solution : Let X denote the number of heads occurring in tosses of two fair coins. The sample space S is

$$S = \{H, T\} \times \{H, T\} = \{HH, HT, TH, TT\}$$

$$\text{Probability of all 2 heads} = P(X = 2) = \frac{1}{4}$$

$$\text{Probability of all 2 tails} = P(X = 0) = \frac{1}{4}$$

$$\text{Probability of one head} = P(X = 1) = \frac{2}{4}$$

Discrete probability distribution is

$X = x_i$	0	1	2
$P(x_i)$	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{1}{4}$

$$\begin{aligned} \text{Expected value of the game} &= 100 \times \frac{2}{4} + 200 \times \frac{1}{4} - 500 \times \frac{1}{4} \\ &= \frac{200 + 200 - 500}{4} = \frac{-100}{4} = -25 \text{ rupees} \end{aligned}$$

$\therefore$ Game is not favourable to the player since $E < 0$

Example 21 : A discrete random variable X has the following distribution function :

$$F(x) = \begin{cases} 0, & \text{for } x < 1 \\ 1/3, & \text{for } 1 \leq x < 4 \\ 1/2, & \text{for } 4 \leq x < 6 \\ 5/6, & \text{for } 6 \leq x < 10 \\ 1, & \text{for } x \geq 10 \end{cases}$$

Find (i) $P(2 < X \leq 6)$ (ii) $P(X = 5)$ (iii) $P(X = 4)$ (iv) $P(X \leq 6)$ (v) $P(X = 6)$

[JNTU (K) Nov. 2009 (Set No. 3)]

Solution :

$$(i) P(2 < X \leq 6) = F(6) - F(2) = P(X \leq 6) - P(X \leq 2) = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

$$(ii) P(X = 5) = P(X \leq 5) - P(X < 5) = F(5) - P(X < 5) = \frac{1}{2} - \frac{1}{2} = 0$$

$$(iii) P(X = 4) = P(X \leq 4) - P(X < 4) = F(4) - P(X < 4) = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

$$(iv) P(X \leq 6) = F(6) = \frac{5}{6}$$

$$(v) P(X = 6) = F(6) - P(X < 6) = \frac{5}{6} - \frac{1}{2} = \frac{2}{6} = \frac{1}{3}$$

Example 24 : Find the distribution function which corresponds to the probability distribution defined by $f(x) = \frac{x}{15}$ for $x = 1, 2, 3, 4, 5$. [JNTU (K) May 2013 (Set No.3)]

Solution :

Given $f(x) = \frac{x}{15}$. So $f(1) = \frac{1}{15}, f(2) = \frac{2}{15}, f(3) = \frac{3}{15}, f(4) = \frac{4}{15}$ and $f(5) = \frac{5}{15}$

Now $F(1) = f(1) = \frac{1}{15}, F(2) = F(1) + f(2) = \frac{1}{15} + \frac{2}{15} = \frac{3}{15}$

$F(3) = F(2) + f(3) = \frac{3}{15} + \frac{3}{15} = \frac{6}{15} = \frac{2}{5}, F(4) = F(3) + f(4) = \frac{6}{15} + \frac{4}{15} = \frac{10}{15} = \frac{2}{3}$

and $F(5) = F(4) + f(5) = \frac{10}{15} + \frac{5}{15} = 1$

Example 21 : A sample of 3 items is selected at random from a box containing 10 items of which 4 are defective. Find the expected number of defective items?

Solution : The probability of defective is $p = \frac{4}{10} = \frac{2}{5}$

Number of items chosen, $n = 3$

The expected number of defective is

$$E(X) = \mu = np = 3 \left(\frac{2}{5} \right) = \frac{6}{5} = 1.2 \approx 1$$

Alternative Method :

$$\text{No. of Exhaustive cases} = {}^{10}C_3 = \frac{10!}{3!7!} = 120$$

$$\text{Probability that there are no defective items} = p(x=0) = \frac{{}^6C_3}{{}^{10}C_3} = \frac{1}{120}$$

$$\text{Probability that there is one defective item} = p(x=1) = \frac{{}^6C_2 \cdot {}^4C_1}{{}^{10}C_3} = \frac{15}{120} = \frac{1}{8}$$

$$\text{Probability that there are two defective items} = p(x=2) = \frac{{}^6C_1 \cdot {}^4C_2}{{}^{10}C_3} = \frac{6}{120} = \frac{1}{20}$$

$$\text{Probability that there are three defective items} = p(x=3) = \frac{{}^4C_3}{{}^{10}C_3} = \frac{1}{120}$$

x	0	1	2	3
$p(x)$	$\frac{1}{120}$	$\frac{15}{120}$	$\frac{6}{120}$	$\frac{1}{120}$

$\therefore$ Expected number of defective items $= E(x) = \sum p_i x_i$

$$\begin{aligned}
 &= 0 \cdot \frac{1}{6} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{3}{10} + 3 \cdot \frac{1}{30} \\
 &= 0 + \frac{1}{2} + \frac{3}{5} + \frac{1}{10} = \frac{12}{10} = 1.2
 \end{aligned}$$

Example 24 : Let X denote the sum of the two numbers that appear when a pair of fair dice is tossed. Determine the (i) Distribution function, (ii) mean and (iii) variance.

[JNTU (K) May 2010 (Set No.4)]

(or) Two dice are rolled at random. Find the probability distribution of the sum of the numbers on them. Also find the mean of the distribution ? [JNTU (K) Nov. 2011 (Set No.4)]

Solution : (i) If two unbiased dice are thrown, then the sum X of the two numbers which turn up must be an integer between 2 and 12.

For $X=2$, there is only one favourable point (1, 1) and hence $P(X=2) = 1/36$, since there are 36 sample points in all.

For $X=3$, there are two favourable sample points (1, 2) and (2, 1) and hence $P(X=3) = 2/36$.

Similarly for $X=4$, there are three favourable sample points (1, 3), (2, 2), (3, 1) and hence $P(X=4) = 3/36$ and so on.

Now the probability distribution in this case is given by the following table :

$X = x_i$	2	3	4	5	6	7	8	9	10	11	12
$P(X = x_i)$	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

(ii) Mean, $\mu = \sum x_i p(x_i)$

$$\begin{aligned}
 &= 2 \times \frac{1}{36} + 3 \times \frac{2}{36} + 4 \times \frac{3}{36} + \dots + 12 \times \frac{1}{36} \\
 &= \frac{1}{36} (2 + 6 + 12 + 20 + 30 + 42 + 40 + 36 + 30 + 22 + 12)
 \end{aligned}$$

$$i.e., E(x) = \frac{252}{36} = 7$$

(iii) Variance, $\sigma^2 = E(x^2) - [E(x)]^2$

$$\begin{aligned}
 &= \sum x_i^2 p(x_i) - (7)^2 \\
 &= 2^2 \times \frac{1}{36} + 3^2 \times \frac{2}{36} + 4^2 \times \frac{3}{36} + 5^2 \times \frac{4}{36} + \dots + 12^2 \times \frac{1}{36} \\
 &= \frac{1}{36} [4 + 18 + 48 + 100 + 180 + 294 + 320 + 324 + 300 + 242 + 144] - 49 \\
 &= \frac{1974}{36} - 49 = 54.83 - 49 = 5.83
 \end{aligned}$$

Example 21 : A fair die is tossed. Let the random variable X denote the twice the number appearing on the die :

(i) Write the probability distribution of X

(ii) The mean

(iii) The variance

[JNTU (H) Apr. 2012 (Set No. 2)]

Solution : Let x denote twice the number appearing on the face when a die is thrown. Then x is a discrete random variable whose probability distribution is given by

(i)

x_i	2	4	6	8	10	12
$p(x_i)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

(ii) Mean = $E(X) = \sum p_i x_i$

$$= 2 \times \frac{1}{6} + 4 \times \frac{1}{6} + 6 \times \frac{1}{6} + 8 \times \frac{1}{6} + 10 \times \frac{1}{6} + 12 \times \frac{1}{6}$$

$$= \frac{1}{6}(2 + 4 + 6 + 8 + 10 + 12) = \frac{42}{6} = 7$$

Now $E(X^2) = \sum x_i^2 p(x_i)$

$$= 2^2 \times \frac{1}{6} + 4^2 \times \frac{1}{6} + 6^2 \times \frac{1}{6} + 8^2 \times \frac{1}{6} + 10^2 \times \frac{1}{6} + 12^2 \times \frac{1}{6}$$

$$= \frac{1}{6}(2^2 + 4^2 + 6^2 + 8^2 + 10^2 + 12^2)$$

$$= \frac{1}{6}(4 + 16 + 36 + 64 + 100 + 144) = \frac{364}{6} = 60.67$$

(iii) Variance = $E(X^2) - [E(X)]^2$

$$= 60.67 - (7)^2 = 60.67 - 49 = 11.67$$

Example 28 : A random sample with replacement of size 2 is taken from $S = \{1, 2, 3\}$.

Let the random variable X denote the sum of the two numbers taken:

(i) Write the probability distribution of X .

(ii) Find the mean.

(iii) Find the variance.

[JNTU (H) Apr. 2012 (Set No. 3)]

Solution : Sample space $S = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$

$X(S) = (2, 3, 4, 5, 6)$

$$p(2) = P(X = 2) = p[(1,1)] = \frac{1}{9}$$

$$p(3) = P(X = 3) = p[(1,2), (2,1)] = \frac{2}{9}$$

$$p(4) = P(X = 4) = p[(1, 3), (2, 2), (3, 1)] = \frac{3}{9}$$

$$p(5) = P(X = 5) = p[(2, 3), (3, 2)] = \frac{2}{9}$$

$$p(6) = P(X = 6) = p[(3, 3)] = \frac{1}{9}$$

(i) The probability distribution of X is

x	2	3	4	5	6
$p(x)$	$\frac{1}{9}$	$\frac{2}{9}$	$\frac{3}{9}$	$\frac{2}{9}$	$\frac{1}{9}$

$$(ii) \text{ Mean } = E(x) = \sum p_i x_i = 2 \times \frac{1}{9} + 3 \times \frac{2}{9} + 4 \times \frac{3}{9} + 5 \times \frac{2}{9} + 6 \times \frac{1}{9}$$

$$= \frac{1}{9}(2 + 6 + 12 + 10 + 6) = \frac{1}{9}(36) = 4$$

$$(iii) E(x^2) = \sum p_i x_i^2$$

$$= \frac{1}{9} \times 4 + \frac{2}{9} \times 9 + \frac{3}{9} \times 16 + \frac{2}{9} \times 25 + \frac{1}{9} \times 36$$

$$= \frac{1}{9}(4 + 18 + 48 + 50 + 36) = \frac{156}{9} = 17.33$$

$$\text{Hence variance} = E(x^2) - [E(x)]^2$$

$$= 17.33 - (4)^2 = 17.33 - 16 = 1.33$$

Example 22 : Take 30 slips of paper and label 5 each -4 and 4 , four each -3 and 3 , three each -2 and 2 , and two each -1 , 0 and 1 . If each slip of paper has the same probability of being drawn find the probabilities of getting $-4, -3, -2, -1, 0, 1, 2, 3, 4$ and find the mean and variance of this distribution.

[JNTU (A) Apr. 2012 (Set No. 4)]

Solution : $P(-4) = \text{Probability of getting } (-4) = \frac{\text{Favourable cases}}{\text{Total cases}} = \frac{5}{30} = \frac{1}{6}$

$$P(4) = \frac{5}{30} = \frac{1}{6}, P(-3) = \frac{4}{30} = \frac{2}{15}$$

$$P(3) = \frac{4}{30} = \frac{2}{15}, P(-2) = \frac{3}{30} = \frac{1}{10}, P(2) = \frac{3}{30} = \frac{1}{10}$$

$$P(-1) = \frac{2}{30} = \frac{1}{15}, P(1) = \frac{2}{30} = \frac{1}{15}, P(0) = \frac{2}{30} = \frac{1}{15} \text{ and}$$

$$P(0) = \frac{2}{30} = \frac{1}{15}$$

The discrete probability distribution is

x	-4	4	-3	3	-2	2	-1	1	0
$p(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{2}{15}$	$\frac{2}{15}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{1}{15}$

Mean of the distribution, $\mu = \sum p_i x_i$

$$= \frac{-4}{6} + \frac{4}{6} - \frac{6}{15} + \frac{6}{15} - \frac{2}{10} + \frac{2}{10} - \frac{1}{15} + \frac{1}{15} + 0 = 0$$

Variance of the distribution $\sigma^2 = \sum p_i x_i^2 - \mu^2$

$$\begin{aligned}
 &= \left[\frac{1}{6}(-4)^2 + \frac{1}{6}(4)^2 + \frac{2}{15}(-3)^2 + \frac{2}{15}(3)^2 + \frac{1}{10}(-2)^2 + \right. \\
 &\quad \left. \frac{1}{10}(2)^2 + \frac{1}{15}(-1)^2 + \frac{1}{15}(1)^2 + \frac{1}{15}(0)^2 \right] - 0 \\
 &= \frac{1}{6}(16+16) + \frac{2}{15}(9+9) + \frac{1}{10}(4+4) + \frac{1}{15}(1+1) \\
 &= \frac{32}{6} + \frac{36}{15} + \frac{8}{10} + \frac{2}{15} = \frac{160+72+24+4}{30} \\
 &= \frac{260}{30} = \frac{26}{3}
 \end{aligned}$$

Example 23 : Find the mean of the probability distribution of the number of heads obtained in three flips of a balanced coin. [JNTU (A) Nov. 2011 (Set No. 1)]

Solution : Let X be the number of heads. Then $X = \{0, 1, 2, 3\}$

Sample space, $S = 2^3 = 8$

i.e. $S = \{HHH, HHT, THH, HTH, HTT, THT, TTH, TTT\}$

$$p(X=0) = \frac{1}{8}; \quad p(X=1) = \frac{3}{8}; \quad p(X=2) = \frac{3}{8}; \quad p(X=3) = \frac{1}{8}$$

$\therefore$ The probability distribution of X is

$X = x_i$	0	1	2	3
$p(X = x_i)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Hence the mean of the probability distribution of X is

$$\mu = \sum p_i x_i = 0\left(\frac{1}{8}\right) + 1\left(\frac{3}{8}\right) + 2\left(\frac{3}{8}\right) + 3\left(\frac{1}{8}\right) = \frac{3+6+3}{8} = \frac{12}{8} = 1.5$$

Example 22 : The probability mass function of a random variable X is zero except at the points $x = 0, 1, 2$. At these points, it has the values $P(0) = 3c^3$, $P(1) = 4c - 10c^2$, $P(2) = 5c - 1$ for some $c > 0$.

- (i) Determine the value of c .
- (ii) Find $P(x < 1)$, $P(x < 2)$, $P(1 < x \leq 2)$ and $P(0 < x < 3)$.
- (iii) Find the largest value of X such that $F(x) < 1/2$
- (iv) Find the smallest value of X such that $F(x) > 1/2$. [JNTU (A) Nov. 2011 (Set No. 2)]

Solution : (i) We know that $p(X = x_i) \geq 0, \forall i$ and $\sum_{i=1}^{\infty} p(X = x_i) = 1$

$$\text{i.e., } p(0) + p(1) + p(2) = 1 \text{ i.e., } 3c^3 + 4c - 10c^2 + 5c - 1 = 1$$

$$\text{or } 3c^3 - 10c^2 + 9c - 2 = 0 \text{ or } (c-1)(c-2)(3c-1) = 0$$

$$\therefore c = 1, 2, \frac{1}{3}$$

$$\text{For } c = 1, 2, p(1) < 0$$

$$\text{For } c = \frac{1}{3}, p(0) > 0, p(1) > 0 \text{ and } p(2) > 0$$

$$\therefore \text{ We take } c = \frac{1}{3}$$

$$(ii) \quad p(x < 1) = p(x = 0) = 3c^3 = 3 \times \left(\frac{1}{3}\right)^3 = \frac{1}{9}$$

$$p(x < 2) = p(x = 0) + p(x = 1) = \frac{1}{9} + (4c - 10c^2) = \frac{1}{9} + \frac{4}{3} - \frac{10}{9} = \frac{4}{3} - 1 = \frac{1}{3} \quad \left[\because c = \frac{1}{3} \right]$$

$$p(1 < x \leq 2) = p(x = 2) = 5c - 1 = \frac{5}{3} - 1 = \frac{2}{3}$$

REVIEW QUESTIONS

1. Explain random variable with suitable example. [JNTU (A) Dec. 2015]
2. (i) Write the types of random variables.
(ii) Explain the types of random variables with suitable examples.
3. Write the properties of Probability function.
4. Define Probability distribution function.
5. What is the formula of (i) Mathematical expectation (Mean) (ii) Variance of a discrete random variable.
6. Write the definitions of (i) Random variable (ii) Discrete Random Variable (iii) Continuous Random Variable and (iv) Probability Distribution function

[JNTU (K) Nov. 2011 (Set No. 2)]

7. Define random variable, discrete probability distribution, continuous probability distribution and cumulative distribution. Give an example of each.
[JNTU 2007, 2008S (Set No.4), (K) Nov. 2011 (Set No. 1)]
8. List the properties of probability distribution function.
9. (i) Define mathematical expectation
(ii) Write the properties of Mathematical expectation.
10. Define (i) Probability density function (ii) Probability mass function
(iii) Discrete random variable (iv) Continuous random variable

EXERCISE 6(A)

1. (a) The probability density function of a variate X is as follows :

$X = x$	0	1	2	3	4	5	6
$P(x)$	K	$3K$	$5K$	$7K$	$9K$	$11K$	$13K$

- (i) Find $P(X < 4)$, $P(X \geq 5)$, $P(3 < X \leq 6)$
(ii) What will be the minimum value of K so that $P(X \leq 2) > 0.3$.

- (b) A random variable X has the following probability function :

X	0	1	3	4	5	6	7
$P(X)$	0	K	$2K$	$2K$	$3K$	K^2	$7K^2 + K$

- (i) Find the value of K (ii) Evaluate $P(X < 6)$, $P(X \geq 6)$
(iii) Evaluate $P(0 < X < 5)$

- (c) A discrete random variable X has the following probability distribution

Value of X	0	1	2	3	4	5	6	7	8
$P(X = x)$	k	$3k$	$5k$	$7k$	$9k$	$11k$	$13k$	$15k$	$17k$

- (i) Find the value of ' k ' (ii) Find $P(X \leq 3)$, $P(0 < X < 3)$, $P(X \geq 3)$

- (d) A discrete random variable X has the following probability distribution

Value of X	1	2	3	4	5	6	7	8
$P(X = x)$	$2k$	$4k$	$6k$	$8k$	$10k$	$12k$	$14k$	$4k$

- (i) Find the value of ' k ' (ii) Find $P(X < 3)$ and $P(X \geq 5)$
(iii) Find the distribution function of X .

2. A random variable X has the following probability function :

$X = x$	-2	-1	0	1	2	3
$P(X)$	0.1	K	0.2	$2K$	0.3	K

Find (i) K (ii) Mean (iii) Variance

- (iv) $P(x \geq 2)$ (v) $P(x < 2)$ (vi) $P(-1 < x < 3)$ [JNTU 2007S, (A) May 2012]

3. A random variable X has the following probability function :

$X=x$	4	5	6	8
$P(X)$	0.1	0.3	0.4	0.2

Determine (i) Expectation (ii) Variance (iii) Standard deviation

4. Find mean and variance for the following discrete distribution

$X=x$	-3	-2	-1	0	1	2	3
$P(X)$	K	0.1	K	0.2	$2K$	0.4	$2K$

[JNTU (H) Dec. 2009 (Set No.2)]

5. (a) A random variable X has the following probability function :

$X=x$	1	2	3	4	5	6
$P(X)$	K	$3K$	$5K$	$7K$	$9K$	$11K$

Determine (i) K (ii) Expectation (iii) Variance

- (b) A random variable X has the following probability function :

X	0	1	2	3	4	5	6	7	8
$P(X)$	a	$3a$	$5a$	$7a$	$9a$	$11a$	$13a$	$15a$	$17a$

(i) Determine the value of a ,

(ii) Evaluate $P(X < 3)$, $P(X \geq 3)$, $P(2 \leq X < 5)$ [JNTU (K) Nov. 2012 (Set No. 2)]

- (c) The probability function of a variate X is

X	0	1	2	3	4	5	6
$P(X)$	k	$3k$	$5k$	$7k$	$9k$	$11k$	$13k$

[JNTU (A) June 2018]

Find k , mean and variance

6. Find the expected value of x and standard deviation for the following discrete distribution :

X	8	12	16	20	24
$P(X)$	$\frac{1}{8}$	$\frac{1}{6}$	$\frac{3}{8}$	$\frac{1}{4}$	$\frac{1}{12}$

[JNTU Jan. 2007]

7. A random variable X has the following probability function :

X	0	1	2	3	4	5	6	7	8
$P(X)$	$\frac{K}{45}$	$\frac{K}{15}$	$\frac{K}{9}$	$\frac{K}{5}$	$\frac{2K}{45}$	$\frac{6K}{45}$	$\frac{7K}{45}$	$\frac{8K}{45}$	$\frac{4K}{45}$

Determine (i) K (ii) Mean (iii) Variance (iv) Standard deviation.

[JNTU (K) Nov. 2011 (Set No. 3)]

8. For the following probability distribution

X	-3	-2	-1	0	1	2	3
$P(X)$	0.001	0.01	0.1	?	0.1	0.01	0.001

Find (i) the missing probability (ii) mean

(iii) variance

(iv) $E(X^2 + 2X + 3)$

9. A fair coin is tossed until a head or five tails occurs.

Find (i) the discrete probability distribution (ii) mean of the distribution.

10. A box contains 8 items of which 2 are defective. A man draws 3 items from the box. Find the expected number of defective items he has drawn.
11. A die is thrown at random. What is the expectation of the number on it ?
12. A box contains 4 white and 6 black balls. A man draws 2 balls and is given Rs. 140 for every white ball and Rs. 70 for every black ball. What is his expectation ?
13. Five defective bolts are accidentally mixed with 20 good ones. Find the probability distribution of the number of defective bolts, if four bolts are drawn at random from this lot.
14. Two dice are rolled at random. Find the probability distribution of the sum of the numbers on them. Also find the mean of the distribution.
15. If 3 cars are selected from a lot of 6 cars containing 2 defective cars, find the probability distribution of the number of defective cars. [JNTU 2008S (Set No.1)]
16. Find the expected value and variance of the following distribution :

X	-10	-20	30	75	80
$p(x)$	$\frac{1}{5}$	$\frac{3}{20}$	$\frac{1}{2}$	$\frac{1}{10}$	$\frac{1}{20}$

17. The probability distribution of a random variable X is given below :

X	-2	-1	0	1	2
$p(x)$	0.2	0.1	0.3	0.3	0.1

Find (i) $E(X)$ (ii) $\text{Var}(X)$ (iii) $E(2X-3)$ (iv) $\text{Var}(2X-3)$

18. Let X denotes the minimum of the two numbers that appear when a pair of fair dice is thrown once. Determine the (i) Discrete probability distribution (ii) Expectation (iii) Variance.
19. The probability distribution of a random variable X is given as follows :

$X = x$	0	1	2	3	4
$P(X = x)$	0.4	0.3	0.1	0.1	0.1

Find (i) Mean (ii) Variance

[JNTU (A) June 2019]

20. The probability distribution of a random variable X is given as follows :

$X = x$	1	2	3	4	5
$P(X = x)$	k	$2k$	$3k$	$4k$	$5k$

[JNTU (A) June 2019]

Find the value of k , the mean and variance of X .

ANSWERS

1. (a) (i) $\frac{33}{49}$ (ii) $\frac{1}{30}$ 2. (i) 0.1 (ii) 0.8 (iii) 2.16
3. (i) 5.9 (ii) 1.49 (iii) 1.22 4. 0.8, $\frac{143}{50}$
5. (a) (i) $\frac{1}{36}$ (ii) 4.46 (iii) 2.08 6. 16, $2\sqrt{5}$

7. (i) 1 (ii) 4.622 (iii) 4.9971 (iv) 2.24

8. (i) 0.778 (ii) 0.2 (iii) 0.258 (iv) 3.698

x	1	2	3	4	5
$f(x)$	$1/2$	$1/4$	$1/8$	$1/16$	$1/16$

 9. 1.9 10. $\frac{3}{4}$ 11. $\frac{7}{2}$ 12. 196

x	0	1	2	3	4
$p(x)$	$\frac{969}{2530}$	$\frac{1140}{2530}$	$\frac{380}{2530}$	$\frac{40}{2530}$	$\frac{1}{2530}$

 13.

$X = x_i$	2	3	4	5	6	7	8	9	10	11	12
$p_i = p(x_i)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

 14. ; 7

x	0	1	2
$p(x)$	$\frac{1}{5}$	$\frac{3}{5}$	$\frac{2}{5}$

 15. 16. 21.5, 950.24 17. (i) 0 (ii) 1.6 (iii) -3 (iv) 6.4

6.7 CONTINUOUS PROBABILITY DISTRIBUTION

When a random variable X takes every value in an interval, it gives rise to continuous distribution of X . The distributions defined by the variates like temperature, heights and weights are continuous distributions.

Probability Density Function

[JNTU 2001]

For continuous variable, the probability distribution is called **Probability Density function** because it is defined for every point in the range and not only for certain values.

Consider the small interval $\left[x - \frac{dx}{2}, x + \frac{dx}{2}\right]$ of length dx round the point x . Let $f(x)$ be any continuous function of x so that $f(x) dx$ represents the probability that the variable X falls in the infinitesimal interval $\left[x - \frac{dx}{2}, x + \frac{dx}{2}\right]$. Symbolically it can be expressed as

$P\left(x - \frac{dx}{2} \leq X \leq x + \frac{dx}{2}\right) = f(x) dx$. Then $f(x)$ is called the *probability density function* or simply *density function* of the variate X and the continuous curve $y = f(x)$ is known as the *probability density curve* or simply *probability curve*.

As the probability for a variate value to lie in the interval dx is $f(x) dx$, so the probability for a variate value to fall in the finite interval (a, b) is $\int_a^b f(x) dx$ which represents the area between the curve $y = f(x)$, the x -axis and the ordinates $x = a$ and

$x = b$. Since the total probability is unity, we have $\int_a^b f(x) dx = 1$, where $[a, b]$ is the range of

the variate X . The range of the variable may be finite or infinite. But even when the range is finite, it is convenient to consider it as infinite by supposing the density function to be zero outside the given interval.

Properties of the Probability Density Function $f(x)$:

- (i) $f(x) \geq 0, \forall x \in R$ (ii) $\int_{-\infty}^{\infty} f(x) dx = 1$
- (iii) $P(a < X < b) = \int_a^b f(x) dx = \text{Area under the curve } f(x) \text{ between ordinates}$
 $x = a \text{ and } x = b.$

Note : In the case of continuous random variable, we associate the probabilities with intervals. In this case the probability of the variable at a particular point is always zero.

$$\text{Thus } P(a < X < b) = P(a < X \leq b) = P(a \leq X < b) = P(a \leq X \leq b) = F(b) - F(a) \\ [\because P(X = a) = 0, P(X = b) = 0]$$

That is, inclusion or non-inclusion of end points, does not change the probability, which is not the case in the discrete distributions.

Cumulative Distribution Function of A Continuous Random Variable :

The *cumulative distribution* function or simply the *distribution function* of a continuous random variable X is denoted by $F(x)$ and is defined as

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx$$

Thus $F(x)$ gives the probability that the value of the variable X will be $\leq x$.

Properties of $F(x)$:

- (i) $0 \leq F(x) \leq 1, -\infty < x < \infty.$
- (ii) $F'(x) = f(x) \geq 0$, so that $F(x)$ is a non-decreasing function.
- (iii) $F(-\infty) = 0$
- (iv) $F(\infty) = 1$
- (v) $F(x)$ is a continuous function of x on the right.
- (vi) The discontinuities of $F(x)$ are countable.

(vii) $P(a \leq X \leq b) = \int_a^b f(x) dx = F(b) - F(a)$

(viii) Since $F'(x) = f(x)$, we have $\frac{d}{dx}[F(x)] = f(x) \Rightarrow dF = f(x) dx$

This is known as probability differential of X .

6.8 MEASURES OF CENTRAL TENDENCY FOR CONTINUOUS PROBABILITY DISTRIBUTION

If a variable is continuous then it takes all possible values in its range like height of an individual, life of a car battery, etc., the expectation of the variable is defined as

$$E(x) = \int x f(x) dx \text{ where } f(x) \text{ is the probability function of the variable } x.$$

On replacing p_i by $f(x) dx$, x_i by x and the summation over ' i ' by integration over the specified range of the variate X in the formulae of discrete probability distribution, we obtain the corresponding formulae for continuous probability distribution.

Let $f(x)$ be the probability density function of a continuous random variable X . Then

- (i) **Mean :** Mean of a distribution is given by $\mu = E(X) = \int_{-\infty}^{\infty} x f(x) dx$

If X is defined from a to b , then $\mu = E(X) = \int_a^b x f(x) dx$

In general, mean or expectation of any function $\phi(x)$ is given by

$$E[\phi(x)] = \int_{-\infty}^{\infty} \phi(x) f(x) dx$$

- (ii) **Median:** Median is the point which divides the entire distribution into two equal parts. In case of continuous distribution, median is the point which divides the total area into two equal parts. Thus if X is defined from a to b and M is the median, then

$$\int_a^M f(x) dx = \int_M^b f(x) dx = \frac{1}{2}$$

Solving for M , we get the median.

- (iii) **Mode :** Mode is the value of x for which $f(x)$ is maximum. Mode is thus given by $f'(x) = 0$ and $f''(x) < 0$ for $a < x < b$.
- (iv) **Variance :** Variance of a distribution is given by

$$\sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx \text{ or } \sigma^2 = \int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2$$

Suppose that the variate X is defined from a to b . Then

$$\sigma^2 = \int_a^b (x - \mu)^2 f(x) dx \text{ or } \sigma^2 = \int_a^b x^2 f(x) dx - \mu^2$$

- (v) **Mean deviation :** Mean deviation about the mean (μ) is given by

$$\int_{-\infty}^{\infty} |x - \mu| f(x) dx$$

SOLVED EXAMPLES

Example 1 : If a random variable has the probability density $f(x)$ as

$$f(x) = \begin{cases} 2e^{-2x}, & \text{for } x > 0 \\ 0, & \text{for } x \leq 0 \end{cases}$$

Find the probabilities that it will take on a value

- (i) between 1 and 3 (ii) greater than 0.5.

[JNTU 2001, 2006S(Set No. 4)]

Solution :

- (i) The probability that a variate takes a value between 1 and 3 is given by

$$\begin{aligned}
 P(1 \leq X \leq 3) &= \int_1^3 f(x) dx = \int_1^3 2e^{-2x} dx \\
 &= 2 \left(\frac{e^{-2x}}{-2} \right)_1^3 = -(e^{-6} - e^{-2}) = e^{-2} - e^{-6}
 \end{aligned}$$

(ii) The probability that a variable takes a value greater than 0.5 is

$$\begin{aligned}
 P(X \geq 0.5) &= \int_{0.5}^{\infty} f(x) dx = \int_{0.5}^{\infty} 2e^{-2x} dx \\
 &= 2 \left(\frac{e^{-2x}}{-2} \right)_{0.5}^{\infty} = -(e^{-\infty} - e^{-1}) = -(0 - e^{-1}) = e^{-1}
 \end{aligned}$$

Example 2 : If the probability density of a random variable is given by

$$f(x) = \begin{cases} k(1-x^2), & \text{for } 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

find the value of k and the probabilities that a random variable having this probability density will take on a value (i) between 0.1 and 0.2 (ii) greater than 0.5. [JNTU 1999S]

Solution : Given $f(x) = \begin{cases} k(1-x^2), & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$

We know that $\int_{-\infty}^{\infty} f(x) dx = 1$

$$i.e., \int_{-\infty}^0 f(x) dx + \int_0^1 f(x) dx + \int_1^{\infty} f(x) dx = 1$$

$$i.e., 0 + \int_0^1 k(1-x^2) dx + 0 = 1$$

$$i.e., k \left(x - \frac{x^3}{3} \right)_0^1 = 1 \quad \text{or} \quad k \left(1 - \frac{1}{3} \right) = 1$$

$$\therefore k = \frac{3}{2}$$

(i) The probability that the variate will take on a value between 0.1 and 0.2 is

$$\begin{aligned}
 P(0.1 < X < 0.2) &= \int_{0.1}^{0.2} f(x) dx = \int_{0.1}^{0.2} k(1-x^2) dx \\
 &= \frac{3}{2} \left(x - \frac{x^3}{3} \right)_{0.1}^{0.2} \quad \left(\because k = \frac{3}{2} \right)
 \end{aligned}$$

$$\begin{aligned}
&= \frac{3}{2} \left[\left(0.2 - \frac{0.008}{3} \right) - \left(0.1 - \frac{0.001}{3} \right) \right] \\
&= \frac{3}{2} \left[0.1 - \frac{0.007}{3} \right] = 0.2965
\end{aligned}$$

(ii) The probability that the variate will take on a value greater than 0.5 is

$$\begin{aligned}
P(X > 0.5) &= \int_{0.5}^{\infty} f(x) dx = \int_{0.5}^1 f(x) dx + \int_1^{\infty} f(x) dx \\
&= \frac{3}{2} \int_{0.5}^1 (1-x^2) dx + 0 = \frac{3}{2} \left(x - \frac{x^3}{3} \right)_{0.5}^1 \\
&= \frac{3}{2} \left[\left(1 - \frac{1}{3} \right) - \left(0.5 - \frac{0.125}{3} \right) \right] \\
&= \frac{3}{2} \left(\frac{2}{3} - 0.4583 \right) = 0.3125
\end{aligned}$$

Example 3 : The probability density $f(x)$ of a continuous random variable is given by $f(x) = c e^{-|x|}$, $-\infty < x < \infty$. Show that $c = 1/2$ and find the mean and variance of the distribution. Also find the probability that the variate lies between 0 and 4.

[JNTU Jan. 2007, (A) Nov. 2010 (Set No. 2), (K) May 2013 (Set No. 4)]

Solution : Given $f(x) = c e^{-|x|}$, $-\infty < x < \infty$

We have $\int_{-\infty}^{\infty} f(x) dx = 1$ [since the total probability is unity]

$$i.e., \int_{-\infty}^{\infty} c e^{-|x|} dx = 1 \quad i.e., c \int_{-\infty}^{\infty} e^{-|x|} dx = 1$$

$$i.e., 2c \int_0^{\infty} e^{-|x|} dx = 1 \left[\because e^{-|x|} \text{ is an even function} \right]$$

$$i.e., 2c \int_0^{\infty} e^{-x} dx = 1 \quad [\because \text{in } 0 \leq x \leq \infty, |x| = x]$$

$$\Rightarrow 2c \left(-e^{-x} \right)_0^{\infty} = 1 \Rightarrow -2c (0 - 1) \Rightarrow 2c = 1 \quad \therefore c = \frac{1}{2}$$

$$\text{Hence } f(x) = c e^{-|x|} = \frac{1}{2} e^{-|x|}$$

(i) Mean of the distribution, $E(x) = \mu = \int_{-\infty}^{\infty} x f(x) dx$

$$\therefore \mu = \frac{1}{2} \int_{-\infty}^{\infty} x e^{-|x|} dx = 0, \text{ since integrand is odd.}$$

(ii) Variance of the distribution,

$$\begin{aligned} \sigma^2 &= \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx = \int_{-\infty}^{\infty} (x - 0)^2 \cdot \frac{1}{2} e^{-|x|} dx \\ &= \frac{1}{2} \int_{-\infty}^{\infty} x^2 e^{-|x|} dx \\ &= 2 \cdot \frac{1}{2} \int_0^{\infty} x^2 e^{-x} dx, \text{ since integrand is even} \\ &= \int_0^{\infty} x^2 e^{-x} dx = \left(x^2 \frac{e^{-x}}{-1} - 2x \frac{e^{-x}}{1} + 2 \frac{e^{-x}}{-1} \right)_0^{\infty}, \text{ using Bernoulli's Rule} \\ &= [0 - (-2)] = 2 \end{aligned}$$

(iii) The probability between 0 and 4 = $P(0 \leq X \leq 4)$

$$\begin{aligned} &= \frac{1}{2} \int_0^4 e^{-|x|} dx = \frac{1}{2} \int_0^4 e^{-x} dx \quad [\because \text{in } 0 < x < 4, |x| = x] \\ &= -\frac{1}{2} (e^{-x})_0^4 = -\frac{1}{2} (e^{-4} - 1) \\ &= \frac{1}{2} (1 - e^{-4}) = 0.4908 \text{ (nearly)} \end{aligned}$$

Example 4 : Probability density function of a random variable X is

$$f(x) = \begin{cases} \frac{1}{2} \sin x, & \text{for } 0 \leq x \leq \pi \\ 0, & \text{elsewhere} \end{cases}$$

Find the mean, mode and median of the distribution and also find the probability between 0 and $\pi/2$.

[JNTU 2004, 2008S, (A) Nov. 2010 (Set No. 4)]

Solution : (i) Mean of the distribution = $\int_{-\infty}^{\infty} x f(x) dx$

$$\begin{aligned}
&= \int_{-\infty}^0 x(0) dx + \int_0^{\pi} x \cdot \frac{1}{2} \sin x dx + \int_{\pi}^{\infty} x(0) dx \\
&= \frac{1}{2} \int_0^{\pi} x \sin x dx = \frac{1}{2} (-x \cos x + \sin x)_0^{\pi} \\
&= \frac{\pi}{2}
\end{aligned}$$

(ii) Mode is the value of x for which $f(x)$ is maximum

$$\text{Now } f'(x) = \frac{1}{2} \cos x$$

For $f(x)$ to be maximum, $f'(x) = 0$

$$\text{i.e., } \cos x = 0 \quad \therefore x = \frac{\pi}{2}$$

$$f''(x) = -\frac{1}{2} \sin x. \text{ At } x = \frac{\pi}{2}, f''(x) = -\frac{1}{2} < 0$$

Hence $f(x)$ is maximum at $x = \frac{\pi}{2}$

$\therefore$ Mode of the distribution is given by $x = \frac{\pi}{2}$

(iii) If M is the median of the distribution, then

$$\int_a^M f(x) dx = \int_M^b f(x) dx = \frac{1}{2}$$

$$\text{i.e., } \int_0^M \frac{1}{2} \sin x dx = \int_M^{\pi} \frac{1}{2} \sin x dx = \frac{1}{2}$$

Solving $\int_0^M \frac{1}{2} \sin x dx = \frac{1}{2}$, we get

$$-\frac{1}{2} (\cos x)_0^M = \frac{1}{2} \Rightarrow -\frac{1}{2} (\cos M - 1) = \frac{1}{2}$$

$$\Rightarrow 1 - \cos M = 1 \Rightarrow \cos M = 0 \quad \therefore M = \frac{\pi}{2}$$

$\therefore$ Median of the distribution = $\frac{\pi}{2}$

Thus Mean = Mode = Median = $\frac{\pi}{2}$

$$\begin{aligned}
 (iv) \quad P(0 < x < \frac{\pi}{2}) &= \int_0^{\pi/2} f(x) dx = \int_0^{\pi/2} \frac{1}{2} \sin x dx \\
 &= -\frac{1}{2} (\cos x)_0^{\pi/2} = -\frac{1}{2} (0 - 1) = \frac{1}{2}
 \end{aligned}$$

Example 4 : A continuous random variable has the probability density function

$$f(x) = \begin{cases} k x e^{-\lambda x}, & \text{for } x \geq 0, \lambda > 0 \\ 0, & \text{otherwise} \end{cases}$$

Determine (i) k (ii) Mean (iii) Variance

[JNTU 2003S, 2004, (A) Dec. 2009, Nov. 2010, Dec. 2011, (H) May 2011, (K) May 2013 (Set No. 1)]

Solution : (i) Since the total probability is unity, we have

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$i.e., \int_{-\infty}^0 0 \cdot dx + \int_0^{\infty} kx e^{-\lambda x} dx = 1 \quad i.e., k \int_0^{\infty} x e^{-\lambda x} dx = 1$$

$$i.e., k \left[x \left(\frac{e^{-\lambda x}}{-\lambda} \right) - 1 \cdot \left(\frac{e^{-\lambda x}}{\lambda^2} \right) \right]_0^{\infty} = 1$$

$$i.e., k \left[(0 - 0) - \left(0 - \frac{1}{\lambda^2} \right) \right] = 1 \quad \text{or } k = \lambda^2$$

Now $f(x)$ becomes

$$f(x) = \begin{cases} \lambda^2 x e^{-\lambda x}, & \text{for } x \geq 0, \lambda > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$(ii) \quad \text{Mean of the distribution, } \mu = \int_{-\infty}^{\infty} x f(x) dx$$

$$i.e., \mu = \int_{-\infty}^0 0 \cdot dx + \int_0^{\infty} x \cdot \lambda^2 x e^{-\lambda x} dx = \lambda^2 \int_0^{\infty} x^2 e^{-\lambda x} dx$$

$$= \lambda^2 \left[x^2 \left(\frac{e^{-\lambda x}}{-\lambda} \right) - 2x \left(\frac{e^{-\lambda x}}{\lambda^2} \right) + 2 \left(\frac{e^{-\lambda x}}{-\lambda^3} \right) \right]_0^{\infty}$$

$$= \lambda^2 \left[(0 - 0 + 0) - (0 - 0 - \frac{2}{\lambda^3}) \right] = \frac{2}{\lambda}$$

(iii) Variance of the distribution, $\sigma^2 = \int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2$

$$\begin{aligned}
 \text{i.e., } \sigma^2 &= \int_0^{\infty} x^2 f(x) dx - \left(\frac{2}{\lambda}\right)^2 = \lambda^2 \int_0^{\infty} x^3 e^{-\lambda x} dx - \frac{4}{\lambda^2} \\
 &= \lambda^2 \left[x^3 \left(\frac{e^{-\lambda x}}{-\lambda} \right) - 3x^2 \left(\frac{e^{-\lambda x}}{\lambda^2} \right) + 6x \left(\frac{e^{-\lambda x}}{-\lambda^3} \right) - 6 \left(\frac{e^{-\lambda x}}{\lambda^4} \right) \right]_0^{\infty} - \frac{4}{\lambda^2} \\
 &= \lambda^2 \left[(0 - 0 + 0 - 0) - \left(0 - 0 + 0 - \frac{6}{\lambda^4} \right) \right] - \frac{4}{\lambda^2} \\
 &= \frac{6}{\lambda^2} - \frac{4}{\lambda^2} = \frac{2}{\lambda^2}
 \end{aligned}$$

Example 4 : A continuous random variable X is defined by

$$f(x) = \begin{cases} \frac{1}{16}(3+x)^2, & \text{if } -3 \leq x < -1 \\ \frac{1}{16}(6-2x^2), & \text{if } -1 \leq x < 1 \\ \frac{1}{16}(3-x)^2, & \text{if } 1 \leq x \leq 3 \\ 0, & \text{elsewhere} \end{cases}$$

Verify that $f(x)$ is a density function and also find the mean of X .

[JNTU 2003S, (K) May 2013 (Set No. 3)]

(or) Show that the area under the curve above x -axis is unity. Also find the mean of the distribution.

Solution : $f(x)$ is clearly ≥ 0 for every x in $[-3, 3]$ and

$$\begin{aligned}
 \int_{-\infty}^{\infty} f(x) dx &= \int_{-\infty}^{-3} f(x) dx + \int_{-3}^{-1} f(x) dx + \int_{-1}^1 f(x) dx + \int_1^3 f(x) dx + \int_3^{\infty} f(x) dx \\
 &= \int_{-\infty}^{-3} 0 dx + \int_{-3}^{-1} \frac{1}{16} (3+x)^2 dx + \int_{-1}^1 \frac{1}{16} (6-2x^2) dx + \int_1^3 \frac{1}{16} (3-x)^2 dx + \int_3^{\infty} 0 dx \\
 &= \frac{1}{16} \left[\frac{(3+x)^3}{3} \right]_{-3}^{-1} + \frac{1}{16} \cdot 2 \int_0^1 (6-2x^2) dx + \frac{1}{16} \cdot \left[\frac{(3-x)^3}{-3} \right]_1^3
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{48} (8-0) + \frac{1}{8} \left(6x - \frac{2x^3}{3} \right)_0^1 - \frac{1}{48} (0-8) \\
 &= \frac{1}{6} + \frac{1}{8} \left(6 - \frac{2}{3} \right) + \frac{1}{6} = \frac{1}{3} + \frac{1}{4} \left(3 - \frac{1}{3} \right) = \frac{1}{3} + \frac{2}{3} = 1
 \end{aligned}$$

Hence the function $f(x)$ satisfies the requirements for a density function.

$$\text{Mean of } f(x) = \int_{-\infty}^{\infty} x f(x) dx$$

$$\begin{aligned}
 &= \frac{1}{16} \int_{-3}^{-1} x(3+x)^2 dx + \frac{1}{16} \int_{-1}^1 x(6-2x^2) dx + \frac{1}{16} \int_1^3 x(3-x)^2 dx \\
 &= \frac{1}{16} \int_{-3}^{-1} x(9+x^2+6x) dx + 0 + \frac{1}{16} \int_1^3 x(9-6x+x^2) dx
 \end{aligned}$$

[$\because$ the integrand of the second integral is odd function]

$$\begin{aligned}
 &= \frac{1}{16} \int_{-3}^{-1} (9x+6x^2+x^3) dx + \frac{1}{16} \int_1^3 (9x-6x^2+x^3) dx \\
 &= \frac{1}{16} \left(\frac{9x^2}{2} + \frac{6x^3}{3} + \frac{x^4}{4} \right)_{-3}^{-1} + \frac{1}{16} \left(\frac{9x^2}{2} - \frac{6x^3}{3} + \frac{x^4}{4} \right)_1^3 \\
 &= \frac{1}{16} \left[\left(\frac{9}{2} - 2 + \frac{1}{4} \right) - \left(\frac{81}{2} - 54 + \frac{81}{4} \right) \right] + \frac{1}{16} \left[\left(\frac{81}{2} - 54 + \frac{81}{4} \right) - \left(\frac{9}{2} - 2 + \frac{1}{4} \right) \right] \\
 &= 0
 \end{aligned}$$

Example 3 : Is the function defined by

$$f(x) = \begin{cases} 0, & x < 2 \\ \frac{1}{18}(2x+3), & 2 \leq x \leq 4 \\ 0, & x > 4 \end{cases}$$

a probability density function? Find the probability that a variate having $f(x)$ as density function will fall in the interval $2 \leq x \leq 3$.

[JNTU (K) Nov. 2009, May 2013 (Set No. 2)]

Solution : (i) For all points x in $-\infty \leq x \leq \infty$, $f(x) \geq 0$ and

$$\begin{aligned}
 \int_{-\infty}^{\infty} f(x) dx &= \int_{-\infty}^2 0 \cdot dx + \int_2^4 \frac{1}{18}(2x+3) dx + \int_4^{\infty} 0 \cdot dx \\
 &= \frac{1}{18} \int_2^4 (2x+3) dx = \frac{1}{18} \left[\frac{(2x+3)^2}{4} \right]_2^4 = \frac{1}{72} (121 - 49) = 1
 \end{aligned}$$

Hence $f(x)$ is a probability density function.

(ii) The probability that the density will fall in the interval $2 \leq x \leq 3$ is

$$\begin{aligned}
 P(2 \leq x \leq 3) &= \int_2^3 f(x) dx = \frac{1}{18} \int_2^3 (2x+3) dx \\
 &= \frac{1}{18} (x^2 + 3x)_2^3 = \frac{1}{18} (18 - 10) = \frac{8}{18} = \frac{4}{9}
 \end{aligned}$$

Example 8 : A random variable X gives measurements x between 0 and 1 with a probability

$$\text{function } f(x) = \begin{cases} 12x^3 - 21x^2 + 10x, & \text{for } 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

(i) Find $P\left(X \leq \frac{1}{2}\right)$ and $P\left(X > \frac{1}{2}\right)$

(ii) Find a number k such that $P(X \leq k) = \frac{1}{2}$

[JNTU 2003]

$$\begin{aligned}
 \text{Solution : (i) } P\left(X \leq \frac{1}{2}\right) &= \int_0^{1/2} f(x) dx = \int_0^{1/2} (12x^3 - 21x^2 + 10x) dx \\
 &= \left(12 \cdot \frac{x^4}{4} - 21 \cdot \frac{x^3}{3} + 10 \cdot \frac{x^2}{2} \right)_0^{1/2} = (3x^4 - 7x^3 + 5x^2)_0^{1/2} \\
 &= \left(\frac{3}{16} - \frac{7}{8} + \frac{5}{4} \right) - 0 = \frac{1}{16} (3 - 14 + 20) = \frac{9}{16}
 \end{aligned}$$

$$\text{Now } P\left(X > \frac{1}{2}\right) = 1 - P\left(X \leq \frac{1}{2}\right) = 1 - \frac{9}{16} = \frac{7}{16}$$

(ii) Given $P(X \leq k) = \frac{1}{2}$

$$\text{i.e., } \int_0^k f(x) dx = \frac{1}{2} \text{ i.e., } \int_0^k (12x^3 - 21x^2 + 10x) dx = \frac{1}{2}$$

$$\text{i.e., } (3x^4 - 7x^3 + 5x^2)_0^k = \frac{1}{2} \text{ i.e., } 3k^4 - 7k^3 + 5k^2 = \frac{1}{2} \text{ or } 6k^4 - 14k^3 + 10k^2 - 1 = 0$$

$$\therefore k = 0.452$$

Example 9 : A continuous random variable X has the distribution function

$$F(x) = \begin{cases} 0, & \text{if } x \leq 1 \\ k(x-1)^4, & \text{if } 1 < x \leq 3 \\ 1, & \text{if } x > 3 \end{cases}$$

Determine (i) $f(x)$ (ii) k (iii) Mean [JNTU 2004S, 2007S, (A) Apr. 2012 (Set No. 2)]

Solution : (i) We know that $f(x) = \frac{d}{dx} [F(x)]$

$$\therefore f(x) = \begin{cases} 0, & \text{if } x \leq 1 \\ 4k(x-1)^3, & \text{if } 1 < x \leq 3 \\ 0, & \text{if } x > 3 \end{cases}$$

(ii) Since total probability is unity, we have $\int_1^3 f(x) dx = 1$ i.e., $4k \int_1^3 (x-1)^3 dx = 1$

$$\text{i.e., } 4k \left[\frac{(x-1)^4}{4} \right]_1^3 = 1 \quad \text{i.e., } k(16 - 0) = 1 \quad \text{or } k = \frac{1}{16}$$

$$\text{Hence } f(x) = \begin{cases} 0, & \text{if } x \leq 1 \\ \frac{1}{4}(x-1)^3, & \text{if } 1 < x \leq 3 \\ 0, & \text{if } x > 3 \end{cases}$$

(iii) Mean of X , $\mu = \int_{-\infty}^{\infty} x f(x) dx = \int_{-\infty}^1 x f(x) dx + \int_1^3 x f(x) dx + \int_3^{\infty} x f(x) dx$

$$= 0 + \int_1^3 x \cdot \frac{1}{4}(x-1)^3 dx + 0 = \frac{1}{4} \int_1^3 x(x-1)^3 dx$$

$$= \frac{1}{4} \int_0^2 (t+1)t^3 dt \quad (\text{Putting } x-1 = t)$$

$$= \frac{1}{4} \int_0^2 (t^4 + t^3) dt = \frac{1}{4} \left(\frac{t^5}{5} + \frac{t^4}{4} \right)_0^2$$

$$= \frac{1}{4} \left(\frac{2^5}{5} + \frac{2^4}{4} \right) = \frac{2^4}{4} \left(\frac{2}{5} + \frac{1}{4} \right) = 4 \left(\frac{13}{20} \right) = \frac{13}{5} = 2.6$$

Example 10 : If X is a continuous random variable and $Y = aX + b$, prove that $E(Y) = a E(X) + b$ and $V(Y) = a^2 V(X)$, where V stands for variance and a, b are constants.

[JNTU 2000, (A) Dec. 2009 (Set No. 2), (H), (A) Nov. 2010, Nov. 2011, Apr. 2012 (Set No. 1)]

Solution : (i) By definition,

$$\begin{aligned} E(Y) &= E(aX + b) = \int_{-\infty}^{\infty} (ax + b) f(x) dx \quad \left[\because E[\phi(x)] = \int_{-\infty}^{\infty} \phi(x) f(x) dx \right] \\ &= a \int_{-\infty}^{\infty} x f(x) dx + b \int_{-\infty}^{\infty} f(x) dx \\ &= a E(X) + b (1) \quad [\text{since total probability is unity}] \\ &= a E(X) + b \end{aligned}$$

(ii) From (i), we have

$$E(Y) = a E(X) + b \quad \dots (1)$$

$$\text{where } Y = aX + b \quad \dots (2)$$

$$(2) - (1) \text{ gives } Y - E(Y) = a [X - E(X)]$$

$$\text{Squaring, } [Y - E(Y)]^2 = a^2 [X - E(X)]^2$$

Taking Expectation of both sides, we get

$$E\{[Y - E(Y)]^2\} = a^2 E\{[X - E(X)]^2\}$$

$$\therefore V(Y) = a^2 V(X)$$

Example 11 : If X is a continuous random variable and k is a constant, then prove that

$$(i) \text{Var}(X + k) = \text{Var}(X) \quad (ii) \text{Var}(kX) = k^2 \text{Var}(X) \quad [\text{JNTU 2006, 2007 (Set No. 1)}]$$

Solution : By definition,

$$\begin{aligned} \text{Var}(X) &= \int_{-\infty}^{\infty} x^2 f(x) dx - \left[\int_{-\infty}^{\infty} x f(x) dx \right]^2 \\ (i) \quad \text{Var}(X + K) &= \int_{-\infty}^{\infty} (x + k)^2 f(x) dx - \left[\int_{-\infty}^{\infty} (x + k) f(x) dx \right]^2 \\ &= \int_{-\infty}^{\infty} (x^2 + 2kx + k^2) f(x) dx - \left[\int_{-\infty}^{\infty} x f(x) dx + k \int_{-\infty}^{\infty} f(x) dx \right]^2 \\ &= \int_{-\infty}^{\infty} x^2 f(x) dx + 2k \int_{-\infty}^{\infty} x f(x) dx + k^2 \int_{-\infty}^{\infty} f(x) dx - \left[\int_{-\infty}^{\infty} x f(x) dx + k \right]^2 \\ &\quad \left[\because \int_{-\infty}^{\infty} f(x) dx = 1 \right] \end{aligned}$$

$$\begin{aligned}
&= E(X^2) + 2k E(X) + k^2 - [E(X) + k]^2 \\
&= E(X^2) + 2k E(X) + k^2 - [E(X)]^2 - 2k E(X) - k^2 \\
&= E(X^2) - [E(X)]^2 \\
&= \text{Var}(X)
\end{aligned}$$

$$\begin{aligned}
(ii) \text{ Var}(kX) &= \int_{-\infty}^{\infty} k^2 x^2 f(x) dx - \left[\int_{-\infty}^{\infty} kx f(x) dx \right]^2 \\
&= k^2 \int_{-\infty}^{\infty} x^2 f(x) dx - k^2 \left[\int_{-\infty}^{\infty} f(x) dx \right]^2 \\
&= k^2 [E(X)^2 - \{E(X)\}^2] = k^2 \text{Var}(X)
\end{aligned}$$

Example 12 : For the continuous random variable X whose probability density function is given by $f(x) = \begin{cases} cx(2-x), & \text{if } 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$ where c is a constant.

Find c , mean and variance of X .

[JNTU 2003, 2004 (Set No.3)]

(OR) The frequency function of a continuous random variable X is given by $f(x) = y_0 x(2-x), 0 \leq x \leq 2$. Find the value of y_0 , mean and variance of X .

[JNTU (A) Dec. 2009 (Set No.3)]

Solution: (i) Since the total probability is unity, we have

$$\int_{-\infty}^{\infty} f(x) dx = 1. \text{ So } \int_0^2 f(x) dx = 1$$

$$\text{i.e., } \int_0^2 cx(2-x) dx = 1 \text{ i.e., } c \left(x^2 - \frac{x^3}{3} \right)_0^2 = 1$$

$$\text{i.e., } c \left(4 - \frac{8}{3} \right) = 1 \text{ or } \frac{4c}{3} = 1 \text{ or } c = \frac{3}{4}$$

$$\therefore f(x) = \begin{cases} \frac{3x}{4}(2-x), & \text{if } 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

$$(ii) \text{ Mean of } X = E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

$$\begin{aligned}
 \text{i.e., } \mu &= \int_0^2 x \cdot \frac{3x}{4} (2-x) dx = \frac{3}{4} \int_0^2 (2x^2 - x^3) dx \\
 &= \frac{3}{4} \left(\frac{2x^3}{3} - \frac{x^4}{4} \right)_0^2 = \frac{3}{4} \left(\frac{2^4}{3} - \frac{2^4}{4} \right) = \frac{3}{4} (2^4) \left(\frac{1}{3} - \frac{1}{4} \right) = \frac{12}{12} = 1
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii) Variance of } X &= V(X) = \int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2 = \int_0^2 x^2 \cdot \frac{3x}{4} (2-x) dx - (1)^2 \\
 &= \frac{3}{4} \int_0^2 (2x^3 - x^4) dx - 1 = \frac{3}{4} \left(2 \cdot \frac{x^4}{4} - \frac{x^5}{5} \right)_0^2 - 1 \\
 &= \frac{3}{4} \left[\frac{32}{4} - \frac{32}{5} \right] - 1 = 24 \left(\frac{1}{4} - \frac{1}{5} \right) - 1 = \frac{6}{5} - 1 = \frac{1}{5}
 \end{aligned}$$

Example 11 : For the continuous probability function $f(x) = k x^2 e^{-x}$ when $x \geq 0$, find

(i) k (ii) Mean (iii) Variance

[JNTU 2005, 2005S, 2007, (K) Dec. 2013 (Set No. 4), (A) June 2016]

Solution : (i) We have $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\therefore \int_0^{\infty} kx^2 e^{-x} dx = 1 \quad (\because x \geq 0)$$

$$\text{i.e., } k \left[x^2 (-e^{-x}) - 2x (e^{-x}) + 2(-e^{-x}) \right]_0^{\infty} = 1, \text{ using Bernoulli's Rule}$$

$$\text{i.e., } k \left\{ -e^{-x} (x^2 + 2x + 2) \right\}_0^{\infty} = 1$$

$$\text{i.e., } k (0 + 2) = 1 \text{ or } k = \frac{1}{2}$$

$$\begin{aligned}
 \text{(ii) Mean} &= E(x) = \int_{-\infty}^{\infty} x f(x) dx = \int_0^{\infty} kx^3 e^{-x} dx \\
 &= k \left[x^3 (-e^{-x}) - 3x^2 (e^{-x}) + 6x (-e^{-x}) - 6e^{-x} \right]_0^{\infty} \\
 &= k \left[-e^{-x} (x^3 + 3x^2 + 6x + 6) \right]_0^{\infty} = k [0 + 6] = 6k \\
 \therefore \mu &= 6 \left(\frac{1}{2} \right) = 3 \quad \left(\because k = \frac{1}{2} \right)
 \end{aligned}$$

$$\begin{aligned}
\text{(iii) Variance} &= E(x^2) - [E(x)]^2 \\
&= \int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2 = \int_0^{\infty} x^2 \cdot kx^2 e^{-x} dx - (3)^2 \\
&= k \int_0^{\infty} x^4 e^{-x} dx - 9 \\
&= k \left[x^4(-e^{-x}) - 4x^3(e^{-x}) + 12x^2(-e^{-x}) - 24x(-e^{-x}) + 24e^{-x} \right]_0^{\infty} - 9 \\
&= \frac{1}{2} \left[-e^{-x}(x^4 + 4x^3 + 12x^2 + 24x + 24) \right]_0^{\infty} - 9 \\
&= \frac{1}{2} [0 + 24] - 9 = 12 - 9 = 3
\end{aligned}$$

Example 14 : The trouble shooting capability of an IC chip in a circuit is a random variable X whose distribution function is given by

$$F(X) = \begin{cases} 0, & \text{for } x \leq 3 \\ 1 - \frac{9}{x^2}, & \text{for } x > 3 \end{cases}$$

where x denote the number of years.

Find the probability that the IC chip will work properly

- (i) Less than 8 years (ii) Beyond 8 years
 (iii) Between 5 to 7 years (iv) Anywhere from 2 to 5 years

[JNTU (K) Nov. 2009 (Set No.4)]

Solution : We have $F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$

$$\therefore F(X) = \int_0^x f(t) dt \quad (\because x > 0) = \begin{cases} 0, & \text{if } x \leq 3 \\ 1 - \frac{9}{x^2}, & \text{if } x > 3 \end{cases}$$

$$(i) \quad P(X \leq 8) = \int_0^8 f(t) dt = 1 - \frac{9}{8^2} = 0.8594$$

$$(ii) \quad P(X > 8) = 1 - P(X \leq 8) = 1 - 0.8594 = 0.1406$$

$$\begin{aligned}
(iii) \quad P(5 \leq x \leq 7) &= F(7) - F(5) = \left(1 - \frac{9}{7^2}\right) - \left(1 - \frac{9}{5^2}\right) \\
&= 9 \left(\frac{1}{25} - \frac{1}{49}\right) = \frac{24 \times 9}{25 \times 49} = 0.1763
\end{aligned}$$

$$(iv) \quad P(2 \leq x \leq 5) = F(5) - F(2) = \left(1 - \frac{9}{5^2}\right) - 0 = \frac{16}{25} = 0.64$$

Example 11 : If the probability density function of a random variable X is given by

$$f(x) = \begin{cases} 2k x e^{-x^2}, & \text{for } x > 0 \\ 0, & \text{for } x \leq 0 \end{cases}$$

Determine (i) k (ii) the distribution function for X .

Solution: (i) We have $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\text{i.e., } \int_{-\infty}^0 f(x) dx + \int_0^{\infty} f(x) dx = 1 \quad \text{i.e., } 0 + \int_0^{\infty} 2k x e^{-x^2} dx = 1$$

$$\text{i.e., } k \int_0^{\infty} 2x e^{-x^2} dx = 1 \quad \text{i.e., } k \int_0^{\infty} e^{-t} dt = 1 \quad (\text{Putting } x^2 = t)$$

$$\text{or } k \left(-e^{-t} \right)_0^{\infty} = 1 \quad \text{or } -k(0 - 1) = 1$$

$$\therefore k = 1$$

(ii) The distribution function is

$$F(X) = P(X \leq x) = \int_{-\infty}^x f(t) dt = 0 \quad \text{if } x \leq 0$$

$$\text{and } F(X) = \int_{-\infty}^0 f(t) dt + \int_0^x f(t) dt, \quad \text{if } x > 0$$

$$= 0 + \int_0^x 2t e^{-t^2} dt = \int_0^{x^2} e^{-u} du \quad (\text{Putting } t^2 = u)$$

$$= - \left(e^{-u} \right)_0^{x^2} = - (e^{-x^2} - 1) = 1 - e^{-x^2}$$

$$\therefore F(x) = \begin{cases} 1 - e^{-x^2}, & \text{if } x > 0 \\ 0, & \text{otherwise} \end{cases}$$

Example 12 : Suppose a continuous random variable X has the probability density $f(x) = K(1 - x^2)$ for $0 < x < 1$, and $f(x) = 0$ otherwise. Find (i) K (ii) Mean (iii) Variance
[JNTU 2007S(Set No.1)]

Solution: (i) Since the total probability is unity, we have

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\text{i.e., } \int_{-\infty}^0 f(x) dx + \int_0^1 f(x) dx + \int_1^{\infty} f(x) dx = 1$$

$$\text{i.e., } 0 + \int_0^1 K(1-x^2) dx + 0 = 1$$

$$\text{i.e., } K \left(x - \frac{x^3}{3} \right)_0^1 = 1 \text{ or } K \left(1 - \frac{1}{3} \right) = 1$$

$$\therefore K = \frac{3}{2}$$

$$(ii) \text{ Mean of } X = E(X) = \int_{-\infty}^{\infty} x.f(x) dx = \int_0^1 x.f(x) dx$$

$$= \int_0^1 x.K(1-x^2) dx = K \int_0^1 (x-x^3) dx$$

$$\text{i.e., } \mu = K \left(\frac{x^2}{2} - \frac{x^4}{4} \right)_0^1 = K \left(\frac{1}{2} - \frac{1}{4} \right) = \frac{K}{4} = \left(\frac{3}{2} \right) \left(\frac{1}{4} \right) = \frac{3}{8} \left[\because K = \frac{3}{2} \right]$$

$$(iii) \text{ Variance of } X = E(X^2) - [E(X)]^2 = \int_0^1 x^2 f(x) dx - \mu^2$$

$$= \int_0^1 x^2 \cdot K(1-x^2) dx - \left(\frac{3}{8} \right)^2$$

$$= K \int_0^1 (x^2 - x^4) dx - \frac{9}{64} = K \left(\frac{x^3}{3} - \frac{x^5}{5} \right)_0^1 - \frac{9}{64}$$

$$\text{i.e., } \sigma^2 = K \left(\frac{1}{3} - \frac{1}{5} \right) - \frac{9}{64} = \frac{2K}{15} - \frac{9}{64} = \frac{2}{15} \left(\frac{3}{2} \right) - \frac{9}{64}$$

$$= \frac{1}{5} - \frac{9}{64} = \frac{19}{320} = 0.06$$

Example 11 : The daily consumption of electric power (in millions of kW-hours) is a random variable having the probability density function (p.d.f)

$$f(x) = \begin{cases} \frac{1}{9} x e^{-x/3}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

If the total production is 12 million KW-hours, determine the probability that there is power cut (shortage) on any given day. **[JNTU Jan. 2008, (K) May 2010S]**

Solution : Probability that the power consumed is between 0 to 12 is

$$P(0 \leq x \leq 12 \text{ million KW-hours}) = \int_0^{12} f(x) dx = \frac{1}{9} \int_0^{12} x e^{-x/3} dx$$

$$\begin{aligned}
&= \frac{1}{9} \left[x \frac{e^{-x/3}}{(-1/3)} - 1 \cdot \frac{e^{-x/3}}{1/9} \right] \\
&= \frac{1}{9} [-36 e^{-4} - 9 e^{-4} + 9] \\
&= \frac{1}{9} (9 - 45 e^{-4}) = 1 - 5 e^{-4}
\end{aligned}$$

Power supply is inadequate if daily consumption exceeds 12 million kW, i.e.,

$$\begin{aligned}
P(x > 12) &= 1 - P(0 \leq x \leq 12) = 1 - (1 - 5 e^{-4}) \\
&= 5 e^{-4} = 0.0915781
\end{aligned}$$

Example 14 : If probability density function is

$$f(x) = \begin{cases} Kx^3 & \text{in } 0 \leq x \leq 3 \\ 0 & \text{elsewhere} \end{cases}$$

Find the value of K and find the probability between $x = 1/2$ and $x = 3/2$.

[JNTU Nov. 2008 (Set No. 1)]

Solution : Given $f(x) = \begin{cases} Kx^3 & \text{in } 0 \leq x \leq 3 \\ 0 & \text{elsewhere} \end{cases}$

$$\text{We have } \int_{-\infty}^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_0^3 Kx^3 dx = 1 \Rightarrow K \left(\frac{x^4}{4} \right)_0^3 = 1 \Rightarrow \frac{81K}{4} = 1 \therefore K = \frac{4}{81}$$

$$\begin{aligned}
\text{Now } P\left(\frac{1}{2} < X < \frac{3}{2}\right) &= \int_{1/2}^{3/2} f(x) dx = \int_{1/2}^{3/2} \frac{4}{81} x^3 dx \\
&= \frac{4}{81} \left(\frac{x^4}{4} \right)_{1/2}^{3/2} = \frac{1}{81} \left[\left(\frac{3}{2} \right)^4 - \left(\frac{1}{2} \right)^4 \right] \\
&= \frac{1}{81 \times 16} [81 - 1] = \frac{80}{81 \times 16} = \frac{5}{81}
\end{aligned}$$

Example 12 : Find the constant K such that

$$f(x) = \begin{cases} Kx^2, & \text{if } 0 < x < 3 \\ 0, & \text{otherwise} \end{cases}$$

is a probability function.

(i) Find the distribution function $F(x)$

(ii) $P(1 < X \leq 2)$

[JNTU Nov. 2008 (Set No. 2)]

Solution : (i) Since the total probability is unity,

$$\int_{-\infty}^{\infty} f(x) dx = 1 \Rightarrow K \int_0^3 x^2 dx = 1 \Rightarrow K \left(\frac{x^3}{3} \right)_0^3 = 1 \Rightarrow K = \frac{1}{9}$$

The distribution function is

$$F(X) = P(X \leq x) = \int_{-\infty}^x f(t) dt = 0 \text{ if } x \leq 0$$

$$\begin{aligned} \text{and } F(X) &= \int_{-\infty}^0 f(t) dt + \int_0^x f(t) dt \text{ if } x > 0 \\ &= 0 + \int_0^x \frac{1}{9} t^2 dt = \frac{1}{9} \left(\frac{t^3}{3} \right)_0^x = \frac{1}{27} x^3 \end{aligned}$$

$$\therefore F(x) = \begin{cases} \frac{1}{27} x^3, & \text{if } x > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{aligned} \text{(ii) } P(1 < X \leq 2) &= \int_1^2 f(x) dx = \int_1^2 Kx^2 dx \\ &= \frac{1}{9} \int_1^2 x^2 dx \quad \left[\because K = \frac{1}{9} \right] \\ &= \frac{1}{9} \left(\frac{x^3}{3} \right)_1^2 = \frac{1}{27} (8 - 1) = \frac{7}{27} \end{aligned}$$

Example 20 : If the probability density function of X is given by

$$f(x) = \begin{cases} x/2, & \text{for } 0 < x \leq 1 \\ 1/2, & \text{for } 1 < x \leq 2 \\ (3-x)/2, & \text{for } 2 < x < 3 \\ 0, & \text{elsewhere} \end{cases}$$

Find the expected value of $f(x) = x^2 - 5x + 3$

[JNTU Nov. 2008 (Set No. 2)]

Solution : The expectation of any function $\phi(x)$ is given by

$$E[\phi(x)] = \int_{-\infty}^{\infty} \phi(x) f(x) dx$$

$$\therefore E(x^2 - 5x + 3) = \int_{-\infty}^{\infty} (x^2 - 5x + 3) f(x) dx$$

$$\begin{aligned}
&= \int_0^1 (x^2 - 5x + 3) \cdot \frac{x}{2} dx + \int_1^2 (x^2 - 5x + 3) \cdot \frac{1}{2} dx + \int_2^3 (x^2 - 5x + 3) \cdot \left(\frac{3-x}{2}\right) dx \\
&= \frac{1}{2} \int_0^1 (x^3 - 5x^2 + 3x) dx + \frac{1}{2} \int_1^2 (x^2 - 5x + 3) dx + \frac{1}{2} \int_2^3 (-x^3 + 8x^2 - 18x + 9) dx \\
&= \frac{1}{2} \left(\frac{x^4}{4} - \frac{5x^3}{3} + \frac{3x^2}{2} \right)_0^1 + \frac{1}{2} \left(\frac{x^3}{3} - \frac{5x^2}{2} + 3x \right)_1^2 + \frac{1}{2} \left(-\frac{x^4}{4} + \frac{8x^3}{3} - \frac{18x^2}{2} + 9x \right)_2^3 \\
&= \frac{1}{24} - \frac{13}{12} - \frac{19}{24} = \frac{-44}{24} = \frac{-11}{6}
\end{aligned}$$

Example 21 : Let the continuous random variable X have the probability density function,

$$f(x) = \begin{cases} 2/x^3, & \text{if } 1 < x < \infty \\ 0, & \text{otherwise} \end{cases}$$

Find $F(x)$.

[JNTU Apr. 2009 (Set No. 3)]

Solution : We are given

$$f(x) = \begin{cases} 2/x^3, & \text{if } 1 < x < \infty \\ 0, & \text{otherwise} \end{cases}$$

$$\text{i.e., } f(x) = \begin{cases} 2/x^3, & \text{if } x > 1 \\ 0, & \text{if } x \leq 1 \end{cases}$$

$$\text{We have } F(x) = \int_{-\infty}^x f(x) dx$$

$$\begin{aligned}
\text{If } x > 1, \quad F(x) &= \int_1^x f(x) dx \\
&= \int_1^x \frac{2}{x^3} dx = 2 \left(\frac{x^{-2}}{-2} \right)_1^x = - \left(\frac{1}{x^2} \right)_1^x \\
&= - \left(\frac{1}{x^2} - 1 \right) = 1 - \frac{1}{x^2}
\end{aligned}$$

If $x < 1$, $F(x) = 0$

$$\therefore F(x) = \begin{cases} 0, & \text{if } x \leq 1 \\ 1 - \frac{1}{x^2}, & \text{if } 1 < x < \infty \end{cases}$$

Example 21 : The probability density function is

$$y = \begin{cases} k(3x^2 - 1), & -1 \leq x \leq 2 \\ 0, & \text{else where} \end{cases}$$

Find the value of k and find $P(-1 \leq x \leq 0)$.

[JNTU Apr. 2009 (Set No. 4)]

Solution : Since $\int_{-\infty}^{\infty} f(x)dx = 1$, we have $\int_{-\infty}^{-1} f(x)dx + \int_{-1}^2 f(x)dx + \int_2^{\infty} f(x)dx = 1$

$$\text{i.e., } \int_{-\infty}^{-1} 0 \cdot dx + \int_{-1}^2 K(3x^2 - 1)dx + \int_2^{\infty} 0 \cdot dx = 1$$

$$\text{i.e., } K \left(3 \cdot \frac{x^3}{3} - x \right)_{-1}^2 = 1 \quad \text{i.e., } K(x^3 - x)_{-1}^2 = 1$$

$$\text{i.e., } K[(8-2) - (-1+1)] = 1 \text{ or } 6K = 1. \quad \therefore K = \frac{1}{6}.$$

$$\text{Thus } y = \begin{cases} \frac{1}{6}(3x^2 - 1), & -1 \leq x \leq 2 \\ 0, & \text{else where} \end{cases}$$

$$\begin{aligned} \text{Hence } P(-1 \leq x \leq 0) &= \int_{-1}^0 f(x)dx = \frac{1}{6} \int_{-1}^0 (3x^2 - 1)dx = \frac{1}{6} \left[3 \cdot \frac{x^3}{3} - x \right]_{-1}^0 \\ &= \frac{1}{6} (x^3 - x)_{-1}^0 = \frac{1}{6} [(0-0) - (-1+1)] = 0 \end{aligned}$$

Example 23 : The density function of a random variable X is $f(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$.

Find $E(X)$, $E(X^2)$, $\text{Var}(X)$.

[JNTU (K) Nov. 2009 (Set No. 1)]

Solution : Given $f(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$

$$\begin{aligned} (i) \quad E(X) &= \int_{-\infty}^{\infty} x \cdot f(x)dx = \int_{-\infty}^0 x(0)dx + \int_0^{\infty} xe^{-x}dx = \int_0^{\infty} xe^{-x}dx \\ &= \left[-(x+1)e^{-x} \right]_0^{\infty} = (-1) \lim_{x \rightarrow \infty} \frac{(x+1)}{e^x} + (0+1) = 0 + 1 = 1. \end{aligned}$$

$$(ii) \quad E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)dx = \int_0^{\infty} x^2 e^{-x}dx$$

$$\begin{aligned}
 &= \left[x^2(-e^{-x}) - 2x(e^{-x}) + 2(-e^{-x}) \right]_0^\infty = \left[-e^{-x}(x^2 + 2x + 2) \right]_0^\infty \\
 &= (-1) \lim_{x \rightarrow \infty} \frac{x^2 + 2x + 2}{e^x} + 2 = 2
 \end{aligned}$$

$$(iii) \quad \text{Var}(X) = E(X^2) - \{E(X)\}^2 = 2 - (1)^2 = 2 - 1 = 1$$

Example 24 : Is the function defined as follows a density function

$$f(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

If so determine the probability that the variate having this density will fall in the interval (1,2)? Find the cumulative probability F(2)? [JNTU (K) Nov. 2009 (Set No. 2)]

Solution : Given $f(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$

(i) Clearly $f(x) \geq 0, \forall x$ in $(1,2)$ and

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^0 0 \cdot dx + \int_0^{\infty} e^{-x} dx = \int_0^{\infty} e^{-x} dx = -(e^{-x})_0^{\infty} = -(0 - 1) = 1$$

Hence the function $f(x)$ is a density function.

$$\begin{aligned}
 (ii) \quad \text{Required probability} &= P(1 \leq x \leq 2) = \int_1^2 f(x) dx = \int_1^2 e^{-x} dx = -(e^{-x})_1^2 \\
 &= -(e^{-2} - e^{-1}) = e^{-1} - e^{-2} = 0.368 - 0.135 = 0.233.
 \end{aligned}$$

(iii) Cumulative probability function

$$\begin{aligned}
 F(2) &= \int_{-\infty}^2 f(x) dx = \int_{-\infty}^0 0 \cdot dx + \int_0^2 e^{-x} dx \\
 &= -(e^{-x})_0^2 = -(e^{-2} - 1) = 1 - e^{-2} \\
 &= 1 - 0.135 = 0.865.
 \end{aligned}$$

Example 23 : The cumulative distribution function for a continuous random variable X

$$\text{is } F(x) = \begin{cases} 1 - e^{-2x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Find (i) the density function $f(x)$, (ii) mean and (iii) variance of the density function.

[JNTU (K) May 2010 (Set No. 2), (A) Dec. 2017]

Solution : (i) The density function $f(x)$ is given by $f(x) = \frac{d}{dx}[F(x)]$

$$\therefore f(x) = \begin{cases} \frac{1}{2}e^{-2x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$(ii) \text{ Mean} = E(x) = \int_{-\infty}^{\infty} x f(x) dx = \int_{-\infty}^0 x f(x) dx + \int_0^{\infty} x f(x) dx$$

$$\begin{aligned} i.e. \mu &= 0 + \int_0^{\infty} x \cdot \frac{1}{2} e^{-2x} dx = \frac{1}{2} \int_0^{\infty} x e^{-2x} dx \\ &= \frac{1}{2} \left[x \left(\frac{e^{-2x}}{-2} \right) - 1 \cdot \left(\frac{e^{-2x}}{4} \right) \right]_0^{\infty} = -\frac{1}{8} [e^{-2x} (2x + 1)]_0^{\infty} \\ &= -\frac{1}{8} [0 - (0 + 1)] = \frac{1}{8} \end{aligned}$$

$$\begin{aligned} (iii) \text{ Variance} &= \int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2 = \int_{-\infty}^0 x^2 f(x) dx + \int_0^{\infty} x^2 f(x) dx - \mu^2 \\ &= 0 + \int_0^{\infty} x^2 \cdot \frac{1}{2} e^{-2x} dx - \left(\frac{1}{8} \right)^2 = \frac{1}{2} \int_0^{\infty} x^2 e^{-2x} dx - \frac{1}{64} \\ &= \frac{1}{2} \left[x^2 \left(\frac{e^{-2x}}{-2} \right) - 2x \left(\frac{e^{-2x}}{4} \right) + 2 \left(\frac{e^{-2x}}{-8} \right) \right]_0^{\infty} - \frac{1}{64} \\ &= -\frac{1}{2} \left[e^{-2x} \left(\frac{1}{2} x^2 - \frac{1}{2} x + \frac{1}{4} \right) \right]_0^{\infty} - \frac{1}{64} \\ &= -\frac{1}{8} [e^{-2x} (2x^2 - 2x + 1)]_0^{\infty} - \frac{1}{64} \\ &= -\frac{1}{8} [0 - (0 - 0 + 1)] - \frac{1}{64} = \frac{1}{8} - \frac{1}{64} = \frac{8-1}{64} = \frac{7}{64} \end{aligned}$$

Example 23 : If X is a continuous random variable with p.d.f $f(x) = \begin{cases} x^2, & 0 \leq x \leq 1, \\ 0, & \text{elsewhere} \end{cases}$

If $p(a \leq x \leq 1) = \frac{19}{81}$, find the value of 'a'.

[JNTU (K) May 2010 (Set No. 3)]

Solution : Given $p(a \leq x \leq 1) = \frac{19}{81} \Rightarrow \int_a^1 f(x) dx = \frac{19}{81} \Rightarrow \int_a^1 x^2 dx = \frac{19}{81}$

$$\Rightarrow \left(\frac{x^3}{3} \right)_a^1 = \frac{19}{81} \Rightarrow \frac{1-a}{3} = \frac{19}{81}$$

$$\Rightarrow 1 - a = \frac{19}{27}.$$

$$\therefore a = 1 + \frac{19}{27} = \frac{46}{27}$$

Example 23 : If a random variable has the probability density function

$$f(x) = \begin{cases} k(x^2 - 1), & -1 \leq x \leq 3 \\ 0, & \text{elsewhere} \end{cases}$$

Find the value of 'k' and $p\left(\frac{1}{2} \leq x \leq \frac{5}{2}\right)$ [JNTU (K) May 2010 (Set No. 4)]

Solution : In order that $f(x)$ should be a probability density function

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$i.e., \int_{-\infty}^{-1} f(x) dx + \int_{-1}^3 f(x) dx + \int_3^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_{-1}^3 k(x^2 - 1) dx = 1 \Rightarrow k \left(\frac{x^3}{3} - x \right)_{-1}^3 = 1$$

$$\Rightarrow k \left[(9 + 1) - \left(-\frac{1}{3} + 1 \right) \right] = 1 \Rightarrow k \left(9 + \frac{1}{3} \right) = 1$$

$$\Rightarrow \frac{28}{3} k = 1. \quad \therefore k = \frac{3}{28}$$

$$\text{Now } p\left(\frac{1}{2} \leq x \leq \frac{5}{2}\right) = \int_{1/2}^{5/2} \frac{3}{28} (x^2 - 1) dx$$

$$= \frac{3}{28} \left(\frac{x^3}{3} - x \right)_{1/2}^{5/2} = \frac{3}{28} \left[\left(\frac{(5/2)^3}{3} - \frac{5}{2} \right) - \left(\frac{(1/2)^3}{3} - \frac{1}{2} \right) \right]$$

$$= \frac{3}{28} \left[\frac{65}{24} + \frac{11}{24} \right]$$

$$= \frac{3}{28} \times \frac{76}{24} = \frac{19}{56}$$

Example 28 : If X is the continuous random variable whose density function is

$$f(x) = \begin{cases} x & \text{if } 0 < x < 1 \\ 2 - x & \text{if } 1 \leq x < 2 \\ 0, & \text{elsewhere} \end{cases}$$

Find $E(25X^2 + 30X - 5)$.

[JNTU (H) Nov. 2010 (Set No. 2)]

Solution: Given $f(x) = \begin{cases} x & \text{if } 0 < x < 1 \\ 2 - x & \text{if } 1 \leq x < 2 \\ 0, & \text{elsewhere} \end{cases}$

$$\begin{aligned} \therefore E(X) &= \int_{-\infty}^{\infty} x f(x) dx \\ &= \int_0^1 x \cdot x dx + \int_1^2 x(2-x) dx = \int_0^1 x^2 dx + \int_1^2 (2x - x^2) dx \\ &= \left(\frac{x^3}{3} \right)_0^1 + \left(2 \cdot \frac{x^2}{2} - \frac{x^3}{3} \right)_1^2 = \frac{1}{3} + \left(x^2 - \frac{x^3}{3} \right)_1^2 \\ &= \frac{1}{3} + \left(4 - \frac{8}{3} \right) - \left(1 - \frac{1}{3} \right) = 4 + \frac{1}{3} - \frac{8}{3} - \frac{2}{3} \\ &= 4 - 1 = 3 \end{aligned}$$

$$\begin{aligned} E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^1 x^2 \cdot x dx + \int_1^2 x^2 (x-2) dx \\ &= \int_0^1 x^3 dx + \int_1^2 (x^3 - 2x^2) dx = \left(\frac{x^4}{4} \right)_0^1 + \left(\frac{x^4}{4} - 2 \cdot \frac{x^3}{3} \right)_1^2 \\ &= \frac{1}{4} + \left(4 - \frac{16}{3} \right) - \left(\frac{1}{4} - \frac{2}{3} \right) = -\frac{2}{3} \end{aligned}$$

$$\begin{aligned} \therefore E(25X^2 + 30X - 5) &= 25 \cdot E(X^2) + 30 \cdot E(X) - 5 \\ &= 25 \left(-\frac{2}{3} \right) + 30(3) - 5 = -\frac{50}{3} + 85 \\ &= \frac{-50 + 255}{3} = \frac{205}{3} = 68.33 \end{aligned}$$

Example 23 : Find the value of K and the distribution function $F(x)$ given the probability density function of a random variable X as :

$$f(x) = \frac{K}{x^2 + 1}, -\infty < x < \infty$$

[JNTU(K) March 2014 (Set No. 3)]

Solution : Since total probability is unity, we have $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\Rightarrow \int_{-\infty}^{\infty} \frac{K}{x^2 + 1} dx = 1 \Rightarrow K \int_{-\infty}^{\infty} \frac{1}{1 + x^2} dx = 1 \Rightarrow K (\tan^{-1} x)_{-\infty}^{\infty} = 1$$

$$\Rightarrow K (\tan^{-1} \infty - \tan^{-1}(-\infty)) = 1 \Rightarrow K \left[\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right] = 1 \Rightarrow K \pi = 1.$$

$$\therefore K = \frac{1}{\pi}.$$

By definition, the distribution function is given by

$$\begin{aligned} F(x) &= \int_{-\infty}^x f(x) dx = \int_{-\infty}^x \frac{K}{1+x^2} dx = \frac{1}{\pi} \int_{-\infty}^x \frac{1}{1+x^2} dx \quad \left[\because K = \frac{1}{\pi} \right] \\ &= \frac{1}{\pi} (\tan^{-1} x)_{-\infty}^x = \frac{1}{\pi} [\tan^{-1} x - \tan^{-1}(-\infty)] = \frac{1}{\pi} \left(\tan^{-1} x + \frac{\pi}{2} \right). \end{aligned}$$

Example 20 : The length of time (in minutes) that a certain lady speaks on the telephone is found to be random phenomenon, with a probability function specified by the function

$$f(x) = \begin{cases} A e^{-x/5}, & \text{for } x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

(i) Find the value of A that makes $f(x)$ a probability density function.

(ii) What is the probability that the number of minutes that she will take over the phone is more than 10 minutes? [JNTU(K) Dec. 2013 (Set No. 2)]

Solution : (i) In order that $f(x)$ should be a probability density function,

$$\int_{-\infty}^{\infty} f(x) dx = 1 \quad \text{i.e.,} \quad \int_0^{\infty} f(x) dx = 1 \quad \text{or} \quad \int_0^{\infty} A e^{-x/5} dx = 1 \Rightarrow A \left(\frac{e^{-x/5}}{-1/5} \right)_0^{\infty} = 1$$

$$\Rightarrow -5A (e^{-\infty} - e^{-0}) = 1 \Rightarrow -5A (0 - 1) = 1 \Rightarrow 5A = 1 \quad \text{or} \quad A = \frac{1}{5}.$$

$$\begin{aligned} \text{(ii)} \quad P(X > 10) &= \int_{10}^{\infty} f(x) dx = \int_{10}^{\infty} A e^{-x/5} dx = A \left(\frac{e^{-x/5}}{-1/5} \right)_{10}^{\infty} \\ &= -5A (e^{-\infty} - e^{-2}) = -5 \left(\frac{1}{5} \right) (0 - e^{-2}) = \frac{1}{e^2}. \end{aligned}$$

Example 21 : The diameter of an electric cable say X is assumed to be a continuous random variable with p.d.f of $f(x) = kx(1-x^2)$ in $0 \leq x \leq 1 = 0$ elsewhere find the value of k and $P(0 \leq x \leq (1/2)), P(x \geq (1/4))$. [JNTU(A) Apr. 2012 (Set No. 3)]

Solution : The probability density function of a continuous random variable is given by

$$f(x) = \begin{cases} kx(1-x^2), & 0 \leq x \leq 1 \\ 0, & \text{else where} \end{cases}$$

We know that $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\text{i.e., } \int_{-\infty}^0 f(x) dx + \int_0^1 f(x) dx + \int_1^{\infty} f(x) dx = 1$$

$$\text{i.e., } 0 + \int_0^1 kx(1-x^2) dx + 0 = 1$$

$$\text{i.e., } k \int_0^1 (x-x^3) dx = 1 \text{ or } k \left(\frac{x^2}{2} - \frac{x^4}{4} \right)_0^1 = 1$$

$$\text{or } k \left(\frac{1}{2} - \frac{1}{4} \right) = 1 \text{ or } \frac{k}{4} = 1 \therefore k = 4$$

Thus the density function is $f(x) = \begin{cases} 4x(1-x^2), & 0 \leq x \leq 1 \\ 0, & \text{else where} \end{cases}$

$$\begin{aligned} (i) \quad P\left(0 \leq x \leq \frac{1}{2}\right) &= \int_0^{1/2} f(x) dx = \int_0^{1/2} 4x(1-x^2) dx \\ &= 4 \int_0^{1/2} (x-x^3) dx = 4 \left(\frac{x^2}{2} - \frac{x^4}{4} \right)_0^{1/2} \\ &= 4 \left(\frac{1}{8} - \frac{1}{64} \right) = 4 \left(\frac{8-1}{64} \right) = \frac{7}{16} \end{aligned}$$

$$(ii) \quad P(x \geq 4) = \int_4^{\infty} f(x) dx = 0$$

Example 23 : A continuous random variable x has a probability density function $f(x) = 3x^2, 0 \leq x \leq 1$, Find a and b such that (i) $P\{x \leq a\} = P\{x > a\}$ and (ii) $P\{x \geq b\} = 0.05$

[JNTU(A) Apr. 2012 (Set No. 4)]

Solution : The probability density function of a continuous random variable is given by

$$f(x) = 3x^2, 0 \leq x \leq 1$$

(i) Since the total probability is always one, we have

$$P(X \leq a) = P(x \geq a) = \frac{1}{2}$$

$$\text{Now } P(X \leq a) = \frac{1}{2} \Rightarrow \int_0^a f(x) dx = \frac{1}{2}$$

$$\Rightarrow \int_0^a 3x^2 dx = \frac{1}{2} \Rightarrow 3 \left(\frac{x^3}{3} \right)_0^a = \frac{1}{2} \Rightarrow a^3 = \frac{1}{2} \text{ or } a = \left(\frac{1}{2} \right)^{1/3} = \frac{1}{2^{1/3}}$$

$$(ii) \quad P(X \geq b) = 0.05 \Rightarrow \int_b^1 f(x) dx = 0.05 \Rightarrow \int_b^1 3x^2 dx = 0.05$$

$$\Rightarrow 3 \left(\frac{x^3}{3} \right)_b^1 = 0.05 \Rightarrow 1 - b^3 = \frac{1}{20} \Rightarrow b^3 = 1 - \frac{1}{20} = \frac{19}{20}$$

$$\therefore b = \left(\frac{19}{20} \right)^{1/3}. \text{ Hence } a = \frac{1}{2^{1/3}} \text{ and } b = \left(\frac{19}{20} \right)^{1/3}$$

REVIEW QUESTIONS

1. Define Continuous Probability distribution.
2. Define Cumulative distribution of a continuous random variable.
3. Define expectation (mean) and variance for continuous random variables.

EXERCISE 6(B)

1. The probability density function $f(x)$ of a continuous random variable is given by

$$f(x) = \begin{cases} kx^3, & \text{for } 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find the value of k and the probability that the random variable takes on a value

- (i) between $\frac{1}{4}$ and $\frac{3}{4}$ (ii) greater than $\frac{2}{3}$.

2. (a) If the probability density of a random variable is given by

$$f(x) = \begin{cases} x, & \text{for } 0 < x < 1 \\ 2 - x, & \text{for } 1 \leq x < 2 \\ 0, & \text{elsewhere} \end{cases}$$

Find the probabilities that a random variable having this probability density will take on a value (i) between 0.2 and 0.8 (ii) between 0.6 and 1.2.

- (b) If the probability density of a random variable is given by $f(x) = \begin{cases} kx^2, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$

Find the value of k and the probabilities that a random variable having this probability density will take on a value (i) between 0.25 and 0.75, (ii) greater than $2/3$.

3. Show that the function $f(x) = \frac{1}{\pi(1+x^2)}$, $-\infty \leq x \leq \infty$ is a probability density function.

[JNTU (K) Nov. 2011 (Set No. 4)]

4. Find the constant k so that the function $f(x)$ defined by

$$(i) f(x) = \begin{cases} \frac{1}{k}, & \text{if } a \leq x \leq b \\ 0, & \text{elsewhere} \end{cases} \quad (ii) f(x) = \begin{cases} k e^{-3x}, & x > 0 \\ 0, & \text{elsewhere} \end{cases} \quad \text{is a density function.}$$

5. "The diameter of an electric cable is assumed to be continuous random variable with probability density function $f(x) = 6x(1-x)$, $0 \leq x \leq 1$ ". Justify.

Find the mean and variance of the distribution.

6. The probability density function of the continuous random variable X is given by

$$(i) f(x) = \begin{cases} \frac{1}{4}, & \text{for } 4 \leq x \leq 8 \\ 0, & \text{elsewhere} \end{cases} \quad (ii) f(x) = \begin{cases} \frac{1}{16}(3+x)^2, & -3 \leq x \leq -1 \\ \frac{1}{16}(2-6x^2), & -1 \leq x < 1 \\ \frac{1}{16}(3-x)^2, & 1 \leq x \leq 3 \\ 0, & \text{elsewhere} \end{cases}$$

Show that the area under the curve above x-axis is unity. Also find the mean of the distribution.

7. X is a continuous random variable with probability density function given by

$$(i) f(x) = \begin{cases} \frac{1}{4}e^{-x/4}, & \text{for } x > 0 \\ 0, & \text{elsewhere} \end{cases} \quad (ii) f(x) = \begin{cases} \frac{1}{b}e^{-x/b}, & x > 0, b > 0 \\ 0, & \text{elsewhere} \end{cases}$$

Find the mean and variance of X .

8. X is a continuous random variable with probability density function given by

$$f(x) = \begin{cases} kx, & \text{for } 0 \leq x < 2 \\ 2k, & \text{for } 2 \leq x < 4 \\ k(6-x), & \text{for } 4 \leq x < 6 \end{cases}$$

Find k and mean of the density function.

[JNTU 2003, 2004]

9. Find the standard deviation of the probability density function

$$(i) f(x) = \begin{cases} x^3, & 0 \leq x \leq 1 \\ (2-x)^3, & 1 \leq x \leq 2 \end{cases} \quad (ii) f(x) = \begin{cases} 30x^4(1-x), & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

10. Find the mean and variance of the density function $f(x) = \frac{2x}{9}$, $0 \leq x \leq 3$.

11. The frequency function of a continuous random variable X is given by

$$f(x) = y_0 x(2-x), 0 \leq x \leq 2.$$

Find mean, median and variance of X . [Hint : Refer Solved Example 12]

12. X is a continuous random variable with probability density function given by

$$f(x) = \begin{cases} k x^{\alpha-1} (1-x)^{\beta-1}, & \text{for } 0 < x < 1, \alpha > 0, \beta > 0 \\ 0, & \text{otherwise} \end{cases}$$

Find k and mean value of X .

[JNTU (K) Nov. 2011 (Set No. 3)]

13. Let $f(x) = 3x^2$, when $0 \leq x \leq 1$ be the probability density function of a continuous random variable X . Determine 'a' and 'b' such that

$$(i) P(X \leq a) = P(X > a) \quad (ii) P(X \geq b) = 0.05.$$

[JNTU 2004 (Set No. 2), (K) Nov. 2011 (Set No. 1)]

14. If $f(x) = \begin{cases} \frac{1}{2}(x+1), & -1 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$ represents the density of a random variable X , find

the mean and standard deviation of X .

[JNTU 2004S (Set No. 4)]

15. The probability density function of a random variable X is given by

$$f(x) = \begin{cases} kx(x-1), & \text{for } 1 \leq x \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

Given that $P(1 \leq X \leq 3) = \frac{28}{3}$. Find the value of k .

16. Let X be a continuous random variable with distribution : $f(x) = \begin{cases} \frac{1}{8}, & \text{if } 0 \leq x \leq 8 \\ 0, & \text{elsewhere} \end{cases}$

Find (i) $P(2 \leq X \leq 5)$ (ii) $P(3 \leq X \leq 7)$ (iii) $P(X \leq 6)$ (iv) Determine and plot the graph of the cumulative distribution function F of X .

[JNTU (K) Nov. 2012 (Set No. 2)]

ANSWERS

- | | |
|--|---|
| 1. (i) 4 (ii) $\frac{5}{16}$ (iii) $\frac{65}{81}$ | 2. (a) (i) 0.3 (ii) 0.5 |
| 4. (i) $b-a$ (ii) 3 | 5. $\frac{1}{2}, \frac{1}{20}$ |
| 6. (i) 6 (ii) 0 | 7. (i) 4, 80 (ii) b, b^2 |
| 8. $\frac{1}{8}, 3$ | 9. (i) $\frac{1}{\sqrt{15}}$ (ii) $\frac{\sqrt{5}}{14}$ |

10. $2, \frac{1}{2}$

11. $1, \frac{1}{5}$

12. $\frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \Gamma(\beta)}, \frac{\alpha}{\alpha + \beta}$

13. $\left(\frac{1}{2}\right)^{1/3}, \left(\frac{19}{20}\right)^{1/3}$

14. $\frac{1}{3}, \frac{\sqrt{2}}{3}$

15. 2

QUIZ

- Maximum value of probability is _____.
- The function $f(x) = kx$ in $0 < x < 1$ is a valid probability density function, if $k =$ _____.
- The function $f(x) = kx^3$ in $0 < x < 1$ is a valid probability density function, if $k =$ _____.
- If $f(x) = \frac{k}{1+x^2}$, $-\infty < x < \infty$ is a valid probability density function, then $k =$ _____.
- $f(x) = kx(1-x)$ in $0 < x < 1$ is a valid probability density function, if $k =$ _____.
- If $f(x) = \begin{cases} kx^2, & 0 < x < 3 \\ 0, & \text{elsewhere} \end{cases}$ is a probability density function, then $k =$ _____.
- If $f(x) = \begin{cases} kxe^{-x}, & x > 0 \\ 0, & \text{elsewhere} \end{cases}$ is the probability density function, then $k =$ _____.
- If $f(x) = x + \frac{2}{k}$, $x = 1, 2, 3, 4, 5$ is the probability distribution of a discrete random variable, then $k =$ _____.
- If $f(x) = Ax^2$ in $0 \leq x \leq 1$ is a probability function, then $A =$ _____.
- The probability density function $f(x)$ of a continuous random variable X is defined by $f(x) = \begin{cases} \frac{k}{x^3}, & \text{for } 5 \leq x \leq 10 \\ 0, & \text{elsewhere} \end{cases}$. Then $k =$ _____.
- For the density function $f(x) = \frac{2x}{9}$, $0 \leq x \leq 3$, the mean is _____.
- For the density function $f(x) = 6x(1-x)$, $0 \leq x \leq 1$, the mean is _____.
- If $f(x) = k(2x+3)$ in $0 < x < 2$ is a probability function, then $k =$ _____.
- If probability density function $f(x) = \frac{x^2}{9}$ in $0 < x < 3$, then mean of the distribution = _____.

15. The mean of the density function $f(x) = \lambda e^{-\lambda x}$ is _____.

16. If $f(x) = \begin{cases} Ax, & 0 \leq x \leq 5 \\ A(10-x), & 5 \leq x \leq 10 \\ 0, & \text{otherwise} \end{cases}$

is a valid probability density function, then $A =$ _____.

17. If $f(x) = (1-k)k^x$, $x = 0, 1, 2, 3, \dots$ is a probability distribution of a random variable, then $k =$ _____.

18. If $f(x) = \frac{2x}{k(k+1)}$, $x = 1, 2, 3, \dots$, k serves as a probability density function, then $k =$ _____.

19. The mean of the probability distribution of the number of heads obtained in two flips of a balanced coin is a _____.

20. The mean of the probability distribution of the numbers on face of die in throwing a die is _____.

21. If X and Y are independent random variables, then $E(XY) =$ _____.

22. X is a continuous random variable with probability density function given by

$$f(x) = \begin{cases} \frac{1}{8}(x+1), & \text{for } 2 < x < 4 \\ 0, & \text{otherwise} \end{cases}$$

Then $E(X) =$ _____.

23. If X_1, X_2 are random variables, and a, b are constants, then $E(aX_1 + bX_2) =$ _____.

24. If $f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, then mean of the distribution is _____ and standard deviation is _____.

25. If k is a constant, then $\text{Var}(k) =$ _____.

26. If k is a constant, then $V(X + k) =$ _____.

27. If X is a random variable and a, b are constants, then $V(aX + b) =$ _____ $V(x)$.

28. If X is a random variable and $V(X) = 2$, then $V(2X + 3) =$ _____.

29. If $f(x) = \begin{cases} \frac{1}{5}, & \text{for } 2 < x < 7 \\ 0, & \text{elsewhere} \end{cases}$

serves as a probability function, then $P(-3 < X < 3) =$ _____.

30. If the density function of the random variable X is given by $f(x) = \begin{cases} \frac{k}{\sqrt{x}}, & \text{for } 4 \leq x \leq 9 \\ 0, & \text{otherwise} \end{cases}$
then $k =$ _____.
31. If the distribution function for X is defined by $F(X) = \begin{cases} 0, & \text{for } x < 0 \\ 1 - \frac{1}{4}e^{-x}, & \text{for } x \geq 0 \end{cases}$
then $P(X = 0)$ is _____.
32. If $f(x) = k e^{-x/5}$, $x \geq 0$ is a probability density function, then $k =$ _____.
33. If $f(x) = 2 e^{-2x}$, $x > 0$ is a probability density function, then $P(X \geq 0.5) =$ _____.
34. If X is uniformly distributed in $[a, b]$, then $E(X) =$ _____.
35. The relation between probability density function and cumulative density function of a random variable is _____.

ANSWERS

- | | | | |
|---------------------------------------|---------------------|---------------------------|--------------------|
| 1. 1 | 2. 2 | 3. 4 | 4. $\frac{1}{\pi}$ |
| 5. 6 | 6. $\frac{1}{9}$ | 7. 1 | 8. $-\frac{5}{7}$ |
| 9. 3 | 10. $\frac{200}{3}$ | 11. 2 | 12. $\frac{1}{2}$ |
| 13. $\frac{1}{10}$ | 14. $\frac{9}{4}$ | 15. $\frac{1}{\lambda}$ | 16. $\frac{1}{25}$ |
| 17. 1 | 18. 1 | 19. 1 | 20. $\frac{7}{2}$ |
| 21. $E(X) E(Y)$ | 22. 3.083 | 23. $a E(X_1) + b E(X_2)$ | |
| 24. 0, 1 | 25. $V(X)$ | 26. 0 | 27. a^2 |
| 28. 8 | 29. $\frac{4}{5}$ | 30. $\frac{1}{2}$ | 31. $\frac{3}{4}$ |
| 32. $\frac{1}{5}$ | 33. $\frac{1}{e}$ | 34. $\frac{1}{2}(a+b)$ | |
| 35. $F(x) = \int_{-\infty}^x f(x) dx$ | | | |

PREVIOUS GATE QUESTIONS

1. Let X be a real - valued random variable with $E[X]$ and $E[X^2]$ denoting the mean values of X and X^2 , respectively. The relation which always holds true is
 (A) $(E[X])^2 > E[X^2]$ (B) $E[X^2] \geq (E[X])^2$ [GATE 2014 (EC)]
 (C) $E[X^2] = (E[X])^2$ (D) $E[X^2] > (E[X])^2$
2. Let X be a random variable which is uniformly chosen from the set of positive odd numbers less than 100. The expectation, $E[X]$ is _____ [GATE 2014 (EC)]
3. Consider a dice with the property that the probability of a face with n dots showing up is proportional to n . The probability of the face with three dots showing up is _____ [GATE 2014 (EE)]
4. Let X be a random variable with probability density function

$$f(x) = \begin{cases} 0.2, & \text{for } |x| \leq 1 \\ 0.1, & \text{for } 1 < |x| \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

The probability $P(0.5 < X < 5)$ is _____ [GATE 2014 (EE)]

5. In the following table, x is a discrete random variable and $p(x)$ is the probability density. The standard deviation of x is

x	1	2	3
$p(x)$	0.3	0.6	1

[GATE 2014 (ME)]

- (A) 0.18 (B) 0.36 (C) 0.54 (D) 0.6
6. The probability density function of evaporation E on any day during a year in a watershed is given by

$$f(E) = \begin{cases} \frac{1}{5}, & 0 \leq E \leq 5 \text{ mm/day} \\ 0, & \text{otherwise} \end{cases}$$

The probability that E lies in between 2 and 4 mm/day in a day in the watershed is (in decimal) _____ [GATE 2014 (CE)]

7. Let the random variable X represent the number of times a fair coin needs to be tossed till two consecutive heads appear for the first time. The expectation of X is _____

[GATE 2015 (EC)]

8. The variance of the random variable X with probability function $f(x) = \frac{1}{2}|x|e^{-|x|}$ is _____

[GATE 2015 (EC)]

9. A random variable X has probability density function $f(x)$ as given below :

$$f(x) = \begin{cases} a + bx, & \text{for } 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

If the expected value $E(X) = \frac{2}{3}$ then $P(X < 0.5)$ is _____ [GATE 2015 (EE)]

10. Consider the following probability mass function (p.m.f) of a random variable X :

$$p(x, q) = \begin{cases} q, & \text{if } X = 0 \\ 1 - q, & \text{if } X = 1 \\ 0, & \text{otherwise} \end{cases}$$

If $q = 0.4$, the variance of X is _____ [GATE 2015 (CE)]

11. For probability density function of a random variable X is

$$f(x) = \begin{cases} \frac{x}{4}(4 - x^2) & \text{for } 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

The mean μ of the random variable X is _____ [GATE 2015 (CE)]

12. The probability density function of a random variable X is $P_X(x) = e^{-x}$, $x \geq 0$ and 0 otherwise. The expected value of the function $g_X(x) = e^{3x/4}$ is _____

[GATE 2015 (IN)]

13. Let the probability density function of a random variable X be given as :

$$f_x(x) = \frac{3}{2}e^{-3x}u(x) + ae^{4x}u(-x) \text{ where } u(x) \text{ is the unit step function.}$$

Then the value of ' a ' and Prob $[X \leq 0]$, respectively are, [GATE 2016 (EE)]

- (A) $2, \frac{1}{2}$ (B) $4, \frac{1}{2}$ (C) $2, \frac{1}{4}$ (D) $4, \frac{1}{4}$

14. Probability density function of a random variable X is given below.

$$f(x) = \begin{cases} 0.25 & \text{if } 1 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

$P(X \leq 4)$ is

[GATE 2016 (CE)]

- (A) $\frac{3}{4}$ (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{1}{8}$

15. If $f(x)$ and $g(x)$ are two probability density functions,

$$f(x) = \begin{cases} \frac{x}{a} + 1, & -a \leq x < 0 \\ -\frac{x}{a} + 1, & 0 \leq x \leq a \\ 0, & \text{otherwise} \end{cases} \quad \text{and} \quad g(x) = \begin{cases} -\frac{x}{a}, & a \leq x < 0 \\ \frac{x}{a} + 1, & 0 \leq x \leq a \\ 0, & \text{otherwise} \end{cases}$$

which one of the following statements is true ?

[GATE 2016 (CE)]

- (A) Mean of $f(x)$ and $g(x)$ are same ; variance of $f(x)$ and $g(x)$ are same
 (B) Mean of $f(x)$ and $g(x)$ are same; variance of $f(x)$ and $g(x)$ are different.
 (C) Mean of $f(x)$ and $g(x)$ are different; variance of $f(x)$ and $g(x)$ are same.
 (D) Mean of $f(x)$ and $g(x)$ are different; variance of $f(x)$ and $g(x)$ are different.

16. A probability density function on the interval $[a, 1]$ is given by $\frac{1}{x^2}$ and outside this interval the value of the function is zero. The value of a is _____

[GATE 2016 (CS)]

17. A six-face fair dice is rolled a large number of times. The mean value of the outcomes is _____

[GATE 2017 (ME)]

18. For the function $f(x) = a + bx, 0 \leq x \leq 1$, to be a valid probability density function, which one of the following statements is correct.

[GATE 2017 (CE)]

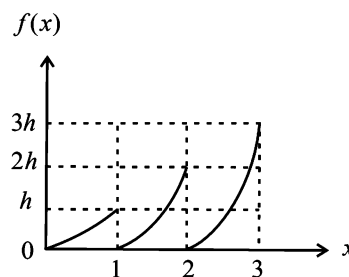
- (A) $a = 1, b = 4$ (B) $a = 0.5, b = 1$ (C) $a = 0, b = 1$ (D) $a = 1, b = -1$

19. The graph of a function $f(x)$ is shown in the figure

For $f(x)$ to be a valid probability density function,

the value of h is [GATE 2018 (CE)]

- (A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) 1 (D) 3



20. X and Y are two independent random variables with variance 1 and 2, respectively. Let $z = X - Y$. The variance of z is _____ [GATE 2018 (IN)]

(A) 0 (B) 1 (C) 2 (D) 3

21. The probability distribution for a discrete random variable X is given below

X	1	2	3	4
$P(X)$	0.3	0.4	0.2	0.1

The expectation value of X is _____ [GATE 2018 (BT)]

22. If X and Y are random variables such that $E(2X + Y) = 0$ and $E(X + 2Y) = 33$, then $E(X) + E(Y) =$ _____ [GATE 2019 (EC)]

23. The mean square of a zero-mean random process is $\frac{KT}{c}$ where K is Boltzmann's temperature, and c is a capacitance. The standard deviation of the random process is

(A) $\frac{KT}{c}$ (B) $\frac{\sqrt{KT}}{c}$ (C) $\sqrt{\frac{KT}{c}}$ (D) $\frac{c}{KT}$ [GATE 2019 (EE)]

24. If x is the mean of data 3, x , 2 and 4, then the mode is _____ [GATE 2019 (ME-Set 2)]

25. Let T be a full binary tree with 8 leaves (A full binary tree has every level full). Suppose two leaves a and b of T are chosen uniformly and independently at random. The expected value of the distance between a and b in T (i.e. the number of edges in the unique path between a and b) is _____ [GATE 2019 (CS)]

26. The function $p(x)$ is given by $p(x) = \frac{A}{x^\mu}$ where A and μ are constants with $\mu > 1$ and $1 \leq x < \infty$ and $p(x) = 0$ for $-\infty < x < 1$. For $p(x)$ to be a probability density function, the value of A should be equal to [GATE 2019 (IN)]

(A) $\mu - 1$ (B) $\mu + 1$ (C) $\frac{1}{\mu - 1}$ (D) $\frac{1}{\mu + 1}$

ANSWERS

- | | | | | |
|------------|---------|--------|---------|----------|
| 1. B | 2. 50 | 3. 1/7 | 4. 0.4 | 5. 0.6 |
| 6. 0.4 | 7. 1.5 | 8. 6 | 9. 0.25 | 10. 0.24 |
| 11. 1.0667 | 12. 4 | 13. A | 14. A | 15. B |
| 16. 0.5 | 17. 3.5 | 18. B | 19. A | 20. D |
| 21. 2.1 | 22. 11 | 23. C | 24. 3 | 25. 4.86 |
| 26. A | | | | |

UNIT - III

PROBABILITY DISTRIBUTIONS

7. *Probability Distributions*

- ♦ *Binomial Distribution*
- ♦ *Poisson Distribution*
- ♦ *Normal Distribution*



CHAPTER - 7

PROBABILITY DISTRIBUTIONS

(Binomial, Poisson, Normal Distributions)

7.1 INTRODUCTION

We have already learnt frequency distributions which are based on the actual observations or experiment. In this chapter, we shall discuss theoretical distributions in which variates are distributed as per some definite law which can be expressed mathematically. That is, theoretical distributions (or Probability distributions) are mathematical models. So, it is possible to deduce mathematically what the frequency distribution of certain population should be. Such distributions are called Theoretical distributions or Frequency Distributions. Hence a theoretical distribution is the frequency distribution of a certain event in which frequencies are obtained by mathematical computation.

The value of the random variables and the corresponding probabilities are arranged such a way that one can suit a mathematical function for the probabilities in terms of the value of the random variable. A good model requires sufficiently large number of observations.

Just, as in the case of actual frequency distribution, the various statistical measures such as mean, standard deviation etc., are computed, so also such statistical measures may easily be calculated from a theoretical distribution and they give a very close approximation to such statistical measures as calculated from the actual frequency distribution.

There are two types of theoretical distributions (or Probability distributions)

(i) Discrete Theoretical Distributions

- (a) Binomial distribution
- (b) Poisson distribution
- (c) Rectangular distribution
- (d) Negative Binomial distribution
- (e) Geometric distribution

(ii) Continuous Theoretical Distributions

- (a) Normal distribution
- (b) Student's t-distribution
- (c) Chi-Square distribution
- (d) F-distribution

In this chapter, we shall study only the three distributions namely, Binomial, Poisson and Normal distributions. Before we discuss these distributions, we first give a brief introduction of uniform and Bernoulli's distributions.

7.2 DISCRETE UNIFORM DISTRIBUTION

A random variable X has a discrete uniform distribution if and only if its probability distribution is given by

$$p(x) = \frac{1}{k}, \text{ for } x = x_1, x_2, \dots, x_k.$$

The random variable X is then called discrete uniform random variable.

For example, the distributions

x	0	1
$p(x)$	$\frac{1}{2}$	$\frac{1}{2}$

and

x	0	1	2	3
$p(x)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

are discrete uniform distributions.

7.3 BERNOULLI'S DISTRIBUTION

A random variable X which takes two values 0 and 1 with probability q and p respectively i.e., $P(X=0) = q$ and $P(X=1) = p$, $q = 1 - p$ is called a *Bernoulli's discrete random variable* and is said to have a *Bernoulli's distribution*. The probability function of Bernoulli's distribution can be written as

$$p(x) = p^x q^{1-x} = p^x (1-p)^{1-x}, x = 0, 1.$$

Note : (i) Mean of the Bernoulli's discrete random variable X is

$$\mu = E(X) = \sum x_i \cdot p(x_i) = 0 \times q + 1 \times p = p$$

(ii) Variance of X is

$$\begin{aligned} V(X) &= E(X^2) - [E(X)]^2 = \sum x_i^2 \cdot p(x_i) - \mu^2 = (0^2 \times q + 1^2 \times p) - p^2 \\ &= p - p^2 = p(1-p) = pq \end{aligned}$$

Hence the standard deviation is $\sigma = \sqrt{pq}$.

Definition : Suppose, associated with a random trial there is an event called 'success' and the complementary event called 'failure'. Let the probability for success be p and probability for failure be q . Suppose the random trials are prepared n times under identical conditions. These are called Bernoullian trials.

Bernoulli's Theorem :

If the probability of the occurrence of an event (success) in a single trial is p , then the probability that it will occur exactly r times out of n independent Bernoullian trials is

$${}^nC_r p^r q^{n-r} \text{ where } p + q = 1$$

Proof : In a series of n independent trials, the number of r successes and $(n-r)$ failures can happen in nC_r mutually exclusive and exhaustive ways, since that is equal to the number of combinations of n things taken r at a time. Now consider a particular set of r successes and $(n-r)$ failures. Let the probability of success in a single trial be p . Then the probability of its failure is $q (= 1 - p)$ and that of success in all the r trials is p^r , by the theorem of compound probability. Also, by the same theorem, the probability of failure in all the $(n-r)$ trials is q^{n-r} . Hence by the multiplication theorem, the probability of getting r successes and $(n-r)$ failures in the particular set considered is equal to $p^r q^{n-r}$. Thus the probability of r successes in n trials with probability p in each trial is ${}^nC_r p^r q^{n-r}$.

Note : 1. By Binomial Theorem,

$$(q + p)^n = q^n + {}^nC_1 q^{n-1} p + {}^nC_2 q^{n-2} p^2 + \dots + {}^nC_r q^{n-r} p^r + \dots + p^n.$$

We notice that the probabilities of getting 0, 1, 2, ..., n successes are given respectively by the successive terms of the above binomial expansion. The probability of exactly r successes and $n - r$ failures is ${}^nC_r p^r q^{n-r}$ and it is the $(r + 1)^{\text{th}}$ term in the above binomial expansion.

2. The probability of getting at least one success in a series of n trials is

$$= {}^nC_1 q^{n-1} p + {}^nC_2 q^{n-2} p^2 + \dots + p^n$$

$$= (q + p)^n - q^n = 1 - q^n$$

3. The probability of no success in n Bernoullian trials is given by q^n , and that of all successes by p^n .

BINOMIAL DISTRIBUTION

7.4 BINOMIAL DISTRIBUTION

Binomial distribution was discovered by James Bernoulli in the year 1700 and it is a discrete probability distribution.

Let us visualise a conceptual or practical situation where a trial or an experiment results in only two outcomes, say 'success' and 'Failure'. Further, the result of one trial does not influence the result of next trial, and the probability of success at each trial is the same from trial to trial.

Some of such situations are:

1. Tossing a coin - Head or Tail
2. Birth of a baby - Girl or Boy
3. Auditing a Bill - contains an error or not
4. An advertisement on TV - Recalled by viewer or not.

The conditions for the applicability of a Binomial distribution are as follows:

- (i) There are n independent trials
- (ii) Each trial has only two possible outcomes
- (iii) The probabilities of two outcomes remain constant.

The Binomial distribution can be described with the help of a function and is derived as follows:

Let the number of trials be n . The trials be independent *i.e.*, the success or failure at one trial doesnot affect the outcome of the other trials. Thus the probability of success remains the same from trial to trial. Also let ' p ' be the probability of success and ' q ' be the probability of failure. Then we have $p + q = 1$ or $q = 1 - p$.

One could be interested in finding the probability of getting r successes, which implies getting $(n - r)$ failures.

One of the ways in which one can get r successes and $(n-r)$ failure is to get r continuous successes first and $(n-r)$ successive failures.

The probability of getting such a sequence of r successes and $(n-r)$ failures is,

$$= p^r q^{n-r} \text{ (using Multiplication Theorem of Probability)}$$

However, the number of ways in which one can get r successes and $n-r$ failures is nC_r and the probability for each of these ways is $p^r q^{n-r}$.

Thus, if x is the random variable representing the number of successes, the probability of getting r successes and $n-r$ failures, in n trials, is given by the probability function

$$P(x=r) = {}^nC_r p^r q^{n-r}, \quad r=0, 1, 2, \dots, n$$

This probability function is popularly known as the Binomial Distribution.

The probabilities of 0, 1, 2, ..., r , ..., n successes are therefore given by

$$q^n, {}^nC_1 p q^{n-1}, {}^nC_2 p^2 q^{n-2}, \dots, {}^nC_r p^r q^{n-r}, \dots, p^n.$$

The probability of the number of successes so obtained is called the *Binomial Probability Distribution*, because these probabilities are the successive terms in the expansion of the binomial $(q+p)^n$. This distribution contains two independent constants namely n and p (or q). They are called parameters of the Binomial Distribution. Sometimes, n is also known as the degree of the distribution.

Definition : A random variable X has a Binomial distribution if it assumes only non-negative values and its probability density function is given by

$$P(X=r) = p(r) = \begin{cases} {}^nC_r p^r q^{n-r}; & r=0, 1, 2, \dots, n; q=1-p \\ 0, & \text{otherwise} \end{cases}$$

Note :1. An alternate notation for binomial distribution is as follows :

$$P(X=x) = p(x) = b(x; n, p) = \begin{cases} {}^nC_x p^x q^{n-x}; & x=0, 1, 2, \dots, n \\ 0, & \text{otherwise} \end{cases}$$

2. The Binomial distribution function is given by

$$F_X(x) = P(X \leq x) = \sum_{r=0}^x {}^nC_r p^r q^{n-r}$$

Examples of Binomial Distribution :

- (i) The number of defective bolts in a box containing ' n ' bolts.
- (ii) The number of machines lying idle in a factory having ' n ' machines.
- (iii) The number of post-graduates in a group of ' n ' men.
- (iv) The number of oil wells yielding natural gas in a group of ' n ' wells test drilled.

Conditions of Binomial Distribution :

The Binomial Distribution holds under the following conditions :

1. Trials are repeated under identical conditions for a fixed number of times, say n times.
2. There are only two possible outcomes, *e.g.* success or failure for each trial.
3. The probability of success in each trial remains constant and does not change from trial to trial.
4. The trials are independent *i.e.*, the probability of an event in any trial is not affected by the results of any other trial.

7.5 CONSTANTS OF BINOMIAL DISTRIBUTION**1. Mean of the Binomial Distribution :****[JNTU 2000, 2002]**

The Binomial probability distribution is given by

$$p(r) = {}^nC_r p^r q^{n-r}, r = 0, 1, 2, \dots, n \text{ and } q = 1 - p$$

$$\text{Mean of } X, \mu = E(X) = \sum_{r=0}^n r p(r)$$

$$\begin{aligned} &= 0 \times q^n + 1 \times {}^nC_1 p q^{n-1} + 2 \times {}^nC_2 p^2 q^{n-2} + \dots + n p^n \\ &= np q^{n-1} + 2 \cdot \frac{n(n-1)}{2!} p^2 q^{n-2} + 3 \cdot \frac{n(n-1)(n-2)}{3!} p^3 q^{n-3} + \dots + np^n \\ &= np \left[q^{n-1} + (n-1)p q^{n-2} + \frac{(n-1)(n-2)}{2!} p^2 q^{n-3} + \dots + p^{n-1} \right] \\ &= np (q + p)^{n-1} \quad [\text{using Binomial Theorem}] \\ &= np (1) = np \quad [\because p + q = 1] \end{aligned}$$

Hence the Arithmetic mean of the Binomial distribution = np .

2. Variance of the Binomial Distribution :**[JNTU 2000, 2002]**

Variance,

$$\begin{aligned} V(X) &= E(X^2) - [E(X)]^2 \\ &= \sum_{r=0}^n r^2 p(r) - \mu^2 = \sum_{r=0}^n [r(r-1) + r] \cdot p(r) - \mu^2 \\ &= \sum_{r=0}^n r(r-1) {}^nC_r p^r q^{n-r} + \sum_{r=0}^n r p(r) - \mu^2 \\ &= \left[2 \cdot {}^nC_2 p^2 q^{n-2} + 3 \cdot 2 \cdot {}^nC_3 p^3 q^{n-3} + \dots + n(n-1) p^n \right] + \mu - \mu^2 \\ &= \left[2 \cdot \frac{n(n-1)}{2!} p^2 q^{n-2} + 6 \cdot \frac{n(n-1)(n-2)}{3!} p^3 q^{n-3} + \dots + n(n-1) p^n \right] + \mu - \mu^2 \\ &= n(n-1)p^2 \left[q^{n-2} + (n-2)p q^{n-3} + \frac{(n-2)(n-3)}{2!} p^2 q^{n-4} + \dots + p^{n-2} \right] + \mu - \mu^2 \end{aligned}$$

$$\begin{aligned}
 \therefore V(X) &= n(n-1)p^2(q+p)^{n-2} + \mu - \mu^2 \quad (\text{by Binomial Theorem}) \\
 &= n(n-1)p^2(1)^{n-2} + \mu - \mu^2 \quad (\text{since } q+p=1) \\
 &= n(n-1)p^2 + np - (np)^2 \quad (\because \mu = np) \\
 &= n^2p^2 - np^2 + np - n^2p^2 = -np^2 + np = np(1-p) = npq
 \end{aligned}$$

i.e., Variance of the Binomial Distribution = npq .

Hence the Standard Deviation of the Binomial Distribution = $\sqrt{npq}$.

3. Mode of the Binomial Distribution :

Mode of the binomial distribution is the value of x at which $p(x)$ has maximum value.

$$\text{Mode} = \begin{cases} \text{integral part of } (n+1)p, & \text{if } (n+1)p \text{ is not an integer} \\ (n+1)p \text{ and } (n+1)p - 1, & \text{if } (n+1)p \text{ is an integer} \end{cases}$$

Recurrence Relation for the Binomial Distribution :

$$\text{We know that } P(r) = {}^nC_r p^r q^{n-r} \quad \dots (1)$$

$$\therefore P(r+1) = {}^nC_{r+1} p^{r+1} q^{n-r-1} \quad \dots (2)$$

(2) $\div$ (1) gives

$$\frac{P(r+1)}{P(r)} = \frac{{}^nC_{r+1}}{{}^nC_r} \cdot \frac{p^{r+1}q^{n-r-1}}{p^r q^{n-r}} = \frac{n-r}{r+1} \cdot \frac{p}{q}$$

$$\text{or } P(r+1) = \frac{(n-r)p}{(r+1)q} \cdot P(r)$$

7.6 BINOMIAL FREQUENCY DISTRIBUTION

If n independent trials constitute one experiment and this experiment is repeated N times, then the frequency of r successes is $N \cdot {}^nC_r p^r q^{n-r}$. Since the probabilities of 0, 1, 2, ..., r , ..., n successes in n trials are given by the terms of the binomial expansion of $(q+p)^n$, therefore in N sets of n trials the theoretical frequencies of 0, 1, 2, ..., r , ..., n successes will be given by the terms of expansion of $N(q+p)^n$. The possible number of successes and their frequencies is called a Binomial Frequency Distribution.

SOLVED EXAMPLES

Example 1 : A fair coin is tossed six times. Find the probability of getting four heads.

Solution : We have p = Probability of getting a head = $1/2$

q = Probability of not getting head = $1/2$

and $n = 6, r = 4$

We know that $P(r) = {}^nC_r p^r q^{n-r}$

$$\therefore P(4) = {}^6C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^{6-4} = \frac{6!}{4!2!} \cdot \left(\frac{1}{2}\right)^6 = \frac{6 \times 5}{2} \cdot \frac{1}{2^6} = \frac{15}{64} = 0.2344.$$

Example 2 : Determine the probability of getting the sum 6 exactly 3 times in 7 throws with a pair of fair dice.

Solution : In a single throw of a pair of fair dice, a sum of 6 can occur in 5 ways :

(1, 5), (5, 1), (2, 4), (4, 2) and (3, 3) out of $6 \times 6 = 36$ ways.

Thus $p = \text{Probability of occurrence of 6 in one throw} = \frac{5}{36}$

$$q = 1 - p = 1 - \frac{5}{36} = \frac{31}{36}$$

$n = \text{Number of trials} = 7$

$\therefore$ Probability of getting 6 exactly thrice in 7 throws

$$= {}^7C_3 p^3 q^{7-3} = {}^7C_3 \left(\frac{5}{36}\right)^3 \left(\frac{31}{36}\right)^4 = \frac{35(125)(31)^4}{(36)^7} = 0.0516 \text{ (nearly).}$$

Example 2 : A die is thrown 6 times. If getting an even number is a success, find the probabilities of (i) at least one success (ii) ≤ 3 successes (iii) 4 successes.

Solution : In a single throw of a die, an even number can occur in 3 ways out of 6 ways. Thus

$$p = \text{Probability of occurrence of an even number in one throw} = \frac{3}{6} = \frac{1}{2}$$

$n = \text{Number of trials} = 6$

$$(i) \quad P(r \geq 1) = 1 - P(r = 0)$$

$$= 1 - \left(\frac{1}{2}\right)^6 = 1 - \frac{1}{64} = \frac{63}{64} = 0.9844$$

$$(ii) \quad P(r \leq 3) = P(r = 0) + P(r = 1) + P(r = 2) + P(r = 3)$$

$$= \left(\frac{1}{2}\right)^6 ({}^6C_0 + {}^6C_1 + {}^6C_2 + {}^6C_3) = \frac{21}{32} = 0.6562$$

$$(iii) \quad P(r = 4) = {}^6C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^2 = \frac{15}{64} = 0.2344$$

Example 4 : Ten coins are thrown simultaneously. Find the probability of getting at least (i) seven heads [JNTU 1999, 2007, 2008S, (K) May 2013 (Set No. 2)]

(ii) six heads [JNTU (A) Dec, 2009, (K) May 2013 (Set No. 2)]

Solution : $p = \text{Probability of getting a head} = 1/2$

$q = \text{Probability of not getting a head} = 1/2$

The probability of getting r heads in a throw of 10 coins is

$$P(X = r) = p(r) = {}^{10}C_r \left(\frac{1}{2}\right)^r \left(\frac{1}{2}\right)^{10-r} ; r = 0, 1, 2, \dots, 10$$

(i) Probability of getting at least seven heads is given by

$$P(X \geq 7) = P(X = 7) + P(X = 8) + P(X = 9) + P(X = 10)$$

$$\begin{aligned}
&= {}^{10}C_7 \left(\frac{1}{2}\right)^r \left(\frac{1}{2}\right)^{10-r} \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^3 + {}^{10}C_8 \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^2 + {}^{10}C_9 \left(\frac{1}{2}\right)^9 \left(\frac{1}{2}\right) + {}^{10}C_{10} \left(\frac{1}{2}\right)^{10} \\
&= \frac{1}{2^{10}} [{}^{10}C_7 + {}^{10}C_8 + {}^{10}C_9 + {}^{10}C_{10}] = \frac{1}{2^{10}} (120 + 45 + 10 + 1) \\
&= \frac{176}{1024} = 0.1719
\end{aligned}$$

(ii) This is left as an exercise to the student.

Example 4 : Two dice are thrown five times. Find the probability of getting 7 as sum

(i) at least once (ii) Two times (iii) $P(1 < X < 5)$. [JNTU(H) Nov. 2009 (Set No. 2)]

Solution : p = The probability of getting a sum 7 in a single throw of a pair of dice

$$= \frac{6}{36} = \frac{1}{6}$$

q = The probability of not getting a sum 7 in a throw of a pair of dice

$$= 1 - p = 1 - \frac{1}{6} = \frac{5}{6}$$

Number of trials, $n = 5$.

(i) $P(\text{at least once}) = P(X \geq 1) = 1 - P(X = 0)$

$$= 1 - {}^5C_0 \left(\frac{1}{6}\right)^0 \left(\frac{5}{6}\right)^5 = 1 - \left(\frac{5}{6}\right)^5$$

(ii) $P(\text{exactly two times}) = P(X = 2) = {}^5C_2 \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^4$

$$= 10 \cdot \frac{5^4}{6^6} = \frac{2}{6} \cdot \left(\frac{5}{6}\right)^5 = \frac{1}{3} \cdot \left(\frac{5}{6}\right)^5$$

(iii) $P(1 < X < 5) = P(X = 2) + P(X = 3) + P(X = 4)$

$$\begin{aligned}
&= {}^5C_2 \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^4 + {}^5C_3 \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^3 + {}^5C_4 \left(\frac{1}{6}\right)^4 \left(\frac{5}{6}\right)^2 \\
&= 10 \cdot \frac{5^4}{6^6} + 10 \cdot \frac{5^3}{6^6} + 5 \cdot \frac{5^2}{6^6} = \frac{5^3}{6^6} (300 + 10 + 1) = 311 \left(\frac{5^3}{6^6}\right)
\end{aligned}$$

Example 6 : It has been claimed that in 60% of all solar heat installations the utility bill is reduced by at least one-third. Accordingly, what are the probabilities that the utility bill will be reduced by at least one-third in (i) four of five installations (ii) at least four of five installations.

[JNTU 1999S, (A) Apr. 2012 (Set No. 4)]

Solution : p = The probability that in the solar heat installations the utility bill is reduced by one-third

$$= 60\% = 0.6$$

$$q = 1 - p = 1 - 0.6 = 0.4$$

(i) $r = 4$, $n = 5$

$$P(X = r) = {}^nC_r p^r q^{n-r}$$

$$\therefore P(X = 4) = {}^5C_4(0.6)^4(0.4)^{5-4} = 5(0.6)^4(0.4) = 0.2592$$

$$\begin{aligned} \text{(ii)} \quad P(X \geq 4) &= P(X = 4) + P(X = 5) \\ &= 0.2592 + {}^5C_5(0.6)^5 \quad [\text{From (i)}] \\ &= 0.2592 + 0.07776 = 0.337 \end{aligned}$$

Example 2 : If 3 of 20 tyres are defective and 4 of them are randomly chosen for inspection, what is the probability that only one of the defective tyre will be included ?

[JNTU 2000S]

Solution : Let p = Probability of a defective tyre (success) = $3/20$
Then q = Probability of a non-defective tyre

$$= 1 - p = 1 - \frac{3}{20} = \frac{17}{20}$$

Given $n = 4$

The probability that exactly one tyre will be defective

$$= P(r = 1) = {}^4C_1 \left(\frac{3}{20} \right)^1 \left(\frac{17}{20} \right)^{4-1} = \frac{4 \cdot 3 \cdot 17^3}{(20)^4} = 0.3685$$

Example 8 : The incidence of an occupational disease in an industry is such that the workers have a 20% chance of suffering from it. What is the probability that out of 6 workers chosen at random, four or more will suffer from the disease ?

[JNTU 2001 S, (K) Nov. 2011 (Set No. 4)]

Solution : The probability of a worker suffering from disease, $p = 20\% = 0.2$

$\therefore$ The probability that no worker suffering from disease, $q = 1 - p = 0.8$

No. of workers, $n = 6$

The probability that four or more workers will suffer from disease

$$\begin{aligned} &= P(X \geq 4) \\ &= P(X = 4) + P(X = 5) + P(X = 6) \\ &= {}^6C_4(0.2)^4(0.8)^2 + {}^6C_5(0.2)^5(0.8) + {}^6C_6(0.2)^6 \\ &= (0.2)^4 [15(0.8)^2 + 6(0.2)(0.8) + (0.2)^2] \\ &= (0.2)^4 [9.6 + 0.96 + 0.4] = 0.0175 \end{aligned}$$

Example 9 : Determine the binomial distribution for which the mean is 4 and variance 3

[JNTU 1999]

Solution : Given mean of the distribution = 4 i.e., $np = 4$... (1)

and variance of the distribution = 3 i.e., $npq = 3$... (2)

(2) $\div$ (1) gives $q = 3/4$

$$\therefore p = 1 - q = 1 - \frac{3}{4} = \frac{1}{4}$$

$$\text{From (1), } n = \frac{4}{p} = 16$$

$\therefore$ The given Binomial Distribution has parameters $n = 16$ and $p = 1/4$.

Note : To find mode of the distribution, we have $(n + 1)p = \frac{17}{4} = 4.25$, which is not an integer.

Hence the unique mode of the distribution = integral part of $(n + 1)p$ i.e., $4.25 = 4$

Example 10 : The mean and variance of a binomial distribution are 4 and $\frac{4}{3}$ respectively. Find $P(X \geq 1)$. [JNTU 2004, (A) Dec. 2009 (Set No. 3), June 2018]

Solution : Mean of the binomial distribution = 4 i.e., $np = 4$... (1)

and variance of the binomial distribution = $\frac{4}{3}$ i.e., $npq = \frac{4}{3}$... (2)

$$(2) \div (1) \text{ gives } q = \frac{1}{3} \quad \therefore p = 1 - q = 1 - \frac{1}{3} = \frac{2}{3}$$

$$\text{From (1), } n = \frac{4}{p} = 4 \left(\frac{3}{2} \right) = 6$$

$$\begin{aligned} \therefore P(X \geq 1) &= 1 - P(X < 1) \\ &= 1 - P(X = 0) = 1 - {}^6C_0 p^0 q^{6-0} \\ &= 1 - 1 \cdot \left(\frac{2}{3} \right)^0 \cdot \left(\frac{1}{3} \right)^6 = 1 - \left(\frac{1}{3} \right)^6 = 0.9986 \end{aligned}$$

Example 11 : The mean and variance of a binomial variable X with parameters n and p are 16 and 8. Find $P(X \geq 1)$ and $P(X > 2)$. [JNTU 2004 (Set No. 3)]

Solution : The mean of the binomial distribution = 16 i.e., $np = 16$... (1)

The variance of the binomial distribution = 8 i.e., $npq = 8$... (2)

$$(2) \div (1) \text{ gives } \frac{npq}{np} = \frac{8}{16} \Rightarrow q = \frac{1}{2}$$

$$\therefore p = 1 - q = 1 - \frac{1}{2} = \frac{1}{2}$$

$$\text{From (1), } n = \frac{16}{p} = \frac{16}{1/2} = 32$$

$$\text{Hence } P(X \geq 1) = 1 - P(X < 1) = 1 - P(X = 0)$$

$$\begin{aligned} &= 1 - {}^{32}C_0 p^0 q^{32} = 1 - {}^{32}C_0 \left(\frac{1}{2} \right)^0 \cdot \left(\frac{1}{2} \right)^{32} \\ &= 1 - 1 \cdot \left(\frac{1}{2} \right)^{32} = 1 - \frac{1}{2^{32}} \end{aligned}$$

$$\begin{aligned}
\text{and } P(X > 2) &= 1 - P(X \leq 2) = 1 - [P(X = 0) + P(X = 1) + P(X = 2)] \\
&= 1 - \left[{}^{32}C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{32} + {}^{32}C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{31} + {}^{32}C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{30} \right] \\
&= 1 - \left(\frac{1}{2}\right)^{32} [{}^{32}C_0 + {}^{32}C_1 + {}^{32}C_2] \\
&= 1 - \left(\frac{1}{2}\right)^{32} [1 + 32 + 496] = 0.9999
\end{aligned}$$

Example 12 : If the probability of a defective bolt is $1/8$, find :

- (i) The mean (ii) The variance for the distribution of defective bolts of 640.

[JNTU (H) Nov. 2009 (Set No. 4)]

Solution : We are given

p = The probability of a defective bolt = $1/8$ and $n = 640$.

$\therefore$ Mean of the distribution, $\mu = np = \frac{640}{8} = 80$

Also $q = 1 - p = 1 - \frac{1}{8} = \frac{7}{8}$.

Hence variance of the distribution = $npq = (np)q = \mu q = 80 \times \frac{7}{8} = 70$

Example 11 : A die is tossed thrice. A success is getting 1 or 6 on a toss. Find the mean and variance of the number of successes.

Solution : The probability of success, $p = \frac{2}{6} = \frac{1}{3}$

$\therefore$ The probability of failure, $q = 1 - p = 1 - \frac{1}{3} = \frac{2}{3}$

No. of trials, $n = 3$

Mean = $np = 3\left(\frac{1}{3}\right) = 1$

Variance = $npq = (np)q = (1)q = q = \frac{2}{3}$

Example 10 : In 256 sets of 12 tosses of a coin, in how many cases one can expect 8 heads and 4 tails. [JNTU 2003, 2004 (Set No. 4)]

Solution : The probability of getting a head, $p = 1/2$

$\therefore$ The probability of getting a tail, $q = 1/2$

Here $n = 12$

The probability of getting 8 heads and 4 tails in 12 trials

$$\begin{aligned}
&= P(X = 8) = {}^{12}C_8 \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^4 \\
&= \frac{12!}{8!4!} \cdot \left(\frac{1}{2}\right)^{12} = \frac{495}{(2)^{12}}
\end{aligned}$$

∴ The expected number of such cases in 256 sets

$$= 256 \times P(X = 8) = 2^8 \times \frac{495}{2^{12}} = \frac{495}{16} = 30.9375 \sim 31$$

Example 11 : If the probability of success is $1/20$, how many trials are necessary in

order that the probability of at least one success is just greater than $1/2$.

Solution : Here $p = \frac{1}{20}$. So $q = 1 - p = 1 - \frac{1}{20} = \frac{19}{20}$

Let n be the required number of trails so that the probability of at least one success

$$\text{is } 1 - q^n = 1 - \left(\frac{19}{20}\right)^n \quad [\because p(r \geq 1) = 1 - p(r = 0) = 1 - {}^nC_0 p^0 q^{n-0} = 1 - q^n]$$

$$\text{Suppose } 1 - \left(\frac{19}{20}\right)^n > \frac{1}{2} \quad \text{i.e., } \left(\frac{19}{20}\right)^n < \frac{1}{2}$$

$$\text{Take } \left(\frac{19}{20}\right)^n = \frac{1}{2} \Rightarrow n \log \left(\frac{19}{20}\right) = \log \left(\frac{1}{2}\right)$$

$$\Rightarrow n = \frac{-0.6931}{-0.0513} = 13.51$$

We have to take $n > 13.51$

$$\therefore n = 14$$

Example 12 : Six dice are thrown 729 times. How many times do you expect at least three dice to show a 5 or 6 ?

Solution : p = Probability of occurrence of 5 or 6 in one throw = $\frac{2}{6} = \frac{1}{3}$

$$\therefore q = 1 - p = 1 - \frac{1}{3} = \frac{2}{3} \quad \text{and } n = 6$$

The probability of getting at least three dice to show a 5 or 6

$$= P(X \geq 3) = P(X = 3) + P(X = 4) + P(X = 5) + P(X = 6)$$

$$= {}^6C_3 \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^3 + {}^6C_4 \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^2 + {}^6C_5 \left(\frac{1}{3}\right)^5 \left(\frac{2}{3}\right) + {}^6C_6 \left(\frac{1}{3}\right)^6$$

$$= \frac{1}{(3)^6} [160 + 60 + 12 + 1] = \frac{233}{729}$$

$$\therefore \text{The expected number of such cases in 729 times} = 729 \left(\frac{233}{729}\right) = 233$$

Example 17 : Two dice are thrown 120 times. Find the average number of times in which, the number on the first die exceeds the number on the second die?

[JNTU (K) 2009 (Set No. 2)]

Solution : If a pair of dice is tossed then $S = \{1,2,3,4,5,6\} \times \{1,2,3,4,5,6\}$ and $n(S) = 36$. Let X be the random variable. The favourable cases are (2,1), (3,1), (3,2), (4,1), (4,2), (4,3), (5,1), (5,2), (5,3), (5,4), (6,1), (6,2), (6,3), (6,4), (6,5).

$\therefore p$ = probability of occurrence with the number on first dice exceeds the number on the

second dice in one throw $= \frac{15}{36} = \frac{5}{12}$

Hence $E(X) = \text{mean} = np = 120 \left(\frac{5}{12} \right) = 50$.

Example 10 : If the probability that a man aged 60 will live to be 70 is 0.65, what is the probability that out of 10 men, now 60, at least 7 will live to be 70 ?

Solution : p = The probability that a man aged 60 will live to be 70 = 0.65

$\therefore q = 1 - p = 1 - 0.65 = 0.35$

n = No. of men = 10

$\therefore$ Required probability $= P(X \geq 7) = P(X = 7) + P(X = 8) + P(X = 9) + P(X = 10)$

$$= {}^{10}C_7 (0.65)^7 (0.35)^3 + {}^{10}C_8 (0.65)^8 (0.35)^2 + {}^{10}C_9 (0.65)^9 (0.35) + (0.65)^{10}$$

$$= (0.65)^7 [120 \cdot (0.35)^3 + 45(0.35)^2(0.65) + 10 \times (0.35)(0.65)^2 + (0.65)^3]$$

$$= (0.65)^7 [5.145 + 3.583 + 1.479 + 0.275]$$

$$= (0.65)^7 (10.482) = 0.5139$$

Example 11 : If the probability of a defective bolt is 0.2, find (i) mean (ii) standard deviation for the distribution of bolts in a total of 400.

Solution : Given $n = 400$, $p = 0.2$. $\therefore q = 1 - p = 1 - 0.2 = 0.8$

(i) Mean $= np = 400 (0.2) = 80$... (1)

(ii) S.D. $= \sqrt{npq} = \sqrt{80(0.8)} = \sqrt{64} = 8$ [by (1)]

Example 10 : If 10% of the rivets produced by a machine are defective, find the probability that out of 5 rivets chosen at random (i) none will be defective (ii) one will be defective, and (iii) at most two rivets will be defective.

Solution : Probability of defective rivets $= p = 10\% = 0.1$

Probability of non defective rivets $= q = 1 - p = 1 - 0.1 = 0.9$

Total number of rivets $= n = 5$

(i) Probability that none is defective = Probability of 0 defective bolt

$$= P(0) = {}^5C_0 (0.1)^0 (0.9)^5 = (0.9)^5 = 0.5905$$

(ii) Probability of 1 defective rivet $= P(1) = {}^5C_1 (0.1)^1 (0.9)^4 = 0.32805$

(iii) Probability of at most 2 defective $= P(0) + P(1) + P(2)$

$$= 0.5905 + 0.32805 + {}^5C_2 (0.1)^2 (0.9)^3$$

$$= 0.5905 + 0.32805 + 0.0729 = 0.99145$$

Example 11 : (i) Out of 800 families with 5 children each, how many would you expect to have (a) 3 boys (b) 5 girls (c) either 2 or 3 boys (d) atleast one boy ? Assume equal probabilities for boys and girls. [JNTU 2005, 2005S, 2008, (K) Dec. 2009, (A) Nov. 2010 (Set No. 1)]

(ii) Out of 800 families with 4 children each, how many families would be expected to have (a) 2 boys and 2 girls (b) atleast one boy (c) no girl (d) at most two girls ? Assume equal probabilities for boys and girls ? [JNTU (K) Nov. 2012 (Set No. 2)]

Solution : (i) Let the number of boys in each family = x

p = The probability of each boy = $\frac{1}{2}$ (since equal probability for boys and girls)

Number of children, $n = 5$

The probability distribution is

$$p(r) = {}^nC_r p^r q^{n-r} = {}^5C_r \left(\frac{1}{2}\right)^r \left(\frac{1}{2}\right)^{5-r} = \frac{1}{2^5} \cdot {}^5C_r \text{ per family}$$

$$(a) \quad P(3 \text{ boys}) = P(r=3) = p(3) = \frac{1}{2^5} \cdot {}^5C_3 = \frac{10}{32} = \frac{5}{16} \text{ per family}$$

Thus for 800 families the probability of number of families having 3 boys

$$= \frac{5}{16}(800) = 250 \text{ families}$$

$$(b) \quad P(5 \text{ girls}) = P(\text{no boys}) = P(r=0) = p(0) = \frac{1}{2^5} \cdot {}^5C_0 = \frac{1}{32} \text{ per family}$$

Thus for 800 families, the probability of number of families having 5 girls

$$= \frac{1}{32}(800) = 25 \text{ families}$$

$$(c) \quad P(\text{either 2 or 3 boys}) = P(r=2) + P(r=3) = p(2) + p(3)$$

$$= \frac{1}{2^5} \cdot {}^5C_2 + \frac{1}{2^5} \cdot {}^5C_3 = \frac{1}{2^5}(10 + 10)$$

$$= \frac{20}{32} = \frac{5}{8} \text{ per family}$$

$$\therefore \text{Expected number of families with 2 or 3 boys} = \frac{5}{8}(800) = 500 \text{ families}$$

$$(d) \quad P(\text{at least one boy}) = P(r=1) + P(r=2) + P(r=3) + P(r=4) + P(r=5)$$

$$= \frac{1}{2^5}({}^5C_1 + {}^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_5) = \frac{1}{2^5}(5 + 10 + 10 + 5 + 1) = \frac{31}{32}$$

$$\therefore \text{Expected number of families with atleast one boy} = \frac{31}{32}(800) = 775$$

(ii) This is left as an exercise to the reader.

Example 12 : A discrete random variable X has the mean 6 and variance 2. If it is assumed that the distribution is binomial find the probability that $5 \leq x \leq 7$. [JNTU 2008 (Set No. 4)]

Solution : We are given

The mean of the Binomial distribution = 6

$$\text{i.e., } np = 6 \quad \dots (1)$$

The variance of the Binomial distribution = 2

$$\text{i.e., } npq = 2 \quad \dots (2)$$

$$\Rightarrow q = \frac{2}{np} = \frac{2}{6}, \text{ using (1) } \Rightarrow q = \frac{1}{3} \quad [\text{or (2) } \div (1) \text{ gives } q = 1/3]$$

$$\therefore p = 1 - q = 1 - \frac{1}{3} = \frac{2}{3}$$

$$\text{From (1), } n = \frac{6}{p} = 6 \times \frac{3}{2} = 9$$

$$\therefore \text{The Binomial distribution is } \left(\frac{2}{3} + \frac{1}{3}\right)^9$$

$$\text{Now } P(5 \leq X \leq 7) = P(5) + P(6) + P(7)$$

$$= {}^9C_5 p^5 q^4 + {}^9C_6 p^6 q^3 + {}^9C_7 p^7 q^2$$

$$= \frac{9!}{5!4!} \left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right)^4 + \frac{9!}{6!3!} \left(\frac{2}{3}\right)^6 \left(\frac{1}{3}\right)^3 + \frac{9!}{2!7!} \left(\frac{2}{3}\right)^7 \left(\frac{1}{3}\right)^2$$

$$= 126 \left(\frac{2^5}{3^9}\right) + 84 \left(\frac{2^6}{3^9}\right) + 36 \left(\frac{2^7}{3^9}\right)$$

$$= \frac{32}{3^9} (126 + 168 + 144) = \frac{14016}{3^9} = 0.7121$$

Example 12 : If X has the binomial distribution with mean 25 and probability of success $1/5$, find $P(X < \mu - 2\sigma)$, where μ and σ^2 are the mean and variance of the distribution?

[JNTU (K) Nov. 2009 (Set No. 3)]

Solution : Given mean, $\mu = 25 = np \quad \dots (1)$

$$\text{and } p = \frac{1}{5}$$

$$(1) \Rightarrow n \left(\frac{1}{5}\right) = 25 \quad \therefore n = 125$$

$$\text{Now } q = 1 - p = 1 - \frac{1}{5} = \frac{4}{5}$$

$$\text{Variance, } \sigma^2 = npq = 125 \times \frac{1}{5} \times \frac{4}{5} = 20$$

$$\therefore \sigma = \sqrt{20} = 4.47$$

$$\text{Now } \mu - 2\sigma = 25 - 8.94 = 16.06$$

$$\begin{aligned} \therefore P(X < \mu - 2\sigma) &= P(X < 16.06) = P\left(z < \frac{x - \mu}{\sigma}\right) = P\left(z < \frac{16.06 - 25}{4.47}\right) \\ &= P(z < -2) = 0.5 - A(2) = 0.5 - 0.4772 \text{ (from normal tables)} \\ &= 0.0228 \end{aligned}$$

Example 24 : In eight throws of a die 5 or 6 is considered a success. Find the mean number of success and the standard deviation. [JNTU 2004 (Set No. 3)]

Solution : p = The probability of success $= p(5) + p(6) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$

$$q = \text{The probability of failure} = 1 - p = 1 - \frac{1}{3} = \frac{2}{3}$$

$$n = \text{Number of throws} = 8$$

$$\therefore \text{Mean} = np = 8 \left(\frac{1}{3}\right) = \frac{8}{3}$$

$$\text{Variance} = npq = (np) q = \left(\frac{8}{3}\right) \left(\frac{2}{3}\right) = \frac{16}{9}$$

$$\text{Hence standard deviation} = \sqrt{\text{variance}} = \sqrt{\frac{16}{9}} = \frac{4}{3}$$

Example 12 : In a binomial distribution consisting of 5 independent trials, probabilities of 1 and 2 success are 0.4096 and 0.2048 respectively. Find the parameter p of the distribution. [JNTU 2004 (Set No. 4)]

Solution : Given n = Number of trials = 5
and $P(X = 1) = 0.4096$, $P(X = 2) = 0.2048$

$$\therefore \frac{P(X=1)}{P(X=2)} = \frac{0.4096}{0.2048}$$

$$\text{i.e., } \frac{{}^5C_1 p q^4}{{}^5C_2 p^2 q^3} = \frac{0.4096}{0.2048}$$

$$\text{i.e., } \frac{5q}{10p} = \frac{0.4096}{0.2048}$$

$$\text{i.e., } 1.024 q = 4.096 p$$

$$\text{i.e., } 1.024 (1 - p) = 4.096 p \quad (\because p + q = 1)$$

$$\text{i.e., } 1.024 = (4.096 + 1.024) p \text{ or } p = \frac{1.024}{5.12} = 0.2$$

Note : $q = 1 - p = 1 - 0.2 = 0.8$

Example 24 : 20% of items produced from a factory are defective. Find the probability that in a sample of 5 chosen at random

- (i) none is defective (ii) one is defective (iii) p ($1 < x < 4$)

[JNTU 2005, 2005S, 2007, (H) Nov. 2009, (A) Nov. 2010 (Set No. 4)]

Solution : Probability of defective items = $p = 20\% = 0.2$

Probability of non defective items = $q = 1 - p = 1 - 0.2 = 0.8$

Total number of items, $n = 5$

- (i) Probability that none is defective = Probability of 0 defective item

$$= p(0) = {}^5C_0 (0.2)^0 (0.8)^5 = (0.8)^5 = 0.32768$$

- (ii) Probability of 1 defective item = $p(1) = {}^5C_1 (0.2)^1 (0.8)^4 = 5(0.2) (0.4096) = 0.4096$

- (iii) p ($1 < x < 4$) = $p(2) + p(3)$

$$= {}^5C_2 (0.2)^2 (0.8)^3 + {}^5C_3 (0.2)^3 (0.8)^2$$

$$= (0.2)^2 (0.8)^2 [10(0.8) + 10(0.2)]$$

$$= 0.0256 (8 + 2) = 0.256$$

Example 12 : Find the maximum n such that the probability of getting no head in tossing a fair coin n times is greater than 0.1.

[JNTU 2004, 2007S (Set No. 4)]

Solution : p = The probability of getting a head = $\frac{1}{2}$

q = The probability of not getting a head = $1 - p = \frac{1}{2}$

Given that $P(X=0) > 0.1$

$$\text{i.e., } {}^nC_0 p^0 q^n > 0.1$$

$$\text{i.e., } q^n > 0.1$$

$$\text{i.e., } \left(\frac{1}{2}\right)^n > 0.1 \Rightarrow 2^n < 10 \Rightarrow n < 4$$

$$[(\text{or}) \text{ For } n = 1, \frac{1}{2} = 0.5 > 0.1]$$

$$\text{For } n = 2, \frac{1}{4} = 0.25 > 0.1$$

$$\text{For } n = 3, \frac{1}{8} = 0.125 > 0.1$$

$$\text{For } n = 4, \frac{1}{16} = 0.0625 < 0.1]$$

Hence the required maximum $n = 3$.

Example 28 : Find the probability that at most 5 defective components will be found in a lot of 200. Experience shows that 2% of such components are defective. Also find the probability of more than five defective components.

[JNTU 2005 (Set No. 4)]

Solution : p = The probability of defective component = 2% = 0.02

n = Number of components in a lot = 200

q = The probability of non-defective component
 $= 1 - p = 1 - 0.02 = 0.98$

- (i) Probability of getting at most 5 defective components = $P(X \leq 5)$
 $= P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5)$
 $= {}^{200}C_0 (0.02)^0 (0.98)^{200} + {}^{200}C_1 (0.02)^1 (0.98)^{199}$
 $+ {}^{200}C_2 (0.02)^2 (0.98)^{198} + {}^{200}C_3 (0.02)^3 (0.98)^{197}$
 $+ {}^{200}C_4 (0.02)^4 (0.98)^{196} + {}^{200}C_5 (0.02)^5 (0.98)^{195}$
 $= (0.98)^{195} [(0.98)^5 + 200 (0.02) (0.98)^4 + 19900 (0.02)^2 (0.98)^3$
 $+ (6600) (199) (0.98)^2 (0.02)^3 + 1650 (199) (197) (0.98) (0.02)^4$
 $+ 330 (199) (197) (196) (0.02)^5]$
 $= (0.98)^{195} [0.904 + 3.689 + 7.4918 + 10.0911 + 10.1426 + 8.1140]$
 $= (0.98)^{195} (40.4325) = 0.7867$
- (ii) Probability of more than 5 defective components
 $= P(X > 5) = 1 - P(X \leq 5) = 1 - 0.7867 = 0.2133$

Example 12 : Assume that 50% of all engineering students are good in Mathematics. Determine the probabilities that among 18 engineering students

- (i) exactly 10 [JNTU (A) Dec. 2017] (ii) at least 10 (iii) at most 8
 (iv) at least 2 and at most 9 are good in mathematics. [JNTU 2005 (Set No.2)]

Solution : Let x be the number of engineering students who are good in Mathematics.

p = The probability of students good in mathematics = 50% = $\frac{1}{2}$

$q = 1 - p = \frac{1}{2}$

n = Number of students = 18

The probability distribution is

$$p(x) = {}^nC_x p^x q^{n-x} = {}^{18}C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{18-x} = {}^{18}C_x \left(\frac{1}{2}\right)^{18}$$

- (i) $P(\text{exactly } 10) = p(10) = \left(\frac{1}{2}\right)^{18} {}^{18}C_{10} = \left(\frac{1}{2}\right)^{18} (43758) = 0.1669235 \approx 0.1670$
- (ii) $P(\text{at least } 10) = P(X \geq 10) = \sum_{x=10}^{18} {}^{18}C_x \left(\frac{1}{2}\right)^{18} = \left(\frac{1}{2}\right)^{18} [{}^{18}C_{10} + {}^{18}C_{11} + \dots + {}^{18}C_{18}]$
- (iii) $P(\text{at most } 8) = P(X \leq 8) = \sum_{x=0}^8 {}^{18}C_x \left(\frac{1}{2}\right)^{18} = \left(\frac{1}{2}\right)^{18} [{}^{18}C_0 + {}^{18}C_1 + \dots + {}^{18}C_8]$

$$\begin{aligned}
 (iv) \quad P(\text{at least 2 and at most 9}) &= P(2 \leq X \leq 9) = \sum_{x=2}^9 {}^{18}C_x \left(\frac{1}{2}\right)^{18} \\
 &= \left(\frac{1}{2}\right)^{18} [{}^{18}C_2 + {}^{18}C_3 + \dots + {}^{18}C_9]
 \end{aligned}$$

Example 10 : The probability of a man hitting a target is $\frac{1}{3}$

(i) If he fires 5 times, what is the probability of his hitting the target at least twice ?

(ii) How many times must he fire so that the probability of his hitting the target at least once is more than 90% ? [JNTU 2006S (Set No. 1)]

Solution : p = The probability of hitting a target = $\frac{1}{3}$

q = The probability of not hitting a target = $1 - \frac{1}{3} = \frac{2}{3}$

n = Number of trials = 5

$$(i) \quad P(\text{at least twice}) = P(X \geq 2) = 1 - P(X < 2) = 1 - [P(X = 0) + P(X = 1)]$$

$$= 1 - \left[{}^5C_0 \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^5 + {}^5C_1 \left(\frac{1}{3}\right)^1 \left(\frac{2}{3}\right)^4 \right]$$

$$= 1 - \left[\left(\frac{2}{3}\right)^5 + \frac{5}{3} \left(\frac{2}{3}\right)^4 \right] = 1 - \left(\frac{2}{3}\right)^4 \left[\frac{2}{3} + \frac{5}{3} \right]$$

$$= 1 - \frac{7}{3} \left(\frac{2}{3}\right)^4 = 0.5391$$

(ii) Given that $P(\text{at least once}) > 90\%$

i.e., $P(X \geq 1) > 0.9$

i.e., $1 - P(X = 0) > 0.9$

i.e., $1 - {}^nC_0 \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^n > 0.9$

or $1 - \left(\frac{2}{3}\right)^n < 0.9$

This is satisfied for $n = 6$ because $1 - \left(\frac{2}{3}\right)^6 = 0.9122 > 0.9$

Example 12 : Determine the probability of getting a sum of 9 exactly twice in 3 throws with a pair of fair dice. [JNTU 2006S, 2007 (Set No.1)]

Solution : In a single throw of a pair of fair dice, a sum of 9 can occur in 4 ways : (3, 6), (4, 5), (5, 4), (6, 3) out of $6 \times 6 = 36$ ways. Thus

$$p = \text{Probability of occurrence of a sum of 9 in one throw} = \frac{4}{36} = \frac{1}{9}$$

$$q = 1 - p = 1 - \frac{1}{9} = \frac{8}{9}$$

$$n = \text{Number of throws} = 3$$

$\therefore$ Probability of getting a sum of 9 exactly twice in 3 throws

$$= {}^3C_2 \left(2; 3, \frac{1}{9}\right) = {}^3C_2 \left(\frac{1}{9}\right)^2 \left(\frac{8}{9}\right)^1 = 3 \times \frac{1}{81} \times \frac{8}{9} = \frac{8}{243} = 0.033$$

Example 32 : A coin is biased in a way that a head is twice as likely to occur as a tail. If the coin is tossed 3 times, find the probability of getting 2 tail and 1 head

[JNTU (H) Dec. 2011 (Set No. 2)]

Solution : Given $P(H) = 2P(T)$. We know that $P(H) + P(T) = 1$

$$\text{Solving } 3P(T) = 1 \Rightarrow P(T) = \frac{1}{3}$$

$$\therefore P(H) = 2P(T) = \frac{2}{3}$$

Let getting a tail is a success and getting a head is a failure. Then $p = 1/3, q = 2/3$.

By Binomial distribution, $P(X = x) = {}^nC_x p^x q^{n-x}$

Here $n = 3, x = 2$

$$\therefore \text{Required probability} = {}^3C_2 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right) = \frac{2}{9}$$

Example 32 : Fit a Binomial distribution to the following frequency distribution :

x	0	1	2	3	4	5	6
f	13	25	52	58	32	16	4

Solution : Here $n = \text{number of trials} = 6$ and $N = \text{total frequency} = \sum f_i = 200$

$$\therefore \text{Mean} = \frac{\sum f_i x_i}{\sum f_i} = \frac{25 + 104 + 174 + 128 + 80 + 24}{200} = \frac{535}{200} = 2.675$$

Now mean of the binomial distribution $= np$

$$\text{i.e., } np = 6p = 2.675 \quad \therefore p = \frac{2.675}{6} = 0.446$$

$$\text{and } q = 1 - p = 1 - 0.446 = 0.554$$

Hence the binomial distribution to be fitted is

$$\begin{aligned} N(q + p)^n &= 200 (0.554 + 0.446)^6 \\ &= 200 [(0.554)^6 + {}^6C_1 (0.554)^5 (0.446) + {}^6C_2 (0.554)^4 (0.446)^2 \\ &\quad + {}^6C_3 (0.554)^3 (0.446)^3 + {}^6C_4 (0.554)^2 (0.446)^4 \\ &\quad + {}^6C_5 (0.554) (0.446)^5 + {}^6C_6 (0.446)^6] \end{aligned}$$

$$\begin{aligned}
 &= 200 [0.02891 + 0.1396 + 0.2809 + 0.3016 + 0.1821 \\
 &\quad + 0.05864 + 0.007866] \\
 &= 5.782 + 27.92 + 56.18 + 60.32 + 36.42 + 11.728 + 1.5732
 \end{aligned}$$

The expected frequencies can be rounded off to the nearest integer to get expected frequencies as whole numbers.

∴ The successive terms in the expansion give the expected or theoretical frequencies which are

x	0	1	2	3	4	5	6
f	13	25	52	58	32	16	4
Expected or Theoretical frequency	6	28	56	60	36	12	2

(since frequencies are always integers).

It may be noted that the expected frequencies are close to observed frequencies. A measure of closeness called 'goodness of fit'.

Example 34 : Fit a Binomial distribution to the following data

x	0	1	2	3	4	5
f	2	14	20	34	22	8

[JNTU 2004S, (A) Nov. 2011(Set No.1)]

Solution : Here $n = 5$ and $N = \sum f_i = 2 + 14 + 20 + 34 + 22 + 8 = 100$

$$\therefore \text{Mean} = \frac{\sum f_i x_i}{\sum f_i} = \frac{0 + 14 + 40 + 102 + 88 + 40}{100} = \frac{284}{100} = 2.84$$

Now the mean of the binomial distribution = np

$$i.e., \quad np = 5p = 2.84 \quad \therefore p = \frac{2.84}{5} = 0.568$$

$$\text{and } q = 1 - p = 0.432$$

Hence the Binomial Distribution to be fitted is

$$\begin{aligned}
 N(q + p)^n &= 100 (0.432 + 0.568)^5 \\
 &= 100 [{}^5C_0(0.432)^5 + {}^5C_1(0.432)^4(0.568) + {}^5C_2(0.432)^3(0.568)^2 \\
 &\quad + {}^5C_3(0.432)^2(0.568)^3 + {}^5C_4(0.432)(0.568)^4 + {}^5C_5(0.568)^5] \\
 &= 100 [0.015 + 0.0989 + 0.260 + 0.341 + 0.224 + 0.059] \\
 &= 1.5 + 9.89 + 26 + 34.1 + 22.4 + 5.9
 \end{aligned}$$

The respective terms of the binomial give the theoretical frequencies. Since frequencies are always integers, therefore, by converting them to the nearest integers, we get 1, 10, 26, 34, 22, 6.

x	0	1	2	3	4	5
f	2	14	20	34	22	8
Expected or Theoretical frequency	2	10	26	34	22	6

Example 35 : Four coins are tossed 160 times. The number of times x heads occur is given below.

x	0	1	2	3	4
No. of times	8	34	69	43	6

[JNTU 2004 S (Set No. 3)]

Fit a binomial distribution to this data on the hypothesis that coins are unbiased.

Solution : The coin is unbiased.

$$\therefore p = \frac{1}{2}, q = \frac{1}{2} \text{ and } n = 4.$$

$$N = \sum f_i = 8 + 34 + 69 + 43 + 6 = 160$$

By Binomial distribution, $p(x) = {}^nC_x p^x q^{n-x}$

We have the recurrence relation

$$p(x+1) = \frac{(n-x)p}{(x+1)q} \cdot p(x) = \frac{4-x}{x+1} \cdot p(x) \left[\because n=4, \frac{p}{q}=1 \right] \quad \dots (1)$$

$$\therefore p(0) = {}^4C_0 p^0 q^4 = {}^4C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^4 = \left(\frac{1}{2}\right)^4$$

No. of heads x	Observed frequency f	Probability $p(x)$	Expected or Theoretical frequency $f(x) = N \cdot p(x)$
0	8	$p(0) = \left(\frac{1}{2}\right)^4$	$f(0) = 160 \cdot p(0)$ $= 160 \times \frac{1}{16} = 10$
1	34	$p(1) = \frac{1}{4}$	$f(1) = 160 \cdot p(1)$ $= 160 \left(\frac{1}{4}\right) = 40$
2	69	$p(2) = \frac{3}{8}$	$f(2) = 160 \cdot p(2)$ $= 160 \left(\frac{3}{8}\right) = 60$
3	43	$p(3) = \frac{1}{4}$	$f(3) = 160 \cdot p(3)$ $= 160 \left(\frac{1}{4}\right) = 40$
4	6	$p(4) = \frac{1}{16}$	$f(4) = 160 \left(\frac{1}{16}\right) = 10$

Example 36 : The probability that John hits a target is $\frac{1}{2}$. He fires 6 times. Find the probability that he hits the target (i) exactly 2 times (ii) more than 4 times (iii) at least once.

[JNTU 2007S, (H) May 2011 (Set No. 4), (A) June 2016]

Solution : Probability of hitting a target = $p = \frac{1}{2}$

Probability of no hit (or failure) = $q = \frac{1}{2}$

Number of trials = $n = 6$

Number of hits (successes) = X

(i) $P(\text{exactly 2 times}) = P(X = 2)$

$$= {}^6C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^4 = \frac{15}{2^6} = 0.2344$$

(ii) $P(\text{more than 4 times}) = P(X > 4) = P(X = 5) + P(X = 6)$

$$\begin{aligned} &= {}^6C_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right) + {}^6C_6 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^0 \\ &= \frac{6}{2^6} + \frac{1}{2^6} = \frac{7}{2^6} = 0.1094 \end{aligned}$$

(iii) $P(\text{at least once}) = P(X \geq 1) = 1 - P(X = 0)$

$$= 1 - {}^6C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^6 = 1 - \frac{1}{2^6} = 0.9844$$

Example 37 : The mean of Binomial distribution is 3 and the variance is $\frac{9}{4}$. Find

(i) the value of n (ii) $P(X \geq 7)$ (iii) $P(1 \leq X < 6)$.

[JNTU 2007, 2008S, (H) Dec. 2009 (Set No. 2)]

Solution : (i) We have

$$\text{Mean of the Binomial distribution} = 3 \Rightarrow np = 3 \quad \dots (1)$$

$$\text{Variance of the Binomial distribution} = \frac{9}{4} \Rightarrow npq = \frac{9}{4} \quad \dots (2)$$

(2) $\div$ (1) gives

$$\frac{npq}{np} = \frac{9}{4 \times 3} \Rightarrow q = \frac{3}{4}$$

$$\therefore p = 1 - q = 1 - \frac{3}{4} = \frac{1}{4}$$

$$\text{From (1), } n = \frac{3}{p} = 12$$

$$\begin{aligned}
(ii) \quad P(X \geq 7) &= 1 - P(X < 7) \\
&= 1 - [P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) \\
&\quad + P(X = 4) + P(X = 5) + P(X = 6)] \\
&= 1 - \left[{}^{12}C_0 \left(\frac{1}{4}\right)^0 \cdot \left(\frac{3}{4}\right)^{12} + {}^{12}C_1 \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^{11} + {}^{12}C_2 \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^{10} \right. \\
&\quad \left. + {}^{12}C_3 \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^9 + {}^{12}C_4 \left(\frac{1}{4}\right)^4 \left(\frac{3}{4}\right)^8 + {}^{12}C_5 \left(\frac{1}{4}\right)^5 \left(\frac{3}{4}\right)^7 + {}^{12}C_6 \left(\frac{1}{4}\right)^6 \left(\frac{3}{4}\right)^6 \right] \\
&= 1 - \left[\left(\frac{3}{4}\right)^{12} + 12 \cdot \frac{1}{4} \cdot \left(\frac{3}{4}\right)^{11} + 66 \cdot \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^{10} + 220 \cdot \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^9 \right. \\
&\quad \left. + 495 \cdot \left(\frac{1}{4}\right)^4 \left(\frac{3}{4}\right)^8 + 72 \cdot \left(\frac{1}{4}\right)^5 \left(\frac{3}{4}\right)^7 + 84 \cdot \left(\frac{1}{4}\right)^6 \left(\frac{3}{4}\right)^6 \right] \\
&= 1 - \left[\frac{3^7}{4^{12}} (243 + 972 + 1782 + 1980 + 1485 + 72 + 28) \right] \\
&= 1 - 0.8554 = 0.1446
\end{aligned}$$

$$\begin{aligned}
(iii) \quad P(1 \leq X < 6) &= P(X = 1) + \dots + P(X = 5) \\
&= {}^{12}C_1 \cdot \frac{1}{4} \cdot \left(\frac{3}{4}\right)^{11} + {}^{12}C_2 \left(\frac{1}{4}\right)^2 \cdot \left(\frac{3}{4}\right)^{10} + {}^{12}C_3 \left(\frac{1}{4}\right)^3 \cdot \left(\frac{3}{4}\right)^9 \\
&\quad + {}^{12}C_4 \left(\frac{1}{4}\right)^4 \cdot \left(\frac{3}{4}\right)^8 + {}^{12}C_5 \left(\frac{1}{4}\right)^5 \left(\frac{3}{4}\right)^7 \\
&= \frac{3^7}{4^{12}} (972 + 1782 + 1980 + 1485 + 72) \\
&= \frac{3^7}{4^{12}} (6291) = 0.82
\end{aligned}$$

Example 38 : The probability that the life of a bulb is 100 days is 0.05. Find the probability that out of 6 bulbs (i) At least one (ii) greater than four (iii) none, will be having a life of 100 days.

[JNTU 2007S, (K) May 2013 (Set No. 3)]

Solution : We have

$$\begin{aligned}
p &= \text{Probability that a bulb is of 100 days life} = 0.05 = \frac{1}{20} \text{ and } n = 6 \\
\therefore q &= 1 - p = 1 - \frac{1}{20} = \frac{19}{20}
\end{aligned}$$

By Binomial distribution,

$$P(X=x) = {}^nC_x p^x q^{n-x} = {}^6C_x \left(\frac{1}{20}\right)^x \left(\frac{19}{20}\right)^{6-x}$$

$$(i) \quad P(x \geq 1) = 1 - P(x=0) = 1 - {}^6C_0 \left(\frac{1}{20}\right)^0 \left(\frac{19}{20}\right)^6 = 1 - \left(\frac{19}{20}\right)^6 = 0.265$$

$$(ii) \quad P(x > 4) = P(x=5) + P(x=6) \\ = {}^6C_5 \left(\frac{1}{20}\right)^5 \cdot \frac{19}{20} + {}^6C_6 \left(\frac{1}{20}\right)^6 \\ = \left(\frac{1}{20}\right)^6 [114 + 1] = \frac{115}{(20)^6}$$

$$(iii) \quad P(x=0) = {}^6C_0 \left(\frac{1}{20}\right)^0 \left(\frac{19}{20}\right)^6 = \left(\frac{19}{20}\right)^6 = 0.7351$$

Example 39 : If the chance that any of the 10 telephone lines is busy at an instant is 0.2. What is the most probable number of busy lines and what is the probability of this number.

[JNTU 2008 (Set No. 4)]

Solution : Given the probability that a line is busy is $p = 0.2$

$\therefore$ The probability that a line is not busy is $q = 1 - p = 1 - 0.2 = 0.8$

On average $np = 10(0.2) = 5$ lines are busy at an instant.

The probability that ' r ' lines are busy is

$$P(r) = {}^{10}C_r (0.2)^r (0.8)^{10-r}$$

$$\therefore P(5 \text{ lines are busy}) = P(5) = {}^{10}C_5 (0.2)^5 (0.8)^{10-5} \\ = {}^{10}C_5 (0.2)^5 (0.8)^5$$

Example 40 : Seven coins are tossed and the number of heads are noted. The experiment is repeated 128 times and the following distribution is obtained.

No. of heads	0	1	2	3	4	5	6	7	Total
Frequency	7	6	19	35	30	23	7	1	128

Fit a binomial distribution assuming. (a) the coin is unbiased.

(b) the nature of the coin is not known.

[JNTU 2009 (Set No. 1)]

Solution : (a) The coin is unbiased.

$$\therefore p = \frac{1}{2}, q = \frac{1}{2} \text{ and } n = 7$$

$$N = \sum f_i = 7 + 6 + 19 + 35 + 30 + 23 + 7 = 128$$

By Binomial distribution, $p(x) = {}^nC_x p^x q^{n-x}$

We have the recurrence relation

$$p(x+1) = \frac{(n-x)p}{(x+1)q} \cdot p(x) = \frac{7-x}{x+1} \cdot p(x) \left[\because n=7, \frac{p}{q}=1 \right]$$

$$\therefore p(0) = {}^7C_0 p^0 q^7 = {}^7C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^7 = \frac{1}{2^7}$$

No. of heads x	Observed frequency	Probability $p(x)$	Expected or Theoretical frequency $f(x) = N \cdot p(x)$
0	7	$p(0) = \frac{1}{2^7}$	$f(0) = 128, p(0) = 128 \times \frac{1}{2^7} = 1$
1	6	$p(1) = 7, p(0) = \frac{7}{2^7}$	$f(1) = 128, p(1) = 128 \times \frac{7}{2^7} = 7$
2	10	$p(2) = 3, p(1) = \frac{21}{2^7}$	$f(2) = 128, p(2) = 128 \times \frac{21}{2^7} = 21$
3	35	$p(3) = \frac{35}{2^7}$	$f(3) = 128, p(3) = 35$
4	30	$p(4) = \frac{35}{2^7}$	$f(4) = 128, p(4) = 35$
5	23	$p(5) = \frac{21}{2^7}$	$f(5) = 128, p(5) = 21$
6	7	$p(6) = \frac{7}{2^7}$	$f(6) = 128, p(6) = 7$
7	1	$p(7) = \frac{1}{2^7}$	$f(7) = 128, p(7) = 1$

(b) The nature of the coin is not known.

We are given $n = 7$ and $N = \sum f_i = 128$

$$\therefore \text{Mean} = \frac{\sum f_i x_i}{\sum f_i} = \frac{6 + 38 + 105 + 120 + 115 + 42 + 7}{128}$$

$$= \frac{433}{128} = 3.383$$

$$\text{i.e., } np = 3.383$$

$$\Rightarrow 7p = 3.383 \quad \therefore p = \frac{3.383}{7} = 0.4833$$

$$\text{and } q = 1 - p = 1 - 0.4833 = 0.5167$$

Hence the Binomial Distribution to be fitted is given by the terms of

$$N(q + p)^n = 128 (0.5167 + 0.4833)^7$$

This information can also be tabulated as in the earlier table.

Example 39 : 30% of items from a factory are defective. Find the probability that in a sample of 8 (i) one (ii) At least two (iii) $p(1 < x < 6)$ are defective.

[JNTU (H) 2010 (Set No. 1)]

Solution : Given probability of a defective item = $30\% = \frac{30}{100} = \frac{3}{10} = 0.3$

$$\therefore q = 1 - p = 1 - 0.3 = 0.7$$

n = Sample size = 8.

For a Binomial Distribution, the probability function is

$$P(X = r) = f(x) = {}^nC_r p^r q^{n-r} = {}^8C_r (0.3)^r (0.7)^{8-r}$$

$$(i) \quad P(X = 1) = {}^8C_1 (0.3) (0.7)^{8-1} = 8(0.3) (0.7)^7 = 0.1976$$

$$\begin{aligned} (ii) \quad P(X \geq 2) &= 1 - P(X < 2) = 1 - [P(X = 0) + P(X = 1)] \\ &= 1 - [{}^8C_0 (0.3) (0.7)^8 + 0.1976] = 1 - [(0.3) (0.7)^8 + 0.1976] \\ &= 1 - [0.0173 + 0.1976] = 1 - 0.3706 = 0.6294. \end{aligned}$$

$$\begin{aligned} (iii) \quad P(1 < X < 6) &= P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5) \\ &= {}^8C_2 (0.3)^2 (0.7)^{8-2} + {}^8C_3 (0.3)^3 (0.7)^{8-3} + {}^8C_4 (0.3)^4 (0.7)^{8-4} + {}^8C_5 (0.3)^5 (0.7)^{8-5} \\ &= \frac{8!}{2!6!} (0.3)^2 (0.7)^6 + \frac{8!}{3!5!} (0.3)^3 (0.7)^5 + \frac{8!}{4!4!} (0.3)^4 (0.7)^4 + \frac{8!}{3!5!} (0.3)^5 (0.7)^3 \\ &= 0.0267 + 0.2541 + 0.1361 + 0.0467 = 0.4636. \end{aligned}$$

Example 42 : The probability that a man hitting a target is $1/3$. If he fires 6 times, find the probability that he fires (i) At the most 5 times (ii) Exactly once (iii) At least two times

[JNTU (H) Nov. 2010 (Set No. 2)]

Solution : Given p = Probability of hitting the target = $\frac{1}{3}$ and $n = 6$.

$$\therefore q = 1 - p = 1 - \frac{1}{3} = \frac{2}{3}$$

For a Binomial Distribution, the probability function is

$$f(x) = P(X = x) = {}^nC_x \cdot p^x \cdot q^{n-x} = {}^6C_x \cdot p^x q^{6-x}$$

$$(i) \quad P(x \leq 5) = 1 - P(x = 6) = 1 - {}^6C_6 \left(\frac{1}{3}\right)^6 \left(\frac{2}{3}\right)^{6-6} = 1 - \frac{1}{729} = \frac{728}{729} = 0.9986$$

$$(ii) \quad P(x = 1) = {}^6C_1 \left(\frac{1}{3}\right) \left(\frac{2}{3}\right)^{6-1} = 2 \left(\frac{2}{3}\right)^5 = 0.2634$$

$$\begin{aligned} (iii) \quad P(x \geq 2) &= 1 - [P(x = 0) + P(x = 1)] = 1 - \left[{}^6C_0 \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^{6-0} + 0.2634 \right] \text{ [from (i)]} \\ &= 1 - \left[6 \left(\frac{2}{3}\right)^6 + 0.2634 \right] = 1 - [0.5267 + 0.2634] = 1 - 0.7901 = 0.2099. \end{aligned}$$

Example 42 : A coin is biased in a way that a head is twice as likely to occur as a tail. If the coin is tossed 3 times, find the probability of getting 2 tail and 1 head

[JNTU (H) Nov. 2010 (Set No. 2)]

Solution : When a coin is tossed, two mutually exclusive events H, T can occur.

We have $P(H) + P(T) = 1$

$$P(H) = 2 P(T) \Rightarrow 3 P(T) = 1$$

$$\Rightarrow P(T) = \frac{1}{3} = p \quad (\text{say probability of success})$$

$$\Rightarrow P(H) = \frac{2}{3} = q \quad (\text{say probability of failure})$$

In a Binomial distribution $n = 3$, Number of successes $= x = 2$.

$\therefore$ Required probability is

$$P(X = x) = {}^nC_x p^x q^{n-x} = {}^3C_2 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^1 = 3 \times \frac{1}{9} \times \frac{2}{3} = \frac{2}{9}.$$

Example 39 : Traffic control engineer reports that 75% of the vehicles passing through a checkpost are from within state. What is the probability that fewer than 4 of the 9 are from out of the state?

[JNTU (H) Nov. 2010 (Set No. 4)]

Solution : The probability that vehicle is from within state $= \frac{75}{100} = \frac{3}{4}$

Considering as binomial distribution, $n = 9$

The probability more than three vehicles are from within state $= P(X \geq 3) = 1 - P(X < 3)$

$$= 1 - P(X = 0) - P(X = 1) - P(X = 2)$$

$$= 1 - {}^9C_0 \left(\frac{3}{4}\right)^0 \left(\frac{1}{4}\right)^9 - {}^9C_1 \left(\frac{3}{4}\right)^1 \left(\frac{1}{4}\right)^8 - {}^9C_2 \left(\frac{3}{4}\right)^2 \left(\frac{1}{4}\right)^7$$

Example 45 : Find the probability of getting an even number 3 or 4 or 5 times in throwing 10 dice. using Binomial Distribution.

[JNTU (H) Apr. 2012 (Set No. 2)]

Solution : Let p be the probability of getting an even number in a throw of a die.

$$\text{Then } p = \frac{3}{6} = \frac{1}{2} \text{ and } q = 1 - p = 1 - \frac{1}{2} = \frac{1}{2}$$

Here $n = 10$

The probability of getting x even numbers in ten throws of a die is

$$P(X=x) = {}^{10}C_x p^x q^{10-x}, x=0,1,2,\dots,10$$

$$= {}^{10}C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{10-x} = {}^{10}C_x \left(\frac{1}{2}\right)^{10}; x=0,1,2,\dots,10$$

$$\therefore P(X=3) = {}^{10}C_3 \left(\frac{1}{2}\right)^{10} = \frac{120}{1024} = 0.112$$

$$P(X=4) = {}^{10}C_4 \left(\frac{1}{2}\right)^{10} = \frac{210}{1024} = 0.2$$

$$P(X=5) = {}^{10}C_5 \left(\frac{1}{2}\right)^{10} = \frac{252}{1024} = 0.246$$

Example 46 : A die is thrown 8 times. If getting a 2 or 4 is a success. Find the probability of (i) 4 success (ii) $P(x \leq 3)$ (iii) $P(x \geq 2)$ [JNTU (A) Nov. 2011 (Set No. 3)]

Solution : We have

$$p = \text{The probability of obtaining a 2 or 4 in one throw (success)} = \frac{2}{6} = \frac{1}{3}$$

$$\therefore q = 1 - p = 1 - \frac{1}{3} = \frac{2}{3}$$

n = Number of trials = 8.

The probability of x successes is given by

$$P(X=x) = p(x) = {}^8C_x p^x q^{8-x}$$

$$(i) \quad p(x=4) = {}^8C_4 p^4 q^{8-4} = {}^8C_4 p^4 q^4$$

$$= \frac{8!}{4!4!} \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^4 = 0.17$$

$$(ii) \quad p(x \leq 3) = p(x=0) + p(x=1) + p(x=2) + p(x=3)$$

$$= q^8 + 8pq^7 + {}^8C_2 p^2 q^6 + {}^8C_3 p^3 q^5$$

$$= \left(\frac{2}{3}\right)^8 + 8\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^7 + 28\left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^6 + 56\left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^5$$

$$= \frac{1}{3^8} [2^8 + 8(2)^7 + 28(2^6) + 56(2^5)]$$

$$= \frac{2^5}{3^8} [8 + 32 + 56 + 56] = \frac{32}{6561} (152) = 0.74$$

$$(iii) \quad p(x \geq 2) = 1 - p(x < 2)$$

$$\begin{aligned} &= 1 - [p(x=0) + p(x=1)] = 1 - [q^8 + 8pq^7] \\ &= 1 - \left[\left(\frac{2}{3}\right)^8 + 8\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^7 \right] = 1 - \frac{1}{3^8} [2^8 + 8 \cdot 2^7] \\ &= 1 - \frac{2^8}{3^8} (1 + 4) = 1 - \left(\frac{2}{3}\right)^8 (5) = 1 - 0.1951 = 0.8049 \end{aligned}$$

Example 47 : In a family of 5 children find the probability that there are

(i) 2 boys (ii) At least one boy (iii) All are boys (iv) No boys.

[JNTU(A) Nov. 2011 (Set No. 4)]

Solution : Here n = number of children = 5

p = The probability of getting a boy = $\frac{1}{2}$ [$\because$ Equal probability for boy and girl]

$$\therefore q = 1 - p = \frac{1}{2}$$

The probability $P(x)$ of x successes in a series of n independent trials is given by

$$P(x) = {}^nC_x p^x q^{n-x}$$

$$\therefore P(x) = {}^5C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{5-x}$$

$$(i) \quad P(2 \text{ boys}) = p(x=2) = {}^5C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{5-2} = \frac{10}{2^5} = \frac{5}{16}$$

$$(ii) \quad P(\text{at least one boy}) = p(x \geq 1) = 1 - p(x=0)$$

$$= 1 - {}^5C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{5-0} = 1 - \frac{1}{32} = \frac{31}{32}$$

$$(iii) \quad P(\text{all are boys}) = p(x=5) = {}^5C_5 \left(\frac{1}{2}\right)^5 = \frac{1}{32}$$

$$(iv) \quad P(\text{no boy}) = p(x=0) = {}^5C_0 \left(\frac{1}{2}\right)^5 = \frac{1}{32}$$

Example 48 : A box contains 9 cards numbered 1 to 9. If four cards are drawn with replacement. What is the probability that none is 1 ?

[JNTU(A) Apr 2012 (Set No. 2)]

Solution : Given $n = 4$

Let $p =$ The probability of getting '1' $= \frac{1}{9}$

Then $q = 1 - p = 1 - \frac{1}{9} = \frac{8}{9}$

The probability p of x successes is given by $P(X = x) = {}^nC_x p^x q^{n-x}$

Hence the probability of getting none is 1 is given by

$$P(X = 0) = {}^4C_0 \left(\frac{1}{9}\right)^0 \left(\frac{8}{9}\right)^4 = \left(\frac{8}{9}\right)^4 = 0.624$$

Example 45 : It has been claimed that in 60% of all solar heat installations the utility bill is reduced by at least one third. Accordingly what are the probabilities that the utility bill will be reduced by at least one - third in (i) fourth of five installations. (ii) at least four of five installations.

[JNTU(A) Apr 2012 (Set No. 4)]

Solution : Let $p =$ The probability that in the solar heat installations the utility bill is reduced by one - third.

$$= 60\% = 0.6$$

Then $q = 1 - p = 1 - 0.6 = 0.4$

- (i) The probability that the utility bill will be reduced by atleast one - third in four of five installations is given by

$$\begin{aligned} P(X = 4) &= {}^5C_4 (0.6)^4 (0.4)^{5-4} \quad [\because P(X = r) = {}^nC_r p^r q^{n-r}] \\ &= 5(0.6)^4 (0.4) = 2(0.6)^4 = 0.2592 \end{aligned}$$

- (ii) The probability that the utility bill will be reduced by atleast one-third in atleast four of five installations is given by

$$\begin{aligned} P(X \geq 4) &= P(X = 4) + P(X = 5) \\ &= 0.2592 + {}^5C_5 (0.6)^5 (0.4)^{5-5}, \text{ using (i)} \\ &= 0.2592 + (0.6)^5 = 0.2592 + 0.07776 = 0.33696 \end{aligned}$$

Example 50 : The probability of a man hitting a target is $\frac{1}{4}$. If he fixes 7 times, the probability of hitting the target at least twice is _____ [JNTU (A) Dec. 2014]

Solution : The probability of a man hitting a target is $p = \frac{1}{4}$

$$\therefore q = 1 - p = 1 - \frac{1}{4} = \frac{3}{4}$$

Also we are given $n = 7$

∴ Required probability = $P(X \geq 2) = 1 - P(X < 2)$

$$= 1 - [P(X = 0) + P(X = 1)] = 1 - {}^nC_0 p^0 q^7 - {}^nC_1 p q^6$$

$$= 1 - {}^7C_0 \left(\frac{3}{4}\right)^7 - {}^7C_1 \frac{1}{4} \left(\frac{3}{4}\right)^6 = 1 - \left(\frac{3}{4}\right)^7 - \frac{7}{4} \left(\frac{3}{4}\right)^6$$

$$= 1 - \frac{3^6}{4^7} [3 + 7] = 1 - \frac{7290}{16384} = \frac{4547}{8192}$$

REVIEW QUESTIONS

1. Define Binomial Distribution.
2. Show that the mean of the Binomial distribution is the product of the parameter p and the number of times n .
3. What is the formula of mean and variance of a Binomial distribution.
4. Find the mean and variance of the Binomial distribution .
5. Prove that the variance of the Binomial distribution is npq .

EXERCISE 7(A)

1. A box contains 9 cards numbered 1 to 9. If four cards are drawn with replacement, find the probability that none is 1. [JNTU (A) Apr. 2012 (Set No. 1)]
2. In tossing ten coins, what is the probability of having exactly 5 heads ?
3. A die is thrown five times. If getting an odd number is a success, find the probability of getting at least four successes.
4. The mean and variance of a binomial distribution are 4 and $4/3$ respectively. Find $P(X \geq 2)$.
5. For a binomial distribution, prove that the variance is less than the mean.
6. The mean and variance of a binomial distribution are 6 and 3 respectively. Find the mode of the binomial distribution.
7. 10% of screws produced in a factory turn out to be defective. Find the probability that in a sample of 10 screws chosen at random, exactly two will be defective.
8. If 10% of bolts produced by a machine are defective, find the probability that out of 10 bolts chosen at random (i) none will be defective (ii) one will be defective, and (iii) at most two bolts will be defective.
9. Eight fair coins are thrown simultaneously. Find the probability of getting (i) exactly six heads (ii) at most six heads.
10. Find the probability of (i) at least 2 heads (ii) at most 4 heads, in a single toss of 6 fair coins.
11. A die is thrown 8 times. Find the probability that 3 will appear (i) exactly 2 times (ii) at least seven times (iii) at least once.

12. The probability that a pen manufactured by a company will be defective is $\frac{1}{10}$. If 12 such pens are manufactured, find the probability that (i) exactly two will be defective (ii) at least two will be defective (iii) none will be defective.
[JNTU (K) Nov. 2012 (Set No. 3)]
13. If the chance that one of the ten telephone lines is busy at an instant is 0.2,
(i) What is the chance that 5 of the lines are busy ?
(ii) What is the most probable number of busy lines and what is the probability of this number ?
(iii) What is the probability that all the lines are busy ?
14. A box contains 100 transistors, 20 of which are defective and 10 are selected at random, find the probability that
(i) all are defective (ii) at least one is defective
(iii) all are good (iv) at most 3 are defective.
15. In a certain city only 50% of the students are capable of doing college work actually go to college. Assuming that this claim is true, find the probability that among 18 such a capable students
(i) exactly 10 will go to college (ii) at least 2 will go to college
(iii) at most 17 will go to college.
16. The probability of a newly generated virus attacked to the computer system will corrupt the 4 files out of 20 files which are opened in an hour. If 20 files are opened in an hour, find the probability that
(i) at least 10 files are corrupted (ii) exactly 3 files are corrupted
(iii) all files are corrupted (iv) all files are safe
(v) more than 2 but not more than 5 files are corrupted.
17. The overall percentage of failures in a certain examination is 20. If six candidates appear in the examination, what is the probability that at least five pass the examination ?
18. A pair of dice is thrown. Find the probability of getting a sum 11 (i) once (ii) twice.
19. Find the probability that in tossing a fair coin 6 times there will appear (i) 0 (ii) 1 (iii) 2 (iv) 3 (v) 4 (vi) 5 and (vii) 6 heads.
20. 5 coins are tossed at a time, 3200 times. Find the frequencies of the distribution of heads and tails and tabulate the results. Also determine the mean number of success and standard deviation.

21. Fit a Binomial Distribution to the following data

x	0	1	2	3	4	5
f	38	144	342	287	164	25

[JNTU 2004S (Set No.2)]

22. The following data show the number of seeds germinating out of 10 on damp filter for 80 set of seeds. Fit a Binomial Distribution to the data.

x	0	1	2	3	4	5	6	7	8	9	10
f	6	20	28	12	8	6	0	0	0	0	0

[JNTU (K) May 2012 (Set No. 4)]

23. 7 coins are tossed at a time, 128 times. The number of heads observed at each throw is recorded and the results are given below :

No. of heads	0	1	2	3	4	5	6	7
Frequency	7	6	19	35	30	23	7	1

Fit a binomial distribution to the data assuming that the coins are (i) biased (ii) unbiased ?

[JNTU 2000 S]

24. Fit a Binomial Distribution to the following frequency data.

x	0	1	2	3	4
f	28	62	46	10	4

[JNTU Jan. 2007]

25. In 800 families with 5 children each, how many families would be expected to have (i) 3 boys and 2 girls (ii) 2 boys and 3 girls (iii) no girl (iv) at the most two girls.

[JNTU Nov. 2012 (Set No. 4)]

26. In a binomial distribution consisting of 5 independent trails, probabilities of 1 and 2 successes are 0.4096 and 0.2048 respectively. Find the parameter 'p' of the distribution.

[JNTU (A) June 2018]

ANSWERS

1. 0.6243 2. 0.2461 3. $\frac{3}{16}$ 4. 0.9822 6. 6 7. 0.1937
8. (i) 0.3487 (ii) 0.3874 (iii) 0.9298 9. (i) 0.109375 (ii) 0.9648
10. (i) 0.890625 (ii) 0.65625 11. (i) 0.26 (ii) 0.0000244 (iii) 0.7674
12. (i) 0.2301 (ii) 0.3412 (iii) 0.2833
13. (i) 0.02642 (ii) 2, 0.0457 (iii) 1.023×10^{-7}
14. (i) $\frac{1}{5^{10}}$ (ii) 0.8926 (iii) 0.1074 (iv) 0.879128
15. (i) 0.1669 (ii) 0.9999 (iii) 0.999999
16. (i) 0.000004526 (ii) 0.00107374 (iii) $\frac{1}{5^{12}}$ (iv) 0.999999 (v) 0.422249
17. 0.65536 18. (i) 0.105 (ii) 0.00308 19. $\frac{1}{2^6}, \frac{6}{2^6}, \frac{15}{2^6}, \frac{20}{2^6}, \frac{15}{2^6}, \frac{6}{2^6}, \frac{1}{2^6}$
20.

x	0	1	2	3	4	5
$p(x)$	100	500	1000	1000	500	100

 ; $\mu = 2.5, \sigma = 1.118$
21. 33, 162, 316, 309, 151, 29 22. $80 [0.8725 + 0.2175]^{10}$
23. (i) $128 [0.517 + 0.483]^7$ (ii) 1, 7, 21, 35, 35, 21, 7, 1

POISSON DISTRIBUTION

7.7 POISSON DISTRIBUTION

Poisson Distribution due to French mathematician Simeon Denis Poisson (1837) is a discrete probability distribution.

S. D. Poisson introduced Poisson distribution as a rare distribution of rare events *i.e.*, the events whose probability of occurrence is very small but the number of trials which could lead to the occurrence of the event, are very large.

Suppose that, in a large number of parts manufactured by a machine, samples of 100 parts are taken from a product which is 2% defective. Then the probabilities of 0, 1, 2, ..., 99, 100 defectives in a single sample are given by the successive terms of the Binomial distribution $(0.98 + 0.02)^{100}$. Expanding the binomial in full will be a tedious process. This difficulty can be avoided by the Poisson distribution.

DERIVATION OF THE POISSON DISTRIBUTION

[JNTU 2004, 2006S, (A) Dec. 2009, (K) 2010S, (H) May 2011, (K) Nov. 2011 (Set No. 3)]

The Poisson distribution can be derived as a limiting case of the Binomial Distribution under the following conditions:

- (i) p , the probability of the occurrence of the event is very small.
- (ii) n is very very large, where n is number of trials *i.e.*, $n \rightarrow \infty$.
- (iii) np is a finite quantity, say $np = \lambda$, then λ is called the parameter of the Poisson distribution.

In the Binomial distribution, the probability $P(r)$ or $P(X = r)$ of r successes in a series of independent trials is given by

$$\begin{aligned} P(r) &= {}^nC_r p^r q^{n-r} = {}^nC_r p^r (1-p)^{n-r} \\ &= \frac{n(n-1)(n-2)\dots(n-r+1)}{r!} \cdot p^r \cdot \frac{(1-p)^n}{(1-p)^r} \end{aligned} \quad \dots (1)$$

Put $np = \lambda$. Then $n = \frac{\lambda}{p}$

Hence (1) becomes,

$$\begin{aligned} P(r) &= \frac{\frac{\lambda}{p} \cdot \left(\frac{\lambda}{p} - 1\right) \left(\frac{\lambda}{p} - 2\right) \dots \left(\frac{\lambda}{p} - r + 1\right)}{r!} \cdot p^r \cdot \frac{(1-p)^n}{(1-p)^r} \\ &= \frac{\lambda(\lambda-p)(\lambda-2p)\dots[\lambda-(r-1)p]}{r! p^r} \cdot p^r \cdot \frac{(1-p)^n}{(1-p)^r} \\ &= \frac{\lambda(\lambda-p)(\lambda-2p)\dots[\lambda-(r-1)p]}{r!} \cdot \frac{\left(1 - \frac{\lambda}{n}\right)^n}{(1-p)^r} \end{aligned} \quad \dots (2)$$

As $n \rightarrow \infty$, $p \rightarrow 0$, so that $np = \lambda$, we have

$$\begin{aligned}
 P(r) &= \frac{\lambda \cdot \lambda \dots r \text{ factors}}{r!} \cdot \lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n \cdot \lim_{p \rightarrow 0} \frac{1}{(1-p)^r} \\
 &= \frac{\lambda^r}{r!} \cdot \lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n \left[\because \lim_{p \rightarrow 0} (1-p)^r = 1 \text{ for a given } r \right] \\
 &= \frac{\lambda^r}{r!} e^{-\lambda} \left[\because \lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n = \lim_{n \rightarrow \infty} \left[\left(1 - \frac{\lambda}{n}\right)^{-n/\lambda}\right]^{-\lambda} = e^{-\lambda} \right]
 \end{aligned}$$

$$\therefore P(r) = \text{Probability of } r \text{ successes} = \frac{e^{-\lambda} \cdot \lambda^r}{r!}$$

This is known as Poisson distribution.

Putting $r = 0, 1, 2, \dots$ etc., the probabilities of 0, 1, 2, ..., r , ... successes are given by $e^{-\lambda}, \lambda e^{-\lambda}, \frac{\lambda^2}{2!} e^{-\lambda}, \dots, \frac{\lambda^r}{r!} e^{-\lambda}, \dots$ respectively, where $\lambda (> 0)$ is a parameter.

The sum of these probabilities is unity as it should be.

7.8 DEFINITION. POISSON DISTRIBUTION

[JNTU 2006S, 2008S, (A), (H) Dec. 2009 (Set No. 4)]

A random variable X is said to follow a Poisson distribution if it assumes only non-negative values and its probability density function is given by

$$p(x, \lambda) = P(X = x) = \begin{cases} \frac{e^{-\lambda} \cdot \lambda^x}{x!}; & x = 0, 1, 2, 3, \dots \\ 0, & \text{otherwise} \end{cases} \quad \dots (1)$$

Here $\lambda > 0$ is called the parameter of the distribution.

It is a distribution suitable for rare events for which the probability of occurrence p is very small and the number of trials n is very large where np is finite. For example, consider the number of persons born blind per year in a large city. Individually, being born blind is a rare event and its probability is very small. But in a large city the total number of births every year is high. As a result, we may get some blind births.

Note : 1. It should be noted that

$$\sum_{x=0}^{\infty} P(X = x) = \sum_{x=0}^{\infty} \frac{e^{-\lambda} \cdot \lambda^x}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = e^{-\lambda} \cdot e^{\lambda} = 1$$

Hence equation (1) is a probability function.

2. The Poisson Distribution function is

$$F_X(x) = P(X \leq x) = \sum_{r=0}^x p(r) = e^{-\lambda} \sum_{r=0}^x \frac{\lambda^r}{r!}, \quad x = 0, 1, 2, \dots$$

Examples of Poisson Distribution :

- (i) The number of defective electric bulbs manufactured by a reputed company.
- (ii) The number of telephone calls per minute at a switch board.
- (iii) The number of cars passing a certain point in one minute.
- (iv) The number of printing mistakes per page in a large text.
- (v) The number of particles emitted by a radio-active substance.
- (vi) The number of persons born blind per year in a large city.

Conditions of Poisson Distribution :

The Poisson Distribution is used under the following conditions :

1. The variable (number of occurrences) is a discrete variable.
2. The occurrences are rare.
3. The number of trials (n) is large.
4. The probability of success (p) is very small (very close to zero)
5. $np = \lambda$ is finite.

7.9 CONSTANTS OF THE POISSON DISTRIBUTION**1. The Mean of the Poisson Distribution**

[JNTU (H) Nov. 2009 (Set No. 4), (K) May 2012, Dec. 2013, Mar. 2014, (A) Dec. 2014]

$$\begin{aligned}
 \text{Mean} = E(X) &= \sum_{x=0}^{\infty} x \cdot p(x) = \sum_{x=0}^{\infty} x \cdot \frac{e^{-\lambda} \cdot \lambda^x}{x!} = e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!} \quad [\because x! = x(x-1)!] \\
 &= e^{-\lambda} \sum_{y=0}^{\infty} \frac{\lambda^{y+1}}{y!} \quad (\text{Putting } y = x-1) \\
 &= \lambda e^{-\lambda} \sum_{y=0}^{\infty} \frac{\lambda^y}{y!} = e^{-\lambda} \cdot \lambda \cdot e^{\lambda} \left[\because \sum_{y=0}^{\infty} \frac{\lambda^y}{y!} = e^{\lambda} \right] \\
 &= \lambda (= np)
 \end{aligned}$$

Thus the parameter λ is the Arithmetic Mean of the Poisson distribution.

2. Variance of Poisson Distribution

[JNTU (H) Nov. 2009 (Set No. 4), (K) May 2012, Dec. 2013, Mar. 2014 (Set No. 1), (A) Dec. 2014]

$$\begin{aligned}
 V(X) &= E(X^2) - [E(X)]^2 \\
 &= \sum_{x=0}^{\infty} x^2 p(x) - \lambda^2 \quad [\because \lambda = \text{mean of P.D.}] \\
 &= \sum_{x=0}^{\infty} x^2 \cdot \frac{e^{-\lambda} \cdot \lambda^x}{x!} - \lambda^2 = e^{-\lambda} \sum_{x=1}^{\infty} \frac{x \cdot \lambda^x}{(x-1)!} - \lambda^2 \\
 &= e^{-\lambda} \sum_{x=1}^{\infty} [(x-1) + 1] \cdot \frac{\lambda^x}{(x-1)!} - \lambda^2
 \end{aligned}$$

$$\begin{aligned}
&= e^{-\lambda} \left[\sum_{x=1}^{\infty} (x-1) \cdot \frac{\lambda^x}{(x-1)(x-2)!} + \sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!} \right] - \lambda^2 \\
&= e^{-\lambda} \left[\sum_{x=2}^{\infty} \frac{\lambda^x}{(x-2)!} + \sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!} \right] - \lambda^2 \\
&= e^{-\lambda} \left[\sum_{y=0}^{\infty} \frac{\lambda^{y+2}}{y!} + \sum_{z=0}^{\infty} \frac{\lambda^{z+1}}{z!} \right] - \lambda^2 \quad (\text{Putting } y = x-2, z = x-1) \\
&= e^{-\lambda} \left[\lambda^2 \sum_{y=0}^{\infty} \frac{\lambda^y}{y!} + \lambda \sum_{z=0}^{\infty} \frac{\lambda^z}{z!} \right] - \lambda^2 = e^{-\lambda} [\lambda^2 \cdot e^{\lambda} + \lambda e^{\lambda}] - \lambda^2 = (\lambda^2 + \lambda) - \lambda^2 = \lambda
\end{aligned}$$

Thus Variance = λ

Hence the Variance of the distribution = Mean of the distribution = λ

Further, Standard deviation of the Poisson distribution, $\sigma = \sqrt{\lambda}$

3. Mode of the Poisson Distribution

Mode is the value of x for which the probability $p(x)$ is maximum.

$$\therefore p(x) \geq p(x+1) \text{ and } p(x) \geq p(x-1)$$

Now $p(x) \geq p(x+1)$

$$\Rightarrow \frac{e^{-\lambda} \cdot \lambda^x}{x!} \geq \frac{e^{-\lambda} \cdot \lambda^{x+1}}{(x+1)!}$$

$$\Rightarrow 1 \geq \frac{\lambda}{x+1} \text{ or } \frac{\lambda}{x+1} \leq 1 \Rightarrow \lambda \leq x+1 \text{ or } x+1 \geq \lambda$$

$$\Rightarrow x \geq \lambda - 1 \quad \dots (1)$$

$$\text{Similarly, } p(x) \geq p(x-1) \Rightarrow x \leq \lambda \quad \dots (2)$$

Combining (1) and (2), we have

$$\lambda - 1 \leq x \leq \lambda$$

Hence mode of the Poisson distribution lies between $\lambda - 1$ and λ .

Case 1 : If λ is an integer then $\lambda - 1$ is also an integer. So we have two maximum values and the distribution is bimodal and the two modes are $(\lambda - 1)$ and λ .

Case 2 : If λ is not an integer, the mode of Poisson distribution is integral part of λ .

4. Recurrence Relation for the Poisson Distribution

$$\text{We have } p(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

$$\therefore p(x+1) = \frac{e^{-\lambda} \cdot \lambda^{x+1}}{(x+1)!} = \frac{\lambda}{x+1} \cdot \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{\lambda}{x+1} p(x)$$

Thus $p(x+1) = \left(\frac{\lambda}{x+1}\right) p(x)$

or $p(x) = \frac{\lambda}{x} \cdot p(x-1)$

which is the required relation. With this formula we can find $p(1), p(2), p(3), \dots$ if $p(0)$ is given.

Note : 1. On successive application of the formula, we get

$$p(x+1) = \frac{\lambda^{x+1}}{(x+1)!} \cdot p(0)$$

$$2. P(X \leq t) = \left[1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \dots + \frac{\lambda^t}{t!} \right] \cdot p(0)$$

$$3. P(a \leq X \leq b) = \left[\frac{\lambda^a}{a!} + \frac{\lambda^{a+1}}{(a+1)!} + \dots + \frac{\lambda^b}{b!} \right] \cdot p(0), \text{ for all } 0 \leq a, b, a, b \in Z^+$$

Properties of Poisson Distribution

1. Range of the variable is from 0 to ∞ .
2. Mean and Variance are equal.
3. Distribution gets more and more symmetrical about the mean as λ increases and tends to normal distribution, described in the next section.

SOLVED EXAMPLES

Example 1 : If the probability that an individual suffers a bad reaction from a certain injection is 0.001, determine the probability that out of 2000 individuals (i) exactly 3 (ii) more than 2 individuals (iii) none (iv) more than one individual suffer a bad reaction.

Solution : Given $p = 0.001$ and $n = 2000$

$\therefore$ Mean, $\lambda = np = 2000(0.001) = 2$

$$\text{Thus } P(r) = \frac{e^{-\lambda} \cdot \lambda^r}{r!} = \frac{e^{-2} \cdot 2^r}{r!} = \frac{2^r}{e^2 r!}$$

$$(i) \quad P(3) = \frac{1}{e^2} \cdot \frac{2^3}{3!} = \frac{8}{6(2.718)^2} = 0.1804$$

$$(ii) \quad P(\text{more than } 2) = P(r \geq 2) = P(3) + P(4) + \dots + P(2000)$$

$$= 1 - [P(0) + P(1) + P(2)] = 1 - \left[\frac{1}{e^2} + \frac{2}{e^2} + \frac{4}{2e^2} \right]$$

$$= 1 - \frac{1}{e^2} (1 + 2 + 2) = 1 - \frac{5}{e^2} = 1 - 0.67667 = 0.3233$$

$$(iii) \quad P(\text{none}) = P(0) = \frac{2^0}{e^2 0!} = \frac{1}{e^2} = 0.1353$$

$$(iv) \quad P(\text{more than one}) = P(r > 1) = P(2) + P(3) + \dots + P(2000) = 1 - [P(0) + P(1)]$$

$$= 1 - \left(\frac{1}{e^2} + \frac{2}{e^2} \right) = 1 - \frac{3}{e^2} = 1 - 0.406 = 0.594$$

Example 2 : A car-hire firm has two cars which it hires out day by day. The number of demands for a car on each day is distributed as a Poisson distribution with mean 1.5. Calculate the proportion of days (i) on which there is no demand (ii) on which demand is refused. [JNTU 2003, (K) May 2013 (Set No. 4)]

Solution : Given mean, $\lambda = 1.5$

$$\text{We have } P(r) = \frac{e^{-\lambda} \cdot \lambda^r}{r!}$$

$$(i) \quad P(\text{no demand}) = P(0) = \frac{e^{-1.5}(1.5)^0}{0!} = e^{-1.5} = 0.2231$$

Note : Number of days in a year there is no demand of car = $365 (0.2231) = 81$ days

(ii) Some demand is refused if the number of demands is more than two i.e., $r > 2$

$$P(\text{demand refused}) = P(r > 2) = 1 - [P(0) + P(1) + P(2)]$$

$$= 1 - \left[e^{-1.5} + \frac{e^{-1.5}(1.5)}{1!} + \frac{e^{-1.5}(1.5)^2}{2!} \right]$$

$$= 1 - e^{-1.5} [1 + 1.5 + 1.125]$$

$$= 1 - 3.625 (e^{-1.5}) = 1 - 0.8088 = 0.1913$$

Note : Number of days in a year when some demand is refused
 $= 365 \times 0.1913 = 69.82 = 70$ days

Example 3 : A hospital switch board receives an average of 4 emergency calls in a 10 minute interval. What is the probability that (i) there are at most 2 emergency calls in a 10 minute interval (ii) there are exactly 3 emergency calls in a 10 minute interval.

[JNTU 2000, (K) May 2010S (Set No. 3)]

Solution : Mean, $\lambda = (4 \text{ calls} / 10 \text{ minutes}) = 4$ calls

$$\therefore P(X = x) = p(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-4} \cdot 4^x}{x!} = \frac{1}{e^4} \cdot \frac{4^x}{x!}$$

$$(i) \quad P(\text{at most 2 calls}) = P(X \leq 2)$$

$$= P(X = 0) + P(X = 1) + P(X = 2)$$

$$= \frac{1}{e^4} + \frac{1}{e^4} \cdot 4 + \frac{1}{e^4} \cdot \frac{4^2}{2!}$$

$$= \frac{1}{e^4} (1 + 4 + 8) = 13 e^{-4} = 0.2381$$

$$(ii) \quad P(\text{exactly 3 calls}) = P(X = 3) = \frac{1}{e^4} \cdot \frac{4^3}{3!} = \frac{32}{3} e^{-4} = 0.1954$$

Example 4 : A manufacturer knows that the condensers he makes contain on average 1% defectives. He packs them in boxes of 100. What is the probability that a box picked at random will contain 3 or more faulty condensers ?

Solution : Here p = Probability of defective condensers = 1% = 0.01

n = Total number of condensers = 100

$\therefore \lambda$ = mean = np = 100 (0.01) = 1

From Poisson Distribution,

$$P(X = x) = p(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-1} \cdot 1^x}{x!} = \frac{e^{-1}}{x!}$$

$$P(X \geq 3) = 1 - P(X < 3) = 1 - [P(X = 0) + P(X = 1) + P(X = 2)]$$

$$= 1 - [e^{-1} + e^{-1} + \frac{e^{-1}}{2}] = 1 - e^{-1} \cdot \frac{5}{2}$$

$$= 1 - 0.9197 = 0.0803$$

Example 4 : If a bank received on the average 6 bad cheques per day, find the probability that it will receive 4 bad cheques on any given day.

Solution : We have $P(X = x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$

Here $\lambda = 6$

$$\therefore P(X = 4) = \frac{e^{-6} 6^4}{4!} = \frac{54}{e^6} = 0.1339$$

Example 6 : A manufacturer of cotter pins knows that 5% of his product is defective. Pins are sold in boxes of 100. He guarantees that not more than 10 pins will be defective. What is the approximate probability that a box will fail to meet the guaranteed quality ?

[JNTU 2000S, (K) May 2013 (Set No. 3)]

Solution : The probability of cotter pins to be defective = p = 5% = 0.05

Total number of cotter pins, n = 100

$\therefore$ Mean, λ = np = 100 (0.05) = 5

We have $P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$

$$\therefore P(X = x) = \frac{e^{-5} 5^x}{x!}$$

$P(\text{a box will fail to meet the guarantee}) = P(X > 10)$

$$= 1 - P(X \leq 10)$$

$$= 1 - [P(X = 0) + P(X = 1) + \dots + P(X = 10)]$$

$$= 1 - \left[\frac{e^{-5}(5)^0}{0!} + \frac{e^{-5}(5)^1}{1!} + \frac{e^{-5}(5)^2}{2!} + \frac{e^{-5}(5)^3}{3!} + \dots + \frac{e^{-5}(5)^{10}}{10!} \right]$$

$$= 1 - 0.9863 = 0.0137$$

Example 3 : If the probability is 0.05 that a certain wide-flange column will fail under a given axial load. What are the probability that among 16 such columns

(i) at most two will fail.

(ii) at least four will fail.

[JNTU 2008 (Set No. 4)]

Solution : Probability that a certain column will fail under a given axial load = .05 = p

Number of columns = 16 = n

Now $\lambda = np = 16 (.05) = 0.80$

$$\therefore P(r \text{ columns will fail}) = \frac{e^{-\lambda} \lambda^r}{r!}$$

$$(i) P(\text{at most two will fail}) = P(0) + P(1) + P(2)$$

$$= e^{-\lambda} \left(1 + \lambda + \frac{\lambda^2}{2!} \right) \text{ where } \lambda = 0.8$$

$$(ii) P(\text{at least four will fail}) = P(4) + P(5) + \dots + P(16)$$

$$= 1 - [P(0) + P(1) + P(2) + P(3)]$$

$$= 1 - e^{-\lambda} \left[1 + \lambda + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} \right]$$

(It is assumed that the probability distribution is Poisson; as p is very small).

Example 8 : It has been found that 2% of the tools produced by a certain machine are defective. What is the probability that in a shipment of 400 such tools

(a) 3% or more

(b) 2% or less will prove defective.

[JNTU (A) May 2010S]

Solution : We are given $n = 400$.

Let p = Probability of a defective tool = 2% = 0.02 .

$\therefore \lambda$ = Mean number of defective tools in a shipment of 400

$$= np = 400 \times 0.02 = 8$$

Since ' p ' is small, we may use Poisson distribution.

Probability of x defective tools in a shipment of 400 is

$$P(X = x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-8} 8^x}{x!}, x = 0, 1, 2, \dots$$

$$(a) P(X \geq 3\%) = P(X \geq 12) = 1 - P(X < 12)$$

$$= 1 - [P(X = 0) + P(X = 1) + \dots + P(X = 11)]$$

$$\begin{aligned}
&= 1 - \left[e^{-8} + \frac{e^{-8} \cdot 8}{1!} + \frac{e^{-8} \cdot 8^2}{2!} + \dots + \frac{e^{-8} \cdot 8^{11}}{11!} \right] \\
&= 1 - e^{-8} \left[1 + 8 + 32 + \frac{8^3}{3!} + \frac{8^4}{4!} + \frac{8^5}{5!} + \frac{8^6}{6!} + \frac{8^7}{7!} + \frac{8^8}{8!} + \frac{8^9}{9!} + \frac{8^{10}}{10!} + \frac{8^{11}}{11!} \right] \\
&= 1 - (e^{-8}) [41 + 85.33 + 170.66 + 273.06 + 364.09 \\
&\quad + 416.10 + 416.10 + 369.87 + 295.89 + 215.19] \\
&= 1 - (e^{-8}) (2647.29) = 0.1119
\end{aligned}$$

Example 9 : If a random variable has a poisson distribution such that $P(1) = P(2)$,
 find (i) mean of the distribution [JNTU (A) June 2016]
 (ii) $P(4)$ (iii) $P(x \geq 1)$ (iv) $P(1 < x < 4)$ [JNTU 2007S (Set No. 2, 3)]
 (or) A random variable x takes on the values 0, 1 and 2. If $P(x=1) = P(x=2)$ and
 $P(x=0) = 0.4$ then find the mean of the random variable x . [JNTU (A) June 2019]

Solution : Given $P(1) = P(2) \Rightarrow \frac{P(2)}{P(1)} = 1 \quad \dots (1)$

By recurrence relation, we have $p(x+1) = \frac{\lambda}{x+1} p(x)$

Put $x = 1$. Then $p(2) = \frac{\lambda}{2} p(1)$

$\Rightarrow \frac{p(2)}{p(1)} = \frac{\lambda}{2} \therefore \frac{\lambda}{2} = 1$, using (1)

or $\lambda = 2$

Hence mean of the distribution = 2

Alternate Method :

$P(1) = P(2)$

Hence $\frac{e^{-\lambda} \lambda^1}{1!} = e^{-\lambda} \frac{\lambda^2}{2!}$

$\therefore \lambda^2 = 2\lambda \Rightarrow \lambda = 0 \text{ or } 2$

But $\lambda \neq 0 \therefore \lambda = 2$

(ii) $p(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-2} \cdot 2^x}{x!}$

$P(X=4) = p(4) = \frac{e^{-2} \cdot 2^4}{4!} = \frac{2}{3e^2} = 0.09022$

(iii) $P(x \geq 1) = 1 - P(x < 1) = 1 - P(x = 0)$

$= 1 - \frac{e^{-2} \cdot 2^0}{0!} = 1 - \frac{1}{e^2} = 0.8647$

(iv) $P(1 < x < 4) = P(x = 2) + P(x = 3)$

$= \frac{e^{-2} \cdot 2^2}{2!} + \frac{e^{-2} \cdot 2^3}{3!} = 2e^{-2} + \frac{4}{3} e^{-2} = e^{-2} \left(\frac{10}{3} \right) = 0.4511$

Example 10 : Using Poisson's distribution, find the probability that the ace of spades will be drawn from a pack of well shuffled cards at least once in 104 consecutive trials.

Solution : Here $p = \frac{1}{52}$ and $n = 104$

$\therefore$ Mean of the distribution, $\lambda = np = \frac{104}{52} = 2$

$P(\text{at least once}) = P(X \geq 1) = 1 - P(X = 0)$

$$= 1 - \frac{e^{-2}(2)^0}{0!} = 1 - 0.1353 = 0.8647$$

Example 11 : The average number of phone calls / minute coming into a switch board between 2 p.m and 4 p.m is 2.5. Determine the probability that during one particular minute there will be (i) 4 or fewer (ii) more than 6 calls. [JNTU 2006 S, 2007 (Set No. 1)]

Solution : Let x be the number of phone calls / minute coming into a switch board.

Given mean, $\lambda = 2.5$

Now the Poisson distribution is

$$p(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-2.5} \cdot (2.5)^x}{x!}$$

$$(i) \quad p(x \leq 4) = p(x = 0) + p(x = 1) + p(x = 2) + p(x = 3) + p(x = 4)$$

$$= e^{-2.5} \left[\frac{(2.5)^0}{0!} + \frac{(2.5)^1}{1!} + \frac{(2.5)^2}{2!} + \frac{(2.5)^3}{3!} + \frac{(2.5)^4}{4!} \right]$$

$$= e^{-2.5} [1 + 2.5 + 3.125 + 2.6042 + 1.6276]$$

$$= (0.080209) (10.8568) = 0.8912$$

$$(ii) \quad p(x > 6) = 1 - p(x \leq 6) = 1 - e^{-2.5} \left[\frac{(2.5)^0}{0!} + \frac{(2.5)^1}{1!} + \dots + \frac{(2.5)^6}{6!} \right]$$

$$= 1 - e^{-2.5} [1 + 2.5 + 3.125 + 2.6042 + 1.6276 + 0.8138 + 0.3391]$$

$$= 1 - (0.080209) (12.01) = 0.01416$$

Example 11 : 2% of the items of a factory are defective. The items are packed in boxes. What is the probability that there will be

(i) 2 defective items (ii) at least three defective items

in a box of 100 items ?

[JNTU 2006S, (K) Nov. 2011 (Set No. 4)]

Solution : Given $n = 100$

and $p =$ The probability of defective items $= 2\% = 0.02$

$\lambda =$ Mean number of defective items in a box of 100 $= np = (0.02) (100) = 2$

Since p is small, we may use Poisson distribution. Probability of x defective items in a box of 100 is

$$P(X = x) = p(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-2} \cdot 2^x}{x!};$$

$$(i) \quad p(2 \text{ defective items}) = p(2) = \frac{e^{-2} \cdot 2^2}{2!} = \frac{2}{e^2} = 0.2706$$

$$(ii) \quad p(\text{at least } 3) = p(x \geq 3) = 1 - [p(x = 0) + p(x = 1) + p(x = 2)]$$

$$= 1 - e^{-2} \left[\frac{2^0}{0!} + \frac{2^1}{1!} + \frac{2^2}{2!} \right]$$

$$= 1 - e^{-2} [1 + 2 + 2] = 1 - 5e^{-2} = 0.3233$$

Example 11 : Average number of accidents on any day on a national highway is 1.8. Determine the probability that the number of accidents are (i) at least one (ii) at most one
[JNTU 2004S, 2005, 2007(Set No. 3)]

Solution : Mean, $\lambda = 1.8$

$$\text{We have } P(X=x) = p(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-1.8} \cdot (1.8)^x}{x!}$$

$$(i) \quad P(\text{at least one}) = P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-1.8} = 1 - 0.1653 = 0.8347$$

$$(ii) \quad P(\text{at most one}) = P(X \leq 1) = P(X = 0) + P(X = 1) \\ = e^{-1.8} + e^{-1.8} (1.8) = e^{-1.8} (2.8) = 0.4628$$

Example 14 : Average number of accidents on any day on a national highway is 1.6. Determine the probability that the number of accidents are

i) at least one ii) at most one

[JNTU 2008 (Set No. 2)]

Solution : Given Mean, $\lambda = 1.6$. We have

$$p(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-1.6} (1.6)^x}{x!}$$

$$(i) \quad P(\text{at least one}) = P(X \geq 1) = 1 - P(X = 0) \\ = 1 - e^{-1.6} = 1 - 0.2019 = 0.7981$$

$$(ii) \quad P(\text{at most one}) = P(X \leq 1) = P(X = 0) + P(X = 1) \\ = e^{-1.6} + e^{-1.6} (1.6) = e^{-1.6} (2.6) \\ = (0.2019) (2.6) = 0.5249$$

Example 15 : Number of monthly breakdowns of a computer is a random variable having Poisson distribution with mean equal to 1.8. Find the probability that the computer will function for a month (i) without a breakdown (ii) with only one breakdown and (iii) with at least one.
[JNTU (K) 2009 (Set No. 3)]

Solution : Given mean, $\lambda = 1.8$

$$\text{From Poisson distribution, } P(X=x) = p(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{(e^{-1.8})(1.8)^x}{x!}$$

$$(i) \quad p(X=0) = \frac{(e^{-1.8})(1.8)^0}{0!} = e^{-1.8} = 0.1653$$

$$(ii) \quad p(X=1) = (e^{-1.8})(1.8) = 0.2975$$

$$(iii) \quad p(X \geq 1) = 1 - p(X < 1) = 1 - p(X = 0) = 1 - 0.1653 = 0.8347$$

Example 11 : If X is a Poisson variate such that $p(x=0) = p(x=1)$, find $p(x=0)$ and using recurrence formula find the probabilities at $x = 1, 2, 3, 4$, and 5.

[JNTU 2004S (Set No. 4)]

Solution : Probability distribution is $p(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$

Given $p(x = 0) = p(x = 1)$

$$\Rightarrow \frac{e^{-\lambda} \cdot \lambda^0}{0!} = \frac{e^{-\lambda} \cdot \lambda}{1!} \quad \text{i.e., } e^{-\lambda} = \lambda e^{-\lambda} \quad \therefore \lambda = 1$$

(i) Now $p(x = 0) = \frac{e^{-\lambda} \cdot \lambda^0}{0!} = e^{-1} = 0.3678$

(ii) By Recurrence formula, we have

$$p(x + 1) = \frac{\lambda}{x + 1} p(x)$$

$$\therefore p(x + 1) = \frac{1}{x + 1} p(x) \quad (\because \lambda = 1) \quad \dots (1)$$

Now $p(1) = p(0) = 0.3678$

Put $x = 1$ in (1). Then

$$p(2) = \frac{1}{2} p(1) = \frac{1}{2} (0.3678) = 0.1839$$

Put $x = 2$ in (1). Then

$$p(3) = \frac{1}{3} p(2) = \frac{1}{3} (0.1839) = 0.0613$$

Put $x = 3$ in (1). Then

$$p(4) = \frac{1}{4} p(3) = \frac{1}{4} (0.0613) = 0.015325$$

Put $x = 4$ in (1). Then

$$p(5) = \frac{1}{5} p(4) = \frac{1}{5} (0.015325) = 0.003065$$

Example 17 : Using recurrence formula find the probabilities when $x = 0, 1, 2, 3, 4$ and 5; if the mean of Poisson distribution is 3. [JNTU 2008, (A) Nov. 2010 (Set No. 3)]

Solution : Given mean of the Poisson distribution is 3 $\Rightarrow \lambda = 3$. Now the Poisson distribution is

$$p(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-3} \cdot 3^x}{x!}$$

$$\therefore p(x = 0) = \frac{e^{-3} \cdot 3^0}{0!} = e^{-3} = 0.0498$$

By recurrence formula, we have

$$p(x + 1) = \frac{\lambda}{x + 1} p(x)$$

$$\Rightarrow p(x+1) = \frac{3}{x+1} p(x) \quad [\because \lambda = 3] \quad \dots (1)$$

Put $x = 0$ in (1). Then

$$p(1) = 3p(0) = 3(0.0498) = 0.1494$$

Put $x = 1$ in (1). Then

$$p(2) = \frac{3}{2} p(1) = \frac{3}{2} (0.1494) = 0.2241$$

Put $x = 2$ in (1). Then

$$p(3) = \frac{3}{3} p(2) = p(2) = 0.2241$$

Put $x = 3$ in (1). Then

$$p(4) = \frac{3}{4} p(3) = \frac{3}{4} (0.2241) = 0.1681$$

Put $x = 4$ in (1). Then

$$p(5) = \frac{3}{5} p(4) = \frac{3}{5} (0.1681) = 0.1008$$

Example 10 : A distributor of bean seeds determines from extensive tests that 5% of large batch of seeds will not germinate. He sells the seeds in packets of 200 and guarantees 90% germination. Determine the probability that a particular packet will violate the guarantee.

[JNTU 2005, (A) Dec. 2009 (Set No. 4)]

Solution : p = The probability of a seed not germinating = 5% = 0.05

λ = Mean number of seeds in a sample of 200

$$= np = 200 \times 0.05 = 10$$

Let x be the number of seeds that do not germinate. Then

$$p(x) = \frac{e^{-10} \cdot 10^x}{x!}$$

A packet will violate guarantee if it contains more than 20 germination seeds.

Probability that the guarantee is violated = $P(X > 20) = 1 - P(X \leq 20)$

$$= 1 - \sum_{x=0}^{20} \frac{e^{-10} 10^x}{x!} = 1 - 0.9984 = 0.0016$$

Example 11 : Suppose 2% of the people on the average are left handed. Find (i) the probability of finding 3 or more left handed (ii) the probability of finding none or one left handed.

[JNTU 2005 (Set No.3)]

Solution : Let x be the number of left handed.

Given mean, $\lambda = 2\% = 0.02$

$$\text{We have } p(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-0.02} (0.02)^x}{x!}$$

- (i) $p(x \geq 3) = 1 - p(x < 3) = 1 - [p(0) + p(1) + p(2)]$

$$= 1 - e^{-0.002} \left[1 + 0.02 + \frac{(0.02)^2}{2} \right]$$

$$= 1 - e^{-0.002} (1.0202) = 1 - 0.9802 (1.0202) = 1.3077 \times 10^{-6}$$
- (ii) $p(x \leq 1) = p(0) + p(1)$

$$= e^{-0.02} + e^{-0.002} (0.02) = e^{-0.02} (1.02) = 0.9998$$

Example 10 : If a Poisson distribution is such that $P(X = 1) \cdot \frac{3}{2} = P(X = 3)$, find

- (i) $P(X \geq 1)$ (ii) $P(X \leq 3)$ (iii) $P(2 \leq X \leq 5)$ [JNTU 2005 S (Set Nos. 1, 2, 3)]

Solution : Given $\frac{3}{2} P(X = 1) = P(X = 3)$

$$i.e., \frac{3}{2} \cdot \frac{e^{-\lambda} \cdot \lambda^1}{1!} = \frac{e^{-\lambda} \cdot \lambda^3}{3!} \quad i.e., \frac{3\lambda}{2} = \frac{\lambda^3}{6}$$

$$i.e., \lambda^3 - 9\lambda = 0 \quad \text{or } \lambda(\lambda^2 - 9) = 0 \quad \text{or } \lambda(\lambda - 3)(\lambda + 3) = 0$$

$$\therefore \lambda = 0, 3, -3$$

$$\Rightarrow \lambda = 3 \quad (\because \lambda > 0)$$

$$\text{Hence } P(X = x) = p(x) = \frac{e^{-3} 3^x}{x!}$$

- (i) $P(X \geq 1) = 1 - P(x < 1) = 1 - P(X = 0) = 1 - \frac{e^{-3} \cdot 3^0}{0!}$

$$= 1 - e^{-3} = 0.950213$$
- (ii) $P(X \leq 3) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$

$$= e^{-3} \left[\frac{3^0}{0!} + \frac{3}{1!} + \frac{3^2}{2!} + \frac{3^3}{3!} \right] = e^{-3} \left(1 + 3 + \frac{9}{2} + \frac{9}{2} \right) = 13 e^{-3} = 0.6472318$$
- (iii) $P(2 \leq X \leq 5) = P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5)$

$$= e^{-3} \left[\frac{3^2}{2!} + \frac{3^3}{3!} + \frac{3^4}{4!} + \frac{3^5}{5!} \right]$$

$$= 9e^{-3} \left(\frac{1}{2} + \frac{1}{2} + \frac{3}{8} + \frac{9}{40} \right) = 9e^{-3} (1.6) = 0.7169337$$

Example 11 : If the variance of a Poisson variate is 3, then find the probability that
 (i) $x = 0$ (ii) $0 < x \leq 3$ (iii) $1 \leq x < 4$. [JNTU 2007, 2007S, (Set No. 4)]

Solution : Given variance of the Poisson distribution = 3

For a Poisson distribution, we have mean = variance

Hence mean of the Poisson distribution = 3 i.e., $\lambda = 3$

$$\text{So } P(X = x) = p(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-3} \cdot 3^x}{x!}$$

$$(i) \quad p(x = 0) = \frac{e^{-3} \cdot 3^0}{0!} = e^{-3} = 0.04979$$

$$\begin{aligned} (ii) \quad p(0 < x \leq 3) &= p(x = 1) + p(x = 2) + p(x = 3) \\ &= e^{-3} \cdot 3 + e^{-3} \cdot \frac{3^2}{2!} + e^{-3} \cdot \frac{3^3}{3!} \\ &= e^{-3} \left(3 + \frac{9}{2} + \frac{9}{2} \right) = e^{-3} (12) = 0.5974 \end{aligned}$$

$$\begin{aligned} (iii) \quad p(1 \leq x < 4) &= p(x = 1) + p(x = 2) + p(x = 3) \\ &= 0.5974, \text{ as in (ii)} \end{aligned}$$

Example 11 : If two cards are drawn from a pack of 52 cards which are diamonds, using Poisson distribution find the probability of getting two diamonds at least 3 times in 51 consecutive trials of two cards drawing each time. [JNTU (A) Dec. 2009 (Set No. 3)]

Solution : The probability of getting two diamonds from a pack of 52 cards =

$$p = \frac{{}^{13}C_2}{{}^{52}C_2} = \frac{3}{51}$$

Number of trials, $n = 51$

$$\therefore \lambda = \text{Mean} = np = \frac{3}{51} (51) = 3$$

Thus the Poisson distribution is

$$p(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-3} \cdot 3^x}{x!}$$

$$\begin{aligned} P(\text{at least 3 times}) &= P(X \geq 3) = 1 - P(X < 3) \\ &= 1 - [P(X = 0) + P(X = 1) + P(X = 2)] \\ &= 1 - [e^{-3} + 3e^{-3} + e^{-3} \cdot \frac{3^2}{2!}] \\ &= 1 - e^{-3} \left(1 + 3 + \frac{9}{2} \right) = 1 - e^{-3} \left(\frac{17}{2} \right) \\ &= 1 - 0.42319 = 0.57681 \end{aligned}$$

Example 23 : Prove that the three successive values of a Poisson variate can not have equal probability of success.

Solution : Let $x - 1$, x and $x + 1$ be three successive Poisson variates.

$$\text{Suppose that } p(x - 1) = p(x) = p(x + 1) \quad \dots (1)$$

Then by recurrence formula,

$$p(x + 1) = \frac{\lambda}{x + 1} p(x)$$

$$(1) \text{ gives } \frac{\lambda}{x} p(x) = p(x) = \frac{\lambda}{x+1} p(x)$$

$$\text{This implies } \frac{\lambda}{x} = 1 = \frac{\lambda}{x+1} \Rightarrow \lambda = x \text{ and } \lambda = x+1.$$

But as x and $x+1$ are distinct and mean λ is unique for a poisson distribution, this is not admissible. Hence three successive values of a poisson variate can not be have equal probability of success.

Example 14 : If X is a Poisson Variate such that $3p(x=4) = \frac{1}{2} p(x=2) + p(x=0)$, find (i) the mean of x (ii) $p(x \leq 2)$ [JNTU 2008S, (A) Nov. 2010, (H) May 2011 (Set No. 1)]

Solution : (i) If X is a Poisson Variate with parameter λ , then

$$P(X=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}, x = 0, 1, 2, \dots, \lambda > 0$$

$$\text{Here we are given } 3p(x=4) = \frac{1}{2} p(x=2) + p(x=0)$$

$$\Rightarrow 3 \cdot \frac{e^{-\lambda} \lambda^4}{4!} = \frac{1}{2} \cdot \frac{e^{-\lambda} \cdot \lambda^2}{2!} + e^{-\lambda}$$

$$\Rightarrow \lambda^4 - 2\lambda^2 - 8 = 0 \Rightarrow (\lambda^2 - 4)(\lambda^2 + 2) = 0$$

$$\Rightarrow \lambda = \pm 2 \Rightarrow \lambda = 2 \quad (\because \lambda > 0)$$

$\therefore$ Mean of the Poisson variate i.e., $\lambda = 2$

$$(ii) \text{ Now we have } P(X=x) = \frac{e^{-2} \cdot 2^x}{x!}$$

$$p(x \leq 2) = p(x=0) + p(x=1) + p(x=2)$$

$$= e^{-2} + \frac{1}{4} e^{-1} + \frac{e^{-2} \cdot 2^2}{2!}$$

$$= 3e^{-2} + \frac{1}{4} e^{-1} = 3(0.135) + \frac{1}{4} (0.3679)$$

$$= 0.498$$

Example 23 : If 2% of light bulbs are defective, find

(i) At least one is defective.

(ii) Exactly 7 are defective.

(iii) $P(1 < x < 8)$ in a sample of 100

[JNTU (H) Dec. 2011 (Set No. 1)]

Solution : We are given $n=100$

Let p = The probability of a defective bulb = $2\% = 0.02$

$\therefore \lambda = \text{Mean number of defective bulbs in a sample of 100}$

$$= np = 100 \times 0.02 = 2$$

Since 'p' is small, we may use Poisson distribution.

Probability of x defective bulbs in a sample of 100 is

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-2} 2^x}{x!}; x = 0, 1, 2, \dots$$

$$(i) \quad P(\text{At least one is defective}) = p(x \geq 1)$$

$$= 1 - p(x = 0)$$

$$= 1 - \frac{e^{-2} \cdot 2^0}{0!} = 1 - e^{-2}$$

$$= 1 - 0.1353 = 0.8647$$

$$(ii) \quad P(\text{Exactly 7 are defective}) = P(x = 7)$$

$$= \frac{e^{-2} 2^7}{7!} = \frac{0.1353 \times 128}{5040} = 0.0034$$

$$(iii) \quad P(1 < x < 8) = P(x = 2) + P(x = 3) + P(x = 4) + P(x = 5) + P(x = 6) + P(x = 7)$$

$$= \frac{e^{-2} \cdot 2^2}{2!} + \frac{e^{-2} \cdot 2^3}{3!} + \frac{e^{-2} \cdot 2^4}{4!} + \frac{e^{-2} \cdot 2^5}{5!} + \frac{e^{-2} \cdot 2^6}{6!} + \frac{e^{-2} \cdot 2^7}{7!}$$

$$= \frac{e^{-2}}{2} \left(4 + \frac{8}{3} + \frac{16}{12} + \frac{32}{60} + \frac{64}{360} + \frac{128}{2520} \right)$$

$$= \frac{0.1353}{2} (4 + 2.6666 + 1.3333 + 0.5333 + 0.1777 + 0.0508)$$

$$= \frac{0.1353}{2} (8.7617) = 0.593$$

Example 26 : Given that $p(x = 2) = 9p(x = 4) + 90p(x = 6)$ for a Poisson variate X .

Find :

$$(i) \quad P(x < 2) \quad (ii) \quad P(x > 4) \quad (iii) \quad P(x \geq 1) \quad [\text{JNTU (H) Dec. 2011 (Set No. 2)}]$$

Solution : If X is a Poisson variate with parameter λ , then

$$P(X = x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}, x = 0, 1, 2, \dots; \lambda > 0$$

Since $P(X = 2) = 9P(X = 4) + 90P(X = 6)$, we have

$$\begin{aligned}\frac{e^{-\lambda} \cdot \lambda^2}{2!} &= 9 \cdot \frac{e^{-\lambda} \cdot \lambda^4}{4!} + 90 \cdot \frac{e^{-\lambda} \cdot \lambda^6}{6!} \\ &= \frac{e^{-\lambda} \cdot \lambda^2}{8} [3\lambda^2 + \lambda^4]\end{aligned}$$

$$\Rightarrow 4 = 3\lambda^2 + \lambda^4 \text{ or } \lambda^4 + 3\lambda^2 - 4 = 0 \text{ or } (\lambda^2 + 4)(\lambda^2 - 1) = 0$$

$\therefore \lambda^2 + 4 = 0$ or $\lambda^2 - 1 = 0$. But λ cannot be imaginary.

[(or) This is a quadratic equation in λ^2

$$\therefore \lambda^2 = \frac{-3 \pm \sqrt{9+16}}{2} = \frac{-3 \pm 5}{2} = 1, -4$$

Since $\lambda > 0$, we get

$$\lambda^2 = 1 \Rightarrow \lambda = 1, \text{ taking positive sign}$$

$$\text{Hence } P(X = x) = \frac{e^{-1} \cdot 1^x}{x!}, x = 0, 1, 2, \dots$$

$$(i) \quad P(x < 2) = P(x = 0) + P(x = 1)$$

$$= e^{-1} + e^{-1} = \frac{2}{e} = 0.7358$$

$$(ii) \quad P(x > 4) = 1 - P(x \leq 4)$$

$$= 1 - [P(x = 0) + P(x = 1) + P(x = 2) + P(x = 3) + P(x = 4)]$$

$$= 1 - \left[e^{-1} + e^{-1} + \frac{e^{-1}}{2} + \frac{e^{-1}}{3!} + \frac{e^{-1}}{4!} \right]$$

$$= 1 - e^{-1} \left[2 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} \right] = 1 - e^{-1} \left[\frac{65}{24} \right]$$

$$= 1 - 0.9963 = 0.0037$$

$$(iii) \quad P(x \geq 1) = 1 - p(x = 0) = 1 - e^{-1} = 1 - 0.3679 = 0.6321$$

Example 23 : If the Mean of a Poisson variable is 1.8, then find

$$(i) \quad p(x > 1) \quad (ii) \quad p(x = 5) \quad (iii) \quad p(0 < x < 5) \quad [\text{JNTU (H) Dec. 2011 (Set No.3)}]$$

Sol. Given mean of a Poisson variate, $\lambda = 1.8$

For Poisson distribution, the probability of x successes is

$$P(X = x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-1.8} (1.8)^x}{x!}; x = 0, 1, 2, \dots$$

$$(i) \quad p(x > 1) = 1 - p(x \leq 1)$$

$$= 1 - [p(x = 0) + p(x = 1)]$$

$$= 1 - [e^{-1.8} + e^{-1.8}(1.8)] = 1 - e^{-1.8}(1 + 1.8) = 1 - 0.4628 = 0.5372$$

$$(ii) \quad p(x = 5) = \frac{e^{-1.8}(1.8)^5}{5!} = \frac{3.1234}{120} = 0.026$$

$$(iii) \quad p(0 < x < 5) = p(x = 1) + p(x = 2) + p(x = 3) + p(x = 4)$$

$$= e^{-1.8} + \frac{e^{-1.8}(1.8)^2}{2!} + \frac{e^{-1.8}(1.8)^3}{3!} + \frac{e^{-1.8}(1.8)^4}{4!}$$

$$= e^{-1.8} \left[1 + \frac{(1.8)^2}{2} + \frac{(1.8)^3}{6} + \frac{(1.8)^4}{24} \right] = e^{-1.8} [1 + 1.62 + 0.972 + 0.4374]$$

$$= e^{-1.8}(4.0294) = 0.666$$

Example 28 : If 2% of light bulbs are defective, find

(i) At least one is defective. (ii) Exactly 7 are defective (iii) $p(1 < x < 8)$

in a sample of 100

[JNTU (H) Apr. 2012 (Set No. 3)]

Solution : Given $n = 100$ and

p = The probability of defective light bulbs = 0.02

$\therefore$ Mean $= \lambda$ = Mean number of defective light bulbs in a sample of 100.

$$= np = 100(0.02) = 2$$

Since p is small, we use Poisson distribution.

Probability of x defective light bulbs in a sample of 100 is

$$P(X = x) = p(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-2} \cdot 2^x}{x!}; x = 0, 1, 2, \dots$$

$$(i) \quad p(\text{at least one is defective}) = p(x \geq 1) = 1 - p(x = 0)$$

$$= 1 - \frac{e^{-2} \cdot 2^0}{0!} = 1 - e^{-2} = 0.8646$$

$$(ii) \quad p(\text{exactly 7 are defective}) = p(x = 7)$$

$$= \frac{e^{-2} \cdot 2^7}{7!} = \frac{128 e^{-2}}{5040} = 0.0034$$

$$(iii) \quad p(1 < x < 8) = p(x = 2) + p(x = 3) + p(x = 4) + p(x = 5) + p(x = 6) + p(x = 7)$$

$$= e^{-2} \left[\frac{2^2}{2!} + \frac{2^3}{3!} + \frac{2^4}{4!} + \frac{2^5}{5!} + \frac{2^6}{6!} + \frac{2^7}{7!} \right]$$

$$\begin{aligned}
&= e^{-2} \left[\frac{4}{2} + \frac{8}{6} + \frac{16}{24} + \frac{32}{120} + \frac{64}{720} + \frac{128}{5040} \right] \\
&= e^{-2} (2 + 1.33 + 0.66 + 0.27 + 0.09 + 0.02) \\
&= e^{-2} (4.37) = 0.59
\end{aligned}$$

Example 23 : Poisson variable has a double mode at $x = 2$ and $x = 3$, find the maximum probability and also find $p(x \geq 2)$. [JNTU (H) Apr. 2012 (Set No. 4)]

Solution : We know that the Poisson distribution is bimodal and two modes are at the points $x = \lambda - 1$ and $x = \lambda$.

Since we are given that the two modes are at the points $x = 1$ and $x = 2$, we find that $\lambda = 2$.

$$\therefore P(X = x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-2} 2^x}{x!}; x = 0, 1, 2, \dots$$

$$\text{Now } P(X = 1) = 2e^{-2} \text{ and } P(X = 2) = \frac{e^{-2} \cdot 2^2}{2!} = 2e^{-2}$$

$$\text{Hence the required probability} = P(X = 1) + P(X = 2) = 2e^{-2} + 2e^{-2} = 4e^{-2} = 0.5413$$

$$\begin{aligned}
\text{Now } p(x \geq 2) &= 1 - p(x < 2) \\
&= 1 - [p(x = 0) + p(x = 1) + p(x = 2)] \\
&= 1 - [e^{-2} + 2e^{-2} + 2e^{-2}] = 1 - 5e^{-2} = 1 - 0.676 \\
&= 0.3233
\end{aligned}$$

Example 23 : A sample of 3 items is selected at random from a box containing 10 items of which 4 are defective. Find the expected number of defective items?

[JNTU (K) May 2010 (Set No. 2)]

Solution : The probability of defective is $p = \frac{4}{10} = \frac{2}{5}$

Number of items chosen, $n = 3$

$$\text{The expected number of defective is } E(X) = \mu = np = 3 \left(\frac{2}{5} \right) = \frac{6}{5} = 1.2 \approx 1$$

Alternative Method :

$$\text{No. of Exhaustive cases} = {}^{10}C_3 = \frac{10!}{3!7!} = 120$$

$$\text{Probability that there are no defective items} = p(x = 0) = \frac{{}^6C_3}{120} = \frac{1}{6}$$

$$\text{Probability that there is one defective item} = p(x=1) = \frac{{}^6C_2 \cdot {}^4C_1}{120} = \frac{1}{2}$$

$$\text{Probability that there are two defective items} = p(x=2) = \frac{{}^6C_1 \cdot {}^4C_2}{120} = \frac{3}{10}$$

$$\text{Probability that there are three defective items} = p(x=3) = \frac{{}^4C_3}{120} = \frac{1}{30}$$

x	0	1	2	3
$p(x)$	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{3}{10}$	$\frac{1}{30}$

$$\begin{aligned} \therefore \text{Expected number of defective items} &= E(x) = \sum p_i x_i \\ &= 0 \cdot \frac{1}{6} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{3}{10} + 3 \cdot \frac{1}{30} \\ &= 0 + \frac{1}{2} + \frac{3}{5} + \frac{1}{10} = \frac{12}{10} = 1.2 \end{aligned}$$

Example 31 : Wireless sets are manufactured with 25 soldered joints each. On the average 1 joint in 500 is defective. How many sets can be expected to be free from defective joints in a consignment of 10,000 sets. [JNTU Nov. 2008 (Set No. 3)]

Solution : On average, one joint in 500 joints is defective

$$\text{Number of soldered joints} = 25 = n$$

$$\text{Probability that a joint is defective} = \frac{1}{500}$$

$$\therefore \text{Mean} = np = 25 \times \frac{1}{500} = \frac{1}{20} = 0.05$$

$$\text{Thus } \lambda = 0.05$$

$$P(\text{there are } r \text{ defective joints}) = \frac{e^{-\lambda} \cdot \lambda^r}{r!}$$

$$\therefore P(r=0) = e^{-\lambda} = e^{-0.05}$$

$$\begin{aligned} \text{Hence expected number of sets free of defective joints among 10000 sets} \\ = 10000 \cdot e^{-0.05} = 10000 (0.951229) = 9512.29 \approx 9512 \end{aligned}$$

Example 32 : If a Poisson variate $2P(x=0) = P(x=2)$ find the probability that

$$(i) P(x \leq 3) \quad (ii) P(2 < x \leq 5) \quad (iii) P(x \geq 3) \quad \text{[JNTU (A) Nov. 2011 (Set No. 2)]}$$

Solution : For Poisson distribution, the probability function is $p(x, \lambda) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$, where λ is the parameter.

$$\text{By the given condition, } 2p(x=0) = p(x=2).$$

$$\Rightarrow 2e^{-\lambda} = e^{-\lambda} \cdot \frac{\lambda^2}{2!} \therefore \lambda^2 = 4. \text{ Hence } \lambda = 2.$$

$$(i) \quad p(x \leq 3) = p(x=0) + p(x=1) + p(x=2) + p(x=3)$$

$$= e^{-2} + e^{-2} \cdot 2 + e^{-2} \cdot \frac{2^2}{2!} + e^{-2} \cdot \frac{2^3}{3!}$$

$$= e^{-2} \left(1 + 2 + 2 + \frac{4}{3} \right) = \frac{19}{3} e^{-2}$$

$$= \frac{19}{3} (0.1353) = \frac{2.5714}{3} = 0.857$$

$$(ii) \quad p(2 < x \leq 5) = p(x=3) + p(x=4) + p(x=5)$$

$$= e^{-2} \cdot \frac{2^3}{3!} + e^{-2} \cdot \frac{2^4}{4!} + e^{-2} \cdot \frac{2^5}{5!} = e^{-2} \left(\frac{8}{6} + \frac{16}{24} + \frac{32}{120} \right)$$

$$= e^{-2} \left(\frac{4}{3} + \frac{2}{3} + \frac{4}{15} \right) = e^{-2} \left(2 + \frac{4}{15} \right)$$

$$= \frac{34}{15} e^{-2} = \frac{34}{15} (0.1353) = \frac{4.6002}{15} = 0.3067$$

$$(iii) \quad p(x \geq 3) = 1 - [p(x=0) + p(x=1) + p(x=2)]$$

$$= 1 - \left[e^{-2} + 2e^{-2} + e^{-2} \cdot \frac{2^2}{2!} \right]$$

$$= 1 - (5e^{-2}) = 1 - 5(0.1353) = 1 - 0.6785$$

$$= 0.3235$$

Example 31 : An insurance agent accepts policies of 5 men all of identical age and good in health. The probability that a man of this age will be alive 30 years is $\frac{2}{3}$. Find the probability that in 30 years. (i) All five men (ii) At least one man. (iii) Almost three will be alive.

[JNTU (A) Apr. 2012 (Set No. 2)]

Solution : Probability that a man will be alive 30 years $= p = \frac{2}{3}$

Number of men = 5.

$$\therefore \lambda = \text{mean} = np = 5 \left(\frac{2}{3} \right) = \frac{10}{3}$$

$$\therefore P(x \text{ men will be alive}) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-10/3} \left(\frac{10}{3} \right)^x}{x!}; x = 0, 1, 2, \dots$$

$$(i) \quad P(x=5) = \frac{e^{-\frac{10}{3}} \left(\frac{10}{3}\right)^5}{5!} = \frac{14.6806}{120} = 0.12$$

$$(ii) \quad P(\text{at least one man}) = P(x \geq 1) = 1 - P(x=0)$$

$$= 1 - \frac{e^{-10/3} \left(\frac{10}{3}\right)^0}{0!} = 1 - e^{-(10/3)}$$

$$= 1 - 0.0357 = 0.9643$$

$$(iii) \quad P(\text{almost three will be alive}) = P(x \leq 3)$$

$$= 1 - [P(x=4) + P(x=5)]$$

$$= 1 - \left[\frac{e^{-(10/3)} \left(\frac{10}{3}\right)^4}{4!} + 0.12 \right]$$

$$= 1 - [0.1835 + 0.12] = 1 - 0.3035 = 0.6965$$

Example 32 : Fit a Poisson distribution for the following data and calculate the expected frequencies

x	0	1	2	3	4
$f(x)$	109	65	22	3	1

[JNTU 2004S]

Solution : Here $N = \text{total frequency} = \sum f_i = 109 + 65 + 22 + 3 + 1 = 200$

$$\text{Mean} = \frac{\sum f_i x_i}{\sum f_i} = \frac{0 + 65 + 44 + 9 + 4}{200} = \frac{122}{200} = 0.61$$

$\therefore$ Mean of Poisson distribution, $\lambda = 0.61$

Hence the theoretical frequencies are given by $N \cdot P(x)$, where $x = 0, 1, 2, 3, 4$

$$i.e., \quad 200 \cdot \frac{e^{-0.61} (0.61)^x}{x!}, \text{ where } x = 0, 1, 2, 3, 4$$

$$i.e., \quad 200 e^{-0.61}, 200 e^{-0.61} (0.61), 200 e^{-0.61} \cdot \frac{(0.61)^2}{2!}, 200 e^{-0.61} \cdot \frac{(0.61)^3}{3!}$$

$$\text{and } 200 e^{-0.61} \cdot \frac{(0.61)^4}{4!}$$

$$i.e., \quad 108.67, 66.29, 20.22, 4.11, 0.63$$

Since frequencies are always integers, therefore by converting them to nearest integers, we get

Observed frequency	109	65	22	3	1
Expected frequency	109	66	20	4	1

Example 35 : Fit a poisson distribution to the following data

x	0	1	2	3	4	5	Total
f	142	156	69	27	5	1	400

[JNTU (K) Nov. 2009, May 2013 (Set No. 1)]

Solution : Mean = $\frac{\sum f_i x_i}{\sum f_i} = \frac{0+156+138+81+20+5}{400} = \frac{400}{400} = 1$

$\therefore$ Mean of Poisson distribution *i.e.*, $\lambda = 1$

Hence the theoretical frequency for x successes is given by

$N.P(x)$ where $x = 0, 1, 2, 3, 4, 5$

i.e., $400 \cdot \frac{e^{-1}(1)^x}{x!}$, where $x = 0, 1, 2, 3, 4, 5$

i.e., $400(e^{-1})$, $400(e^{-1})$, $200(e^{-1})$, $66.67(e^{-1})$, $16.67(e^{-1})$, $3.33(e^{-1})$

i.e., 147.15, 147.15, 73.58, 24.53, 6.13, 1.23

$\therefore$ The theoretical frequencies are

x	0	1	2	3	4	5
f	147	147	74	25	6	1

Example 36 : The distribution of typing mistakes committed by a typist is given below.

Assuming the distribution to be Poisson, find the expected frequencies.

x	0	1	2	3	4	5
$f(x)$	42	33	14	6	4	1

Solution : $N = \sum f_i = 100$

Mean of the distribution, $\lambda = \frac{\sum f_i x_i}{\sum f_i}$

$$= \frac{0+33+28+18+16+5}{100} = \frac{100}{100} = 1$$

By Poisson distribution, $p(x) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-1}}{x!}$

$\therefore f(0) = N.p(0) = \frac{100}{e} = 36.79$, $f(1) = N.p(1) = \frac{100}{e} = 36.79$

$f(2) = N.p(2) = \frac{100}{2e} = 18.39$, $f(3) = N.p(3) = \frac{100}{6e} = 6.13$

$$f(4) = \frac{100}{24e} = 1.53, \quad f(5) = \frac{100}{120e} = 0.31$$

Since frequencies are always integers, by converting them to nearest integers, we get

x	0	1	2	3	4	5
Observed frequency	42	33	14	6	4	1
Expected frequency	37	37	18	6	2	0

Note : In the above example we observe that the frequencies of first two values are same. Is it true for all the data ? Observe. Moreover, is it possible to get more than two same frequencies.

Poisson Distribution as an Approximation to Binomial Distribution

When n is large, say greater than 30, and p is quite small, say less than 0.1, the Binomial distribution can be approximated by the Poisson distribution with mean np and variance also np .

We know that the variance of the Binomial distribution is npq or $np(1-p)$, so that when p approaches to zero, then q or $(1-p)$ will approach to 1. That is why the variance npq of the Binomial distribution is approximated by np , in the Poisson distribution.

Relationship with Normal Distribution

When n is very large, say >30 , the Poisson distribution is well approximated by the Normal distribution, discussed in the next section.

REVIEW QUESTIONS

1. Define Poisson distribution. List the properties of Poisson distribution.
2. Compute the mean and variance of a random variable X which follow a Poisson distribution $P(x, \lambda) = \frac{e^{-\lambda} \lambda^x}{x!}, \lambda \geq 0, x = 0, 1, \dots$
 (or) Show that the mean and variance are equal for Poisson distribution.
 (or) Derive mean and variance of a Poisson distribution. [JNTU (A) Dec. 2014]
3. Prove that the Poisson distribution can be approximated by Binomial Distribution.
 [JNTU (K) Nov. 2011 (Set No. 3)]
 (or) Prove that Poisson distribution is limiting case of Binomial distribution.
 [JNTU (K) May 2010S (Set No. 1)]
4. Write the formula of mean and variance of a Poisson distribution.

EXERCISE 7(B)

1. On an average, 2 vehicles pass by a road per minute. Find the probability of 0, 1, 2, 3, 4, 5 vehicles per minute.
2. The fraction defective in a bulk is 4%. Determine the probability of not more than one defective in a sample of size 10.
3. If 3% of the electric bulbs manufactured by a company are defective, find the probability that in a sample of 100 bulbs, there are 5 defective bulbs.
4. Find the probability of finding at most 5 defective fuses in a box of 200 fuses if the experience shows that 2% of such fuses will be found defective.
5. Suppose that a book of 200 pages contains 20 printing mistakes. Assume that these errors are randomly distributed throughout the book and x , the number of errors per page has a poisson distribution. Find the probability that 30 pages selected at random will be free of errors.
6. A manufacturer known that the razor blades he makes contain on an average 0.5% of defectives. He packs them in packets of 5. What is the probability that a packet picked at random will contain 3 or more faulty blades.
7. Suppose 3% of bolts made by a machine are defective, the defects occurring at random during production. If bolts are packed 50 per box, find the probability that a given box will contain 5 defectives.
8. Assume that the probability of an individual coal miner being killed in a mine accident during a year is $\frac{1}{2400}$. What is the probability that in a mine employing 200 miners, there will be at least one fatal accident in a year.
9. In a city, 6% of all drivers get at least one parking ticket per year. Determine the probabilities that among 80 drivers
 - (i) 4 drivers will get at least one parking ticket in a year.
 - (ii) at least 3 drivers will get at least one parking ticket in a year.
10. Using Poisson distribution find the probability that the ace of spades will be drawn from a pack of well shuffled cards at least once in 104 consecutive trials.
11. The number of arrivals of customers during any day follows Poisson distribution with mean of 5. What is the probability that the total number of customers on two days selected at random is less than 2 ?
12. An insurance company found that only 0.01% of the population is involved in a certain type of accident each year. If its 1000 policy holders were randomly selected from the population, what is the probability that not more than two of its clients are involved in such an accident next year ?
13. What is the probability that an ace of spade will not be drawn even once from a deck of cards in 100 consecutive trials.
14. Out of 100 balls 50 are red and the rest white. If 60 balls are picked up at random, what is the probability of picking up (i) 3 red balls (ii) not more than 3 red balls in the sample.

15. A person receives an average of 4 telephone calls a day. Assuming Poisson distribution what is the probability that on a given day he will receive (i) no telephone call (ii) exactly only call (iii) at least two telephone calls.
16. The random variable X has a Poisson distribution. If $P(X = 1) = 0.01487$ and $P(X = 2) = 0.04461$, then find $P(X = 3)$.
17. If X is a Poisson variate such that $P(X = 0) = P(X = 2) + 3 P(X = 4)$, find (i) the mean of X (ii) $P(X \leq 2)$. **[JNTU 2004S (Set No. 3)]**
18. Using recurrence formula find the probabilities when $x = 0, 1, 2, 3, 4$ and 5 , if the mean of Poisson distribution is 3 . **[JNTU 2005 (Set No. 4)]**
19. Suppose that X has a Poisson distribution. If $P(X = 2) = \frac{2}{3} P(X = 1)$, find (i) $P(X = 0)$ and (ii) $P(X = 3)$.
20. If X and Y are independent Poisson variates such that $P(X = 1) = P(X = 2)$ and $P(Y = 2) = P(Y = 3)$. Find the variance of (i) X (ii) Y (iii) $X - 2Y$.
21. If X follows a poisson law such that $2P(X = 0) = P(X = 2)$, find (i) $p(X \leq 3)$ (ii) $p(2 < X \leq 5)$ (iii) $p(X \geq 3)$. **[JNTU (A) Nov. 2011 (Set No. 2)]**
22. If X is a poisson variate such that $P(X = 2) = 9P(X = 4) + 90 P(X = 6)$, find the mean and variance of X . **[JNTU (K) May 2012 (Set No. 1)]**
23. If X is a poisson variate such that $P(X = 2) = 45 P(X = 6) - 3 \cdot P(X = 4)$, find (i) $P(X \geq 1)$ (ii) $P(X < 2)$.
24. A Poisson distribution has a double mode at $x = 1$ and $x = 2$. What is the probability that x will have one or the other of these two values ?

[JNTU (H) May 2012 (Set No. 4)]

25. If X is a Poisson variate such that $P(X = 0) = P(X = 1) = k$, determine k .
26. Fit a Poisson distribution for the following data and calculate the expected frequencies.

x	0	1	2	3	4	5	6	7	8
f	56	156	132	92	37	22	4	0	1

27. The distribution of typing mistakes committed by a typist is given below assuming the distribution to be Poisson, find the expected frequencies

x	0	1	2	3	4	5
$f(x)$	125	95	49	20	8	3

28. In 1000 sets of trials for an event of small probability the frequencies f of the numbers of x of successes are

x	0	1	2	3	4	5	6	7
f	305	365	210	80	28	9	2	1

find the expected frequencies.

- (OR) Fit a Poisson distribution to the following data

x	0	1	2	3	4	5	6	7
f	305	365	210	80	28	9	2	1

[JNTU (K) May 2010S (Set No. 4)]

29. If 0.8 % of the fuses delivered to an arsenal are defective, use the Poisson approximation to determine the probability that 4 fuses will be defective in a random sample of 400.

[JNTU (A) Dec. 2018]

30. At a checkout counter customers arrive at an average of 1.5 per minute. Find the probabilities that in any given minute of time : (i) at most 4 will arrive. (ii) Exactly 4 will arrive (iii) At least 4 will arrive.

[JNTU (A) June 2018]

ANSWERS

1. 0.1353, 0.2706, 0.2706, 0.1804, 0.0902, 0.0361
2. 0.9384
3. 0.1008
4. 0.7845
5. 0.05
6. 0.000002555
7. 0.01412
8. 0.08
9. (i) 4.8 (ii) 0.8575
10. 0.865
11. 4.994×10^{-4}
12. 0.9998
13. 0.15
14. (i) 0.2241 (ii) 0.5976
16. 0.08922
17. (i) 1.21 (ii) 0.8773
18. 0.147, 0.2205, 0.2205, 0.1653, 0.099, 0.0495
19. (i) 0.264
20. (i) 2 (ii) 3 (iii) 4
21. (i) 0.0861 (ii) 0.0308 (iii) 0.32
22. 1, 1
23. (i) 0.864 (ii) 0.408
24. 0.542
25. $1/e$
26. 70, 137, 135, 89, 44, 17, 6, 20
27. 110, 110, 55, 18, 5, 1
28. 301, 361, 217, 87, 26, 6, 1, 0

NORMAL DISTRIBUTION

The normal law of error stands out in the mankind as one of the broadest generalisations of natural philosophy. It serves as the guiding instrument in researches in the physical and social sciences and in engineering, medicine and agriculture. It is an indispensable tool for the analysis and interpretation of the basic data obtained by Observation and experience.

- W.J. Youden

Normal distribution is applicable in the following situations:

1. Life of items subjected to wear and tear like tyres, batteries, bulbs, currency notes, etc.
2. Length and diameter of certain products like pipes, screws and discs.
3. Height and weight of baby at birth.
4. Aggregate marks obtained by students in an examination.
5. Weekly sales of an item in a store.

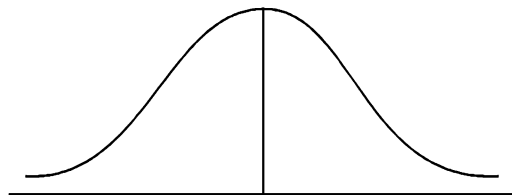
The most common pattern of distribution of a continuous variable, found in nature, is of the following type:



Distribution of Weights of Men

Maximum number of men will be around the average weight, and as one goes to the left or right, the number of persons will keep on reducing. In fact, even other human characteristics like age, intelligence, etc. follow this pattern or distribution.

This type of distribution was discovered by Karl Gauss. For this reason, it is called Gaussian Distribution. But, it is found so often in real life that it is called Normal Distribution. The name which is commonly used. If the curve is drawn in a continuous manner, it would look as follows.



Shape of Normal Distribution

7.10 NORMAL DISTRIBUTION

The distributions discussed so far are discrete distributions *i.e.*, distributions in which the variate can take only integral values. Examples of such distributions are the numbers of defective items manufactured in a factory, the numbers of persons suffering from a disease, etc. Now we shall consider a continuous distribution, namely the *Normal Distribution*. A

continuous distribution is a distribution in which the variate can take all values within a given range. Examples of continuous distributions are the heights of persons, the speed of a vehicle, etc.

The Normal distribution was first discovered by English Mathematician De-Moivre (1667-1745) in 1733 and further refined by French Mathematician Laplace (1749-1827) in 1774 and independently by Karl Friedrich Gauss (1777-1855). Normal distribution is also known as Gaussian Distribution. It is another limiting form of the Binomial Distribution for large values of n when neither p nor q is very small. The Normal Distribution is, therefore, derived from the Binomial Distribution by increasing the number of trials indefinitely.

Definition : A random variable X is said to have a Normal Distribution, if its density function or probability distribution is given by **[JNTU (H) Dec. 2009 (Set No. 2)]**

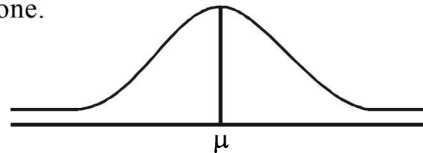
$$f(x; \mu, \sigma) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}; -\infty < x < \infty, -\infty < \mu < \infty, \sigma > 0 \quad \dots (1)$$

where μ is the Mean and σ is the Standard deviation of x . As can be seen, the function, called probability density function of the normal distribution, depends on two values μ and σ . These are referred as the two parameters of the normal distribution.

The curve has maximum value at $x = \mu$ and tapers off on either side but never touches the horizontal line. The curve on the left side goes upto $-\infty$, and on the right side it goes upto $+\infty$. However, as much as 99.73% of the area under the curve lies between $(\mu - 3\sigma)$ and $(\mu + 3\sigma)$ and only 0.277% of the area lies beyond these points.

The random variable X is then said to be a normal random variable or normal variate. The curve representing the Normal Distribution (1) is called the normal curve and the total area bounded by the curve and the x -axis is one.

$$i.e., \int_{-\infty}^{\infty} f(x) dx = 1$$



The area under the curve between the ordinates $x = a$ and $x = b$, where $a < b$, represents the probability that x lies between a and b . i.e., $P(a < x < b)$.

Thus $P(a < x < b) =$ area under the normal curve between the vertical lines $x = a$ and $x = b$, which is $\int_a^b f(x) dx$.

Note : A random variable X with mean μ and variance σ^2 and following the probability law (1) is expressed by $X \sim N(\mu, \sigma^2)$.

7.11 NORMAL DISTRIBUTION AS A LIMITING FORM OF BINOMIAL DISTRIBUTION

[JNTU (A) Dec. 2009, (K) Nov. 2011 (Set No. 1)]

Normal distribution is a limiting case of the Binomial distribution under the following conditions :

(i) n , the number of trials is indefinitely large i.e., $n \rightarrow \infty$.

(ii) neither p nor q is very small.

The probability of x successes in a series of n independent trials is

$$p(x) = {}^nC_x p^x q^{n-x}; \quad x = 0, 1, 2, \dots, n \quad \dots (1)$$

Let us now consider the variate

$$Z = \frac{X - np}{\sqrt{npq}}, \quad X = 0, 1, 2, \dots, n \quad \dots (2)$$

$$\text{When } X = 0, Z = \frac{-np}{\sqrt{npq}} = -\sqrt{\frac{np}{q}} \quad \text{and}$$

$$\begin{aligned} \text{When } X = n, Z &= \frac{n - np}{\sqrt{npq}} = \frac{n(1-p)}{\sqrt{npq}} \\ &= \frac{nq}{\sqrt{npq}} = \sqrt{\frac{nq}{p}} \end{aligned}$$

Thus in the limit as $n \rightarrow \infty$, Z takes the values from $-\infty$ to ∞ . Hence the distribution of X will be a continuous distribution over the range $-\infty$ to ∞ .

We want the limiting form of (1) under the above conditions.

$$\text{Hence } \lim_{n \rightarrow \infty} p(x) = \lim_{n \rightarrow \infty} \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

Using Stirling's approximation for $n!$ as $n \rightarrow \infty$, $\lim_{n \rightarrow \infty} n! \approx \sqrt{2\pi} e^{-n} n^{n+\frac{1}{2}}$, we get

$$\begin{aligned} \lim_{n \rightarrow \infty} p(x) &= \lim_{n \rightarrow \infty} \left[\frac{\sqrt{2\pi} e^{-n} n^{n+\frac{1}{2}} \cdot p^x q^{n-x}}{\left(\sqrt{2\pi} e^{-x} x^{x+\frac{1}{2}}\right) \cdot \left(\sqrt{2\pi} e^{-(n-x)} (n-x)^{n-x+\frac{1}{2}}\right)} \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{1}{\sqrt{2\pi}} \cdot \frac{n^{n+\frac{1}{2}}}{x^{x+\frac{1}{2}} (n-x)^{n-x+\frac{1}{2}}} \cdot p^x q^{n-x} \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{1}{\sqrt{2\pi}} \cdot \frac{1}{\sqrt{npq}} \cdot \frac{(np)^{x+\frac{1}{2}} (nq)^{n-x+\frac{1}{2}}}{x^{x+\frac{1}{2}} (n-x)^{n-x+\frac{1}{2}}} \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{1}{\sqrt{2\pi}} \cdot \frac{1}{\sqrt{npq}} \cdot \left(\frac{np}{x}\right)^{x+\frac{1}{2}} \cdot \left(\frac{nq}{n-x}\right)^{n-x+\frac{1}{2}} \right] \quad \dots (3) \end{aligned}$$

$$\text{Now, substitute } N = \left(\frac{x}{np}\right)^{x+\frac{1}{2}} \cdot \left(\frac{n-x}{nq}\right)^{n-x+\frac{1}{2}}$$

$$\text{Then (3) becomes } \lim_{n \rightarrow \infty} p(x) = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{\sqrt{npq}} \cdot \frac{1}{N} \quad \dots (4)$$

From (2), we have

$$X = np + Z \cdot \sqrt{npq} \quad \dots (4.1)$$

$$\Rightarrow \frac{X}{np} = 1 + Z \cdot \sqrt{\frac{q}{np}} \quad \dots (4.2)$$

$$\begin{aligned} \text{Also } n - X &= n - np - Z \cdot \sqrt{npq} = n(1 - p) - Z \cdot \sqrt{npq} \\ &= nq - Z \cdot \sqrt{npq} \end{aligned} \quad \dots (4.3)$$

$$\therefore \frac{n - X}{nq} = 1 - Z \cdot \sqrt{\frac{p}{nq}} \quad \dots (4.4)$$

$$\begin{aligned} \text{Now } \log N &= \left(x + \frac{1}{2}\right) \log\left(\frac{x}{np}\right) + \left(n - x + \frac{1}{2}\right) \cdot \log\left(\frac{n - x}{nq}\right) \\ &= \left(np + Z \cdot \sqrt{npq} + \frac{1}{2}\right) \cdot \log\left(1 + Z \cdot \sqrt{\frac{q}{np}}\right) + \left(nq - Z \cdot \sqrt{npq} + \frac{1}{2}\right) \cdot \log\left(1 - Z \cdot \sqrt{\frac{p}{nq}}\right) \\ &\quad \text{using (4.1), (4.2), (4.3), (4.4)} \\ &= \left(np + Z \cdot \sqrt{npq} + \frac{1}{2}\right) \left[Z \cdot \sqrt{\frac{q}{np}} - \frac{1}{2} Z^2 \left(\frac{q}{np}\right) + \frac{1}{3} Z^3 \left(\frac{q}{np}\right)^{3/2} - \dots \right] \\ &\quad + \left(nq - Z \cdot \sqrt{npq} + \frac{1}{2}\right) \left[-Z \sqrt{\frac{p}{nq}} - \frac{1}{2} Z^2 \left(\frac{p}{nq}\right) - \frac{1}{3} Z^3 \left(\frac{p}{nq}\right)^{3/2} - \dots \right] \\ &= \left[\left\{ Z \sqrt{npq} - \frac{1}{2} q Z^2 + \frac{1}{3} Z^3 \frac{q^{3/2}}{\sqrt{np}} + Z^2 q - \frac{1}{2} Z^3 \frac{q^{3/2}}{\sqrt{np}} + \frac{1}{2} Z \cdot \sqrt{\frac{q}{np}} - \frac{1}{4} Z^2 \cdot \frac{q}{np} + \dots \right\} \right. \\ &\quad \left. + \left\{ -Z \sqrt{npq} - \frac{1}{2} Z^2 p - \frac{1}{3} Z^3 \cdot \frac{p^{3/2}}{\sqrt{nq}} + Z^2 p + \frac{1}{2} Z^3 \frac{p^{3/2}}{\sqrt{nq}} - \frac{1}{2} Z \cdot \sqrt{\frac{p}{nq}} - \frac{1}{4} Z^2 \cdot \frac{p}{nq} + \dots \right\} \right] \\ \text{i.e., } \log N &= \left[-\frac{1}{2} Z^2 (p + q) + Z^2 (p + q) + \frac{Z}{2\sqrt{n}} \left\{ \sqrt{\frac{q}{p}} + \sqrt{\frac{p}{q}} \right\} + O\{n^{-1/2}\} \right] \\ &= \frac{Z^2}{2} + O(n^{-1/2}) \rightarrow \frac{Z^2}{2} \text{ as } n \rightarrow \infty \end{aligned}$$

$$\therefore \lim_{n \rightarrow \infty} \log N = \frac{Z^2}{2} \Rightarrow \lim_{n \rightarrow \infty} N = e^{Z^2/2}$$

Now, from (4) we have

$$\lim_{n \rightarrow \infty} p(x) = \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{\sqrt{npq}} e^{-Z^2/2}$$

$$\text{or } P(z \leq Z \leq z + dz) = \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{\sqrt{npq}} e^{-Z^2/2} \quad \dots (5)$$

$$\text{But } Z = \frac{X - np}{\sqrt{npq}} = \frac{x - \mu}{\sigma}$$

To obtain the distribution of X , we put $Z = \frac{x - \mu}{\sigma}$ in (5). We get

$$P(x \leq X \leq x + dx) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < x < \infty$$

7.12 CONSTANTS OF NORMAL DISTRIBUTION

1. Mean of Normal Distribution [JNTU 2006S, (H) Dec. 2009 (Set No.2)]

Consider the Normal Distribution with b, σ as the parameters.

$$\text{Then } f(x; b, \sigma) = \frac{e^{-\frac{(x-b)^2}{2\sigma^2}}}{\sigma\sqrt{2\pi}}$$

The mean $\mu = E(X)$ is given by

$$\begin{aligned} \mu &= \int_{-\infty}^{\infty} x f(x) dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} x e^{-\frac{1}{2}\left(\frac{x-b}{\sigma}\right)^2} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (\sigma z + b) e^{-\frac{z^2}{2}} dz \quad \left[\text{Putting } z = \frac{x-b}{\sigma} \text{ so that } dz = \frac{dx}{\sigma} \right] \\ &= \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z e^{-\frac{z^2}{2}} dz + \frac{b}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}} dz \\ &= 0 + \frac{b}{\sqrt{2\pi}} \cdot 2 \int_0^{\infty} e^{-\frac{z^2}{2}} dz \quad \left[\because z e^{-\frac{z^2}{2}} \text{ is odd function and } e^{-\frac{z^2}{2}} \text{ is even function} \right] \\ &= \frac{2b}{\sqrt{2\pi}} \cdot \frac{\sqrt{\pi}}{\sqrt{2}} \quad \left[\because \int_0^{\infty} e^{-\frac{x^2}{2}} dx = \sqrt{\frac{\pi}{2}} \right] \\ &= b \end{aligned}$$

$\therefore$ **Mean, $\mu = b$**

2. Variance of Normal Distribution

By definition,

$$\begin{aligned} \text{Variance} &= E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx \\ &= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (x - b)^2 e^{-\frac{1}{2}\left(\frac{x-b}{\sigma}\right)^2} dx \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (\sigma z)^2 e^{-\frac{z^2}{2}} dz \quad \left[\text{Putting } z = \frac{x-b}{\sigma} \text{ so that } dz = \frac{dx}{\sigma} \right] \\
&= \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^2 e^{-\frac{z^2}{2}} dz \\
&= \frac{\sigma^2}{\sqrt{2\pi}} \cdot 2 \int_0^{\infty} z^2 e^{-\frac{z^2}{2}} dz \quad [\because \text{Integrand is even function}] \\
&= \frac{2\sigma^2}{\sqrt{2\pi}} \int_0^{\infty} 2t e^{-t} \cdot \frac{dt}{\sqrt{2t}} \quad \left[\text{Putting } \frac{z^2}{2} = t \text{ so that } dz = \frac{dt}{\sqrt{2t}} \right] \\
&= \frac{2\sigma^2}{\sqrt{\pi}} \int_0^{\infty} e^{-t} \sqrt{t} dt = \frac{2\sigma^2}{\sqrt{\pi}} \int_0^{\infty} e^{-t} t^{\frac{3}{2}-1} dt = \frac{2\sigma^2}{\sqrt{\pi}} \Gamma\left(\frac{3}{2}\right) \\
&= \frac{2\sigma^2}{\sqrt{\pi}} \cdot \frac{1}{2} \cdot \Gamma\left(\frac{1}{2}\right) = \frac{2\sigma^2}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = \sigma^2
\end{aligned}$$

Hence variance = σ^2

Thus the Standard deviation of the Normal Distribution is σ .

3. Mode of Normal Distribution

[JNTU 2004 S, (A) Dec. 2017]

Mode is the value of x for which $f(x)$ is maximum. That is, mode is the solution of $f'(x) = 0$ and $f''(x) < 0$.

By definition, we have

$$f(x) = \frac{1}{\sigma \cdot \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2}$$

Differentiating w.r.t. x , we get

$$f'(x) = \frac{1}{\sigma \cdot \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2} \left[-\left(\frac{x-\mu}{\sigma} \right) \right] = -\frac{x-\mu}{\sigma} \cdot f(x)$$

Now $f'(x) = 0 \Rightarrow x - \mu = 0$ i.e., $x = \mu$

$$\begin{aligned}
f''(x) &= -\frac{1}{\sigma^2} [(x-\mu) \cdot f'(x) + f(x)] \\
&= -\frac{1}{\sigma^2} \left[(x-\mu) \cdot -\frac{(x-\mu)}{\sigma^2} \cdot f(x) + f(x) \right] \\
&= \frac{-f(x)}{\sigma^2} \left[1 - \frac{(x-\mu)^2}{\sigma^2} \right]
\end{aligned}$$

At the point $x = \mu$, we have

$$f''(x) = -\left[\frac{f(x)}{\sigma^2} \right]_{x=\mu} = -\frac{1}{\sigma^2 \cdot \sqrt{2\pi} \cdot \sigma} < 0$$

Hence $x = \mu$ is the mode of the Normal Distribution.

4. Median of Normal distribution

[JNTU (A) 2004 S, Dec. 2017]

If M is the median of the normal distribution, we have

$$\int_{-\infty}^M f(x) dx = \frac{1}{2}$$

$$i.e. \quad \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^M e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = \frac{1}{2}$$

$$i.e. \quad \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\mu} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx + \frac{1}{\sigma\sqrt{2\pi}} \int_{\mu}^M e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = \frac{1}{2} \quad \dots (1)$$

$$\text{Consider } \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\mu} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

Put $\frac{x-\mu}{\sigma} = z$. Then $dx = \sigma dz$

$$\begin{aligned} \therefore \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\mu} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx &= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^0 e^{-\frac{z^2}{2}} \cdot \sigma dz = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^0 e^{-\frac{z^2}{2}} dz \\ &= \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-\frac{z^2}{2}} dz \text{ (by symmetry)} = \frac{1}{\sqrt{2\pi}} \cdot \sqrt{\frac{\pi}{2}} = \frac{1}{2} \quad \dots (2) \end{aligned}$$

$\therefore$ From (1) and (2), we have

$$\begin{aligned} \frac{1}{2} + \frac{1}{\sigma\sqrt{2\pi}} \int_{\mu}^M e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx &= \frac{1}{2} \\ \Rightarrow \frac{1}{\sigma\sqrt{2\pi}} \int_{\mu}^M e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx &= 0 \Rightarrow \int_{\mu}^M e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = 0 \\ \Rightarrow \mu &= M \left[\because \text{if } \int_a^b f(x) dx = 0 \text{ then } a = b, \text{ where } f(x) > 0 \right] \end{aligned}$$

Hence for the Normal Distribution, Mean = Median

Note : From above, we notice that for the Normal distribution mean, median and mode coincide.

i.e., mean = median = mode

[JNTU 2004S, 2008S (Set No.1,2)]

Hence the distribution is symmetrical.

5. Mean deviation from the Mean for Normal Distribution

[JNTU 2001S, (A) Dec. 2009 (Set No. 4)]

By definition,

Mean deviation (about mean)

$$\begin{aligned}
 &= \int_{-\infty}^{\infty} |x - \mu| f(x) dx \\
 &= \frac{1}{\sigma \sqrt{2\pi}} \int_{-\infty}^{\infty} |x - \mu| \cdot e^{-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2} dx \\
 &= \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} |z| e^{-z^2/2} dz \quad [\text{Putting } \frac{x - \mu}{\sigma} = z] \\
 &= \frac{\sigma}{\sqrt{2\pi}} \cdot 2 \int_0^{\infty} |z| \cdot e^{-\frac{z^2}{2}} dz \quad [\because \text{Integrand is an even function}] \\
 &= \frac{\sqrt{2}}{\pi} \cdot \sigma \int_0^{\infty} z e^{-\frac{z^2}{2}} dz \quad (\because |z| = z \text{ in } [0, \infty]) \\
 &= \frac{\sqrt{2}}{\pi} \cdot \sigma \int_0^{\infty} e^{-t} dt \quad [\text{Putting } \frac{z^2}{2} = t] \\
 &= \frac{\sqrt{2}}{\pi} \cdot \sigma (-e^{-t})_0^{\infty} = \frac{\sqrt{2}}{\pi} \cdot \sigma (0 + 1) \\
 &= \sigma \cdot \sqrt{\frac{2}{\pi}} = \frac{4\sigma}{5} \text{ (approximately)}
 \end{aligned}$$

Hence the mean deviation from the Mean for Normal distribution is equal to $\frac{4}{5}$ times of Standard deviation approximately.

7.13 CHIEF CHARACTERISTICS OF THE NORMAL DISTRIBUTION [JNTU 2004]

1. The graph of the Normal distribution $y = f(x)$ in the xy -plane is known as the normal curve.

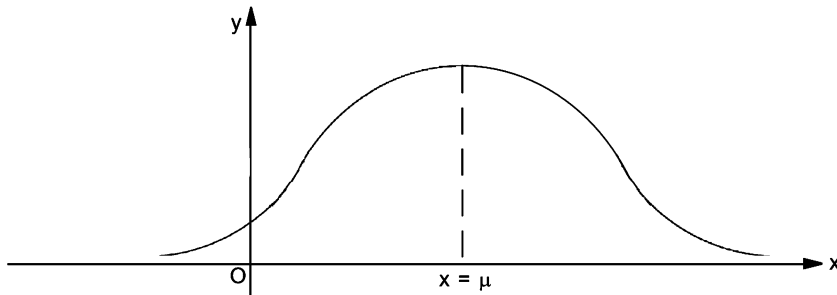
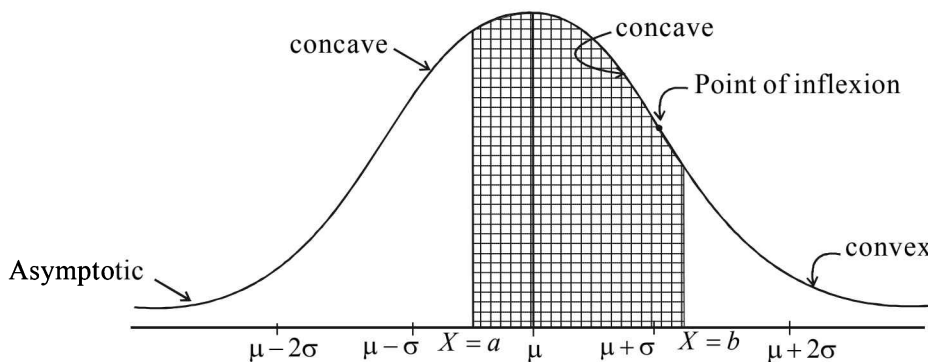


Fig.

2. The curve is a bell shaped curve and symmetrical with respect to mean *i.e.*, about the line $x = \mu$ and the two tails on the right and the left sides of the mean (μ) extends to infinity. The top of the bell is directly above the mean μ .
3. Area under the normal curve represents the total population.
4. Mean, median and mode of the distribution coincide at $x = \mu$ as the distribution is symmetrical. So normal curve is unimodal (has only one maximum point).
5. x -axis is an asymptote to the curve.
6. Linear combination of independent normal variates is also a normal variate.
7. The points of inflexion of the curve are at $x = \mu \pm \sigma$ and the curve changes from concave to convex at $x = \mu + \sigma$ to $x = \mu - \sigma$.

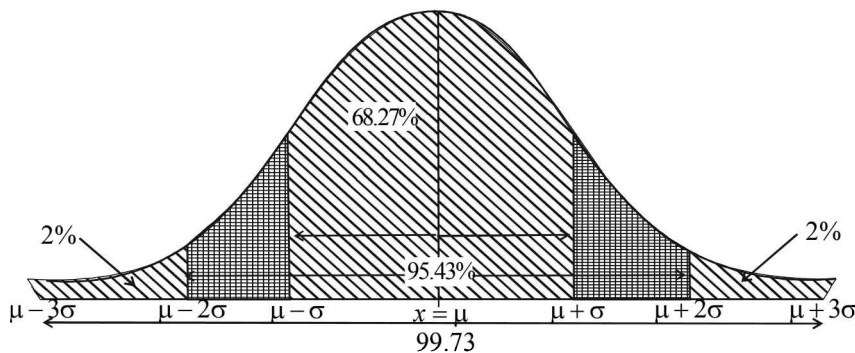


8. The probability that the normal variate X with mean μ and standard deviation σ lies between x_1 and x_2 is given by

$$P(x_1 \leq X \leq x_2) = \frac{1}{\sigma\sqrt{2\pi}} \int_{x_1}^{x_2} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \quad \dots (1)$$

Since (1) depends on the two parameters μ and σ , we get different normal curves for different values of μ and σ and it is an impracticable task to plot all such normal curves. Instead, by putting $z = \frac{x-\mu}{\sigma}$, the R.H.S. of equation (1) becomes independent of the two parameters μ and σ . Here z is known as the standard variable.

9. Area under the normal curve is distributed as follows :



- (i) Area of normal curve between $\mu - \sigma$ and $\mu + \sigma$ is 68.27%
i.e., $P(\mu - \sigma < X < \mu + \sigma) = 0.6826$.
- (ii) Area of normal curve between $\mu - 2\sigma$ and $\mu + 2\sigma$ is 95.43%.
- (iii) Area of normal curve between $\mu - 3\sigma$ and $\mu + 3\sigma$ is 99.73%.

Standard Normal Distribution :

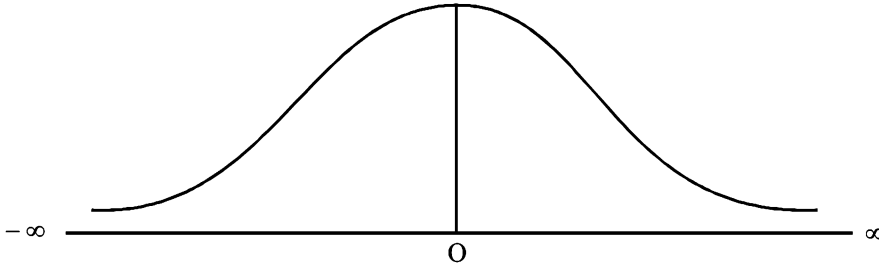
Definition : The Normal distribution with mean $(\mu) = 0$ and S. D. $(\sigma) = 1$, is known as *Standard Normal Distribution*.

The random variable that follows this distribution is denoted by z . If a variable x follows normal distribution with mean μ and *s.d.* σ , the variable z defined as

$$z = \frac{x - \mu}{\sigma}$$

has standard normal distribution with mean 0 and *s.d.* as 1. This is also referred as *z-score*.

The shape of this distribution is as follows:



It may be noted that the curve has the same shape as the normal distribution but, with the special properties that $\mu = 0$ and $\sigma = 1$ and has maximum at $z = 0$, and extends from $-\infty$ to $+\infty$ like normal distribution.

This is not the case with Binomial or Poisson distributions. They have different skewness (asymmetry) for different parameters.

We can transform any normal distribution to the standardised form because the normal distribution has the same shape whatever its parameter (μ and σ) may be.

If we take $\frac{x - \mu}{\sigma} = z$ in $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$, we obtain the Standard Normal

Distribution with mean zero and S. D. $(\sigma) = 1$ defined by the density function

$$p(z) = \frac{1}{\sqrt{2\pi}} e^{-(z^2/2)}, \quad -\infty < z < \infty$$

Here z is called the standard normal variate with mean 0 and standard deviation 1 and in short, we say that $p(z)$ is $N(0,1)$.

Area under any normal curve is found from the table of standard normal probability distribution showing the area between the mean and any value of the normally distributed random variable. For given values of μ and σ , and a specific value, x , of the random variable, the standardized variate z is derived from the formula $z = \frac{x - \mu}{\sigma}$. The purpose of standardisation of the normal distribution is that we can make use of the tables of the area of the standard curve $f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ (representing probability) for the various points along the x - axis.

(The standard normal tables are given at the end of the book).

The standard normal distribution is also known as Unit Normal Distribution or Z - Distribution.

The standard normal curve helps us to find the areas within two assigned limits under the curve. The areas between the standard normal curve drawn at two assigned limits a and b will give the proportion of cases for which the values of z lie between a and b . Thus the area between two assigned limits a and b under the standard normal curve will represent the probability that z will lie between a and b . It is denoted by $P(a \leq z \leq b)$.

Uses of Normal Distribution:

1. The normal distribution can be used to approximate Binomial and Poisson distributions.
2. It has extensive use in sampling theory. It helps us to estimate parameter from statistic and to find confidence limits of the parameter.
3. It has a wide use in testing Statistical Hypothesis and Tests of significance in which it is always assumed that the population from which the samples have been drawn should have normal distribution.
4. It serves as a guiding instrument in the analysis and interpretation of statistical data.

Some other uses and situations where normal distribution is applicable are mentioned in the section 7.15.

7.14 AREA UNDER THE NORMAL CURVE (NORMAL PROBABILITY INTEGRAL) CONVERSION INTO STANDARD NORMAL FORM

By taking $z = \frac{x - \mu}{\sigma}$, the standard normal curve is formed. The total area under this curve is 1. The area under the curve is divided into two equal parts by $z = 0$. Left hand side area and right hand side area to $z = 0$ is 0.5. The area between the ordinate $z = 0$ and any other ordinate can be noted from the table given in the subsequent pages.

The probability that random variable X will lie between $X = \mu$ and $X = x_1$ is given by

$$P(\mu < X < x_1) = \int_{\mu}^{x_1} f(x) dx = \frac{1}{\sigma \sqrt{2\pi}} \int_{\mu}^{x_1} e^{-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2} dx = \frac{1}{\sigma \sqrt{2\pi}} \int_0^{z_1} e^{-\frac{z^2}{2}} \cdot \sigma dz$$

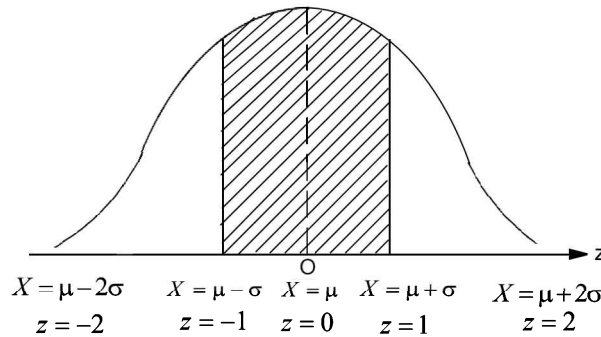
$$[\text{Putting } \frac{x - \mu}{\sigma} = z \text{ and } \frac{x_1 - \mu}{\sigma} = z_1]$$

$$\text{i.e., } P(\mu < X < x_1) = P(0 < z < z_1) = \frac{1}{\sqrt{2\pi}} \int_0^{z_1} e^{-\frac{z^2}{2}} dz = A(z_1) \text{ say.}$$

$$\text{i.e., } A(z_1) = \frac{1}{\sqrt{2\pi}} \int_0^{z_1} e^{-\frac{z^2}{2}} dz.$$

The function $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$ is the probability function of standard normal variate.

The definite integral $\int_0^{z_1} e^{-\frac{z^2}{2}} dz$ is known as the *normal probability integral* and gives the area under standard normal curve between $z = 0$ and $z = z_1$.



In particular, the probability that a random variable X lies in the interval $(\mu - \sigma, \mu + \sigma)$ is given by

$$P(\mu - \sigma < X < \mu + \sigma) = \int_{\mu - \sigma}^{\mu + \sigma} f(x) dx$$

$$\Rightarrow P(-1 < z < 1) = \frac{1}{\sqrt{2\pi}} \int_{-1}^1 e^{-\frac{z^2}{2}} dz \left(z = \frac{x - \mu}{\sigma} \right)$$

$$= \frac{2}{\sqrt{2\pi}} \int_0^1 e^{-\frac{z^2}{2}} dz = 2 \int_0^1 \phi(z) dz \text{ where } \phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

$$= 2(0.3413) \text{ (from tables)}$$

$$= 0.6826$$

Note : Since the integrand is an even function, it follows that $A(-z_1) = A(z_1)$.

$$\begin{aligned} \text{Similarly } P(\mu - 2\sigma < X < \mu + 2\sigma) &= P(-2 < z < 2) = \int_{-2}^2 \phi(z) dz \\ &= 2 \int_0^2 \phi(z) dz = 2(0.4772) = 0.9544 \end{aligned}$$

How to find Probability Density of normal curve

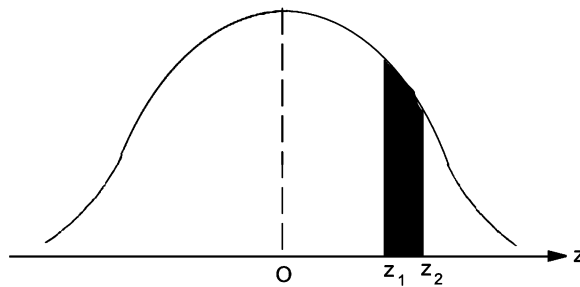
The probability that the normal variate X with mean μ and standard deviation σ lies between two specific values x_1 and x_2 with $x_1 \leq x_2$ can be obtained using area under the standard normal curve as follows :

Step 1 : Perform the change of scale $z = \frac{x - \mu}{\sigma}$ and find z_1 and z_2 corresponding to the values of x_1 and x_2 respectively.

Step 2 (a) : To find $P(x_1 \leq X \leq x_2) = P(z_1 \leq z \leq z_2)$

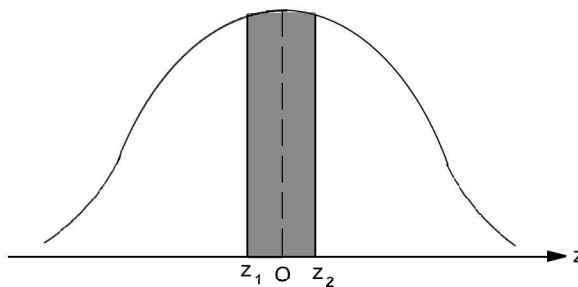
Case 1 : If both z_1 and z_2 are positive (or both negative), then

$$\begin{aligned} P(x_1 \leq X \leq x_2) &= |A(z_2) - A(z_1)| \\ &= (\text{Area under the normal curve from } 0 \text{ to } z_2) \\ &\quad - (\text{Area under the normal curve from } 0 \text{ to } z_1) \end{aligned}$$

**Fig.**

Case 2 : If $z_1 < 0$ and $z_2 > 0$, then

$$P(x_1 \leq X \leq x_2) = A(z_2) + A(z_1)$$

**Fig.**

Step 2 (b) : To find $P(z > z_1)$

Case 1 : If $z_1 > 0$ then

$$P(z > z_1) = 0.5 - A(z_1) \quad [\because P(z < 0) = P(z > 0) = \frac{1}{2}]$$

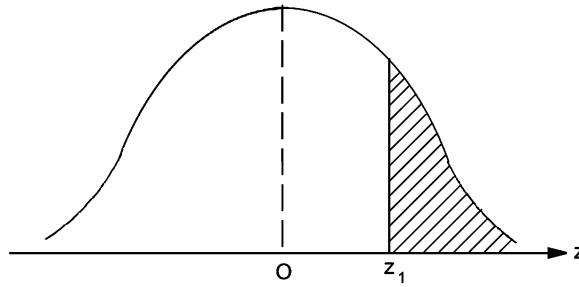


Fig.

Case 2 : If $z_1 < 0$, then $P(z > z_1) = 0.5 + A(z_1)$

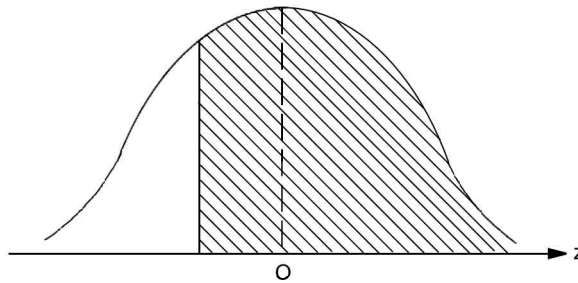


Fig.

Step 2 (c) : To find $P(z < z_1) = 1 - P(z > z_1)$

Case 1 : If $z_1 > 0$, then

$$P(z < z_1) = 1 - P(z > z_1) = 1 - [0.5 - A(z_1)] = 0.5 + A(z_1)$$

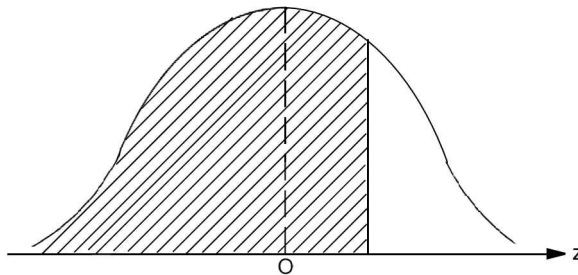


Fig.

Case 2 : If $z_1 \leq 0$, then

$$P(z < z_1) = 1 - P(z > z_1) = 1 - [0.5 + A(z_1)] = 0.5 - A(z_1)$$

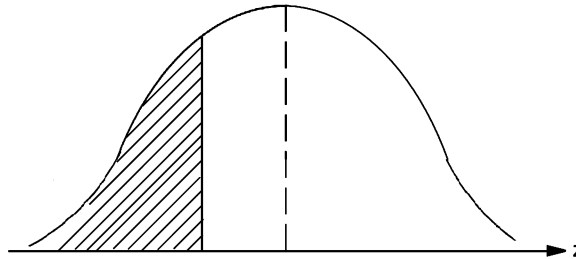


Fig.

7.15 THE IMPORTANCE AND APPLICATIONS OF THE NORMAL DISTRIBUTION

[JNTU 2000, 200S, 2000S (Set No. 4)]

Normal distribution plays a very important role in statistical theory because of the following reasons :

1. Data obtained from Psychological, Physical and Biological measurements approximately follow Normal distribution. I.Q. scores, heights and weights of individuals etc., are examples of measurements which are normally distributed or nearly so.
2. Most of the distributions that are encountered in practice, for example, Binomial, Poisson, Hypergeometric, etc. can be approximated to Normal distribution. If the number of trials n is indefinitely large and neither p nor q is very small, then Binomial distribution tends to Normal distribution. If the parameter $\lambda \rightarrow \infty$, then Poisson distribution tends to Normal distribution.
3. Since the Normal distribution is a limiting case of the Binomial distribution for exceptionally large numbers, it is applicable to many applied problems in kinetic theory of gases and fluctuations in the magnitude of an electric current.
4. Even if a variable is normally distributed, it can sometimes be brought to normal form by simple transformation of the variable.
5. For large samples, any statistic (i.e., sample mean, sample S.D., etc.) approximately follows Normal distribution and as such it can be studied with the help of normal curve.
6. Normal curve is used to find confidence limits of the population parameters.
7. The proofs of all the tests of significance in sampling are based upon the fundamental assumption that the population from which the samples have been drawn is normal.
8. Normal distribution finds large applications in Statistical Quality Control in industry for finding control limits.

	Table : Area under standard normal curve from 0 to $\frac{x - \mu}{\sigma}$									
$\frac{x - \mu}{\sigma}$	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.0000	.0040	.0080	.0120	.0159	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1084	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1916	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2258	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2518	.2549
0.7	.2580	.2611	.2642	.2671	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4430	.4441
1.6	.4452	.4463	.4474	.4485	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4564	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4762	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4865	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4892
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4980	.4980	.4981
2.9	.4981	.4982	.4983	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990
3.1	.4990	.4991	.4991	.4991	.4992	.4992	.4992	.4992	.4993	.4993

SOLVED EXAMPLES

Example 1 : For a normally distributed variate with mean 1 and standard deviation 3, find the probabilities that

(i) $3.43 \leq x \leq 6.19$ (ii) $-1.43 \leq x \leq 6.19$

Solution : Given $\mu = 1$ and $\sigma = 3$

(i) When $x = 3.43$,

$$z = \frac{x - \mu}{\sigma} = \frac{3.43 - 1}{3} = \frac{2.43}{3} = 0.81 = z_1 \text{ (say)}$$

When $x = 6.19$,

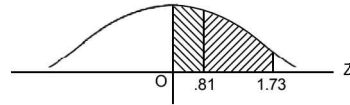
$$z = \frac{x - \mu}{\sigma} = \frac{6.19 - 1}{3} = \frac{5.19}{3} = 1.73 = z_2 \text{ (say)}$$

$$\therefore P(3.43 \leq x \leq 6.19) = P(0.81 \leq z \leq 1.73)$$

$$= |A(z_2) - A(z_1)|$$

$$= |A(1.73) - A(0.81)| = 0.4582^* - 0.2910 \text{ (from normal tables)}$$

$$= 0.1672 \text{ (Cross hatched area in the figure)}$$



(ii) When $x = -1.43$,

$$z = \frac{x - \mu}{\sigma} = \frac{-1.43 - 1}{3} = -0.81 = z_1 \text{ (say)}$$

When $x = 6.19$,

$$z = \frac{x - \mu}{\sigma} = \frac{6.19 - 1}{3} = 1.73 = z_2 \text{ (say)}$$

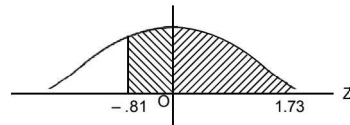
$$\therefore P(-1.43 \leq x \leq 6.19) = P(-0.81 \leq z \leq 1.73)$$

$$= A(z_2) + A(z_1)$$

$$= A(1.73) + A(-0.81)$$

$$= A(1.73) + A(0.81) \quad [\because A(-z) = A(z)]$$

$$= 0.4582 + 0.2910 = 0.7492 = \text{shaded area in the figure.}$$



Example 2 : If X is a normal variate with mean 30 and standard deviation 5.

Find (i) $P(26 \leq X \leq 40)$ (ii) $P(X \geq 45)$ [JNTU 2000, 2000S, (K) Nov. 2009 (Set No. 1)]

Solution : Given mean, $\mu = 30$ and S.D., $\sigma = 5$

(i) When $x = 26$, $z = \frac{x - \mu}{\sigma} = \frac{26 - 30}{5} = -0.8 = z_1 \text{ (say)}$

When $x = 40$, $z = \frac{x - \mu}{\sigma} = \frac{40 - 30}{5} = 2 = z_2 \text{ (say)}$

$$\therefore P(26 \leq X \leq 40) = P(-0.8 \leq z \leq 2)$$

$$= A(z_2) + A(z_1) = A(2) + A(-0.8)$$

$$= 0.4772 + 0.2881 = 0.7653 \text{ (from normal distribution tables)}$$

* See Normal Distribution Table given, read the data in row 1.7 and column 0.03.

$$(ii) \text{ When } x = 45, z = \frac{x - \mu}{\sigma} = \frac{45 - 30}{5} = 3 = z_1 \text{ (say)}$$

$$\therefore P(X \geq 45) = P(z_1 \geq 3)$$

$$= 0.5 - A(z_1) = 0.5 - A(3) = 0.5 - 0.49865 = 0.00135$$

Example 1 : In a Normal distribution, 7% of the items are under 35 and 89% are under 63. Determine the mean and variance of the distribution.

[JNTU Jan. 2007, (H) Dec. 2011, (K) Nov. 2012 (Set No. 2)]

(OR) Find the mean and standard deviation of a normal distribution in which 7% of items are under 35 and 89 % are under 63. [JNTU (H) Nov., 2009, (A) Nov. 2010 (Set No. 4), Dec. 2015]

(OR) In a distribution of normal, 7% of items are under 35 and 89% are under 63. What are the mean and standard deviation of the distribution ? [JNTU (A) Dec. 2018]

Solution : Let μ be the mean (at $z = 0$) and σ the standard deviation of the normal curve. 7% of the items are under 35 means the area to the left of the ordinate $x = 35$.

$$\text{Given } P(X < 35) = 0.07 \text{ and } P(X < 63) = 0.89$$

$$\therefore P(X > 63) = 1 - P(X < 63) = 1 - 0.89 = 0.11$$

The points $X = 35$ and $X = 63$ are shown in the following Figure.

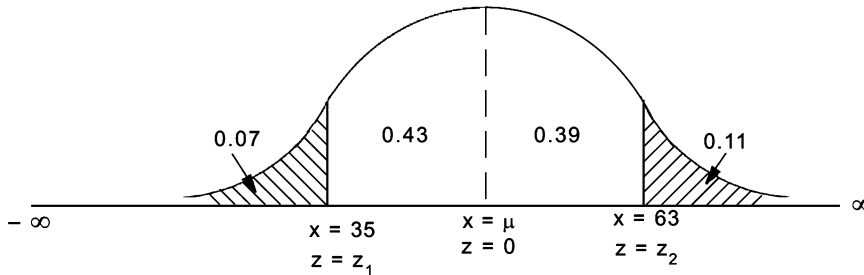


Fig.

$$\text{When } x = 35, z = \frac{x - \mu}{\sigma} = \frac{35 - \mu}{\sigma} = -z_1 \text{ (say)} \quad \dots (1)$$

$$\text{When } x = 63, z = \frac{x - \mu}{\sigma} = \frac{63 - \mu}{\sigma} = z_2 \text{ (say)} \quad \dots (2)$$

From the figure, we have

$$P(0 < z < z_2) = 0.39 \Rightarrow z_2 = 1.23 \quad (\text{from normal tables})$$

$$\text{and } P(0 < z < z_1) = 0.43 \Rightarrow z_1 = 1.48$$

$$\text{From (1), we have } \frac{35 - \mu}{\sigma} = -1.48 \quad \dots (3)$$

$$\text{From (2), we have } \frac{63 - \mu}{\sigma} = 1.23 \quad \dots (4)$$

(4) - (3) gives

$$\frac{28}{\sigma} = 2.71 \Rightarrow \sigma = \frac{28}{2.71} = 10.332$$

From (3), $35 - \mu = -1.48 (\sigma) = (-1.48) (10.332) = -15.3$

$$\therefore \mu = 35 + 15.3 = 50.3$$

and variance $= \sigma^2 = 106.75$

Example 4 : In a normal distribution 31% of the items are under 45 and 8% are over 64. Find the mean and variance of the distribution.

[JNTU (A) Dec. 2009, (K) Nov. 2012, Mar. 2014 (Set No. 2)]

Solution : Let X be the continuous random variable.

Let μ be the mean and σ the standard deviation.

Given $P(X < 45) = 0.31$ and $P(X > 64) = 0.08$

Standard variable, $z = \frac{X - \mu}{\sigma}$

When $X = 45$, let $z = z_1$

$$\text{so that } z_1 = \frac{45 - \mu}{\sigma} \quad \dots (1)$$

$$\therefore \int_{-\infty}^{z_1} \phi(z) dz = 0.31 \quad \text{or} \quad \int_{-\infty}^0 \phi(z) dz - \int_{z_1}^0 \phi(z) dz = 0.31$$

$$\therefore \int_{z_1}^0 \phi(z) dz = \int_{-\infty}^0 \phi(z) dz - 0.31 = 0.5 - 0.31 = 0.19$$

$$\text{Hence } P(0 < z < z_1) = 0.19 \Rightarrow z_1 = -0.5 \quad (\text{from table}) \quad \dots (2)$$

$$\text{When } X = 64, \quad z = \frac{64 - \mu}{\sigma} = z_2 \quad (\text{say}) \quad \dots (3)$$

$$\therefore \int_{z_2}^{\infty} \phi(z) dz = 0.08 \quad \text{or} \quad \int_0^{\infty} \phi(z) dz - \int_0^{z_2} \phi(z) dz = 0.08$$

$$\text{Hence } \int_0^{z_2} \phi(z) dz = \int_0^{\infty} \phi(z) dz - 0.08 = 0.5 - 0.08 = 0.42$$

$$\text{Thus } P(0 < z < z_2) = 0.42 \Rightarrow z_2 = 1.4 \quad (\text{from tables}) \quad \dots (4)$$

$$\text{From (1) and (2), we have } \frac{45 - \mu}{\sigma} = -0.5 \Rightarrow 45 - \mu = -0.5\sigma \quad \dots (5)$$

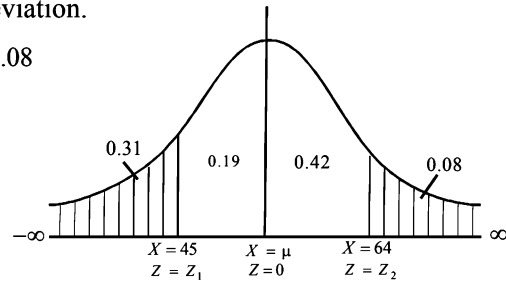
$$\text{From (3) and (4), we have } \frac{64 - \mu}{\sigma} = 1.4 \Rightarrow 64 - \mu = 1.4\sigma \quad \dots (6)$$

$$(5) - (6) \text{ gives } (45 - \mu) - (64 - \mu) = -0.5\sigma - 1.4\sigma$$

$$\Rightarrow -19 = -1.9\sigma \quad \therefore \sigma = \frac{19}{1.9} = 10$$

$$\text{From (5), } \mu = 45 + 0.5\sigma = 45 + 0.5(10) = 50$$

Hence mean = 50 and standard deviation = 10



Example 1 : Suppose 10 percent of the probability for a normal distribution $N(\mu, \sigma^2)$ is below 35 and 5 percent above 90. What are the values of μ and σ ? [JNTU (K) 2009(Set No.2)]

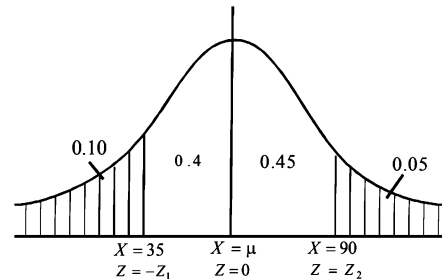
Solution : If $X \sim N(\mu, \sigma^2)$ then we are given $P(X < 35) = 0.10$ and $P(X > 90) = 0.05$. The points $X = 35$ and $X = 90$ are located as shown in Figure.

$$\text{Standard variable, } z = \frac{X - \mu}{\sigma}$$

$$\text{When } X = 35, z = \frac{35 - \mu}{\sigma} = -z_1 \text{ (say)}$$

and when $X = 90$,

$$z = \frac{90 - \mu}{\sigma} = z_2 \text{ (say)}$$



Thus we have $P(0 < z < z_1) = 0.4$ and $P(0 < z < z_2) = 0.45$.

Hence from Normal tables, we have $z_1 = 1.3$ and $z_2 = 1.65$

$$\therefore \frac{35 - \mu}{\sigma} = -1.3 \text{ and } \frac{90 - \mu}{\sigma} = 1.65$$

$$\text{Subtracting, we get } \frac{55}{\sigma} = 2.95 \Rightarrow \sigma = \frac{55}{2.95} = 18.64$$

$$\therefore \mu = 35 + 1.3\sigma = 35 + 1.3(18.64) = 35 + 24.24 = 59.24$$

Hence $\mu = 59.24$ and $\sigma = 18.64$

Example 1 : The mean and standard deviation of the marks obtained by 1000 students in an examination are respectively 34.5 and 16.5. Assuming the normality of the distribution, find the approximate number of students expected to obtain marks between 30 and 60.

[JNTU (A) Nov. 2010 (Set No. 1)]

Solution : Given mean, $\mu = 34.5$ and S.D., $\sigma = 16.5$

$$\text{When } x = 30, z = \frac{x - \mu}{\sigma} = \frac{30 - 34.5}{16.5} = -0.27 = z_1 \text{ (say)}$$

$$\text{When } x = 60, z = \frac{x - \mu}{\sigma} = \frac{60 - 34.5}{16.5} = 1.54 = z_2 \text{ (say)}$$

$$\begin{aligned} \therefore P(30 \leq x \leq 60) &= P(z_1 \leq z \leq z_2) \\ &= A(z_2) + A(z_1) \quad [\because z_1 < 0 \text{ and } z_2 > 0] \\ &= A(1.54) + A(-0.27) = A(1.54) + A(0.27) \\ &= 0.4382 + 0.1084 \quad [\text{From normal distribution tables}] \\ &= 0.5916 \end{aligned}$$

$\therefore$ The number of students who get marks between 30 and 60

$$= 0.5916 \times 1000 = 591.6$$

Hence 592 students get marks between 30 and 60.

Example 2 : The marks obtained in mathematics by 1000 students is normally distributed with mean 78% and standard deviation 11%. Determine

- (i) How many students got marks above 90%
- (ii) What was the highest mark obtained by the lowest 10% of the students
- (iii) Within what limits did the middle of 90% of the students lie.

[JNTU 2006S, 2007, (A) Nov. 2010, Apr. 2012, (K) Nov. 2011 (Set No. 2)]

Solution : Given mean, $\mu = 0.78$ and S.D., $\sigma = 0.11$

(i) When $x = 0.9$, $z = \frac{x - \mu}{\sigma} = \frac{0.9 - 0.78}{0.11} = 1.09 = z_1$ (say)

Hence the number of students with marks more than 90%
 $= 0.1379 \times 1000$
 $= 137.9 \simeq 138$

- (ii) The 0.1 area to the left of z corresponds to the lowest 10% of the students

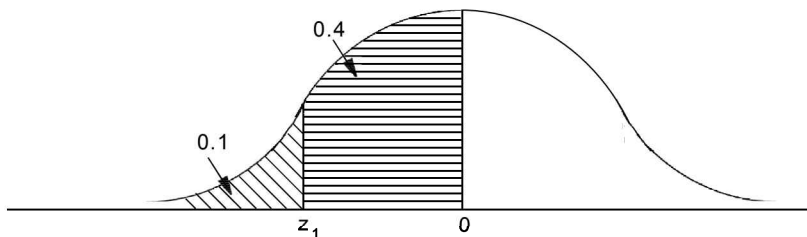


Fig.

From Figure,

$$0.4 = 0.5 - 0.1 = 0.5 - \text{Area from } 0 \text{ to } z_1$$

$$\therefore z_1 = -1.28 \text{ (from normal tables)}$$

$$\text{Thus } -1.28 = \frac{x - \mu}{\sigma} = \frac{x - 0.78}{0.11} \Rightarrow x = 0.78 - 1.28(0.11) = 0.6392$$

Hence the highest mark obtained by the lowest 10% of students
 $= 0.6392 \times 1000 \simeq 64\%$

- (iii)

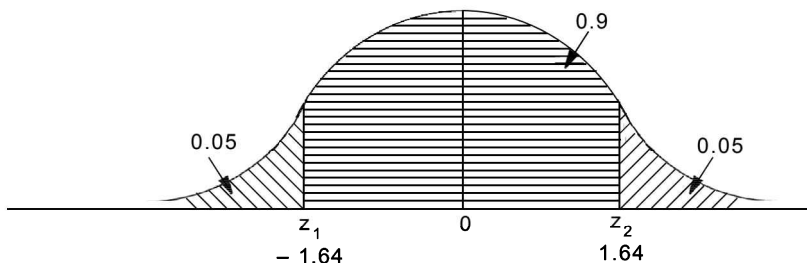


Fig.

Middle 90% correspond to 0.9 area, leaving 0.05 area on both sides. Then the corresponding z 's are ± 1.64 [since if the area from 0 to z is 0.45, then $z = 1.64$]

$$\therefore -1.64 = z_1 = \frac{x_1 - \mu}{\sigma} = \frac{x_1 - 0.78}{0.11}$$

$$\Rightarrow x_1 = 0.78 - 1.64(0.11) = 0.5996 \text{ or } 59.96\%$$

$$\text{and } 1.64 = z_2 = \frac{x_2 - \mu}{\sigma} = \frac{x_2 - 0.78}{0.11}$$

$$\Rightarrow x_2 = 1.64(0.11) + 0.78 = 0.9604 \text{ or } 96.04\%$$

Thus the middle 90% have marks in between 60 to 96.

Example 8 : Suppose the weights of 800 male students are normally distributed with mean $\mu = 140$ pounds and standard deviation 10 pounds. Find the number of students whose weights are (i) between 138 and 148 pounds (ii) more than 152 pounds

[JNTU 2006 S (Set No.2). (A) June 2016]

Solution : Let μ be the mean and σ be the standard deviation. Then $\mu = 140$ pounds and $\sigma = 10$ pounds.

$$(i) \quad \text{When } x = 138, z = \frac{x - \mu}{\sigma} = \frac{138 - 140}{10} = -0.2 = z_1 \text{ (say)}$$

$$\text{When } x = 148, z = \frac{x - \mu}{\sigma} = \frac{148 - 140}{10} = 0.8 = z_2 \text{ (say)}$$

$$\begin{aligned} \therefore P(138 \leq x \leq 148) &= P(-0.2 \leq z \leq 0.8) \\ &= A(z_2) + A(z_1) = A(0.8) + A(-0.2) \\ &= A(0.8) + A(0.2) \\ &= 0.2881 + 0.0793 = 0.3674 \end{aligned}$$

Hence the number of students whose weights are between 138 pounds and 148 pounds $= 0.3674 \times 800 \approx 294$

$$(ii) \quad \text{When } x = 152, z = \frac{x - \mu}{\sigma} = \frac{152 - 140}{10} = 1.2 = z_1 \text{ (say)}$$

$$\begin{aligned} \therefore P(x > 152) &= P(z > z_1) \\ &= 0.5 - A(z_1) = 0.5 - A(1.2) \\ &= 0.5 - 0.3849 = 0.1151 \end{aligned}$$

$$\begin{aligned} \therefore \text{Number of students whose weights are more than 152 pounds} \\ &= 800 \times 0.1151 = 92 \end{aligned}$$

Example 9 : The marks obtained in Statistics in a certain examination found to be normally distributed. If 15% of the students ≥ 60 marks, 40% < 30 marks, find the mean and standard deviation. [JNTU 2006S (Set No.3)]

Solution : Let μ be the mean (at $z = 0$) and σ the standard deviation of the normal distribution.

Let the variable X denote the marks in statistics.

Then we are given

$$P(X < 30) = 0.4 \text{ and } P(X \geq 60) = 0.15$$

$$\text{When } X = 30, z = \frac{X - \mu}{\sigma} = \frac{30 - \mu}{\sigma} = -z_1 \text{ (say)} \quad \dots (1)$$

$$\text{When } X = 60, z = \frac{X - \mu}{\sigma} = \frac{60 - \mu}{\sigma} = z_2 \text{ (say)} \quad \dots (2)$$

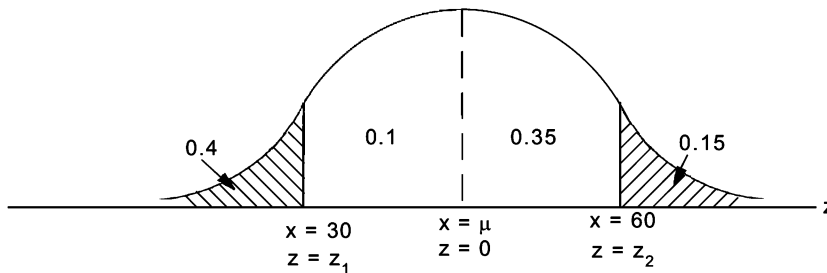


Fig.

$$\therefore P(0 < z < z_2) = 0.5 - 0.15 = 0.35$$

$$\text{and } P(0 < z < z_1) = P(-z_1 < z < 0) \text{ (By symmetry)} \\ = 0.5 - 0.4 = 0.1$$

$\therefore$ From normal tables, we get

$$z_1 = 0.25 \text{ and } z_2 = 1.04$$

$$\text{Hence } \frac{30 - \mu}{\sigma} = -0.25 \text{ or } \frac{\mu - 30}{\sigma} = 0.25, \text{ using (1)} \quad \dots (3)$$

$$\text{and } \frac{60 - \mu}{\sigma} = 1.04, \text{ using (2)} \quad \dots (4)$$

(3) + (4) gives

$$\frac{30}{\sigma} = 1.29 \quad \therefore \sigma = \frac{30}{1.29} = 23.26$$

$$\therefore \mu = 0.25 \sigma + 30 \text{ [From (3)]} \\ = 0.25 (23.26) + 30 = 35.81$$

Hence mean = 35.81 and standard deviation = 23.26

Example 10 : If X is a normal variate, find the area **A**

(i) to the left of $z = -1.78$

(ii) to the right of $z = -1.45$

(iii) corresponding to $-0.8 \leq z \leq 1.53$

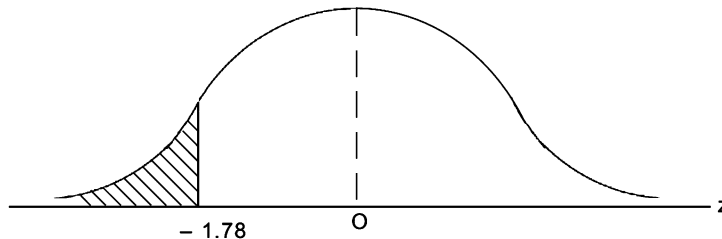
(iv) to the left of $z = -2.52$ and to the right of $z = 1.83$

[JNTU 2004S, 2005S, 2007(Set No.3)]

Solution :

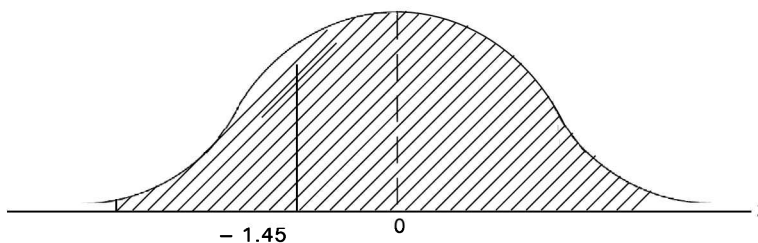
(i) Required area, A

$$\begin{aligned}
 &= 0.5 - \text{Area (0 to } -1.78) = 0.5 - \text{Area (0 to } 1.78) \text{ (By symmetry)} \\
 &= 0.5 - 0.4625 \text{ (from tables)} = 0.0375
 \end{aligned}$$

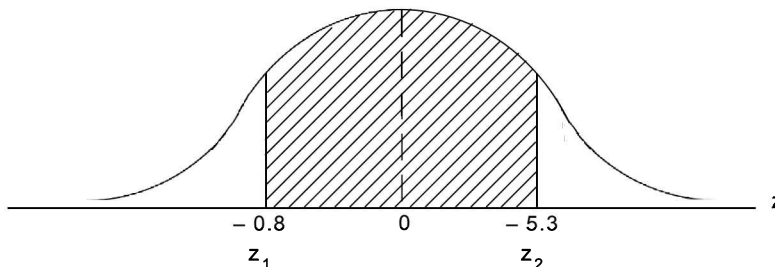
**Fig.**

(ii) Required area, A = 0.5 + Area from 0 to -1.45

$$\begin{aligned}
 &= 0.5 + \text{Area from 0 to } 1.45 \text{ (By symmetry)} \\
 &= 0.5 + 0.4265 = 0.9265
 \end{aligned}$$

**Fig.**(iii) Required area = A(z₂) + A(z₁)

$$\begin{aligned}
 &= A(1.53) + A(-0.8) = A(1.53) + A(0.8) \\
 &= 0.437 + 0.2881 = 0.7251
 \end{aligned}$$

**Fig.****Aliter :**

$$\begin{aligned}
 \text{Required area} &= \text{Area from (0 to } -0.8) + \text{Area from (0 to } 1.53) \\
 &= \text{Area from (0 to } 0.8) + \text{Area from (0 to } 1.53) \\
 &= 0.4370 + 0.2881 = 0.7251
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv) Required area} &= [0.5 - \text{Area from 0 to 2.52}] + [0.5 - \text{Area from 0 to 1.83}] \\
 &= (0.5 - 0.4941) + (0.5 - 0.4664) = 0.0059 + 0.0336 = 0.0395
 \end{aligned}$$

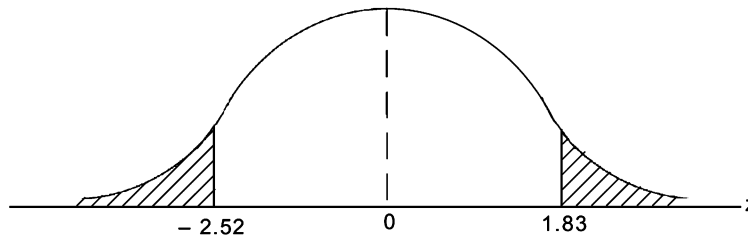


Fig.

Example 11 : In a sample of 1000 cases, the mean of a certain test is 14 and standard deviation is 2.5. Assuming the distribution to be normal, find

(i) how many students score between 12 and 15 ?

(ii) how many score above 18 ?

(iii) how many score below 18 ?

[JNTU 2004 S, 2007S (Set No. 2,3)]

Solution : Let μ be the mean and σ the standard deviation of the normal distribution.

Then we are given

$$\mu = 14 \text{ and } \sigma = 2.5$$

Let the variable X denote the score in a test.

$$(i) \text{ When } X = 12, z = \frac{X - \mu}{\sigma} = \frac{12 - 14}{2.5} = -0.8 = z_1 \text{ (say)}$$

$$\text{When } X = 15, z = \frac{X - \mu}{\sigma} = \frac{15 - 14}{2.5} = 0.4 = z_2 \text{ (say)}$$

$$\begin{aligned}
 \therefore P(12 < X < 15) &= P(-0.8 < z < 0.4) \\
 &= A(z_2) + A(z_1) = A(0.4) + A(-0.8) \\
 &= A(0.4) + A(0.8) \quad (\text{due to symmetry}) \\
 &= 0.1554 + 0.2881 \\
 &= 0.4435
 \end{aligned}$$

Hence number of students score between 12 and 15

$$= 1000 \times 0.4435 = 443 \text{ (approximately)}$$

$$(ii) \text{ When } X = 18, z = \frac{X - \mu}{\sigma} = \frac{18 - 14}{2.5} = 1.6$$

$$\begin{aligned}
 \therefore P(X > 18) &= P(z > 1.6) \\
 &= 0.5 - A(1.6) = 0.5 - 0.4452 \\
 &= 0.0548
 \end{aligned}$$

Hence number of students score above 18 = 1000×0.0548

$$= 54.8 = 55 \text{ (approximately)}$$

$$(iii) \quad P(X < 18) = P(z < 1.6)$$

$$= 0.5 + A(1.6) = 0.5 + 0.4452 = 0.9452$$

$$\text{Aliter : } P(X < 18) = 1 - P(X > 18) = 1 - 0.0548 = 0.9452$$

$$\therefore \text{ Number of students score below 18} = 1000 \times 0.9452 = 945$$

Example 12 : A sales tax officer has reported that the average sales of the 500 business that he has to deal with during a year is Rs. 36,000 with a standard deviation of 10,000. Assuming that the sales in these business are normally distributed, find

(i) the number of business as the sales of which are Rs. 40,000.

(ii) the percentage of business the sales of which are likely to range between Rs. 30,000 and Rs. 40,000. [JNTU 2005 (Set No. 1, 2, 3)]

Solution : Let μ be the mean and σ the standard deviation of the sales. Then we are given that $\mu = 36000$ and $\sigma = 10000$

Let the variable X denote the sales in the business

$$\text{When } X = 40000, z = \frac{X - \mu}{\sigma} = \frac{40000 - 36000}{10000} = 0.4$$

$$\text{When } X = 30000, z = \frac{X - \mu}{\sigma} = \frac{30000 - 36000}{10000} = -0.6$$

$$(i) \quad P(X > 40000) = P(z > 0.4)$$

$$= 0.5 - A(0.4) = 0.5 - 0.1554 = 0.3446$$

Number of business as the sales of which are Rs. 40,000

$$= 500 \times 0.3446 = 172 \text{ (approximately)}$$

$$(ii) \quad P(30000 < X < 40000) = P(-0.6 < z < 0.4)$$

$$= A(0.4) + A(-0.6) = A(0.4) + A(0.6)$$

$$= 0.1554 + 0.2257 = 0.3811$$

$\therefore$ The required percentage of business = 38.11%

Example 12 : If the masses of 300 students are normally distributed with mean 68 kgs and standard deviation 3 kgs, how many students have masses

(i) Greater than 72 kgs.

(ii) Less than or equal to 64 kgs.

(iii) Between 65 and 71 kgs. inclusive.

[JNTU 2005, 2008, (A) Nov. 2010, (H) Dec. 2011, (K) May 2013 (Set No. 2)]

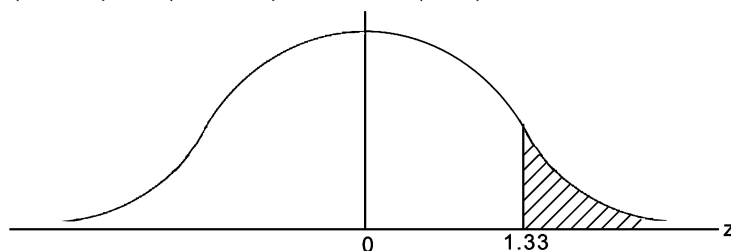
Solution : Let μ be the mean and σ the standard deviation of the distribution. Then

$$\mu = 68 \text{ kgs and } \sigma = 3 \text{ kgs}$$

Let the variable X denote the masses of students

$$(i) \quad \text{When } X = 72, \quad z = \frac{X - \mu}{\sigma} = \frac{72 - 68}{3} = 1.33$$

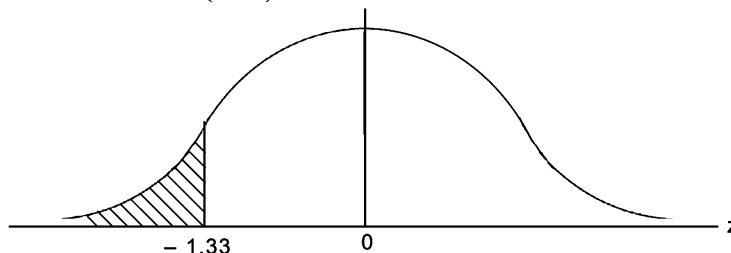
$$\therefore P(X > 72) = P(z > 1.33) = 0.5 - A(1.33) = 0.5 - 0.4082 = 0.0918$$

**Fig.**

Number of students with more than 72 kgs = $300 \times 0.0918 = 28$ (approximately)

$$(ii) \text{ When } X = 64, z = \frac{X - \mu}{\sigma} = \frac{64 - 68}{3} = -1.33$$

$$\begin{aligned} \therefore P(X \leq 64) &= P(z \leq -1.33) \\ &= 0.5 - A(1.33) = 0.5 - 0.4082 = 0.0918 \end{aligned}$$

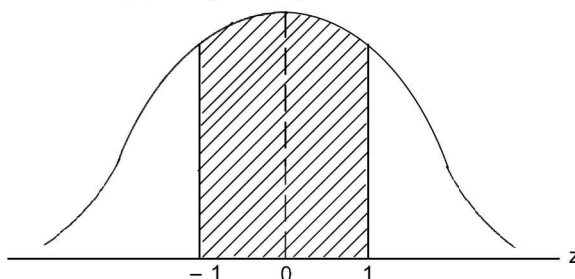
**Fig.**

Number of students have masses less than or equal to 64 Kg
 $= 300 \times 0.0918 = 28$ (approximately)

$$(iii) \text{ When } X = 65, z = \frac{X - \mu}{\sigma} = \frac{65 - 68}{3} = -1 = z_1 \text{ (say)}$$

$$\text{When } x = 71, z = \frac{X - \mu}{\sigma} = \frac{71 - 68}{3} = 1 = z_2 \text{ (say)}$$

$$\begin{aligned} \therefore P(65 \leq X \leq 71) &= P(-1 \leq z \leq 1) \\ &= A(z_2) + A(z_1) = A(1) + A(-1) = A(1) + A(1) \\ &= 2.A(1) = 2(0.3413) = 0.6826 \end{aligned}$$

**Fig.**

$\therefore$ Required number of students = $300 \times 0.6826 = 205$ (approximately)

Example 14 : In an examination it is laid down that a student passes if he secures 40 percent or more. He is placed in the first, second and third division according as he secures 60% or more marks, between 50% and 60% marks and marks between 40% and 50% respectively. He gets a distinction in case he secures 75% or more. It is noticed from the results that 10% of the students failed in the examination, whereas 5% of them obtained distinction. Calculate the percentage of students placed in the second division. (Assume normal distribution of marks). [JNTU (K) May 2013 (Set No. 1)]

Solution : Let μ be the mean and σ the standard deviation of the normal distribution of marks.

Let the variable X denote the marks in the examination. Then we are given

$$P(X < 40) = 0.10 \text{ and } P(X \geq 75) = 0.05$$

$$\text{When } X = 40, z = \frac{X - \mu}{\sigma} = \frac{40 - \mu}{\sigma} = -z_1 \text{ (say)}$$

$$\text{When } X = 75, z = \frac{X - \mu}{\sigma} = \frac{75 - \mu}{\sigma} = z_2 \text{ (say)}$$

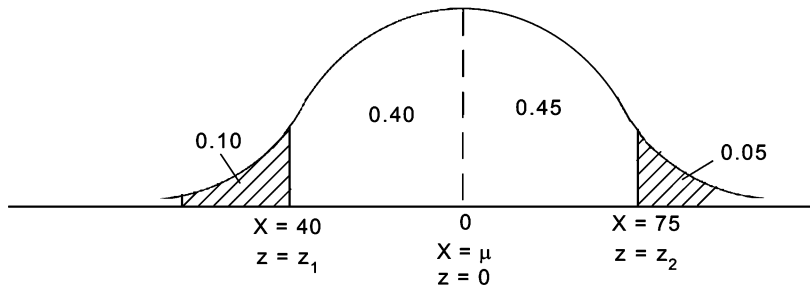


Fig.

$$\therefore P(0 < z < z_2) = 0.5 - 0.05 = 0.45$$

$$\text{and } P(0 < z < z_1) = P(-z_1 < z < 0) = 0.5 - 0.10 = 0.40$$

$$\therefore \text{From normal tables, we have } z_1 = 1.28 \text{ and } z_2 = 1.64$$

$$\text{Hence } \frac{40 - \mu}{\sigma} = -1.28 \Rightarrow \frac{\mu - 40}{\sigma} = 1.28 \dots (1) \quad \text{and} \quad \frac{75 - \mu}{\sigma} = 1.64 \dots (2)$$

$$(1) + (2) \text{ gives } \frac{35}{\sigma} = 2.92 \Rightarrow \sigma = \frac{35}{2.92} = 11.99 \simeq 12$$

$$\text{From (1), } \mu = 40 + \sigma (1.28) = 40 + 12 (1.28) \simeq 55$$

The probability ' p ' that a candidate is placed in the second division is equal to the probability that his score lies between 50 and 60. That is,

$$\begin{aligned} p &= P(50 < X < 60) = P(-0.42 < Z < 0.42) \quad \left[\because Z = \frac{X - 55}{12} \right] \\ &= A(0.42) + A(-0.42) \\ &= A(0.42) + A(0.42) \quad (\text{due to symmetry}) \\ &= 2A(0.42) = 2(0.1628) = 0.3256 = 0.32 \text{ (approximately)} \end{aligned}$$

Hence 32% candidates get second division in the examination.

Example 11 : A manufacturer knows from experience that the resistance of resistors he produces is normal with mean 100 ohms and standard deviation 2 ohms. What percentage of resistors will have resistance between 98 ohms and 102 ohms ?

Solution : Let μ be the mean and σ the standard deviation of the normal distribution.

Then we are given that

$$\mu = 100 \text{ ohms and } \sigma = 2 \text{ ohms}$$

Let the variable X represents the resistance of resistors

$$\text{When } X = 98 \text{ ohms, } z = \frac{X - \mu}{\sigma} = \frac{98 - 100}{2} = -1$$

$$\text{and when } X = 102 \text{ ohms, } z = \frac{X - \mu}{\sigma} = \frac{102 - 100}{2} = 1$$

$$\begin{aligned} \therefore P(98 < X < 102) &= P(-1 < z < 1) \\ &= A(1) + A(-1) = A(1) + A(1) = 2A(1) \\ &= 2(0.3413) = 0.6826 \end{aligned}$$

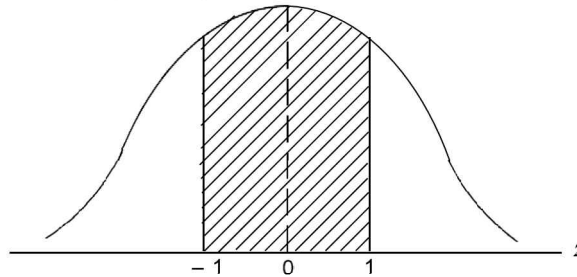


Fig.

$\therefore$ Percentage of resistors having resistance between 98 ohms and 102 ohms = 68.26 = 68 (approximately).

Example 16 : Given that the mean height of students in a class is 158 cms with standard deviation of 20 cms. Find how many students heights lie between 150 cms and 170 cms, if there are 100 students in the class. [JNTU 2007, 2007S, (H) Dec. 2011 (Set No. 4)]

Solution : We have

Mean, $\mu = 158$ cm and standard deviation, $\sigma = 20$ cms

$$\therefore z = \frac{X - \mu}{\sigma} = \frac{X - 158}{20}$$

$$\text{When } X = 150, z = \frac{150 - 158}{20} = \frac{-8}{20} = -0.4$$

$$\text{When } X = 170, z = \frac{170 - 158}{20} = \frac{12}{20} = 0.6$$

$$\begin{aligned} \therefore P(150 \leq X \leq 170) &= P(-0.4 \leq z \leq 0.6) \\ &= P(-0.4 \leq z \leq 0) + P(0 \leq z \leq 0.6) \\ &= P(0 \leq z \leq 0.4) + P(0 \leq z \leq 0.6), \text{ due to symmetry} \\ &= 0.1554 + 0.2257 = 0.3811 \end{aligned}$$

Number of students whose height lie between 150 cms and 170 cms

$$= \text{Probability} \times \text{total no. of students}$$

$$= 0.3811 \times 100 = 38 \quad (\because \text{Number of students should be integer.})$$

Example 16 : Suppose 2% of the people on the average are left handed. Find

(i) The probability of finding 3 or more left handed

(ii) The probability of finding ≤ 1 left handed

(b) The mean and standard deviation of a normal variable are 8 and 4 respectively.

Find (i) $p(5 \leq X \leq 10)$ (ii) $p(x \geq 5)$

[JNTU (H) 2009 (Set No. 1)]

Solution : Given mean, $\mu = 8$ and standard deviation, $\sigma = 4$.

$$(i) \text{ When } x = 5, z = \frac{x - \mu}{\sigma} = \frac{5 - 8}{4} = \frac{-3}{4} = -0.75 = z_1 \text{ (say)}$$

$$\text{When } x = 10, z = \frac{x - \mu}{\sigma} = \frac{10 - 8}{4} = \frac{2}{4} = \frac{1}{2} = 0.5 = z_2 \text{ (say)}$$

$$\therefore P(5 \leq X \leq 10) = P(-0.75 \leq z \leq 0.5)$$

$$= A(z_2) + A(z_1) = A(-0.75) + A(0.5)$$

$$= A(0.75) + A(0.5) \quad [\because A(-z) = A(z)]$$

$$= 0.2734 + 0.1916 = 0.465$$

(ii) When $x = 5$, $z_1 = -0.75$, from above

$$\therefore P(X \geq 5) = P(z_1 \geq -0.75) = 0.5 - A(z_1)$$

$$= 0.5 - A(-0.75) = 0.5 - 0.2734 = 0.2266$$

Example 14 : In a test on 2000 electrical bulbs, it was found that the life of a particular make, was normally distributed with an average life of 2040 hours and S.D. of 40 hrs. Estimate the number of bulbs likely to burn for

(i) more than 2140 hrs. (ii) between 1920 and 280 hrs.

(iii) less than 1960 hrs.

Solution : Given mean, $\mu = 2040$ hrs and S.D., $\sigma = 40$ hrs

$$(i) \text{ When } x = 2140, z = \frac{x - \mu}{\sigma} = \frac{2140 - 2040}{40} = \frac{100}{40} = 2.5$$

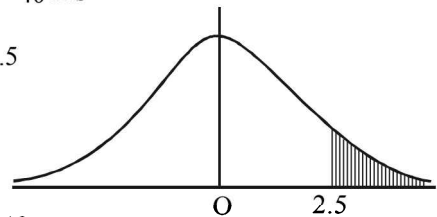
$$P(x > 2140) = P(z > 2.5) = 0.5 - P(0 \leq z \leq 2.5)$$

$$= 0.5 - 0.4938 = 0.0062$$

Hence the number of bulbs likely to burn

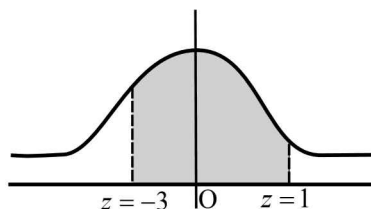
for more than 2140 hours $= 0.0062 \times 2000 = 12.4 \approx 12$.

$$(ii) \text{ When } x_1 = 1920, z = \frac{x_1 - \mu}{\sigma} = \frac{1920 - 2040}{40} = \frac{-120}{40} = -3$$



$$\text{When } x_2 = 2080, z = \frac{x_2 - \mu}{\sigma} = \frac{2080 - 2040}{40} = \frac{40}{40} = 1$$

$$\begin{aligned}\therefore P(1920 \leq x \leq 2080) &= P(-3 \leq z \leq 1) \\ &= P(-3 \leq z \leq 0) + P(0 \leq z \leq 1) \\ &= P(0 \leq z \leq 3) + P(0 \leq z \leq 1) \quad (\text{by symmetry}) \\ &= 0.4986 + 0.3413 = 0.8399\end{aligned}$$

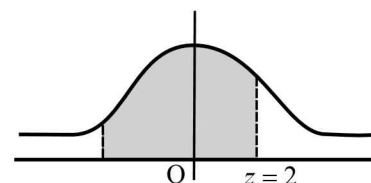


Hence number of bulbs which are likely to burn between 1920 hrs and 2080 hrs.

$$= 0.8399 \times 2000 = 1679.8 \approx 1680$$

$$(iii) \text{ When } x = 1960, z = \frac{x - \mu}{\sigma} = \frac{1960 - 2040}{40} = \frac{80}{40} = 2$$

$$\begin{aligned}\therefore P(x \leq 1960) &= P(z \leq 2) \\ &= 0.5 + P(0 \leq z \leq 2) = 0.5 + 0.4772 = 0.9772\end{aligned}$$



Hence number of bulbs likely to burn less than 1960 hrs. $= 0.9772 \times 2000 = 1954$

Example 16 : If the marks obtained by a number of students in a certain subject are approximately normally distributed with mean 65 and standard deviation 5. If three students are selected at random from this group what is the probability that at least one of them would have scored above 75? [JNTU (K) May 2010 (Set No. 2)]

Solution : Here $\mu = \text{mean} = 65$ and $\sigma = \text{S.D.} = 5$

We have $z = \frac{x - \mu}{\sigma}$ when

$$x = 75, z = \frac{75 - 65}{5} = \frac{10}{5} = 2$$

$$\therefore p = P(x > 75) = P(z > 2) = 0.5 - P(0 \leq z \leq 2) = 0.5 - 0.4772 = 0.0228$$

Hence the required probability that out of 3 students, at least one of them will have marks over 75 is given by

$$\begin{aligned}{}^3C_1 p q^2 + {}^3C_2 p^2 q^{3-2} + {}^3C_3 p^3 q^{3-3} &= 3p(1-p)^2 + 3p^2(1-p) + p^3 \quad [\because q = 1 - p] \\ &= 3(0.0228)(1 - 0.0228)^2 + 3(0.0228)^2(1 - 0.0228) + (0.0228)^3 = 0.0668.\end{aligned}$$

Example 20 : If X is normally distributed with mean 2 and variance 0.1, then find $P(|X - 2| \geq 0.01)$? [JNTU (K) May 2010 (Set No. 3)]

Solution : Here $\mu = 2, \sigma = 0.1$

$$\text{when } X = 1.99, z = \frac{x - \mu}{\sigma} = \frac{1.99 - 2}{0.1} = -0.1$$

$$\text{when } X = 2.01, z = \frac{2.01 - 2}{0.1} = \frac{0.01}{0.1} = 0.1$$

$$\therefore P(|X - 2| < 0.01) = P(1.99 < X < 2.01)$$

$$\begin{aligned}
 &= P(-0.1 < z < 0.1) = P(-0.1 < z < 0) + P(0 < z < 0.1) \\
 &= 2P(0 < z < 0.1) \text{ [By symmetry]} \\
 &= 2 \times 0.0398 = 0.0796
 \end{aligned}$$

$$\therefore P(|X - 2| \geq 0.01) = 1 - P(|X - 2| < 0.01) = 1 - 0.0796 = 0.9204$$

Example 11 : It is claimed that a random sample of 100 tires with a mean life of 15269 is drawn from a population of tires which has a mean life of 15200 km and a S. D. of 1248 km. Test the claim at $\alpha = 0.05$. [JNTU (A) Nov. 2011 (Set No. 3)]

Solution : Let the Null Hypothesis is $H_0 : \mu = 15200$

Then the Alternative Hypothesis is $H_1 : \mu \neq 15200$

Since H_1 is two - tailed (*i.e.*, two-sided), we are to use two - tailed test.

Level of significance, $\alpha = 0.05$

Let us assume that H_0 is true.

The test statistic is $z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$

Here $\bar{x}$ = sample mean = 15269, $n = 100 (> 30)$, $\mu = 15200$, $\sigma = 1248$

Since the sample size is large, $\bar{x}$ is approximately normally distributed with mean μ and

$$\text{S. D.} = \frac{\sigma}{\sqrt{n}}$$

$$\therefore z = \frac{15269 - 15200}{1248 / \sqrt{100}} = \frac{690}{1248} = 0.553$$

The table value of z at 5% level of significance for two-tailed test is 1.96.

Since $|z| = 0.553 < 1.96$, we accept the Null hypothesis H_0 at 5% level of significance and conclude that the claim is true.

Example 12 : When the mean of marks was 50% and S. D. 5% then 60% of the students failed in an examination ? Determine the grace marks to be awarded in order to show that 70% of the students passed. Assume that the marks are normally distributed.

[JNTU (A) Apr. 2012 (Set No. 1)]

Solution : Let x be the marks obtained in the examination.

Given $\mu = 0.5$, S. D., $\sigma = 0.05$

Before grace marks were added, 60% failed. That is, 40% passed the examination.

The area $0.4 \left(< \frac{1}{2} \right)$ corresponds pass students.

Find the value of z_1 such that the area to its right is 0.4.

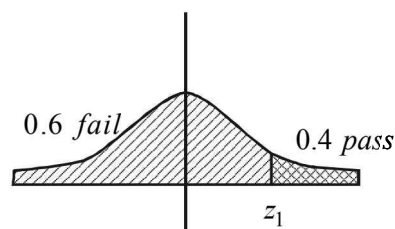
The area from 0 to z_1 is $0.5 - 0.4 = 0.1$.

From normal table for the area 0.1, the value of z is 0.25.

The standard normal variate is $z = \frac{x - \mu}{\sigma}$.

$$\Rightarrow 0.25 = z_1 = \frac{x_1 - 0.5}{0.05}$$

$$\Rightarrow x_1 = (0.25)(0.05) + 0.5 = 0.5125$$



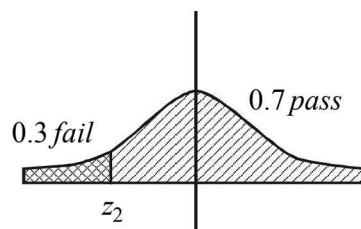
After grace marks were added, 70% passed the examination. The area $0.70 \left(> \frac{1}{2} \right)$

corresponds pass students. So determine z_2 such that the area to its right is 0.7. z_2 must be negative. The area from 0 to z_2 is $0.5 - 0.3 = 0.2$.

From table $z_2 = -0.52$.

$$\therefore z_2 = -0.52 = \frac{x_2 - 0.5}{0.05}$$

$$\begin{aligned} \Rightarrow x_2 &= (-0.52)(0.05) + 0.5 \\ &= -0.026 + 0.5 = 0.474 \end{aligned}$$



$\therefore$ The minimum pass mark for a student is 51.25 before grace while the minimum pass mark is 47.40 after grace. So grace marks awarded is $51.25 - 47.40 = 3.85$

Example 11 : The mean height of students in a college is 155 cms and standard deviation is 15. What is the probability that the mean height of 36 students is less than 157 cms.

[JNTU (H) Dec. 2011 (Set No. 4)]

Solution : Let X be the continuous random variable denoting the height of students. Then mean, $\mu = 155$ and S. D., $\sigma = 15$.

We know that the standardized value of X is $z = \frac{X - \mu}{\sigma}$

$$\text{when } X = 157, z = \frac{157 - 155}{15} = \frac{2}{15} = 0.13$$

Probability that the height of student is less than 157 cm.

$$= P(X < 157) = P(z < 0.13)$$

$$= \text{area under the normal curve to the right of } z = 0.13$$

$$= 0.0517$$

$$\therefore \text{Required probability} = 0.0517 \times 36 = 1.86$$

Example 14 : The mean inside diameter of a sample of 200 washers produced by a Machine is 500 cms with standard deviation 0.005 cms. The purpose of which these washers are intended a maximum tolerance in the diameter 0.495 to 0.505 cms. otherwise the washers are considered defective. Determine the percentage of defective washers produced by the machine, assuming the diameters are normally distributed. [JNTU (H) Apr. 2012 (Set No. 3)]

Solution : We are given

μ = Mean of the inside diameters = 0.500 cm

σ = Standard deviation = 0.005 cm

The tolerance limits of non - defective tubes are 0.495 cm and 0.505 cm.

When $x = 0.495$ (standardized value of 0.495),

$$z = \frac{x - \mu}{\sigma} = \frac{0.495 - 0.500}{0.005} = -1$$

When $x = 0.505$ (i.e. standardized value of 0.505),

$$z = \frac{x - \mu}{\sigma} = \frac{0.505 - 0.500}{0.005} = 1$$

$\therefore$ Proportion of non-defective washers

= Area under the standard normal curve between $z = -1$ and $z = +1$

= 2 (Area between $z = 0$ and $z = 1$)

= $2 \times 0.3413 = 0.6826 = 68.26\%$

Hence the required percentage of defective washers = $100\% - 68.26\% = 31.74\%$

[(or) Probability of defective washers = 1 – probability of non-defective washers

$$= 1 - p(0.495 \leq x \leq 0.505)$$

$$= 1 - p(-1 \leq z \leq 1)$$

$$= 1 - 2 \times p(0 \leq z \leq 1)$$

$$= 1 - 2(0.3413) = 0.3174$$

$\therefore$ Percentage of defective washers = $31.74 \approx 32\%$

Example 16 : 1000 students have written an examination the mean of test is 35 and standard deviation is 5. Assuming the distribution to be normal, find:

(i) How many students marks lie between 25 and 40?

(ii) How many students get more than 40?

(iii) How many students get below 20?

(iv) How many students get more than 50?

Solution : Here $\mu = 35$ and $\sigma = 5$

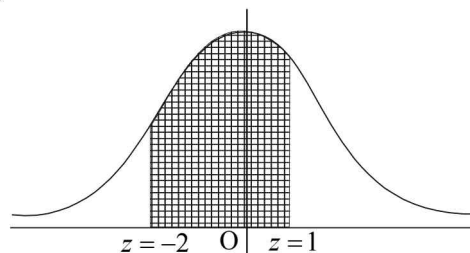
We know, standardised variate, $z = \frac{x - \mu}{\sigma} = \frac{x - 35}{5}$

(i) When $x_1 = 25$, then

$$z = \frac{25 - 35}{5} = \frac{-10}{5} = -2$$

When $x_2 = 40$, then

$$z = \frac{40 - 35}{5} = \frac{5}{5} = 1$$

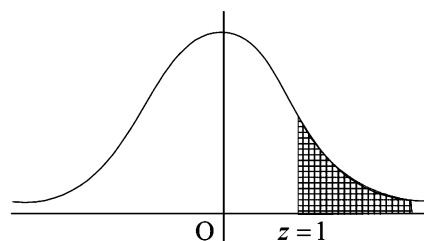


$$\begin{aligned} \therefore \text{Probability of students whose marks lie between 25 and 40} &= P(25 \leq X \leq 40) \\ &= P(-2 \leq z \leq 1) = P(-2 \leq z \leq 0) + P(0 \leq z \leq 1) \\ &= P(0 \leq z \leq 2) + P(0 \leq z \leq 1), \text{ by symmetry} \\ &= 0.4772 + 0.3415 \text{ [From Tables]} \\ &= 0.8185 \end{aligned}$$

Number of students whose marks lie between 25 and 40 $= 1000 \times 0.8185 = 819$ (approximately)

(ii) When $x = 40$, then $z = \frac{40 - 35}{5} = \frac{5}{5} = 1$

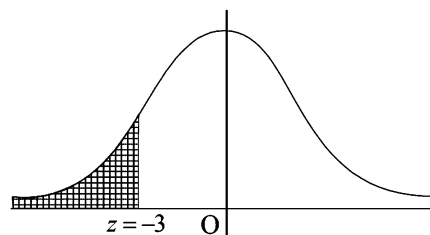
$$\begin{aligned} P(X > 40) &= P(z > 1) = 0.5 - P(z \leq 1) \\ &= 0.5 - 0.3415 \\ &= 0.1585 \text{ [From Tables]} \end{aligned}$$



$\therefore$ Number of students whose marks are greater than 40 $= 1000 \times 0.1585 = 159$ (approximately)

(iii) When $x = 20$, then $z = \frac{20 - 35}{5} = \frac{-15}{5} = -3$

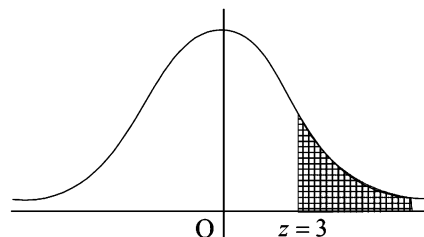
$$\begin{aligned} P(X < 20) &= P(z \leq -3) = 0.5 - P(-3 \leq z \leq 0) \\ &= 0.5 - P(0 \leq z \leq 3), \text{ by symmetry} \\ &= 0.5 - 0.499 = 0.001 \text{ [From Tables]} \end{aligned}$$



$\therefore$ Number of students whose marks are less than 20 $= 1000 \times 0.001 = 1$

(iv) When $x = 50$, $z = \frac{50 - 35}{5} = \frac{15}{5} = 3$

$$\begin{aligned} \therefore P(X > 50) &= P(z > 3) = 0.5 - P(0 \leq z \leq 3) \\ &= 0.5 - 0.499 \\ &= 0.001 \text{ [From Tables]} \end{aligned}$$



Hence number of students whose marks are greater than 50 $= 1000 \times 0.001 = 1$.

Example 16 : An electrical firm manufactures light bulbs that have a life, before burnout, which is normally distributed with mean equal to 800h and a standard deviation of 40 h. Find the probability that a bulb burns between 778h and 834 h. [JNTU(A) Dec. 2014]

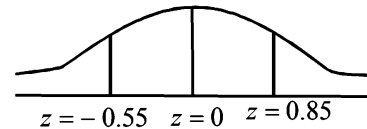
Solution : Let X be a random normal variate measuring the life time of bulbs.

Here mean $\mu = 800$ and $\sigma = 40$

$$\therefore Z = \frac{X - \mu}{\sigma} = \frac{X - 800}{40} \text{ is a standard normal variate } N(0,1)$$

$$\text{When } X = 778, \text{ then } Z = \frac{778 - 800}{40} = \frac{-22}{40} = -0.55$$

$$\text{When } X = 834, \text{ then } Z = \frac{834 - 800}{40} = \frac{34}{40} = 0.85$$



When X lies between 778 and 834, then Z lies between -0.55 and 0.85

$$\begin{aligned} \therefore \text{Required probability} &= \text{Area under the curve from } z = -0.55 \text{ to } z = 0.85 \\ &= \text{Area under the curve from } (z = -0.55 \text{ to } z = 0) \\ &\quad + \text{Area under the curve from } (z = 0 \text{ to } z = 0.85) \\ &= \text{Area under the curve from } (z = 0 \text{ to } z = 0.55) \\ &\quad + \text{Area under the curve from } (z = 0 \text{ to } z = 0.85) \\ &\quad \text{(symmetric property)} \\ &= 0.2088 + 0.3023 = 0.5111 \end{aligned}$$

7.16 NORMAL APPROXIMATION TO THE BINOMIAL DISTRIBUTION

The Normal distribution can be used to approximate the Binomial distribution. Suppose the number of successes x ranges from x_1 to x_2 . Then the probability of getting x_1 to x_2 successes is given by

$$\sum_{r=x_1}^{x_2} n_c p^r q^{n-r}$$

For large n , the calculation of binomial probabilities is very difficult. In such cases the binomial curve can be replaced by the normal curve and the required probability is computed. We consider two cases.

Case 1 : When $p = q = \frac{1}{2}$

Even when n is not large, Binomial distribution (B.D) can be approximated by Normal distribution (N.D).

$$\text{Mean of the B.D.} = np \quad \text{and} \quad \text{S.D.} = \sqrt{npq}$$

Hence for the corresponding Normal distribution, μ and σ are known.

We know that $z = \frac{x - \mu}{\sigma}$

Let z_1 and z_2 be the values of z , corresponding to x_1 and x_2 of x respectively. Then

$$P(x_1 < x < x_2) = P(z_1 < z < z_2) = \int_{z_1}^{z_2} \phi(z) dz$$

and this can be determined by using the normal tables.

Case 2 : When $p \neq q$

For large n , we can approximate the binomial curve by the normal curve and calculate the probability.

For any success x , real class interval is $\left(x - \frac{1}{2}, x + \frac{1}{2}\right)$. Hence z_1 must correspond to the lower limit of the x_1 class and z_2 , to the upper limit of the x_2 class.

$$\text{Hence } z_1 = \frac{\left(x_1 - \frac{1}{2}\right) - \mu}{\sigma} = \frac{x_1 - \frac{1}{2} - np}{\sqrt{npq}}$$

$$\text{and } z_2 = \frac{\left(x_2 + \frac{1}{2}\right) - \mu}{\sigma} = \frac{x_2 + \frac{1}{2} - np}{\sqrt{npq}}$$

$\therefore$ The required probability $= \int_{z_1}^{z_2} \phi(z) dz$ and this can be evaluated using normal tables.

SOLVED EXAMPLES

Example 1 : Find the probability that out of 100 patients between 84 and 95 inclusive will survive a heart-operation given that the chances of survival is 0.9.

Solution : Here $p = 0.9$ and $n = 100$

$$\therefore q = 1 - p = 1 - 0.9 = 0.1$$

$$\text{Standard deviation} = \sqrt{npq} = \sqrt{90(0.1)} = \sqrt{9} = 3$$

$$\text{The required probability} = \sum_{r=84}^{95} {}^{100}C_r (0.9)^r (0.1)^{100-r}$$

We have to sum up a large number of terms of the binomial. To avoid it, we can replace the binomial distribution by a normal distribution.

Here $x_1 = 84$ and $x_2 = 95$. Hence while calculating z_1, z_2 corresponding to 84 and 95, we take $P(84 \leq x \leq 95) = P(83.5 < x < 95.8)$ and calculate z_1, z_2 correspondingly,

$$\text{Hence } z_1 = \frac{84 - (1/2) - 90}{3} = \frac{-13}{6} \text{ and } z_2 = \frac{95 + (1/2) - 90}{3} = \frac{11}{6}$$

$$\begin{aligned}
 \text{Hence the required probability} &= \int_{-13/6}^{11/6} \phi(z) dz \\
 &= P\left(-\frac{13}{6} \leq z \leq \frac{11}{6}\right) \\
 &= A\left(\frac{11}{6}\right) + A\left(\frac{-13}{6}\right) = A\left(\frac{11}{6}\right) + A\left(\frac{13}{6}\right) \\
 &= A(1.83) + A(2.17) \\
 &= 0.4664 + 0.4850 = 0.9514
 \end{aligned}$$

Example 2 : Eight coins are tossed together. Find the probability of getting 1 to 4 heads in a single toss.

Solution : Here $p = q = \frac{1}{2}$

$$\text{Mean, } \mu = np = 8 \times \frac{1}{2} = 4 \text{ and standard deviation, } \sigma = \sqrt{npq} = \sqrt{4 \times \frac{1}{2}} = \sqrt{2}$$

Even though $n = 8$ is not sufficiently large, we can approximate the symmetric binomial by a normal distribution.

Here $x_1 = 1$ and $x_2 = 4$

$$\text{Thus } z_1 = \frac{x_1 - \frac{1}{2} - \mu}{\sigma} = \frac{1 - \frac{1}{2} - 4}{\sqrt{2}} = \frac{-7}{2\sqrt{2}} = -2.47$$

$$\text{and } z_2 = \frac{x_2 + \frac{1}{2} - \mu}{\sigma} = \frac{4 + \frac{1}{2} - 4}{\sqrt{2}} = \frac{1}{2\sqrt{2}} = 0.35$$

$$\begin{aligned}
 \text{The required probability} &= \int_{x_1}^{x_2} \phi(z) dz = \int_{-2.47}^{0.35} \phi(z) dz \\
 &= P(-2.47 \leq z \leq 0.35) = A(0.35) + A(-2.47) \\
 &= A(0.35) + A(2.47) \\
 &= 0.4932 + 0.1368 = 0.63
 \end{aligned}$$

Example 2 : Find the probability that by guess-work a student can correctly answer 25 to 30 questions in a multiple-choice quiz consisting of 80 questions. Assume that in each question with four choices, only one choice is correct and student has no knowledge of the subject.

Solution : Here $p = \frac{1}{4}$. So $q = 1 - p = 1 - \frac{1}{4} = \frac{3}{4}$

$$\text{Mean, } \mu = np = 80 \left(\frac{1}{4}\right) = 20 \text{ and } \sigma = \sqrt{npq} = \sqrt{20 \times \frac{3}{4}} = \sqrt{15}, x_1 = 25 \text{ and } x_2 = 30$$

$$\text{Thus } z_1 = \frac{\left(x_1 - \frac{1}{2}\right) - \mu}{\sigma} = \frac{\left(25 - \frac{1}{2}\right) - 20}{3.87} = \frac{4.5}{3.87} = 1.16$$

$$\text{and } z_2 = \frac{\left(x_2 + \frac{1}{2}\right) - \mu}{\sigma} = \frac{\left(30 + \frac{1}{2}\right) - 20}{3.87} = \frac{10.5}{3.87} = 2.71$$

Hence the required probability = $P(25 \leq X \leq 30)$

$$= P(1.16 \leq z \leq 2.71) = |A(2.71) - A(1.16)|$$

$$= 0.4966 - 0.3770 = 0.1196$$

Example 4 : Find the probability of getting an even number on face 3 to 5 times in throwing 10 dice together. [JNTU (H) May 2012 (Set No. 2)]

Solution : p = Probability of getting an even number on face

$$= \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$$

$$\therefore q = 1 - p = \frac{1}{2}$$

Here $n = 10$

$$\text{Mean, } \mu = np = 10 \left(\frac{1}{2}\right) = 5$$

$$\text{and } \sigma = \sqrt{npq} = \sqrt{\frac{5}{2}} = 1.58$$

Also $x_1 = 3$ and $x_2 = 5$

$$\text{Hence } z_1 = \frac{\left(x_1 - \frac{1}{2}\right) - \mu}{\sigma} = \frac{(3 - 0.5) - 5}{1.58} = \frac{-2.5}{1.58} = -1.58$$

$$z_2 = \frac{\left(x_2 + \frac{1}{2}\right) - \mu}{\sigma} = \frac{(5 + 0.5) - 5}{1.58} = \frac{0.5}{1.58} = 0.32$$

$$\text{Hence the required probability} = \int_{x_1}^{x_2} \phi(z) dz = \int_{-1.58}^{0.32} \phi(z) dz$$

$$= P(-1.58 \leq z \leq 0.32)$$

$$= A(0.32) + A(1.58)$$

$$= 0.1256 + 0.4429 = 0.5685$$

7.17 FITTING OF NORMAL DISTRIBUTION

Consider the following frequency distribution

Class-interval (x)	Frequency (f)
$l_1 - u_1$	f_1
$l_2 - u_2$	f_2
$l_3 - u_3$	f_3
...	...
...	...
$l_i - u_i$	f_i
...	...
$l_n - u_n$	f_n

Then the normal curve (*i.e.*, Normal Distribution) can be fitted to it using

$$f(x) = \frac{1}{\sigma \cdot \sqrt{2\pi}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < x < \infty$$

Working Procedure :

1. Calculate the mean μ and standard deviation σ from the given frequency distribution.
2. Compute the standard normal variate corresponding to the lower limits of each of the class interval. That is, find $z_i = \frac{x_i' - \mu}{\sigma}$, where x_i' is the lower limit of the i th class interval.
3. Compute the areas under the normal curve from 0 to z_i using normal tables.
4. Areas for the successive class intervals are obtained by taking the difference between the successive areas calculated in Step 3 (when z_i 's are of opposite sign, add the successive areas).
5. Expected or Theoretical normal frequencies are obtained by multiplying the areas in Step 4 by $N (= \Sigma f_i)$.

SOLVED EXAMPLES

Example 2 : Fit a normal distribution to the following data :

Class-interval	60-62	63-65	66-68	69-71	72-74
Frequency (f)	5	18	42	27	8

Solution : Here $N = \Sigma f_i = 100$, mean = $\mu = 67.45$ and standard deviation = $\sigma = 2.92$. The calculations are arranged as follows :

S. No.	Class interval	Observed frequency (f)	True lower class limit x_i	Standard variate $z_i = (x_i - 67.45)/2.92$	Area from 0 to z_i	Area for each class (A)	Expected frequency = NA = 100A
1	60-62	5	59.5	- 2.72	0.4967	0.0413	4.13
2.	63-65	18	62.5	- 1.70	0.4554	0.2068	20.68
			65.5	- 0.67	0.2486	0.3892	38.92
3.	66-68	42	68.5	0.36	0.1406	0.2771	27.71
4.	69-71	27	71.5	1.39	0.4177	0.0743	7.43
5.	72-74	8	74.5	2.41	0.4920		

Hence the expected frequencies corrected to nearest integers are 4, 21, 39, 28, 7 which agree with the observed frequencies.

REVIEW QUESTIONS

- (i) Define Normal Distribution.
(ii) Define Normal Distribution and derive Normal Distribution as a limiting form of Binomial distribution. [JNTU (K) Nov. 2011 (Set No. 1)]
- Write the uses of Normal Distribution
- List the properties of Normal Distribution.
- Write about the importance and applications of Normal distribution.
- (i) Derive Mean and Variance of normal distribution
(ii) Derive Mode and Median of normal distribution. [JNTU (A) Dec. 2017]

EXERCISE 7(C)

- The mean and standard deviation of a normal variate are 8 and 4 respectively, find
(i) $P(5 \leq x \leq 10)$ (ii) $P(x \geq 5)$ [JNTU 2005 S, (H) Nov. 2009 (Set No. 1)]
(iii) $P(10 \leq x \leq 15)$ (iv) $P(x \leq 15)$ [JNTU 2004 (Set No.2)]
- If the mean and standard deviation of a normal distribution are 70 and 16, find $P(38 < x < 46)$. [JNTU 2005S (Set No.4)]
- If X is normally distributed with mean 70 and standard deviation 16, find
(i) $P(38 \leq X \leq 46)$ (ii) $P(82 \leq X \leq 94)$ (iii) $P(62 \leq X \leq 86)$
- Find the probability that a random variable having standard normal distribution will take on a value between 0.87 and 1.28. [JNTU 2001]
- If X is a normally distributed with mean 30 and standard deviation 5, find $P(|X-30|>5)$.
- Find the mean and standard deviation of a normal distribution in which 31% of items are under 45 and 8% are over 64. [Hint : Refer Example 4]
[JNTU 2003, (A) Dec 2009 (Set No.2), (K) Nov. 2012, Dec. 2014 (Set No. 2)]
- (i) The mean height of 500 students is 151 cm. and the standard deviation is 15 cm. Assuming that the heights are normally distributed, find how many students heights lie between 120 and 155 cm.

- (ii) The heights of 1000 students are normally distributed with a mean of 174.5 cm and a standard deviation of 6.9 cm. Assuming that the heights are recorded to the nearest half-cm, how many of these students would you expect to have heights :
- (a) Less than 160.0 cms ? (b) Between 171.5 and 182.0 cms inclusive ?
 (c) Greater than or equal to 188.0 cms. **[JNTU (A) June 2018]**
8. The life of electronic tubes of a certain type may be assumed to be normally distributed with mean 155 hours and standard deviation 19 hours. Determine the probability that the life of a randomly chosen tube
- (i) is between 136 hours and 174 hours.
 (ii) less than 117 hours (iii) will be more than 395 hours.
9. In a test on 2000 electric bulbs, it was found that the life of a particular make, was normally distributed with an average life of 2040 hours and standard deviation of 60 hours. Find the number of bulbs likely to burn for
- (i) more than 2150 hours (ii) less than 1950 hours and
 (iii) more than 1920 hours and but less than 2160 hours.
10. Estimate the minimum marks a student must get in order to receive an 'A' grade if the top 10% of the students are awarded 'A' grade in an examination where the mean of the marks is 72 and standard deviation is 9.
11. When the mean of marks was 50% and standard deviation 5% then 60% of the students failed in an examination. Determine the 'grace' marks to be awarded in order to show that 70% of the students passed. Assume that the marks are normally distributed, **[JNTU (A) May 2012 (Set No. 1)]**
12. The mean and standard deviation of marks scored by a group of students in an examination are 15 and 1.5 respectively. If only 25% of the students are promoted, determine the minimum marks required for the promotion.
13. Find the probability of getting at least 55 heads when 100 coins are tossed using normal distribution.
14. Twelve coins are tossed together. Find the probability of getting 8,9 or 10 heads in a single toss using normal distribution.
15. Ten cards are drawn from a deck of 52 cards. Find the probability of getting 2 to 5 diamonds using normal distribution.
16. In a referendum 60% of voters voted in favour. A random sample of 200 voters was selected. Find the probability by using normal distribution that in the sample
- (i) more that 130 voted in favour ?
 (ii) between 105 and 130 inclusive voted in favour ?
 (iii) 120 voted in favour ?
17. Determine the probability of getting at least 60 heads when 100 coins are tossed.
18. The weekly wages of 1000 workers are normally distributed around a mean of Rs. 70 and S.D. of Rs. 5/-. Estimate the number of workers whose weekly wages will be
- (i) between Rs. 70 and Rs. 72
 (ii) between Rs. 69 and Rs. 72. **[JNTU 2007S (Set No.1)]**

- 19.** A sample of 100 dry battery cells tested to find the length of life produced the following results : $\bar{x} = 12$ hours, $\sigma = 2$ hours

Assuming the data to be normally distributed, what percentage of battery cells are expected to have life

(i) More than 15 hours (ii) Less than 6 hours

(iii) Between 10 and 15 hours ?

- 20.** Fit a normal distribution to the following data :

x	2	4	6	8	10
f	1	4	6	4	1

- 21.** Fit a normal curve to the following distribution :

Class	5 – 9	10 – 14	15 – 19	20 – 24	25 – 29	30 – 34	35 – 39
Frequency	1	10	37	36	13	2	1

- 22.** Fit a normal distribution to the following data :

x	2	4	6	8	10
f	1	4	6	4	1

ANSWERS

1. (i) 0.465 (ii) 0.7734 (iii) 0.6514 (iv) 0.0401
2. 0.044
3. (i) 0.044 (ii) 0.1598 (iii) 0.5328
4. 0.0919
5. 0.3174
6. $\mu = 50, \sigma = 10$
7. 294
8. (i) 0.6826 (ii) 0.0228 (iii) 0
9. (i) 67 (ii) 184 (iii) 1909
10. 84
11. 3.85
12. 28
13. 0.1841
14. 0.1887
15. 0.7530
16. (i) 0.0648 (ii) 0.8964 (iii) 0.0575
17. 0.02872
18. 155, 235
20. 1, 4, 6, 4, 1
21. 2, 1, 3, 36, 16, 3, 0
22. 1, 4, 6, 4, 1

QUIZ

1. The probability of getting 2 heads in tossing 5 coins is _____.
2. If a coin is tossed 6 times in succession, the probability of getting at least one head is _____.
3. The mean of the binomial distribution is _____.
4. If n and p are the parameters of a binomial distribution, the standard deviation of this distribution is _____.
5. The probability of having at least one tail in four throws with a coin is _____.
6. If mean of the binomial distribution is 8 and variance is 6, the mode of this distribution is _____.
7. If mean of the binomial distribution is 6 and variance is 2, then mode of this distribution is _____.
8. If mean of the binomial distribution is 4 and variance is 2 then $p =$ _____.
9. The probability of getting one boy in a family of 4 children is _____.
10. If the mean and variance of a binomial variate are 12 and 4, then the distribution is _____.
11. In a binomial distribution the sum and product of the mean and variance are $\frac{25}{3}$ and $\frac{50}{3}$ respectively. The distribution is _____.
12. If the probability of a defective bolt is 0.1, the mean and standard deviation for the distribution bolts in a total of 400 are _____ and _____ respectively.
13. A coin is tossed 3 times. The probability of obtaining two heads will be _____.
14. The probability of producing a defective bolt is 0.1. The probability that out of 5 bolts one will be defective is _____.
15. The binomial distribution whose mean is 5 and variance is $\frac{10}{3}$ is _____.
16. The probability of getting four heads in six tosses of a fair coin is _____.
17. A die is thrown 8 times. The probability that 3 will show exactly 2 times is _____.
18. If the mean of a Poisson distribution is m , then standard deviation of this distribution is _____.
19. The mean and variance of a Poisson distribution are _____.
20. If the mean of a Poisson variate X is one then $P(X = 1)$ is _____.
21. The variance of a Poisson distribution with parameter $\lambda = 2$ is _____.
22. If $P(1) = P(2)$, then the mean of the Poisson distribution is _____.
23. Given $X = B(n, p)$ then the conditions under which X tends to a Poisson distribution are _____.

24. In a Poisson distribution if $2P(x = 1) = P(x = 2)$ then the variance is _____.
25. For a Poisson variate x , $P(x = 1) = P(x = 2)$, then the mean of x is _____.
26. Let X be the Poisson random variable such that $2 P(X = 0) = P(X = 2)$.
Then standard deviation of X is _____.
27. If the variance of a poisson variate is 3 then $P(x = 0)$ is _____.
28. If X is a Poisson variate such that $P(X = 1) = 0.3$ and $P(X = 2) = 0.2$ then $P(X = 0)$ is _____.
29. Among the items manufactured in a factory, 5% are defective. The probability of getting one defective blade in a pack of 5 blades is _____.
30. If X is a poisson variate such that $P(X = k) = P(X = k + 1)$ for some positive integer k then mean is _____.
31. X is a poisson variate. If $P(X = 3) = \frac{1}{6}$ and $P(X = 2) = \frac{1}{3}$ then $P(X = 0)$ is _____.
32. If X is a poisson variate such that $P(X = 1) = 24 P(X = 3)$ then $P(X = 0)$ is _____.
33. The probability of the king of hearts drawn from a pack of cards once in 52 trials is _____.
34. If X is a Poisson variate such that $P(X = 2) = P(X = 3)$, then $P(X = 0) =$ _____.
35. For a Poisson variate, probability of getting at least one success is _____.
36. The distribution in which mean, median and mode are equal is _____.
37. The mean, median and mode of a Normal distribution are _____.
38. If $f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ then its mean is _____ and standard deviation is _____.
39. In the standard normal curve the area between $z = -1$ and $z = 1$ is nearly _____.
40. If $\mu = 5$ and $\sigma = 2$, the equation of the normal distribution is _____.
41. If X be a normal variate with mean 10 and variance 4 then $P(X < 11) =$ _____.
42. The area under the whole normal curve is _____.
43. In a normal distribution, mean deviation : standard deviation = _____.
44. The probability density function of a binomial distribution is _____.
45. The probability density function for the standard normal variable is _____.
46. If the mean of a poisson distribution is 8, then its variance is _____.
47. The graph of the normal distribution is symmetric with respect to the line _____.
48. The probable error is _____ times S. D. approximately.

ANSWERS

1. $\frac{5}{256}$
2. $\frac{63}{64}$
3. np
4. $\sqrt{npq}$
5. $\frac{15}{16}$
6. 8
7. 6
8. $\frac{1}{2}$
9. $\frac{1}{4}$
10. $\left(\frac{1}{3} + \frac{2}{3}\right)^{18}$
11. $\left(\frac{2}{3} + \frac{1}{3}\right)^{15}$
12. 40, 6
13. $\frac{3}{8}$
14. $\frac{1}{2} (0.9)^4$
15. $15 {}_c r \left(\frac{1}{3}\right)^r \left(\frac{2}{3}\right)^{15-r}$
16. $\frac{15}{64}$
17. $\frac{28 \times 5^6}{6^8}$
18. $\sqrt{m}$
19. equal
20. e^{-1}
21. 2
22. 2
23. $n \rightarrow \infty, p \rightarrow 0$ such that np is fixed.
24. 4
25. 2
26. $\sqrt{2}$
27. e^{-3}
28. $e^{-4/3}$
29. 0.0244
30. $k + 1$
31. $e^{-\frac{3}{2}}$
32. $e^{-\frac{1}{2}}$
33. $\frac{1}{e}$
34. e^{-3}
35. $1 - e^{-\lambda}$
36. normal distribution
37. zero
38. 1
39. 68%
40. $f(x) = \frac{1}{2\sqrt{2\pi}} \cdot e^{-\frac{(x-5)^2}{8}}$
41. 0.6915
42. unity
43. 4 : 5
44. $(q + p)^n$
45. $\frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$
46. 8
47. $x = \mu$
48. 2/3

PREVIOUS GATE QUESTIONS

1. An unbiased coin is tossed an infinite number of times. The probability that the fourth head appears at the tenth toss is [GATE 2014 (EC)]
 (A) 0.067 (B) 0.073 (C) 0.082 (D) 0.091
2. A fair coin is tossed n times. The probability that the difference between the number of heads and tails is $(n - 3)$ is [GATE 2014 (EE)]
 (A) 2^{-n} (B) 0 (C) ${}^nC_{n-3}2^{-n}$ (D) 2^{-n+3}
3. Consider an unbiased cubic dice with opposite faces coloured identically and each face coloured red, blue or green such that each colour appears only two times on the dice. If the dice is thrown thrice, the probability of obtaining red colour on top face of the dice atleast twice is _____ [GATE 2014 (ME)]
4. The number of accidents occurring in a plant in a month follows Poisson distribution with mean as 5.2. The probability of occurrence of less than 2 accidents in the plant during a randomly selected month is [GATE 2014 (ME)]
 (A) 0.029 (B) 0.034 (C) 0.039 (D) 0.044
5. A traffic office imposes on an average 5 number of penalties daily on traffic violators. Assume that the number of penalties on different days is independent and follows a Poisson distribution. The probability that there will be less than 4 penalties in a day is _____ [GATE 2014 (CE)]
6. If $\{x\}$ is a continuous, real valued random variable defined over the interval $(-\infty, +\infty)$ and its occurrence is defined by the density function given as :

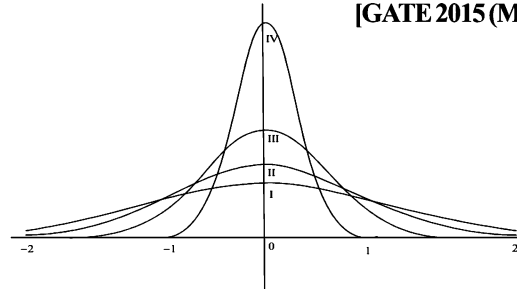
$$f(x) = \frac{1}{\sqrt{2\pi} * b} e^{-\frac{1}{2} \left(\frac{x-a}{b} \right)^2} \text{ where 'a' and 'b' are the statistical attributes of the random}$$

variable $\{x\}$. The value of the integral $\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi} * b} e^{-\frac{1}{2} \left(\frac{x-a}{b} \right)^2} dx$ is [GATE 2014 (CE)]

- (A) 1 (B) 0.5 (C) π (D) $\pi / 2$
7. An observer counts 240 veh/hr at a specific highway location. Assume that the vehicle arrival at the location is Piosson distributed, the probability of having one vehicle arriving over a 30-second time interval is _____ [GATE 2014 (CE)]

8. Among the four normal distributions with probability density function as shown below, which one has the lowest variance ? [GATE 2015 (ME)]

- (A) I
(B) II
(C) III
(D) IV



9. The probability of obtaining atleast two "SIX" in throwing a fair dice 4 times is

[GATE 2015 (ME)]

- (A) $\frac{425}{432}$ (B) $\frac{19}{144}$ (C) $\frac{13}{144}$ (D) $\frac{125}{432}$

10. The probability that a thermistor randomly picked up from a production unit is defective is 0.1. The probability that out of 10 thermistors randomly picked up, 3 are defective is

[GATE 2015 (IN)]

- (A) 0.001 (B) 0.057 (C) 0.0107 (D) 0.3

11. Consider a Poisson distribution for the tossing of a biased coin. The mean for this distribution is μ . The standard deviation for this distribution is given by

- (A) $\sqrt{\mu}$ (B) μ^2 (C) μ (D) $1/\mu$ [GATE 2016 (ME)]

12. The probability that a screw manufactured by a company is defective is 0.1. The company sells screws in packets containing 5 screws and gives a guarantee of replacement if one or more screws in the packet are found to be defective. The probability that a packet would have to be replaced is _____ [GATE 2016 (ME)]

13. The number of parameters in the univariate exponential and Gaussian distributions, respectively, are [GATE 2017 (CE)]

- (A) 2 and 2 (B) 1 and 2 (C) 2 and 1 (D) 1 and 1

14. Vehicles arriving at an intersection from one of the approach road follow the Poisson distribution. The mean rate of arrival is 900 vehicles per hour. If a gap is defined as the time difference between two successive vehicle arrivals (with vehicles assumed to be points), the probability (upto four decimal places) that the gap is greater than 8 seconds is _____ [GATE 2017 (CE)]

15. For a construction project, the mean and standard deviation of the completion time are 200 days and 6.1 days, respectively. Assume normal distribution and use the value of standard normal deviate $z = 1.64$ for the 95% confidence level. The maximum time required (in days) for the completion of the project would be _____ [GATE 2017 (CE)]

16. If a random variable X has a Poisson distribution with mean 5, then the expectation $E[(X+2)^2]$ equals _____ [GATE 2017 (CS)]
17. Let X_1, X_2, X_3 and X_4 be independent normal random variables with zero mean and unit variance. The probability that X_4 is the smallest among the four is _____ [GATE 2018 (EC)]
18. A six- faced fair dice is rolled five times. The probability (in %) of obtaining “ONE” at least four times is _____ [GATE 2018 (ME)]
(A) 33.3 (B) 3.33 (C) 0.33 (D) 0.0033
19. Let X_1, X_2 be two independent normal random variables with means μ_1, μ_2 and standard deviations σ_1, σ_2 respectively. Consider $Y = X_1 - X_2; \mu_1 = \mu_2 = 1, \sigma_1 = 1, \sigma_2 = 2$. Then
(A) Y is normally distributed with mean 0 and variance 1 [GATE 2018 (ME)]
(B) Y is normally distributed with mean 0 and variance 5
(C) Y has mean 0 and variance 5, but is NOT normally distributed
(D) Y has mean 0 and variance 1, but is NOT normally distributed
20. Consider a sequence of tossing of a fair coin where the outcomes of tosses are independent. The probability of getting the head for the third time in the fifth toss is _____ [GATE 2018 (IN)]
(A) $\frac{5}{16}$ (B) $\frac{3}{16}$ (C) $\frac{3}{5}$ (D) $\frac{9}{16}$
21. Let z be an exponential random variable with mean 1. That is, the cumulative distribution function of z is given by _____
$$F_z(x) = \begin{cases} 1 - e^{-x}, & \text{if } x \geq 0 \\ 0, & \text{if } x < 0 \end{cases}$$

Then $P_r(z > 2 | z > 1)$, rounded off to two decimal places, is equal to _____ [GATE 2019 (EC)]
22. The probability of a resistor being defective is 0.02. There are 50 such resistors in a circuit. The probability of two or more defective resistors in the circuit (rounded off to two decimal places is _____ [GATE 2019 (EE)]
23. The lengths of a large stock of titanium rods follow a normal distribution with a mean (μ) of 440 mm and a standard deviation (σ) of 1 mm. What is the percentage of rods whose lengths lie between 438 mm and 441 mm ? [GATE 2019 (ME-Set 1)]
(A) 81.85 % (B) 68.4 % (C) 99.75% (D) 86.64 %

24. The probability that a part manufactured by a company will be defective is 0.05. If 15 such parts are selected randomly and inspected, then the probability that at least two parts will be defective is ____ (rounded off 2 decimal places) [GATE 2019 (ME-Set 2)]
25. The probability density function of a continuous random variable distributed uniformly between x and y (for $y > x$) is _____ [GATE 2019 (CE-AN)]
- (A) $\frac{1}{y-x}$ (B) $\frac{1}{x-y}$ (C) $x-y$ (D) $y-x$
26. Suppose Y is distributed uniformly in the open interval $(1,6)$. The probability that the polynomial $3x^2 + 6xy + 3y + 6$ has only one real root is (rounded off to 1 decimal place)

[GATE 2019 (CS)]

ANSWERS

- | | | | | |
|----------|----------|-------------------|------------|----------|
| 1. C | 2. B | 3. $\frac{7}{27}$ | 4. B | 5. 0.265 |
| 6. B | 7. 0.27 | 8. D | 9. B | 10. B |
| 11. A | 12. 0.41 | 13. B | 14. 0.3154 | 15. 210 |
| 16. 54 | 17. 0.25 | 18. C | 19. B | 20. B |
| 21. 0.37 | 22. 0.26 | 23. A | 24. 0.17 | 25. A |
| 26. 0.8 | | | | |

UNIT - IV

ESTIMATION & TESTING OF HYPOTHESIS (LARGE SAMPLE TESTS)

8. *Estimation*

- ♦ *Point Estimation*
- ♦ *Interval Estimation*
- ♦ *Bayesian Estimation*

9. *Testing of Hypothesis (Large Sample Tests)*

- ♦ *Population and Sample*
- ♦ *One - Tail Test*
- ♦ *Two - Tail Test*
- ♦ *Type I and Type II Errors*
- ♦ *Hypothesis Concerning One and Two Means*
- ♦ *Hypothesis concerning One and Two Proportions*



CHAPTER - 8

ESTIMATION

8.1 INTRODUCTION

Consider a random sample of n observations, say $x_1, x_2, \dots, x_n$ is selected from a population whose probability distribution function is $f(x; \theta_1, \theta_2, \dots, \theta_k)$ where $\theta_1, \theta_2, \dots, \theta_k$ are the population parameters. Then there will always be an infinite number of functions of sample values, called statistic, which may be proposed as estimates of one or more of the parameters. The statistic whose distribution concentrates as closely as possible near the true value of the parameter may be regarded as the best estimate. In other words, the problem of estimation can be written as “determining the functions of sample observations $\hat{\theta}_1(x_1, x_2, \dots, x_n)$, $\hat{\theta}_2(x_1, x_2, \dots, x_n)$, $\dots, \hat{\theta}_k(x_1, x_2, \dots, x_n)$ such that their distribution is concentrated as closely as possible near the true value of the parameter”. These estimating functions are also referred to as estimators.

8.2 DEFINITIONS

Quantities appearing in distributions, such as p in the binomial distribution and μ and σ in the normal distribution are called parameters.

Estimate [JNTU (H) Nov. 2009, (A) 2010S, Nov. 2011 (Set No. 1), May 2012]

An estimate is a statement made to find an unknown population parameter.

Estimator [JNTU (H) Nov. 2009, (A) 2010S, Nov. 2011 (Set No. 1), May 2012]

The procedure or rule to determine an unknown population parameter is called an estimator. For instance, sample mean is an estimator of population mean because sample mean is a method of determining the population mean. Remember that an estimator must be a statistic and it must depend only on the sample and not on the parameter to be estimated. So an estimator is a statistic which for all practical purposes, can be used in place of unknown parameter of the population.

A parameter can have one or two or many estimators.

Types of Estimation :

Basically, there are two kinds of estimates to determine the statistic of the population parameters namely, (a) **Point Estimation** and (b) **Interval Estimation**.

Statistical Estimation

W. A. Spur and C. P. Bonini defined **Statistical Inference** as “the process by which we draw a conclusion about some measure of a population based on a sample value. The measure might be a variable, such as the mean, S. D., etc. The purpose of sampling is to estimate some characteristics for the population from which the sample is selected”.

Basically, there are two types of problems under statistical inference :

(1) Hypothesis testing - to test some hypothesis about parent population from which the sample is drawn.

(2) Estimation - to use the statistics obtained from the sample as estimate of the unknown parameter of the population from which the sample is drawn.

An important problem of Statistical Inference is the estimation of population parameters (*i.e.*, population mean, population S.D, etc.) from the corresponding sample statistics (*i.e.*, sample mean, sample S.D. etc)

Point Estimation and Interval Estimation

If an estimate of the population parameter is given by a single value, then the estimate is called a **Point Estimation** of the parameter. But if an estimate of a population parameter is given by two different values between which the parameter may be considered to lie, then the estimate is called an interval estimation of the parameter.

Example : If the height of a student is measured as 162 cms, then the measurement gives a point estimation. But if the height is given as (163 ± 3.5) cms, then the height lies between 159.5 cms and 166.5 cms and the measurement gives an interval estimation.

The sample mean $\bar{x}$ is a point estimate of population mean μ , sample variance s^2 is a point estimate of population variance σ^2 .

Definition : A point estimate of a parameter θ is a single numerical value, which is computed from a given sample and serves as an approximation of the unknown exact value of the parameter.

Definition : A point estimator is a statistic for estimating the population parameter θ and will be denoted by $\hat{\theta}$ (read as theta hat).

Properties of Estimation : An estimator is not expected to estimate the population parameter without error. An estimator should be close to the true value of unknown parameter.

Unbiased and Biased Estimates

A statistic is said to be an *unbiased estimator* of the corresponding parameter if the mean of the sampling distribution of the statistic is equal to the corresponding population parameter. Otherwise the statistic is called a *biased estimator* of the corresponding parameter. The values of statistics in the above two cases are called *unbiased* and *biased estimates* respectively.

If t be a statistic and θ be the corresponding parameter and $E(t) = \theta$, then t is an unbiased estimator of θ . Otherwise t is a biased estimator of θ and the bias is $E(t) - \theta$.

For example, sample mean $\bar{x}$ is an unbiased estimator of population mean μ [since $E(\bar{x}) = \mu$].

[JNTU(A) Nov. 2011 (Set No. 2)]

Proof : Let $x_1, x_2, \dots, x_n$ be a random sample drawn from a given population with mean μ and variance σ^2 . Then

$$\begin{aligned}
 E(\bar{x}) &= E\left(\frac{1}{n} \sum_{i=1}^n x_i\right) = \frac{1}{n} E(x_1 + x_2 + \dots + x_n) \\
 &= \frac{1}{n} [E(x_1) + E(x_2) + \dots + E(x_n)] \\
 &= \frac{1}{n} [\mu + \mu + \dots + \mu] \quad [\because E(x_i) = \mu]
 \end{aligned}$$

Hence the sample mean $\bar{x}$ is an unbiased estimator of the population mean μ .

Unbiased Estimator : Let $\hat{\theta}$ be an estimator of θ . The statistic θ is said to be an unbiased estimator, or its value an unbiased estimate, if, and only if the mean or expected value of $\hat{\theta}$ is equal to θ . This is equivalent to say that the mean of the probability distribution of $\hat{\theta}$ (or the mean of the sampling distribution of $\hat{\theta}$) is equal to θ . An estimator possessing this property is said to be unbiased.

Definition. Unbiased estimator : A statistic or point estimator $\hat{\theta}$ is said to be an *unbiased* estimator of the parameter θ if $E(\hat{\theta}) = \theta$.

In other words, if $E(\text{statistic}) = \text{parameter}$, then statistic is said to be an unbiased estimator of the parameter.

Example 1 : Show that S^2 is an unbiased estimator of the parameter σ^2 .

Solution : Let us write

$$\begin{aligned}
 \sum_{i=1}^n (x_i - \bar{x})^2 &= \sum_{i=1}^n [(x_i - \mu) - (\bar{x} - \mu)]^2 \\
 &= \sum_{i=1}^n (x_i - \mu)^2 - 2(\bar{x} - \mu) \sum_{i=1}^n (x_i - \mu) + n(\bar{x} - \mu)^2 \\
 &= \sum_{i=1}^n (x_i - \mu)^2 - n(\bar{x} - \mu)^2 \quad \dots (1)
 \end{aligned}$$

$$\begin{aligned}
 \text{Now } E(S^2) &= E\left[\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}\right] \\
 &= \frac{1}{n-1} \left[\sum_{i=1}^n E(x_i - \mu)^2 - n E(\bar{x} - \mu)^2 \right], \text{ using (1)} \\
 &= \frac{1}{n-1} \left(\sum_{i=1}^n \sigma_{X_i}^2 - n \sigma_{\bar{X}}^2 \right)
 \end{aligned}$$

However, $\sigma_{X_i}^2 = \sigma^2$ for $i = 1, 2, \dots, n$ and $\sigma_{\bar{X}}^2 = \frac{\sigma^2}{n}$

$$\therefore E(S^2) = \frac{1}{n-1} \left(n\sigma^2 - n \cdot \frac{\sigma^2}{n} \right) = \sigma^2$$

Although S^2 is an unbiased estimator of σ^2 , S , on the other hand, is a biased estimator of σ with the bias becoming insignificant for large samples. This example illustrates why we divide by $(n-1)$ rather than n when the variance is estimated.

Example 2 : Prove that for a random sample of size n , $X_1, X_2, \dots, X_n$ taken from an infinite population $s^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$ is not unbiased estimator of the parameter σ^2 but

$$\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \text{ is unbiased.}$$

[JNTU (H) Nov. 2010]

Solution : Let μ be the population mean. Then

$$E(x_i) = \mu \text{ and } \text{var}(x_i) = E[(x_i - \mu)^2] = \sigma^2 \text{ for } i = 1, 2, \dots, n$$

$$\text{If } s \text{ be the sample S.D., then } s^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \quad \dots (1)$$

$$\text{Now } s^2 = \frac{1}{n} \sum_{i=1}^n (x_i^2 - 2x_i\bar{x} + \bar{x}^2)$$

$$= \frac{1}{n} \sum x_i^2 - 2\bar{x} \cdot \frac{\sum x_i}{n} + \frac{1}{n} \sum \bar{x}^2$$

$$= \frac{\sum x_i^2}{n} - 2\bar{x}^2 + \bar{x}^2 = \frac{\sum x_i^2}{n} - \bar{x}^2$$

$$= \frac{\sum (x_i - \mu)^2}{n} - (\bar{x} - \mu)^2$$

$$\left[\because \frac{\sum (x_i - \mu)^2}{n} - (\bar{x} - \mu)^2 = \frac{\sum x_i^2}{n} - 2\mu \frac{\sum x_i}{n} + n \cdot \frac{\mu^2}{n} - (\bar{x}^2 + \mu^2 - 2\bar{x}\mu) = \frac{\sum x_i^2}{n} - \bar{x}^2 \right]$$

$$\therefore E(s^2) = E \left[\frac{\sum (x_i - \mu)^2}{n} - (\bar{x} - \mu)^2 \right] = E \left[\frac{\sum (x_i - \mu)^2}{n} \right] - E(\bar{x} - \mu)^2$$

$$= \frac{\sum E(x_i - \mu)^2}{n} - E(\bar{x} - \mu)^2 = \frac{\sum \text{var}(x_i)}{n} - \text{var}(\bar{x})$$

$$= \frac{\sum \sigma^2}{n} - \frac{\sigma^2}{n} \left(\because \bar{x} = \frac{\sum x_i}{n} \right)$$

$$= \frac{n\sigma^2}{n} - \frac{\sigma^2}{n} = \sigma^2 \left(1 - \frac{1}{n} \right) = \frac{n-1}{n} \sigma^2$$

Thus $E(s^2) \neq \sigma^2$

Hence, s^2 is a biased estimator of σ^2

Let us consider $S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{n}{n-1} s^2$ [by (1)]

Then $E(S^2) = E\left(\frac{n}{n-1} s^2\right) = \frac{n}{n-1} E(s^2) = \frac{n}{n-1} \times \frac{n-1}{n} \sigma^2 = \sigma^2$

Hence $S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ is an unbiased estimator of σ^2 .

Variance of a Point Estimator : If $\hat{\theta}_1$ and $\hat{\theta}_2$ are two unbiased estimators of the same population parameter θ , we would choose the estimator whose sampling distribution has the smaller variance. Hence, if $\sigma_{\hat{\theta}_1}^2 < \sigma_{\hat{\theta}_2}^2$, we say that $\hat{\theta}_1$ is a more efficient estimator of θ than $\hat{\theta}_2$.

Most Efficient Estimator : If we consider all possible unbiased estimators of some parameter θ , the one with the smallest variance is called the most efficient estimator of θ .

[JNTU (A) Nov. 2011 (Set No. 2)]

In Figure, we illustrates the sampling distribution of three different estimators $\hat{\theta}_1$ and $\hat{\theta}_2$ and $\hat{\theta}_3$ all estimating θ . It is clear that only $\hat{\theta}_1$ and $\hat{\theta}_2$ are unbiased, since their distributions are centered at θ . The estimator $\hat{\theta}_1$ has a smaller variance than $\hat{\theta}_2$ and is therefore more efficient. Hence our choice for an estimator of θ , among the three considered would be $\hat{\theta}_1$.

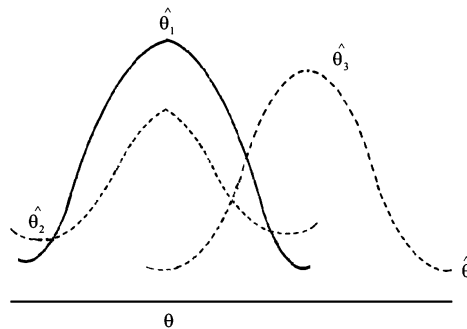


Fig. Sampling Distribution of Different Estimation of θ

Properties of Estimators :

A good estimator is one which is as close to the true value of the parameter as possible.

The important properties of a good estimator are :

(i) Consistency (ii) Unbiasedness (iii) Efficiency and (iv) Sufficiency

(i) An estimator $\hat{\theta}_n$ of a parameter θ is consistent if it converges to θ , as $n \rightarrow \infty$.

(ii) A statistic $\hat{\theta}$ is said to be an unbiased estimate of θ if $E(\hat{\theta}_n) = \theta$ for all θ .

(iii) A statistic $\hat{\theta}_1$ is said to be a more efficient unbiased estimator of the parameter θ than the statistic $\hat{\theta}_2$ if

(a) $\hat{\theta}_1$ and $\hat{\theta}_2$ are both unbiased estimators of θ

(b) $V(\hat{\theta}_1) < V(\hat{\theta}_2)$

(iv) An estimator is said to be sufficient for a parameter, if it contains all the information in the sample regarding the parameter.

8.3 INTERVAL ESTIMATION

[JNTU (A) Nov. 2011 (Set No. 3,4)]

Point estimates rarely coincide with quantities they are intended to estimate. So instead of point estimation where the quantity to be estimated is replaced by a single value a better way of estimation is interval estimation, which determines an interval in which the parameter lies.

Interval Estimate : Even the most efficient unbiased estimator cannot estimate the population parameter exactly. It is true that our accuracy increases with large samples. But there is still no reason why we should expect a **point estimate** from a given sample to be exactly equal to the population parameter, it is supposed to estimate.

Therefore in many situations it is preferable to determine an *interval* within which we would expect to find the value of the parameter. Such an interval is called an *interval estimate*. Thus an interval estimate is an interval (confidence interval) obtained from a sample.

Interval Estimation : An interval estimate of a population parameter θ is an interval of the form $\hat{\theta}_L < \theta < \hat{\theta}_u$, where $\hat{\theta}_L$ and $\hat{\theta}_u$ depend on the value of the statistic $\hat{\theta}$ for a particular sample and also on the sampling distribution of $\hat{\theta}$.

Since different samples will generally yield different values of $\hat{\theta}$ and, therefore, different values $\hat{\theta}_L$ and $\hat{\theta}_u$. These end points of the interval are values of corresponding random variables $\hat{\theta}_L$ and $\hat{\theta}_u$.

From the sampling distribution of $\hat{\theta}$ we shall be able to determine $\hat{\theta}_L$ and $\hat{\theta}_u$ such that the $P(\hat{\theta}_L < \theta < \hat{\theta}_u)$ is equal to any positive fractional value we care to specify. If, for instance, we find $\hat{\theta}_L$ and $\hat{\theta}_u$ such that $P(\hat{\theta}_L < \theta < \hat{\theta}_u) = 1 - \alpha$ for $0 < \alpha < 1$, then we have a probability of $1 - \alpha$ of selecting a random sample that will produce an interval containing θ . The interval $\hat{\theta}_L < \theta < \hat{\theta}_u$, computed from the selected sample, is then called a $(1 - \alpha)$ 100% **confidence interval**. The fraction $1 - \alpha$ called the confidence coefficient or the degree of confidence, and the end points, $\hat{\theta}_L$ and $\hat{\theta}_u$, are called the lower and upper confidence limits.

Thus, when $\alpha = 0.05$, we have a 95% confidence interval, and when $\alpha = 0.01$, we obtain a wider 99% confidence interval.

8.4 CONFIDENCE INTERVAL - ESTIMATES OF PARAMETERS

In an interval estimation of the population parameter θ , if we can find two quantities t_1 and t_2 based on sample observations drawn from the population such that the unknown parameter θ is included in the interval $[t_1, t_2]$ in a specified percentage of cases, then this interval is called a *confidence interval* for the parameter θ . The confidence interval has a specified confidence or probability of correctly estimating the true value of the population parameter. In this case, $P(t_1 \leq \theta \leq t_2) = c$ where c is a given (specified) probability.

The computation of confidence interval and confidence limits is based on the sampling distribution of a statistic. By central limit theorem, we know that the sampling distribution of the sample mean $\bar{x}$ of a random sample of size n ($n \geq 30$) drawn from a population having mean μ and S.D. σ is approximately a normal distribution with mean μ and

$$\text{S.D} = \text{S.E. of } \bar{x} = \frac{\sigma}{\sqrt{n}}$$

$$\therefore z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \text{ has a standard normal distribution with mean } = 0 \text{ and S.D.} = 1.$$

From the standard normal curve, we know that 95.45% of the area lies between the ordinates $z = \pm 2$ i.e., $P(-2 < z < 2) = 95.45\% = 0.9545$, where $z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$.

$$\text{This shows that the inequality } -2 \leq \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \leq 2 \quad \dots (1)$$

holds in 95.45% of cases and it does not hold in only 4.55 % of the cases.

$$\text{From (1), we have } \frac{-2\sigma}{\sqrt{n}} \leq \bar{x} - \mu \leq \frac{2\sigma}{\sqrt{n}}$$

$$\text{or } \bar{x} - \frac{2\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + \frac{2\sigma}{\sqrt{n}}$$

$$\text{or } \bar{x} - 2 (\text{S.E. of } \bar{x}) \leq \mu \leq \bar{x} + 2 (\text{S.E. of } \bar{x}) \left[\because \text{S.E. of } \bar{x} = \frac{\sigma}{\sqrt{n}} \right]$$

$$\text{or } [\bar{x} - 2 (\text{S.E. of } \bar{x}), \bar{x} + 2 (\text{S.E. of } \bar{x})]$$

This interval is called the 95.45% *confidence interval* for the population mean μ and $\bar{x} \pm 2 (\text{S.E. of } \bar{x})$ are called the 95.45% *confidence limits* for μ . Here 2 is called the confidence coefficient or z-value.

Again 99.73% of the area under the standard normal curve lies between the ordinates $z = \pm 3$ i.e., $P(-3 \leq z \leq 3) = 0.9973$. Proceeding as above, 99.73% *confidence interval* and *confidence limits* for the population μ are respectively

$$[\bar{x} - 3 (\text{S.E. of } \bar{x}), \bar{x} + 3 (\text{S.E. of } \bar{x})] \text{ and } \bar{x} \pm 3 (\text{S.E. of } \bar{x})$$

Also 95% of the area under the standard normal curve lies between the ordinates $z = \pm 1.96$ i.e., $P(-1.96 \leq z \leq 1.96) = 0.95$. Proceeding as above, 95% *confidence interval* and *confidence limits* for the population mean μ are respectively

$$[\bar{x} - 1.96 (\text{S.E. of } \bar{x}), \bar{x} + 1.96 (\text{S.E. of } \bar{x})] \text{ and } \bar{x} \pm 1.96 (\text{S.E. of } \bar{x})$$

Similarly, using the table for the area under standard normal curve, we can find the confidence interval and confidence limits corresponding to any specified (or given) percentage.

By central limit theorem, the sampling distribution of the sample proportion p of a large random sample of size n ($n \geq 30$) drawn from a population having population proportion P is approximately a normal distribution with mean P and S. D. = S. E. of p . Using this theorem, we can find the confidence interval and confidence limits for the population proportion P , proceeding as above.

For small samples ($n < 30$) drawn from a normal population, t -distribution (discussed later) is used to find the confidence interval of the population mean if σ is not known.

The formulae for confidence limits of some well-known statistic for large random samples are given below.

I. Confidence limits for Population Mean μ

- (i) 95% confidence limits are $\bar{x} \pm 1.96 (\text{S.E. of } \bar{x})$
- (ii) 99% confidence limits are $\bar{x} \pm 2.58 (\text{S.E. of } \bar{x})$
- (iii) 99.73% confidence limits are $\bar{x} \pm 3 (\text{S.E. of } \bar{x})$
- (iv) 90% confidence limits are $\bar{x} \pm 1.64 (\text{S.E. of } \bar{x})$

II. Confidence limits for Population Proportion P

- (i) 95% confidence limits are $p \pm 1.96 (\text{S.E. of } p)$
- (ii) 99% confidence limits are $p \pm 2.58 (\text{S.E. of } p)$
- (iii) 99.73% confidence limits are $p \pm 3 (\text{S.E. of } p)$
- (iv) 90% confidence limits are $p \pm 1.64 (\text{S. E. of } p)$

III. Confidence limits for the difference $\mu_1 - \mu_2$ of two Population Means μ_1 and μ_2

- (i) 95% confidence limits are $(\bar{x}_1 - \bar{x}_2) \pm 1.96 (\text{S.E. of } (\bar{x}_1 - \bar{x}_2))$
- (ii) 99% confidence limits are $(\bar{x}_1 - \bar{x}_2) \pm 2.58 (\text{S.E. of } (\bar{x}_1 - \bar{x}_2))$
- (iii) 99.73% confidence limits are $(\bar{x}_1 - \bar{x}_2) \pm 3 (\text{S.E. of } (\bar{x}_1 - \bar{x}_2))$
- (iv) 90% confidence limits are $(\bar{x}_1 - \bar{x}_2) \pm 1.64 (\text{S.E. of } (\bar{x}_1 - \bar{x}_2))$

IV. Confidence limits for the difference $P_1 - P_2$ of two population proportions

- (i) 95% confidence limits are $(p_1 - p_2) \pm 1.96 (\text{S.E. of } (p_1 - p_2))$

(ii) 99% confidence limits are $(p_1 - p_2) \pm 2.58$ (S.E. of $(p_1 - p_2)$)

(iii) 99.73% confidence limits are $(p_1 - p_2) \pm 3$ (S.E. of $(p_1 - p_2)$)

(iv) 90% confidence limits are $(p_1 - p_2) \pm 1.64$ (S.E. of $(p_1 - p_2)$)

Note. If for a statistic, $P(-3 < z < 3)$ i.e., confidence limits cover 99.73% of the area under the standard normal curve, then we say that the confidence limits for the statistic are almost sure limits without mentioning the degree of confidence.

In calculating S.E. if population parameters are unknown, corresponding sample statistics are used to find an approximate value of S. E. For example, if the population S. D., σ is not given, the S.E. of $\bar{x}$ is calculated by using sample S.D. s in place of σ .

8.5 DETERMINATION OF PROPER SAMPLE SIZE

Determination of proper sample size is important for testing of hypothesis in business problems. The size of sample should neither be too small nor too large. If the size of the sample is too small, then it may not give a valid conclusion. On the other hand, if the size of the sample is too large, then there may be loss of time and money without getting the required results.

1. Sample Size for Estimating Population Mean

Let $\bar{x}$ be the mean of a random sample drawn from a population having mean μ and S.D. σ . Let the sampling distribution of the sample mean $\bar{x}$ be approximately a normal distribution with mean μ and S. D. σ . If E be the permissible sampling error, then $E = \bar{x} - \mu$.

The confidence interval for the population mean μ is $\bar{x} \pm z$ (S.E. of $\bar{x}$) $= \bar{x} \pm E$ where z = confidence coefficient or z - value (which is 1.96 at 5% level of significance)

and $E = z$ (S.E. of $\bar{x}$) $= z \cdot \frac{\sigma}{\sqrt{n}}$, n being the sample size and σ = population S.D.

$$\text{Thus } E = \frac{2\sigma}{\sqrt{n}} \text{ or } \sqrt{n} = \frac{z\sigma}{E}$$

$$\therefore n = \left(\frac{z\sigma}{E} \right)^2, \text{ which gives the required sample size for estimating population mean.}$$

2. Sample Size for Estimating Population Proportion

Let p be the sample proportion and P be the population proportion. If E be the permissible sampling error, then $E = p - P$ and the confidence interval for the population

proportion P is $p \pm z$ (S.E. of p) $= p \pm E$ where $E = z$ (S.E. of p) $= z \cdot \sqrt{\frac{PQ}{n}}$, n being the

sample size, or, $E^2 = \frac{z^2 PQ}{n}$ or $n = \frac{z^2 PQ}{E^2}$ where $Q = 1 - P$.

Proposition : If $x_1, x_2, \dots, x_n$ be a random sample from an infinite population with variance σ^2 and sample mean $\bar{x}$, then $\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ is a biased estimate of σ^2 , but

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \text{ is an unbiased estimate of } \sigma^2.$$

3. Maximum Error of Estimate E for Large Samples : [JNTU (A) Dec. 2009 (Set No. 2)]

Since the sample mean estimate very rarely equals to the mean of population μ , a point estimate is generally accompanied with a statement of error which gives difference between estimate and the quantity to be estimated, the estimator.

Thus error is $|\bar{x} - \mu|$.

For large n , the random variable $\frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$ is normal variate approximately.

Then $P(-z_{\alpha/2} < z < z_{\alpha/2}) = 1 - \alpha$ where $z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$

Hence $P\left(-z_{\alpha/2} < \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} < z_{\alpha/2}\right) = 1 - \alpha$

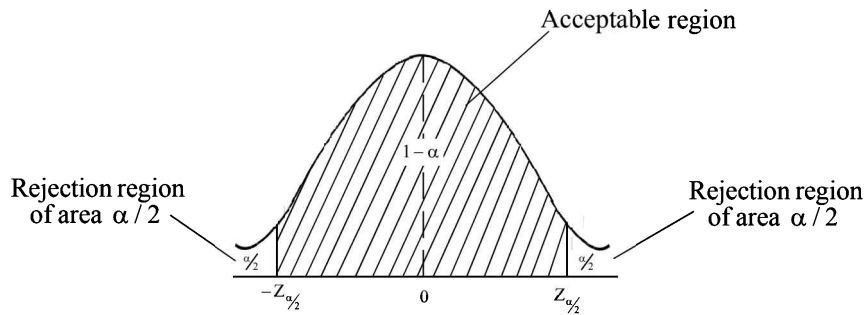


Fig. $P(-z_{\alpha/2} < z < z_{\alpha/2}) = 1 - \alpha$

Multiplying each term in the inequality by $\sigma/\sqrt{n}$, and then subtracting $\bar{x}$ from each term and multiplying by -1 .

$$\therefore P\left(\bar{x} - z_{\alpha/2} \cdot \left(\frac{\sigma}{\sqrt{n}}\right) < \mu < \bar{x} + z_{\alpha/2} \cdot \left(\frac{\sigma}{\sqrt{n}}\right)\right) = 1 - \alpha$$

Confidence Interval for μ, σ known :

[JNTU (A) Nov. 2011 (Set No. 3)]

If $\bar{x}$ is the mean of a random sample of size n from the population with known variance σ^2 , $(1 - \alpha)$ 100% confidence interval for μ is given by

$$\bar{x} - z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) < \mu < \bar{x} + z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

where $z_{\alpha/2}$ is the z -value leaving an area of $\alpha/2$ to the right.

So, the maximum error of estimate E with $(1 - \alpha)$ probability is given by

$$E = z_{\alpha/2} \cdot \left(\frac{\sigma}{\sqrt{n}} \right)$$

Thus in the point estimation of population mean μ with sample mean $\bar{x}$ for a large random sample ($n \geq 30$), one can assert with probability $(1 - \alpha)$ that the error

$|\bar{x} - \mu|$ will not exceed $z_{\alpha/2} \cdot \left(\frac{\sigma}{\sqrt{n}} \right)$.

Sample Size : When α , E , σ are known, the sample size n is given by

$$n = \left(\frac{z_{\alpha/2} \sigma}{E} \right)^2$$

When σ is unknown : In this case, σ is replaced by s , the standard deviation of sample to determine E .

Thus the maximum error estimate $E = t_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$ with $(1 - \alpha)$ probability.

4. Maximum Error of Estimate E for Small Samples :

When $n < 30$, small sample, we use S , the S.D of sample to determine E . When σ is not known, t can be used to construct a confidence interval as μ .

The procedure is the same as that with known σ except that σ is replaced by S and the standard normal distribution is replaced by the t -distribution.

$$P(-t_{\alpha/2} < t < t_{\alpha/2}) = 1 - \alpha$$

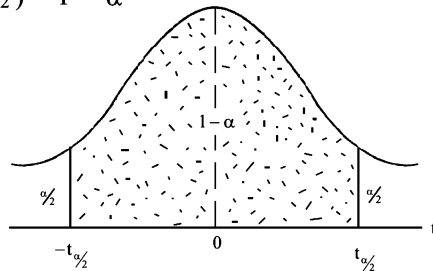


Fig. $P(-t_{\alpha/2} < t < t_{\alpha/2}) = 1 - \alpha$

where $t_{\alpha/2}$ is the t -value within $(n - 1)$ degrees of freedom above which we find an area of $\alpha/2$.

Because of symmetry, an equal area of $\alpha/2$ will fall to the left of $-t_{\alpha/2}$.

$$\text{Substituting for } t, \text{ we write } P\left(-t_{\alpha/2} < \frac{\bar{x} - \mu}{S / \sqrt{n}} < t_{\alpha/2}\right) = 1 - \alpha$$

Multiplying each term in the inequality by $S/\sqrt{n}$, and then subtracting $\bar{x}$ from each term and multiplying by -1 , we obtain.

$$P(\bar{x} - t_{\alpha/2}(S/\sqrt{n}) < \mu < \bar{x} + t_{\alpha/2}(S/\sqrt{n})) = 1 - \alpha$$

Confidence interval for μ ; σ unknown :

[JNTU (A) Nov. 2011 (Set No. 3)]

If $\bar{x}$ and s are the mean and standard deviation of a random sample from a normal population with unknown variance σ^2 , a $(1-\alpha)100\%$ confidence interval for μ is

$$\bar{x} - t_{\alpha/2}(s/\sqrt{n}) < \mu < \bar{x} + t_{\alpha/2}(s/\sqrt{n})$$

where $t_{\alpha/2}$ is the t -value with $v = n - 1$ degrees of freedom, leaving an area of $\alpha/2$ to the right.

$\therefore$ The maximum error of estimate for small samples is given by

$$E = t_{\alpha/2} \cdot s/\sqrt{n}, \text{ where } n = \text{Sample size, } s = \text{S.D of sample}$$

8.6 BAYESIAN ESTIMATION

[JNTU (A), (H) Dec 2009, (A) Nov. 2011 (Set No.4)]

The new concept introduced in Bayesian methods is personal or subjective probability. Also, parameters are considered as random variable in Bayesian method. To estimate the mean of a population, μ is treated as a random variable whose distribution is indicative of the “strong feelings” or assumption of a person about the possible value of μ . Let μ_0 and σ_0 be the mean and standard deviation of such a subjective “prior distribution”.

Bayesian Estimation :

Combining the prior feelings about the possible values of μ with direct sample evidence, the “posterior” distribution of μ in Bayesian estimation is approximated by normal distribution with

$$\mu_1 = \frac{n\bar{x}\sigma_0^2 + \mu_0\sigma^2}{n\sigma_0^2 + \sigma^2}$$

$$\text{and } \sigma_1 = \sqrt{\frac{\sigma_0^2 \sigma^2}{n\sigma_0^2 + \sigma^2}}$$

where n = sample size, $\bar{x}$ = sample mean and S is the standard deviation of sample.

Use $S = \sigma$.

Here μ_1 and σ_1 are known as the mean and standard deviation of the posterior distribution. In the computation of μ_1 and σ_1 , σ^2 is assumed to be known, when σ^2 is unknown, which is generally the case, is replaced by sample variance S^2 provided $n \geq 30$ (large sample).

Bayesian interval for μ :

$(1 - \alpha) 100\%$ Bayesian interval for μ is given by

$$\mu_1 - z_{\alpha/2} \cdot \sigma_1 < \mu < \mu_1 + z_{\alpha/2} \cdot \sigma_1$$

SOLVED EXAMPLES

Example 1 : In a study of an automobile insurance a random sample of 80 body repair costs had a mean of ₹. 472.36 and the S.D of ₹. 62.35. If $\bar{x}$ is used as a point estimate to the true average repair costs, with what confidence we can assert that the maximum error doesn't exceed ₹.10. [JNTU Dec 2004S, (K) May 2010S (Set No. 2)]

Solution : Size of a random sample, $n = 80$

The mean of random sample, $\bar{x} = ₹ 472.36$

Standard deviation, $\sigma = ₹ 62.35$

Maximum error of estimate, $E_{\max} = ₹ 10$

$$\text{We have } E_{\max} = z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \Rightarrow z_{\alpha/2} = \frac{E_{\max} \cdot \sqrt{n}}{\sigma} = \frac{10\sqrt{80}}{62.35} = \frac{89.4427}{62.35} = 1.4345$$

$$\Rightarrow z_{\alpha/2} = 1.43$$

The area when $z = 1.43$ from tables is 0.4236.

$$\therefore \frac{\alpha}{2} = 0.4236 \Rightarrow \alpha = 0.8472$$

$$\therefore \text{Confidence} = (1 - \alpha) 100\% = 84.72\%$$

Hence we are 84.72% confidence that the maximum error is ₹ 10.

Example 2 : What is the size of the smallest sample required to estimate an unknown proportion to within a maximum error of 0.06 with atleast 95% confidence.

[JNTU Dec. 2004 S, April 2006, (K) Nov. 2009, (A) Nov. 2010]

Solution : We are given

The maximum error, $E = 0.06$

Confidence limit = 95%

$$\text{i.e. } (1 - \alpha) 100 = 95$$

$$\Rightarrow 1 - \alpha = 0.95 \Rightarrow \alpha = 0.05 \Rightarrow \alpha/2 = 0.025$$

$$\therefore z_{\alpha/2} = 1.96$$

Here P is not given. So we take $P = \frac{1}{2}$. Thus $Q = \frac{1}{2}$

$$\text{Hence } n = \left(\frac{z_{\alpha/2}}{E} \right)^2 (PQ)$$

$$\text{When } P \text{ is unknown, sample size } n = \frac{1}{4} \left[\frac{z_{\alpha/2}}{E} \right]^2$$

$$\therefore n = \frac{1}{4} \left[\frac{1.96}{0.06} \right]^2 = 266.78 \approx 267$$

Example 1 : If we can assert with 95% that the maximum error is 0.05 and $P = 0.2$, find the size of the sample. [JNTU Dec 2005, (H) Nov. 2010 (Set No. 1)]

Solution : Given $P = 0.2, E = 0.05$

We have $Q = 0.8$ and $z_{\alpha/2} = 1.96$ (for 95%)

We know that maximum error, $E = z_{\alpha/2} \sqrt{\left(\frac{PQ}{n}\right)}$

$$\Rightarrow 0.05 = 1.96 \sqrt{\frac{0.2 \times 0.8}{n}}$$

$$\Rightarrow \text{Sample size, } n = \frac{0.2 \times 0.8 \times (1.96)^2}{(0.05)^2} = 246$$

Example 4 : Assuming that $\sigma = 20.0$, how large a random sample be taken to assert with probability 0.95 that the sample mean will not differ from the true mean by more than 3.0 points ? [JNTU Apr. 05, Dec 05, (A) Nov. 2010 (Set No. 1)]

Solution : Given maximum error $E = 3.0$ and $\sigma = 20.0$

We have $z_{\alpha/2} = 1.96$

We know that, $n = \left(\frac{z_{\alpha/2} \cdot \sigma}{E}\right)^2$

$$\Rightarrow n = \left(\frac{1.96 \times 20}{3}\right)^2 = 170.74$$

$$\therefore n \cong 171$$

Example 4 : It is desired to estimate the mean number of hours of continuous use until a certain computer will first require repairs. If it can be assumed that $\sigma = 48$ hours, how large a sample be needed so that one will be able to assert with 90% confidence that the sample mean is off by at most 10 hours.

[JNTU Dec 2004, April 2006, JNTU (K) Nov. 2009, (H) Nov. 2010 (Set No. 3)]

Solution : It is given that

Maximum error, $E = 10$ hours

$\sigma = 48$ hours

and $z_{\alpha/2} = 1.645$ (for 90%)

$$\therefore n = \left(\frac{z_{\alpha/2} \cdot \sigma}{E}\right)^2 = \left(\frac{1.645 \times 48}{10}\right)^2 = 62.3 \approx 62$$

Hence sample size = 62

Example 2 : What is the maximum error one can expect to make with probability 0.90 when using the mean of a random sample of size $n = 64$ to estimate the mean of population with $\sigma^2 = 2.56$. [JNTU 2004S, (K) Nov. 2009, (H) May 2011]

Solution : Here $n = 64$

The probability = 0.90

$$\sigma^2 = 2.56 \Rightarrow \sigma = \sqrt{2.56} = 1.6$$

Confidence limit = 90%

$$\therefore z_{\alpha/2} = 1.645$$

$$\text{Hence maximum error } E = z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} = 1.645 \times \frac{1.6}{\sqrt{64}} = 0.329.$$

Example 2 : A random sample of size 100 has a standard deviation of 5. What can you say about the maximum error with 95% confidence. [JNTU May 2005, (A) Apr. 2012 (Set No. 4)]

Solution : Given $s = 5$, $n = 100$

$$z_{\alpha/2} \text{ for 95\% confidence} = 1.96$$

We know that

$$\text{Maximum error, } E = z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

$$\therefore E = (1.96) \cdot \frac{5}{\sqrt{100}} = 0.98$$

Example 8 : The efficiency expert of a computer company tested 40 engineers to estimate the average time it takes to assemble a certain computer component, getting a mean of 12.73 minutes and S.D of 2.06 minutes.

- If $\bar{x} = 12.73$ is used as a point estimate of the actual average time required to perform the task, determine the maximum error with 99% confidence.
- Construct 98% confidence intervals for the true average time it takes to do the job.
- With what confidence can we assert that the sample mean does not differ from the true mean by more than 30 seconds. [JNTU (K) Nov. 2009 (Set No. 1)]

(OR)

To estimate the average time it takes to assemble a certain computer component, the industrial engineer at an electronic firm timed 40 technicians in the performance of the task, getting a mean of 12.73 min. and a S. D. of 2.06 min.

(i) What can we say with 99% confidence about the maximum error if $\bar{x} = 12.73$ is used as a point estimate of the actual average time required to do the job ?

(ii) Use the given data to construct 98% confidence interval.

(iii) With what confidence we can assert that the sample mean does not differ from the true mean by more than 30 sec. [JNTU (A) Apr. 2012 (Set No. 1)]

Solution : Here $\bar{x} = 12.73$, $S = 2.06$, $n = 40$. For 99%, $z_{\alpha/2} = 2.575$

$$(a) \text{ Maximum error of estimate } E = z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

$$\therefore E = (2.575) \cdot \frac{(2.06)}{\sqrt{40}} = 0.8387$$

$$(b) \text{ For 98\% confidence, } E = (2.33) \cdot \frac{(2.06)}{\sqrt{40}} = 0.758915$$

98% confidence interval limits are

$$\bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} = \bar{x} \pm E = 12.73 \pm 0.7589$$

i.e., confidence interval is (11.97, 13.4889)

$$(c) \frac{30}{60} \text{ minutes} = \frac{1}{2} \text{ minutes} \Rightarrow E = z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} = z_{\alpha/2} \cdot \frac{2.06}{\sqrt{40}}$$

$$\therefore z_{\alpha/2} = 1.5350.$$

From normal distribution table, the area corresponding to $z_{\alpha/2} = 1.5350$ is 0.4370.

Then the area between $z_{-\alpha/2}$ to $z_{\alpha/2}$ is $2(0.4370) = 0.8740$

Thus we have 87.4% confidence.

Example 9 : The mean and standard deviation of a population are 11,795 and 14,054 respectively. What can one assert with 95% confidence about the maximum error if $\bar{x} = 11,795$ and $n = 50$. And also construct 95% confidence interval for the true mean.

[JNTU (A) Apr. 2012 (Set No. 2)]

(OR) The mean and the standard deviation of a population are 11,795 and 14054 respectively. If $n = 50$, find 95% confidence interval for the mean.

[JNTU 2006, 2008S, 2008 (Set No. 3), (K) May 2013 (Set No. 2)]

Solution : Here

Mean of population, $\mu = 11795$

S.D of population, $\sigma = 14054$

$$\bar{x} = 11795$$

$n =$ sample size $= 50$

$$\text{maximum error} = z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

$$z_{\alpha/2} \text{ for } 95\% \text{ confidence} = 1.96$$

$$\text{Max. error, } E = z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} = 1.96 \cdot \frac{(14054)}{\sqrt{50}} = 3899$$

$$\begin{aligned} \therefore \text{Confidence interval} &= (\bar{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}) \\ &= (11795 - 3899, 11795 + 3899) \\ &= (7896, 15694) \end{aligned}$$

Example 10 : A sample of 10 cam shafts intended for use in gasoline engines has an average eccentricity of 1.02 and a standard deviation of 0.044 inch. Assuming the data may be treated a random sample from a normal population, determine a 95% confidence interval for the actual mean eccentricity of the cam shaft ?

[JNTU 2004, (K) May 2010S (Set No. 4)]

Solution : We know that confidence interval is $\bar{x} \pm E$ where $E = z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$

Given $n = 10$, $z_{\alpha/2} = 1.96$, $\sigma = 0.044$, $\bar{x} = 1.02$

$$\text{Confidence interval is } 1.02 \pm (1.96) \left(\frac{0.044}{\sqrt{10}} \right) = 1.02 \pm 0.027$$

$$\therefore \text{Confidence interval} = (0.993, 1.047)$$

Example 11 : A random sample of size 81 was taken whose variance is 20.25 and mean is 32, construct 98% confidence interval. [JNTU Dec 2005 (Set No. 2)]

Solution : Given $\bar{x}$ = sample mean = 32, n = 81,

$$\sigma^2 = 20.25 \Rightarrow \sigma = 4.5$$

$$\text{and } z_{\alpha/2} = 2.33 \text{ (for 98\%)}$$

We know that 98% confidence interval is $\left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$

$$\text{Now } z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} = (2.33) \frac{4.5}{\sqrt{81}} = 1.165$$

$$\therefore \text{Confidence interval} = (32 - 1.29, 32 + 1.29) \\ = (30.835, 33.165)$$

Example 12 : A research worker wants to determine the average time it takes a mechanic to rotate the tyres of a car and he wants to be able to assert with 95% Confidence that the mean of his sample is off by atmost 0.5 minutes. If he can presume from past experience that $\sigma = 1.6$ minutes, how large a sample will have to take? [JNTU 1999]

Solution : Max. error, $E = 0.5$

n = sample size = ?

σ = standard deviation of the population = 1.6 minutes

$$z_{\alpha/2} = 1.96$$

$$\text{Hence } n = \left[\frac{z_{\alpha/2} \cdot \sigma}{E} \right]^2 = \left[\frac{1.96 \times 1.6}{0.5} \right]^2 \approx 40$$

$\therefore$ 40 mechanics will have to perform the task.

Example 12 : It is desired to estimate the mean time of continuous use until an answering machine will first require service. If it can be assumed that $\sigma = 60$ days, how large a sample is needed so that one will be able to assert with 90% confidence that the sample mean is off by at most 10 days.

Solution : We have

Maximum error = 10 days = E , $\sigma = 60$ days and $z_{\alpha/2} = 1.645$

$$\therefore n = \left[\frac{z_{\alpha/2} \cdot \sigma}{E} \right]^2 = \left[\frac{1.645 \times 60}{10} \right]^2 = 97$$

Example 10 : The dean of a college wants to use the mean of a random sample to estimate the average amount of time students take to get from one class to the next and she wants to be able to assert with 99% confidence that the error is at most 0.25 minute. If it can be presumed from experience that $\sigma = 1.40$ minutes. How large a sample will she have to take ? [JNTU 2000]

$$E(\bar{x}) = 6 \cdot \frac{1}{10} + 8 \cdot \frac{1}{10} + 12 \cdot \frac{1}{10} + 10 \cdot \frac{1}{10} + \frac{14}{10} + 9 \cdot \left(\frac{1}{10} \right) + 13 \cdot \frac{1}{10} \\ + 17 \cdot \frac{1}{10} + 16 \cdot \frac{1}{10} + 15 \cdot \frac{1}{10} = \frac{1}{10} \times 120 = 12 = \theta$$

$$\therefore E(\bar{x}) = \theta$$

$\therefore \bar{x}$ is an unbiased estimate of θ .

i.e., the mean of a random sample is an unbiased estimator of the mean of the population.

Example 12 : Suppose that we observe a random variable having the binomial distribution and get x successes in n trials.

(a) Show that $\frac{x}{n}$ is an unbiased estimate of the binomial parameter p .

(b) Show that $\frac{x+1}{n+2}$ is not an unbiased estimate of the binomial parameter p .

Solution : $\hat{\theta}$ is an unbiased estimator of θ if $E(\hat{\theta}) = \theta$

$$(a) \quad E\left(\frac{x}{n}\right) = \frac{1}{n} E(x) = \frac{np}{n} = p \quad [\because E(x) = np]$$

$\therefore \frac{x}{n}$ is an unbiased estimator of p .

$$(b) \quad E\left(\frac{x+1}{n+2}\right) = E\left(\frac{x}{n+2} + \frac{1}{n+2}\right) = \frac{1}{n+2} E(x) + \frac{1}{n+2} \quad [\because E(ax+b) = aE(x) + b] \\ = \frac{p}{n+2} + \frac{1}{n+2} \quad [\because E(x) = p] \\ = \frac{p+1}{n+2} \neq p$$

$\therefore \frac{x+1}{n+2}$ is not an unbiased estimator of p .

Example 10 : Find 95% confidence limits for the mean of a normality distributed population from which the following sample was taken

15, 17, 10, 18, 16, 9, 7, 11, 13, 14.

[JNTU (H) Nov. 2009, (A) Nov. 2010]

Solution : We have $\bar{x} = \frac{15+17+10+18+16+9+7+11+13+14}{10} = 13$

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} \\ = \frac{1}{9} [(15-13)^2 + (17-13)^2 + (10-13)^2 + (18-13)^2 + (16-13)^2 \\ + (9-13)^2 + (7-13)^2 + (11-13)^2 + (13-13)^2 + (14-13)^2] = \frac{40}{3}$$

Since $z_{\alpha/2} = 1.96$, we have

$$z_{\alpha/2} \cdot \frac{s}{\sqrt{n}} = 1.96 \cdot \frac{\sqrt{40}}{\sqrt{10} \cdot \sqrt{3}} = 2.26$$

$$\therefore \text{Confidence limits are } \bar{x} \pm z_{\alpha/2} \cdot \frac{s}{\sqrt{n}} = 13 \pm 2.26 = (10.74, 15.26)$$

Example 12 : Determine a 95% confidence interval for the mean of a normal distribution with variance 0.25, using a sample of $n = 100$ values with mean 212.3.

Solution : We have $n = 100$, $\bar{x} = 212.3$, S.D. = $\sigma = \sqrt{0.25}$ and $z_{\alpha/2} = 1.96$ (for 95%)
We know that 95% confidence interval is

$$\left(\bar{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \right)$$

$$\text{Now } z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} = \frac{1.96 \times \sqrt{0.25}}{\sqrt{100}} = \frac{1.96 \times 0.5}{10} = \frac{0.98}{10} = 0.098$$

$$\therefore \text{Confidence interval} = (212.3 - 0.098, 212.3 + 0.098) \\ = (212.202, 212.398).$$

Example 10 : A professor's feelings about the mean mark in the final examination in "Probability" of a large group of students is expressed subjectively by normal distribution with $\mu_0 = 67.2$ and $\sigma_0 = 1.5$.

- If the mean mark lies in the interval (65.0, 70.0) determine the prior probability the professor should assign to the mean mark.
- Find the professor mean μ_1 and the posterior S.d σ_1 if the examinations are conducted on a random sample of 40 students yielding mean 74.9 and S.D. 7.4. Use $S = 7.4$ as an estimate σ .
- Determine the posterior probability which he will thus assign to the mean mark being in the interval (65.0, 70.0) using results obtained in (b).
- Construct a 95% Bayesian interval for μ . [JNTU(K) Nov. 2011 (Set No. 4)]

Solution : (a) Here $\mu_0 = 67.2$, $\sigma_0 = 1.5$ and $n = 40$

Standard variable corresponding to 65.0 is

$$z_1 = \frac{65.0 - 67.2}{1.5} = -1.466$$

$$\text{Similarly, } z_2 = \frac{70 - 67.2}{1.5} = 1.866$$

Let X be the mean mark obtained in the final examination

$$\text{Prior probability} = P(65 < x < 70) = P(-1.47 < z < 1.87) \\ = 0.4292 + 0.4693 = 0.8985$$

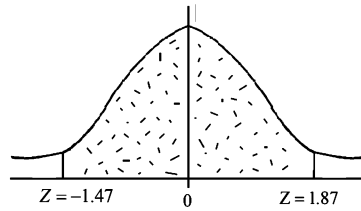


Fig.

- (b) Here $\bar{x} = 74.9$, $\sigma = S = 7.4$

$$\text{Posterior mean, } \mu_1 = \frac{n\bar{x}\sigma_0^2 + \mu_0\sigma^2}{n\sigma_0^2 + \sigma^2}$$

$$\text{i.e., } \mu_1 = \frac{40(74.9)(1.5)^2 + (67.2)(7.4)^2}{40(1.5)^2 + (7.4)^2} = 71.987 \approx 72$$

$$\text{Posterior S.D, } \sigma_1 = \sqrt{\frac{\sigma^2\sigma_0^2}{n\sigma_0^2 + \sigma^2}} = \sqrt{\frac{(7.4)^2(1.5)^2}{40(1.5)^2}} = 0.922568 \approx 0.923$$

- (c) Here $\mu_1 = 72$, $\sigma_1 = 0.923$

Standard variable corresponding to 65 is

$$Z_1 = \frac{65 - 72}{0.923} = -7.5839$$

Similarly, corresponding to 70.0 is

$$Z_2 = \frac{70 - 72}{0.923} = -2.16684$$

Posterior probability = $P(65 < x < 70)$

$$= P(-7.584 < Z < -2.167)$$

$$= 0.5 - 0.4850 = 0.0150$$

- (d) 95% Bayesian Interval limits are

$$\mu_1 \pm Z_{\alpha/2} \cdot \sigma_1 = 71.987 \pm (1.96)(0.922568)$$

Thus the Bayesian interval is (70.17876, 72.909568)

Example 12 : A random sample of 100 teachers in a large metropolitan area revealed a mean weekly salary of Rs. 487 with a standard deviation Rs 48. With what degree of confidence can we assert that the average weekly salary of all teachers in the metropolitan area is between 472 to 502 ?

[JNTU 2006S, 2007, (A) Nov. 2010, Apr. 2012, (K) Nov. 2011]

Solution : Given $\mu = 487$, $\sigma = 48$, $n = 100$

$$\therefore z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{\bar{x} - 487}{\frac{48}{\sqrt{100}}} = \frac{\bar{x} - 487}{4.8}$$

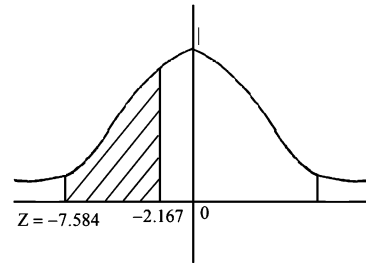


Fig.

Standard variable corresponding to Rs. 472 is

$$z_1 = \frac{472 - 487}{4.8} = -3.125$$

Standard variable corresponding to Rs. 502 is

$$z_2 = \frac{502 - 487}{4.8} = 3.125$$

Let X be the mean salary of teacher. Then

$$\begin{aligned} P(472 < x < 502) &= P(-3.125 < z < 3.125) \\ &= 2 (0 < z < 3.125) \\ &= 2 \int_0^{3.125} \phi(z) dz \\ &= 2 (0.4991) = 0.9982 \end{aligned}$$

Thus we can ascertain with 99.82% confidence.

Example 11 : A population random variable has mean 100 and standard deviation 16. What are the mean and standard deviation of the sample mean for the random samples of size 4 drawn with replacement ?
[JNTU April 2004, (A) Apr. 2012]

Solution : Given $\mu = 100$, $\sigma = 16$, $n = 4$

Since the sampling is done with replacements, the population may be considered as infinite.

We have to find $\mu_{\bar{x}}$ and $\sigma_{\bar{x}}$.

$$\therefore \mu_{\bar{x}} = \mu = 100 \quad \text{and} \quad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{16}{\sqrt{4}} = 8$$

Example 11 : A random sample of 400 items is found to have mean 82 and S.D. of 18. Find the maximum error of estimation at 95% confidence interval. Find the confidence limits for the mean if $\bar{x} = 82$.
[JNTU Nov. 2008, (A) Nov. 2011, (H) May 2011 (Set No. 2)]

Solution : Given standard deviation $= \sigma = 18$

Sample size, $n = 400$

$Z_{\alpha/2}$ for 95% confidence = 1.96 (from tables)

Sample mean $= \bar{x} = 82$

$$\text{Maximum error, } E = Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} = \frac{1.96 \times 18}{20} = 1.764$$

The limits for the confidence are

$$\bar{x} - Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

$\therefore$ Confidence limits are 80.236 and 83.764

Example 24 : Measurements of the weights of a random sample of 200 ball bearing made by a certain machine during one week showed a mean of 0.824 and a standard deviation of 0.042. Find maximum error at 95% confidence interval ? Find the confidence limits for the mean if $x = 32$. [JNTU Nov. 2008, (A) Nov. 2011 (Set No. 4)]

Solution : We are given

Mean of the sample = $\bar{x} = 0.824$

$Z_{\alpha/2}$ = Z value for 95% level = 1.96 (From Tables)

σ = Standard deviation = 0.042

n = size of the sample = 200

$$\text{Maximum error, } E = Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} = \frac{1.96 \times 0.042}{\sqrt{200}} = 0.0059$$

$$\text{Now } \bar{x} - Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} = 0.824 - 0.0059 = 0.8181$$

$$\text{and } \bar{x} + Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} = 0.824 + 0.0059 = 0.8299$$

Hence the limits for the confidence are

$$\bar{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

$\therefore$ Confidence limits are 0.8181 and 0.8299

Example 16 : A sample of size 300 was taken whose variance is 225 and mean 54. Construct 95% confidence interval for the mean. [JNTU (H) Nov. 2010 (Set No. 1)]

Solution : Since the sample size 300 is large (> 30), normal distribution is used as the sampling distribution.

Here $n = 300$, $\bar{x}$ = sample mean = 54, $\sigma = \sqrt{225} = 15$

$$\therefore \text{S.E. of } \bar{x} = \frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{300}} = 0.866$$

95% confidence limits for the population mean are

$$\bar{x} \pm 1.96 (\text{S.E. of } \bar{x}) = 54 \pm 1.96 (0.866) = 54 \pm 1.697 = 55.697 \text{ and } 52.3$$

$\therefore$ The required confidence interval is (52.3, 55.7).

Example 24 : In a random sample of 100 packages shipped by air freight 13 had some damage. Construct 95% confidence interval for the true proportion of damage package.

$$\text{Solution : Here } p = \text{Sample proportion of damage packages} = \frac{13}{100} = 0.13$$

$$\therefore q = 1 - p = 1 - 0.13 = 0.87$$

$$\begin{aligned}\text{S.E. of } p &= \sqrt{\frac{PQ}{n}} = \sqrt{\frac{pq}{n}} \quad (\because P \text{ is not known, we take } p \text{ for } P) \\ &= \sqrt{\frac{0.13 \times 0.87}{100}} = 0.034\end{aligned}$$

$\therefore$ 95% confidence limits for the population proportion of P of damage packages are $p \pm 1.96 (\text{S.E. of } p) = 0.13 \pm 1.96 (0.034) = 0.13 \pm 0.067$

i.e., the confidence limits are 0.063 and 0.197

Hence the 95% confidence interval for the true proportion of damage packages is (0.063, 0.197).

Example 11 : It is desired to estimate the mean number of hours of continuous use until a certain computer will first require repairs. If it can be assumed that $\sigma = 48$ hours. How large the sample will be needed so that one will be able to assert with 90% confidence that the sample mean is off by at most 10 hours. [JNTU(H) Nov. 2010 (Set No. 3)]

Solution : Sampling error, $E = 10$ hours, $\sigma = 48$ hours, $z_{\alpha/2} = 1.645$

$$\text{We have } E = \frac{z\sigma}{\sqrt{n}} \text{ or } \sqrt{n} = \frac{z\sigma}{E}$$

$$\therefore n = \left(\frac{z_{\alpha/2} \sigma}{E} \right)^2 = \left(\frac{1.645 \times 48}{10} \right)^2 = (7.896)^2 = 62.35$$

Hence, sample size, $n = 62$

Example 10 : Among 100 fish caught in a large lake, 18 were inedible due to the pollution of the environment. With what confidence can we assert that the error of this estimate is at most .065? [JNTU(H) Dec. 2011 (Set No. 3)]

Solution : We are given

$n = \text{Sample size} = 100$

and Max. error of estimate, $E = 0.065$

$$\text{Here } p = \text{sample proportion of inedible fish} = \frac{18}{100} = 0.18$$

$$\therefore q = 1 - p = 1 - 0.18 = 0.82$$

$$\text{Maximum error of estimate for true proportion} = E = z_{\alpha/2} \cdot \sqrt{\frac{pq}{n}}$$

[$\because$ P is not known, we take p for P]

$$\Rightarrow 0.065 = z_{\alpha/2} \cdot \sqrt{\frac{0.18 \times 0.82}{100}} = z_{\alpha/2} (0.038)$$

$$\therefore z_{\alpha/2} = \frac{0.065}{0.038} = 1.71$$

When $z = 1.71$, the probability = 0.4564 (from normal tables)

$$P(z > z_{\alpha}) = 0.4564 \Rightarrow 2 \times P(z > z_{\alpha}) = 2(0.4564) = 0.9128 = 91.28\% = 91\%$$

Hence with 91% confidence we can assert that the error of this estimate is at most 0.065.

Example 11 : A random sample of 500 points on a heated plate resulted in an average temperature of 73.54 degrees Fahrenheit with a standard deviation of 2.79 degree Fahrenheit. Find a 99% confidence interval for the average temperature of the plate.

[JNTU(H) Dec. 2011 (Set No. 4)]

Solution : We are given

$$n = 500, \bar{x} = 73.54 \text{ and } \sigma = 2.79$$

We have $z_{\alpha/2} = 2.58$ (for 99%)

We know that 99% confidence interval is $\left(\bar{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \right)$

$$\text{Now } z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} = 2.58 \times \frac{2.79}{\sqrt{500}} = \frac{7.1982}{\sqrt{500}} = 0.32$$

$$\therefore 99\% \text{ confidence interval} = (73.54 - 0.32, 73.54 + 0.32) \\ = (73.22, 73.86)$$

Example 10 : In a study of an automobile insurance a random sample of 80 body repair costs had a mean of Rs.472.36 and a standard deviation of Rs.62.35. If $\bar{x}$ is used as point estimate to the true average repair costs, with what confidence we can assert that the maximum error does not exceed Rs.10?

[JNTU(H) Apr. 2012 (Set No. 4)]

Solution : We are given

$$n = \text{sample size} = 80, \sigma = \text{S. D.} = 62.35$$

Maximum error, $E = 10$

$$\text{We have } E = z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \Rightarrow 10 = z_{\alpha/2} \cdot \frac{62.35}{\sqrt{80}}$$

$$\therefore z_{\alpha/2} = \frac{10\sqrt{80}}{62.35} = 1.43$$

From normal table, the area corresponding to $z_{\alpha/2} = 1.43$ is 0.4236.

Then the area between $z_{-\alpha/2}$ to $z_{\alpha/2}$ is $2(0.4236) = 0.8472$

Thus we can ascertain with 84.7% confidence.

Example 11 : The mean mark in mathematics in common entrance that will vary from year to year. If this variation of the mean mark is expressed subjectively by a normal distribution with mean $\mu_0 = 72$ and variance $\sigma_0^2 = 5.76$.

- (i) What probability can we assign to the actual mean mark being somewhere between 71.8 and 73.4 for the next years test?
- (ii) Construct a 95% Bayesian interval for μ if the test is conducted for a random sample of 100 students from the next incoming class yielding a mean mark of 70 with s.d. of 8.
- (iii) What posterior probability should we assign to the event of part (i)

[JNTU(K) May 2010 (Set No. 1)]

Solution : (a) Here $\mu_0 = \text{mean} = 72$ and $\sigma_0 = \text{S.D.} = \sqrt{5.76} = 2.4, n = 100$

$$\text{We have } z = \frac{x - 72}{2.4}$$

$$\text{When } x = 71.8, z_1 = \frac{71.8 - 72}{2.4} = -0.0833$$

$$\text{When } x = 73.4, z_2 = \frac{73.4 - 72}{2.4} = 0.5833$$

Let X be the mean mark obtained in the common entrance test. Then

$$\begin{aligned} \text{Prior probability} &= P(71.8 < X < 73.4) \\ &= P(-0.083 < z < 0.583) \\ &= 0.0319 + 0.2190 = 0.2509 \end{aligned}$$

(b) Here $n = 100, \bar{x} = 70, \sigma_0 = 2.4, \mu_0 = 72, \sigma = 8$

$$\begin{aligned} \therefore \text{Posterior mean, } \mu_1 &= \frac{n\bar{x}\sigma_0^2 + \mu_0\sigma^2}{n\sigma_0^2 + \sigma^2} \\ &= \frac{100(70)(2.4)^2 + 72(8)^2}{100(2.4)^2 + (8)^2} = \frac{44928}{640} = 70.2 \end{aligned}$$

$$\begin{aligned} \text{Posterior standard deviation, } \sigma_1 &= \sqrt{\frac{\sigma^2\sigma_0^2}{n\sigma_0^2 + \sigma^2}} \\ &= \sqrt{\frac{(8)^2(2.4)^2}{100(2.4)^2 + (8)^2}} = \sqrt{\frac{368.64}{640}} \\ &= \sqrt{0.576} = 0.7589 \approx 0.76 \end{aligned}$$

$\therefore$ 95% Bayesian interval for μ is given by

$$\mu_1 \pm z_{\alpha/2}(\sigma_1) = 70.2 \pm 1.96(0.7589) = 70.2 \pm 1.4874$$

i.e. $68.71 < \mu < 71.69$

(c) Here $\mu_1 = 70.2, \sigma_1 = 0.76$

When $x = 71.8, z_1 = \frac{71.8 - 70.2}{0.76} = 2.105$

When $x = 73.4, z_2 = \frac{73.4 - 70.2}{0.76} = 4.2105 \approx 4.21$

$\therefore$ Posterior probability $= P(71.8 < X < 73.4) = P(2.105 < z < 4.21)$
 $= 0.5 - 0.4821 = 0.0179$

Example 12 : If sample size $n = 144$, standard deviation $\sigma = 4$ and the mean $= 150$, then 95% confidence interval for μ is _____ [JNTU(A) Dec. 2014]

Solution : We are given $n = 144$, $\sigma = 4$, $\bar{x} = 150$ and 95% confidence level $\Rightarrow z_{\alpha/2} = 1.96$

$$\begin{aligned}\therefore \text{Confidence interval} &= \left(\bar{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \right) \\ &= \left(150 - 1.96 \left(\frac{4}{\sqrt{144}} \right), 150 + 1.96 \left(\frac{4}{\sqrt{144}} \right) \right) \\ &= \left(150 - \frac{1.96}{3}, 150 + \frac{1.96}{3} \right) = (150 - 0.65, 150 + 0.65) \\ &= (149.35, 150.65)\end{aligned}$$

REVIEW QUESTIONS

1. Define estimate, estimator and estimation. [JNTU (A) Nov. 2011 (Set No. 1), May 2012]
2. Define unbiased estimator. What is the more efficient unbiased estimator, explain briefly? [JNTU (A) Nov. 2011 (Set No. 2)]
3. In how many ways the estimation can be done and what are they? Explain in detail [JNTU (A) Nov. 2011 (Set No. 1), May 2012]
4. Explain the ways in which an estimation can be done. Also explain the Bayesian estimation [JNTU (A) Nov. 2010 (Set No. 1)]
5. Give the difference between the interval estimation and Bayesian estimation [JNTU (A) Nov. 2011]
6. What is the "interval estimation"? Give the relations used to find the confidence interval of large and small samples. [JNTU (A) Nov. 2011 (Set No. 3)]
7. Write a short note on interval estimation and Bayesian estimation. [JNTU (A) Nov. 2011 (Set No. 4)]
8. Show that $\bar{x}$ is an unbiased estimator of the population mean μ [JNTU (A) Nov. 2011 (Set No. 2)]
9. Explain briefly Point and Interval estimations.

10. Define the problem of estimation. Show that if $\hat{\theta}$ is an unbiased estimator of θ then show that $\hat{\theta}^2$ is a biased estimator of θ^2 .
11. List the properties of estimators with the examples.

EXERCISE

1. To estimate the average amount of time visitors take to move from one building to another in an office complex, the mean of a random sample of size n is used. Given $\sigma = 1.40$ minutes, determine how large should be the sample size if it is ascertained with 99% confidence that the error E is at most 0.25.
2. Find the degree of confidence to assert that the average salary of school teachers is between Rs. 272 and Rs. 302 if a random sample of 100 such teachers revealed a mean salary of Rs. 287 with S.D of Rs. 48.
3. Using the mean of a random sample of size 150 to estimate the mean mechanical aptitude of mechanics of a large workshop and assuming $\sigma = 6.2$, what can we assert with 0.99 probability about the maximum size of the error.
4. The pulse rate of 50 yoga practitioners decreased on the average by 20.2 beats/minute with S.D of 3.5.
 - (a) If $\bar{x} = 20.2$ is used as a point estimate of the true average decrease in pulse rate. What can we assert with 95% confidence about the maximum error E .
 - (b) Construct 99% confidence intervals for the true average decrease in pulse rate.
5. Determine 99% confidence interval for the mean of contents of soft drink bottles if contents of 7 such soft drink bottles are 10.2, 10.4, 9.8, 10.0, 9.8, 10.2, 9.6 ml.

[JNTU (K) May 2010S (Set No. 4)]
6. A random sample of 20 fuses subjected to overload has mean time for blow of 10.63 minutes with S.d of 2.48 mt. What can we assert with 95% confidence about the maximum error if we use $\bar{x} = 10.63$ mts as a point estimate of true average it takes such fuses for blow when subjected to overload.
7. Construct a 99% confidence interval for the true mean weight loss if 16 persons on diet control after one month had a mean weight loss of 3.42 kgs with S.D of 0.68 kgs.
8. Calculate μ_1, σ_1 , for the posterior distribution if the random sample size is 80. $\bar{x} = 18.85, S = 5.55$ using S for S.D of population σ .
9. The mean mark in mathematics in common entrance that will vary from year to year. If this variation of the mean mark is expressed subjectively by a normal distribution with mean $\mu_0 = 72$ and variance $\sigma_0^2 = 5.76$.
 - (a) What probability can we assign to the actual mean mark being some where between 71.8 and 73.4 for the next year's test ?

- (b) Construct a 95% Bayesian interval for μ if the test is conducted for a random sample of 100 students from the next incoming class yielding a mean mark of 70 with S.d of 8.
- (c) What posterior probability should we assign to the event of part (a).
- [JNTU (K) May 2010S, (A) Apr. 2012 (Set No. 3)]**
- 10.** A random sample of 200 measurements from a large population gave a mean value of 50 and S. D. of 9. Determine the 95% confidence interval for the mean of the population.
- [JNTU (K) Nov. 2011 (Set No. 2)]**
- 11.** Determine a 99% confidence interval for the mean of a normal distribution with variance $\sigma^2 = 4$, using a sample of $n = 200$ values with mean $\bar{x} = 10$.
- [Hint : Confidence interval = $\bar{x} \pm 2.58 \cdot \frac{\sigma}{\sqrt{n}}$]
- 12.** Determine a 95% confidence interval for the mean of a normal distribution with variance $\sigma^2 = 0.25$, using a sample of $n = 100$ values with mean $\bar{x} = 212.3$.

ANSWERS

- | | |
|----------------------------|--|
| 1. $n = 207.98 \simeq 208$ | 2. 99.82% confidence |
| 3. $E = 1.30$ | 4. (a) $E = 0.97015$ (b) $19.229 < \mu < 21.17015$ |
| 5. $t = 3.707$ with 6 d.f | 6. $E = 1.16$ mt |
| 7. $2.92 < \mu < 3.92$ | 8. $\mu_1 = 18.77, \sigma_1 = 0.60$ |
| 9. (a) 0.2509 | (b) $68.71 < \sigma < 71.69$ (c) 0.0179 |

QUIZ

1. If $n = 40, \sigma = 2.06$ then the maximum error with 99% confidence is []
 (a) 0.7377 (b) 0.8387
 (c) 0.6387 (d) 0.536
2. If $n = 100, \sigma = 5$ then the maximum error with 95% confidence is []
 (a) 0.98 (b) 1.2875
 (c) 3.92 (d) 1.16
3. If the size of the sample is 25 and the maximum error with 95% confidence is 0.1 then the standard deviation of the sample is []
 (a) 2.55 (b) 2.12
 (c) 0.255 (d) 0.025
4. If the maximum error with probability 0.95 is 1.2 and the standard deviation of population is 10, then sample size is []
 (a) 26 (b) 266
 (c) 267 (d) 269

5. A sample of size n is taken from a population whose variance is 9. The maximum error of estimate for μ with 95% confidence is 0.5. Then $n =$ []
(a) 12 (b) 68
(c) 128 (d) 139
6. If the maximum error with 99% confidence is 0.25 and the sample size is 400 then $\sigma =$ []
(a) 0.93 (b) 1.93
(c) 2.93 (d) 2.55
7. A sample of size 64 is taken from a population whose variance is 2 with probability 0.99. Then the maximum error is []
(a) 0.456 (b) 0.35
(c) 0.24 (d) 0.58
8. If the maximum error with 99% confidence is 0.86 and size of the sample is 144, then the variance of the population is []
(a) 2 (b) 4
(c) 8 (d) 16
9. If $n=144, \sigma=4$ and $\bar{x}=150$ then 95% confidence interval for mean is []
(a) (149.35,150.65) (b) (149.14,150.86)
(c) (149.15,150.85) (d) None
10. If $n=81, \sigma=4.5$ and $\bar{x}=32$ then 99% confidence interval for mean is []
(a) (30.71,33.29) (b) (30.835,33.165)
(c) (31.02,32.98) (d) None
11. A random sample of size 169 was taken from a population whose variance is 25 and mean is 50. Then 99% confidence interval is []
(a) (49,51) (b) (49.25,50.75)
(c) (48,50) (d) None
12. A random sample of size $n=100$ is taken from a population with $\sigma=5.1$. Given that the sample mean $\bar{x}=21.6$. Then the 95% confidence interval for the population mean μ is []
(a) (20.6,22.6) (b) (21.6,22.6)
(c) (20.28,22.9) (d) None
13. If a random sample of size $n=20$ from a normal population with variance $\sigma^2=144$ has the mean $\bar{x}=34.3$, then the 95% confidence interval for the population mean μ is []
(a) (29,39.56) (b) (29.8,39.5)
(c) (28.64,38.34) (d) None
14. In a campus interview 64 students are attended for a test and taken an average of 28.5 minutes to complete the answers with a variance of 9.3 square minutes.

- The 99% confidence interval for the true average time it takes a freshman to complete the test is []
- (a) (27.2,29.4) (b) (27.5,29.5)
(c) (27.8,29.2) (d) None
15. The length of the time required to execute 100 programs has a mean of 5.68 sec and a standard deviation of 0.29 sec. Assuming that such measurements are normally distributed, the 99% confidence interval for the mean time length of execution is []
- (a) (4.98,5.12) (b) (4.95,5.05)
(c) (5.6,5.75) (d) None
16. The mean blood pressure of 25 patients in a certain hospital is 140 mm of Hg. If those data can be looked upon as a random sample from a normal population with standard deviation $\sigma = 10$ mm of Hg, then 95% confidence interval of the population mean μ is []
- (a) (22.8,25.24) (b) (23.1,26.89)
(c) (23.34,26.66) (d) None
17. In a random sample of 60 sections of a pipe in a chemical plant, 8 showed signs of serious corrosion. The maximum error with 95% confidence for the true proportion of pipe sections is []
- (a) 0.85 (b) 0.085
(c) 0.082 (d) None
18. A random sample of 500 apples were taken and out of which 60 were found to be bad. The maximum error with 95% confidence is []
- (a) 0.016 (b) 0.028
(c) 0.037 (d) 0.045
19. In a sample of 100 ceramic pistons made for an experimental designed engine 18 were found to be cracked. The maximum error with 95% confidence is []
- (a) 0.075 (b) 0.098
(c) 0.089 (d) None
20. In a sample of 500 people in Maharashtra 300 are wheat eaters. The maximum error with 99% confidence is []
- (a) 0.056 (b) 0.042
(c) 0.067 (d) None
21. If we can assert with 95% that the maximum error is 0.5 and $P = 0.2$, then the sample size is []
- (a) 122 (b) 244
(c) 256 (d) 269

ANSWERS

- | | | | |
|---------|---------|---------|---------|
| 1. b | 2. a | 3. c | 4. c |
| 5. d | 6. b | 7. a | 8. d |
| 9. a | 10. a | 11. a | 12. a |
| 13. a | 14. b | 15. c | 16. c |
| 17. b | 18. b | 19. a | 20. a |
| 21. b | | | |

CHAPTER - 9

TESTING OF HYPOTHESIS (LARGE SAMPLE TESTS)

9.1 INTRODUCTION

The outcome of a statistical experiment may be recorded either as a numerical value or as a descriptive representation.

When a pair of dice are tossed and the sum of the numbers on the faces is the outcome of interest, we record a numerical value. However, if the students of a certain school are given blood tests and the type of blood is of interest, then a descriptive representation might be most useful. A person's blood can be classified in 8 ways. It must be AB , A , B or O , with a plus or minus sign, depending on the presence or absence of the Rh antigen.

The statistician is primarily concerned with the analysis of numerical data.

For the classification of blood types, it may be convenient to use numbers from 1 to 8 to represent the blood types and then record the appropriate number for each student.

In any particular study, the number of possible observations may be small, large but finite, or infinite. For example, in the classification of blood types we can only have as many observations as there are students in the school. The project, therefore results in a finite number of observations.

On the other hand, if we could toss a pair of dice indefinitely and record the sums that occur, we would obtain an infinite set of values, each value representing the result of a single toss of a pair of dice.

In this chapter we focus on sampling from distributions or populations and study such important quantities as the **sample mean** and **sample variance**.

9.2 POPULATION AND SAMPLE

[JNTU (A) Dec. 2018]

We begin this section by discussing the notions of population and sample.

The totality of observations with which we are concerned, whether this number be finite or infinite, constitutes what we call *population*. There was a time when the word *population* referred to observations obtained from statistical studies about people. Today, the statistician uses the term to refer to groups of people, animals, or all possible outcomes from some complicated biological or engineering system or numerical data. Thus the population does not imply living beings alone, e.g., we speak of the population of births, heights, weights, prices of vegetables and so on.

Definition : Population (or universe) is the aggregate or totality of statistical data forming a subject of investigation. For example,

- (i) the population of the heights of Indians,
- (ii) the population of Nationalised Banks in India, etc.

Definition : The number of observations in the population is defined to be the *size of the population*. It may be *finite* or *infinite*. *Size of population* is denoted by N .

Examples : If there are 600 students in the school that we classified according to blood type, we say that we have a population of size 600. The numbers on the cards in a deck, the heights of residents in a certain city, and the lengths of fish in a particular lake are all examples of population with *finite* size.

The observations obtained by measuring the atmospheric pressure every day from the past on into the future, or all measurements on the depth of a lake from any conceivable position, are examples of population whose sizes are infinite.

In the field of statistical inference the statistician is interested in arriving at conclusions concerning a population when it is impossible or impractical to observe the entire set of observations that make up the population.

For example, in attempting to determine the average length of life of a certain brand of light bulb, it would be impossible to test all such bulbs if we are to have any left to sell.

Therefore, we must depend on a subset of observations from the population to help us make inferences concerning the same population. This brings us to consider the notion of sampling.

Sampling : Most of the times, study of entire population may not be possible to carry out and hence a part alone is selected from the given population. A portion of the population which is examined with a view to determining the population characteristics is called a sample. *i.e.*, a sample is a subset of population and the number of objects in the sample is called the size of the sample. Size of the sample is denoted by n .

The process of selection of a sample is called sampling. It is quite often used in our day-to-day practical life. For example, (i) to assess the quality of a bag of rice, sugar, wheat or any other commodity, we examine only a portion of it by taking a handful of it from the bag and then decide to purchase it or not. The portion selected from the bag is called a sample, while the whole quantity of rice, sugar or wheat in the bag is the population, (ii) to estimate the proportion of defective articles in a large consignment, only a portion (*i.e.*, a few of them) is selected and examined. The portion selected is a sample (iii) cars produced in India is the population and the Nano cars is the sample.

When information is collected in respect of every individual item, the enquiry is said to be done by *Complete Enumeration* or *Census*. For example, during the *Census of Population* (which is done every ten years in India), information in respect of each individual person residing in India is collected. This method gives information for each and every unit of the population with greater accuracy. But this method involves multiplicity of causes viz., administrative and financial implications, time factor, etc. In most cases of statistical inquiry, because of limitations of time and cost, only a portion (*i.e.*, a sample) of the available source of information is examined, and the data collected from them. This process of partial enumeration is known as *Sample Survey*. The results are then generalised and made applicable to the whole field of inquiry. This is known as *Sampling*.

Thus in estimating the characteristics of the population, instead of enumerating entire population, only the individuals in the sample are examined. Then the sample characteristics are utilised to approximately estimate the population. The error involved in such approximation is known as *sampling error* and is inherent and unavoidable in any and

every sampling scheme. However the advantage of sampling process lies in considerable gains in time, cost and handling of large data.

The population is thus a universal set for the sample. The statistical constants like mean, standard deviation, correlation coefficient, etc. obtained for the population are called parameters.

Similarly, constants for the sample drawn from the given population *i.e.*, mean ($\bar{x}$), standard deviation (S) etc., are called the statistic.

If our inferences from the sample to the population are to be valid, we must obtain samples that are representative of the population. All too often, we are tempted to choose a sample by selecting the most convenient members of the population. Such a procedure may lead to erroneous inferences concerning the population. Any sampling procedure that produces inferences that consistently over estimate (or) consistently under estimate some characteristics of the population is said to be biased. To eliminate any possibility of bias in the sampling procedure it is desirable to choose a random sample in the sense that the observations are made independently and at random.

9.3 DIFFERENT METHODS OF SAMPLING

Some important methods of sampling are discussed below.

I. Probability Sampling Methods

1. Random Sampling (or Probability Sampling)

It is the process of drawing a sample from a population in such a way that each member of the population has an equal chance of being included in the sample. The sample obtained by the process of random sampling is called a random sample.

For example : (i) A handy of cards dealt from a well - shuffled pack of cards is a random sample.

(ii) Selecting randomly 20 words from a dictionary is a random sample.

(iii) Choosing 10 patients from a hospital in order to test the efficacy of a certain newly - invented drug.

If each element of a population may be selected more than once then it is called **sampling with replacement** whereas if the element cannot be selected more than once, it is called **sampling without replacement**.

Note : If N is the size of a population and n is the sample size, then

(i) The number of samples with replacement $= N^n$

(ii) The number of samples without replacement $= {}^N C_n$

2. Stratified Sampling (or Stratified Random Sampling)

This method is useful when the population is heterogeneous. In this type of sampling, the population is first sub-divided into several parts (or small groups) called *strata* according to some relevant characteristics so that each stratum is more or less homogeneous. Each stratum is called a sub-population. Then a small sample (called sub-sample) is selected from each stratum at random. All the sub-samples are combined together to form the stratified sample which represents the population properly. The process of obtaining and examining a stratified sample with a view to estimating the characteristic of the population is known as *Stratified Sampling*.

For example, let us select a stratified sample of 500 families from a city having 50,000 families, with a view to studying their economic condition. For this purpose, the city area is divided into a number of strata, according to economic condition of their inhabitants, as measured by annual income (say). Thus, localities mostly inhabited by people with more or less similar annual income may be included under one stratum. A few families are then chosen at random from each so that the sum total of all the families from all the strata is 500.

3. Systematic Sampling (or Quasi - Random Sampling)

As the name suggests this means forming the sample in some systematic manner by taking items at regular intervals. In this method, all the units of the population are arranged in some order. If the population size is finite, all the units of the population are arranged in some order. Then from the first k items, one unit is selected at random. This unit and every k th unit of the serially listed population combined together constitute a systematic sample. This type of sampling is known as *Systematic Sampling*.

The difference between random sampling and systematic sampling lies in the fact that in the case of a random sample all the members have to be chosen randomly, whereas in the case of a systematic sample only the first member has to be chosen at random.

II. Non-Probability Sampling Methods.

4. Purposive Sampling (or Judgement Sampling)

When the choice of the individual items of a sample entirely depends on the individual judgement of the investigator (or sampler), it is called a *Purposive or Judgement Sampling*.

In this method, the members constituting the sample are chosen not according to some definite scientific procedure, but according to convenience and personal choice of the individual, who selects the sample. Two or more such independent purposive samples may give widely different estimates of the same population. In this type, the investigator must have a good deal of experience and a thorough knowledge of the population. Purposive selection is always subject to some kind of bias. This method is suitable when the sample is small.

For example, if a sample of 20 students is to be selected from a class of 100 to analyse the extra-curricular activities of the students, the investigator would select 20 students who, in his judgement, would represent the class.

5. Sequential Sampling

It consists of a sequence of sample drawn one after another from the population depending on the results of previous samples. If the result of the first sample leads to a decision which is not acceptable, the lot from which the sample was drawn is rejected. But if the result of the first sample is acceptable, no new sample is drawn. But if the first sample leads to no clear decision, a second sample is drawn and, as before, if required, a third sample is drawn to arrive at a final decision to accept or reject the lot. It is widely used in Statistical Quality Control in factories engaged in mass production and other areas.

9.4 CLASSIFICATION OF SAMPLES

[JNTU (A) June 2018]

Samples are classified in two ways.

1. Large sample : If the size of the sample (n) ≥ 30 , the sample is said to be large sample.

2. Small sample : If the size of the sample (n) < 30 , the sample is said to be small sample (or) exact sample.

In sampling *with replacement*, each member of the population may be chosen more than once, since the member is replaced in the population.

Thus sampling from finite population *with replacement* can be considered theoretically as sampling from *infinite population*. Whereas, in sampling *without replacement*, an element of the population cannot be chosen more than once, as it is not replaced. Therefore the sampling from finite population *without replacement* can be considered theoretically as sampling from finite population only.

Definition : Let $X_1, X_2, \dots, X_n$ be n independent random variables each having the same probability distribution $f(x)$. We then define $X_1, X_2, \dots, X_n$ to be a random sample of size n from the population $f(x)$ and write its joint probability distribution as

$$f(x_1, x_2, \dots, x_n) = f(x_1), f(x_2), \dots, f(x_n)$$

Our main purpose in selecting random sample is to elicit information about the unknown population parameters.

Statistic : It is a real valued function of the random sample. So statistic is a function of one or more random variables not involving any unknown parameter. Therefore statistic is a random variable and has a probability or frequency distribution.

9.5 PARAMETERS AND STATISTICS

Parameter is a statistical measure based on all the units (or observations) of a population. Statistic (or sample statistic) is a statistical measure based only on all the units selected in a sample. For example, the statistical constants of the population namely mean μ , variance σ^2 are usually referred to as *parameters*, statistical measures computed from the sample observations alone e.g., sample mean ($\bar{x}$), sample variance (s^2) etc., are usually referred to as *statistics*.

In other words, the mean, median, mode, standard deviation, variance measures of the population are called **parameters** and the measures obtained from the sample of the population are called **statistics**.

Thus *parameters* refer to population while *statistics* refer to sample.

Note : As the units selected in two or more samples drawn from a population are not the same, the value of a statistic varies from sample to sample, but the parameter always remains constant (since all the units in a population remain the same). This variation in the value of a statistic is called *sampling fluctuation*. A parameter has no sampling fluctuation. Usually, statistic is used to estimate the value of an unknown parameter obtained from the sample, is a function of the sample values only.

1. The Sample Mean :

Definition : If $X_1, X_2, \dots, X_n$ represent a random sample of size n , then the sample mean is defined by the statistic

$$\bar{X} = \sum_{i=1}^n \frac{X_i}{n}$$

Note that the statistic $\bar{x}$ has the value $\bar{x} = \sum_{i=1}^n \frac{x_i}{n}$ when x assumes the values $x_1, x_2, \dots$. For instance, the term sample mean is applied to both the statistic $\bar{X}$ and its computed value $\bar{x}$.

2. The Sample Variance :

Definition : If $X_1, X_2, \dots, X_n$ represent a random of size n , then the sample variance

is defined by the statistic $s^2 = \sum_{i=1}^n \frac{(X_i - \bar{X})^2}{n-1} = \frac{n \sum X_i^2 - (\sum X_i)^2}{n(n-1)}$

(measure of variability of data about the mean)

Note that s^2 is essentially defined to be the average of the squares of the deviations of the observations from their mean. With the change that the sum of the squared deviations is divided by $n-1$ and not n .

Ex 1 : If a sample size 5 results in the sample values of 8, 5, 9, 6 and 2, find the sample mean. [JNTU (A) June 2019]

Solution : We know that mean $= \bar{x} = \frac{\sum_{i=1}^n x_i}{n}$. Here $n = 5$.

$$\therefore \text{Sample mean} = \frac{x_1 + x_2 + x_3 + x_4 + x_5}{5} = \frac{8 + 5 + 9 + 6 + 2}{5} = \frac{30}{5} = 6$$

Ex 2 : A comparison of coffee prices at 4 randomly selected grocery stores in a city showed increases from the previous month of 12, 15, 17 and 20 rupees for 1 kg. Find the variance of this sample of price increases.

Solution : Calculating the sample mean, we get

$$\bar{x} = \frac{12 + 15 + 17 + 20}{4} = \text{Rs. } 16/-$$

$$\begin{aligned} \therefore s^2 &= \sum_{i=1}^4 \frac{(x_i - 16)^2}{4-1} \\ &= \frac{(12-16)^2 + (15-16)^2 + (17-16)^2 + (20-16)^2}{3} \\ &= \frac{(-4)^2 + (-1)^2 + (1)^2 + (4)^2}{3} = \frac{34}{3} \end{aligned}$$

3. The Sample Standard Deviation :

The sample S. D., denoted by s , is the positive square root of the sample variance.

Note : In the above example 2 S.D., $s = \sqrt{34/3}$.

9.6 SAMPLING DISTRIBUTION

Sampling theory is the study of relationships between a population and samples drawn from the population, and it is applicable to random samples only. We need to obtain maximum information about the population with the help of samples, i.e., to determine the true value of the population parameters (population mean, population S. D., population proportion, etc.) by using sample statistics like sample mean, sample S. D., sample

proportion, etc., and to find the limits of accuracy of estimates based on samples. It helps us to determine whether the differences between two samples are actually due to chance variation or whether they are really significant. Sampling theory also useful in testing of hypothesis and significance which is important in the theory of decisions.

9.7 SAMPLING DISTRIBUTION OF A STATISTIC

Sampling distribution of a statistic is the frequency distribution which is formed with various values of a statistic computed from different samples of the same size drawn from the same population. We can draw a large number of samples of same size from a population of fixed size, each sample containing different population members. Any statistic (statistical measure of sample) like mean, median, variance, etc. may be computed for each of these samples. As a result a series of various values of that statistic may be obtained. These various values can be arranged into a frequency distribution table, which is known as the sampling distribution of the statistic.

Sampling may be done with replacement or without replacement. Sampling with replacement means that the same unit of the population may be included in each sample more than once. Sampling without replacement means that the same unit of population may not be included in each sample more than once.

Let us consider a finite population of size N and let us draw all possible random samples each of the same size n . Then we get ${}^N C_n = \frac{N!}{n!(N-n)!} = k$ (say) samples.

Compute a statistic t (such as sample mean, S.D., etc) for each of these samples. The value of a statistic t may vary from sample to sample. Let $t_1, t_2, \dots, t_k$ be the values of statistic k for the k samples. Each of these values occur with a definite probability. Thus we can construct a table showing the set of values $t_1, t_2, \dots, t_k$ of t with their respective probabilities. This probability distribution of t is known as the Sampling Distribution of t . Thus sampling distribution describes how a statistic t will vary from one sample to the other of the same size. Although all the k samples are drawn from the given population, the members included in different samples are different.

If N is large, then the number k of all possible samples is also large (*i.e.*, sampling without replacement). In such a case, the values $t_1, t_2, \dots, t_k$ of t may be arranged in the form of a relative frequency distribution and the limiting form of this relative frequency distribution when $k \rightarrow \infty$ is called the sampling distribution of statistic t . A statistic (*i.e.*, sample mean, sample S.D, etc.) has always a sampling distribution, but a parameter (*i.e.* population mean, population S.D.) has no sampling distribution.

If the statistic t is mean, then the corresponding distribution of the statistic is known as sampling distribution of means.

The sample mean can be regarded as a random variable $\bar{x}$ and each sample mean then constitute as the observed value of this new random variable $\bar{x}$. Let these values be: $\bar{x}_1, \bar{x}_2, \bar{x}_3, \dots, \bar{x}_n$. These mean values $\bar{x}_1, \bar{x}_2, \bar{x}_3, \dots, \bar{x}_n$ can be used to form a frequency distribution. Then this frequency distribution of the statistic $\bar{x}$ is known as the sampling

distribution of the sample mean. Similarly the sampling distribution of standard deviation or variance may be constructed with the various values of standard deviation or variance respectively.

The main characteristic of the Sampling distribution of a statistic is that it approaches normal distribution even when the population distribution is not normal provided the sample size is sufficiently large (greater than 30). Another important feature of the sampling distribution of statistic is that the mean and the standard deviation of the sampling distribution of sample mean bear a definite relation to the corresponding parameters i.e., mean and standard deviation of parent population. These characteristics of the sampling distribution help us:

- (i) To estimate the unknown population parameter from the known statistic
- (ii) To set the confidence limits of the parameter within which the parameter values are expected to lie.
- (iii) To test a hypothesis and to draw a statistical inference from it.

9.8 CENTRAL LIMIT THEOREM

If $\bar{x}$ be the mean of a random sample of size n drawn from a population having mean μ and S.D. σ , then the sampling distribution of the sample mean $\bar{x}$ is approximately a normal distribution with mean μ and S.D. = S.E. of $\bar{x} = \frac{\sigma}{\sqrt{n}}$ provided the sample size n is large ($n \geq 30$). This is established by central limit theorem stated below (without proof).

Theorem : If $\bar{x}$ be the mean of a sample size n drawn from a population with mean μ and S. D. σ then the standardized sample mean

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

is a random variable whose distribution function approaches that of the standard normal distribution $N(z; 0, 1)$ as $n \rightarrow \infty$.

9.9 STANDARD ERROR (S. E.) OF A STATISTIC

The standard error of a statistic t (i.e., S. E. of sample mean, or sample S.D.) is the standard deviation of the sampling distribution of the statistic. Thus S.E. of sample mean $\bar{x}$ is the S.D. of the sampling distribution of sample mean. It is used for assessing the difference between the expected value and observed value. It plays an important role in large sample theory and forms the basis in tests of hypothesis or tests of significance. It gives an idea about the reliability and precision of a sample. S. E. enables us to determine the confidence limits within which the parameters are expected to lie. For example, the probable limits for population proportion P are given by $p \pm 3 \cdot \sqrt{pq/n}$.

The standard errors (S.E.) of some of the well-known statistics are given below (without proof), where μ the population mean, σ^2 the population variance, P the population proportion, n is the sample size, $\bar{x}$ the sample mean, s^2 the sample variance and p the sample proportion.

Note : S.E. of a statistic may be reduced by increasing sample size n , but this results in corresponding increase in cost, time and labour, etc.

Formulae for S.E.

1. S.E. of sample mean $\bar{x} = \frac{\sigma}{\sqrt{n}}$. It is written as S. E. $(\bar{x}) = \frac{\sigma}{\sqrt{n}}$

2. S.E. of sample proportion $p = \sqrt{\frac{PQ}{n}}$, where $Q = 1 - P$

3. S.E. of sample S. D $(s) = \frac{\sigma}{\sqrt{2n}}$

4. S.E. of the difference of two sample means $\bar{x}_1$ and $\bar{x}_2$

i.e., S.E. of $(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$ where $\bar{x}_1$ and $\bar{x}_2$ are the means of two random samples of sizes n_1 and n_2 drawn from two populations with S.D. σ_1 and σ_2 respectively.

5. S.E. of $(p_1 - p_2) = \sqrt{\frac{P_1Q_1}{n_1} + \frac{P_2Q_2}{n_2}}$, where p_1 and p_2 are the proportions of two random samples of sizes n_1 and n_2 drawn from two populations with proportions P_1 and P_2 respectively.

6. S.E. of $(s_1 - s_2) = \sqrt{\frac{\sigma_1^2}{2n_1} + \frac{\sigma_2^2}{2n_2}}$

For a finite population of size N , when a sample is drawn without replacement, we have

(i) S.E. of sample mean $= \frac{\sigma}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}}$

(ii) S.E. of sample proportion $= \sqrt{\frac{PQ}{n}} \cdot \sqrt{\frac{N-n}{N-1}}$

For an infinite population when the sample is drawn without replacement, Formulae 1 and 2 remain the same.

Proposition (Statement without proof) : The mean and S. E. of sample mean $\bar{x}$ (*i.e.*, mean and S.D. of the sampling distribution of $\bar{x}$) when samples of the same size n are drawn from a population having mean μ and S.D. σ are given by Mean of $\bar{x} = E(\bar{x}) = \mu$

and S.E. of $\bar{x} = \frac{\sigma}{\sqrt{n}}$.

9.10 SAMPLING DISTRIBUTION OF MEAN (σ KNOWN)

The probability distribution of $\bar{X}$ is called the sampling distribution of means.

The sampling distribution of a statistic depends on the size of the population, the size of the samples, and the method of choosing the samples.

Let $X_1, X_2, X_3, \dots, X_n$ be the n random samples drawn from a population of size N with mean μ and variance σ^2 and $\bar{X}$ is the mean of samples.

Infinite Population : Suppose the samples are drawn from an infinite population *i.e.*, $N \rightarrow \infty$ (or) sampling is done with replacement, then

(i) The mean of the sampling distribution of means, $\mu_{\bar{X}} = \mu$ *i.e.*, $E(\bar{X}) = \mu$

Proof: We have $\bar{X} = \frac{X_1 + X_2 + X_3 + \dots + X_n}{n}$

Since $\bar{X}$ is a linear combination of the random variables $X_1, X_2, X_3, \dots, X_n$, therefore it is also a random variable.

$$\begin{aligned} \therefore \text{Mean of } \bar{X} &= E(\bar{X}) = \frac{1}{n} E(X_1 + X_2 + X_3 + \dots + X_n) \\ &= \frac{1}{n} [E(X_1) + E(X_2) + E(X_3) + \dots + E(X_n)] \\ &= \frac{1}{n} [\mu + \mu + \mu + \dots + \mu] = \frac{n\mu}{n} = \mu. \end{aligned}$$

(ii) Variance, $\sigma_{\bar{X}}^2 = \frac{\sigma^2}{n}$ *i.e.*, $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$

$$\therefore \text{S.D of mean, } \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

Proof: If the sample is drawn with replacement, then $X_1, X_2, X_3, \dots, X_n$ are independent random variables. Thus we have $\text{Var}(\bar{X}) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$.

The sampling distribution of $\bar{x}$ will be approximately *normal* with mean μ and variance $\frac{\sigma^2}{n}$ provided that the sample size is large.

$$\text{Standardized sample mean, } z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Finite Population :

Consider a finite population of size N with mean μ and standard deviation σ . Draw all possible samples of size n without replacement, from this population. Then

(i) The mean of the sampling distribution of means (for $N > n$) is $\mu_{\bar{X}} = \mu$.

$$\text{i.e., } E(\bar{X}) = \mu$$

$$(ii) \quad \text{The variance is } \sigma_{\bar{x}}^2 = \frac{\sigma^2}{n} \cdot \left(\frac{N-n}{N-1} \right)$$

$$i.e., \text{Var}(\bar{X}) = \frac{N-n}{N-1} \cdot \frac{\sigma^2}{n} \quad (\text{if the sample is drawn without replacement})$$

$$\text{and S.D is } \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}}$$

Here, the factor $\left(\frac{N-n}{N-1} \right)$, often called the finite population **correction factor**.

We note that this term tends to become closer and closer to unity as population size becomes larger and larger.

Normal Population (Small Sample) :

Sampling distribution of $\bar{X}$ is normally distributed even for small samples of size $n < 30$ provided sampling is from normal population.

Non - Normal Population (Large Sample) :

Consider a population with unknown (non-normal) distribution. Let the population mean μ and population variance σ be both finite. Let the population be finite or infinite. In case the population is finite assume that the population size N is atleast twice the sample size n . Draw all possible samples of size n . Then the sampling distribution of $\bar{X}$ is approximately normally distributed with mean $\mu_{\bar{x}} = \mu$ and variance

$$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n} \quad \text{provided the sample size is large (i.e., } n \geq 30 \text{)}.$$

SOLVED EXAMPLES

Example 1 : If $\bar{x} = 31.45, \mu = 32.3, \sigma = 1.6$ and $n = 40$, find z-test statistic.

Solution : We have $z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \therefore z = \frac{31.45 - 32.3}{1.6 / \sqrt{40}} = -6.32$ [JNTU (A) June 2019]

Example 2 : (i) What is the value of correction factor if $n = 5$ and $N = 200$.

(ii) The size of the population is 2000 and the size of the sample is 200. Find the correction factor in the population. [JNTU (A) June 2019]

Solution : (i) Given $N =$ Size of the finite population = 200

$n =$ Size of the sample = 5

$$\therefore \text{Correction factor} = \frac{N-n}{N-1} = \frac{200-5}{200-1} = \frac{195}{199} = 0.98$$

(ii) This is left as an exercise to the reader.

Example 2 : Find the value of the finite population correction factor for $n = 10$ and $N = 100$. [JNTU 1999, 2000S, (A) Dec. 2009, Nov. 2010, Apr. 2012 (Set No. 3)]

Solution : Given $N =$ Size of the finite population = 1000

$n =$ Size of the sample = 10

$$\therefore \text{Correction factor} = \frac{N-n}{N-1} = \frac{1000-10}{1000-1} = \frac{990}{999} = 0.991$$

Example 4 : How many different samples of size two can be chosen, from a finite population of size 25.

Solution : We can take ${}^N C_n$ samples of size n from the population of size N .

Here $N = 25$, $n = 2$

$\therefore$ We can take ${}^{25} C_2 = 300$ samples of size 2 from finite population of size 25.

9.11 TEST OF HYPOTHESIS

We need to decide whether to accept or reject a statement about the parameter. This statement is called a *hypothesis* and the decision-making procedure about the hypothesis is called *hypothesis testing*. This is one of the most useful aspects of statistical inference, since many types of decision-making problems, tests or experiments in the engineering world can be formulated as hypothesis - testing problems.

The main object of the Sampling Theory is the study of the Tests of Hypothesis or Tests of Significance.

In many circumstances, we are to make decisions about population on the basis of only sample information. For example, on the basis of sample data,

- (i) a drug chemist is to decide whether a new drug is really effective in curing a disease,
- (ii) a quality control manager is to determine whether a process is working properly,
- (iii) a statistician has to decide whether a given coin is biased, etc.

Such decisions are called statistical decisions (or simply decisions)

Definition : Statistical Hypothesis

In many circumstances, to arrive at decisions about the population on the basis of sample information, we make assumptions (or guesses) about the population parameters involved. Such an assumption (or statement) is called a *Statistical Hypothesis* which may or may not be true. The procedure which enables us to decide on the basis of sample results whether a hypothesis is true or not, is called Test of Hypothesis or Test of Significance.

Since we use probability distributions to represent population, a statistical hypothesis may also be thought of as a statement about the probability distribution of a random variable. The hypothesis will usually involve one or more parameters of this distribution.

Example : 1. The majority of men in the city are smokers.

2. The teaching methods in both the schools are effective.

9.12 PROCEDURE FOR TESTING A HYPOTHESIS [JNTU NOV. 2008 (SET NO. 2)]

Test of Hypothesis involves the following steps :

Step 1 : Statement (or assumption) of Hypothesis

There are two types of hypothesis :

- (i) Null Hypothesis
- (ii) Alternative Hypothesis.

(i) **Null Hypothesis :** For applying the tests of significance, we first set up a hypothesis - a definite statement about the population parameter. Such a hypothesis is usually a hypothesis of no-difference, is called Null Hypothesis.

It is in the form $H_0 : \mu = \mu_0$

μ_0 is the value which is assumed or claimed for the population characteristic. It is the reference point against which the Alternative Hypothesis is set up, as explained in the next step.

Definition : A null hypothesis is the hypothesis which asserts that there is no significant difference between the statistic and the population parameter and whatever observed difference is there, is merely due to fluctuations in sampling from the same population. It is always denoted by H_0 . To test whether one procedure is better than another, we assume that there is no difference between the procedures. Similarly to test whether there is a relationship between two variates, we take H_0 that there is no relationship.

For example, in case of a single statistic, H_0 will be that the sample statistic does not differ significantly from the hypothetical parameter value and in the case of two statistics (H_0) will be that the sample statistics do not differ significantly.

(ii) **Alternative Hypothesis :** Any hypothesis which contradicts the Null Hypothesis is called an *Alternative Hypothesis*, usually denoted by H_1 . The two hypothesis H_0 and H_1 are such that if one is true, the other is false and vice versa. For example, if we want to test the null hypothesis that the population has a specified mean μ_0 (say) i.e., $H_0 : \mu = \mu_0$, then the Alternative Hypothesis would be

(i) $H_1 : \mu \neq \mu_0$ (i.e., either $\mu > \mu_0$ or $\mu < \mu_0$)

or (ii) $H_1 : \mu > \mu_0$

or (iii) $H_1 : \mu < \mu_0$

The Alternative Hypothesis (i) is known as a two-tailed alternative and the Alternative Hypothesis in (ii) is known as right-tailed and in (iii) is known as left-tailed.

The setting of alternative hypothesis is very important to decide whether we have to use a single-tailed (right or left) or two-tailed test.

Alternate Hypothesis is in one of the following forms.

$H_1 : \mu \neq \mu_0$

or $H_1 : \mu > \mu_0$

or $H_1 : \mu < \mu_0$

One has to choose from the above three forms depending on the situation posed, as explained below.

In the example relating to the heights of students, the situation involved only testing the statement made about the average height of a class of students. Therefore, the Alternative Hypothesis was of the form $\mu \neq \mu_0$.

In the example about the life of electric bulbs, the company manager wanted to test whether the average life was more than 560 hrs. Therefore the Alternative Hypothesis was of the form $\mu > \mu_0$.

Consider another example, the manufacturer of cigarettes ordered that the nicotine content in a cigarette should not exceed the stipulated level of 25 mgs. In order to check this, he selected a sample of 100 cigarettes from the lot to be packed, and found that the average nicotine content was 23.5 mgs. Could he be reasonably sure that the stipulations laid down by him were being met? In this case, the manufacturer wanted the nicotine content to be less than a particular value, and therefore, the Alternative Hypothesis is of the form $\mu < \mu_0$.

Step 2 : Specification of the Level of Significance

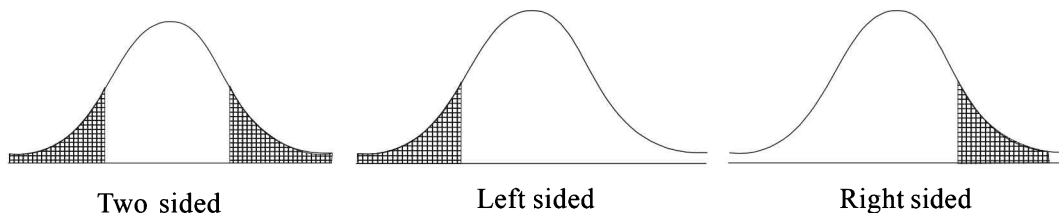
The level of significance denoted by α is the confidence with which we reject or accepts the Null hypothesis H_0 i.e., it is the maximum possible probability with which we are willing to risk an error in rejecting H_0 when it is true. The level of significance is generally specified before a test procedure so that the results obtained may not influence our decision. In practice, we take either 5% (i.e., 0.05) or 1% (i.e., 0.01) level of significance, although other levels such as 2%, 1/2% etc. may also be used. 5% Level of significance in a test procedure indicates that there are about 5 cases in 100 that we would reject the null hypothesis H_0 when it is true i.e., we are about 95% confident that we have made the right decision. Similarly, in 1% Level of significance, there is only 1 case in 100 that the null hypothesis H_0 is rejected when it is true i.e., we are about 99% confident that we have made the right decision. Level of significance is also known as the size of the test.

Step 3 : Identification of the Test Statistic

There are several tests of significance, viz., z, t, F etc. First we have to select the right test depending on the nature of the information given in the problem. Then we construct the test criterion and select the appropriate probability distribution.

Step 4: Critical Region: The critical region is formed based on following factors.

- (a) Distribution of the Statistic i.e., whether the statistic follows the normal, ' t ', χ^2 or ' F ' distribution (will be discussed later).
- (b) **Form of Alternative Hypothesis:** If the form has $\neq$ sign, the critical region is divided equally in the left and right tails, sides of the distribution.



If the form of alternative hypothesis has $<$ sign, the entire critical region is taken in the left tail of the distribution.

If the form of alternative hypothesis has $>$ sign, the entire critical region is taken on the right side of the distribution.

Step 5 : Making Decision

We compute the value of the appropriate statistic and ascertain whether the computed value falls in acceptance or rejection region depending on the specified Level of significance. In finding the acceptance or rejection region we have to use critical values given in Statistical Tables. By comparing the computed value and the critical value decision is taken for accepting or rejecting H_0 . If the computed value $<$ critical value, we accept H_0 , otherwise we reject H_0 .

9.13 ERRORS OF SAMPLING

[JNTU (H) Dec. 2009 (Set No. 2)]

The main objective in sampling theory is to draw valid inferences about the population parameters on the basis of the sample results. In practice we decide to accept or to reject the lot after examining a sample from it. As such we have two types of errors.

(i) **Type I error** : Reject H_0 when it is true. [JNTU (A) Dec. 2009 (Set No. 3)]

If the Null Hypothesis H_0 is true but it rejected by test procedure, then the error made is called Type I error or α error.

(ii) **Type II error** : Accept H_0 when it is wrong *i.e.*, accept H_0 when H_1 is true.

If the Null Hypothesis is false but it is accepted by test, then error committed is called Type II error or β error.

If we write

$$P(\text{Reject } H_0 \text{ when it is true}) = P(\text{Type I error}) = \alpha$$

$$\text{and } P(\text{Accept } H_0 \text{ when it is wrong}) = P(\text{Type II error}) = \beta$$

then α and β are called sizes of Type I and Type II errors respectively

$$\text{i.e., } \alpha = P(\text{Rejecting a good lot})$$

$$\beta = P(\text{Accepting a bad lot})$$

The sizes of Type I and Type II errors are also known as *producer's risk* and *consumer's risk* respectively.

The statistical testing of hypothesis aims at limiting the Type I error to a preassigned value (say : 1% or 5%) and to minimize the type II error. The only way to reduce both types of errors is to increase the sample size, if possible.

1. Critical Region

Under a given hypothesis let the sampling distribution of a statistic t is approximately a normal distribution with mean $E(t)$ and S. D. $= \sigma_t = \text{S. E. of } t$.

$$\text{Then } z = \frac{t - E(t)}{\text{S.E. of } t} = \frac{\text{Observed value} - \text{Expected value}}{\text{S.E. of } t} \text{ is called the standardized}$$

normal variate or test statistic or z -score and its distribution is the standard normal distribution with mean 0 and S.D. 1 when the sample is large.

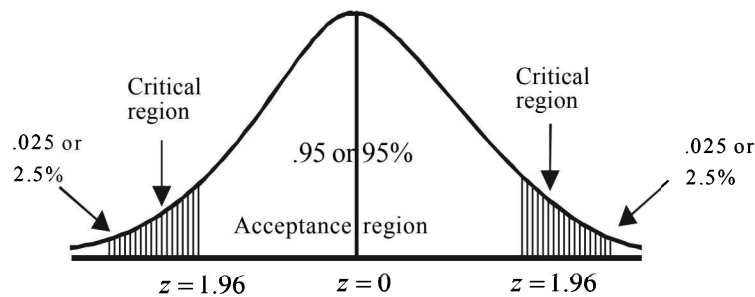


Fig.

From above figure, we see that if z lies between -1.96 and 1.96 then we are 95% confident that the hypothesis is true, since the area under the standard normal curve between $z = -1.96$ and $z = 1.96$ is 0.95 i.e., 95% of the total area i.e. $P(-1.96 \leq z \leq 1.96) = 0.95$. But if for a simple random sample we find that z lies outside the range -1.96 to 1.96 i.e., if $|z| > 1.96$, then we say that such an event occurs with probability of only 0.05 when the given hypothesis is true. In this case, we say that z -score differ significantly from the value expected under the hypothesis and hence the hypothesis H_0 is to be rejected at 5% level of significance. Thus if $|z| > 1.96$, the hypothesis H_0 is rejected at 5% level of significance. The set of z -scores outside the range -1.96 to 1.96 i.e., $|z| > 1.96$ constitutes the *critical region* or *region of rejection* of the hypothesis or the *region of significance*. On the otherhand, the set of z -scores inside the range -1.96 to 1.96 is called the region of acceptance of the hypothesis. The values -1.96 and 1.96 are called *critical values* at 5% level of significance.

Similarly, we can define critical region at any other level of significance.

Definition. A region corresponding to a statistic ' t ', in the sample space S which leads to the rejection of H_0 is called *Critical Region* or *Rejection Region*. Those region which lead to the acceptance of H_0 give us a region called *Acceptance Region*.

In other words, the region in which a sample value falling is rejected is known as the critical region. In general we take two critical regions which cover 5% and 1 % areas of the normal curve.

2. Critical Values or Significant Values :

[JNTU 2004S]

The value of the test statistic, which separates the critical region (or rejection region) and the acceptance region is called the *critical value* or *significant value*. This value is dependent on

- (i) the level of significance used, and
- (ii) the alternative hypothesis, whether it is one-tailed or two-tailed.

For larger samples, corresponding to the statistic t , the variable $z = \frac{t - E(t)}{\text{S.E of } t}$ is normally distributed with mean 0 and variance 1.

The value of z given above under the null hypothesis is known as test statistic. The critical value z_α of the test statistic at level of significance α for a two-tailed test is given by

$$P(|z| > z_\alpha) = \alpha \quad \dots(1)$$

That is, z_α is the value of Z so that the total area of the critical region on both tails is α . Since the normal curve is a symmetrical one, equation (1) implies,

$$P(z > z_\alpha) + P(z < -z_\alpha) = \alpha$$

$$\text{i.e. } 2 P(z > z_\alpha) = \alpha$$

$$\text{or } P(z > z_\alpha) = \alpha/2$$

That is, the area of each tail is $\alpha/2$.

The critical value z_α is that value such that the area to the right of z_α is $\alpha/2$ and the area to the left of $-z_\alpha$ is $\alpha/2$. (Refer figure).

3. Two-tailed test at level of significance ' α ' :

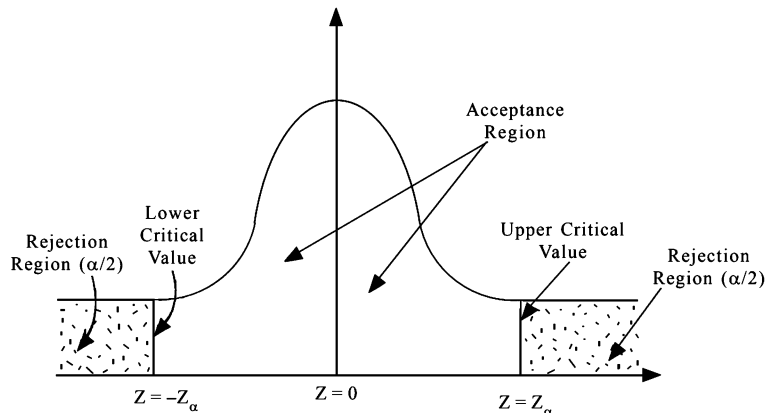


Fig.

In the case of one-tailed alternative,

$$P(Z > Z_\alpha) = \alpha \text{ if it is one-tailed (right)}$$

$$P(Z < -Z_\alpha) = \alpha \text{ if it is one-tailed (left)}$$

For Level of Significance ' α '

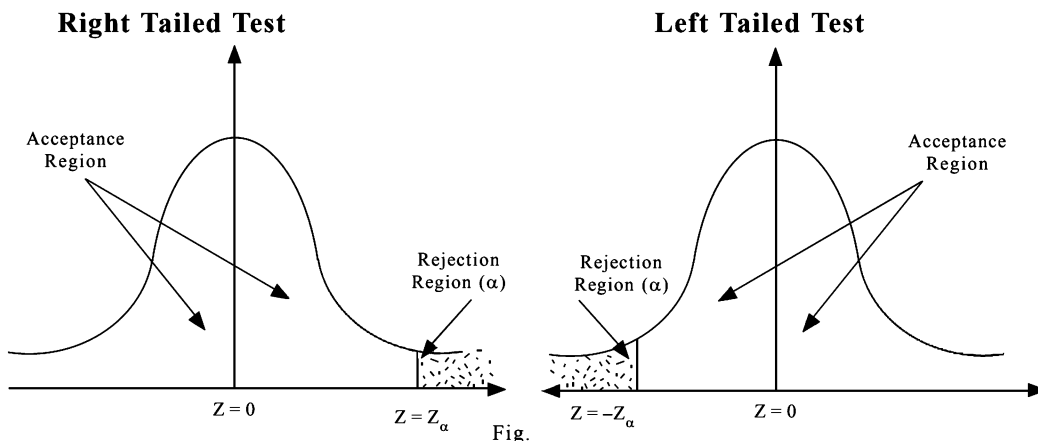


Fig.

From the above figures, it is clear, that the critical value of Z for a single-tailed test (right or left) at level of significance ' α ' is same as the critical value of Z for two-tailed test at level of significance ' 2α '.

The critical values of Z at different level of significance (α) for both single tailed and two-tailed tests are calculated from equations

$$P(|Z| > Z_{\alpha}) = \alpha$$

$$P(Z > Z_{\alpha}) = \alpha$$

$$P(Z < -Z_{\alpha}) = \alpha$$

using the normal tables. They are listed below.

Critical values (Z_{α}) of Z

	Level of Significance		
	1% (.01)	5% (.05)	10% (.1)
Two-tailed test	$ Z_{\alpha} = 2.58$	$ Z_{\alpha} = 1.96$	$ Z_{\alpha} = 1.645$
Right-tailed test	$Z_{\alpha} = 2.33$	$Z_{\alpha} = 1.645$	$Z_{\alpha} = 1.28$
Left-tailed test	$Z_{\alpha} = -2.33$	$Z_{\alpha} = -1.645$	$Z_{\alpha} = -1.28$

9.14 ONE - TAILED AND TWO - TAILED TESTS

[JNTU (A) Dec. 2009, (K) Nov. 2011, (H) May 2012 (Set No. 1)]

If we have to test whether the population mean μ has a specified value μ_0 , then the Null Hypothesis is $H_0 : \mu = \mu_0$ and the Alternative Hypothesis may be (i) $H_1 : \mu \neq \mu_0$ (i.e., $\mu > \mu_0$ or $\mu < \mu_0$) or (ii) $H_1 : \mu > \mu_0$ or (iii) $H_1 : \mu < \mu_0$. The Alternative Hypothesis in (i) is known as a two-tailed (i.e., both right and left tail) alternatives and the alternative hypothesis in (ii) and (iii) are known as right - tailed and left - tailed alternatives respectively.

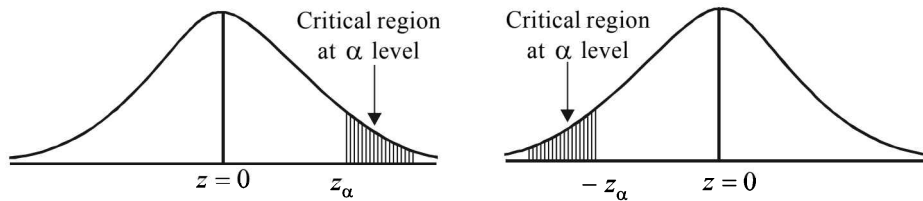


Fig.

If the Alternative Hypothesis H_1 in a test of a statistical hypothesis be one - tailed (i.e., either right - tailed or left - tailed but not both), then the test is called a *one-tailed test*. For example, to test whether the population mean $\mu = \mu_0$, we have $H_0 : \mu = \mu_0$ against the alternative hypothesis H_1 given by

(i) $H_1 : \mu > \mu_0$ (right tailed) or (ii) $H_1 : \mu < \mu_0$ (left - tailed) and the corresponding test is a single - tailed or one-tailed or one sided. In the right - tail test $H_1 : \mu > \mu_0$, the

critical region (or rejection region) $z > z_\alpha$ lies entirely in the right tail of the sampling distribution of sample mean $\bar{x}$ with area equal to the level of significance α (see figure). Similarly, in the left - tailed test ($H_1 : \mu < \mu_0$), the critical region $z < -z_\alpha$ lies entirely in the left tail of the sampling distribution of the sample mean $\bar{x}$ with area equal to the level of significance α (see figure).

Two-Tailed Test :

Suppose we want to test the Null Hypothesis $H_0 : \mu = \mu_0$ against the Alternative Hypothesis $H_1 : \mu \neq \mu_0$.

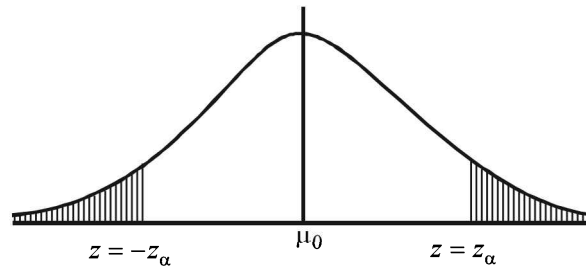


Fig.

Since H_1 is two - tailed alternative hypothesis, the critical region under the curve is equally distributed on both sides of the mean.

Thus, the critical area under the right - tail = The critical area under the left - tail
= Half of the total area

= $\frac{1}{2}$ probability of rejection

= $\frac{\alpha}{2}$ with critical statistic $Z_{\alpha/2}$,

where α is the level of significance.

The critical region is then, $z \leq -z_{\alpha/2}$ or $z_{\alpha/2} \leq z$.

Critical values of x for both two-tailed and one-tailed tests at 1%, 5% and 10% level of significance are given in the following table :

Table : Critical Values of z			
Level of significance α	1%	5%	10%
Critical values for two-tailed test	$ z_\alpha = 2.58$	$ z_\alpha = 1.96$	$ z_\alpha = 1.645$
Critical values for Right-tailed test	$z_\alpha = 2.33$	$z_\alpha = 1.645$	$z_\alpha = 1.28$
Critical values for Left-tailed test	$z_\alpha = -2.33$	$z_\alpha = -1.645$	$z_\alpha = -1.28$

Applying one-tailed or two-tailed test for a particular problem depends entirely on the nature of the Alternative Hypothesis. If the alternative test is two-tailed we apply two-tailed test and if Alternative Hypothesis is one-tailed, we apply one-tailed test.

For example, to test whether a coin is biased or not, two - tailed test should be used, since a biased coin gives either more number of heads than tails (which corresponds to right tail) or more number of tails than heads (which corresponds to left tail)

Example : Consider two population brands of bulbs one manufactured by routine process (mean μ_1) and the other manufactured by new technique (mean μ_2). If we want to test if the bulbs differ significantly then the hypothesis is $H_0 : \mu_1 = \mu_2$ and the alternative hypothesis will be $H_0 : \mu_1 \neq \mu_2$. This gives us a two-tailed test. Suppose if we want to test if the bulbs produced by new process (μ_2) have higher average life than those produced by standard process (μ_1), then we have

$$H_0 : \mu_1 = \mu_2 \text{ and } H_1 : \mu_1 < \mu_2$$

In this case we have to adopt a left-tailed test.

If we want to test whether the product of new process (μ_2) is inferior to that of standard process (μ_1), then we have

$$H_0 : \mu_1 = \mu_2 \text{ and } H_1 : \mu_1 > \mu_2$$

which gives a right-tailed test.

Hence the decision about applying a two-tail test or a single-tail (left or right) test will mainly depend on the problem under study.

PROCEDURE FOR TESTING OF HYPOTHESIS :

[JNTU (H) Nov. 2009 (Set No.2,4), (K) Nov. 2011]

Various steps involved in testing of Hypothesis are given below : Infact, the same steps are followed for conducting all tests of significance.

Step 1 : Null Hypothesis : Define or set up a Null Hypothesis H_0 taking into consideration the nature of the problem and data involved.

Step 2 : Alternative Hypothesis : Set up the Alternative Hypothesis H_1 so that we could decide whether we should use one-tailed or two-tailed test.

Step 3 : Level of Significance : Select the appropriate level of significance(α) depending on the reliability of the estimates and permissible risk. That is, a suitable α is selected in advance if it is not given in the problem.
(Usually we choose 5% level of significance)

Step 4 : Test Statistic: Compute the test statistic $Z = \frac{t - E(t)}{\text{S.E of } t}$ under the null hypothesis.

Here t is a sample statistic and S. E. is the standard error of t .

Step 5 : Conclusion : We compare the computed value of the test statistic Z with the critical value Z_α at given level of significance (α).

If $|Z| < Z_\alpha$, (that is, if the absolute value of the calculated value of Z is less than the critical value Z_α) we conclude that it is not significant. We accept the null hypothesis.

If $|Z| > Z_\alpha$ then the difference is significant and hence the null hypothesis is rejected at the level of significance α .

Clearly,

For Two-Tailed Test :

If $|Z| < 1.96$ accept H_0 at 5% level of significance.

If $|Z| > 1.96$ reject H_0 at 5% level of significance.

If $|Z| < 2.58$, accept H_0 at 1% level of significance.

If $|Z| > 2.58$ reject H_0 at 1% level of significance.

For Single-Tailed (Right or Left) Test :

If $|Z| < 1.645$, accept H_0 at 5% level of significance.

If $|Z| > 1.645$, reject H_0 at 5% level of significance.

If $|Z| < 2.33$ accept H_0 at 1% level of significance.

If $|Z| > 2.33$ reject H_0 at 1% level of significance.

9.15 TEST OF SIGNIFICANCE FOR LARGE SAMPLES

If the sample size $n \geq 30$, then we consider such samples as large samples. The tests of significance used in large samples are different from those used in small samples because small samples fail to satisfy the assumptions under which large sample analysis is done. If n is large, the distributions, such as Binomial, Poisson, Chi-square etc. are closely approximated by normal distributions. Therefore, for large samples, the sampling distribution of a statistic is approximately a normal distribution.

Suppose we wish to test the hypothesis that the probability of success in such trial is p . Assuming it to be true, the mean μ and the standard deviation σ of the sampling distribution of number of successes are np and $\sqrt{npq}$ respectively.

If x be the observed number of successes in the sample and Z is the standard normal variate then $Z = \frac{x - \mu}{\sigma}$.

Thus we have the following test of significance :

- (i) If $|Z| < 1.96$, the difference between the observed and expected number of successes is not significant.
- (ii) If $|Z| > 1.96$, the difference is significant at 5% level of significance.
- (iii) If $|Z| > 2.58$, the difference is significant at 1% level of significance.

Assumptions for Large Samples :

The following are the assumptions under which significance tests are applied :

1. The random sampling distribution of statistic has the properties of the normal curve. This may not hold good in case of small samples.
2. Values (*i.e.*, statistic) given by the samples are sufficiently close to the population values (*i.e.*, parameters) and can be used in its place for calculating the standard error (S. E.) of the estimate.

For example, for computing the S. E., if the S. D. of the universe is not known, it can be replaced by the S. D. of the sample. In case of small samples, this is not possible.

SOLVED EXAMPLES

Example 1 : A coin was tossed 960 times and returned heads 183 times. Test the hypothesis that the coin is unbiased. Use a 0.05 level of significance.

Solution : Here $n = 960$, $p = \text{Probability of getting head} = 1/2$

$$\therefore q = 1 - p = \frac{1}{2}; \quad \mu = np = 960 \left(\frac{1}{2} \right) = 480$$

$$\sigma = \sqrt{npq} = \sqrt{(np)q} = \sqrt{480 \times \frac{1}{2}} = \sqrt{240} = 15.49$$

$x = \text{number of successes} = 183$

1. **Null Hypothesis** H_0 : The coin is unbiased

2. **Alternative Hypothesis** H_1 : The coin is biased

3. **Level of significance** : $\alpha = 0.05$

4. **The test statistic is** $Z = \frac{x - \mu}{\sigma} = \frac{183 - 480}{15.49} = \frac{-297}{15.49} = -19.17$

$$\therefore |Z| = 19.17$$

As $|Z| > 1.96$, the null hypothesis H_0 has to be rejected at 5% level of significance and we conclude that the coin is biased.

Example 2 : A coin was tossed 400 times and returned heads 216 times. Test the hypothesis that the coin is unbiased. Use a 0.05 Level of significance.

[JNTU (K) Nov. 2009 (Set No. 3)]

Solution : Here $n = 400$

$$p = \text{Probability of getting head} = \frac{1}{2}$$

$$q = 1 - p = 1 - \frac{1}{2} = \frac{1}{2}, \quad \mu = np = 400 \left(\frac{1}{2} \right) = 200$$

$$\sigma = \sqrt{npq} = \sqrt{np(q)} = \sqrt{200 \times \frac{1}{2}} = \sqrt{100} = 10$$

$x = \text{number of successes} = 216$

1. **Null Hypothesis** H_0 : The coin is unbiased

2. **Alternative Hypothesis** H_1 : The coin is biased

3. **Level of significance** : $\alpha = 0.05$

4. **The test statistic is** $z = \frac{x - \mu}{\sigma}$

$$\therefore z = \frac{216 - 200}{10} = \frac{16}{10} = 1.6$$

As $|z| < 1.96$, the null hypothesis H_0 has to be accepted and we conclude that the coin is unbiased.

Example 1 : A die is tossed 960 times and it falls with 5 upwards 184 times. Is the die unbiased at a level of significance of 0.01 ? [JNTU (K) Nov. 2009, May 2013 (Set No. 2)]

Solution : Here $n = 960$, $p =$ The probability of throwing 5 with one die $= 1/6$

$$\therefore q = 1 - \frac{1}{6} = \frac{5}{6}; \quad \mu = np = 960 \left(\frac{1}{6} \right) = 160$$

$$\sigma = \sqrt{npq} = \sqrt{160 \times \frac{5}{6}} = 11.55$$

$$x = \text{number of successes} = 184$$

1. **Null Hypothesis** H_0 : The coin is unbiased

2. **Alternative Hypothesis** H_1 : The coin is biased

3. **Level of significance** : $\alpha = 0.01$

4. **The test statistic is** $Z = \frac{x - \mu}{\sigma} = \frac{184 - 160}{11.55} = \frac{24}{11.55} = 2.08$

As $|Z| < 2.58$, the null hypothesis H_0 has to be accepted at 1 % level of significance and we conclude that the die is unbiased.

Note : As $|Z| > 1.96$, the null hypothesis H_0 has to be rejected at 5% level of significance and we conclude that the die is biased.

Example 4 : A die is tossed 256 times and it turns up with an even digit 150 times. Is the die biased ? [JNTU (K) Nov. 2011 (Set No. 3)]

Solution : Here $n = 256$, $p =$ The probability of getting an even digit (2 or 4 or 6) $= \frac{3}{6} = \frac{1}{2}$

$$q = 1 - p = \frac{1}{2}; \quad \mu = np = 256 \left(\frac{1}{2} \right) = 128$$

$$\sigma = \sqrt{npq} = \sqrt{np(q)} = \sqrt{128 \times \frac{1}{2}} = \sqrt{64} = 8$$

$$x = \text{number of successes} = 150$$

1. **Null Hypothesis** H_0 : The die is unbiased

2. **Alternative Hypothesis** H_1 : The die is biased

3. **Level of significance** : $\alpha = 0.05$

4. **The test statistic is** $Z = \frac{x - \mu}{\sigma}$ i.e., $Z = \frac{150 - 128}{8} = \frac{22}{8} = 2.75$

As $|Z| > 1.96$, the null hypothesis H_0 has to be rejected at 5% level of significance and we conclude that the die is biased.

Under large sample tests, we will see four important tests to test the significance.

1. Testing of significance for single proportion.
2. Testing of significance for difference of proportions.
3. Testing of significance for single mean.
4. Testing of significance for difference of means.

9.16 TEST OF SIGNIFICANCE OF A SINGLE MEAN - LARGE SAMPLES

Let a random sample of size n ($n \geq 30$) has the sample mean $\bar{x}$, and μ be the population mean. Also the population mean μ has a specified value μ_0 .

Working Rule :

1. **The Null Hypothesis** is $H_0 : \bar{x} = \mu$ i.e., “there is no significance difference between the sample mean and population mean” or the sample has been drawn from the parent population.

2. **The Alternative Hypothesis** is

$$(i) H_1 : \bar{x} \neq \mu \quad (\mu \neq \mu_0)$$

$$\text{or } (ii) H_1 : \bar{x} > \mu \quad (\mu > \mu_0)$$

$$\text{or } (iii) H_1 : \bar{x} < \mu \quad (\mu < \mu_0)$$

Since n is large, the sampling distribution of $\bar{x}$ is approximately normal.

3. **Level of Significance.** Set the level of significance α .

4. **The Test Statistic :**

We have the following two cases.

Case I : When the standard deviation σ of population is known.

In this case, standard Error of Mean, S. E. ($\bar{x}$) = $\frac{\sigma}{\sqrt{n}}$, where n = sample size,

σ = s.d. of population.

$\therefore$ The test statistic is

$$z = \frac{\bar{x} - \mu}{\text{S.E.}(\bar{x})} = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}, \text{ where } \mu \text{ is the sample mean.}$$

Case II : When the standard deviation σ of population is not known.

In this case, we take s , the s.d. of sample to compute the S. E. of mean

$$\therefore \text{S.E.}(\bar{x}) = \frac{s}{\sqrt{n}}$$

Hence the test statistic is $z = \frac{\bar{x} - \mu}{\text{S.E.}(\bar{x})} = \frac{\bar{x} - \mu}{s / \sqrt{n}}$

5. Find the critical value z_α of z at the level of significance α from the normal table.
6. **Decision :**
 - (a) If $|z| < z_\alpha$ we accept the Null Hypothesis H_0 .
 - (b) If $|z| > z_\alpha$, we reject the Null Hypothesis H_0 .

The rejection rule for $H_0 : \bar{x} = \mu$ (or $\mu = \mu_0$) is given below :

Table : Critical Values of z			
Level of significance α	1%	5%	10%
Critical region for $\mu \neq \mu_0$	$ z > 2.58$	$ z > 1.96$	$ z > 1.645$
Critical region for $\mu > \mu_0$	$z > 2.33$	$z > 1.645$	$z > 1.28$
Critical region for $\mu < \mu_0$	$z < -2.33$	$z < -1.645$	$z < -1.28$

Note: 1. We reject Null Hypothesis H_0 when $|z| > 3$ without mentioning any level of significance.

The test statistic is, $z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$ where σ is the S.D of the population

If the population S.D is not known, then use the statistic

$$z = \frac{\bar{x} - \mu}{S / \sqrt{n}} \text{ where } S \text{ is the sample S.D.}$$

2. The values $\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}$ are called 95% fiducial limits or confidence limits for the mean of the population corresponding to the given sample.

Similarly, $\bar{x} \pm 2.58 \frac{\sigma}{\sqrt{n}}$ (or) $[\bar{x} - 2.58 (\text{S.E. of } \bar{x}), \bar{x} + 2.58 (\text{S.E. of } \bar{x})]$ are called

99% confidence limits and $\bar{x} \pm 2.33 \frac{\sigma}{\sqrt{n}}$ are called 98% confidence limits.

SOLVED EXAMPLES

Example 1 : According to the norms established for a mechanical aptitude test, persons who are 18 years old have an average height of 73.2 with a standard deviation of 8.6. If 4 randomly selected persons of that age averaged 76.7, test the hypothesis $\mu = 73.2$ against the alternative hypothesis $\mu > 73.2$ at the 0.01 level of significance. [JNTU 2004, 2005S (Set No. 1)]

Solution : Given $n = 4$, $\mu = 73.2$, $\bar{x}$ = mean of the sample = 76.7
and σ = S.D of population = 8.6

1. **Null Hypothesis** $H_0 : \mu = 73.2$
2. **Alternative Hypothesis** $H_1 : \mu > 73.2$ (Right-tailed test)
3. **Level of significance** : $\alpha = 99\%$ (or probability is 0.01)
4. **The test statistic** is $z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{76.7 - 73.2}{\frac{8.6}{\sqrt{4}}} = \frac{3.5}{4.3} = 0.814$

Tabulated value of z at 99% level of significance is 2.33.

Hence calculated $z <$ tabulated z

$\therefore$ The null hypothesis H_0 is accepted. That is, $\bar{x}$ and μ do not differ significantly.

Example 1 : A sample of 64 students have a mean weight of 70 kgs. Can this be regarded as a sample from a population with mean weight 56 kgs and standard deviation 25 kgs with level of significance 0.05 ? [JNTU 06, (A) Nov. 10 (Set No. 2), Dec. 2015]

Solution :

Given $\bar{x}$ = Mean of the sample = 70 kgs

μ = Mean of the population = 56 kgs

σ = S.D of population = 25 kgs

and n = Sample size = 64

1. **Null Hypothesis** H_0 : A sample of 64 students with mean weight of 70 kgs can be regarded as a sample from a population with mean weight 56 kgs and standard deviation 25 kgs.
2. **Alternative Hypothesis** H_1 : Sample cannot be regarded as one coming from the population.
3. **Level of significance** : $\alpha = 0.05$ (Given)
4. **The test statistic** is $z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{70 - 56}{\left(\frac{25}{\sqrt{64}}\right)} = 4.48$

5. The null hypothesis H_0 is rejected, since $|Z| > 1.645$

Note : The null hypothesis can be rejected even at 1% level of significance.

Example 1 : An oceanographer wants to check whether the depth of the ocean in a certain region is 57.4 fathoms, as had previously been recorded. What can he conclude at the 0.05 level of significance, if readings taken at 40 random locations in the given region yielded a mean of 59.1 fathoms with a standard deviation of 5.2 fathoms.

[JNTU 2003, 2003 S, 2004]

Solution : Given $n = 40$, $\bar{x} = 59.1$ and $\sigma = 5.2$

1. **Null Hypothesis H_0 :** $\mu = 57.4$
2. **Alternative hypothesis H_1 :** $\mu \neq 57.4$
3. **Level of significance :** $\alpha = 0.05$
4. **The test statistic is $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{59.1 - 57.4}{5.2/\sqrt{40}} = 2.067$**

Tabulated value of Z at 5% level of significance is 1.96

Hence calculated $Z >$ tabulated Z .

$\therefore$ The null hypothesis H_0 is rejected.

Example 4 : In a random sample of 60 workers, the average time taken by them to get to work is 33.8 minutes with a standard deviation of 6.1 minutes. Can we reject the null hypothesis $\mu = 32.6$ minutes in favour of alternative null hypothesis $\mu > 32.6$ at $\alpha = 0.025$ level of significance. [JNTU 2005 (Set No. 1)]

Solution : Given $n = 60$, $\bar{x} = 33.8$, $\mu = 32.6$ and $\sigma = 6.1$

1. **Null Hypothesis H_0 :** $\mu = 32.6$
2. **Alternative Hypothesis H_1 :** $\mu > 32.6$
3. **Level of significance :** $\alpha = 0.025$
4. **The test statistic is $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{33.8 - 32.6}{6.1/\sqrt{60}} = \frac{1.2}{0.7875} = 1.5238$**

Tabulated value of Z at 0.025 level of significance is 2.58.

Hence calculated $Z <$ tabulated Z

$\therefore$ The null hypothesis H_0 is accepted.

Example 5 : A sample of 900 members has a mean of 3.4 cms and S.D 2.61 cms. Is this sample has been taken from a large population of mean 3.25 cm and S.D 2.61 cms. If the population is normal and its mean is unknown find the 95% fiducial limits of true mean.

[JNTU (H) May 2011 (Set No. 1), (K) Nov. 2011 (Set No. 1)]

Solution :

Given $n = 900$, $\mu = 3.25$

$\bar{x} = 3.4$ cm, $\sigma = 2.61$

and $S = 2.61$

1. **Null Hypothesis H_0 :** Assume that the sample has been drawn from the population with mean $\mu = 3.25$.
2. **Alternative Hypothesis H_1 :** $\mu \neq 3.25$.

3. The test statistic is, $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{3.4 - 3.25}{2.61/\sqrt{900}} = 1.724$

i.e., $Z = 1.724 < 1.96$

$\therefore$ We accept the null hypothesis H_0 .

i.e., The sample has been drawn from the population with mean $\mu = 3.25$.

95% confidence limits are given by

$$\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}} = 3.4 \pm 1.96 \times \frac{2.61}{\sqrt{900}} = 3.4 \pm 0.1705$$

i.e., 3.57 and 3.2295

Example 1 : A sample of 400 items is taken from a population whose standard deviation is 10. The mean of the sample is 40. Test whether the sample has come from a population with mean 38. Also calculate 95% confidence interval for the population.

[JNTU 2005, (H) Dec. 2011 (Set No. 1), (K) May 2013 (Set No. 4)]

Solution : Given $n = 400$, $\bar{x} = 40$, $\mu = 38$ and $\sigma = 10$

1. **Null Hypothesis H_0 :** $\mu = 38$

2. **Alternative Hypothesis H_1 :** $\mu \neq 38$

3. **Level of significance :** $\alpha = 0.05$

4. **The test statistic is,** $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{40 - 38}{10/\sqrt{400}} = 4$ *i.e.*, $Z = 4 > 1.96$

$\therefore$ We reject the null hypothesis H_0 .

i.e., The sample is not from the population whose mean is 38.

95% confidence interval is $\left(\bar{x} - 1.96 \cdot \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \cdot \frac{\sigma}{\sqrt{n}} \right)$

i.e., $\left(40 - \frac{1.96(10)}{\sqrt{400}}, 40 + \frac{1.96(10)}{\sqrt{400}} \right)$

or $\left(40 - \frac{1.96}{20}, 40 + \frac{1.96}{20} \right)$ or $(40 - 0.98, 40 + 0.98)$

i.e., (39.02, 40.98)

Example 1 : (i) An ambulance service claims that it takes on the average less than 10 minutes to reach its destination in emergency calls. A sample of 36 calls has a mean of 11 minutes and the variance of 16 minutes. Test the claim at 0.05 level significance.

[JNTU 2005, (H) May 2012 (Set No. 4)]

(ii) An ambulance service claims that it takes on the average 8.9 minutes to reach its destination in emergency calls. To check this claim, the agency which licenses ambulance services has them timed on 50 emergency calls, getting a mean of 9.3 minutes with a standard deviation of 1.6 minutes. What can they conclude at the level of significance of 0.05 ?

[JNTU (A) June 2018]

Solution : (i) Given $n = 36$, $\bar{x} = 11$, $\mu = 10$ and $\sigma = \sqrt{16} = 4$

1. Null Hypothesis $H_0 : \mu = 10$

2. Alternative Hypothesis $H_1 : \mu < 10$

3. Level of significance : $\alpha = 0.05$

4. The test statistic is, $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{11 - 10}{4/\sqrt{36}} = \frac{6}{4} = 1.5$

Tabulated value of Z at 5% level of significance is 1.645.

Hence calculated $Z < \text{tabulated } Z$

$\therefore$ We accept the null hypothesis H_0 .

(ii) This is left as an exercise to the reader.

Example 8 : It is claimed that a random sample of 49 tyres has a mean life of 15200 km. This sample was drawn from a population whose mean is 15150 kms and a standard deviation of 1200 km. Test the significance at 0.05 level.

[JNTU 2005, 2006S, (K) May 2013 (Set No. 2)]

Solution : Given $n = 49$, $\bar{x} = 15200$, $\mu = 15150$ and $\sigma = 1200$

1. Null Hypothesis $H_0 : \mu = 15150$

2. Alternative Hypothesis $H_1 : \mu \neq 15150$

3. Level of significance : $\alpha = 0.05$

4. Critical region : Accept the null hypothesis if $-1.96 < Z < 1.96$

5. The test statistic is, $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{15200 - 15150}{1200/\sqrt{49}} = 0.2917$

Since $|Z| < 1.96$ therefore, we accept the null hypothesis.

Example 9 : An insurance agent has claimed that the average age of policy holders who issue through him is less than the average for all agents which is 30.5 years. A random sample of 100 policy holders who had issued through him gave the following age distribution.

Age	16-20	21-25	26-30	31-35	36-40
No. of persons	12	22	20	30	16

Calculate the Arithmetic mean and Standard deviation of this distribution and use these values to test his claim at 5% level of significance.

Solution : Take $A = 28$, $d_i = x_i - A$

$$\therefore \text{A.M.} = \bar{x} = A + \frac{h \sum f_i d_i}{N} = 28 + \frac{5 \times 16}{100} = 28.8$$

$$\begin{aligned} S.D : S &= h \sqrt{\frac{\sum f d^2}{N} - \left(\frac{\sum f d}{N}\right)^2} = 5 \cdot \sqrt{\frac{164}{100} - \left(\frac{16}{100}\right)^2} \quad [\because h = 5] \\ &= 6.35 \end{aligned}$$

1. **Null Hypothesis H_0 :** The sample is drawn from a population with mean μ i.e. $\bar{x}$ and μ do not differ significantly where $\mu = 30.5$ years.
2. **Alternative Hypothesis H_1 :** $\mu < 30.5$ years (left tail test)

Now, $\bar{x} = 28.8$, $S = 6.35$, $\mu = 30.5$ years and $n = 100$

$$3. \text{ The test statistic is, } Z = \frac{\bar{x} - \mu}{S/\sqrt{n}} = \frac{28.8 - 30.5}{6.35/\sqrt{100}} = -2.677$$

$$\therefore |Z| = 2.68.$$

Tabulated value of Z at 5% level of significance is 1.645 (left tail test).

Here calculated $Z >$ tabulated Z .

$\therefore$ The Null hypothesis H_0 is rejected.

i.e., $\bar{x}$ and μ differ significantly.

i.e., The sample is not drawn from a population with mean $\mu = 30.5$ years

Example 10 : The mean life time of a sample of 100 light tubes produced by a company is found to be 1560 hrs with a population S.D of 90 hrs. Test the hypothesis for $\alpha = 0.05$ that the mean life time of the tubes produced by the company is 1580 hrs.

[JNTU (A) Dec. 2009 (Set No. 1)]

Solution : Given $\bar{x}$ = Mean of the sample = 1560 hrs

μ = Mean of the population = 1580 hrs

n = Sample size = 100

σ = Standard deviation = 90 hrs

1. **Null Hypothesis H_0 :** $\mu = 1580$
2. **Alternative Hypothesis H_1 :** $\mu \neq 1580$
3. **Level of significance :** $\alpha = 0.05$

$$4. \text{ The test statistic is } z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{1560 - 1580}{90/\sqrt{100}} = \frac{-20}{9}$$

$$\therefore |z| = \frac{20}{9} = 2.22$$

$z_{\alpha/2}$ for $\alpha = 0.05$ is 1.96. Since $|z| > 1.96$, the null Hypothesis H_0 is rejected.

$\therefore \mu \neq 1580$.

Note: $z_{\alpha/2}$ for $\alpha = 0.01$ is 2.58

Since $|z| < 2.58$, the null Hypothesis H_0 can be accepted at 1% level of significance.

Example 11 : If the distribution of the weights of all men travelling by air between Delhi and Mumbai has a mean of 163 pounds and a standard deviation of 18 pounds. What is the probability that the consigned gross weight of 36 men travelling between these two cities is more than 6000 pounds. [JNTU (A) Dec. 2009 (Set No. 2)]

Solution : Given μ = The mean of the population = 163 pounds

σ = The standard deviation of the population = 18 pounds

Gross weight of 36 men = 6000 pounds

$$\therefore \bar{x} = \text{Average weight of a man} = \frac{6000}{36} = 166.7 \text{ pounds}$$

Also we have n = sample size = 36

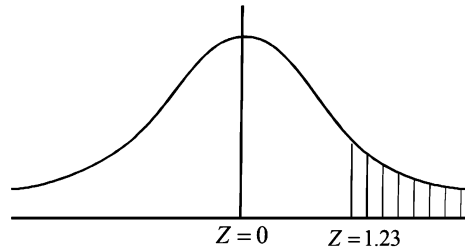
$$\therefore z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{166.7 - 163}{18 / \sqrt{36}} = \frac{3.7}{3} = 1.23$$

Hence $P(\bar{x} \geq 166.7) = P(Z \geq 1.23)$

$$= 0.5 - P(Z < 1.23)$$

$$= 0.5 - 0.3907 \text{ (from normal tables)}$$

$$= 0.1093$$



Thus the probability that the consigned gross weight of 36 men travelling between these two cities is more than 6000 pounds is 0.1093.

Example 11 : The length of life X of certain computers is approximately normally distributed with mean 800 hours and standard deviation 40 hours. If a random sample of 30 computers has an average life of 788 hours, test the null hypothesis that $\mu = 800$ hours against the alternative that $\mu \neq 800$ hours at (i) 0.5% (ii) 1% (iii) 4% (iv) 5% (v) 10% (vi) 15% level. [JNTU (K) May 2010 (Set No. 1)]

Solution : The Null Hypothesis is $H_0 = 800$ hours.

The Alternative Hypothesis is $H_1 : \mu \neq 800$ hours.

Since H_1 is two-tailed (*i.e.* two sided), we are to use two -tailed test.

Let us assume that H_0 is true.

The test statistic is $z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$

Here $\bar{x}$ = sample mean = 788 hours

$n = \text{sample size} = 30$

$\sigma = \text{standard deviation} = 40$

$$\therefore z = \frac{788 - 800}{40 / \sqrt{30}} = -1.643$$

and $|z| = 1.643$

- (i) Since $|z| < 2.81$, we accept the Null Hypothesis H_0 at 0.5% level of significance.
- (ii) Since $|z| < 2.58$, we accept the Null Hypothesis H_0 at 1% level of significance.
- (iii) Since, $|z| < 2.06$, we accept H_0 at 4% level of significance.
- (iv) Since, $|z| < 1.96$, we accept H_0 at 5% level of significance.
- (v) Since, $|z| < 1.645$, we accept H_0 at 10% level of significance.
- (vi) Since, $|z| > 1.44$, we reject H_0 at 15% level of significance.

Example 13 : In 64 randomly selected hours of production, the mean and the standard deviation of the number of acceptance pieces produced by an automatic stamping machine are $\bar{x} = 1.038$ and $\sigma = .146$

At the .05 level of significance does this enable us to reject the null hypothesis $\mu = 1.000$ against the alternative hypothesis $\mu > 1.000$? [JNTU (H) Nov. 2010 (Set No. 4)]

Solution : Let the Null Hypothesis be $H_0 : \mu = 1.000$

Then the Alternative Hypothesis is $H_1 : \mu > 1.000$

Here $\bar{x} = \text{Mean of the sample} = 1.038$

$\mu = \text{Mean of the population} = 1.000$

$\sigma = \text{S.D. of the population} = 0.146$

and $n = \text{Sample size} = 64$

$$\begin{aligned} \therefore \text{The test statistic is } z &= \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \\ &= \frac{1.038 - 1.000}{0.146 / \sqrt{64}} = \frac{0.038}{0.146 / 8} = 2.082 \end{aligned}$$

Thus we see that $z = 2.082 > 1.645$

Hence, we reject the Null Hypothesis H_0 at 5% level of significance and conclude that the mean of the population $\mu > 1.000$

Example 14 : A trucking rm suspects the claim that average life of certain tyres is atleast 28,000 miles. To check the claim the rm puts 40 of these tyres on its trucks and gets a mean life time of 27463 miles with a standard deviation of 1348 miles. Can the claim be true? [JNTU (H) Apr. 2012 (Set No. 2)]

Solution : The Null Hypothesis is $H_0 : \mu = 28,000$ miles

The Alternative Hypothesis is $H_1 : \mu \neq 28,000$

Since H_1 is two - tailed (*i.e.*, two - sided), we are to use two - tail test.

Let us assume that H_0 is true.

The test statistic is $z = \frac{\bar{x} - \mu}{\text{S.E. of } \bar{x}} = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$

Here $\bar{x}$ = sample mean = 27463 miles, $n = 40$ and $\sigma = s = 1348$ miles

[$\because$ population S. D. σ is not known]

$$\therefore z = \frac{27463 - 28000}{1348 / \sqrt{40}} = \frac{-537(\sqrt{40})}{1348} = -2.52$$

$$\text{and } |z| = |-2.52| = 2.52 > 1.96$$

Hence, the Null Hypothesis H_0 is rejected at 5% level of significance and we conclude that the mean life of tyres, cannot be taken as 28000 miles.

9.17 TEST FOR EQUALITY OF TWO MEANS - LARGE SAMPLES (TEST OF SIGNIFICANCE FOR DIFFERENCE OF MEANS OF TWO LARGE SAMPLES)

Let $\bar{x}_1$ and $\bar{x}_2$ be the sample means of two independent large random samples sizes n_1 and n_2 drawn from two populations having means μ_1 and μ_2 and standard deviations σ_1 and σ_2 . To test whether the two population means are equal.

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

S. E. of $(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$, where σ_1 and σ_2 are the S. D. of the two populations

To test whether there is any significant difference between $\bar{x}_1$ and $\bar{x}_2$, we have to use the statistic

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - \delta}{\text{S.E. of } (\bar{x}_1 - \bar{x}_2)} = \frac{(\bar{x}_1 - \bar{x}_2) - \delta}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

where $\delta = \mu_1 - \mu_2$ (= given constant.)

If $\delta = 0$, the two populations have the same means.

If $\delta \neq 0$ the two populations are different.

Under $H_0 : \mu_1 = \mu_2$, the test statistic becomes
$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

is approximately normally distributed with mean 0 and S. D. 1.

Note : If the samples have been drawn from the population with common S. D. σ , then $\sigma_1^2 = \sigma_2^2 = \sigma^2$

$$\text{Hence } z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}}} = \frac{\bar{x}_1 - \bar{x}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

is normally distributed with mean zero and standard deviation one.

If σ is not known we can use an estimate of σ^2 given by $\sigma^2 = \frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2}$

Rejection Rule for $H_0 : \mu_1 = \mu_2$

- (i) If $|z| > 1.96$, reject H_0 at 5% level of significance.
- (ii) If $|z| > 2.58$, reject H_0 at 1% level of significance.
- (iii) If $|z| > 1.645$, reject H_0 at 10% level of significance.
- (iv) If $z > 3$ then either the samples have not been drawn from the same population or the sampling is not simple.

Otherwise accept H_0 .

Note : If the two samples are drawn from two populations with unknown Standard deviations σ_1^2 and σ_2^2 , then σ_1^2 and σ_2^2 can be replaced by sample variances S_1^2 and S_2^2 provided both the samples n_1 and n_2 are large.

In this case, the test statistic is
$$z = \frac{(\bar{x}_1 - \bar{x}_2) - \delta}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

Write the formula for testing the hypothesis concerning “Two Means”.

[JNTU Nov. 2008 (Set No.1)]

Solution : Let $\bar{x}_1$ be the mean of a random sample of size n_1 drawn from a population with mean μ_1 and variance σ_1^2 . Let $\bar{x}_2$ be the mean of an independent random sample of size n_2 drawn from another population with mean μ_2 and variance σ_2^2 . To test the hypothesis for difference of means, the statistic is given by

$$z = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

If $|z| < 1.96$, it is accepted that the samples have come from two populations with the same mean, at 5% level of significance. Otherwise, at this level of significance, we claim that the difference in means is significant.

SOLVED EXAMPLES

Example 1 : The means of two large samples of sizes 1000 and 2000 members are 67.5 inches and 68.0 inches respectively. Can the samples be regarded as drawn from the same population of S.D 2.5 inches. [JNTU (A) Nov. 2010 (Set No. 1)]

Solution : Let μ_1 and μ_2 be the means of the two populations.

Given $n_1 = 1000$, $n_2 = 2000$ and $\bar{x}_1 = 67.5$ inches, $\bar{x}_2 = 68$ inches

Population S.D, $\sigma = 2.5$ inches

1. **Null Hypothesis H_0 :** The samples have been drawn from the same population of S.D 2.5 inches

i.e., $\mu_1 = \mu_2$ and $\sigma = 2.5$ inches

2. **Alternative Hypothesis H_1 :** $\mu_1 \neq \mu_2$

3. **The test statistic is,**
$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}}} = \frac{67.5 - 68}{\sqrt{(2.5)^2 \left(\frac{1}{1000} + \frac{1}{2000} \right)}}$$

$$\Rightarrow z = \frac{-0.5}{0.0968} = -5.16$$

$\therefore |z| = 5.16 > 1.96$ *i.e.*, the calculated value of $z >$ the table value of z .

Hence the null hypothesis H_0 is rejected at 5% level of significance and we conclude that the samples are not drawn from the same population of S.D. 2.5 inches.

Example 1 : The mean yield of wheat from a district A was 210 pounds with S.D. 10 pounds per acre from a sample of 100 plots. In another district the mean yield was 220 pounds with S.D. 12 pounds from a sample of 150 plots. Assuming that the S.D of yield in the entire state was 11 pounds, test whether there is any significant difference between the mean yield of crops in the two districts.

Solution : Let μ_1 and μ_2 be the means of the two populations.

Given $\bar{x}_1 = 210$, $\bar{x}_2 = 200$, and $n_1 = 100$, $n_2 = 150$,

Population S.D., $\sigma = 11$

1. **Null Hypothesis H_0 :** $\mu_1 = \mu_2$ *i.e.* there is no difference between μ_1 and μ_2

2. **Alternative Hypothesis H_1 :** $\mu_1 \neq \mu_2$

3. **The test statistic is**
$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}}} = \frac{210 - 200}{\sqrt{11^2 \left(\frac{1}{100} + \frac{1}{150} \right)}} = 7.04178$$

$\therefore |z| = 7.041 > 1.96$ *i.e.* the calculated value of $z >$ the table value of z .

Hence we reject the Null Hypothesis H_0 at 5% level of significance and conclude that there is a significant difference between the mean yield of crops in the two districts.

Example 1 : In a survey of buying habits, 400 women shoppers are chosen at random in super market 'A' located in a certain section of the city. Their average weekly food expenditure is Rs. 250 with a S.D. of Rs. 40. For 400 women shoppers chosen at random in super market 'B' in another section of the city, the average weekly food expenditure is Rs. 220 with a S.D. of Rs. 55. Test at 1% level of significance whether the average weekly food expenditure of the two populations of shoppers are equal.

Solution : Let μ_1 and μ_2 be the means of the two populations.

Given $n_1 = 400$, $\bar{x}_1 = \text{Rs. } 250$, $S_1 = \text{Rs. } 40$

and $n_2 = 400$, $\bar{x}_2 = \text{Rs. } 220$, $S_2 = \text{Rs. } 55$

1. **Null Hypothesis H_0 :** Assume that the average weekly food expenditure of the two populations of shoppers are equal i.e., $H_0 : \mu_1 = \mu_2$

2. **Alternative Hypothesis H_1 :** $\mu_1 \neq \mu_2$

3. **Test statistic** is, $z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} = \frac{250 - 220}{\sqrt{\frac{(40)^2}{400} + \frac{(55)^2}{400}}} = \frac{30}{3.4} = 8.82$

i.e. $z = 8.82 > 2.58$

Hence we reject the Null Hypothesis H_0 at 1% level of significance and conclude that the average weekly food expenditure of the two populations of shoppers are not equal.

Example 4 : Samples of students were drawn from two universities and from their weights in kilograms, mean and standard deviations are calculated and shown below. Make a large sample test to test the significance of the difference between the means.

	Mean	S.D	Size of the sample
University A	55	10	400
University B	57	15	100

[JNTU 2005 (Set No. 3)]

Solution :

Given $\bar{x}_1 = 55$, $\bar{x}_2 = 57$, $n_1 = 400$, $n_2 = 100$, $S_1 = 10$ and $S_2 = 15$

1. **Null Hypothesis H_0 :** $\bar{x}_1 = \bar{x}_2$ i.e., there is no difference

2. **Alternative Hypothesis H_1 :** $\bar{x}_1 \neq \bar{x}_2$

3. **Level of significance :** $\alpha = 0.05$ (assumed)

4. **Critical region :** Accept H_0 if $-1.96 < Z < 1.96$

5. **The test statistic** is $Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} = \frac{55 - 57}{\sqrt{\frac{100}{400} + \frac{225}{100}}} = \frac{-2}{\sqrt{\frac{1}{4} + \frac{9}{4}}} = -1.26$

$$\therefore |Z| = 1.26 < 1.96$$

Hence, we accept the Null Hypothesis H_0 at 5% level of significance and conclude that there is no significant difference between the means.

Example 1 : The research investigator is interested in studying whether there is a significant difference in the salaries of MBA grades in two metropolitan cities. A random sample of size 100 from Mumbai yields on average income of Rs. 20,150. Another random sample of 60 from Chennai results in an average income of Rs. 20,250. If the variances of both the populations are given as $\sigma_1^2 = \text{Rs. } 40,000$ and $\sigma_2^2 = \text{Rs. } 32,400$ respectively.

[JNTU Nov. 2008 (Set No. 1)]

Solution : Given $n_1 = 100$, $\bar{x}_1 = 20,150$ and $n_2 = 60$, $\bar{x}_2 = 20250$

and $\sigma_1^2 = 40,000$, $\sigma_2^2 = 32400$

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$. That is, the difference of means is significant.

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

$$\begin{aligned} \text{The test statistic is } z &= \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{20150 - 20250}{\sqrt{\frac{40000}{100} + \frac{32400}{60}}} \\ &= \frac{100}{\sqrt{400 + 540}} = \frac{100}{30.66} = 3.26 \end{aligned}$$

Since $Z = 3.26 > 1.96 = Z_{0.05}$, we reject the Null Hypothesis at 5% level of significance and conclude that there is a significant difference in the salaries of MBA grades in two cities.

Example 1 : A researcher wants to know the intelligence of students in a school. He selected two groups of students. In the first group there 150 students having mean IQ of 75 with a S.D. of 15 in the second group there are 250 students having mean IQ of 70 with S.D. of 20. Is there a significant difference between the means of two groups ?

[JNTU Nov. 2008 (Set No. 4)]

Solution : Given $n_1 = 150$, $\bar{x}_1 = 75$, $\sigma_1 = 15$

and $n_2 = 250$, $\bar{x}_2 = 70$, $\sigma_2 = 20$

1. **Null Hypothesis H_0 :** The groups have been came from the same population

i.e., $\mu_1 = \mu_2$

2. **Alternative Hypothes** is $H_1 : \mu_1 \neq \mu_2$

$$\begin{aligned} \text{3. The test statistic is, } z &= \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{75 - 70}{\sqrt{\frac{225}{150} + \frac{400}{250}}} \\ &= \frac{5}{\sqrt{\frac{9}{5} + \frac{8}{5}}} = \frac{5\sqrt{5}}{\sqrt{17}} = 2.7116 \end{aligned}$$

Tabulated value of z at 1% level of significance is 2.33

$\therefore$ Calculated $z >$ tabulated z .

Hence we reject the Null Hypothesis H_0 at 1% level of significance and conclude that the groups have not been taken from the same population.

Example 1 : Two types of new cars produced in U.S.A. are tested for petrol mileage, one sample is consisting of 42 cars averaged 15 kmpl while the other sample consisting of 80 cars averaged 11.5 kmpl with population variances as $\sigma_1^2 = 2.0$ and $\sigma_2^2 = 1.5$ respectively. Test whether there is any significance difference in the petrol consumption of these two types of cars. (use $\alpha = 0.01$). [JNTU Apr. 2009 (Set No. 1)]

Solution : Let the types of the cars be named as A and B.

Number of cars of type A = 42

Average mileage for A = $\bar{x}_1 = 15$, Variance = $\sigma_1^2 = 2.0$

Number of cars of type B = 80

Average mileage for B = $\bar{x}_2 = 11.5$

Variance = $\sigma_2^2 = 1.5$

1. Null Hypothesis $H_0 : \mu_1 = \mu_2$

2. Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

$$\begin{aligned} \text{3. The test statistic is } z &= \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{|15 - 11.5|}{\sqrt{\frac{2}{42} + \frac{1.5}{80}}} \\ &= \frac{3.5}{\sqrt{0.0476 + 0.01875}} = \frac{3.5}{\sqrt{0.06635}} = 13.587 \end{aligned}$$

Tabulated value of z at 1% significance level is 2.58 (two-tailed)

Since $(z)_{\text{calculated}} > 2.58$ (z table), we reject Null Hypothesis H_0 at 1% level of significance and conclude that there is a significant difference in petrol consumption.

Example 8 : A simple sample of the height of 6400 Englishmen has a mean of 67.85 inches and a S. D of 2.56 inches while a simple sample of heights of 1600 Austrains has a mean of 68.55 inches and S. D. of 2.52 inches. Do the data indicate the Austrains are on the average taller than the Englishmen? (Use α as 0.01). [JNTU (A) 2009 (Set No. 2)]

Solution : We are given

n_1 = Size of the first sample = 6400

n_2 = Size of the second sample = 1600

$\bar{x}_1$ = Mean of the first sample = 67.85

$\bar{x}_2$ = Mean of the second sample = 68.55

σ_1 = Standard deviation of the first sample = 2.56

σ_2 = Standard deviation of the second sample = 2.52

1. **Null Hypothesis** $H_0 : \mu_1 = \mu_2$
2. **Alternative Hypothesis** $H_1 : \mu_1 < \mu_2$
3. **Level of significance** : $\alpha = 0.05$
4. **The test statistic is**

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\left(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)}} = \frac{67.85 - 68.55}{\sqrt{\frac{(2.56)^2}{6400} + \frac{(2.52)^2}{1600}}} = \frac{-0.7}{\sqrt{\frac{6.5536}{6400} + \frac{6.35}{1600}}}$$

$$= \frac{-0.7}{\sqrt{0.001 + 0.004}} = \frac{-0.7}{0.0707} = -9.9$$

$$\therefore |z| = 9.9 > 1.96$$

Hence, we reject the Null Hypothesis H_0 at 5% level of significance and conclude that the Australians are taller than Englishmen.

Example 9 : The mean life of a sample of 10 electric bulbs (or motors) was found to be 1456 hours with S.D. of 423 hours. A second sample of 17 bulbs (motors) chosen from a different batch showed a mean life of 1280 hours with S.D. of 398 hours. Is there a significant difference between the means of two batches? [JNTU (K) 2009, Nov. 2012 (Set No. 1)]

Solution : It is given that

n_1 = Sample size of first batch = 10

n_2 = Sample size of second batch = 17

$\bar{x}_1$ = Mean life of first batch = 1456

$\bar{x}_2$ = Mean life of second batch = 1280

σ_1 = Standard deviation of first batch = 423

σ_2 = Standard deviation of second batch = 398

1. **Null Hypothesis** $H_0 : \mu_1 = \mu_2$
2. **Alternative Hypothesis** $H_1 : \mu_1 \neq \mu_2$
3. **Level of significance** : $\alpha = 0.05$

4. **The test statistic is**

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{1456 - 1280}{\sqrt{\frac{(423)^2}{10} + \frac{(398)^2}{17}}}$$

$$= \frac{176}{\sqrt{17892.9 + 9317.88}} = \frac{176}{164.96} = 1.067$$

Since $z < z_{\alpha} = 1.96$, we accept the null hypothesis H_0 i.e., there is no difference between the mean life of electric bulbs of two batches.

Example 10 : The average marks scored by 32 boys is 72 with a S.D. of 8. While that for 36 girls is 70 with a S.D. of 6. Does this indicate that the boys perform better than girls at level of significance 0.05? [JNTU (K) May 2010 (Set No. 3)]

Solution : Let μ_1 and μ_2 be the means of the two populations

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 > \mu_2$

Let us assume that H_0 is true, *i.e.*, there is no difference between μ_1 and μ_2

Since the sample sizes are large we use the test statistic

$$z = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Here $\bar{x} = 72, \bar{y} = 70, \sigma_1 = 8, \sigma_2 = 6, n_1 = 32, n_2 = 36$

$$\therefore z = \frac{72 - 70}{\sqrt{\frac{64}{32} + \frac{36}{36}}} = \frac{2}{\sqrt{2+1}} = \frac{2}{\sqrt{3}} = 1.1547 < 1.96$$

Since the computed value of z is less than the table value, we cannot reject the Null Hypothesis at 5% level and conclude that the performance of boys and girls is the same.

Example 13 : At a certain large university a sociologist speculates that male students spend considerably more money on junk food than do female students. To test her hypothesis, the sociologist randomly selects from the registrar's records the names of 200 students. Of these, 125 are men and 75 are women. The sample mean of the average amount spent on junk food per week by the men is Rs. 400 and standard deviation is 100. For the women the sample mean is Rs. 450 and the sample standard deviation is Rs.150. Test the difference between the mean at .05 level. [JNTU (H) Nov. 2010 (Set No. 2)]

Solution : Let μ_1 and μ_2 be the means of the two populations.

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

Let us assume that H_0 is true *i.e.*, there is no difference between μ_1 and μ_2 .

We are given

$$n_1 = \text{Number of men} = 125$$

$$n_2 = \text{Number of women} = 75$$

$$\bar{x}_1 = \text{Mean of men} = 400$$

$$\bar{x}_2 = \text{Mean of women} = 450$$

$\sigma_1 = \text{S.D. of men} = 100$

$\sigma_2 = \text{S.D. of women} = 150$

Level of significance, $\alpha = 0.05$

$$\begin{aligned} \text{The test statistic is } z &= \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{400 - 450}{\sqrt{\frac{(100)^2}{125} + \frac{(150)^2}{75}}} \\ &= \frac{-50}{\sqrt{80 + 300}} = \frac{-50}{\sqrt{380}} = \frac{-50}{19.49} = -2.5654 \end{aligned}$$

$\therefore |z| = |-2.5654| = 2.5654 > 1.96$ i.e., the difference is highly significant.

Hence, we reject the Null Hypothesis at 5% level of significance and conclude that the two population means are not equal.

Example 11 : A company claims that its bulbs are superior to those of its main competitor. If a study showed that a sample of 40 of its bulbs have a mean life time of 647 hrs of continuous use with a S.D of 27 hrs. While a sample of 40 bulbs made by its main competitor had a mean life time of 638 hrs of continuous use with a S.D of 31 hrs. Test the significance between the difference of two means at 5% level.

[JNTU (H) Nov. 2010 (Set No. 4)]

Solution : Let μ_1 and μ_2 be the means of the two populations.

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 > \mu_2$

Since the sample sizes are large, the test statistic

$$z = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

is approximately normally distributed with mean 0 and S.D. 1.

Here $\bar{x} = 647, \bar{y} = 638, \sigma_1 = 27, \sigma_2 = 31, n_1 = n_2 = 40$

$$\begin{aligned} \therefore z &= \frac{647 - 638}{\sqrt{\frac{(27)^2}{40} + \frac{(31)^2}{40}}} \\ &= \frac{9}{\sqrt{\frac{729 + 961}{40}}} = \frac{9}{6.5} = 1.38 \end{aligned}$$

i.e., $|z| = 1.38 < 1.645$

Since the computed value of z is less than the table value, we cannot reject the Null Hypothesis at 5% level and conclude that the difference between the two sample means is not significant.

Example 13 : Studying the flow of traffic at two busy intersections between 4 p.m and 6 p.m to determine the possible need for turn signals. It was found that on 40 week days there were on the average 247.3 cars approaching the first intersection from the south which made left turn, while on 30 week days there were on the average 254.1 cars approaching the first intersection from the south made left turns. The corresponding sample standard deviations are 15.2 and 12. Test the significance between the difference of two means at 5% level. [JNTU (H) Dec. 2011 (Set No.1)]

Solution : Let the average cars in two places be μ_1 and μ_2 respectively.

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

Let us assume that H_0 is true *i.e.*, there is no significant difference between μ_1 and μ_2

Since the sample sizes are large, the test statistic is

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Here $n_1 = 40, n_2 = 30, \bar{x}_1 = 247.3, \bar{x}_2 = 254.1, \sigma_1 \approx s_1 = 15.2$ and $\sigma_2 \approx s_2 = 12$.

$$\begin{aligned} \therefore z &= \frac{247.3 - 254.1}{\sqrt{\frac{(15.2)^2}{40} + \frac{(12)^2}{30}}} = \frac{-6.8}{\sqrt{5.776 + 4.8}} \\ &= \frac{-6.8}{\sqrt{10.576}} = \frac{-6.8}{3.2521} = -2.091 \end{aligned}$$

$$\therefore |z| = 2.091 > 1.96$$

Since the computed value of z is greater than the table value, we reject the Null Hypothesis at 5% level and conclude that the two average cars are significantly different *i.e.*, they are not the same in the two busy intersections.

Example 14 : In a certain factory there are two independent processes for manufacturing the same item. The average weight in a sample of 700 items produced from one process is found to be 250 gms with a standard deviation of 30 gms while the corresponding figures in a sample of 300 items from the other process are 300 and 40. Is there significant difference between the mean at 1% level. [JNTU (H) Apr. 2012 (Set No.1)]

Solution : Let the average weight in the two independent processes be μ_1 and μ_2 respectively.

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

Let us assume that H_0 is true

i.e., there is no significant difference between μ_1 and μ_2 .

The test statistic is
$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Here $\bar{x}_1 = 250, \bar{x}_2 = 300, \sigma_1 = 30, \sigma_2 = 40$ and $n_1 = 700, n_2 = 300$

$$\therefore z = \frac{250 - 300}{\sqrt{\frac{900}{700} + \frac{1600}{300}}} = \frac{-50}{\sqrt{\frac{9}{7} + \frac{16}{93}}} = -19.43$$

Thus $|z| = 19.43 > 2.58$

Since the computed value of z is greater than the table value, we reject the Null Hypothesis at 1% level and conclude that there is a significant difference between the means.

Example 14 : The mean height of 50 male students who participated in sports is 68.2 inches with a S.D of 2.5. The mean height of 50 male students who have not participated in sport is 67.2 inches with a S.D of 2.8. Test the hypothesis that the height of students who participated in sports is more than the students who have not participated in sports.

[JNTU (H) Apr. 2012 (Set No. 2)]

Solution : Let the mean height in the two cases be μ_1 and μ_2 respectively.

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

Let us assume that H_0 is true, *i.e.*, there is no significant difference between μ_1 and μ_2

The test statistic is
$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Here $\bar{x}_1 = 68.2, \bar{x}_2 = 67.2, \sigma_1 = 2.5, \sigma_2 = 2.8, n_1 = 50, n_2 = 50$

$$\therefore z = \frac{68.2 - 67.2}{\sqrt{\frac{(2.5)^2}{50} + \frac{(2.8)^2}{50}}} = \frac{1}{\sqrt{\frac{6.5 + 7.84}{50}}} = 1.88$$

$$\therefore |z| = 1.88 < 1.96$$

Since the computed value of z is less than the tabulated value of z , we accept the Null Hypothesis at 5% level of significance and conclude that there is no significant difference in the heights.

Example 16 : The nicotene in milligrams of two samples of tobacco were found to be as follows. Find the standard error and confidential limits for the difference between the means at 0.05 level.

Sample A	24	27	26	23	25	
Sample B	29	30	30	31	24	36

[JNTU (H) Apr. 2012 (Set No. 4)]

Solution :

Calculations for means and s_1^2, s_2^2					
Sample A			Sample B		
x_1	$x_1 - \bar{x}_1$ $= x_1 - 25$	$(x_1 - \bar{x}_1)^2$	x_2	$x_2 - \bar{x}_2$ $= x_2 - 30$	$(x_2 - \bar{x}_2)^2$
24	-1	1	29	-1	1
27	2	4	30	0	0
26	1	1	30	0	0
23	-2	4	31	1	1
25	0	0	24	-6	36
			36	6	36
$125 = \sum x_1$		10	$180 = \sum x_2$		74

$$\bar{x}_1 = \frac{\sum x_1}{n_1} = \frac{125}{5} = 25, \quad \bar{x}_2 = \frac{\sum x_2}{n_2} = \frac{180}{6} = 30$$

$$\text{Hence variance, } s_1^2 = \frac{\sum (x_1 - \bar{x}_1)^2}{n_1 - 1} = \frac{10}{4} = 2.5$$

$$\text{and variance, } s_2^2 = \frac{\sum (x_2 - \bar{x}_2)^2}{n_2 - 1} = \frac{74}{5} = 14.8$$

$$\therefore \text{S. E. of } (\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{2.5}{5} + \frac{14.8}{6}} = 1.72$$

Hence the 95% confidence limits are

$$(\bar{x}_1 - \bar{x}_2) \pm 1.96 (\text{S. E. of } \bar{x}_1 - \bar{x}_2)$$

$$\text{i.e., } (25 - 30) \pm 1.96 (1.72) \text{ i.e., } -5 \pm 3.37 \text{ or } (-8.37, -1.63)$$

REVIEW QUESTIONS

1. Define population and sample. [JNTU (A) Dec. 2018]
3. Define small and large samples. [JNTU (A) June 2018]
3. What is testing ? What is the purpose of testing ? [JNTU (A) Dec. 2018]
4. (a) Define statistical Hypothesis
(b) Define (i) Test of statistical hypothesis
(ii) Type-I and Type II error.
5. (a) Write about (i) Null hypothesis (ii) Type I and Type II errors (iii) Alternative hypothesis. [JNTU 2005, 2008S (Set No. 1)]
(OR) Explain briefly the following :
(i) Type I error (ii) Type II error (iii) Critical Region
(b) Explain the types of errors in sampling.
(c) Define Null hypothesis and Alternative hypothesis. [JNTU (A) June 2016]
6. What is meant by level of significance. [JNTU (H) Nov. 2009 (Set No. 3)]
(or) Explain the level of significance. [JNTU (A) Dec. 2018]
7. Explain Alternative Hypothesis. [JNTU (A) Dec. 2015]
8. What is the formula of confidence interval ? [JNTU (A) Dec. 2015]
9. (a) Write about (i) Critical region (ii) Left tailed test (iii) Right tailed test
(iv) Two tailed test. (or) Write about one tailed and two tailed tests.
[JNTU 2004, (H) May 2012 (Set No. 1)]
(b) What is meant by Level of significance, one tailed and two tailed tests ?
[JNTU (K) Nov. 2011 (Set No. 4)]
(c) Write about (i) Null hypothesis (ii) Critical region (iii) Level of significance.
(d) Explain the critical region and accepting region [JNTU (A) June 2018]
10. Write about null hypothesis and testing of null hypothesis.
[JNTU 2005 (Set No. 4)]
11. Explain the procedure generally followed in testing of hypothesis.
[JNTU (H) Nov. 2009 (Set No. 2,4), (K) Nov. 2011]
(OR) Explain working rule for testing of Hypothesis.
[JNTU (K) Nov. 2011 (Set No. 1)]
12. Write the test statistic in case of single mean for large samples.
[JNTU (A) Dec. 2017]
13. Write short notes on type I and type II Error. [JNTU (H) Nov. 2009 (Set No. 2)]
(OR) Explain the types of errors in sampling [JNTU (A) Nov. 2010 (Set No. 4)]
(OR) Explain type I and type II of errors in testing of hypothesis
14. Differentiate two-tailed test of hypothesis from one-tailed test.

9.18 TEST OF SIGNIFICANCE FOR SINGLE PROPORTION - LARGE SAMPLES

[JNTU (H) Nov. 2009 (Set - 4)]

Suppose a large random sample of size n has a sample proportion p of members possessing a certain attribute (*i.e.*, proportion of successes). To test the hypothesis that the proportion P in the population has a specified value P_0 .

Let us set the Null Hypothesis be $H_0 : P = P_0$ (P_0 is a particular value of P)

Then the Alternative Hypothesis is

(i) $H_1 : P \neq P_0$ (*i.e.*, $P > P_0$ or $P < P_0$)

(ii) $H_1 : P > P_0$ or

(iii) $H_1 : P < P_0$

Since n is large, the sampling distribution of p is approximately normal.

$\therefore$ If H_0 is true, the test statistic

$$z = \frac{p - P_0}{\text{S.E. of } p} \text{ or}$$

$$z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}, \text{ where } p \text{ is the sample proportion}$$

is approximately normally distributed.

The critical region for z depending on the nature of H_1 and level of significance α is given in the following table :

Table 9.1 : Rejection Rule for $H_0 : P = P_0$			
Level of significance	1%	5%	10%
Critical region for $P \neq P_0$ (Two-tailed test)	$ z > 2.58$	$ z > 1.96$	$ z > 1.645$
Critical region for $P > P_0$ (Right-tailed test)	$z > 2.33$	$z > 1.645$	$z > 1.28$
Critical region for $P < P_0$ (Left-tailed test)	$z < -2.33$	$z < -1.645$	$z < -1.28$

Note 1 : Without any reference to the level of significance, we may reject the Null Hypothesis H_0 when $|z| > 3$.

Note 2 : (i) Limits for population proportion P are given by $p \pm 3\sqrt{\frac{pq}{n}}$ where $q = 1 - p$.

(ii) Confidence interval for proportion P for large sample at α level of significance is

$$P - z_{\alpha/2} \cdot \sqrt{\frac{PQ}{n}} < P < P + z_{\alpha/2} \cdot \sqrt{\frac{PQ}{n}} \text{ where } Q = 1 - P \text{ and}$$

$$z_{\alpha/2} = 1.96 \text{ (for 95\%)}, \quad z_{\alpha/2} = 2.33 \text{ (for 98\%) and}$$

$$z_{\alpha/2} = 2.58 \text{ (for 99\%).}$$

SOLVED EXAMPLES

Example 1 : A manufacturer claimed that atleast 95% of the equipment which he supplied to a factory conformed to specifications. An examination of a sample of 200 pieces of equipment revealed that 18 were faulty. Test his claim at 5% level of significance.

[JNTU (K), (H) Nov. 2009, (H) Dec. 2011 (Set No. 1)]

Solution : Given sample size, $n = 200$

Number of pieces confirming to specification = $200 - 18 = 182$

$\therefore p$ = Proportion of pieces confirming to specifications

$$= \frac{182}{200} = 0.91$$

and P = Population proportion = $\frac{95}{100} = 0.95$

1. Null Hypothesis H_0 : The proportion of pieces confirming to specifications *i.e.* $P = 95\%$.

2. Alternative Hypothesis H_1 : $P < 0.95$ (left - tail test)

3. The test statistic is $z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.91 - 0.95}{\sqrt{\frac{0.95 \times 0.05}{200}}} = \frac{-0.04}{0.0154} = -2.59$.

Since alternative hypothesis is left tailed, the tabulated value of Z at 5% level of significance is 1.645.

Since calculated value of $|z| = 2.6$ is greater than 1.645, we reject the Null Hypothesis H_0 at 5% level of significance and conclude that the manufacture's claim is rejected.

Example 2 : In a sample of 1000 people in Karnataka 540 are rice eaters and the rest are wheat eaters. Can we assume that both rice and wheat are equally popular in this state at 1% level of significance ? [JNTU (K), (H) Nov. 2009, (A) Nov. 2010 (Set No. 3)]

Solution :

Given $n = 1000$

$$p = \text{Sample proportion of rice eaters} = \frac{540}{1000} = 0.54$$

$$P = \text{Population proportion of rice eaters} = \frac{1}{2} = 0.5$$

$$\therefore Q = 0.5$$

Null Hypothesis H_0 : Both rice and wheat are equally popular in the state.

Alternative Hypothesis H_1 : $P \neq 0.5$ (two - tailed alternative)

$$\text{Test statistic is } z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.54 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{1000}}} = 2.532$$

The calculated value of $z = 2.532$

The tabulated value of z at 1% level of significance for two-tailed test is 2.58.

Since calculated $z <$ tabulated z , we accept the Null Hypothesis H_0 at 1% level of significance and conclude that both rice and wheat are equally popular in the state.

Example 1 : In a big city 325 men out of 600 men were found to be smokers. Does this information support the conclusion that the majority of men in this city are smokers ?

[JNTU (K) Nov. 2009, (A) Nov. 2010, (K) May 2013 (Set No. 1)]

Solution : Given $n = 600$

Number of smokers = 325

$$p = \text{Sample proportion of smokers} = \frac{325}{600} = 0.5417$$

$$P = \text{Population proportion of smokers in the city} = \frac{1}{2} = 0.5$$

$$Q = 1 - P = 1 - 0.5 = 0.5$$

1. Null Hypothesis H_0 : The number of smokers and non-smokers are equal in the city.

2. Alternative Hypothesis : $P > 0.5$ (Right tailed)

$$\text{3. The Test statistic is } z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.5417 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{600}}} = 2.04$$

$\therefore$ Calculated value of $z = 2.04$

Tabulated value of z at 5% level of significance for right tail test is 1.645.

Since, calculated value of $z >$ tabulated value of z , we reject the Null Hypothesis and conclude that the majority of men in the city are smokers.

Example 4 : A die was thrown 9000 times and of these 3220 yielded a 3 or 4. Is this consistent with the hypothesis that the die was unbiased ? [JNTU (K) Nov. 2009 (Set No. 3)]

Solution : Given $n = 9000$

$$p = \text{Proportion of successes of getting 3 or 4 in 9000 throws} = \frac{3220}{9000} = 0.3578$$

$P =$ Population proportion of successes

$$= P(\text{getting a 3 or 4}) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3} = 0.3333$$

$$\therefore Q = 1 - P = 1 - \frac{1}{3} = \frac{2}{3} = 0.6667$$

1. Null Hypothesis H_0 : The die is unbiased.

2. Alternative Hypothesis H_1 : $P \neq \frac{1}{3}$ (two-tailed)

$$3. \text{ The test statistic is } z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.3578 - 0.3333}{\sqrt{\frac{0.3333 \times 0.6667}{9000}}} = 4.94$$

Calculated $z = 4.94$

As $z > 2.58$, the null hypothesis H_0 has to be rejected at 1% level of significance and we conclude that the die is biased.

Example 4 : A random sample of 500 pineapples was taken from a large consignment and 65 were found to be bad. Find the percentage of bad pineapples in the consignment.

Solution :

Given $n = 500$

$$p = \text{Proportion of bad pineapples in the sample} = \frac{65}{500} = 0.13$$

$$q = 1 - p = 0.87$$

We know that the limits for population proportion P are given by

$$p \pm 3 \sqrt{\frac{pq}{n}} = 0.13 \pm 3 \sqrt{\frac{0.13 \times 0.87}{500}} = 0.13 \pm 0.045 = (0.085, 0.175)$$

$\therefore$ The percentage of bad pineapples in the consignment lies between 8.5 and 17.5.

Example 6 : In a random sample of 125 cool drinkers, 68 said they prefer thumsup to pepsi. Test the null hypothesis $P = 0.5$ against the alternative hypothesis $P > 0.5$.

[JNTU 2006S (Set No. 2)]

Solution : We have $n = 125$, $x = 68$ and $p = \frac{x}{n} = \frac{68}{125} = 0.544$

1. Null Hypothesis H_0 : $P = 0.5$

2. Alternative Hypothesis H_1 : $P > 0.5$

3. **Level of significance :** $\alpha = 0.05$ (assumed)

$$4. \quad \text{The test statistic is } z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.544 - 0.5}{\sqrt{\frac{(0.5)(0.5)}{125}}} \\ = \frac{0.044}{0.045} = 0.9839$$

Since calculated value of $|Z|$ is less than 1.645, we accept the Null Hypothesis H_0 at 5% level of significance.

Example 2 : Experience had shown that 20% of a manufactured product is of the top quality. In one day's production of 400 articles only 50 are of top quality. Test the hypothesis at 0.05 level. [JNTU 2005 S, 2008 S (Set No. 2), (K) May 2013 (Set No. 4)]

Solution : We have $n = 400$, $x = 50$ and $p = \frac{x}{n} = \frac{50}{400} = 0.125$

1. **Null Hypothesis H_0 :** $P = 0.2$
2. **Alternative Hypothesis H_1 :** $P \neq 0.2$
3. **Level of significance :** $\alpha = 0.05$

$$4. \quad \text{The test statistic is } z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} \\ \text{i.e., } z = \frac{0.125 - 0.2}{\sqrt{\frac{(0.2)(0.8)}{400}}} = \frac{-0.075}{0.02} = -3.75$$

Since $|z| = 3.75 > 1.96$, we reject the Null Hypothesis H_0 at 5% level of significance.

i.e., $P = 20\%$ is not correct.

Example 3 : A social worker believes that fewer than 25% of the couples in a certain area have ever used any form of birth control. A random sample of 120 couples was contacted. Twenty of them said that they have used. Test the belief of the social worker at 0.05 level. [JNTU 2005 (Set No. 1)]

Solution : We have

$$n = 120, x = 20, p = \frac{x}{n} = \frac{20}{120} = \frac{1}{6} \text{ and } P = 0.25, Q = 1 - P = 0.75$$

1. **Null Hypothesis H_0 :** $P = 0.25$
2. **Alternative Hypothesis H_1 :** $P < 0.25$ (left tailed test)
3. **Level of significance, α :** $\alpha = 0.05$
4. **The test statistic is**

$$z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{\frac{1}{6} - 0.25}{\sqrt{\frac{(0.25)(0.75)}{120}}} = \frac{-0.0833}{0.0395} = -2.107$$

Since $|z| = 2.107 < 2.33 = z_{0.05}$, we accept the null hypothesis H_0 . That is, the claim or belief of social worker is true.

Example 9 : A manufacturer claims that only 4% of his products are defective. A random sample of 500 were taken among which 100 were defective. Test the hypothesis at 0.05 level. [JNTU 2005 (Set No. 2)]

Solution : We have

$$x = 100, n = 500, p = \frac{x}{n} = 0.2 \text{ and } P = 0.04, Q = 1 - P = 0.96$$

1. Null Hypothesis $H_0 : P = 0.04$
2. Alternative Hypothesis $H_1 : P > 0.04$ (Right tailed test)
3. Level of significance : $\alpha = 0.05$
4. The test statistic is

$$z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.2 - 0.04}{\sqrt{\frac{(0.04)(0.96)}{500}}} = \frac{-0.16}{0.00876} = -18.26$$

Since $|z| = 18.26 > 1.645 = z_{0.05}$, we reject the null hypothesis H_0 .

Example 10 : In a sample of 500 from a village in Rajasthan, 280 are found to be wheat eaters and the rest rice eaters. Can we assume that the both articles are equally popular.

[JNTU (K) 2009, May 2012 (Set No. 4)]

Solution : Given $n = 500$

$$p = \text{Sample proportion of wheat eaters} = \frac{280}{500} = 0.56$$

$$P = \text{Population proportion of wheat eaters} = \frac{1}{2} = 0.5$$

$$\therefore Q = 1 - P = 1 - 0.5 = 0.5$$

1. Null Hypothesis H_0 : Both rice and wheat are equally popular in Rajasthan.
2. Alternative Hypothesis $H_1 : P \neq 0.5$
3. Level of significance : $\alpha = 0.01$
4. The test statistic is $z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.56 - 0.5}{\sqrt{(0.5)(0.5)/500}} = \frac{0.06}{0.022} = 2.68$

$\therefore$ The calculated value of $z = 2.68$

The tabulated value of z at 1% level of significance for two-tailed test is 2.58.

Since calculated value of $z >$ tabulated z , we reject the null hypothesis H_0 . i.e., Both rice and wheat are not equally popular in Rajasthan at 1% level of significance.

Example 11 : Among 900 people in a state 90 are found to be chapati eaters. Construct 99% confidence interval for the true proportion. [JNTU 2005 S, (K) May 2010S (Set No. 2)]

Solution : Given $x = 90$, $n = 900$

$$\therefore P = \frac{x}{n} = \frac{90}{900} = \frac{1}{10} = 0.1 \text{ and } Q = 1 - P = 0.9$$

$$\text{Now } \sqrt{\frac{PQ}{n}} = \sqrt{\frac{(0.1)(0.9)}{900}} = 0.01$$

Confidence interval is

$$\left(P - z_{\alpha/2} \cdot \sqrt{\frac{PQ}{n}}, P + z_{\alpha/2} \cdot \sqrt{\frac{PQ}{n}} \right) \quad \text{or} \quad \left(P - 3 \sqrt{\frac{PQ}{n}}, P + 3 \cdot \sqrt{\frac{PQ}{n}} \right)$$

$$i.e., (0.1 - 0.03, 0.1 + 0.03)$$

$$i.e., (0.07, 0.13).$$

Example 12 : In a random sample of 160 workers exposed to a certain amount of radiation, 24 experienced some ill effects. Construct a 99% confidence interval for the corresponding true percentage. [JNTU 2004, 2006S (Set No. 1)]

Solution : We have $x = 24$, $n = 160$ and $P = \frac{24}{160} = 0.15$, $Q = 1 - P = 0.85$

$$\text{Now } \sqrt{\frac{PQ}{n}} = \sqrt{\frac{0.15 \times 0.85}{160}} = 0.028$$

Confidence interval at 99% level of significance is

$$\left(P - 3 \sqrt{\frac{PQ}{n}}, P + 3 \sqrt{\frac{PQ}{n}} \right)$$

$$i.e., (0.15 - 3 \times 0.028, 0.15 + 3 \times 0.028)$$

$$i.e., (0.065, 0.234).$$

Example 12 : In a random sample of 400 industrial accidents, it was found that 231 were due atleast partially to unsafe working conditions. Construct a 99% confidence interval for the corresponding true proportion. [JNTU 2006S, (K) May 2010S (Set No. 2)]

Solution : We have $x = 231$, $n = 400$ and $P = \frac{x}{n} = \frac{231}{400} = 0.5775$

$$\therefore Q = 1 - P = 1 - 0.577 = 0.4225$$

$$\text{Now } \sqrt{\frac{PQ}{n}} = \sqrt{\frac{0.5775 \times 0.4225}{400}} = 0.0247$$

Confidence interval is

$$\left(P - 3 \cdot \sqrt{\frac{PQ}{n}}, P + 3 \cdot \sqrt{\frac{PQ}{n}} \right)$$

$$\text{i.e., } (0.5775 - 3 \times 0.0247, 0.5775 + 3 \times 0.0247)$$

$$\text{i.e., } (0.5034, 0.6516).$$

Example 14 : 20 people were attacked by a disease and only 18 survived. Will you reject the hypothesis that the survival rate if attacked by this disease is 85% in favour of the hypothesis that is more at 5% level. [JNTU 2008S, (K) May 2010S, (H) April 2012 (Set No. 1)]

Solution :

$$n = \text{Sample size} = 20$$

$$X = \text{Number of survived people} = 18$$

$$p = \text{Proportion of survived people} = \frac{X}{n} = \frac{18}{20} = 0.9$$

$$P = 0.85$$

$$\therefore Q = 1 - P = 1 - 0.85 = 0.15$$

1. **Null Hypothesis H_0 :** $P = 0.85$
2. **Alternative Hypothesis H_1 :** $P > 0.85$ (Right tailed test)
3. **Level of significance :** $\alpha = 0.5$
4. **The test statistic is**

$$z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.9 - 0.85}{\sqrt{\frac{0.85 \times 0.15}{20}}} = \frac{0.05}{0.8} = 0.625$$

$$\therefore \text{Calculated } z = 0.625$$

$$\text{Tabulated } z \text{ at 5\% level of significance, } z_\alpha = 1.645$$

Since calculated $z <$ tabulated z , we accept the Null Hypothesis H_0

i.e., The proportion of the survived people is 0.85.

Example 11 : If 80 patients are treated with an antibiotic 59 got cured. Find a 99% confidence limits to the true population of cure. [JNTU 2004 S (Set No. 2)]

$$\textbf{Solution : } n = 80, x = 59 \text{ and } P = \frac{x}{n} = \frac{59}{80} = 0.7375$$

$$\therefore Q = 1 - P = 1 - 0.7375 = 0.2625$$

$$\text{Now } \sqrt{\frac{PQ}{n}} = \sqrt{\frac{0.7375 \times 0.2625}{80}} = 0.049$$

Confidence interval is $\left(P - z_{\alpha/2} \cdot \sqrt{\frac{PQ}{n}}, P + z_{\alpha/2} \cdot \sqrt{\frac{PQ}{n}} \right)$ where $z_{\alpha/2} = 2.58$

$$\text{or } \left(P - 3 \cdot \sqrt{\frac{PQ}{n}}, P + 3 \cdot \sqrt{\frac{PQ}{n}} \right)$$

i.e., $(0.7375 - 3 \times 0.049, 0.7375 + 3 \times 0.049)$ *i.e.*, $(0.59, 0.88)$

Example 12 : In a random sample of 100 packages shipped by air freight 13 had some damage. Construct 95% confidence interval for the true proportion of damage package.

[JNTU (H) Nov. 2010 (Set No. 2)]

Solution : Here p = Sample proportion of damage packages = $\frac{13}{100} = 0.13$

$$\therefore q = 1 - p = 1 - 0.13 = 0.87$$

$$\begin{aligned} \text{S.E. of } p &= \sqrt{\frac{PQ}{n}} = \sqrt{\frac{pq}{n}} \quad (\because P \text{ is not known, we take } p \text{ for } P) \\ &= \sqrt{\frac{0.13 \times 0.87}{100}} = 0.034 \end{aligned}$$

$\therefore$ 95% confidence limits for the population proportion of P of damage packages are $p \pm 1.96 (\text{S.E. of } p) = 0.13 \pm 1.96 (0.034) = 0.13 \pm 0.067$

i.e., the confidence limits are 0.063 and 0.197

Hence the 95% confidence interval for the true proportion of damage packages is $(0.063, 0.197)$.

Example 12 : In a hospital 480 females and 520 male babies were born in a week. Do these figures confirm the hypothesis that males and females are born in equal number?

[JNTU (K) May 2010 (Set No. 1)]

Solution : Null Hypothesis H_0 : The probability of equal proportion *i.e.* $P = 0.5$

Alternative Hypothesis H_1 : The proportion is not equal *i.e.*, $P \neq 0.5$

$$n = \text{Total number of births} = 480 + 520 = 1000$$

$$p = \text{Proportion of females born} = \frac{480}{1000} = 0.48$$

$$\text{Population proportion} = 0.5 = P, Q = 1 - P = 1 - 0.5 = 0.5$$

The test statistic is

$$z = \frac{p - P}{\sqrt{PQ/n}} = \frac{0.48 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{1000}}} = \frac{-0.02}{0.0158} = -1.265$$

$\therefore |z| = 1.265$. We use two-tailed test.

Since $|z| = 1.265 < 1.96$, we accept the Null Hypothesis H_0 at 5% level of significance and conclude that the males and females are born in equal proportions.

Example 10 : A manufacturer claimed that at least 95% of the equipment which he supplied to a factory conformed to specifications. An examination of a sample of 200 pieces of equipments received 80 were faulty. Test the claim at 0.05 level.

[JNTU (H) Nov. 2010 (Set No. 3)]

Solution : We have $P = 95\% = 0.95, n = 200$

$\therefore Q = 1 - P = 1 - 0.95 = 0.05$

$p = \text{Observed proportion of faulty pieces} = \frac{80}{200} = 0.4$

Let the Null Hypothesis be $H_0 : P = 0.95$

The Alternative Hypothesis is $H_1 : P < 0.95$

Since H_1 is one-tailed (left-tailed), we use one-tailed (left-tailed) test.

Let us assume that H_0 is true.

$$\begin{aligned} \text{The test statistic is } z &= \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.4 - 0.95}{\sqrt{\frac{0.95 \times 0.05}{200}}} \\ &= \frac{-0.55}{0.0154} = -0.0085 \end{aligned}$$

Thus we see that $z = -0.0085 > -1.645$

Hence, we accept the Null Hypothesis H_0 at 5% level of significance and conclude that the claim of the manufacturer is justified.

Example 11 : A random sample of 500 Apples was taken from a large consignment of 60 were found to be bad, obtain the 98% confidence limits for the percentage number of bad apples in the consignment.

[JNTU (H) Dec. 2011 (Set No. 2), (A) Dec. 2017]

Solution : We are given

$n = \text{Sample size} = 500$

$x = \text{Number of bad apples} = 60$

$\therefore p = \text{Proportion of bad apples} = \frac{x}{n} = \frac{60}{500} = 0.12$

Hence $q = 1 - p = 1 - 0.12 = 0.88$

$\therefore \text{Standard error of } p = \sqrt{\frac{pq}{n}} = \sqrt{\frac{0.12 \times 0.88}{500}} = 0.014$

The 98% confidence limits for p are

$$\left(p - z_{\alpha/2} \cdot \sqrt{\frac{pq}{n}}, p + z_{\alpha/2} \cdot \sqrt{\frac{pq}{n}} \right) \text{ where } z_{\alpha/2} = 2.33$$

$$= (0.12 - 2.33 \times 0.014, 0.12 + 2.33 \times 0.014)$$

$$= (0.087, 0.153)$$

Thus the confidence limits for percentage of bad apples

$$= (0.087 \times 100, 0.153 \times 100) = (8.7, 15.3)$$

Example 20 : In a study designed to investigate whether certain detonators used with explosives in coal mining meet the requirement that at least 90% will ignite the explosive when charged. It is found that 174 of 200 detonators function properly. Test the null hypothesis $P = 0.9$ against the alternative hypothesis $P \neq 0.9$ at the 0.05 level of significance.

[JNTU (H) Apr. 2012 (Set No. 2)]

Solution : Let the Null Hypothesis be $H_0 : P = 0.9$

Then the Alternative Hypothesis is $H_1 : P \neq 0.9$

Since H_1 is two - tailed, we use two - tailed test.

Let us assume that H_0 is true.

The test statistic is $z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$

If p is the observed proportion of detonators functioning properly, then $p = \frac{174}{200} = 0.87$

$$Q = 1 - P = 1 - 0.9 = 0.1 \text{ and } n = 200$$

$$\therefore z = \frac{0.87 - 0.9}{\sqrt{\frac{0.9 \times 0.1}{200}}} = \frac{-0.03}{\sqrt{\frac{0.09}{200}}} = -1.41$$

$$\therefore |z| = 1.41 < 1.645$$

Hence, we accept the Null Hypothesis at 5% level of significance and conclude that there is no significant evidence to say that the given kind of detonators fails to meet the required standard.

9.19 TEST FOR EQUALITY OF TWO PROPORTIONS (OR TEST OF SIGNIFICANCE OF DIFFERENCE BETWEEN TWO SAMPLE PROPORTIONS - LARGE SAMPLES)

Let p_1 and p_2 be the sample proportions in two large random samples of sizes n_1 and n_2 drawn from two populations having proportions P_1 and P_2 .

To test whether the two samples have been drawn from the same population.

1. The Null Hypothesis $H_0 : P_1 = P_2$

2. The Alternative Hypothesis $H_1 : P_1 \neq P_2$

3. The Test Statistic : There are two ways of computing a test statistic z .

(a) When the population proportions P_1 and P_2 are known.

In this case, $Q_1 = 1 - P_1$ and $Q_2 = 1 - P_2$ and p_1, p_2 are sample proportions.

$$\therefore \text{Standard Error of Difference} = \text{S. E. } (p_1 - p_2) = \sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}$$

Hence the test statistic is

$$z = \frac{p_1 - p_2}{\text{S.E.}(p_1 - p_2)} = \frac{p_1 - p_2}{\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}}$$

(b) When the population proportions P_1 and P_2 are not known but sample proportions p_1 and p_2 are known.

In this case we have two methods to estimate P_1 and P_2 .

(i) Method of Substitution : In this method, sample proportions p_1 and p_2 are substituted for P_1 and P_2 .

$$\therefore \text{S. E. } (p_1 - p_2) = \sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}$$

Hence the test statistic is

$$z = \frac{p_1 - p_2}{\text{S. E.}(p_1 - p_2)} = \frac{p_1 - p_2}{\sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}}$$

(ii) Method of Pooling : In this method, the estimated value for the two population proportions is obtained by pooling the two sample proportions p_1 and p_2 into a single proportion p by the formula given below.

Sample proportion of two samples or estimated value of p is given by

$$p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}, \text{ so that } q = 1 - p$$

$$\therefore \text{S. E. } (p_1 - p_2) = \sqrt{\frac{pq}{n_1} + \frac{pq}{n_2}} = \sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

Hence the test statistic is

$$z = \frac{p_1 - p_2}{\text{S.E.}(p_1 - p_2)} = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

Rejection Rule for $H_0 : P_1 = P_2$.

- (i) If $|z| > 1.96$, then reject H_0 at 5% level of significance.
- (ii) If $|z| > 2.58$, then reject H_0 at 1% level of significance.
- (iii) If $|z| > 1.645$, then reject H_0 at 10% level of significance.

SOLVED EXAMPLES

Example 6 : Random samples of 400 men and 600 women were asked whether they would like to have a flyover near their residence. 200 men and 325 women were in favour of the proposal. Test the hypothesis that proportions of men and women in favour of the proposal are same, at 5% level. [JNTU (H) May 2011 (Set No. 2)]

Solution : Given sample sizes, $n_1 = 400$, $n_2 = 600$

Proportion of men, $p_1 = \frac{200}{400} = 0.5$

Proportion of women, $p_2 = \frac{325}{600} = 0.541$

1. **Null hypothesis H_0 :** Assume that there is no significant difference between the option of men and women as far as proposal of flyover is concerned.

i.e., $H_0 : p_1 = p_2 = p$

2. **Alternative hypothesis H_1 :** $p_1 \neq p_2$ (two tailed)

3. **The test statistic is $z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$** [Method of pooling]

where $p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$

$$= \frac{400 \times \frac{200}{400} + 600 \times \frac{325}{600}}{400 + 600} = \frac{525}{1000} = 0.525$$

and $q = 1 - p = 1 - 0.525 = 0.475$

$$\therefore z = \frac{0.5 - 0.541}{\sqrt{0.525 \times 0.475 \left(\frac{1}{400} + \frac{1}{600} \right)}} = \frac{-0.041}{0.032} = -1.28$$

$$|z| = 1.28$$

Since $|z| < 1.96$, we accept the null hypothesis H_0 at 5% level of significance.

i.e., there is no difference of opinion between men and women as far as proposal of flyover is concerned.

Example 2 : Random samples of 400 men and 200 women in a locality were asked whether they would like to have a bus stop near their residence. 200 men and 40 women in favour of the proposal. Test the significance between the difference of two proportions at 5% level.

[JNTU (H) Dec. 2011 (Set No. 1)]

Solution : Let P_1 and P_2 be the population proportions in a locality who favour the bus stop.

Let the Null Hypothesis be $H_0 : P_1 = P_2$

Then the Alternative Hypothesis is $H_1 : P_1 \neq P_2$

Here n_1 = Number of men in I sample = 400

n_2 = Number of women in II sample = 200

x_1 = Number of men in favour of the proposal = 200

x_2 = Number of women in favour of the proposal = 40

$$\therefore p_1 = \frac{x_1}{n_1} = \frac{200}{400} = \frac{1}{2} \text{ and } p_2 = \frac{x_2}{n_2} = \frac{40}{200} = \frac{1}{5}$$

$$\text{We have } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{200 + 40}{400 + 200} = \frac{240}{600} = \frac{2}{5}$$

$$\therefore q = 1 - p = 1 - \frac{2}{5} = \frac{3}{5}$$

Assuming that H_0 is true, the test statistic is

$$\begin{aligned} z &= \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{\frac{1}{2} - \frac{1}{5}}{\sqrt{\frac{2}{5} \times \frac{3}{5} \left(\frac{1}{400} + \frac{1}{200} \right)}} \\ &= \frac{0.3}{0.0424} = 7.07 > 1.96 \end{aligned}$$

Since $z > 1.96$, we reject the Null Hypothesis at 5% level of significance and conclude that the difference between the two proportions is highly significant *i.e.*, there is difference between the men and women in their attitude towards the bus stop near their residence.

Example 2 : A manufacturer of electronic equipment subjects samples of two completing brands of transistors to an accelerated performance test. If 45 of 180 transistors of the first kind and 34 of 120 transistors of the second kind fail the test, what can he conclude at the level of significance $\alpha = 0.05$ about the difference between the corresponding sample proportions ?

[JNTU 2003S, 2004 (Set No. 1)]

Solution : We have $n_1 = 180$, $x_1 = 45$, $x_2 = 34$, $n_2 = 120$

$$\text{and } p_1 = \frac{x_1}{n_1} = \frac{45}{180} = 0.25, p_2 = \frac{x_2}{n_2} = \frac{34}{120} = 0.283$$

$$\therefore p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{45 + 34}{180 + 120} = \frac{79}{300} = 0.263$$

$$q = 1 - p = 1 - 0.263 = 0.737$$

1. **Null Hypothesis H_0 :** $p_1 = p_2$ i.e., there is no difference
2. **Alternative Hypothesis H_1 :** $p_1 \neq p_2$ i.e., there is a difference
3. **Level of significance :** $\alpha = 0.05$
4. **The test statistic is**

$$\begin{aligned} z &= \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \quad [\text{Method of pooling}] \\ &= \frac{0.25 - 0.283}{\sqrt{(0.263)(0.737) \left(\frac{1}{180} + \frac{1}{120} \right)}} \\ &= \frac{-0.033}{\sqrt{(0.194)(0.0134)}} = \frac{-0.033}{0.051} = -0.647 \end{aligned}$$

$$\therefore |z| = 0.647$$

Since $|z| < 1.96$, we accept the null hypothesis H_0 at 5% level of significance i.e., the difference between the proportions is not significant.

Example 4 : On the basis of their total scores, 200 candidates of a civil service examination are divided into two groups, the upper 30% and the remaining 70%. Consider the first question of the examination. Among the first group, 40 had the correct answer, whereas among the second group, 80 had the correct answer. On the basis of these results, can one conclude that the first question is not good at discriminating ability of the type being examined here ?

[JNTU 2003S, 2004, JNTU (K) Nov. 2009 (Set No. 2)]

Solution : We have $n_1 = 60$, $n_2 = 140$, $x_1 = 40$, $x_2 = 80$

$$\text{and } p_1 = \frac{x_1}{n_1} = \frac{40}{60} = \frac{2}{3} = 0.667, p_2 = \frac{x_2}{n_2} = \frac{80}{140} = 0.571$$

$$\therefore p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{40 + 80}{60 + 140} = \frac{120}{200} = 0.6$$

$$\text{and } q = 1 - p = 1 - 0.6 = 0.4$$

1. **Null Hypothesis** $H_0 : p_1 = p_2$
2. **Alternative Hypothesis** $H_1 : p_1 \neq p_2$
3. **Level of significance** : $\alpha = 0.05$ (assumed)
4. **The test statistic** is

$$z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \quad [\text{Method of pooling}]$$

$$= \frac{0.667 - 0.571}{\sqrt{(0.6)(0.4) \left(\frac{1}{60} + \frac{1}{40} \right)}} = \frac{0.096}{0.0756} = 1.27$$

Since $|z| < 1.96$, we accept the null hypothesis H_0 at 5% level of significance i.e., the first question is good enough in discriminating the ability of the students of both groups.

Note : Suppose the population proportions P_1 and P_2 are given and $P_1 \neq P_2$. If we want to test the hypothesis that the difference $(P_1 - P_2)$ in population proportions is likely to be hidden in simple samples of sizes n_1 and n_2 from the two populations respectively, then

$$z = \frac{(p_1 - p_2) - (P_1 - P_2)}{\sqrt{\left(\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2} \right)}}$$

If the sample proportions are not known, then we use

$$|z| = \frac{|P_1 - P_2|}{\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}}$$

Example 2 : A cigarette manufacturing firm claims that its brand A line of cigarettes outsells its brand B by 8%. If it is found that 42 out of a sample of 200 smokers prefer brand A and 18 out of another sample of 100 smokers prefer brand B, test whether the 8% difference is a valid claim.

Solution : Here $n_1 = 200$, $n_2 = 100$,

$$p_1 = \frac{42}{200} = 0.21, \quad p_2 = \frac{18}{100} = 0.18$$

$$\text{and } P_1 - P_2 = 8\% = \frac{8}{100} = 0.08$$

1. **Null Hypothesis H_0** : Assume that 8% difference in the sale of two brands of cigarettes is a valid claim *i.e.*,

$$H_0 : P_1 - P_2 = 0.08$$

2. **Alternative Hypothesis H_1** : $P_1 - P_2 \neq 0.08$ (Two tail test)

3. **The test statistic** is, $z = \frac{(p_1 - p_2) - (P_1 - P_2)}{\sqrt{pq\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$ [Method of pooling]

$$\text{Now } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{200(0.21) + 100(0.18)}{200 + 100} = \frac{42 + 18}{300} = 0.2$$

$$\text{and } q = 1 - 0.2 = 0.8$$

$$\therefore z = \frac{0.03 - 0.08}{\sqrt{0.2 \times 0.8 \left(\frac{1}{200} + \frac{1}{100}\right)}} = \frac{-0.05}{0.0489} = -1.02$$

$$\therefore |z| = 1.02$$

Since $|z| < 1.96$, we accept the null hypothesis H_0 at 5% level of significance. *i.e.*, 8% difference in the sale of two brands of cigarettes is a valid claim.

Example 6 : In two large populations, there are 30%, and 25% respectively of fair haired people. Is this difference likely to be hidden in samples of 1200 and 900 respectively from the two populations. [JNTU (A) Nov. 2010, (K) Nov. 2009, May 2013 (Set No. 1)]

Solution :

$$\text{Given } n_1 = 1200, n_2 = 900$$

$$P_1 = \text{Proportion of fair haired people in the first population} = \frac{30}{100} = 0.3$$

$$P_2 = \text{Proportion of fair haired people in the second population} = \frac{25}{100} = 0.25$$

1. **Null Hypothesis H_0** : Assume that the sample proportions are equal *i.e.*, the difference in population proportions is likely to be hidden in sampling *i.e.*,

$$H_0 : P_1 = P_2$$

2. **Alternative Hypothesis H_1** : $P_1 \neq P_2$

3. **The test statistic** is, $z = \frac{P_1 - P_2}{\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}}$

$$\text{where } Q_1 = 1 - P_1 = 1 - 0.3 = 0.7$$

$$Q_2 = 1 - P_2 = 1 - 0.25 = 0.75$$

$$\therefore z = \frac{0.3 - 0.25}{\sqrt{\frac{0.3 \times 0.7}{1200} + \frac{0.25 \times 0.75}{900}}} = \frac{0.05}{0.0195} = 2.56$$

$$i.e., z = 2.56$$

Since $z > 1.96$, therefore we reject the Null Hypothesis H_0 at 5% level of significance (Two-tailed test) *i.e.*, the sample proportions are not equal. Thus we conclude that the difference in population proportions is unlikely that the real difference will be hidden.

Example 4 : A company wanted to introduce a new plan of work and a survey was conducted for this purpose. Out of sample of 500 workers in one group 62% favoured the new plan and another group of sample of 400 workers 41% were against the new plan. Is there any significant difference between the two groups in their attitude towards the new plan at 5% level of significance? [JNTU (A) Dec. 2009 (Set No. 1)]

Solution : Given $n_1 = 500, n_2 = 400, p_1 = 0.62$ and $p_2 = 1 - 0.41 = 0.59$.

$$\therefore p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{500(0.62) + 400(0.59)}{500 + 400} = \frac{310 + 236}{900} = \frac{546}{900} = 0.607$$

$$\text{and } q = 1 - p = 1 - 0.607 = 0.393.$$

1. Null Hypothesis $H_0 : p_1 = p_2$ *i.e.*, there is no significant difference between the two groups in their attitude towards the new plan.

2. Alternative Hypothesis $H_1 : p_1 \neq p_2$ (two tailed test)

3. Level of significance : $\alpha = 0.05$

$$\begin{aligned} \text{4. The test statistic is } z &= \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{0.62 - 0.59}{\sqrt{(0.607)(0.393) \left(\frac{1}{500} + \frac{1}{400} \right)}} \\ &= \frac{0.03}{\sqrt{0.2385 \times 0.45}} = \frac{0.03}{0.3276} = 0.0916 \end{aligned}$$

Since $|z| < 1.96$, the null hypothesis is accepted at 5% level of significance. Hence, we conclude that there is no significant difference between the two groups in their attitude towards the new plan.

Example 8 : In a city A, 20% of a random sample of 900 school boys has a certain slight physical defect. In another city B, 18.5% of a random sample of 1600 school boys has the same defect. Is the difference between the proportions significant at 0.05 level of significance. [JNTU (K) Nov. 2009 (Set No. 1)]

Solution : We are given

$$n_1 = \text{First sample size} = 900$$

$$n_2 = \text{Second sample size} = 1600$$

$$x_1 = 20\% \text{ of } 900 = 180$$

$$x_2 = 18.5\% \text{ of } 1600 = 296$$

$$\therefore p_1 = \frac{x_1}{n_1} = \frac{180}{900} = 0.2, \quad p_2 = \frac{x_2}{n_2} = \frac{296}{1600} = 0.185$$

$$\text{Thus } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{180 + 296}{900 + 1600} = \frac{476}{2500} = 0.19$$

$$\text{Now } q = 1 - p = 1 - 0.19 = 0.81$$

1. Null Hypothesis $H_0 : p_1 = p_2$

2. Alternative Hypotheses $H_1 : p_1 \neq p_2$

3. Level of significance : $\alpha = 0.05$

4. The test statistic is $z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$

$$\begin{aligned} \text{i.e., } z &= \frac{0.2 - 0.185}{\sqrt{(0.19)(0.81) \left(\frac{1}{900} + \frac{1}{1600} \right)}} = \frac{-0.015}{\sqrt{(0.1539)(0.001736)}} \\ &= \frac{-0.015}{0.01634} = -0.918 \end{aligned}$$

$$\therefore |z| = 0.918.$$

Since $|z| < 1.96$, we accept the null hypothesis H_0 at 5% level of significance *i.e.*, there is no significant difference between the proportions.

Example 9 : In a random sample of 1000 persons from town A, 400 are found to be consumers of wheat. In a sample of 800 from town B, 400 are found to be consumers of wheat. Do these data reveal a significant difference between town A and town B, so far as the proportion of wheat consumers is concerned? [JNTU (H) Nov. 2009, (K) May 2012 (Set No. 2)]

Solution : We have

$$n_1 = \text{sample size of town A} = 1000$$

$$n_2 = \text{sample size of town B} = 800$$

$$x_1 = \text{Number of consumers of wheat from town A} = 400$$

$$x_2 = \text{Number of consumers of wheat from town B} = 400$$

$$p_1 = \text{Proportion of consumers of wheat in town A} = \frac{x_1}{n_1} = \frac{400}{1000} = 0.4$$

$$p_2 = \text{Proportion of consumers of wheat in town B} = \frac{x_2}{n_2} = \frac{400}{800} = 0.5$$

$$\therefore p = \frac{x_1 + x_2}{n_1 + n_2} = \frac{400 + 400}{1000 + 800} = \frac{800}{1800} = \frac{8}{18} = \frac{4}{9} \text{ and } q = 1 - p = 1 - \frac{4}{9} = \frac{5}{9}.$$

1. Null Hypothesis $H_0 : p_1 = p_2$ i.e., there is no difference.

2. Alternative Hypothesis $H_1 : p_1 \neq p_2$ i.e., there is a difference.

3. Level of significance : $\alpha = 0.05$

$$\begin{aligned} \text{4. The test statistic is } z &= \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{0.4 - 0.5}{\sqrt{\frac{4}{9} \times \frac{5}{9} \left(\frac{1}{1000} + \frac{1}{800} \right)}} \\ &= \frac{-0.1}{\sqrt{\frac{20}{81} \times \frac{1800}{(1000)(800)}}} = -4.242 \end{aligned}$$

$$\therefore |z| = 4.2424$$

$$\text{Also } z_\alpha = 1.96$$

Since $|z| > z_\alpha$, we reject the null hypothesis H_0 at 5% level of significance.

$\therefore$ There is significant difference between town A and town B, as the proportion of wheat consumers is concerned.

Example 10 : In a sample of 600 students of a certain college 400 are found to use ball pens. In another college from a sample of 900 students 450 were found to use ball pens. Test whether 2 colleges are significantly different with respect to the habit of using ball pens.

[JNTU (H) Nov. 2010 (Set No. 1)]

Solution : Let P_1 and P_2 be the population proportions of students who use ball pens in two colleges.

Let the Null Hypothesis be $H_0 : P_1 = P_2$

Then the Alternative Hypothesis is $H_1 : P_1 \neq P_2$

Given x_1 = Number of students who use ball pens in first college = 400

n_1 = Sample size of first sample = 600

x_2 = Number of students who use ball pens in second college = 450

n_2 = Sample size of second sample = 900

$$\therefore p_1 = \text{Proportion of students who use ball pens in first college} = \frac{x_1}{n_1} = \frac{400}{600} = \frac{2}{3}$$

$$p_2 = \text{Proportion of students who use ball pens in second college} = \frac{x_2}{n_2} = \frac{450}{900} = \frac{1}{2}$$

$$\therefore p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{400 + 450}{600 + 900} = \frac{850}{1500} = \frac{17}{30} = 0.57$$

$$q = 1 - p = 1 - \frac{17}{30} = \frac{13}{30} = 0.43$$

Assuming that H_0 is true, the test statistic is

$$\begin{aligned} z &= \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{\frac{2}{3} - \frac{1}{2}}{\sqrt{(0.57)(0.43) \left(\frac{1}{600} + \frac{1}{900} \right)}} \\ &= \frac{1}{6 \cdot \sqrt{0.2451(0.0027)}} = \frac{1}{0.1565} = 6.39 \end{aligned}$$

$$\therefore z = 6.39 > 1.96$$

Hence, we reject the Null Hypothesis at 5% level of significance and conclude that there is a significant difference between the two colleges as far as using ball pens habit is concerned.

Example 11 : 100 articles from a factory are examined and 10 are found to be defective. 500 similar articles from a second factory are found to be 15 defective. Test the significance between the difference of two proportions at 5% level. [JNTU (H) Nov. 2010 (Set No. 2)]

Solution : Let P_1 and P_2 be the proportions of defective articles in the population of articles manufactured by the two factories.

Let the Null Hypothesis be $H_0 : P_1 = P_2$

Then the Alternative Hypothesis is $H_1 : P_1 \neq P_2$

$$\text{Here } n_1 = 100, n_2 = 500, p_1 = \frac{10}{100} = 0.1 \text{ and } p_2 = \frac{15}{500} = 0.03$$

$$\text{We have } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{100(0.1) + 500(0.03)}{100 + 500} = \frac{10 + 15}{600} = \frac{25}{600} = \frac{1}{24} = 0.042$$

$$q = 1 - p = 1 - 0.042 = 0.958$$

$$\text{and S.E. of } (p_1 - p_2) = \sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} = \sqrt{(0.042)(0.958) \left(\frac{1}{100} + \frac{1}{500} \right)} = 0.022$$

Assuming that H_0 is true, the test statistic is

$$z = \frac{p_1 - p_2}{\text{S.E. of } (p_1 - p_2)} = \frac{0.1 - 0.03}{0.022} = \frac{0.07}{0.022} = 3.18$$

$$\therefore z = 3.18 > 1.96$$

Hence, we reject the Null Hypothesis at 5% level of significance and conclude that there is a significant difference between the two proportions.

Example 12 : A machine produced 20 defective articles in a batch of 400. After overhauling it produced 10 defectives in a batch of 300. Has the machine being improved after overhauling? [JNTU(K) Nov.2011 (Set No.4)]

Solution : If p_1 and p_2 be the sample proportions of imperfect items put out by the machine before and after the overhauling and P_1 and P_2 be the corresponding population proportions.

Here $n_1 = 400, x_1 = 20, n_2 = 300$ and $x_2 = 10$.

$$\text{Now } p_1 = \frac{20}{400} = \frac{1}{20}, p_2 = \frac{10}{300} = \frac{1}{30}$$

$$\text{Thus } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{30}{700} = \frac{3}{70}$$

$$\therefore q = 1 - p = 1 - \frac{3}{70} = \frac{67}{70}$$

1. **Null Hypothesis:** $H_0 : p_1 = p_2$

2. **Alternative Hypothesis:** $H_1 : p_1 > p_2$. It leads to a one-tailed test.

3. **Level of significance:** $\alpha = 0.05$

$$\begin{aligned} \text{4. Test Statistic is } z &= \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{\frac{1}{20} - \frac{1}{30}}{\sqrt{\frac{3}{70} \times \frac{67}{70} \left(\frac{1}{400} + \frac{1}{300} \right)}} \\ &= \frac{0.0167}{\sqrt{0.041(0.0058)}} = \frac{0.0167}{0.0154} = 1.0844. \end{aligned}$$

5. **Critical value:** The critical value of z at $\alpha = 0.05$ is $z_{0.05} = 1.645$ for one tailed test

6. **Decision.** Since $|z| (= 1.0844) < z_{0.05} (= 1.645)$, therefore, the Null Hypothesis is accepted

i.e., There is no significant difference.

Hence, the machine has not improved after overhauling.

Example 12 : A machine puts out 9 imperfect articles in a sample of 200 articles. After the Machine is overhauled it puts out 5 imperfect articles in a sample of 700 articles. Test at 5% level whether the Machine is improved? [JNTU (H) Nov. 2010 (Set No. 3)]

Solution : Let P_1 and P_2 be the proportions of imperfect articles in the population of articles manufactured by the machine before and after overhauling, respectively.

Let the Null Hypothesis be $H_0 : P_1 = P_2$

The Alternative Hypothesis is $H_1 : P_1 > P_2$

We have $p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$ and S.E. of $(p_1 - p_2) = \sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$

Here $n_1 = 200, p_1 = \frac{9}{200} = 0.045, n_2 = 700, p_2 = \frac{5}{700} = 0.00714$

$$\therefore p = \frac{0.045 \times 200 + 0.00714 \times 700}{200 + 700} = \frac{9 + 5}{900} = \frac{14}{900} = 0.0156$$

and $q = 1 - p = 1 - 0.0156 = 0.9844$

$$\therefore \text{S.E. of } (p_1 - p_2) = \sqrt{0.0156 \times 0.9844 \left(\frac{1}{200} + \frac{1}{700} \right)} = 0.0099$$

Assuming that H_0 is true, the test statistic is

$$z = \frac{p_1 - p_2}{\text{S.E. of } (p_1 - p_2)} = \frac{0.045 - 0.00714}{0.0099} = \frac{0.03786}{0.0099} = 3.824$$

Since $z = 3.824 > 3$, we reject the Null Hypothesis at 5% level of significance and conclude that the machine is improved.

Example 14 : A machine puts out 16 imperfect articles in a sample of 500 articles. After the Machine is overhauled it puts out 3 imperfect articles in a sample of 100 articles. Has the Machine improved? [JNTU (H) Dec. 2011 (Set No. 3)]

Solution : Let P_1 and P_2 be the proportions of imperfect articles in the population of articles manufactured by the machine before and after overhauling, respectively.

Let the Null Hypothesis be $H_0 : P_1 = P_2$

Then the Alternative Hypothesis is $H_1 : P_1 > P_2$

Here

n_1 = Sample size before the machine overhauling = 500

x_1 = No. of imperfect articles before overhauling = 16

n_2 = Sample size after the machine overhauling = 100

x_2 = No. of imperfect articles after overhauling = 3

$$\therefore p_1 = \frac{x_1}{n_1} = \frac{16}{500} = 0.032 \text{ and } p_2 = \frac{x_2}{n_2} = \frac{3}{100} = 0.03 \text{ and } p_1 > p_2$$

$$\text{We have } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{16 + 3}{500 + 100} = \frac{19}{600} = 0.032$$

and $q = 1 - p = 0.968$

Since H_1 is one - tailed (right - tailed), we use one-tailed test (right-tailed test).

Let us assume that H_0 is true. The test statistic is

$$z = \frac{p_1 - p_2}{\sqrt{pq\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{0.032 - 0.03}{\sqrt{0.032 \times 0.968\left(\frac{1}{500} + \frac{1}{100}\right)}} \\ = \frac{0.002}{0.019} = 0.104$$

Thus we see that $z = 0.104 < 1.645$. Hence we accept the Null Hypothesis H_0 at 5% level of significance and conclude that the machine has improved.

Example 12 : In an investigation on the machine performance the following results are obtained.

	No.of units inspected	No.of defectives
Machine 1	375	17
Machine 2	450	22

Test whether there is any significant performance of two machines at $\alpha = 0.05$

[JNTU (H) Nov. 2010 (Set No. 4)]

Solution : Let P_1 and P_2 be the proportions of defective units in the population of units inspected in Machine 1 and Machine 2 respectively.

Let the Null Hypothesis be $H_0 : P_1 = P_2$

The Alternative Hypothesis is $H_1 : P_1 > P_2$

$$p_1 = \frac{17}{375} = 0.045, \quad p_2 = \frac{22}{450} = 0.049 \quad \text{and} \quad p_1 > p_2$$

$$\text{We have } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} \quad \text{and S.E. of } (p_1 - p_2) = \sqrt{pq\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

$$\text{Here } n_1 = 375, n_2 = 450, p_1 = 0.045, p_2 = 0.049$$

$$\therefore p = \frac{375 \times 0.045 + 450 \times 0.049}{375 + 450} = \frac{17 + 22}{375 + 450} = \frac{39}{825} = 0.047$$

$$q = 1 - p = 1 - 0.047 = 0.953$$

$$\therefore \text{ S.E. of } (p_1 - p_2) = \sqrt{0.047 \times 0.953\left(\frac{1}{375} + \frac{1}{450}\right)} = 0.015$$

Assuming that H_0 is true, the test statistic is

$$z = \frac{p_1 - p_2}{\text{S.E. of } (p_1 - p_2)} = \frac{0.045 - 0.049}{0.015} = -0.267$$

$$\therefore |z| = 0.267 < 1.96$$

Hence, we accept the Null Hypothesis at 5% level of significance and conclude that there is no significant difference in performance of two machines.

Example 14 : Among the items produced by a factory out of 500, 15 were defective, in another sample out of 400, 20 were defective. Test the significance between the differences of two proportions at 5% level. [JNTU (H) Dec. 2011 (Set No. 2)]

Solution : Let P_1 and P_2 be the proportions of defective items in the population of two sample items produced by the factory.

Let the Null Hypothesis be $H_0 : P_1 = P_2$

Then the Alternative Hypothesis is $H_1 : P_1 \neq P_2$.

Here $n_1 = 500, n_2 = 400, x_1 = 15$ and $x_2 = 20$

$$\therefore p_1 = \frac{x_1}{n_1} = \frac{15}{500} = \frac{3}{100} = 0.03 \text{ and } p_2 = \frac{x_2}{n_2} = \frac{20}{400} = \frac{1}{20} = 0.05$$

$$\text{We have } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{15 + 20}{500 + 400} = \frac{35}{900} = 0.039$$

$$\text{Now } q = 1 - p = 1 - 0.039 = 0.961$$

Assuming that H_0 is true, the test statistic is

$$\begin{aligned} z &= \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{0.03 - 0.05}{\sqrt{0.039 \times 0.961 \left(\frac{1}{500} + \frac{1}{400} \right)}} \\ &= \frac{-0.02}{0.013} = -1.54 \end{aligned}$$

$$\therefore |z| = 1.54 < 1.96$$

Hence we accept the Null Hypothesis at 5% level of significance and conclude that there is no significant difference between the two proportions.

Example 12 : Among the items produced by a factory out of 800, 65 were defective in another sample out of 300, 40 were defective. Test the significance between the differences of two proportions at 1% level. [JNTU (H) Apr. 2012 (Set No. 2), (A) Dec. 2017]

Solution : Let P_1 and P_2 be the proportions of defective items in the population of items produced by a factory in two samples respectively.

Let the Null Hypothesis be $H_0 : P_1 = P_2$

The Alternative Hypothesis is $H_1 : P_1 < P_2$

Since H_1 is one - tailed (left - tailed), we use one - tailed test (left - tailed test)

$$\text{Here } n_1 = 800, x_1 = 65, p_1 = \frac{65}{800} = 0.081, n_2 = 300, x_2 = 40, p_2 = \frac{40}{300} = 0.1333$$

and $p_1 < p_2$

$$\therefore p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{65 + 40}{800 + 300} = \frac{105}{1100} = 0.095$$

and $q = 1 - p = 1 - 0.095 = 0.905$

Assuming that H_0 is true, the test statistic is

$$\begin{aligned} z &= \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{0.081 - 0.1333}{\sqrt{0.095 \times 0.905 \left(\frac{1}{800} + \frac{1}{300} \right)}} \\ &= \frac{-0.0523}{\sqrt{0.086 \times 0.00459}} = \frac{-0.0523}{0.01986} = -2.6324 \end{aligned}$$

Thus we see that $z = -2.6324 < -2.33$

Hence, we reject the Null Hypothesis H_0 at 1% level of significance and conclude that there is a significant difference between the two proportions.

Example 10 : Before an increase on excise duty on tea 500 people out of a sample of 900 found to have the habit of having tea. After an increase on excise duty 250 are have the habit of having tea among 1100. Is there any decrease in the consumption of tea. Test at 5% level.

[JNTU (H) Nov. 2010 (Set No. 4)]

Solution : Let P_1 and P_2 be the population proportions of people who have the habit of tea before and after an increase on excise duty on tea respectively.

Let the Null Hypothesis be $H_0 : P_1 = P_2$

The Alternative Hypothesis is $H_1 : P_1 > P_2$

$$\text{Now } p_1 = \frac{500}{900} = 0.556, p_2 = \frac{250}{1100} = 0.227$$

$$\text{We have } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} \text{ and S.E. of } (p_1 - p_2) = \sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

Here $n_1 = 900, n_2 = 1100, p_1 = 0.556, p_2 = 0.227$

$$\therefore p = \frac{900 \times 0.556 + 1100 \times 0.227}{900 + 1100} = 0.375$$

$$q = 1 - p = 1 - 0.375 = 0.625$$

$$\therefore \text{ S.E. of } (p_1 - p_2) = \sqrt{0.375 \times 0.625 \left(\frac{1}{900} + \frac{1}{1100} \right)} = 0.022$$

Assuming that H_0 is true, the test statistic is

$$z = \frac{p_1 - p_2}{\text{S.E. of } (p_1 - p_2)} = \frac{0.556 - 0.227}{0.022} = 14.95$$

Hence, we reject the Null Hypothesis at 5% level of significance and conclude that there is a significant difference in the consumption of tea before and after an increase on excise duty on tea.

Example 14 : During a country wide investigation the incidence of tuberculosis was found to be 1%. In a college of 400 students 3 reported to be affected, where as in another college of 1200 students 10 were affected. Does this indicate any significant difference?

[JNTU (H) Apr. 2012 (Set No. 3)]

Solution : Let P_1 and P_2 be the population proportions of students who are affected tuberculosis in two colleges respectively.

Let the Null Hypothesis be $H_0 : P_1 = P_2$

Then the Alternative Hypothesis is $H_1 : P_1 \neq P_2$

Here $n_1 = 400, x_1 = 3, n_2 = 1200, x_2 = 10$

$$\therefore p_1 = \frac{x_1}{n_1} = \frac{3}{400} = 0.0075 \text{ and } p_2 = \frac{x_2}{n_2} = \frac{10}{1200} = 0.0083$$

Given $p = 0.01$ and $q = 1 - P = 0.99$

Assuming that H_0 is true, the test statistic is $z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$

$$\therefore z = \frac{0.0075 - 0.0083}{\sqrt{0.01 \times 0.99 \left(\frac{1}{400} + \frac{1}{1200} \right)}} = \frac{-0.0008}{0.0057} = -0.14$$

$$\therefore |z| = 0.14 < 1.96$$

Hence, we accept the Null Hypothesis at 5% level of significance and conclude that there is no significant difference between the two colleges as far as the incidence of tuberculosis concerned.

Example 10 : A random sample of 300 shoppers at a supermarket includes 204 who regularly use cents off coupons. In another sample of 500 shoppers at a supermarket includes 75 who regularly use cents off coupons

(a) Construct confidence interval for the probability that any one shopper at the supermarket, selected at random, will regularly use cents off coupons.

(b) Test the significance between the difference of two proportions at 2% level.

[JNTU (H) Apr. 2012 (Set No. 4)]

Solution : Here we are given

$$n_1 = 300, x_1 = 204, n_2 = 500 \text{ and } x_2 = 75$$

$$p_1 = \text{Proportion of shoppers who use cents off coupons in the first sample} = \frac{204}{300} = 0.68$$

$p_2 = \text{Proportion of shoppers who use cents off coupons in the second sample}$

$$= \frac{75}{500} = 0.15$$

- (a) The 98% confidence interval for the probability that any one shopper in sample selected at random is

$$\left(p_1 - z_{\alpha/2} \cdot \sqrt{\frac{p_1 q_1}{n_1}}, p_1 + z_{\alpha/2} \cdot \sqrt{\frac{p_1 q_1}{n_1}} \right) \text{ where } q_1 = 1 - p_1 = 1 - 0.68 = 0.32$$

$$\text{i.e., } \left(0.68 - (2.33) \sqrt{\frac{0.68 \times 0.32}{300}}, 0.68 + (2.33) \sqrt{\frac{0.68 \times 0.32}{300}} \right)$$

$$\text{i.e., } (0.68 - 0.063, 0.68 + 0.063) \text{ or } (0.62, 0.74)$$

- (b) Here we set up the Null Hypothesis H_0 that $p_1 = p_2$

i.e., the sample proportions are equal

i.e., there is no difference in the two proportions.

Then the Alternative Hypothesis is $H_1 : p_1 \neq p_2$

$$\text{Now } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{204 + 75}{300 + 500} = \frac{279}{800} = 0.35$$

$$\therefore q = 1 - p = 1 - 0.35 = 0.65$$

Hence the test statistic is

$$\begin{aligned} z &= \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \\ &= \frac{0.68 - 0.15}{\sqrt{0.35 \times 0.65 \left(\frac{1}{300} + \frac{1}{500} \right)}} \\ &= \frac{0.53}{0.035} = 15.14 \end{aligned}$$

Since $z = 15.14 > 2.58$, we reject the Null Hypothesis at 1% level of significance, and conclude that there is a significant difference in the two proportions.

Example 21 : A sample poll of 300 voters from district A and 200 voters from district B showed that 56% and 48% respectively, were in favour of a given candidate. At a 0.05 level of significance, test the hypothesis that there is a difference in the districts.

[JNTU (A) April 2012 (Set No. 3)]

Solution : We are given $n_1 = 300, n_2 = 200$ and $P_1 = \frac{56}{100} = 0.56, P_2 = \frac{48}{100} = 0.48$

1. Null Hypothesis $H_0 : P_1 = P_2$ i.e., There is no significant difference in the districts.
2. Alternative Hypothesis $H_1 : P_1 \neq P_2$
3. The level of significance, $\alpha = 0.05$
4. The test statistic is

$$z = \frac{P_1 - P_2}{\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}} \quad \text{where } Q_1 = 1 - P_1 = 1 - 0.56 = 0.44$$

$$\text{and } Q_2 = 1 - P_2 = 1 - 0.48 = 0.52$$

$$\therefore z = \frac{0.56 - 0.48}{\sqrt{\frac{0.56 \times 0.44}{300} + \frac{0.48 \times 0.52}{200}}} = \frac{0.08}{0.045} = 1.78$$

Since $z < 1.96$, we accept the Null Hypothesis at 5% level of significance and conclude that there is no significant difference between the two districts towards the favour of a given candidate.

Example 22 : A study shows that 16 of 200 tractors produced on one assembly line required extensive adjustments before they could be shipped, while the same was true for 14 of 400 tractors produced on another assembly line. At the 0.01 level of significance, does this support the claim that the second production line does superior work ?

[JNTU (A) April 2012 (Set No. 3)]

Solution : We are given

$$n_1 = \text{sample size of first sample} = 200$$

$$n_2 = \text{sample size of second sample} = 400$$

$$x_1 = \text{number of tractors on first assembly line which required adjustments} = 16.$$

$$x_2 = \text{number of tractors on second assembly line which required adjustments} = 14.$$

$$\text{Now } p_1 = \frac{x_1}{n_1} = \frac{16}{200} = 0.08 \quad \text{and} \quad p_2 = \frac{x_2}{n_2} = \frac{14}{400} = 0.035$$

$$\therefore p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{16 + 14}{200 + 400} = \frac{30}{600} = \frac{1}{20} = 0.05$$

$$q = 1 - p = 1 - 0.05 = 0.95$$

1. **Null Hypothesis** $H_0 : p_1 = p_2$
2. **Alternative Hypothesis** $H_1 : p_1 < p_2$
3. **Level of significance** $\alpha = 0.05$
4. The test statistic is

$$z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{0.08 - 0.035}{\sqrt{0.05 \times 0.95 \left(\frac{1}{200} + \frac{1}{400} \right)}}$$

$$= \frac{0.045}{\sqrt{0.0475 \times 0.0075}} = \frac{0.045}{0.0189} = 2.384$$

Since $z > 1.96$, we reject the Null Hypothesis at 5% level and conclude that the second production is not superior than first.

Example 22 : If 120 out of 200 patients suffering from a certain disease are cured by allopathy and 240 out of 500 patients are cured by homeopathy, is there reason enough to believe that allopathy is better than homeopathy in curing the disease ? Use $\alpha = 0.05$ LOS.

[JNTU (A) Dec. 2014]

Solution : Let p_1, p_2 respectively be the proportions of patients suffering from a certain disease are cured by allopathy and homeopathy.

We are given $n_1 = 200, x_1 = 120, n_2 = 500$ and $x_2 = 240$

$$\text{Now } p_1 = \frac{x_1}{n_1} = \frac{120}{200} = 0.6, p_2 = \frac{x_2}{n_2} = \frac{240}{500} = 0.48$$

$$\text{Thus } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{120 + 240}{200 + 500} = \frac{360}{700} = 0.51$$

$$\therefore q = 1 - p = 1 - 0.51 = 0.49$$

1. **Null Hypothesis** is $H_0 : p_1 = p_2$
i.e. there is no difference between allopathy and homeopathy
2. **Alternative Hypothesis** $H_1 : p_1 > p_2 \Rightarrow$ It leads to a one-tailed test
i.e. Allopathy is better than homeopathy

3. **Level of significance**, $\alpha = 0.05$

4. **Test statistic** is $z = \frac{p_1 - p_2}{\sqrt{pq\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$

$$\text{i.e. } z = \frac{0.60 - 0.48}{\sqrt{0.51 \times 0.49 \left(\frac{1}{200} + \frac{1}{500}\right)}} = 2.9$$

5. **Critical value** : The critical value of z at $\alpha = 0.05$ is $z_{0.05} = 1.645$ for one-tailed test

6. **Decision** : Since the calculated value of $|z| (= 2.9) > z_{0.05} (= 1.645) \Rightarrow$ that the null hypothesis is rejected
 $\Rightarrow$ Allopathy is better than homeopathy in curing particular disease.

REVIEW QUESTIONS

- Write the Test Statistic :
 - in the test of significance for single mean.
 - for the test of significance for single proportion and
 - for the test of significance for difference of proportion.

EXERCISE

- In a random sample of 400 persons from a large population, 120 are females. Can it be said that males and females are in the ratio 5 : 3 in the population ? use 1% level of significance.
- A coin is tossed 900 times and heads appear 490 times. Does this result support the hypothesis that the coin is unbiased.
 - A coin is tossed 10,000 times and it turns up head 5195 times. Test the hypothesis that the coin is unbiased. Use a 0.01 level of significance.
 - A die is tossed 9000 times and it falls with 5 or 6 upwards 3240 times. Test the hypothesis that the die is unbiased. Use a 0.01 level of significance.
- A wholesaler in apples claims that only 4% of the apples supplied by him are defective. A random sample of 600 apples contained 36 defective apples. Test the claim of the wholesaler.
- In a sample of 500 people in Tamil Nadu 280 are tea drinkers and the rest are coffee drinkers. Can we assume that both coffee and tea are equally popular in this state at 1% level of significance.

5. A manufacturer claimed that atleast 98% of the steel pipes which he supplied to a factory conformed to specifications. An examination of a sample of 500 pieces of pipes revealed that 30 were defective. Test his claim at a significance level 5%.
6. The machine puts out 16 imperfect articles in a sample of 500. After machine is overhauled, it puts out 3 imperfect articles in a batch of 100. Has the machine improved ?
7. In a sample of 600 students of a certain college 400 are found to use ball pens. In another college, from a sample of 900 students 450 were found to use ball pens. Test whether the two colleges are significantly different with respect to the habit of using ball pens. **[JNTU (H) Nov. 2010 (Set No. 1)]**
8. In a certain district *A*, 450 persons were considered regular consumers of tea out of a sample of 1000 persons. In another district *B*, 400 were regular consumers of tea out of a sample of 800 persons. Do these facts reveal a significant difference between the two districts as far as tea drinking habit is concerned ?
9. (i) A machine produced 10 defective articles in a batch of 200. After overhauling it produced 4 defectives in a batch of 100. Has the machine been improved.
(ii) A machine puts out 21 defective articles in a sample of 500 articles. Another machine gives 3 defective articles in a sample of 100. Are the two machines significantly different in their performance ?
(iii) A machine puts out 10 imperfect articles in a sample of 200. After the machine is overhauled it puts out 4 imperfect articles in a batch of 100. Has the machine been improved ? **[JNTU (K) May 2012 (Set No. 1)]**
10. A random sample of 100 articles selected from a batch of 2,000 articles shows that the average diameter of the articles = 0.354 with a *S.D.* = 0.048. Find 95% confidence interval for the average of this batch of 2,000 articles.
11. A sample of 100 iron bars is said to be drawn from a large number of bars whose lengths are normally distributed with mean 4 feet and *S.D.* 0.6 feet. If the sample mean is 4.2 feet, can the sample be regarded as a truly random sample.
12. The mean life of a sample of 100 electric bulbs produced by a company is found to be 1570 hrs with a *S.D.* of 120 hrs. If μ is the mean life time of all the bulbs produced by the company, test the hypothesis $\mu = 1600$ hrs against the alternative hypothesis $\mu \neq 1600$ hrs at 5% level of significance.
13. A sample of 400 male students is found to have a mean height of 171.38 cm. Can it be reasonably regarded as a sample from a large population with mean height of 171.17 cm and *S.D.* 3.30 cm.
14. A sample of 100 workers in a large plant gave a mean assembly time of 294 seconds with a *S.D.* of 12 seconds in a time and motion study. Find a 95% confidence interval for the mean assembly time for all the workers in the plant.

15. The mean breaking strength of the cables supplied by a manufacturer is 1800 with a *S.D.* 100. By a new technique in the manufacturing process, it is claimed that the breaking strength of the cables have increased. In order to test this claim, a sample of 50 cables is tested. It is found that the mean breaking strength is 1850. Can we support the claim at 1% level of significance.
16. An investigation of the relative merits of two kinds of flash light batteries showed that a random sample of 100 batteries of brand A tested on the average 36.5 hrs with a *S.D.* of 1.8 hrs while a random sample of 80 batteries of brand B tested on the average 36.8 hrs with a *S.D.* of 1.5 hrs. Use a level of significance of 0.05 to test whether the observed difference between the average life times is significant.
17. Given the following information relating to two places *A* and *B*. Test whether there is any significant difference between their mean wages :

	<i>A</i>	<i>B</i>
Mean wages (Rs.)	47	49
<i>S.D.</i> (Rs.)	28	40
Number of workers	1000	1500

18. The mean consumption of food grains among 400 sampled middle class consumers is 380 gms per day per person with a *S.D.* of 120 gms. A similar sample survey of 600 working class consumers gave a mean of 410 gms with a *S.D.* of 80 gms. Are we justified in saying that the two classes consume the same quantity of food grains. Use 5% level of significance.
19. In a city 250 men out of 750 were found to be smokers. Does this information support the conclusion that the majority of men in this city are smokers.

[JNTU 2005S, 2006S, 2008S (Set No. 1)]

20. Test the significance of the difference between the means of the sample from the following data :

	Size of sample	Mean	S.D
Sample A	100	61	4
Sample B	200	63	6

21. In a random sample of 100 tube lights produced by company A, the mean life time of tube light is 1190 hours with standard deviation of 90 hours. Also in a random sample of 75 tube lights from company B the mean life time is 1230 hours with standard deviation of 120 hours. Is there a difference between the mean life times of the two brands of tube lights at a significance level of 0.05?
22. 500 articles from a factory are examined and found to be 2% defective. 800 similar articles from a second factory are found to have only 1.5 % defective. Can it reasonably be concluded that the products of the first factory are inferior to those of second ?

[JNTU (K) Nov. 2011 (Set No. 2)]

ANSWERS

1. $Z = -3.125$; The males and females in the population are not in the ratio 5 : 3.
2. (i) $H_0 : P = \frac{1}{2}$ (coin is unbiased), $Z = 2.39$, rejected at 5% level.
 (ii) $H_0 : P = \frac{1}{2}$ (coin is unbiased), $Z = 3.9$, rejected at 1% level.
 (iii) $H_0 : P = \frac{1}{3}$ (die is unbiased), $Z = 5.4$, rejected at 1% level.
3. $H_0 : P = 0.04$; $H_1 : P > 0.04$, $Z = 10$, highly significant.
4. $H_0 : P = \frac{1}{2}$, $H_1 : P \neq \frac{1}{2}$, $Z = 2.68$, significant at 1% level.
5. $H_0 : P = 0.98$, $H_1 : P < 0.98$ (left-tail test), $Z = 6.38$, H_0 rejected.
6. $H_0 : P_1 = P_2$, $H_1 : P_2 < P_1$, $Z = 1.04$, not significant.
7. $H_0 : P_1 = P_2$; $H_1 : P_1 \neq P_2$; $|Z| = 6.48$, significant.
8. $H_0 : P_1 = P_2$, $H_1 : P_1 \neq P_2$, $|Z| = 2.08$, significant at 5% level.
9. (i) $H_0 : P_1 = P_2$, $H_1 : P_1 > P_2$, $|Z| = 0.3846$, not significant.
 (i.e., not improved the performance of the machine)
10. 0.34, 0.36.
11. $H_0 : \mu = 4$ ft, $H_1 : \mu \neq 4$ ft, $Z = 3.33$ significant.
12. $|Z| = 2.5$, significant at 5% level of significance.
14. 291.64, 296.35.
15. $H_0 : \mu = 1800$, $\sigma = 100$, $H_1 : \mu_1 > 1800$, $Z = 3.53$, Accept H_0
16. $H_0 : \bar{x}_1 = \bar{x}_2$; $|Z| = 1.21$, Accept H_0
17. $H_0 : \bar{x}_1 = \bar{x}_2$, $|Z| < 1.96$, Accept H_0
18. $H_0 : \bar{x}_1 = \bar{x}_2$, $H_1 : \bar{x}_1 \neq \bar{x}_2$, $|Z| = 4.39$ significant.
19. $H_0 : p = \frac{1}{3}$, $H_1 : P < 0.5$; $z = -9.128$, H_0 rejected.
21. $|z| = 2.42$

QUIZ

1. A hypothesis is true, but is rejected, this is an error of type []
(a) I (b) II
(c) I and II (d) None
2. A hypothesis is false, but accepted, this is an error of type []
(a) I (b) II
(c) I and II (d) None
3. A random sample of 400 products contains 52 defective items. Standard error of proportion is []
(a) 0.168 (b) 0.0168
(c) 0.0016 (d) 1.68
4. 500 eggs are taken from a large consignment and 50 are found to be bad. Standard error of proportion is []
(a) 1.3 (b) 0.13
(c) 0.013 (d) None
5. Among 900 people in a state 90 are found to be chapati eaters. The 99% confidence interval for the true proportion is []
(a) (0.08,0.12) (b) (0.8,1.2)
(c) (0.07,0.13) (d) None
6. In a random sample of 160 workers exposed to a certain amount of radiation, 24 experienced some ill effects. The 99% confidence interval for the true proportion is []
(a) (0.065,0.234) (b) (0.65,0.23)
(c) (0.56,0.023) (d) None
7. A manufacturer of electric bulbs claims that the percentage defectives in his product does not exceed 6. A sample of 40 bulbs is found to contain 5 defectives. The test statistic $|Z|$ = []
(a) 0.24 (b) 1.24
(c) 0.024 (d) 1.12
8. In a city 250 men out of 750 men were found to be smokers. The test statistic $|Z|$ = []
(a) 2.25 (b) 2.5
(c) 5.25 (d) 5.50
9. A die is thrown 256 times. An even digit turns up 150 times. Then the die is []
(a) biased (b) unbiased
(c) not determined (d) None

10. A single - tailed test is used when _____
11. Null hypothesis is defined as _____

ANSWERS

1. *a* 2. *b* 3. *b* 4. *c*
5. *c* 6. *a* 7. *b* 8. *c*
9. *b*
10. We are testing the hypothesis that one process is better than another.
11. The hypothesis formulated for the sake of rejecting it, under the assumption that it is true.

PREVIOUS GATE QUESTIONS

1. Type II error in hypothesis testing is [GATE 2016 (CE)]
(A) acceptance of the null hypothesis when it is false and should be rejected.
(B) rejection of the null hypothesis when it is true and should be accepted.
(C) rejection of the null hypothesis when it is false and should be rejected.
(D) acceptance of the null hypothesis when it is true and should be accepted.
2. For accepting or rejecting a null hypothesis, which one of the following is NOT used as a significance test method in Statistics ? [GATE 2018 (AG)]
(A) *z*- test
(B) Student's *t*-test
(C) Pearson's correlation
(D) Relative standard deviation
3. Standard error is [GATE 2018 (BT)]
(A) the probability of a type I error in a statistical test
(B) the error in estimating a sample standard deviation
(C) the standard deviation of a variable that follows standard normal distribution.
(D) the standard deviation of distribution of sample means

ANSWERS

1. A 2. D 3. D



UNIT - V

SMALL SAMPLE TESTS

10. *Test of Significance (Small Sample Tests)*

- ♦ *Student t-distribution*
- ♦ *Chi-Square (χ^2) Distribution*
- ♦ *F-distribution*
- ♦ *Students 't'-Test*
- ♦ *F-Test*
- ♦ *Chi-Square (χ^2) Test*



CHAPTER - 10

TEST OF SIGNIFICANCE (SMALL SAMPLE TESTS)

10.1 INTRODUCTION

In the earlier chapter, we considered certain tests of significance based on the theory of the normal distribution. The assumptions made in deriving those tests will be valid only for large samples. When the sample is small ($n < 30$), we can use normal distribution to test for a specified population mean or difference of two population means as in large sample tests only when the sample is drawn from a normal population whose S.D., σ is known. If σ is not known, we cannot proceed as above. If a population is normally distributed, the sampling distribution of the sample mean for any sample size is also normally distributed whether σ is known or not.

10.2 DEGREE OF FREEDOM (d.f.)

[JNTU (K) Dec. 2013 (Set No. 2)]

The number of independent variates which make up the statistic is known as the degree of freedom (d.f.) and it is denoted by ν (the letter 'Nu' of the Greek alphabet). In other words, it is the number of values in a set of data which may be assigned arbitrarily or, it refers to the number of "independent constraints" in a set of data.

Definition : It is a number which indicates how many of the values of a variable may be independently (or freely) chosen. For example, if we have to choose any four numbers freely, then we may choose 11, 6, 14, 28 or any other set of four numbers. As in this case all the four numbers have freedom to vary, we say that the degrees of freedom is 4. If we impose a restriction on the numbers, say the sum is 50, then we can choose first three numbers freely and the fourth number is such that the sum is 50. Thus, the three variables are free and independent choices for finding the fourth. Hence these are the degrees of freedom. In this case, the degrees of freedom is 3, i.e., $4 - 1$.

In general, the number of degrees of freedom is equal to the total number of observations less the number of independent constraints imposed on the observations. For example, in a set of data of n observations, if k is the number of independent constraints then $\nu = n - k$.

10.3 t- DISTRIBUTION (OR) STUDENT'S t - DISTRIBUTION

[JNTU (H) Dec. 2011 (Set No. 1)]

It is used for testing of hypothesis when the sample size is small and population S.D. σ is not known.

Definition : If $\{x_1, x_2, \dots, x_n\}$ be any random sample of size n drawn from a normal (or approximately normal) population with mean μ and variance σ^2 , then the

test statistic t is defined by $t = \frac{\bar{x} - \mu}{S / \sqrt{n}}$ where $\bar{x}$ = sample mean and

$S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ is an unbiased estimate of σ^2 . The test statistic $t = \frac{\bar{x} - \mu}{S/\sqrt{n}}$ is a random variable having the t -distribution with $\nu = n - 1$ degrees of freedom and with probability density function $f(t) = y_0 \left(1 + \frac{t^2}{\nu}\right)^{\frac{\nu+1}{2}}$ where $\nu = n - 1$ and y_0 is a constant got by $\int_{-\infty}^{\infty} f(t) dt = 1$. This is known as “Student’s t -distribution” or simply “ t distribution”.

The above result is more general than the central limit theorem, since it does not require the knowledge of σ , and since population is assumed to be normal it is less general than the central limit theorem.

Properties of t -Distribution :

[JNTU (A) Nov. 2011, (H) Dec. 2011, (K) Dec. 2013 (Set No.2)]

1. The shape of t -distribution is **bell-shaped**, which is similar to that of a normal distribution and is symmetrical about the mean.

2. The t -distribution curve is also asymptotic to the t -axis, i.e., the two tails of the curve on both sides of $t = 0$ extends to infinity.

3. It is symmetrical about the line $t = 0$.

4. The form of the probability curve varies with degrees of freedom i.e., with sample size.

5. It is unimodal with Mean = Median = Mode.

6. The *mean* of standard normal distribution and as well as t -distribution is zero but the variance of t -distribution depends upon the parameter ν which is called the degrees of freedom.

7. The **variance** of t -distribution exceeds 1, but approaches 1 as $n \rightarrow \infty$. Infact the t -distribution with ν -degrees of freedom approaches standard normal distribution as $\nu = (n-1) \rightarrow \infty$.

The selected values of t_α for various values of ν can be obtained from the tables of t -distribution, where t_α denotes the area under t -distribution to its right is equal to α . These values are tabulated in tables. In tables the left-hand column contains values of ν , the column heading are areas α in the right-hand tail of the t -distribution, and the entries are values of t_α . It is not necessary to tabulate values of t_α for $\alpha < 0.50$, as it follows from the symmetry of the t -distribution that $t_{1-\alpha} = -t_\alpha$ i.e., the t -value leaving an area of $1-\alpha$ to the right and therefore an area α to its left, is equal to the negative t -value which leads an area α in the right. The bottom row of the table of t -distribution correspond to the values of Z that cut off right hand tails of area α under the standard normal curve. We use the notation Z_α for such a value of Z , it can be seen for example, that $Z_{0.025} = 1.96 = t_{0.025}$ for $\nu = \infty$.

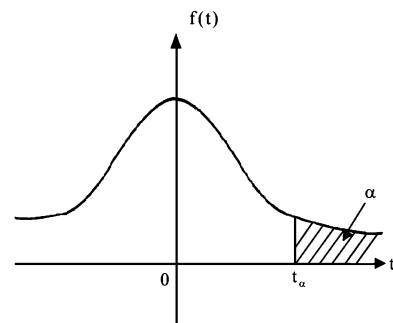


Fig. t -distribution

This result should really have been expected, since the t -distribution approaches the standard normal distribution as $v \rightarrow \infty$. In fact, if we observe, we find that the value of t_α for 29 or more degrees of freedom are close to the corresponding values of Z ; we conclude that the standard normal distribution provides a good approximation to the t -distribution for samples of size $n \geq 30$.

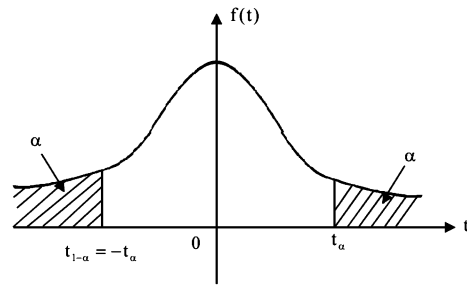


Fig. Symmetry of t -distribution

The t -distribution is extensively used in hypothesis about one mean, or about equality of two means when σ is unknown.

10.4. APPLICATIONS OF THE t - DISTRIBUTION

The t - distribution has a wide number of applications in Statistics, some of them are given below :

- (1) To test the significance of the sample mean, when population variance is not given
- (2) To test the significance of the mean of the sample *i.e.*, to test if the sample mean differs significantly from the population mean.
- (3) To test the significance of the difference between two sample means or to compare two samples.
- (4) To test the significance of an observed sample correlation coefficient and sample regression coefficient.

10.5 CHI-SQUARE (χ^2) DISTRIBUTION

[JNTU (H) May 2012 (Set No. 3), (K) Dec. 2013 (Set No. 3)]

Chi-squared distribution is a continuous probability distribution of a continuous random variable X with probability density function given by

$$f(x) = \begin{cases} \frac{1}{2^{v/2} \Gamma(v/2)} x^{(v/2)-1} \cdot e^{-x/2}, & \text{for } x > 0 \\ 0, & \text{otherwise} \end{cases}$$

where v is a positive integer is the only single parameter of the distribution, also known as “degrees of freedom (dof)”.

χ^2 - distribution was extensively used as a measure of goodness of fit and to test the independence of attributes.

Properties of χ^2 - distribution :

1. χ^2 - distribution curve is not symmetrical, lies entirely in the first quadrant, and hence not a normal curve, since χ^2 varies from 0 to ∞ .
2. It depends only on the degrees of freedom v .

3. If χ_1^2 and χ_2^2 are two - independent distributions with v_1 and v_2 degrees of freedom, the $\chi_1^2 + \chi_2^2$ will be chi-squared distribution with $(v_1 + v_2)$ degrees of freedom. That is, it is additive.

4. Here α denotes the area under the chi-square distribution to the right of χ_α^2 .

So, χ_α^2 represents the χ^2 - value such that the area under the chi-square curve to its (χ_α^2) right is equal to α . Chi-square distribution is very important in estimation and hypothesis testing. χ^2 -distribution is used in sampling distribution, analysis of variance, mainly, it is used as a measure of goodness of fit and in analysis of $r \times c$ tables.

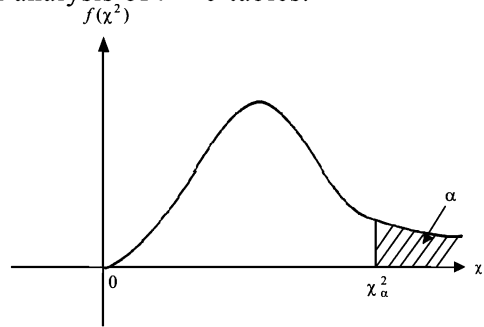


Fig. χ^2 - distribution

For various values of α and v , the values of χ_α^2 are tabulated in tables.

In χ^2 table, the left hand column contains values of v (degrees of freedom), the column headings are areas α in the right hand tail of χ^2 - distribution curve, the entries are χ^2 values. It is necessary to calculate values of χ_α^2 for $\alpha > 0.50$ since χ^2 curve or distribution is not symmetrical.

5. Mean = v and variance = $2v$

Sampling Distribution of Variance S^2 :

The theoretical sampling distribution of the sample variance for random samples from normal population is related to the chi-squared distribution as follows :

Let S^2 be the variance of a random sample of size n , taken from a normal population having the variance σ^2 .

$$\text{Then } \chi^2 = \frac{(n-1)S^2}{\sigma^2} = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{\sigma^2} \left[\because S^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1} \right]$$

is a value of a random variable having the χ^2 -distribution with $v = n - 1$ degrees of freedom.

Exactly 95% of χ^2 -distribution lies between $\chi^2_{0.975}$ and $\chi^2_{0.025}$ and when σ^2 is too small, χ^2 falls to the right of $\chi^2_{0.025}$ and when σ^2 is too large, χ^2 falls to the left of $\chi^2_{0.975}$. Thus when σ^2 is correct, χ^2 -values are to the left of $\chi^2_{0.975}$ or to the right of $\chi^2_{0.025}$.

10.6 APPLICATIONS OF χ^2 DISTRIBUTION [JNTU (H) MAY 2012 (SET NO. 3)]

- (1) To test the goodness of fit.
- (2) To test the independence of attributes.
- (3) To test the homogeneity of independent estimation of the population variance.
- (4) To test the homogeneity of independent estimation of the population correlation coefficient.

10.7 F-DISTRIBUTION

(SAMPLING DISTRIBUTION OF THE RATIO OF TWO SAMPLE VARIANCES)

Another important continuous probability distribution which plays an important role in connection with sampling from normal populations is the F-distribution.

Let S_1^2 be the sample variance of an independent sample of size n_1 drawn from a normal population $N(\mu_1, \sigma_1^2)$. Similarly, let S_2^2 be the sample variance in an independent sample of size n_2 drawn from another normal population $N(\mu_2, \sigma_2^2)$.

Thus S_1^2 and S_2^2 are two variances of two random samples of sizes n_1 and n_2 respectively drawn from the normal population, with the variances σ_1^2 and σ_2^2 . To determine whether the two samples come from two populations having equal variances.

Consider the sampling distribution of the ratio of the variances of the two independent random samples defined by

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} = \frac{\sigma_2^2 S_1^2}{\sigma_1^2 S_2^2}$$

which follows F-distribution with $v_1 = n_1 - 1$ and $v_2 = n_2 - 1$ degrees of freedom.

F-distribution can be used to test the equality of several population means, comparing sample variances and analysis of variance.

Under the assumption (hypothesis) that two normal populations have the same variance *i.e.*, $\sigma_1^2 = \sigma_2^2$, we have $F = \frac{S_1^2}{S_2^2}$.

F determines whether the ratio of two sample variances S_1 and S_2 is too small or too large.

When F is close to 1, the two sample variances S_1 and S_2 are almost same. In practice, it is customary, to take the larger sample variance as the numerator.

F is always a positive number.

The sampling distribution of F is of the form $f(F) = K \frac{F^{(v_1-2)/2}}{(v_1 F + v_2)^{(v_1+v_2)/2}}$ where

v_1 and v_2 are two degrees of freedom and K is determined by $\int_0^{\infty} f(F) dF = 1$.

Properties of F -distribution :

- (i) F - distribution is free from population parameters and depends upon degrees of freedom only.
- (ii) F -distribution curve lies entirely in first-quadrant.
- (iii) The F -curve depends not only on the two parameters v_1 and v_2 but also on the order in which they are stated.
- (iv) $F_{1-\alpha}(v_1, v_2) = \frac{1}{F_{\alpha}(v_2, v_1)}$

where $F_{\alpha}(v_1, v_2)$ is the value of F with v_1 and v_2 degrees of freedom such that the area under the F -distribution curve to the right of F_{α} is α .

- (v) The mode of F - distribution is less than unity.

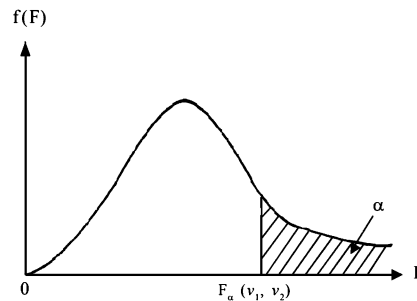


Fig. F - distribution

SOLVED EXAMPLES

Example 1 : Find (a) $t_{0.05}$ when $v = 16$

(b) $t_{-0.01}$ when $v = 10$

(c) $t_{0.995}$ when $v = 7$

Solution :

From tables

(a) When $v = 16$, $t_{0.05} = 1.746$

(b) when $v = 10$, $t_{-0.01} = -2.764$

(c) when $v = 7$, $t_{0.995} = t_{1-0.005} = -t_{0.005} = -3.499$

Example 1 : Find (a) $P(t < 2.365)$ when $\nu = 7$

(b) $P(t > 1.318)$ when $\nu = 24$

(c) $P(-1.356 < t < 2.179)$ when $\nu = 12$

(d) $P(t > -2.567)$ when $\nu = 17$

Solution :

- (a) When $t < 2.365$, $P(t < 2.365)$ is given by the area to the left of $t = 2.365$
From table $t_{\alpha} = 2.365$ for $\nu = 7$ degrees of freedom. Then $\alpha = 0.025$
 $\therefore P(t < 2.365) = 1 - 0.025 = 0.975$

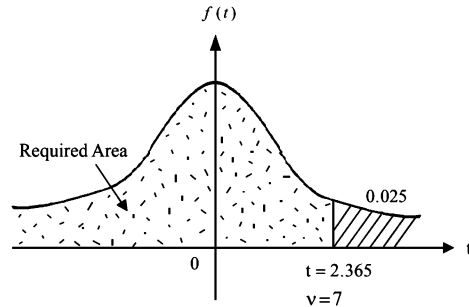


Fig.

- (b) When $t > 1.318$
 $P(t > 1.318)$ is given by the area to the right of $t = 1.318$
From table $t_{\alpha} = 1.318$ with $\nu = 24$ d.o.f. Then $\alpha = 0.10$

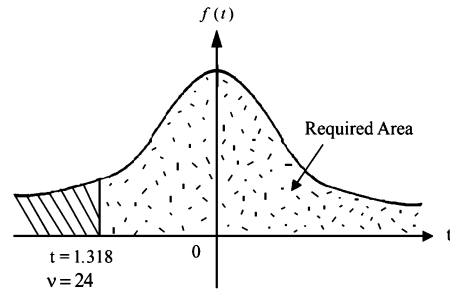


Fig.

- (c) When $t < 2.179$ with $\nu = 12$ d.o.f.
Area to the right of 2.179 is 0.025
When $t > -1.356$ with $\nu = 12$, d.o.f.
Area to the left of 1.356 is 0.10
 $\therefore$ When $-1.356 < t < 2.179$, the area is $1 - 0.10 - 0.025 = 0.875$
 $\therefore P(-1.356 < t < 2.179) = 0.875$

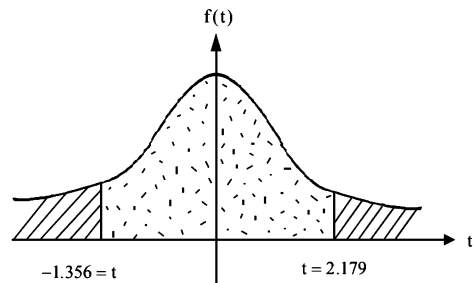


Fig.

- (d) When $t > -2.567$
 $P(t > -2.567)$ is given by the area to the right of $t = -2.567$
From table
 $t_{-\alpha} = -2.567$ with $\nu = 17$ d.o.f.
Then $\alpha = 1 - 0.01 = 0.99$

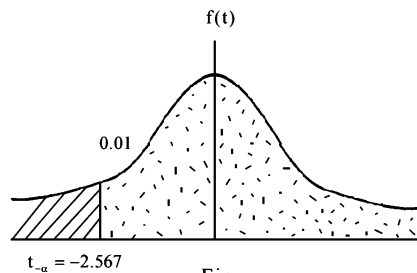


Fig.

Example 1 : A random sample of size 25 from a normal population has the mean $\bar{x} = 47.5$ and the standard deviation $S = 8.4$. Does this information tend to support or refute the claim that the mean of the population is $\mu = 42.5$?

[JNTU (A) Nov. 2010, (H) Dec. 2011, (K) Nov. 2009, May 2012, 2013 (Set No. 1)]

Solution :

Given n = The size of the sample = 25

$\bar{x}$ = The mean of the sample = 47.5

μ = The population mean = 42.5

S = S.D of sample = 8.4

We have t -distribution,

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}} = \frac{47.5 - 42.5}{8.4/\sqrt{25}} = \frac{5\sqrt{25}}{8.4} = 2.98$$

This value of t has 24 degrees of freedom.

From the table of t -distribution for $v = 24$ with $\alpha = 0.005$ is 2.797. We conclude that the information given in the data of this example tend to refuse the claim that the mean of the population is $\mu = 42.5$ (i.e. μ cannot be 42.5).

Example 4 : Ten bearings made by a certain process have a mean diameter of 0.5060 cm with a standard deviation of 0.0040 cm. Assuming that the data may be taken as a random sample from a normal distribution, construct a 95% confidence interval for the actual average diameter of the bearings ?

[JNTU Dec 2004, April 2005 (Set No. 3)]

Solution : Given that sample size $n = 10 < 30$, so use t -distribution.

Maximum error estimate of 95% confidence is

$$E = t_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} = \frac{(2.262)(0.004)}{\sqrt{10}} = 0.00286$$

Since $t_{0.025}$ with $n - 1 = 10 - 1 = 9$ d.o.f. is 2.262, therefore,

95% confidence interval limits are $\bar{x} \pm t_{\alpha/2} \cdot \sigma/\sqrt{n}$

$$\text{i.e., } 0.5060 \pm 2.262 \frac{(0.004)}{\sqrt{10}} = (0.5031, 0.5089)$$

Example 4 : A sample of size 10 was taken from a population S.D of sample is 0.03. Find the maximum error with 99% confidence.

[JNTU (A) Nov. 2010 (Set No. 2), (K) Dec. 2013 (Set No. 2,4)]

Solution :

S = Standard deviation = 0.03

n = Sample size = 10

$t_{\alpha/2}$ for $v = 9$, 99% = 3.25

$$\therefore E = t_{\alpha/2} \cdot \frac{S}{\sqrt{n}} = 3.25 \times \frac{0.03}{3} = 0.0325$$

Example 4 : A sample of 11 rats from a central population had an average blood viscosity of 3.92 with a S.D of 0.61. Estimate the 95% confidence limits for the mean blood viscosity of the population.

Solution : We have $t_{\alpha/2}$ for 95% and $v = n - 1 = 11 - 1 = 10$ is 2.23

Given $S = \text{S.D} = 0.61$, $n = \text{sample size} = 11$ and $\bar{x} = 3.92$

Confidence limits are $\bar{x} \pm t_{\alpha/2} \cdot \frac{S}{\sqrt{n}}$

$$\text{i.e., } 3.92 \pm 2.23 \left(\frac{0.61}{\sqrt{11}} \right)$$

$$\text{or } 3.92 \pm 0.41 = (3.51, 4.33)$$

Example 1 : A process for making certain ball bearings is under control if the diameters of the bearings have a mean of 0.5000 cm. If a random sample of 10 of these bearings has a mean diameter of 0.5060 cm and S.D of 0.0040 cm, is the process under control ? [(or) what can we say about this process] [JNTU (K) May 2012 (Set No. 3)]

Solution :

$n = \text{The size of sample} = 10$

$\bar{x} = \text{Sample mean} = 0.5060$

$\mu = \text{Population mean} = 0.5000$

$S = \text{Sample S.D} = 0.0040$

$\therefore$ We have t -distribution,

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}} = \frac{0.5060 - 0.5000}{(0.0040)/\sqrt{10}} = 4.7334$$

Here $v = n - 1 = 10 - 1 = 9$ degrees of freedom

Since $t = 4.7334 > 3.250 = t_{\alpha}$ with $\alpha = 0.005$ and $v = 9$

The process is not under control.

Example 8 : In 16 one hour test runs, the gasoline consumption of an engine averaged 16.4 gallons with a S.D of 2.1 gallons. Test the claim that the average gasoline consumption of this engine is 12.0 gallons per hour.

Solution :

Given Sample size, $n = 16$

Sample mean, $\bar{x} = 16.4$

Population mean, $\mu = 12.0$

and Sample S.D $S = 2.1$

From t -distribution, we have

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}} = \frac{16.4 - 12.0}{2.1/\sqrt{16}} = \frac{4.4 \sqrt{16}}{2.1} = \frac{17.6}{2.1} = 8.38$$

The table of t -distribution shows that the probability of getting a value of t greater than 2.947 is 0.005 for 15 degrees of freedom, then the probability of getting a value greater than 8 must be negligible. Hence it would seem reasonable to conclude that the true average hourly gasoline consumption of the engine exceeds 12.0 gallons.

Example 9 : A manufacture claims that any of his list of items cannot have variance more than 1 cm². A sample of 25 items has a variance of 1.2 cm². Test whether the claim of the manufacturer is correct.

Solution :

Given sample size, $n = 25$

Sample variance, $S^2 = 1.2$, population variance, $\sigma^2 = 1$

$$\text{Now } \chi^2 = \frac{(n-1) S^2}{\sigma^2} = \frac{24 (1.2)}{1} = 28.8$$

$$\text{For } \alpha = 0.05, \chi^2_{0.05, 24} = 36.41, \chi^2_{0.025, 24} = 39.36$$

$$\text{Clearly } \chi^2 < \chi^2_{\alpha/2}$$

Hence the claim of the manufacturer is correct.

Example 10 : Suppose that the thickness of a part used in a semiconductor is its critical dimension and that the process of manufacturing these parts is considered to be under control if the true variation among the thickness of the parts is given by a S.D not greater than $s = 0.60$ thousandth of an inch. To keep a check on the process, random samples of size $n = 20$ are taken periodically, and it is regarded to be “out of control” if the probability that S^2 will take on a value greater than or equal to the observed sample value is 0.01 or less (even though $s = 0.60$). What can be concluded about the process if the S.D of such a periodic random sample is $S = 0.84$, thousandth of an inch ?

Solution :

Given sample size, $n = 20$

Sample S.D, $S = 0.84$

Population mean, $\sigma = 0.60$ and $\alpha = 0.01$.

The process will be declared out of control if the calculated value of χ^2 for 19 degrees of freedom exceeds the table value *i.e.*, $\chi^2_{0.001, 19} = 36.191$

$$\therefore \chi^2 = \frac{(n-1) S^2}{\sigma^2} = \frac{19 (0.84)^2}{(0.60)^2} = \frac{19 (0.7056)}{(0.3600)} = \frac{13.4064}{0.36} = 37.24$$

Now the probability that χ^2 will exceed 36.191 is 0.01 for 19 degrees of freedom, then the probability of getting a value greater than 37.24 is even less.

Hence the process is said to be out of control.

Example 11 : If two independent random samples of size $n_1 = 13$ and $n_2 = 7$ are taken from a normal population. What is the probability that the variance of the first sample will be atleast four times as large as that of the second sample ? [JNTU (K) May 2013 (Set No. 3)]

Solution : Given $n_1 = 13$. Then

$$v_1 = n_1 - 1 = 13 - 1 = 12 \text{ and } n_2 = 7. \text{ Then}$$

$$v_2 = n_2 - 1 = 7 - 1 = 6 \text{ and } S_1^2 = 4S_2^2.$$

i.e. the variance of the first sample will be at least four times as large as that of the second sample.

$$\text{Now } F = \frac{S_1^2}{S_2^2} = \frac{4S_2^2}{S_2^2} = 4.00$$

$$\therefore F = 4.00 \quad (\because S_1^2 = 4S_2^2)$$

This value of F follows F -distribution with $v_1 = 12$ and $v_2 = 6$ degrees of freedom. Hence from the table we get $F_{0.05, 12, 6} = 4.00$.

Therefore, the required probability is 0.05.

Example 11 : For an F -distribution, find

(a) $F_{0.05}$ with $v_1 = 7$ and $v_2 = 15$

(b) $F_{0.01}$ with $v_1 = 24$ and $v_2 = 19$

(c) $F_{0.95}$ with $v_1 = 19$ and $v_2 = 24$

(d) $F_{0.99}$ with $v_1 = 28$ and $v_2 = 12$ [JNTU (H) Nov. 2010, (H) Dec. 2011 (Set No. 2)]

Solution :

(a) From table, $F_{0.05}$ with $v_1 = 7$ and $v_2 = 15$ is 2.71

(b) $F_{0.01}$ with $v_1 = 24$, $v_2 = 19$ is 2.92

$$(c) F_{0.95}(19, 24) = \frac{1}{F_{0.05}(24, 19)} = \frac{1}{2.11} = 0.473933$$

$$(d) F_{0.99}(28, 12) = \frac{1}{F_{0.01}(12, 28)} = \frac{1}{2.90} = 0.34482$$

Example 13 : In the past a machine has produced washers having a mean thickness of 0.050 inch. To determine whether the machine is in proper working order a sample of 10 washers is chosen for which the mean thickness is 0.053 inch and the standard deviation is 0.003 inch. Test the hypothesis that the machine is in proper working order using a level of significance of 0.05? [JNTU (K) May 2010 (Set No.1)]

Solution : We have $\mu = 0.050, n = 10, \bar{x} = 0.053, s = 0.003$, where s is the sample S.D.

Since the sample size is small and population S.D σ is not known, we use t - test.

Let the Null Hypothesis be

$$H_0 : \mu = 0.050$$

Then the Alternative Hypothesis is

$$H_1 : \mu \neq 0.050$$

The test statistic is

$$z = \frac{\bar{x} - \mu}{s / \sqrt{n-1}} = \frac{0.053 - 0.050}{0.003 / \sqrt{10-1}} = \frac{0.003}{0.001} = 3$$

Degrees of freedom $= n - 1 = 10 - 1 = 9$.

The tabulated value of t for 9 d.f. at 5% level $= 2.262$

Since the calculated value of $t >$ the tabulated value, we reject the Null Hypothesis at 5% level and conclude that the machine is not in proper working order.

REVIEW QUESTIONS

1. What is meant by degree of freedom ?
2. Define number of degrees of freedom. List the properties of t - distribution.
3. Write a note on student's t - distribution.
4. Write the properties of t - distribution. [JNTU (A) Nov. 2011 (Set No. 2)]
5. Write the applications of t - distribution.
6. Explain student t - distribution, its properties and applications.
[JNTU (H) Dec 2011 (Set No. 1)]
7. Define the statistics ' t ' and ' F ' and write down their sampling distributions. State the important assumptions in respect of them. [JNTU 2004 (Set No. 4)]
8. Explain Chi-square distribution, its properties and applications.
[JNTU (H) Apr. 2012 (Set No. 3)]

(or) Write a short note on χ^2 - test for goodness of fit.

9. Write the properties χ^2 - distribution.
10. Write the applications of χ^2 - distribution. [JNTU (H) May 2012 (Set No. 3)]
11. Write F - distribution. List the properties of F - distribution.

EXERCISE 10(A)

1. Find the t -value with $v = 14$ d.o.f that leaves an area 0.025 to the left.
2. Determine (a) $t_{0.01}$ with $v = 18$
 (b) $t_{0.05}$ with $v = 12$
 (c) $t_{-0.10}$ with $v = 15$
3. Fuses produced by a company will blow in 12.40 minutes as the average when overloaded. Suppose the mean blow time of 20 fuses subjected to overload is 10.63 minutes and S.D 2.48 mts. Does this information tend to support or refuse the claim that the population mean blow time is 12.40 mts ?
4. A random sample of size 25 from a normal population has the mean = 47.5 and S.D = 8.4. Does this information tend to support or refute the claim that the mean of the population is 42.1.
5. The process of making certain bearings is under control if the diameter of the bearings have a mean of 0.5000 cm. What can we say about this process if a sample of 10 of these bearings has a mean diameter of 0.5060 cm and a S.D of 0.0040 cm .
6. Find the value of
 (a) $F_{0.05}$ for $v_1 = 15$ and $v_2 = 7$
 (b) $F_{0.95}$ for 12 and 15 d.o.f.
7. Find the value of
 (a) $F_{0.95}$ for $v_1 = 10$ and $v_2 = 20$
 (b) $F_{0.99}$ for $v_1 = 6$ and $v_2 = 20$
8. If independent random samples of size $n_1 = n_2 = 8$ come from normal populations having the same variance, what is the probability that either sample variance will be atleast seven times as large as the other.
9. Can we conclude that the two population variances are equal for the following data of post graduates passed out from a 'state' and 'private' university.
 State : 8350 8260 8130 8340 8070
 Private : 7890 8140 7900 7950 7840 7920
10. The claim that the variance of a normal population is $\sigma^2 = 21.3$ is rejected if the variance of a random sample of size 15 exceeds 39.74. What is the probability that the claim will be rejected even though $\sigma^2 = 21.3$?
11. The claim that the variance of a normal population is $\sigma^2 = 4$ is to be rejected if the variance of a random sample of size 9 exceeds 7.7535. What is the probability that this claim will be rejected even though $\sigma^2 = 4$?

ANSWERS

1. $t_{-0.025} = -2.145$
2. (a) 2.878 (b) 2.179 (c) 1.753
3. $t = -3.19$ at 19 d.o.f. refutes at $\alpha = 0.005$
4. does not support the claim
5. $t = 4.74$, Process out of control
6. (a) 3.51 (b) $F_{0.95}(12,15) = \frac{1}{F_{0.05}(15,12)} = \frac{1}{2.62} = 0.38$
7. (a) $F_{0.95}(10,20) = 0.36$ (b) $F_{0.99}(6,20) = 0.135135$
8. $F_{0.01} = 6.99 \approx 7$ [for $v_1 = v_2 = 7$]
9. $S_1^2 = 15750, S_2^2 = 10920, F = \frac{S_1^2}{S_2^2} = 1.442$, not significantly different
10. $\chi^2 = 26.120, \chi^2_{0.025, 14} = 26.119$; $\chi^2 > \chi^2_{0.05, 14}$ rejected.
11. $\alpha = 0.05$

10.8 TEST OF SIGNIFICANCE FOR SMALL SAMPLES

A very important aspect of the sampling theory is the study of tests of significance, which enable us to decide on the basis of the sample results, if

(i) The deviation between the observed sample statistic and the hypothetical parameter value is significant.

(ii) The deviation between two sample statistics is significant.

The following are some important tests for small samples :

(i) Student's 't' test (ii) F-test (iii) χ^2 -test

10.9 STUDENT'S 't' TEST

Let $\bar{x}$ = Mean of a sample

n = Size of the sample

σ = Standard deviation of the sample

μ = Mean of the population supposed to be normal.

Then the student's t is defined by the statistic $t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$

If s^2 be the sample variance, $s^2 = \frac{\sum_i (x_i - \bar{x})^2}{n}$

If $S^2 = \frac{\sum_i (x_i - \bar{x})^2}{n-1}$, then $\frac{S^2}{s^2} = \frac{n}{n-1}$ or $S = \sqrt{\frac{n}{n-1}} \cdot s$

$$\therefore t = \frac{\bar{x} - \mu}{S/\sqrt{n}} = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$$

S^2 is called the unbiased estimate of population variance σ^2 .

Note : 1. If the standard deviation of a sample is given directly, then the statistic is

$$\text{given by } t = \frac{\bar{x} - \mu}{\text{S.D.}/\sqrt{n-1}}$$

2. If the calculated value of t exceeds the table value of t at 5% level of significance, the null hypothesis H_0 is rejected. If the calculated value of t is *less than* the tabulated value of t at 5% level the null hypothesis is accepted.

Assumptions for Student's t -test :

The following are made in student's t test

- (1) Sample size, $n < 30$
- (2) The parent population from which sample is drawn is normal.
- (3) The population standard deviation is unknown.
- (4) The sample observations are independent, i.e., sample is random.

Uses of t -test :

This test is used

- (1) to test for a specified mean
- (2) to test for equality of two means of two independent samples drawn from two normal populations, S.D. of the populations being unknown, and
- (3) to test the significance of difference between the means of paired data.

10.10 CONFIDENCE OR FIDUCIAL LIMITS FOR μ

Suppose we want to find from the sample data the limits within which the population mean will lie with a probability of 0.95. The limits are called the 95% confidence limits of the population mean for the given sample.

If $t_{0.05}$ is the table value of t for $(n-1)$ degrees of freedom at 5% level of significance, then 95% confidence limits for μ are given by $\bar{x} \pm t_{0.05} \cdot \frac{S}{\sqrt{n}}$.

For $P(|t| > t_{0.05}) = 0.05$ i.e., $P(|t| \leq t_{0.05}) = 0.95$

95% confidence limits for μ are given by

$$|t| \leq t_{0.05} \text{ i.e., } \left| \frac{\bar{x} - \mu}{S/\sqrt{n}} \right| \leq t_{0.05} \text{ or } \frac{|\bar{x} - \mu|}{S/\sqrt{n}} \leq t_{0.05}$$

$$\Rightarrow -t_{0.05} \leq \frac{\bar{x} - \mu}{S/\sqrt{n}} \leq t_{0.05}$$

$$\Rightarrow \bar{x} - t_{0.05} \cdot S/\sqrt{n} \leq \mu \leq \bar{x} + t_{0.05} \cdot S/\sqrt{n}$$

Similarly 99% confidence limits for μ are $\bar{x} \pm t_{0.01} \cdot S/\sqrt{n}$ where $t_{0.01}$ is the tabulated value of t for $(n-1)$ degrees of freedom at 1% level of significance.

10.11 STUDENT'S 't' TEST FOR SINGLE MEAN

Suppose we want to test

- (a) if a random sample x_i of size n has been drawn from a normal population with a specified mean μ .
- (b) if the sample mean differs significantly from the hypothetical value μ of the population mean.

In this case the statistic is given by $t = \frac{\bar{x} - \mu}{S/\sqrt{n}} \sim t_{(n-1)}$ where $\bar{x}$, μ , S , n have usual meanings.

Let a random sample of size n ($n < 30$) has a sample mean $\bar{x}$. To test the hypothesis that the population mean μ has a specified value μ_0 when population S. D. σ is not known.

Let the Null Hypothesis be $H_0 : \mu = \mu_0$

Then the Alternative Hypothesis is $H_1 : \mu \neq \mu_0$.

Assuming that H_0 is true, the test statistic given by $t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$, where s is the sample S. D. follows t -distribution with $v = (n-1)$ d.f.

We calculate the value of $|t|$ and compare this value with the table value of t at α level of significance. If the calculated value of $t >$ the table value of t , we reject H_0 at α level. Otherwise we accept H_0 .

In this case, 95% confidence limits for the population mean μ are $\bar{x} \pm t_{\alpha} \cdot \frac{s}{\sqrt{n-1}}$ where $\alpha = 0.025$ for two-tailed test and s = sample S. D. and 99% confidence limits for μ are $\bar{x} \pm t_{\alpha} \cdot \frac{s}{\sqrt{n-1}}$ where $\alpha = 0.05$.

For a two-tailed test at α level of significance, value of $\alpha/2$ is taken for α .

SOLVED EXAMPLES

Example 1 : A mechanist is making engine parts with axle diameters of 0.700 inch. A random sample of 10 parts shows a mean diameter of 0.742 inch with a S.D. of 0.040 inch. Compute the statistic you would use to test whether the work is meeting the specification at 0.05 level of significance. [JNTU (A) Apr. 2012 (Set No. 2)]

Solution : Here the sample size $n = 10 < 30$

Hence the sample is small sample.

Also sample mean $\bar{x} = 0.742$ inches, the population mean $\mu = 0.700$ inches and $S.D. = 0.040$ inches are given.

$\therefore$ We use student's t -Test

1. **Null Hypothesis H_0 :** The product is confirming to specification.
2. **Alternative Hypothesis H_1 :** $\mu \neq 0.700$
3. **Level of significance, $\alpha = 0.05$**

4. **The test statistic is, $t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$**

Here $\bar{x} = 0.742$ inches, $\mu = 0.700$ inches, $S.D. = 0.040$ inches and $n = 10$.

Degrees of freedom (d.f) $= n - 1 = 10 - 1 = 9$

$$\therefore t = \frac{0.742 - 0.700}{\frac{0.040}{\sqrt{10-1}}} = 3.15$$

Thus the calculated value of $t = 3.15$

The tabulated value of t at 5% level with 9 degrees of freedom is $t_{0.05} = 2.26$

Since calculated value of $t >$ tabulated value of t , therefore, H_0 is rejected.

$\therefore$ The product is not meeting the specification.

Example 4 : A sample of 26 bulbs gives a mean life of 990 hours with a S.D. of 20 hours. The manufacturer claims that the mean life of bulbs is 1000 hours. Is the sample not upto the standard. [JNTU (A) Nov. 2010 (Set No. 2,4)]

Solution : Here sample size, $n = 26 < 30$

$\therefore$ The sample is small sample.

Also given, sample mean, $\bar{x} = 990$

Population mean, $\mu = 1000$ and $S.D., s = 20$

Degrees of freedom $= n - 1 = 26 - 1 = 25$

Here we know $\bar{x}$, μ , $S.D.$ and n .

So we use students ' t ' test.

1. **Null Hypothesis H_0 :** The sample is upto the standard.
2. **Alternative Hypothesis H_1 :** $\mu < 1000$
(The sample is below standard) (left-tail test)

3. **Level of significance :** $\alpha = 0.05$

4. **The test statistic** is $t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}} = \frac{990 - 1000}{20/\sqrt{25}} = -2.5$

$$\therefore |t| = 2.5$$

i.e., Calculated value of $t = 2.5$

Tabulated value of ' t ' at 5% level with 25 degrees of freedom for left-tailed test is 1.708.

Since calculated $t >$ tabulated t , we reject the null hypothesis H_0 and conclude that the sample is not upto the standard.

Example 1 : A machine is designed to produce insulating washers for electrical devices of average thickness of 0.025 cm. A random sample of 10 washers was found to have a thickness of 0.024 cm with a S.D of 0.002 cm. Test the significance of the deviation. Value of t for 9 degrees of freedom at 5% level is 2.262.

Solution : Here the sample size is $10 < 30$

$\therefore$ The sample is small

Also given Sample mean, $\bar{x} = 0.024$ cm

Population mean, $\mu = 0.025$ cm

S.D. = 0.002 cm

Degrees of freedom (d.f) = $n - 1 = 10 - 1 = 9$

1. **Null Hypothesis H_0 :** The difference between $\bar{x}$ and μ is not significant.

2. **Alternative Hypothesis H_1 :** $\mu_1 \neq 0.025$

3. **Level of significance :** $\alpha = 0.05$

4. **The test statistic** is ' t ' = $\frac{\bar{x} - \mu}{s/\sqrt{n-1}} = \frac{0.024 - 0.025}{\frac{0.002}{\sqrt{10-1}}} = -1.5$

$$\Rightarrow |t| = 1.5$$

$\therefore$ Calculated value of $t = 1.5$ for two tailed test.

Tabulated value of t for 9 degrees of freedom at 5% level = 2.262

Since calculated $t <$ tabulated t , we accept the null hypothesis and conclude that the difference between $\bar{x}$ and μ is not significant.

Example 4 : The mean life time of a sample of 25 fluorescent light bulbs produced by a company is computed to be 157 hours with a S.D of 120 hours. The company claims that the average life of the bulbs produced by the company is 1600 hours using the level of significance of 0.05. Is the claim acceptable ?

Solution : Given sample size, $n = 25$

Sample mean, $\bar{x} = 1570$

Population mean, $\mu = 1600$

S.D.(s) = 120

Degrees of freedom = $n - 1 = 24$

1. **Null Hypothesis H_0** : The claim is acceptable. $\mu = 1600$ hrs
2. **Alternative Hypothesis H_1** : $\mu \neq 1600$ hrs
3. **Level of significance** : $\alpha = 0.05$
4. **The test statistic** is $t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}} = \frac{1570 - 1600}{120/\sqrt{24}} = \frac{-30}{24.49} = -1.22$

$$\therefore |t| = 1.22$$

i.e., Calculated $t = 1.22$

The tabulated value of t at 5% level with 24 degrees of freedom for two tailed test is 2.06.

Since the calculated value of $t <$ tabulated value of t , we accept the null hypothesis H_0 *i.e.*, the claim that the average life of the bulbs produced by the company is 1600 hrs is acceptable.

Example 4 : The average breaking strength of the steel rods is specified to be 18.5 thousand pounds. To test this sample of 14 rods were tested. The mean and standard deviations obtained were 17.85 and 1.955 respectively. Is the result of experiment significant ?

[JNTU (H) May 2011 (Set No. 3), (K) Nov. 2011 (Set No. 1)]

Solution : We are given
 Sample size, $n = 14$
 Sample mean, $\bar{x} = 17.85$
 S.D. (s) = 1.955
 Population mean, $\mu = 18.5$
 Degrees of freedom = $n - 1 = 13$

1. **Null Hypothesis H_0** : The result of the experiment is not significant.
2. **Alternative Hypothesis H_1** : $\mu \neq 18.5$
3. **Level of significance** : $\alpha = 0.05$
4. **The test statistic** is, $t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}} = \frac{17.85 - 18.5}{1.955/\sqrt{13}} = \frac{0.65}{0.542} = -1.199$

$$\therefore |t| = 1.199$$

i.e., Calculated $t = 1.199$

Tabulated t at 5% level of significance for 13 d.f. for two tailed test = 2.16

Since calculated $t <$ tabulated t , we accept the Null Hypothesis H_0 at 5% level and conclude that the result of the experiment is not significant.

Example 4 : A random sample of size 16 values from a normal population showed a mean of 53 and a sum of squares of deviations from the mean equals to 150. Can this sample be regarded as taken from the population having 56 as mean ? Obtain 95% confidence limits of the mean of the population.

[JNTU (K) May 2012 (Set No.2)]

Solution : Given sample size, $n = 16$

Sample mean, $\bar{x} = 53$

$$\text{Now } \sum (x_i - \bar{x})^2 = 150$$

$$\therefore S^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{150}{15} = 10 \Rightarrow S = \sqrt{10}$$

$$\text{Degrees of freedom, } v = n - 1 = 16 - 1 = 15$$

(a) 1. **Null Hypothesis H_0** : The sample is taken from the population having 56 as mean *i.e.*, $\mu = 56$

2. **Alternative hypothesis H_1** : $\mu \neq 56$

3. **Level of significance** : $\alpha = 0.05$

4. **The test statistic**, $t = \frac{\bar{x} - \mu}{S/\sqrt{n}}$ [Note that *S.D.* is not given directly]

$$= \frac{53 - 56}{\sqrt{10}/\sqrt{16}} = -3.79$$

$$\therefore |t| = 3.79$$

The tabulated value of t at 5% level of significance for 15 d.f for two tailed test is 2.13.

Since calculated $t >$ tabulated t , the null hypothesis H_0 is rejected *i.e.*, the sample cannot be regarded as taken from the population.

(b) The 95% confidence limits of the mean of the population are given by

$$\begin{aligned} \bar{x} \pm t_{0.05} \frac{S}{\sqrt{n}} &= 53 \pm 2.13 \times 0.79 = 53 \pm 1.6827 \\ &= 54.68 \text{ and } 51.31 \end{aligned}$$

Hence 95% confidence limits are [51.31, 54.68]

Example 7 : A random sample of six steel beams has a mean compressive strength of 58,392 p.s.i (pounds per square inch) with a standard deviation of 648 p.s.i. Use this information and the level of significance $\alpha = 0.05$ to test whether the true average compressive strength of the steel from which this sample came is 58,000 p.s.i. Assume normality.

[JNTU 2005S, 2006S, (H) Nov. 2010, (H) May & Dec. 2011 (Set No. 2), (K) May 2013 (Set No. 1)]

Solution : We have

n = Sample size (number of steel beams) = 6 < 30. $\therefore$ The sample is small.

$\bar{x}$ = Sample mean (average compressive strength) = 58392 psi.

S = Standard deviation of six beams = 648 psi

Degrees of freedom (d.f.) = $n - 1 = 6 - 1 = 5$

In this problem σ is known and $n < 30$. Hence we use t - distribution.

1. **Null Hypothesis is $H_0 : \mu = 58000$**
2. **Alternative Hypothesis $H_1 : \mu \neq 58000$**
3. **Level of Significance : $\alpha = 0.05$**
4. **Critical Region :** Since Alternative Hypothesis is of the type ' $\neq$ ', the test is two-tailed and the critical region is $-3.365 < t < 3.365$
5. **The Test Statistic is $t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}} = \frac{58392 - 58000}{648/\sqrt{5}} = 1.353$**

Since $t = 1.353 < 3.365 = t_{\alpha/2}$, we accept the null hypothesis H_0 .

Hence the average compressive strength of the steel beam is not equal to 58000 psi.

Example 8 : A sample of 100 iron bars is said to be drawn from a large number of bars whose lengths are normally distributed with mean 4 feet and S.D. 6 ft. If the sample mean is 4.2 feet, Can the sample be regarded as a truly random sample?

[JNTU (H) Nov. 2010 (Set No. 2)]

Solution : Let the Null Hypothesis be H_0 : The sample can be regarded as a truly random sample.

Then the Alternative Hypothesis is H_1 : The sample cannot be regarded as a random sample.

Here $n = \text{Sample size} = 100$

$\bar{x} = \text{Sample mean} = 4.2$

$\mu = \text{Population mean} = 4$

$\sigma = \text{S.D. of population} = 0.6$

Let us assume that H_0 is true. The test statistic is

$$t = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{4.2 - 4}{0.6/\sqrt{100}} = \frac{0.2}{0.06} = \frac{2}{0.6} = \frac{10}{3} = 3.33$$

$$\therefore t = 3.33 > 3$$

Hence we reject the Null Hypothesis and conclude that the given sample cannot be regarded as a truly random sample from the given population.

Note : Since $|z| > 3$ and level of significance is not given, we may reject H_0 at 5% or 1% level of significance. In the above problem, the Null Hypothesis is rejected without mentioning any level of significance.

Example 9 : A sample of 155 members has a mean 67 and S.D. 5.2. Is this sample has been taken from a large population of mean 70? [JNTU (H) Nov. 2010 (Set No. 3)]

Solution : Let the Null Hypothesis be H_0 : The sample has been taken from population

Then the Alternative Hypothesis is H_1 : The sample has not been taken from population.

Here n = Sample size = 155

$\bar{x}$ = Mean of the sample = 67

σ = S.D. of sample = 5.2

μ = Mean of the population = 70

$$\begin{aligned}\therefore \text{The test statistic is } t &= \frac{\bar{x} - \mu}{\text{S.D} / \sqrt{n-1}} = \frac{67 - 70}{5.2 / \sqrt{155-1}} \\ &= \frac{-3}{5.2 / \sqrt{154}} = -7.16\end{aligned}$$

$$\therefore |t| = 7.16 > 3$$

We reject the Null Hypothesis at 5% level of significance and conclude that the sample has not been taken from the given population.

Example 10 : In the past a machine has produced washers having a mean thickness of 0.050 inch. To determine whether the machine is in proper working order a sample of 10 washers is chosen for which the mean thickness is 0.053 inch and the standard deviation is 0.003 inch. Test the hypothesis that the machine is in proper working order using a level of significance of 0.05?

Solution : We have $\mu = 0.050, n = 10, \bar{x} = 0.053, s = 0.003$, where s is the sample S.D.

Since the sample size is small and population S.D σ is not known, we use t - test.

Let the Null Hypothesis be $H_0 : \mu = 0.050$

Then the Alternative Hypothesis is $H_1 : \mu \neq 0.050$

$$\text{The test statistic is } t = \frac{\bar{x} - \mu}{s / \sqrt{n-1}} = \frac{0.053 - 0.050}{0.003 / \sqrt{10-1}} = \frac{0.003}{0.001} = 3$$

Degrees of freedom = $n - 1 = 10 - 1 = 9$.

The tabulated value of t for 9 d.f. at 5% level = 2.262

Since the calculated value of $t >$ the tabulated value, we reject the Null Hypothesis at 5% level and conclude that the machine is not in proper working order.

Example 11 : The following are the times between six calls for an ambulance in a city and the patient's arrival at the hospital : 27,15,20,32,18 and 26 minutes. Use these figures to judge the reasonableness of the ambulance services claim that it takes on the average 20 minutes between the call for an ambulance and patient's arrival at the hospital ?

[JNTU (A) Nov. 2011 (Set No. 1)]

Solution : We have $n =$ Sample size $= 6$

$$\bar{x} = \text{mean of the sample} = \frac{27+15+20+32+18+26}{6} = \frac{138}{6} = 23$$

$$\mu = \text{mean of the population} = 20$$

and

$$\begin{aligned} S^2 &= \text{sample variance} = \frac{\sum (x_i - \bar{x})^2}{n-1} \\ &= \frac{1}{5} [(27-23)^2 + (15-23)^2 + (20-23)^2 + (32-23)^2 + (18-23)^2 + (26-23)^2] \\ &= \frac{1}{5} (16+64+9+81+25+9) = \frac{204}{5} = 40.8 \end{aligned}$$

$$\therefore S = \text{Sample S. D.} = \sqrt{40.8} = 6.4$$

Since the sample size is small and population S. D. σ is not known, we use t - test.

Let the Null Hypothesis be $H_0 : \mu = 20$

Then the Alternative Hypothesis is $H_1 : \mu \neq 20$

The test statistic is

$$t = \frac{\bar{x} - \mu}{S / \sqrt{n}} = \frac{23 - 20}{6.4 / \sqrt{6}} = \frac{3\sqrt{6}}{6.4} = \frac{7.348}{6.4} = 1.148$$

$$\text{Degrees of freedom} = n - 1 = 6 - 1 = 5$$

The tabulated value of t for 5 d.f. at 5% level $= 2.571$

[$\because$ for two -tail test, $\alpha / 2$ is taken for α]

$$\therefore |t| = 1.148 < 2.571$$

Since the calculated value of $|t| <$ the tabulated value, we accept the Null Hypothesis at 5% level and conclude that the average time between the call for an ambulance and patient's arrival at the hospital is significant.

Problems related to Student's t Test (When S.D of the sample is not given directly)

SOLVED EXAMPLES

Example 7 : A random sample of 10 boys had the following I.Q's : 70, 120, 110, 101, 88, 83, 95, 98, 107 and 100.

- (a) Do these data support the assumption of a population mean I.Q of 100 ?
 (b) Find a reasonable range in which most of the mean I.Q. values of samples of 10 boys lie. [JNTU 2008, (A) Apr. 2012, (H) May 2011 (Set No. 4), (K) May 2013 (Set No. 2)]

Solution : (a) Here *S.D.* and mean of the sample is not given directly.

We have to determine these *S.D.* and mean as follows.

$$\text{Mean, } \bar{x} = \frac{\sum x}{n} = \frac{972}{10} = 97.2$$

x	$x - \bar{x}$	$(x - \bar{x})^2$
70	-27.2	739.84
120	22.8	519.84
110	12.8	163.84
101	3.8	14.44
88	-9.2	84.64
83	-14.2	201.64
95	-2.2	4.84
98	0.8	0.64
107	9.8	96.04
100	2.8	7.84
972		1833.60

$$\text{We know that } S^2 = \frac{1}{n-1} \sum_i (x_i - \bar{x})^2 = \frac{1833.60}{9}$$

$$\therefore \text{Standard deviation, } S = \sqrt{203.73} = 14.27$$

- 1. Null Hypothesis H_0 :** The data support the assumption of a population mean I.Q of 100 in the population.
- 2. Alternative Hypothesis H_1 :** $\mu \neq 100$
- 3. Level of significance, $\alpha = 0.05$**

$$\text{4. The test statistic is } t = \frac{\bar{x} - \mu}{S/\sqrt{n}} = \frac{97.2 - 100}{14.27/\sqrt{10}} = -0.62$$

$$\therefore |t| = 0.62 \text{ i.e., Calculated value of } t = 0.62$$

Tabulated value of t for $(10 - 1)$ d.f. *i.e.*, 9 d.f. at 5% level of significance is 2.26 (two-tailed list).

Since calculated value of $t <$ tabulated value of t , we accept the null hypothesis H_0 . *i.e.*, the data support the assumption of mean I.Q of 100 in the population.

(b) The 95% confidence limits are given by $\bar{x} \pm t_{0.05} \cdot S/\sqrt{n}$

$$= 97.2 \pm 2.26 \times 4.512 = 97.2 \pm 10.198 = 107.4 \text{ and } 87$$

$\therefore$ The 95% confidence limits within which the mean I.Q values of samples of 10 boys will lie is (87, 107.40).

Example 7 : The heights of 10 males of a given locality are found to be 70, 67, 62, 68, 61, 68, 70, 64, 64, 66 inches. Is it reasonable to believe that the average height is greater than 64 inches? Test at 5% significance level assuming that for 9 degrees of freedom ($t = 1.833$ at $\alpha = 0.05$).

[JNTU (H) May 2011 (Set No. 2)]

Solution : Mean, $\bar{x} = \frac{\sum x}{n} = \frac{660}{10} = 66$

Calculation for sample mean and S.D.

x	$x - \bar{x}$	$(x - \bar{x})^2$
70	4	16
67	1	1
62	-4	16
68	2	4
61	-5	25
68	2	4
70	4	16
64	-2	4
64	-2	4
66	0	0
660		90

We know that $S^2 = \frac{1}{n-1} \sum_i (x_i - \bar{x})^2 = \frac{1}{9} \times 90 = 10$

$\therefore$ Sample S.D.(S) = $\sqrt{10} = 3.16$

- 1. Null Hypothesis H_0 :** The average height is not greater than 64 inches
i.e., $\mu = 64$ inches.
- 2. Alternative Hypothesis H_1 :** $\mu > 64$ inches
- 3. Level of significance :** $\alpha = 0.05$

4. **Test statistic** is $t = \frac{\bar{x} - \mu}{S/\sqrt{n-1}} = \frac{66-64}{3.16/\sqrt{9}} = \frac{2}{1.05} = 1.9$

i.e., Calculated value $t = 1.9$

Tabulated t for $10 - 1 = 9$ d.f at 5% level of significance for single tail test is 1.833 (given).

Since calculated value of $t >$ tabulated value of t , we reject the null hypothesis H_0 . *i.e.*, The average height is greater than 64 inches.

Example 7 : A random sample from a company's very extensive files shows that the orders for a certain kind of machinery were filled, respectively in 10, 12, 19, 14, 15, 18, 11 and 13 days. Use the level of significance $\alpha = 0.01$ to test the claim that on the average such orders are filled in 10.5 days. Choose the Alternative Hypothesis so that rejection of Null Hypothesis $\mu = 10.5$ days implies that it takes longer than indicated.

[JNTU 2004S, (K) May 2010S (Set No. 3)]

Solution : We have

$$n = 8, \quad \bar{x} = \frac{1}{8} (10 + 12 + 19 + 14 + 15 + 18 + 11 + 13) = \frac{112}{8} = 14$$

and sample S.D, S is given by

$$\begin{aligned} S^2 &= \frac{\sum (x_i - \bar{x})^2}{n-1} \\ &= \frac{1}{7} [(10 - 14)^2 + (12 - 14)^2 + (19 - 14)^2 + \dots + (13 - 14)^2] \\ &= \frac{1}{7} (16 + 4 + 25 + 0 + 1 + 16 + 9 + 1) \\ &= \frac{1}{7} (72) = 10.286 \end{aligned}$$

$$\therefore S = \sqrt{10.286} = 3.207$$

1. **Null Hypothesis H_0 :** $\mu = 10.5$ days
2. **Alternative Hypothesis H_1 :** $\mu > 10.5$ days
3. **Level of significance :** $\alpha = 0.01$
4. **Critical region :** $t > t_{0.01}$

Reject H_0 if $t > t_{0.01} = 2.998$

5. **The Test Statistic** is $t = \frac{\bar{x} - \mu}{S/\sqrt{n-1}} = \frac{14 - 10.5}{3.207/\sqrt{7}} = 3.087$

Tabulated value of t for $(8 - 1)$ d.f. at 1% level of significance is 2.998 (two tailed test).

Since calculated value of $t >$ tabulated value of t , we reject the null hypothesis H_0 . *i.e.*, The orders on average are filled in more than 10.5 days.

Example 4 : The life time of electric bulbs for a random sample of 10 from a large consignment gave the following data.

Item :	1	2	3	4	5	6	7	8	9	10
Life in 1000 hrs :	1.2	4.6	3.9	4.1	5.2	3.8	3.9	4.3	4.4	5.6

Can we accept the hypothesis that the average life time of bulbs is 4000 hrs. Use a 0.05 level of significance. [JNTU Nov. 2008 (Set No. 2)]

Solution : Here $n = 10$.

1. Null Hypothesis $H_0 : \mu = 4000$

2. Alternative Hypothesis $H_1 : \mu \neq 4000$

3. Level of significance : $\alpha = 0.05$

4. The test statistic is $t = \frac{\bar{x} - \mu}{S / \sqrt{n}}$

From given data, $\bar{x}$ = average life time of bulbs

$$= \frac{1.2 + 4.6 + 3.9 + 4.1 + 5.2 + 3.8 + 3.9 + 4.3 + 4.4 + 5.6}{10}$$

$$= \frac{1}{10} (41) = 4.1 \approx 4100$$

$$\begin{aligned} \text{Now } S^2 &= \sum_{i=1}^{10} \frac{1}{n-1} (x_i - \bar{x})^2 \\ &= \frac{1}{9} \left[(1.2 - 4.1)^2 + (4.6 - 4.1)^2 + (3.9 - 4.1)^2 + (4.1 - 4.1)^2 + (5.2 - 4.1)^2 \right. \\ &\quad \left. + (3.8 - 4.1)^2 + (3.9 - 4.1)^2 + (4.3 - 4.1)^2 + (4.4 - 4.1)^2 + (5.6 - 4.1)^2 \right] \\ &= \frac{1}{9} (8.41 + 0.25 + 0.04 + 1.21 + 0.09 + 0.04 + 0.03 + 0.09 + 2.25) \\ &= \frac{12.42}{9} = 1.38 \approx 1380 \Rightarrow S = \sqrt{1380} \end{aligned}$$

$$\text{Hence } t = \frac{\bar{x} - \mu}{S / \sqrt{n}} = \frac{4100 - 4000}{\sqrt{1380} / \sqrt{10}} = 8.512.$$

$$\therefore |t| = 8.512$$

No. of degrees of freedom = 9

$$\therefore t_{\alpha/2} = 2.262 \text{ (From tables)}$$

Thus we see that $|t| > t_{\alpha/2}$.

Hence, we reject the Null Hypothesis H_0 at 5% level and conclude that the average life time bulbs is not equal to 4000 hrs.

Example 5 : The manufacturer of a certain make of electric bulbs claims that his bulbs have a mean life of 25 months with a S. D. of 5 months. A random sample of 6 such bulbs gave the following values. Life of months : 24, 26, 30, 20, 20, 18. Can you regard the producers, claims to be valid at 1% level of significance ? [JNTU (A) 2009 (Set No. 2)]

Solution : We have $n = 6$,

$$\text{Sample mean, } \bar{x} = \frac{24 + 26 + 30 + 20 + 20 + 18}{6} = \frac{138}{6} = 23$$

$$\text{and sample S.D, } S \text{ is given by } S^2 = \frac{\sum(x_i - \bar{x})^2}{n-1}$$

$$\begin{aligned} i.e., S^2 &= \frac{1}{5}[(24-23)^2 + (26-23)^2 + (30-23)^2 + (20-23)^2 + (20-23)^2 + (18-23)^2] \\ &= \frac{1}{5}[1 + 9 + 49 + 9 + 9 + 25] = \frac{102}{5} = 20.4 \end{aligned}$$

$$\therefore S = \sqrt{20.4} = 4.52$$

1. Null Hypothesis $H_0 : \mu = 25$

2. Alternative Hypothesis $H_1 : \mu \neq 25$

3. Level of significance : $\alpha = 0.01$

$$\text{4. The test statistic is } t = \frac{\bar{x} - \mu}{S/\sqrt{n}}$$

$$i.e., t = \frac{23 - 25}{4.52/\sqrt{6}} = \frac{-2\sqrt{6}}{4.52} = -1.08$$

$$\therefore |t| = 1.08$$

i.e., calculated value of $t = 1.08$

Tabulated value of t for $(6-1)$ degrees of freedom at 1% level of significance is 3.365.

Since calculated value of $t <$ tabulated value of t , we accept the null hypothesis H_0 and conclude that the claim is valid.

Example 7 : Producer of 'gutkha', claims that the nicotine content in his 'gutkha' on the average is 1.83mg. Can this claim accepted if a random sample of 8 'gutkha' of this type have the nicotine contents of 2.0, 1.7, 2.1, 1.9, 2.2, 2.1, 2.0, 1.6mg? Use a 0.05 level of significance.

[JNTU (K) 2009, (A) Apr. 2012 (Set No. 1)]

Solution : Given $n = 8$ and $\mu = 1.83$ mg

1. Null Hypothesis $H_0 : \mu = 1.83$

2. Alternative Hypothesis $H_1 : \mu \neq 1.83$

3. Level of significance : $\alpha = 0.05$

$$\text{4. The test statistic is } t = \frac{\bar{x} - \mu}{S/\sqrt{n}} \sim t_{(n-1)}$$

where $\bar{x}$ and S^2 are to be computed from the sample values gutkha.

x	$x - \bar{x}$	$(x - \bar{x})^2$
2.0	0.05	0.0025
1.7	-0.25	0.0625
2.1	0.15	0.0225
1.9	-0.05	0.0025
2.2	0.25	0.0625
2.1	0.15	0.0225
2.0	0.05	0.0025
1.6	-0.35	0.1225
Total	15.6	0.3

$$\text{Now } \bar{x} = \frac{15.6}{8} = 1.95 \text{ and } S^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{0.3}{7} \text{ or } S = 0.21$$

$$\therefore t = \frac{1.95 - 1.83}{0.21 / \sqrt{8}} = \frac{(0.12)(\sqrt{8})}{0.21} = 1.62$$

Tabulated $t_{0.05}$ for $(8-1)$ i.e., 7d.f is 1.895.

Since calculated $t <$ tabulated $t_{0.05}$, the null hypothesis H_0 may be accepted at 5% level of significance and we may conclude that the data are consistent with the assumption of gutkha on the average is 1.83mg.

Example 7 : Eight students were given a test in STATISTICS and after one month coaching they were given another test of the similar nature. The following table gives the increase I their marks in the second test over the first.

Student No.	1	2	3	4	5	6	7	8
Increase of Marks	4	-2	6	-8	12	5	-7	2

Do the marks indicate that the students have gained from the coaching.

[JNTU 2008, (A) Nov. 2011 (Set No. 2)]

Solution : We shall compute the mean and S.D. of the increase of marks as follows.

$$\bar{x} = \frac{\sum x_i}{8} = \frac{4 - 2 + 6 - 8 + 12 + 5 - 7 + 2}{8} = \frac{12}{8} = 1.5$$

$$\begin{aligned} \text{We have } S^2 &= \frac{1}{n-1} \sum_i (x_i - \bar{x})^2 \\ &= \frac{1}{7} [(4 - 1.5)^2 + (-2 - 1.5)^2 + (6 - 1.5)^2 + (-8 - 1.5)^2 \\ &\quad + (12 - 1.5)^2 + (5 - 1.5)^2 + (-7 - 1.5)^2 + (2 - 1.5)^2] \\ &= \frac{1}{7} [6.25 + 12.25 + 20.25 + 90.25 + 110.25 + 12.25 + 72.25 + 0.25] \end{aligned}$$

$$= \frac{324}{7} = 46.2857$$

$$\therefore \text{S.D.} = S = \sqrt{46.2857} = 6.8$$

Assuming that the students have not been benefitted by coaching, it implies that the mean of the difference of the two tests is zero i.e., $\mu = 0$.

$$\text{Then } t = \frac{\bar{x} - \mu}{S / \sqrt{n-1}} = \frac{1.5 - 0}{(6.8) / \sqrt{7}} = 0.5836$$

$$\text{No. of degrees of freedom} = 8 - 1 = 7$$

$$\therefore \text{Tabulated } t_{0.05} = 2.36$$

As the calculated value of $t < t_{0.05}$, the value of t is not significant at 5% level significance i.e., the test provides no evidence that the students have been benefitted by coaching.

Example 8 : A random sample of 10 bags of pesticide are taken whose weights are 50, 49, 52, 44, 45, 48, 46, 45, 49, 45 (in kgs). Test whether the average packing can be taken to be 50 kgs.

[JNTU (H) Apr. 2012 (Set No. 1)]

Solution : We have $\mu = 50, n = 10$

Let the Null Hypothesis be $H_0 : \mu = 50$

Then the Alternative Hypothesis is $H_1 : \mu \neq 50$.

$$\text{The test statistic is } t = \frac{\bar{x} - \mu}{s / \sqrt{n-1}}$$

$$\text{Now } \bar{x} = \frac{1}{10}(50 + 49 + 51 + 44 + 45 + 48 + 46 + 45 + 49 + 45) = \frac{472}{10} = 47.2$$

$$\begin{aligned} \text{Variance, } s^2 &= \frac{\sum (x_i - \bar{x})^2}{n-1} \\ &= \frac{1}{9}[(50-47.2)^2 + (49-47.2)^2 + (51-47.2)^2 + (44-47.2)^2 + \\ &\quad (45-47.2)^2 + (48-47.2)^2 + (46-47.2)^2 \\ &\quad + (45-47.2)^2 + (49-47.2)^2 + (45-47.2)^2] \\ &= \frac{1}{9}[7.84 + 3.24 + 14.44 + 10.24 + 4.84 \\ &\quad + 0.64 + 1.44 + 4.84 + 3.24 + 4.84] = \frac{55.6}{9} = 6.1777 \end{aligned}$$

$$\therefore \text{Standard deviation, } s = \sqrt{6.18} = 2.48$$

$$\text{Hence } t = \frac{47.2 - 50}{2.48/3} = \frac{-8.4}{2.48} = -3.39$$

Degrees of freedom = $n - 1 = 10 - 1 = 9$.

The tabulated value of t for 9 d.f. at 5% level = 2.262

$\therefore |t| = 3.39 > 2.262$ [$\because$ for two-tailed test, $\alpha/2$ is taken for α]

Example 9 : A stimulus administered to 12 patients resulted in the following changes in the blood pressure : 5, 2, 8, -1, 3, 0, -2, 1, 5, 0, 4, 6. Can it be concluded that the stimulus will in general be accompanied by an increase in blood pressure? [JNTU(K) May.2012 (Set No.1)]

Solution : Let x denote the increment in blood pressure. We shall compute the mean and standard deviation of blood pressure as follows:

$$\bar{x} = \frac{\sum x_i}{12} = \frac{1}{12} (5 + 2 + 8 - 1 + 3 + 0 - 2 + 1 + 5 + 0 + 4 + 6) = \frac{31}{12} = 2.58$$

$$\begin{aligned} \text{Also } S^2 &= \frac{1}{n^2} \sum_i (x_i - \bar{x})^2 \\ &= \frac{1}{144} [(5 - 2.58)^2 + (2 - 2.58)^2 + (8 - 2.58)^2 + (-1 - 2.58)^2 + (3 - 2.58)^2 + \\ &\quad (0 - 2.58)^2 + (-2 - 2.58)^2 + (1 - 2.58)^2 + (5 - 2.58)^2 + (0 - 2.58)^2 + \\ &\quad (4 - 2.58)^2 + (6 - 2.58)^2] \\ &= 9.53 \\ \therefore S &= 3.08. \end{aligned}$$

1. **Null Hypothesis:** $H_0 : \mu_1 = \mu_2$.

2. **Alternative Hypothesis:** $\mu_1 \neq \mu_2$.

3. **Level of significance:** $\alpha = 0.05$.

4. The test statistic is $t = \frac{\bar{x} - \mu}{S/\sqrt{n}} = \frac{2.58 - 0}{3.08/\sqrt{12}} = \frac{2.58}{0.89} = 2.89$

Tabulated $t_{0.05}$ for $(12-1)$ degrees of freedom = 2.2

Since calculated $t >$ tabulated t , the Null hypothesis is rejected at 5% level of significance and we may conclude that the stimulus will in general be accompanied by an increase in blood pressure.

Example 10 : A new process of producing synthetic diamonds can be operated at a profitable level only if the average weight of the diamonds is greater than 0.5 carat. To test the profitability of the process, 6 diamonds are produced with weights 0.46, 0.60, 0.52, 0.49, 0.58 and 0.54 carat respectively. Do the 6 measurements present sufficient evidence to indicate that the average weight of the diamonds produced by the process is in excess of 0.5 carat?

[JNTU(A) Dec. 2014]

Solution : Here $\bar{x} = \frac{1}{6}(0.45 + 0.60 + 0.52 + 0.49 + 0.58 + 0.54) = \frac{3.18}{6} = 0.53$

Calculation for Sample Mean and S. D.

x	$x - \bar{x}$	$(x - \bar{x})^2$
0.45	-0.08	0.0064
0.60	0.07	0.0049
0.52	-0.01	0.0001
0.49	-0.04	0.0016
0.58	0.05	0.0025
0.54	0.01	0.0001
3.18		0.0156

$$\therefore S^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2 = \frac{1}{5} \times 0.0156 = 0.00312$$

$$\Rightarrow S = \sqrt{0.00312} = 0.056$$

- 1. Null Hypothesis** H_0 : The average weight of the diamond is not greater than 0.5 carat *i.e.* $\mu = 0.5$
- 2. Alternative Hypothesis** H_1 : $\mu > 0.5$
- 3. Level of significance** : $\alpha = 0.05$

- 4. Test statistic** is $t = \frac{\bar{x} - \mu}{S / \sqrt{n}}$

$$\text{i.e. } t = \frac{0.53 - 0.5}{0.056 / \sqrt{6}} = \frac{(0.03)\sqrt{6}}{0.056} = 1.3122$$

- 5. Critical Region** :

Tabulated value of t for $(6-1)$ d.f. at 5% level of significance for one tail test is 2.015

- 6. Decision** : Since calculated value of $t <$ tabulated value of t , we accept the null hypothesis H_0 . *i.e.*, the average weight of the diamond is not greater than 0.5 carat.

The calculated t does not fall within the rejection region. Therefore, we cannot reject H_0 . Hence the data do not give sufficient evidence to indicate that the mean diamond weight exceeds 0.5 carat.

Example 11 : A random sample of 8 envelopes is taken from the letter box of a post office and their weights in grams are found to be: 12.1, 11.9, 12.4, 12.3, 11.5, 11.6, 12.1 and 12.4.

- (i) Does this sample indicate at 1% level that the average weight of envelopes received at their post office is 12.35 grms.
- (ii) Find 95% confidence limits for the mean weight of the envelopes received at that post office. [JNTU(K) March.2014 (Set No.4)]

Solution :

- (i) 1. **Null Hypothesis** is $H_0 : \mu = 12.35$ grms *i.e.*, the samples have been drawn from the same population of mean 12.35 grms.
2. **Alternative Hypothesis** is $H_1 : \mu \neq 12.35$ grms
3. **Level of significance** : $\alpha = 0.01$
4. **The test statistic** is $t = \frac{\bar{x} - \mu}{S/\sqrt{n}}$

Here $n = 8$.

Now, sample mean,

$$\bar{x} = \frac{1}{8} (12.1 + 11.9 + 12.4 + 12.3 + 11.5 + 11.6 + 12.1 + 12.4) = \frac{96.3}{8} = 12.0375.$$

$$\text{Sample variance, } S^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}$$

$$\begin{aligned} &= \frac{1}{7} [(12.1 - 12.0375)^2 + (11.9 - 12.0375)^2 + (12.4 - 12.0375)^2 + (12.3 - 12.0375)^2 + \\ &\quad (11.5 - 12.0375)^2 + (11.6 - 12.0375)^2 + (12.1 - 12.0375)^2 + (12.4 - 12.0375)^2] \\ &= \frac{1}{7} [0.0039 + 0.0189 + 0.1314 + 0.0689 + 0.2889 + 0.1914 + 0.0039 + 0.1314] \\ &= \frac{0.8387}{7} = 0.1198 \end{aligned}$$

$$\therefore S = \sqrt{0.1198} = 0.3461$$

$$\text{Thus } t = \frac{12.0375 - 12.35}{0.3461/\sqrt{8}} = \frac{2.8284(-0.3125)}{0.3461} = \frac{-0.883875}{0.3461} = -2.5538$$

$$\therefore |t| = 2.5538 \text{ i.e., calculated value of } t = 2.5538$$

Tabulated value of t for $(8-1)$ degrees of freedom at $\frac{\alpha}{2} = \frac{1}{2}\%$ level of significance is

$$3.499 \text{ i.e., } t_{\alpha/2} = t_{0.005} = 3.499$$

Since calculated value of $t <$ tabulated value of t , we accept the Null Hypothesis H_0 and conclude that the sample is taken from the population whose mean is 12.35.

(ii) The 95% confidence limits are given by $\bar{x} \pm t_{\alpha/2} \cdot \frac{S}{\sqrt{n}}$

$$i.e., \left(\bar{x} - t_{\alpha/2} \cdot \frac{S}{\sqrt{n}}, \bar{x} + t_{\alpha/2} \cdot \frac{S}{\sqrt{n}} \right)$$

Now $t_{\alpha/2} = t_{0.025} = 2.365$ [From Tables]

$$\therefore t_{\alpha/2} \cdot \frac{S}{\sqrt{n}} = \frac{(2.365)(0.3461)}{\sqrt{8}} = \frac{0.8185}{2.8284} = 0.2894$$

Hence confidence limits = $(12.0375 - 0.2894, 12.0375 + 0.2894)$
 $= (11.7481, 12.3269)$.

10.12 STUDENT'S t - TEST FOR DIFFERENCE OF MEANS

Let $\bar{x}$ and $\bar{y}$ be the means of two independent samples of sizes n_1 and n_2 ($n_1 < 30, n_2 < 30$) drawn from two normal populations having means μ_1 and μ_2 . To test whether the two population means are equal (*i.e.*, to test whether the difference $\mu_1 - \mu_2$ is significant).

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

If $\sigma_1 = \sigma_2 = \sigma$, then an unbiased estimate S^2 of the common variance σ^2 is given by $S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$ where s_1^2, s_2^2 are the two sample variances.

$$\text{Also S. E. of } (\bar{x} - \bar{y}) = S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \text{ where } S = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}}$$

Assuming that H_0 is true, the test statistic t given by $t = \frac{\bar{x} - \bar{y}}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$, follows t -distribution with $(n_1 + n_2 - 2)$ d.f.

$$\text{Here } \bar{x} = \frac{1}{n_1} \sum_{i=1}^{n_1} x_i, \bar{y} = \frac{1}{n_2} \sum_{i=1}^{n_2} y_i \text{ and } S^2 = \frac{1}{n_1 + n_2 - 2} [\sum (x_i - \bar{x})^2 + \sum (y_i - \bar{y})^2]$$

$$\text{or } S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

We calculate the value of $|t|$ and compare this value with the table value at α level of significance. If the calculated value of $|t| >$ the table value, we reject H_0 at α level. Otherwise we accept H_0 .

95% confidence limits for the difference of two population means are

$$(\bar{x} - \bar{y}) \pm t_{\alpha} \cdot S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \quad \text{where } \alpha = 0.025.$$

and 99% confidence limits for the difference of two population means are

$$(\bar{x} - \bar{y}) \pm t_{\alpha} \cdot S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \quad \text{where } \alpha = 0.0005$$

SOLVED EXAMPLES

Example 1 : Samples of two types of electric light bulbs were tested for length of life and following data were obtained

Type I	Type II
Sample number, $n_1 = 8$	$n_2 = 7$
Sample mean, $\bar{x} = 1234$ hours	$\bar{y} = 1036$ hrs
Sample S.D., $s_1 = 36$ hrs	$s_2 = 40$ hrs

Is the difference in the means sufficient to warrant that type I is superior to type II regarding length of life. [JNTU (A) June 2016]

Solution : Since the sample sizes are small and σ_1, σ_2 are not known, we use t -test. Let μ_1 and μ_2 be the two population means.

Null Hypothesis H_0 : The two types I and II of electric bulbs are identical

$$i.e., \mu_1 = \mu_2$$

Alternative Hypothesis H_1 : $\mu_1 \neq \mu_2$

Since two sample means $\bar{x}$ and $\bar{y}$ are given and also sample standard deviations S_1 and S_2 are given, we use the statistic

$$t = \frac{\bar{x} - \bar{y}}{S^2 \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$\text{where } S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

$$= \frac{1}{8 + 7 - 2} [(8(36)^2 + 7(40)^2)] = 1659.08$$

$$\therefore t = \frac{1234 - 1036}{\sqrt{1659.08 \left(\frac{1}{8} + \frac{1}{7}\right)}} = 9.39$$

Degrees of freedom (d.f) = $n_1 + n_2 - 2 = 8 + 7 - 2 = 13$

Tabulated value of t for 13 d.f at 5% level is 2.160 (two-tail test)

Since calculated $t >$ tabulated t , we reject the null hypothesis H_0 and conclude that the two types I and II of electric bulbs are not identical.

Example 2 : The means of two random samples of sizes 9 and 7 are 196.42 and 198.82 respectively. The sum of the squares of the deviations from the mean are 26.94 and 18.73 respectively. Can the sample be considered to have been drawn from the same normal population.

Solution : Given $n_1 = 9$, $n_2 = 7$, $\bar{x} = 196.42$, $\bar{y} = 198.82$

and $\sum(x_i - \bar{x})^2 = 26.94$, $\sum(y_i - \bar{y})^2 = 18.73$

$$\therefore S^2 = \frac{\sum(x_i - \bar{x})^2 + \sum(y_i - \bar{y})^2}{n_1 + n_2 - 2} = \frac{26.94 + 18.73}{9 + 7 - 2} = 3.26$$

$$\Rightarrow S = 1.81$$

1. **Null Hypothesis H_0 :** The two samples are drawn from the same population
i.e. $\mu_1 = \mu_2$

2. **Alternative Hypothesis H_1 :** $\mu_1 \neq \mu_2$

3. **Level of significance :** $\alpha = 0.05$

4. **The test statistic is $t = \frac{\bar{x} - \bar{y}}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{196.42 - 198.82}{(1.81) \sqrt{\frac{1}{9} + \frac{1}{7}}}$**

$$= \frac{-2.4}{0.912} = -2.63$$

i.e., calculated value of $|t| = 2.63$

Tabulated value of t for $9 + 7 - 2 = 14$ d.f at 5% level of significance is 2.15. Since calculated value of $t >$ tabulated value of t , we reject the null hypothesis H_0 and conclude that the two samples are not drawn from the same population.

Example 1 : Below are given the gain in weights (in lbs) of pigs fed on two diets A and B.

Diet A	25	32	30	34	24	14	32	24	30	31	35	25	—	—	—
Diet B	44	34	22	10	47	31	40	30	32	35	18	21	35	29	22

Test, if the two diets differ significantly as regards their effect on increase in weight.

Solution :

1. **Null Hypothesis H_0 :** There is no significant difference between the mean increase in weight due to diets A and B *i.e.*, $\mu_1 = \mu_2$.

2. **Alternative Hypothesis H_1 :** $\mu_1 \neq \mu_2$

3. **Level of significance, $\alpha = 0.05$**

4. Calculation for sample means and S.D's.

Here $n_1 = 12$, $n_2 = 15$, $\Sigma x = 336$, $\Sigma y = 450$

$$\therefore \bar{x} = \frac{336}{12} = 28, \bar{y} = \frac{450}{15} = 30$$

x	$x - \bar{x}$	$(x - \bar{x})^2$	y	$(y - \bar{y})$	$(y - \bar{y})^2$
25	-3	9	44	14	196
32	4	16	34	4	16
30	2	4	22	-8	64
34	6	36	10	-20	400
24	-4	16	47	17	289
14	-14	196	31	1	1
32	4	16	40	10	100
24	-4	16	30	0	0
30	2	4	32	2	4
31	3	9	35	5	25
35	7	49	18	-12	144
25	-3	9	21	-9	81
			35	5	25
			29	-1	1
			22	-8	64
336		380	450		1410

$$\sum_{i=1}^{12} (x_i - \bar{x})^2 = 380, \quad \sum_{i=1}^{15} (y_i - \bar{y})^2 = 1410$$

$$\therefore S^2 = \frac{1}{n_1 + n_2 - 2} \left[\sum_i (x_i - \bar{x})^2 + \sum_i (y_i - \bar{y})^2 \right]$$

$$= \frac{1}{12 + 15 - 2} [380 + 1410] = 71.6$$

5. The test statistic is

$$t = \frac{\bar{x} - \bar{y}}{\sqrt{S^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{30 - 28}{\sqrt{71.6 \left(\frac{1}{12} + \frac{1}{15} \right)}} = \frac{2}{\sqrt{10.74}} = 0.609$$

Degrees of freedom = $n_1 + n_2 - 2 = 12 + 15 - 2 = 25$

Tabulated t for 25 d.f at 5% level = 2.06

Since calculated $t <$ tabulated t , therefore, we accept the null hypothesis H_0 .

That is, the two diets do not differ significantly as regards their effect on increase in weight.

Example 4 : A group of 5 patients treated with medicine A weigh 42, 39, 48, 60 and 41 kgs. Second group of 7 patients from the same hospital treated with medicine B weigh 38, 42, 56, 64, 68, 69 and 62 kgs. Do you agree with the claim that medicine B increases the weight significantly. [JNTU (K) Nov. 2011 (Set No. 3)]

Solution : Calculation for sample means and S.D's

x	$x - \bar{x}$	$(x - \bar{x})^2$	y	$(y - \bar{y})$	$(y - \bar{y})^2$
42	-4	16	38	-19	361
39	-7	49	42	-15	225
48	2	4	56	-1	1
60	14	196	64	7	49
41	-5	25	68	11	21
			69	12	44
			62	5	25
230	0	290	399	0	926

$$\text{Now } \bar{x} = \frac{230}{5} = 46, \quad \bar{y} = \frac{399}{7} = 57,$$

$$\text{and } \sum_{i=1}^5 (x_i - \bar{x})^2 = 290, \quad \sum_{j=1}^7 (y_j - \bar{y})^2 = 926$$

$$\therefore S^2 = \frac{1}{n_1 + n_2 - 2} [\sum (x_i - \bar{x})^2 + \sum (y_j - \bar{y})^2] = \frac{1}{5+7-2} [290 + 926] = 121.6$$

$$\Rightarrow S = 11.03$$

1. **Null Hypothesis H_0 :** There is no significant difference between the medicines A and B as regards their effect on increase in weight i.e., $H_0 : \mu_1 = \mu_2$
2. **Alternative Hypothesis H_1 :** $\mu_1 > \mu_2$
3. **Level of significance, $\alpha = 0.05$**

$$4. \text{ The test statistic is } t = \frac{\bar{x} - \bar{y}}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{46 - 57}{(11.03) \sqrt{\frac{1}{5} + \frac{1}{7}}} = \frac{-11}{6.46} = -1.7$$

$$\therefore \text{Calculated value of } |t| = 1.7$$

Tabulated value of t for $5 + 7 - 2 = 10$ d.f at 5% level of significant is 1.81 (right-tail test).

Since calculated $t <$ tabulated t , we accept H_0 , i.e., The medicines A and B do not differ significantly as regards their effect on increase in weight.

Example 5 : Two horses A and B were tested according to the time (in seconds) to run a particular track with the following results.

Horse A	28	30	32	33	33	29	34
Horse B	29	30	30	24	27	29	

Test whether the two horses have the same running capacity.

[JNTU 2006S, (A) Nov. 2010 (Set No. 1)]

Solution : Given $n_1 = 7$, $n_2 = 6$

We first compute the sample means and standard deviations.

$$\begin{aligned}\bar{x} &= \text{Mean of first sample} = \frac{1}{7}(28 + 30 + 32 + 33 + 33 + 29 + 34) \\ &= \frac{1}{7}(219) = 31.286\end{aligned}$$

$$\begin{aligned}\bar{y} &= \text{Mean of second sample} = \frac{1}{6}(29 + 30 + 30 + 24 + 27 + 29) \\ &= \frac{1}{6}(169) = 28.16\end{aligned}$$

x	$x - \bar{x}$	$(x - \bar{x})^2$	y	$(y - \bar{y})$	$(y - \bar{y})^2$
28	-3.286	10.8	29	0.84	0.7056
30	-1.286	1.6538	30	1.84	3.3856
32	0.714	0.51	30	1.84	3.3856
33	1.714	2.94	24	-4.16	17.3056
33	1.714	2.94	27	-1.16	1.3456
29	-2.286	5.226	29	0.84	0.7056
34	2.714	7.366			
219		31.4358	169		26.8336

$$\text{Now } S^2 = \frac{1}{n_1 + n_2 - 2} [\Sigma(x_i - \bar{x})^2 + \Sigma(y_i - \bar{y})^2]$$

$$= \frac{1}{11} [31.4358 + 26.8336] = \frac{1}{11} (58.2694) = 5.23$$

$$\therefore S = \sqrt{5.23} = 2.3$$

1. Null Hypothesis $H_0 : \mu_1 = \mu_2$
2. Alternative Hypothesis $H_1 : \mu_1 \neq \mu_2$
3. Level of significance : $\alpha = 0.05$

4. The test statistic is $t = \frac{\bar{x} - \bar{y}}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{31.286 - 28.16}{(2.3) \sqrt{\frac{1}{7} + \frac{1}{6}}} = 2.443$

Tabulated value of t for $7 + 6 - 2 = 11$ d.f. at 5% level of significance is 2.2.

Since Calculated $t >$ Tabulated t , we reject the null hypothesis H_0 and conclude that both horses A and B do not have the same running capacity.

Example 4 : To examine the hypothesis that the husbands are more intelligent than the wives, an investigator took a sample of 10 couples and administered them a test which measures the I.Q. The results are as follows :

Husbands	117	105	97	105	123	109	86	78	103	107
Wives	106	98	87	104	116	95	90	69	108	85

Test the hypothesis with a reasonable test at the level of significance of 0.05.

[JNTU 2004, 2005, 2007S, (A) Nov. 2010, (K) May 2013 (Set No. 3)]

Solution : We have $n_1 = 10$, $n_2 = 10$ and

$$\begin{aligned}\bar{x} &= \frac{1}{10}(117 + 105 + 97 + 105 + 123 + 109 + 86 + 78 + 103 + 107) \\ &= \frac{1}{10}(1030) = 103\end{aligned}$$

$$\begin{aligned}\bar{y} &= \frac{1}{10}(106 + 98 + 87 + 104 + 116 + 95 + 90 + 69 + 108 + 85) \\ &= \frac{1}{10}(958) = 95.8\end{aligned}$$

Now we compute the standard deviations of both samples

x	$x - \bar{x}$	$(x - \bar{x})^2$	y	$(y - \bar{y})$	$(y - \bar{y})^2$
117	14	196	106	10.2	104.04
105	2	4	98	2.2	4.84
97	-6	36	87	-8.8	77.44
105	2	4	104	8.2	67.24
123	20	400	116	20.2	408.04
109	6	36	95	-0.8	0.64
86	-17	289	90	-5.8	33.64
78	-25	625	69	-26.8	718.24
103	0	0	108	12.2	148.84
107	4	16	85	-10.8	116.64
1030		1606	958		1679.6

$$\begin{aligned}\text{Now } S^2 &= \frac{1}{n_1 + n_2 - 2} [\sum (x_i - \bar{x})^2 + \sum (y_i - \bar{y})^2] \\ &= \frac{1}{18} [1606 + 1679.6] = \frac{1}{18} (3285.6) = 182.53\end{aligned}$$

$$\therefore S = 13.51$$

1. **Null Hypothesis H_0 :** $\mu_1 = \mu_2$ (i.e., no difference in I.Q.)
2. **Alternative Hypothesis H_1 :** $\mu_1 > \mu_2$
(i.e., husbands are more intelligent than wives) (one tailed test, right)
3. **Level of significance :** $\alpha = 0.05$
4. **The test statistic is $t = \frac{\bar{x} - \bar{y}}{S \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{103 - 95.8}{(13.51) \sqrt{\frac{1}{10} + \frac{1}{10}}} = 1.19168$**

Since $t_{cal} = 1.19168 < t_{tab} = 1.734$, we accept the null hypothesis H_0 i.e., There is no difference in I.Q's.

Example 2 : Find the maximum difference that we can expect with probability 0.95 between the means of samples of sizes 10 and 12 from a normal population if their standard deviations are found to be 2 and 3 respectively.

[JNTU 2004S, (A) Nov. 2010, Apr. 2012 (Set No. 2)]

Solution : We have $n_1 = 10$, $n_2 = 12$, $s_1 = 2$ and $s_2 = 3$

$$\therefore S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2} = \frac{1}{10 + 12 - 2} [10(2)^2 + 12(3)^2] = 7.4$$

$$\Rightarrow S = \sqrt{7.4} = 2.72$$

1. **Null Hypothesis H_0 :** $\mu_1 = \mu_2$
2. **Alternative Hypothesis H_1 :** $\mu_1 \neq \mu_2$
3. **Level of significance :** $\alpha = 0.05$

$$4. \text{ The test statistic is } t = \frac{\bar{x} - \bar{y}}{S \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$\Rightarrow |\bar{x} - \bar{y}| = |t| \cdot S \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = (2.086) (2.72) \sqrt{\frac{1}{10} + \frac{1}{12}} = 2.43$$

[$\because$ Tabulated value of t for $10+12-2=20$ d.f. at 5% level of significance is 2.086 (two-tailed)]

Hence the maximum difference between the means is 2.43.

Example 8 : Two independent samples of 8 and 7 items respectively had the following values.

Sample I	11	11	13	11	15	9	12	14
Sample II	9	11	10	13	9	8	10	–

Is the difference between the means of samples significant ?

[JNTU 2005S (Set No. 1), (A) Dec. 2017]

Solution : Given $n_1 = 8$, $n_2 = 7$ and

$$\bar{x} = \frac{1}{8}(11 + 11 + 13 + 11 + 15 + 9 + 12 + 14) = \frac{96}{8} = 12$$

$$\bar{y} = \frac{1}{7}(9 + 11 + 10 + 13 + 9 + 8 + 10) = \frac{70}{7} = 10$$

x	$x - \bar{x}$	$(x - \bar{x})^2$	y	$(y - \bar{y})$	$(y - \bar{y})^2$
11	-1	1	9	-1	1
11	-1	1	11	1	1
13	1	1	10	0	0
11	-1	1	13	3	9
15	3	9	9	-1	1
9	-3	9	8	-2	4
12	0	0	10	0	0
14	2	4			
96		26	70		16

$$\begin{aligned} \text{Now } S^2 &= \frac{1}{n_1 + n_2 - 2} [\Sigma(x_i - \bar{x})^2 + \Sigma(y_i - \bar{y})^2] \\ &= \frac{1}{8 + 7 - 2} (26 + 16) = \frac{42}{13} = 3.23 \end{aligned}$$

$$\therefore S = 1.8$$

1. **Null Hypothesis H_0 :** $\mu_1 = \mu_2$
2. **Alternative Hypothesis H_1 :** $\mu_1 \neq \mu_2$ (two tailed test)
3. **Level of significance :** $\alpha = 0.05$

$$4. \text{ The test statistic is } t = \frac{\bar{x} - \bar{y}}{S \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{12 - 10}{(1.8) \sqrt{\frac{1}{8} + \frac{1}{7}}} = \frac{2}{0.9316} = 2.15$$

Tabulated value of t for $8 + 7 - 2 = 13$ d.f. at 5% level of significance is 2.16.

Since calculated $t <$ tabulated t , we accept H_0 i.e., the difference between the means of samples is not significant.

Example 9 : Measuring specimens of nylon yarn, taken from two machines, it was found that 8 specimens from first machine had a mean denier of 9.67 with a standard deviation of 1.81 while 10 specimens from second machine had a mean denier of 7.43 with a standard deviation of 1.48. Assuming that the proportions are normal, test the hypothesis $H_0 : \mu_1 - \mu_2 = 1.5$ against $H_1 : \mu_1 - \mu_2 > 1.5$ at 0.05 level of significance.

[JNTU 2005S, 2006S (Set No. 1, 4)]

Solution: We have $n_1 = 8$, $n_2 = 10$, $\bar{x} = 9.67$, $\bar{y} = 7.43$, $s_1 = 1.81$ and $s_2 = 1.48$

$$\begin{aligned} \text{Now } S^2 &= \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2} = \frac{8(1.81)^2 + 10(1.48)^2}{8 + 10 - 2} \\ &= \frac{26.2088 + 21.901}{16} = \frac{48.1128}{16} = 3.007 \end{aligned}$$

$$\therefore S = \sqrt{3.007} = 1.734$$

1. Null Hypothesis $H_0 : \mu_1 - \mu_2 = 1.5$ ($= \delta$)
2. Alternative Hypothesis $H_1 : \mu_1 - \mu_2 > 1.5$
3. Level of significance : $\alpha = 0.05$

$$\begin{aligned} \text{4. The test statistic is } t &= \frac{(\bar{x} - \bar{y}) - \delta}{S \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \\ &= \frac{(9.67 - 7.43) - 1.5}{(1.734) \sqrt{\frac{1}{8} + \frac{1}{10}}} = \frac{0.74}{60.8225} = 0.8997 \end{aligned}$$

Tabulated value of t for $8 + 10 - 2 = 16$ d.f. at 5% level of significance is 1.746 (one-tail test).

Since calculated $t <$ tabulated t , we accept H_0 .

Example 10 : Ten soldiers participated in a shooting competition in the first week. After intensive training they participated in the competition in the second week. Their scores before and after training are given as follows :

Scores before	67	24	57	55	63	54	56	68	33	43
Scores after	70	38	58	58	56	67	68	75	42	38

Do the data indicate that the soldiers have been benefited by the training.

[JNTU 2005S (Set No. 2)]

Solution: We have $n_1 = 10$, $n_2 = 10$

Let $\bar{x}$ = Mean of the scores before training

$$= \frac{1}{10}(67 + 24 + 57 + 55 + 63 + 54 + 56 + 68 + 33 + 43)$$

$$= \frac{1}{10}(520) = 52$$

and $\bar{y}$ = Mean of the scores after training

$$= \frac{1}{10}(70 + 38 + 58 + 58 + 56 + 67 + 68 + 75 + 42 + 38)$$

$$= \frac{1}{10}(570) = 57$$

Now we compute the standard deviations of the two samples.

x	$x - \bar{x}$	$(x - \bar{x})^2$	y	$(y - \bar{y})$	$(y - \bar{y})^2$
67	15	225	70	13	169
24	-28	784	38	-19	361
57	5	25	58	1	1
55	3	9	58	1	1
63	11	121	56	-1	1
54	2	4	67	10	100
56	4	16	68	11	121
68	16	256	75	18	324
33	-19	361	42	-15	225
43	-9	81	38	-19	361
520		1882	570		1664

$$\begin{aligned} \text{Now } S^2 &= \frac{1}{n_1 + n_2 - 2} \left[\sum_i (x_i - \bar{x})^2 + \sum_i (y_i - \bar{y})^2 \right] \\ &= \frac{1}{10 + 10 - 2} [1882 + 1664] = \frac{1}{18}(3546) = 197 \end{aligned}$$

$$\Rightarrow S = 14.0357$$

1. **Null Hypothesis H_0 :** $\mu_1 = \mu_2$ i.e, there is no benefit of training
2. **Alternative Hypothesis H_1 :** $\mu_1 < \mu_2$ i.e, there is a benefit of training.
3. **Level of significance :** $\alpha = 0.05$

$$\begin{aligned} \text{4. The test statistic is } t &= \frac{(\bar{x} - \bar{y})}{S \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{52 - 57}{(14.0357) \cdot \sqrt{\frac{1}{10} + \frac{1}{10}}} \\ &= \frac{-5}{6.277} = -0.796 \end{aligned}$$

Tabulated value of t for $10 + 10 - 2 = 18$ d.f. at 5% level of significance (left tailed test) is -1.734

Since $|t|_{\text{cal}} = 0.796 < 1.734 = |t|_{\text{tab}}$, we accept the null Hypothesis H_0 i.e., the soldiers have been benefited by the training.

Example 11 : To compare two kinds of bumper guards, 6 of each kind were mounted on a car and then the car was run into a concrete wall. The following are the costs of repairs.

Bumper Guard 1	107	148	123	165	102	119
Bumper Guard 2	134	115	112	151	133	129

Use the 0.01 level of significance to test whether the difference between the means of these two samples is significant.

[JNTU 2005, (H) Nov. 2010, (A) Nov. 2011 (Set No. 3), June 2018]

Solution : We have

$$n_1 = 6, n_2 = 6, \bar{x} = \frac{1}{6}(107 + 148 + 123 + 165 + 102 + 119) = \frac{764}{6} = 127.33$$

and $\bar{y} = \frac{1}{6}(134 + 115 + 112 + 151 + 133 + 129) = \frac{774}{6} = 129$

Now we compute the standard deviations of two samples.

x	$x - \bar{x}$	$(x - \bar{x})^2$	y	$(y - \bar{y})$	$(y - \bar{y})^2$
107	-20.33	413.31	134	5	25
148	20.67	427.25	115	-14	196
123	-4.33	18.75	112	-17	289
165	37.67	1419.03	151	22	484
102	-25.33	641.61	133	4	16
119	-8.33	69.39	129	0	0
764		2989.34	774		1010

$$\begin{aligned} \therefore S^2 &= \frac{1}{n_1 + n_2 - 2} [\Sigma(x_1 - \bar{x})^2 + \Sigma(y_1 - \bar{y})^2] \\ &= \frac{1}{16 + 6 - 2} [2989.34 + 1010] = \frac{1}{10} (3999.34) = 399.934 \\ \Rightarrow S &= 19.998 \end{aligned}$$

1. Null Hypothesis $H_0 : \mu_1 = \mu_2$
2. Alternative Hypothesis $H_1 : \mu_1 \neq \mu_2$ (two tailed test)
3. Level of significance : $\alpha = 0.01$
4. The test statistic is

$$t = \frac{\bar{x} - \bar{y}}{S \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{127.33 - 129}{(19.998) \sqrt{\frac{1}{6} + \frac{1}{6}}} = \frac{-1.67}{11.546} = -0.1446$$

$\therefore$ Calculated value of $|t| = 0.1446$

Tabulated value of t for $6 + 6 - 2 = 10$ d.f. at 1% level of significance is 3.169

Since calculated $t <$ tabulated t , we accept the Null Hypothesis H_0 . i.e., there is no significant difference between the sample means.

Example 12 : Random samples of specimens of coal from two mines A and B are drawn and their heat-producing capacity (in millions of calories/ton) were measured yielding the following results:

Mine A	8350	8070	8340	8130	8260	---
Mine B	7900	8140	7920	7840	7890	7950

Is there significant difference between the means of these two samples at 0.01 level of significance.
[JNTU 2005, (K) Nov. 2009 (Set No. 3)]

Solution : Here $n_1 = 5$, $n_2 = 6$

Calculation for mean's and S.D's of the samples:

Mine A			Mine B		
x	$x - \bar{x}$ ($x - 8230$)	$(x - \bar{x})^2$	y	$(y - \bar{y})$ ($y - 7940$)	$(y - \bar{y})^2$
8350	120	14400	7900	-40	1600
8070	-160	25600	8140	200	40000
8340	110	12100	7920	-20	400
8130	-100	10000	7840	100	10000
8260	30	900	7890	-50	2500
			7950	10	100
41150		63000	47640		54600

$$\bar{x} = \frac{\sum x}{n} = \frac{41150}{5} = 8230, \quad \bar{y} = \frac{\sum y}{n} = \frac{47640}{6} = 7940$$

$$\text{Now } S^2 = \frac{1}{n_1 + n_2 - 2} [\sum (x_i - \bar{x})^2 + \sum (y_i - \bar{y})^2]$$

$$= \frac{1}{5 + 6 - 2} [63000 + 54600] = \frac{1}{9} (117600) = 13066.67$$

$$\therefore S = 114.31$$

1. Null Hypothesis $H_0 : \mu_1 = \mu_2$

2. Alternative Hypothesis $H_1 : \mu_1 \neq \mu_2$

3. Level of significance, $\alpha = 0.01$

4. The test statistic is

$$t = \frac{\bar{x} - \bar{y}}{S \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{8230 - 7940}{(114.31) \sqrt{\frac{1}{5} + \frac{1}{6}}} = \frac{290}{(114.31)(0.6055)} = 4.19$$

i.e., calculated value of $|t| = 4.19$

Tabulated value of t for $5 + 6 - 2 = 9$ d.o.f at $\alpha = 0.01$ is 2.821

Since calculated $t > t_{0.005}$, we reject the null hypothesis H_0 .

Thus we conclude that there is significant difference between the means of the two samples.

Example 11 : The IQs (intelligence quotients) of 16 students from one area of a city showed a mean of 107 with a standard deviation of 10, while the IQs of 14 students from another area of the city showed a mean of 112 with a standard deviation of 8. Is there a significant difference between the IQs of the two groups at a 0.05 level of significance?

[JNTU (K) May 2010 (Set No. 2)]

Solution : Let μ_1 and μ_2 be the means of the two populations.

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$.

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$.

We have $n_1 = 16, \bar{x} = 107, s_1 = 10$ and $n_2 = 14, \bar{y} = 112, s_2 = 8$

Since the sample sizes are small and σ_1, σ_2 are not known, we use t -test.

An unbiased estimate S^2 of the common variance is given by

$$S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2} = \frac{16 \times 100 + 14 \times 64}{16 + 14 - 2} = \frac{2496}{28} = 89.14$$

$$\therefore S = \sqrt{89.14} = 9.44$$

Let us assume that H_0 is true, *i.e.*, there is no difference between μ_1 and μ_2 .

The test statistic is

$$t = \frac{\bar{x} - \bar{y}}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{107 - 112}{9.44 \sqrt{\frac{1}{16} + \frac{1}{14}}} = \frac{-5}{9.44 \times 0.366} = -1.447$$

Also d.f. = $16 + 14 - 2 = 28$

$\therefore |t| = 1.447 < 1.701$ for 28 d.f. *i.e.*, the calculated value of $|t| <$ the tabulated value at 5% level with 28 d.f.

Hence, we accept the Null Hypothesis and conclude that the difference between the IQs of the two groups is not significant.

Example 12 : The table gives the biological values of protein from 6 cow's milk and 6 buffalo's milk. Examine whether the differences are significant.

Cow's milk	1.8	2.0	1.9	1.6	1.8	1.5
Buffalo's milk	2	1.8	1.8	2.0	2.1	1.9

[JNTU (H) Dec. 2011 (Set No. 3)]

Solution : Let x and y represent the protein values of cow's milk and buffalo's milk respectively.

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

Calculations for Sample Means and S. D.'s

x	$x - \bar{x}$ $= x - 1.77$	$(x - \bar{x})^2$	y	$y - \bar{y}$ $= y - 1.93$	$(y - \bar{y})^2$
1.8	0.03	0.0009	2	0.07	0.0049
2.0	0.23	0.0529	1.8	-0.13	0.0169
1.9	0.13	0.0169	1.8	-0.13	0.0169
1.6	-0.17	0.0289	2.0	0.07	0.0049
1.8	0.03	0.0009	2.1	0.17	0.0289
1.5	-0.27	0.0729	1.9	-0.03	0.0009
$10.6 = \sum x_i$		0.1734	$11.6 = \sum y_i$		0.0734

$$\bar{x} = \frac{\sum x_i}{n_1} = \frac{10.6}{6} = 1.77 \text{ and } \bar{y} = \frac{\sum y_i}{n_2} = \frac{11.6}{6} = 1.93$$

$$\begin{aligned} \text{Now } S^2 &= \frac{1}{n_1 + n_2 - 2} [\sum (x - \bar{x})^2 + \sum (y - \bar{y})^2] \\ &= \frac{1}{6 + 6 - 2} [0.1734 + 0.0734] = \frac{0.2468}{10} = 0.0247 \end{aligned}$$

$$\therefore S = \sqrt{0.0247} = 0.157$$

$\therefore$ The test statistic is

$$\begin{aligned} t &= \frac{\bar{x} - \bar{y}}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{1.77 - 1.93}{(0.157) \sqrt{\frac{1}{6} + \frac{1}{6}}} \\ &= \frac{(-0.16)\sqrt{3}}{0.157} = \frac{-0.2771}{0.157} = -1.7651 \end{aligned}$$

$$\therefore |t| = 1.7651$$

$$\text{d.f.} = n_1 + n_2 - 2 = 6 + 6 - 2 = 10$$

Tabulated value of t for 10 d.f. at 5% level of significance is 2.228 (two -tailed test).

Since the calculated value of $|t| <$ the table value of t at 5% level with 10 d.f. for two -tailed test, we accept Null Hypothesis H_0 and conclude that there is no significant difference in their means.

10.13 PAIRED - SAMPLE t - TEST

Paired observations arise in many practical situations where each homogeneous experimental unit receives both population conditions. As a result, each experimental unit has a pair of observations, one for each population.

For instance, to test the effectiveness of “drug” some 11 persons blood pressure is measured “before” and “after” the intake of certain drug. Here the individual person is the experimental unit and the two populations are blood pressure “before” and “after” the drug is given. Thus for each observation in one sample, there is a corresponding observation in the other sample pertaining to the same character. Hence the two samples are not independent.

Consider another example. Suppose a business concern is interested to know whether a particular media of promoting sales of a product is really effective or not. In this case we have to test whether the average sales before and after the sales promotion are equal.

If $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ be the pairs of sales data before and after the sales promotion in a business concern, we apply paired t - test to examine the significance of the difference of the two situations .

Let $d_i = x_i - y_i$ or $y_i - x_i$ for $i = 1, 2, 3, \dots, n$

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$ (*i.e.* $\mu = 0$), there is no significant difference between the means in two situations.

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

Assuming that H_0 is true, the test statistic for n paired observations (which are dependent) by taking the differences $d_1, d_2, \dots, d_n$ of the paired data.

$$t = \frac{\bar{d} - \mu}{S / \sqrt{n}} = \frac{\bar{d}}{S / \sqrt{n}} \quad (\because \mu = 0)$$

$$\text{where } \bar{d} = \frac{1}{n} \sum d_i \text{ and } S^2 = \frac{1}{n-1} \sum_{i=1}^n (d_i - \bar{d})^2$$

$$\text{or } S^2 = \frac{\sum d^2 - n(\bar{d})^2}{n-1} \text{ or } \frac{1}{n-1} \left[\sum d^2 - \frac{(\sum d)^2}{n} \right]$$

are the mean and variance of the differences $d_1, d_2, \dots, d_n$ respectively and μ is the mean of the population of differences.

The above statistic follows student's t - distribution with $(n-1)$ degrees of freedom.

SOLVED EXAMPLES

Example 1 : Ten workers were given a training programme with a view to study their assembly time for a certain mechanism. The results of the time and motion studies before and after the training programme are given below.

Workers	1	2	3	4	5	6	7	8	9	10
X_1	15	18	20	17	16	14	21	19	13	22
Y_1	14	16	21	10	15	18	17	16	14	20

X_1 = Time taken for assembling before training.

Y_1 = Time taken for assembling after training.

Test whether there is significant difference in assembly times before and after training.

[JNTU (A) May 2010S]

Solution : From the given paired data, we see that we are to use paired t -test. Let μ be the mean of population of differences.

1. **Null Hypothesis** $H_0 : \mu_1 = \mu_2$ or $\mu = 0$ i.e., training is not useful.
2. **Alternative Hypothesis** $H_1 : \mu_1 \neq \mu_2$ i.e., training is useful in assembly time.
3. **Level of significance** $\alpha = 0.05$
4. **Computation :** Differences d_i 's (before and after training) are

1, 2, -1, 7, 1, -4, 4, 3, -1, 2

$$\bar{d} = \text{mean of differences of sample data} = \frac{\sum d}{n} = \frac{14}{10} = 1.4$$

$$\begin{aligned}
 \text{Now } S^2 &= \frac{1}{n-1} \sum_{i=1}^n (d_i - \bar{d})^2 \\
 &= \frac{1}{9} [(1-1.4)^2 + (2-1.4)^2 + (-1-1.4)^2 + (7-1.4)^2 + (1-1.4)^2 + (-4-1.4)^2 \\
 &\quad + (4-1.4)^2 + (3-1.4)^2 + (-1-1.4)^2 + (2-1.4)^2] \\
 &= \frac{1}{9} [0.16 + 0.36 + 5.76 + 31.36 + 0.16 + 29.16 + 6.76 + 2.56 + 5.76 + 0.36] \\
 &= \frac{82.4}{9} = 9.1555
 \end{aligned}$$

$$\therefore S = 3.026$$

$$\text{5. The test statistic is } t = \frac{\bar{d} - \mu}{S/\sqrt{n}} = \frac{\bar{d}}{S/\sqrt{n}} = \frac{1.4}{3.026/\sqrt{10}} = \frac{(1.4)(3.1623)}{3.026} = 1.46$$

$\therefore$ Calculated $|t| = 1.46$

Tabulated $t_{0.05}$ with $10-1=9$ degrees of freedom is 1.833

Since calculated $t < t_{0.05}$, we accept the Null Hypothesis H_0 and conclude that there is no significant difference in assembly times before and after training.

Example 2 : Scores obtained in a shooting competition by 10 soldiers before and after intensive training are given below :

Before	67	24	57	55	63	54	56	68	33	43
After	70	38	58	58	56	67	68	75	42	38

Test whether the intensive training is useful at 0.05 level of significance.

Solution : Let us apply paired test.

Let μ be the mean of population of differences.

1. Null Hypothesis $H_0 : \mu_1 = \mu_2$ i.e., $\mu = 0$ there is no significant effect of the training.

2. Alternative Hypothesis $H_1 : \mu_1 \neq \mu_2$ intensive training is useful.

3. Level of significance, $\alpha = 0.05$

4. Computation :

Differences d_i 's (before and after training) are $-3, -14, -1, -3, 7, -13, -12, -7, -9, 5$

$$\bar{d} = \frac{1}{10} \sum_{i=1}^{10} d_i = \frac{-50}{10} = -5$$

$$\begin{aligned} \text{Now } S^2 &= \frac{1}{n-1} \sum_{i=1}^n (d_i - \bar{d})^2 = \frac{1}{9} \sum_{i=1}^{10} (d_i - \bar{d})^2 \\ &= \frac{1}{9} [(2)^2 + (-9)^2 + (4)^2 + (2)^2 + (12)^2 + (-8)^2 + (-7)^2 + (-2)^2 + (-4)^2 + (10)^2] \\ &= \frac{1}{9} [4 + 81 + 16 + 4 + 144 + 64 + 49 + 4 + 16 + 100] \\ &= \frac{482}{9} = 53.5555 \end{aligned}$$

$$\therefore S = 7.32$$

$$\text{5. The test statistic is } t = \frac{\bar{d}}{S/\sqrt{n}} = \frac{-5-0}{7.32/\sqrt{10}} = -2.16$$

$$\therefore |t| = 2.16$$

Tabulated $t_{0.05}$ with $10-1=9$ degrees of freedom is 1.83.

Since calculated $t > t$ tabulated t , we reject the Null Hypothesis and conclude that the intensive training is useful.

Example 2 : The Blood Pressure of 5 women before and after intake of a certain drug are given below :

Before	110	120	125	132	125
After	120	118	125	136	121

Test whether there is significant change in Blood Pressure at 1 % level of significance .

[JNTU (H) Nov. 2010 (Set No. 3), (K) May 2013 (Set No. 4)]

Solution : Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$ i.e., there is no significant difference in blood pressure before and after intake of a drug.

Then the Alternative Hypothesis is $H_1 : \mu_1 < \mu_2$

Assuming that H_0 is true, the test statistic is

$$t = \frac{d}{S/\sqrt{n}} \text{ where } \bar{d} = \frac{\sum d}{n}, d = y - x$$

$$\text{and } S^2 = \frac{\sum (d - \bar{d})^2}{n-1} = \frac{\sum d^2 - (\bar{d})^2 \times n}{n-1}$$

Calculations For $\bar{d}$ and S				
Womens	B.P. before intake of drug (x)	B.P. after intake of drug (y)	$d = y - x$	d^2
1	110	120	10	100
2	120	118	-2	4
3	123	125	2	4
4	132	136	4	16
5	125	121	-4	16
Total			$\sum d = 10$	$\sum d^2 = 140$

$$\therefore \bar{d} = \frac{\sum d}{n} = \frac{10}{5} = 2 \text{ and } S^2 = \frac{140 - (2)^2 \times 5}{4} = \frac{140 - 20}{4} = \frac{120}{4} = 30$$

$$\Rightarrow S = \sqrt{30}$$

$$\therefore t = \frac{\bar{d}}{S/\sqrt{n}} = \frac{2}{\sqrt{30}/\sqrt{5}} = \frac{2}{\sqrt{6}} = 0.82$$

$$\text{Degrees of freedom} = n - 1 = 5 - 1 = 4$$

Thus $t = 0.82 < 4.6$ at 1% level with 4 d.f. .

Since the calculated value of $t <$ the tabulated value with 4 d.f. at 1% level, we accept H_0 at 1% level and conclude that there is no significant change in Blood Pressure after intake of a certain drug.

Example 4 : Memory capacity of 10 students were tested before and after training.

State whether the training was effective or not from the following scores.

Before training	12	14	11	8	7	10	3	0	5	6
After training	15	16	10	7	5	12	10	2	3	8

[JNTU (H) Nov. 2010, Apr. 2012 (Set No. 2), (A) Dec. 2018]

Solution : Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

i.e., there is no significant effect of the training

Then the Alternative Hypothesis is $H_1 : \mu_1 < \mu_2$

Assuming that H_0 is true, the test statistic is

$$t = \frac{\bar{d}}{S/\sqrt{n}} \text{ where } \bar{d} = \frac{\Sigma d}{n}, d = x - y \text{ and}$$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (d_i - \bar{d})^2 = \frac{\Sigma d^2 - (\bar{d})^2 \times n}{n-1} \quad \dots (1)$$

Calculations for $\bar{d}$ and S			
Memory capacity before training (x)	Memory capacity after training (y)	$d = x - y$	d^2
12	15	-3	9
14	16	-2	4
11	10	1	1
8	7	1	1
7	5	2	4
10	12	-2	4
3	10	-7	49
0	2	-2	4
5	3	2	4
6	8	-2	4
		$-12 = \Sigma d$	$84 = \Sigma d^2$

$$\therefore \bar{d} = \frac{\Sigma d}{n} = \frac{-12}{10} = -1.2$$

$$\text{Now } S^2 = \frac{84 - (-1.2)^2 \times 10}{10-1} \text{ [using (1)]}$$

$$= \frac{84 - 14.4}{9} = \frac{69.6}{9} = 7.73$$

$$\Rightarrow S = \sqrt{7.73} = 2.78$$

$$\therefore t = \frac{\bar{d}}{S/\sqrt{n}} = \frac{-1.2}{2.78/\sqrt{10}} = -1.365 \text{ and d.f.} = n-1 = 10-1 = 9$$

Thus $|t| = 1.365 < 1.833$ at 5% level with 9 d.f.

Since the calculated value of $t <$ the tabulated value of t with 9 d.f. at 5% level, we accept H_0 at 5% level and conclude that there is no significant change in memory capacity after the training programme.

Example 1 : The average losses of workers, before and after certain program are given below. Use 0.05 level of significance to test whether the program is effective (paired sample t - test). 40 and 35, 70 and 65, 45, -42, 120 and 116, 35 and 55 and 50, 77 and 73.

[JNTU (H) Dec. 2011 (Set No. 3)]

Solution : Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$ i.e., there is no significant effect of the program.

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

Assuming that H_0 is true, the test statistic is $t = \frac{\bar{d}}{S/\sqrt{n}}$

where $\bar{d} = \frac{\sum d}{n}$, $d = y - x$ and $S^2 = \frac{\sum (d - \bar{d})^2}{n-1}$

Calculations for $\bar{d}$ and S

Before program (x)	After program (y)	$d_i = y - x$	$d_i - \bar{d}$ $= d_i - 4$	$(d_i - \bar{d})^2$
40	35	5	1	1
70	65	5	1	1
45	42	3	-1	1
120	116	4	0	0
35	33	2	-2	4
55	50	5	1	1
77	73	4	0	0
		28		8

$$\bar{d} = \frac{\sum d_i}{n} = \frac{28}{7} = 4$$

$$\text{Now } S^2 = \frac{\sum (d_i - \bar{d})^2}{n-1} = \frac{8}{6} = \frac{4}{3} \Rightarrow S = \frac{2}{\sqrt{3}}$$

$$\therefore t = \frac{4}{(2/\sqrt{3})/\sqrt{7}} = 2\sqrt{21} = 9.16$$

$$\text{d.f.} = n - 1 = 7 - 1 = 6$$

Thus $t = 9.16 > 2.447$ at 5% level with 6 d.f.

Since the calculated value of $t >$ the tabulated value with 6 d.f. at 5% level, we reject the Null Hypothesis H_0 and conclude that the program is not effective.

10.14 SNEDECOR'S F -TEST OF SIGNIFICANCE [JNTU (K) DEC. 2013 (SET NO.1)]

When testing the significance of the difference of the means of two samples, we assumed that the two samples came from the same population or from populations with equal variances. If the variances of the populations are not equal, a significant difference in the means may arise. Hence, before we apply the t -test for the significance of the difference of two means, we have to test for the equality of population variances using F -test of significance.

If s_1^2 and s_2^2 are the variances of two samples of sizes n_1 and n_2 respectively, then the population variances are given by

$$n_1 s_1^2 = (n_1 - 1) S_1^2 \quad \text{and} \quad n_2 s_2^2 = (n_2 - 1) S_2^2$$

The quantities $v_1 = n_1 - 1$ and $v_2 = n_2 - 1$ are called degrees of freedom of these estimates. We want to test if these estimates S_1^2 and S_2^2 are significantly different or if the samples may be regarded as drawn from the same population or from two populations with same variance σ^2 .

Test for Equality of Two population variances

Let two independent random samples of sizes n_1 and n_2 be drawn from two normal populations.

To test the hypothesis that the two population variances σ_1^2 and σ_2^2 are equal.

Let the Null Hypothesis be $H_0 : \sigma_1^2 = \sigma_2^2$

Then the Alternative Hypothesis is $H_1 : \sigma_1^2 \neq \sigma_2^2$

The estimates of σ_1^2 and σ_2^2 are given by

$$S_1^2 = \frac{n_1 s_1^2}{n_1 - 1} = \frac{\sum (x_i - \bar{x})^2}{n_1 - 1} \quad \text{and} \quad S_2^2 = \frac{n_2 s_2^2}{n_2 - 1} = \frac{\sum (y_i - \bar{y})^2}{n_2 - 1}, \quad \text{where } s_1^2 \text{ and } s_2^2 \text{ are the}$$

variances of the two samples.

Assuming that H_0 is true, the test statistic $F = \frac{S_1^2}{S_2^2}$ or $\frac{S_2^2}{S_1^2}$ according as $S_1^2 > S_2^2$ or

$S_2^2 > S_1^2$ follows F -distribution with $(n_1 - 1, n_2 - 1)$ degrees of freedom.

Conclusion : If the calculated value of $F >$ the tabulated value of F at 5% level, we reject the Null Hypothesis H_0 and conclude that the variances σ_1^2 and σ_2^2 are not equal. Otherwise, we accept the Null Hypothesis H_0 and conclude that σ_1^2 and σ_2^2 are equal.

Note : 1. In numerical problems, we take the greater of the two variances S_1^2, S_2^2 in the numerator and the other in the denominator.

$$i.e., F = \frac{\text{Greater variance}}{\text{Smaller variance}}$$

2. When F is close to 1, the two sample variances S_1 and S_2 are nearly same.
3. If sample variance S^2 is given, we can obtain population variance σ^2 by using the relation $n \sigma^2 = (n - 1) S^2$ and vice-versa.
4. $F_{\alpha}(v_1, v_2)$ is the value of F with v_1 and v_2 degrees of freedom such that the area under the F - distribution to the right of F_{α} is α . In tables F_{α} tabulated for $\alpha = 0.05$ and $\alpha = 0.01$ for various combinations of the d.o.f v_1 and v_2 . Clearly value of F at 5% significance is lower than at 1%

SOLVED EXAMPLES

Example 2 : In one sample of 8 observations from a normal population, the sum of the squares of deviations of the sample values from the sample mean is 84.4 and in another sample of 10 observations it was 102.6. Test at 5% level whether the populations have the same variance.

(or) In two independent samples of sizes 8 and 10 the sum of squares of deviations of the sample values from the respective sample means were 84.4 and 102.6. Test whether the difference of variances of the population is significant or not. Use a 0.05 level of significance.

[JNTU (A) June 2016]

Solution : Let σ_1^2 and σ_2^2 be the variances of the two normal populations from which the samples are drawn.

Let the Null Hypothesis be $H_0 : \sigma_1^2 = \sigma_2^2$

Then the Alternative Hypothesis is $H_1 : \sigma_1^2 \neq \sigma_2^2$

Here $n_1 = 8, n_2 = 10$

Also $\sum (x_i - \bar{x})^2 = 84.4, \sum (y_i - \bar{y})^2 = 102.6$

If S_1^2 and S_2^2 be the estimates of σ_1^2 and σ_2^2 then

$$S_1^2 = \frac{1}{n_1 - 1} \sum (x_i - \bar{x})^2 = \frac{84.4}{7} = 12.057$$

$$\text{and } S_2^2 = \frac{1}{n_2 - 1} \sum (y_i - \bar{y})^2 = \frac{102.6}{9} = 11.4$$

Let H_0 be true. Since $S_1^2 > S_2^2$, the test statistic is

$$F = \frac{S_1^2}{S_2^2} = \frac{12.057}{11.4} = 1.057$$

i.e. calculated $F = 1.057$.

Degrees of freedom are given by $v_1 = n_1 - 1 = 8 - 1 = 7$ and $v_2 = n_2 - 1 = 10 - 1 = 9$.

Tabulated value of F at 5% level for (7, 9) degrees of freedom is 3.29

i.e., $F_{0.05}(7, 9) = 3.29$

Since calculated $F < \text{tabulated } F$, we accept the Null Hypothesis H_0 and conclude that the populations have the same variance.

Example 1 : In one sample of 10 observations from a normal population, the sum of the squares of the deviations of the sample values from the sample mean is 102.4 and in another sample of 12 observations from another normal population, the sum of the squares of the deviations of the sample values from the sample mean is 120.5. Examine whether the two normal populations have the same variance.

Solution : Let σ_1^2 and σ_2^2 be the variances of the two normal populations from which the samples are drawn.

Let the Null Hypothesis be $H_0 : \sigma_1^2 = \sigma_2^2$

Then the Alternative Hypothesis is $H_1 : \sigma_1^2 \neq \sigma_2^2$

Given $n_1 = 10$, $n_2 = 12$, $\sum (x - \bar{x})^2 = 102.4$, $\sum (y - \bar{y})^2 = 120.5$

If S_1^2 and S_2^2 be the estimates of σ_1^2 and σ_2^2 then

$$S_1^2 = \frac{\sum (x - \bar{x})^2}{n_1 - 1} = \frac{102.4}{9} = 11.37$$

$$S_2^2 = \frac{\sum (y - \bar{y})^2}{n_2 - 1} = \frac{120.5}{11} = 10.95$$

$$\begin{aligned} \text{The test statistic is } F &= \frac{S_1^2}{S_2^2} \quad [\because S_1^2 > S_2^2] \\ &= \frac{11.37}{10.95} = 1.038 \end{aligned}$$

i.e., calculated $F = 1.132$

Tabulated value of F for (9, 11) d.f. at 5% level of significance is 2.91 (approx.)

Since calculated $F < \text{tabulated } F$, we accept the null hypothesis H_0 i.e., the two normal populations have the same variance.

Example 2 : Two random samples reveal the following results :

Sample	Size	Sample Mean	Sum of Squares of Deviations from the Mean
1	10	15	90
2	12	14	108

Test whether the samples came from the same normal population.

Solution :

Null Hypothesis H_0 : The two samples have been drawn from the same normal population.

i.e., $H_0 : \mu_1 = \mu_2$ and $\sigma_1^2 = \sigma_2^2$

Here we have to use two tests (i) To test equality of variances by *F*-test

(ii) To test equality of means by *t*-test.

(i) F-test (equality of variances)

Given $n_1 = 10$, $n_2 = 12$, $\bar{x}_1 = 15$, $\bar{x}_2 = 14$, $\sum(x_i - \bar{x})^2 = 90$, $\sum(y_i - \bar{y})^2 = 108$

$$\therefore S_1^2 = \frac{\sum(x_i - \bar{x})^2}{n_1 - 1} = \frac{90}{9} = 10; S_2^2 = \frac{\sum(y_i - \bar{y})^2}{n_2 - 1} = \frac{108}{11} = 9.82$$

$$\text{Hence } F = \frac{S_1^2}{S_2^2} = \frac{10}{9.82} = 1.018$$

i.e., calculated $F = 1.018$

Tabulated F at 5% level for (9, 11) degrees of freedom is 2.90

i.e., $F_{0.05}(9, 11) = 2.90$

Since calculated $F <$ tabulated F , we accept the hypothesis

$$\therefore \sigma_1^2 = \sigma_2^2$$

i.e., the samples came from the same normal populations with same variance.

(ii) t-test (To test equality of means)

Null Hypothesis H_0 : $\mu_1 = \mu_2$

Given $\bar{x}_1 = 15$, $\bar{x}_2 = 14$, $n_1 = 10$, $n_2 = 12$,

$$\sum(x_i - \bar{x})^2 = 90, \sum(y_i - \bar{y})^2 = 108$$

$$\begin{aligned} \text{Now } S^2 &= \frac{1}{n_1 + n_2 - 2} [\sum(x - \bar{x})^2 + \sum(y - \bar{y})^2] \\ &= \frac{1}{10 + 12 - 2} [90 + 108] = 9.9 \end{aligned}$$

$$\therefore S = 3.15$$

$$\text{The test statistic is } t = \frac{\bar{x} - \bar{y}}{S \sqrt{\frac{1}{n_2} + \frac{1}{n_2}}} = \frac{15 - 14}{3.15 \sqrt{\frac{1}{10} + \frac{1}{12}}} = 0.74$$

$\therefore$ Calculated value of $|t| = 0.74$

Tabulated value of t for 20 d.f. ($n_1 + n_2 - 2$) at 5% level of significance is 2.086.

Since calculated value of $t <$ tabulated value of t , we accept the null hypothesis H_0 : i.e., $\mu_1 = \mu_2$

Hence from (i) and (ii), the given samples have been drawn from the same normal population. Hence we accept the null hypothesis that $\mu_1 = \mu_2$ and $\sigma_1^2 = \sigma_2^2$.

Example 4 : It is known that the mean diameters of rivets produced by two firms A and B are practically the same, but the standard deviations may differ. For 22 rivets produced by firm A, the S.D is 2.9 mm, while for 16 rivets manufactured by firm B, the S.D is 3.8 mm. Compute the statistic you would use to test whether the products of firm A have the same variability as those of firm B and test its significance.

Solution : Let the Null Hypothesis be $H_0 : \sigma_1^2 = \sigma_2^2$
i.e., the variances of sources are equal.

The Alternative Hypothesis is $H_1 : \sigma_1^2 \neq \sigma_2^2$

Given $n_1 = 22$, $n_2 = 16$, $s_1 = 2.9$ mm, $s_2 = 3.8$ mm

Since the S.D's of the samples s_1 and s_2 are given,

$\therefore$ The population variances S_1^2 and S_2^2 are obtained by using the relations

$$n_1 s_1^2 = (n_1 - 1) S_1^2 \text{ and } n_2 s_2^2 = (n_2 - 1) S_2^2$$

$$\text{i.e. } S_1^2 = \frac{n_1 s_1^2}{n_1 - 1} = \frac{22 \times (2.9)^2}{21} = 8.805 \text{ and}$$

$$S_2^2 = \frac{n_2 s_2^2}{n_2 - 1} = \frac{16 \times (3.8)^2}{15} = 15.393$$

Since $S_2^2 > S_1^2$, the test statistic is $F = \frac{S_2^2}{S_1^2}$

$$\text{i.e. } F = \frac{15.393}{8.805} = 1.74 \text{ i.e., calculated } F = 1.74$$

Tabulated value of F with (15, 21) d.f at 5% level of significance is 2.20.

Since calculated value of $F <$ tabulated value of F , we accept the null hypothesis H_0 , i.e., the products of both the firms A and B have the same variability. So we may conclude that the products of firm A are not superior to those of firm B.

Example 1 : Pumpkins were grown under two experimental conditions. Two random samples of 11 and 9 pumpkins, show the sample standard deviations of their weights as 0.8 and 0.5 respectively. Assuming that the weight distributions are normal, test hypothesis that the true variances are equal. [JNTU (A) Nov. 2011, (H) May 2011 (Set No. 4)]

Solution : Let the Null Hypothesis be $H_0 : \sigma_1^2 = \sigma_2^2$ i.e., the variances of sources are equal.

The Alternative Hypothesis is $H_1 : \sigma_1^2 \neq \sigma_2^2$

Given $n_1 = 11$, $n_2 = 9$, $S_1 = 0.8$, $S_2 = 0.5$

Here samples S.D's are given.

We have to determine the population variances S_1^2 and S_2^2 by using the relations

$$n_1 \cdot s_1^2 = (n_1 - 1) S_1^2 \text{ and } n_2 \cdot s_2^2 = (n_2 - 1) S_2^2$$

$$\Rightarrow S_1^2 = \frac{n_1 s_1^2}{n_1 - 1} = \frac{11 \times (0.8)^2}{10} = 0.704$$

$$\text{and } S_2^2 = \frac{n_2 s_2^2}{n_2 - 1} = \frac{9 \times (0.5)^2}{8} = 0.281$$

$$\text{The test statistic is } F = \frac{S_1^2}{S_2^2} = \frac{0.704}{0.281} = 2.5 \quad [\because S_1^2 > S_2^2]$$

$\therefore$ Calculated $F = 2.5$

Tabulated value of F for (10, 8) d.f at 5% level of significance is 3.35.

Since calculated $F <$ tabulated F , we accept the null hypothesis H_0 at 5% level with d.f. (10,8) and conclude that the variances of the two populations is the same, and therefore, the two samples have the same variance

Example 4 : Time taken by the workers in performing a job by method I and method II is given below :

Method I	20	16	26	27	23	22	—
Method II	27	33	42	35	32	34	38

Do the data show that the variances of time distribution from population from which these samples are drawn do not differ significantly ?

[JNTU (K) May 2013 (Set No.1)]

Solution : Let the Null Hypothesis be $H_0 : \sigma_1^2 = \sigma_2^2$ where σ_1^2 and σ_2^2 are the variances of the two populations from with the samples are drawn.

The Alternative Hypothesis is $H_1 : \sigma_1^2 \neq \sigma_2^2$.

Calculation of Sample Variances

x	$x - \bar{x}$	$(x - \bar{x})^2$	y	$y - \bar{y}$	$(y - \bar{y})^2$
20	-2.3	5.29	27	-7.4	54.76
16	-6.3	39.69	33	-1.4	1.96
26	3.7	13.69	42	7.6	57.76
27	4.7	22.09	35	0.6	0.36
23	0.7	0.49	32	-2.4	5.76
22	-0.3	0.09	34	-0.4	0.16
			38	3.6	12.96
134		81.34	241		133.72

Given $n_1 = 6, n_2 = 7$

$$\therefore \bar{x} = \frac{\sum x}{n_1} = \frac{134}{6} = 22.3,$$

$$\bar{y} = \frac{\sum y}{n_2} = \frac{241}{7} = 34.4$$

and $\sum(x_i - \bar{x})^2 = 81.34, \sum(y_i - \bar{y})^2 = 133.72$

If S_1^2 and S_2^2 be the estimates of σ_1^2 and σ_2^2 , then

$$S_1^2 = \frac{\sum(x_i - \bar{x})^2}{n_1 - 1} = \frac{81.34}{5} = 16.26$$

$$\text{and } S_2^2 = \frac{\sum(y_i - \bar{y})^2}{n_2 - 1} = \frac{133.72}{6} = 22.29$$

Let H_0 be true.

Since $S_2^2 > S_1^2$, the statistic is

$$F = \frac{S_2^2}{S_1^2} = \frac{22.29}{16.268} = 1.3699 \approx 1.37$$

Tabulated value of F for $(n_1 - 1, n_2 - 1)$ d.f. i.e., (5, 6) d.f. at 5% level significance is 4.39.

Since calculated $F <$ tabulated F , we accept the null hypothesis H_0 at 5% level of significance i.e., there is no significant difference between the variances of the time distribution by the workers.

Example 2 : The nicotine contents in milligrams in two samples of tobacco were found to be as follows :

Sample A	24	27	26	21	25	—
Sample B	27	30	28	31	22	36

Can it be said that the two samples have come from the same normal population?

[JNTU (A) Nov. 2011 (Set No. 4)]

Solution :

To test whether the two samples came from same normal population, we have to test (i) the equality of variances by using F -test (ii) the equality of means by using student's t -test.

Given $n_1 = 5$, $n_2 = 6$

Calculation for mean's and S.D's of the samples.

x	$x - \bar{x}$	$(x - \bar{x})^2$	y	$y - \bar{y}$	$(y - \bar{y})^2$
24	0.6	0.36	27	-2	4
27	2.4	5.76	30	1	1
26	1.4	1.96	28	-1	1
21	3.6	12.96	31	2	4
25	0.4	0.16	22	-7	49
			36	7	49
123		21.2	174		108

$$\therefore \bar{x} = \frac{\sum x}{n_1} = \frac{123}{5} = 24.6, \quad \bar{y} = \frac{\sum y}{n_2} = \frac{174}{6} = 29$$

$$\sum(x_i - \bar{x})^2 = 21.2, \quad \sum(y_i - \bar{y})^2 = 108$$

$$\therefore S_1^2 = \frac{\sum(x_i - \bar{x})^2}{n_1 - 1} = \frac{21.2}{4} = 5.3$$

$$\text{and } S_2^2 = \frac{\sum(y_i - \bar{y})^2}{n_2 - 1} = \frac{108}{5} = 21.6$$

(i) ***F-Test :***

Null Hypothesis $H_0 : \sigma_1^2 = \sigma_2^2$

$$\text{The test statistic is } F = \frac{S_2^2}{S_1^2} = \frac{21.6}{5.3} = 4.075$$

$$\therefore \text{Calculated } F = 4.075$$

Tabulated value of F for $(n_2 - 1, n_1 - 1) = (5, 4)$ d.f at 5% level of significance is 6.26. Since calculated $F <$ tabulated F , we accept the null hypothesis H_0 i.e., the variances are equal.

$$\text{i.e. } \sigma_1^2 = \sigma_2^2$$

(ii) **Student's *t*-test :** The two means are equal i.e., $\mu_1 = \mu_2$

Given $\bar{x} = 24.6$, $\bar{y} = 29$

$$\sum (x_i - \bar{x})^2 = 21.2, \sum (y_i - \bar{y})^2 = 108$$

$$\text{Now } S^2 = \frac{1}{n_1 + n_2 - 2} [\sum (x_i - \bar{x})^2 + \sum (y_i - \bar{y})^2]$$

$$= \frac{1}{5+6-2} [21.2 + 108] = 14.35$$

$$\therefore S = 3.78$$

$$\text{The test statistic is } t = \frac{\bar{x} - \bar{y}}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{24.6 - 29}{3.78 \sqrt{\frac{1}{5} + \frac{1}{6}}} = -1.92$$

Calculated value of $|t| = 1.92$

Tabulated value of t for 9 d.f at 5% level of significance is 2.26.

Since calculated value of $t <$ tabulated value of t , we accept H_0 i.e., the two means are equal.

$\therefore$ From (i) and (ii) we conclude that the two samples came from same normal population.

Example 8 : The measurements of the output of two units have given the following results. Assuming that both samples have been obtained from the normal populations at 10% significant level, test whether the two populations have the same variance.

Unit - A	14.1	10.1	14.7	13.7	14.0
Unit - B	14.0	14.5	13.7	12.7	14.1

[JNTU 2007, 2008S (Set No. 4)]

Solution : Let the Null Hypothesis be $H_0 : \sigma_1^2 = \sigma_2^2$

i.e., the variances of sources are equal.

The Alternative Hypothesis is $H_1 : \sigma_1^2 \neq \sigma_2^2$

Given $n_1 = 5$, $n_2 = 5$

$$\text{Now } \bar{x} = \frac{\sum x}{5} = \frac{1}{5} (14.1 + 10.1 + 14.7 + 13.7 + 14.0)$$

$$= \frac{66.6}{5} = 13.32$$

$$\text{and } \bar{y} = \frac{\sum y}{5} = \frac{1}{5} (14.0 + 14.5 + 13.7 + 12.7 + 14.1)$$

$$= \frac{69}{5} = 13.8$$

Computing Standard Deviations of the samples.

x	$x - \bar{x}$	$(x - \bar{x})^2$	y	$y - \bar{y}$	$(y - \bar{y})^2$
14.1	0.78	0.6084	14.0	0.2	0.04
10.1	-3.22	10.3684	14.5	0.7	0.49
14.7	1.38	1.9044	13.7	-0.1	0.01
13.7	0.38	0.1444	12.7	-1.1	1.21
14.0	0.68	0.4624	14.1	0.3	0.09
66.6		13.488	69		1.84

If S_1^2 and S_2^2 be the estimates of σ_1^2 and σ_2^2 then

$$S_1^2 = \frac{\sum(x_i - \bar{x})^2}{n_1 - 1} = \frac{13.488}{4} = 3.372$$

$$S_2^2 = \frac{\sum(y_i - \bar{y})^2}{n_2 - 1} = \frac{1.84}{4} = 0.46$$

Let H_0 be true. Since $S_1^2 > S_2^2$, the test statistic is

$$F = \frac{S_1^2}{S_2^2}$$

$$\text{i.e., } F = \frac{3.372}{0.46} = 7.33$$

$\therefore$ Calculated $F = 7.33$

Degrees of freedom are $(n_1 - 1, n_2 - 1)$ i.e., $(5 - 1, 5 - 1)$ or $(4, 4)$

Tabulated value of F for $(4, 4)$ d.f. at 10% level of significance is 6.39.

Since calculated $F >$ tabulated F , we reject the null hypothesis H_0 .

i.e. There is significant difference between the variances.

Example 9 : The following random samples are measurements of the heat-producing capacity (in millions of calories per ton) of specimens of coal from two mines :

Mine 1:	8,260	8,130	8,350	8,070	8,340	---
Mine 2:	7,950	7,890	7,900	8,140	7,920	7,840

Use the 0.02 level of significance to test whether it is reasonable to assume that the variances of the two populations samples are equal. [JNTU (K) 2009 (Set No. 1)]

Solution : Here $n_1 = 5$ and $n_2 = 6$

Calculation for mean's and S.D's of the samples:

Mine1			Mine 2		
x	$(x - \bar{x}) = (x - 8230)$	$(x - \bar{x})^2$	y	$(y - \bar{y}) = (y - 7940)$	$(y - \bar{y})^2$
8260	30	900	7950	10	100
8130	-100	10000	7890	-50	2500
8350	120	14400	7900	-40	1600
8070	-160	25600	8140	200	40000
8340	110	12100	7920	-20	400
			7840	-100	10000
Total	41150	63000	47640		54600

$$\text{Now } \bar{x} = \frac{\sum x}{n_1} = \frac{41150}{5} = 8230, \bar{y} = \frac{\sum y}{n_2} = \frac{47640}{6} = 7940, \sum (x - \bar{x})^2 = 63000$$

$$\text{and } \sum (y - \bar{y})^2 = 54600$$

$$\therefore S_1^2 = \frac{\sum (x - \bar{x})^2}{n_1 - 1} = \frac{63000}{4} = 15750; S_2^2 = \frac{\sum (y - \bar{y})^2}{n_2 - 1} = \frac{54600}{5} = 10920$$

1. Null Hypothesis $H_0 : S_1^2 = S_2^2$

2. Alternative Hypothesis $H_1 : S_1^2 \neq S_2^2$

3. Level of significance : $\alpha = 0.02$

4. The test statistic, $F = \frac{S_1^2}{S_2^2} \quad (\because S_1^2 > S_2^2)$

$$\text{i.e., } F = \frac{15750}{10920} = 1.44$$

$$\therefore \text{Calculated } F = 1.44$$

5. Conclusion : Tabulated value of F for (4,5) degrees of freedom at 5% level of significance is 5.19. So at 2% level of significance $F(4,5) > 5.19$ (value of F at 5% significance is lower than at 2%). Since calculated $F <$ tabulated F, we accept the null hypothesis H_0 i.e., the variances of the two populations samples are equal.

Example 10 : In one sample of 10 observations, the sum of the squares of the deviations of the sample values from sample mean was 120 and in the other sample of 12 observations, it was 314. Test whether the difference is significant at 5% level ?

[JNTU (H) Nov. 2010, Apr. 2012 (Set No. 2)]

Solution : Let the Null Hypothesis be $H_0 : \sigma_1^2 = \sigma_2^2$, where σ_1^2, σ_2^2 are the variances of the two populations from which the samples are drawn.

The Alternative Hypothesis is $H_1 : \sigma_1^2 \neq \sigma_2^2$

We have $S_1^2 = \frac{\Sigma(x - \bar{x})^2}{n_1 - 1}$ and $S_2^2 = \frac{\Sigma(y - \bar{y})^2}{n_2 - 1}$

Here $n_1 = 10$, $\Sigma(x - \bar{x})^2 = 120$, $n_2 = 12$, $\Sigma(y - \bar{y})^2 = 314$

$$\therefore S_1^2 = \frac{120}{10-1} = 13.33 \text{ and } S_2^2 = \frac{314}{12-1} = 28.54$$

Let H_0 be true. Since $S_2^2 > S_1^2$, the test statistic is

$$F = \frac{S_2^2}{S_1^2} = \frac{28.54}{13.33} = 2.14$$

Degrees of freedom are $v_1 = n_1 - 1 = 10 - 1 = 9$ and $v_2 = n_2 - 1 = 12 - 1 = 11$.

The table value of F at 5% level for $v_1 = 9$ and $v_2 = 11$ degrees of freedom is 2.90 and $2.14 < 2.9$. Since the calculated value of F < the table value, we accept the Null Hypothesis H_0 and conclude that the difference is not significant at 5% level.

Example 11 : An instructor has two classes A and B, in a particular subject. Class A has 16 students while class B has 25 students. On the same examination, although there was no significant difference in mean grades, class A has a standard deviation of 9 while class B had a standard deviation of 12. Can conclude at the 0.01 level of significance that the variability of class B is greater than that of A? [JNTU (K) May 2010 (Set No. 2)]

Solution : Let the Null Hypothesis be $H_0 : \sigma_1^2 = \sigma_2^2$ i.e., the variances of sources A and B are equal.

The Alternative Hypothesis is $H_1 : \sigma_1^2 > \sigma_2^2$

We have $S_1^2 = \left(\frac{n_1}{n_1 - 1} \right) s_1^2$ and $S_2^2 = \left(\frac{n_2}{n_2 - 1} \right) s_2^2$

Here $n_1 = 16, s_1 = 9$ and $n_2 = 25, s_2 = 12$

$$\therefore S_1^2 = \left(\frac{16}{16-1} \right) \times 81 = \frac{16}{15} \times 81 = 86.4$$

$$\text{and } S_2^2 = \left(\frac{25}{25-1} \right) \times 144 = \frac{25}{24} \times 144 = 150$$

Assuming that H_0 is true, the test statistic is

$$F = \frac{S_2^2}{S_1^2} \left[\because S_2^2 > S_1^2 \right]$$

$$= \frac{150}{86.4} = 1.736$$

Degrees of freedom are $v_1 = n_1 - 1 = 16 - 1 = 15$ and $v_2 = n_2 - 1 = 25 - 1 = 24$

The table value of F with d.f. $v_1 = 15, v_2 = 24$ at 1% level is 2.89.

Since the calculated value of $F <$ the tabulated value, we accept the Null Hypothesis H_0 at 1% level with $v_1 = 15, v_2 = 24$ and conclude that the two variances of A and B are the same.

Example 11 : Two independent samples of 8 and 7 items respectively had the following values of the variables. **[JNTU (A) April 2012 (Set No. 3)]**

Sample I	9	11	13	11	16	10	12	14
Sample II	11	13	11	14	10	8	10	—

Do the estimates of the population variance differ significantly.

Solution : Let the Null Hypothesis be $H_0 : \sigma_1^2 = \sigma_2^2$ where σ_1^2 and σ_2^2 are the variances of the two populations from which the samples are drawn.

Then the Alternative Hypothesis is $H_1 : \sigma_1^2 \neq \sigma_2^2$

Calculation of Sample Variances

x	$x - \bar{x}$ $= x - 12$	$(x - \bar{x})^2$	y	$y - \bar{y}$ $= y - 11$	$(y - \bar{y})^2$
9	-3	9	11	0	0
11	-1	1	13	2	4
13	1	1	11	0	0
11	-1	1	14	3	9
16	4	16	10	-1	1
10	-2	4	8	-3	9
12	0	0	10	-1	1
14	2	4	—	—	—
96		36	77		24

Given $n_1 = 8$ and $n_2 = 7$

$$\therefore \bar{x} = \frac{\sum x}{n_1} = \frac{96}{8} = 12 \text{ and } \bar{y} = \frac{\sum y}{n_2} = \frac{77}{7} = 11$$

If S_1^2 and S_2^2 be the estimates of σ_1^2 and σ_2^2 , then

$$S_1^2 = \frac{\sum (x - \bar{x})^2}{n_1 - 1} = \frac{36}{8 - 1} = \frac{36}{7} = 5.14; \quad S_2^2 = \frac{\sum (y - \bar{y})^2}{n_2 - 1} = \frac{24}{7 - 1} = \frac{24}{6} = 4$$

The test statistic is $F = \frac{S_1^2}{S_2^2} [\because S_1^2 > S_2^2]$

$$= \frac{5.14}{4} = 1.285$$

The table value of F for $(n_1 - 1, n_2 - 1)$ d.f. i.e., (7,6) d.f. at 5% level of significance is 4.21.

Since, the calculated value of $F <$ the tabulated value of F at 5% level with (7,6) d.f., we accept the Null Hypothesis H_0 and conclude that there is no significant difference between the variances.

10.15 CHI-SQUARE (χ^2) TEST

When a fair coin is tossed 100 times, the theoretical considerations leads us to expect 50 heads and 50 tails. But in practice, these results are rarely achieved i.e., the results obtained in an experiment do not agree exactly with the theoretical results. The magnitude of discrepancy between the theory and observation is given by the quantity χ^2 (a Greek letter, pronounced as “chi-square”). If $\chi = 0$, the observed and expected frequencies completely coincide. As the value of χ^2 increases, the discrepancy between the observed and theoretical frequencies increases. Thus, χ^2 affords a measure of the correspondence between theory and observation.

Definition : If a set of events $A_1, A_2, \dots, A_n$ are observed to occur with frequencies $O_1, O_2, \dots, O_n$ respectively and according to probability rules $A_1, A_2, \dots, A_n$ are expected to occur with frequencies $E_1, E_2, \dots, E_n$ respectively with $O_1, O_2, \dots, O_n$ are called observed frequencies and $E_1, E_2, \dots, E_n$ are called expected frequencies.

If O_i ($i=1, 2, \dots, n$) is a set of observed (experimental) frequencies and E_i ($i=1, 2, \dots, n$) is the corresponding set of expected (theoretical) frequencies, then

χ^2 is defined as $\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$ with $(n-1)$ degrees of freedom.

χ^2 is used to test whether differences between observed and expected frequencies are significant.

Note : If the data is given in a series of ‘ n ’ numbers then degrees of freedom = $n - 1$.

In case of Binomial distribution, d.f. = $n - 1$.

In case of Poisson distribution, d.f. = $n - 2$

In case of Normal distribution, d.f. = $n - 3$

Chi-square Distribution is an important continuous probability distribution and it is used in both large and small tests. In chi-square tests, χ^2 - distribution is mainly used

- (i) to test the goodness of fit,
- (ii) to test the independence of attributes,
- (iii) to test if the population has a specified value of the variance σ^2 .

We shall discuss these tests one by one as applications of χ^2 - distribution.

10.16 χ^2 TEST AS A TEST OF GOODNESS OF FIT

We use this test to decide whether the discrepancy between theory and experiment is significant or not i.e., to test whether the difference between the theoretical and observed values can be attributed to chance or not.

In other words, if the two curves *i.e.*, observed frequency curve and the expected frequency curve, are drawn, then the Chi-square statistic may be used to determine whether the two curves so drawn are fitted good or not.

Thus, the term goodness of fit is used to test the concordance of the fitness of these two curves. In this case we have only one variable

$\therefore$ No. of d.f. *i.e.*, $v = n - 1$.

Let the Null Hypothesis H_0 be that there is no significant difference between the observed values and the corresponding expected values.

The Alternative Hypothesis H_1 is that the above difference is significant.

Let $O_1, O_2, \dots, O_n$ be a set of observed frequencies and $E_1, E_2, \dots, E_n$ the corresponding set of expected frequencies. Then the test statistic χ^2 is given by

$$\begin{aligned}\chi^2 &= \sum_{i=1}^n \left[\frac{(O_i - E_i)^2}{E_i} \right] \\ &= \frac{(O_1 - E_1)^2}{E_1} + \frac{(O_2 - E_2)^2}{E_2} + \dots + \frac{(O_n - E_n)^2}{E_n}\end{aligned}$$

Assuming that H_0 is true, the test statistic χ^2 follows Chi-square distribution with $(n-1)$ d.f., where

$$\sum_{i=1}^n O_i = \sum_{i=1}^n E_i \quad \text{or} \quad \sum_{i=1}^n (O_i - E_i) = 0$$

Conclusion : If the calculated value of $\chi^2 >$ tabulated value of χ^2 at α level, the Null Hypothesis H_0 is rejected. Otherwise, H_0 is accepted.

χ^2 test enables us to ascertain how well the theoretical distributions such as Binomial, Poisson, Normal, etc., fit the distributions obtained from sample data.

10.17 CONDITIONS OF VALIDITY

Following are the conditions which should be satisfied before χ^2 test can be applied.

1. The sample observations should be independent.
2. N, the total frequency is large, *i.e.*, > 50 .
3. The constraints on the cell frequencies, if any, are linear.
4. No theoretical (or expected) frequency should be less than 10. If small theoretical frequencies occur, the difficulty is overcome by grouping 2 or more classes together before calculating $(O - E)$. Note that the degrees of freedom is determined with the number of classes after regrouping.
5. It is wholly dependent on the degrees of freedom.

SOLVED EXAMPLES

Example 1 : The number of automobile accidents per week in a certain community are as follows : 12, 8, 20, 2, 14, 10, 15, 6, 9, 4. Are these frequencies in agreement with the belief that accident conditions were the same during this 10 week period.

[JNTU(H) Dec. 2011 (Set No. 4)]

Solution : Expected frequency of accidents each week = $\frac{100}{10} = 10$.

Null Hypothesis H_0 : The accident conditions were the same during the 10 week period.

Alternative Hypothesis : The accidents conditions are different during the 10 week period.

Observed Frequency (O_i)	Expected Frequency (E_i)	$(O_i - E_i)$	$\frac{(O_i - E_i)^2}{E_i}$
12	10	2	0.4
8	10	-2	0.4
20	10	10	10.0
2	10	-8	6.4
14	10	4	1.6
10	10	0	0.0
15	10	5	2.5
6	10	-4	1.6
9	10	-1	0.1
4	10	-6	3.6
100	100		26.6

$$\text{Now } \chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = 26.6 \text{ i.e., Calculated } \chi^2 = 26.6$$

Here $n = 10$ observations are given

$$\therefore \text{Degrees of freedom (d.f)} = n-1 = 10 - 1 = 9$$

$$\text{Tabulated } \chi^2 = 16.9$$

Since Calculated $\chi^2 > \text{Tabulated } \chi^2$, therefore, the Null Hypothesis is rejected and conclude that the accident conditions were not the same during the 10 week period.

Example 1 : A sample analysis of examination results of 500 students was made. It was found that 220 students had failed, 170 had secured a third class, 90 were placed in second class and 20 got a first class. Do these figures commensurate with the general examination result which is in the ratio of 4 : 3 : 2 : 1 for the various categories respectively.

Solution :

Null Hypothesis H_0 : The observed results commensurate with the general examination results.

Expected frequencies are in the ratio of 4 : 3 : 2 : 1

Total frequency = 500

If we divide the total frequency 500 in the ratio 4 : 3 : 2 : 1, we get the expected frequencies as 200, 150, 100, 50.

Calculations for χ^2

Class	Observed Frequency (O_i)	Expected Frequency (E_i)	$(O_i - E_i)$	$\frac{(O_i - E_i)^2}{E_i}$
Failed	220	200	20	2.00
Third	170	150	20	2.667
Second	90	100	-10	1.000
First	20	50	-30	18.00
	500	500		23.667

$$\text{Calculated } \chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = 23.667$$

$$\text{Degrees of freedom} = 4 - 1 = 3 \text{ i.e. } v = 3$$

$$\text{For } v = 3, \chi_{0.05}^2 = 7.81$$

i.e. tabulated value of χ^2 at 5% level for 3 d.f = 7.81

Since calculated $\chi_{0.05}^2 > \text{tabulated } \chi_{0.05}^2$, we reject the null hypothesis.

i.e. the observed results are not commensurate with the general examination results.

Example 1 : The following figures show the distribution of digits in numbers chosen at random from a telephone directory.

Digits	0	1	2	3	4	5	6	7	8	9
Frequency	1026	1107	997	966	1075	933	1107	972	964	853

Test whether the digits may be taken to occur equally frequently in the directory.

Solution :

Null Hypothesis H_0 : The digits occur equally frequently in the directory.

Alternative Hypothesis H_1 : The digits do not occur equally frequently.

Under this null hypothesis the expected frequencies for each of the digits

$$0,1,2,\dots,9 \text{ is } \frac{10,000}{10} = 1000.$$

We calculate χ^2 from the following table :

Computation Table for χ^2

Digits	Observed Frequency (O_i)	Expected Frequency (E_i)	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
0	1026	1000	676	0.676
1	1107	1000	11449	11.449
2	997	1000	9	0.009
3	966	1000	1156	1.156
4	1075	1000	5625	5.625
5	933	1000	4489	4.489
6	1107	1000	11449	11.449
7	972	1000	784	0.784
8	964	1000	1296	1.296
9	853	1000	21609	21.609
Total	10000	10,000		58.542

$$\text{Calculated } \chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = 58.542$$

$$\text{Degrees of freedom} = n - 1 = 10 - 1 = 9$$

The tabulated value of χ^2 for 9 d.f. at 5% level = 16.919

Since calculated $\chi^2 >$ tabulated χ^2 , we reject the null hypothesis H_0 and conclude that the digits do not occur equally frequently in the directory.

Example 4 : A die is thrown 264 times with the following results. Show that the die is biased. [Given $\chi^2_{0.05} = 11.07$ for 5 d.f] [JNTU (H) Dec. 2011 (Set No. 4)]

No. appeared on the die	1	2	3	4	5	6
Frequency	40	32	28	58	54	52

Solution: Null Hypothesis H_0 : The die is unbiased.

The expected frequency of each of the numbers 1, 2, 3, 4, 5, 6 is $\frac{264}{6} = 44$

Computation Table for χ^2

Observed frequency (O_i)	Expected frequency (E_i)	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
40	44	16	0.3636
32	44	144	3.2727
28	44	256	5.8181
58	44	196	4.4545
54	44	100	2.2727
52	44	64	1.4545
264	264		17.6362

$$\therefore \chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = 17.6362$$

The number of degrees of freedom = $n - 1 = 5$

The tabulated value of χ^2 for 5 d.f at 5% level = 11.07

Since calculated $\chi^2 >$ tabulated χ^2 , we reject the null hypothesis H_0 .

i.e., we reject the hypothesis that the die is unbiased. Hence the die is biased.

Example 4 : 200 digits were chosen at random from a set of tables. The frequencies of the digits are shown below :

Digit	0	1	2	3	4	5	6	7	8	9
Frequency	18	19	23	21	16	25	22	20	21	15

Use the chi-square test to assess the correctness of the hypothesis that the digits were distributed in equal number in the tables from which these were chosen.

Solution:

Null Hypothesis H_0 : The digits were distributed in equal number in the tables.

The expected frequency of each digit = $\frac{200}{10} = 20$

Computation Table for χ^2

Observed frequency (O_i)	Expected frequency (E_i)	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
18	20	4	0.2
19	20	1	0.05
23	20	9	0.45
21	20	1	0.05
16	20	16	0.8
25	20	25	1.25
22	20	4	0.2
20	20	0	0
21	20	1	0.05
15	20	25	1.25
200	200		4.3

$$\therefore \chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = 4.3$$

Thus calculated $\chi^2 = 4.3$

Tabulated χ^2 for $10 - 1 = 9$ d.f. at 5% level of significance is 16.919. Since calculated $\chi^2 < \text{tabulated } \chi^2$, we accept the null hypothesis H_0 . i.e., the digits are distributed in equal number in the tables.

Example 1 : A pair of dice are thrown 360 times and the frequency of each sum is indicated below :

Sum	2	3	4	5	6	7	8	9	10	11	12
Frequency	8	24	35	37	44	65	51	42	26	14	14

Would you say that the dice are fair on the basis of the chi-square test at 0.05 level of significance ?

[JNTU 2006S, (A) Nov. 2010 (Set No. 2)]

Solution :

1. **Null Hypothesis H_0 :** The dice are fair
2. **Alternative Hypothesis H_1 :** The dice are not fair
3. **Level of significance :** $\alpha = 0.05$

The probabilities of getting a sum 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 and 12 are

$X = x_i$	2	3	4	5	6	7	8	9	10	11	12
$P(x_i)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Computation Table for χ^2

Sum	Observed Frequency (O_i)	Expected Frequency (E_i) $E_i = 360 \cdot P(x)$	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
2	8	10	4	0.4
3	24	20	16	0.8
4	35	30	25	0.833
5	37	40	9	0.225
6	44	50	36	0.72
7	65	60	25	0.417
8	51	50	1	0.02
9	42	40	4	0.1
10	26	30	16	0.53
11	14	20	36	1.8
12	14	10	16	1.6
	360	360		7.445

$$\therefore \chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = 7.445$$

The number of degrees of freedom = $n - 1 = 11$

The tabulated value of χ^2 for 11 d.f. at 5% level of significance is 18.3

Since calculated $\chi^2 <$ tabulated χ^2 , we accept the null hypothesis H_0 .

i.e., The dice are fair.

Example 7 : 4 coins were tossed 160 times and the following results were obtained.

No. of Heads :	0	1	2	3	4
Observed Frequencies :	17	52	54	31	6

Under the assumption that coins are balanced, find the expected frequencies of 0, 1, 2, 3 or 4 heads, and test the goodness of fit ($\alpha = 0.05$). [JNTU 2008 (Set No. 3)]

Solution : Given $N = 160$, $n = 4$

$$p = \text{probability of getting a head} = \frac{1}{2}$$

$$q = \text{probability of getting a tail} = \frac{1}{2}$$

The expected frequencies of getting 0, 1, 2, 3, 4 are the successive terms of the binomial expansion $160 (p + q)^4$

$$= 160 [p^4 + 4p^3q + 6p^2q^2 + 4pq^3 + q^4]$$

$$\begin{aligned}
 &= 160 \left[\left(\frac{1}{2}\right)^4 + 4 \cdot \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right) + 6 \cdot \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 + 4 \cdot \left(\frac{1}{2}\right) \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^4 \right] \\
 &= 160 \left[\frac{1}{16} + \frac{4}{16} + \frac{6}{16} + \frac{4}{16} + \frac{1}{16} \right] = 10 + 40 + 60 + 40 + 10
 \end{aligned}$$

Thus the theoretical or expected frequencies are 10, 40, 60, 40, 10.

$$\begin{aligned}
 \text{Hence } \chi^2 &= \sum \frac{(O_i - E_i)^2}{E_i} \\
 &= \frac{(17-10)^2}{10} + \frac{(52-40)^2}{40} + \frac{(54-60)^2}{60} + \frac{(31-40)^2}{40} + \frac{(6-10)^2}{10} \\
 &= \frac{1}{120} (588 + 432 + 72 + 243 + 192) = \frac{1527}{120} = 12.725
 \end{aligned}$$

No. of degrees of freedom = 5 - 1 = 4

$\therefore \chi^2_{0.05} = 9.49$ (From tables)

Since the calculated value of χ^2 is greater than $\chi^2_{0.05}$ the hypothesis that the data follows the binomial law is rejected.

Example 8 : Fit a Poisson distribution to the following data and for its goodness of fit at level of significance 0.05?

x	0	1	2	3	4
f	419	352	154	56	19

[JNTU (K) May 2010 (Set No. 3)]

Solution :

x	f	$f \cdot x$
0	419	0
1	352	352
2	154	308
3	56	168
4	19	76
	$N = 1000$	$\Sigma fx = 904$

$$\text{Mean } \lambda = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{904}{1000} = 0.904$$

Theoretical distribution is given by $N \times p(x) = 1000 \times \frac{e^{-\lambda} \lambda^x}{x!}$

Hence the theoretical frequencies are given by

$$f(x) = \frac{1000 e^{-0.904} (0.904)^x}{x!} = \frac{1000 \times 0.4049 \times (0.904)^x}{x!} \quad \dots (1)$$

Putting $x = 0, 1, 2, 3, 4$, we get

x	0	1	2	3	4	Total
f	404.9 (406.2)	366	165.4	49.8	11.3 (12.6)	997.4

In order that the total observed and expected frequencies may agree, we take first and last theoretical frequencies as 406.2 and 12.6 instead of 404.9 and 11.3 as shown in brackets.

$$\begin{aligned}
 \therefore \chi^2 &= \sum \frac{(O_i - E_i)^2}{E_i} \\
 &= \frac{(419 - 406.2)^2}{406.2} + \frac{(352 - 366)^2}{366} + \frac{(154 - 165.4)^2}{165.4} + \frac{(56 - 49.8)^2}{49.8} + \frac{(19 - 12.6)^2}{12.6} \\
 &= 0.403 + 0.536 + 0.786 + 0.772 + 3.251 \\
 &= 5.748
 \end{aligned}$$

Degrees of freedom $= n - 2 = 5 - 2 = 3$

[$\therefore$ one d.f. being lost because $\Sigma O = \Sigma E$; 1 d.f., is lost because the parameter λ has been estimated]

For $v = 3$, we have, $\chi_{0.05}^2 = 7.82$

Since, the calculated value of $\chi^2 < \chi_{0.05}^2$, this agreement between the fact and theory is good and hence the Poisson distribution can be fitted to the data.

Example 9 : A survey of 240 families with 4 children each revealed the following distribution.

Male Births	4	3	2	1	0
Observed freq.	10	55	105	58	12

[JNTU (H) Apr. 2012 (Set No. 3)]

Sol. Let the Null Hypothesis be H_0 that male and female births are equally probable.

$$i.e., p = q = \frac{1}{2}.$$

The expected frequency x of male births is given by

$$f(x) = N \times {}^n C_x p^x q^{n-x}, \text{ where } N = 240, n = 4, x = 0, 1, 2, 3, 4$$

$$\therefore f(0) = 240 \times {}^4 C_0 \times \left(\frac{1}{2}\right)^0 \times \left(\frac{1}{2}\right)^{4-0} = 240 \times 1 \times 1 \times \frac{1}{16} = 15$$

$$f(1) = 240 \times {}^4 C_1 \times \left(\frac{1}{2}\right)^1 \times \left(\frac{1}{2}\right)^{4-1} = 240 \times 4 \times \frac{1}{2} \times \frac{1}{8} = 60$$

$$f(2) = 240 \times {}^4C_2 \times \left(\frac{1}{2}\right)^2 \times \left(\frac{1}{2}\right)^{4-2} = 240 \times 6 \times \frac{1}{4} \times \frac{1}{4} = 90$$

$$f(3) = 240 \times {}^4C_3 \times \left(\frac{1}{2}\right)^3 \times \left(\frac{1}{2}\right)^{4-3} = 240 \times 4 \times \frac{1}{8} \times \frac{1}{2} = 60$$

$$f(4) = 240 \times {}^4C_4 \times \left(\frac{1}{2}\right)^4 \times \left(\frac{1}{2}\right)^{4-4} = 240 \times 1 \times \frac{1}{16} \times 1 = 15$$

The expected or theoretical (Binomial) frequencies of male births are :

x	4	3	2	1	0
$f(x)$	15	60	90	60	15

Let us now apply χ^2 - test to examine the goodness of fit of the given data to the above Binomial distribution.

Calculations for χ^2

No. of families		$O_i - E_i$	$(O_i - E_i)^2$	$(O_i - E_i)^2 / E_i$
Observed frequency (O)	Expected frequency (E)			
10	15	-5	25	1.67
55	60	-5	25	0.42
105	90	15	225	2.5
58	60	-2	4	0.07
12	15	-3	9	0.6
Total 240	240	—	—	5.26

$$\therefore \chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = 5.26$$

$$\text{Degrees of freedom} = n - 1 = 5 - 1 = 4$$

The tabulated value of χ^2 at 5% level with 4 d.f. is 9.488. Since the calculated value of $\chi^2 < \text{the tabulated value}$, we accept the Null Hypothesis and conclude that the male and female births are equally probable.

10.18 CHI-SQUARE TEST FOR INDEPENDENCE OF ATTRIBUTES

Definition : Literally, an attribute means a quality or characteristic. Examples of attributes are drinking, smoking, blindness, honesty, beauty etc.

An attribute may be marked by its presence (position) or absence in a number of a given population. Let the observations be classified according to two attributes and the frequencies O_i in the different categories be shown in a two-way table,

called contingency table. We have to test on the basis of cell frequencies whether the two attributes are independent or not. We take the Null - Hypothesis H_0 that there is no association between the attributes *i.e.*, we assume that the two attributes are

independent. The expected frequencies (E_i) of any cell = $\frac{\text{Row total} \times \text{Column total}}{\text{Grand total}}$

The test statistic $\chi^2 = \sum_i \left[\frac{(O_i - E_i)^2}{E_i} \right]$ approximately follows Chi-square distribution with d.f. = (No. of rows - 1) $\times$ (No. of columns - 1)

If the calculated value of χ^2 is less than the table value at a specified level (generally 5%) of significance, the hypothesis holds good *i.e.*, the attributes are independent and do not bear any association. On the other hand, if the calculated value of χ^2 is greater than the table value at a specified level of significance, we say that the results of the experiment do not support the hypothesis, in other words, the attributes are associated.

Let us consider two attributes A and B . A is divided into two classes and B is divided into two classes. The various cell frequencies can be expressed in the following table known as 2×2 contingency table.

A	a	b
B	c	d
	$a + b$	
	$c + d$	
$a + c$	$b + d$	$N = a + b + c + d$

The expected frequencies are given by

$E(a) = \frac{(a+c)(a+b)}{N}$	$E(b) = \frac{(b+d)(a+b)}{N}$	$a + b$
$E(c) = \frac{(a+c)(c+d)}{N}$	$E(d) = \frac{(b+d)(c+d)}{N}$	$c + d$
$a + c$	$b + d$	$N = a + b + c + d$

The value of χ^2 is given by $\chi^2 = \frac{N(ad - bc)^2}{(a+b)(c+d)(a+c)(b+d)}$ where

$N = a + b + c + d$ with d.f. = $(2-1)(2-1) = 1$. We use this formula when the expected frequencies are in fractions (or decimals).

SOLVED EXAMPLES

Example 1 : On the basis of information given below about the treatment of 200 patients suffering from a disease, state whether the new treatment is comparatively superior to the conventional treatment.

	<i>Favourable</i>	<i>Not favourable</i>	<i>Total</i>
<i>New</i>	60	30	90
<i>Conventional</i>	40	70	110

Solution :

Null Hypothesis H_0 : No difference between new and conventional treatment (or) New and conventional treatment are independent.

The number of degrees of freedom is $(2 - 1)(2 - 1) = 1$

Expected frequencies are given in the table :

$$\text{Expected frequency} = \frac{\text{Row total} \times \text{Column total}}{\text{Grand total}}$$

$\frac{90 \times 100}{200} = 45$	$\frac{90 \times 100}{200} = 45$	90
$\frac{100 \times 110}{200} = 55$	$\frac{100 \times 110}{200} = 55$	110
100	100	200

Calculation of χ^2 :

Observed Frequency (O_i)	Expected Frequency (E_i)	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
60	45	225	5
30	45	225	5
40	55	225	4.09
70	55	225	4.09
200	200		18.18

$$\therefore \chi^2 = \sum \frac{(O - E)^2}{E} = 18.18$$

Tabulated χ^2 for 1 d.f. at 5% level of significance is 3.841.

Since calculated $\chi^2 >$ tabulated χ^2 we reject the null hypothesis H_0 i.e., new and conventional treatment are not independent. The new treatment is comparatively superior to conventional treatment.

Example 2 : The following table gives the classification of 100 workers according to sex and nature of work. Test whether the nature of work is independent of the sex of the worker.

	<i>Stable</i>	<i>Unstable</i>	<i>Total</i>
<i>Males</i>	40	20	60
<i>Females</i>	10	30	40
<i>Total</i>	50	50	100

[JNTU (K) May 2013 (Set No. 2)]

Solution :

Null Hypothesis H_0 : The nature of work is independent of the sex of the workers.

Expected frequencies are given in the table :

$\frac{50 \times 60}{100} = 30$	$\frac{50 \times 60}{100} = 30$	60
$\frac{50 \times 40}{100} = 20$	$\frac{50 \times 40}{100} = 20$	40
50	50	100

Calculation of χ^2 :

O_i	E_i	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
40	30	100	3.333
20	30	100	3.333
10	20	100	5.000
30	20	100	5.000
100	100	$\Sigma \frac{(O_i - E_i)^2}{E_i}$	16.66

$$\therefore \chi^2 = \Sigma \frac{(O_i - E_i)^2}{E_i} = 16.66$$

$$\therefore \text{Calculated } \chi^2 = 16.66$$

Tabulated value of χ^2 for $(2 - 1) (2 - 1) = 1$ d.f. at 5% level of significance is 3.84.

Since calculated $\chi^2 >$ tabulated χ^2 , we reject the null hypothesis H_0 , i.e., the nature of work is not independent of the sex of the workers.

i.e., there is difference in the nature of work on the basis of sex.

Example 2 : Given the following contingency table for hair colour and eye colour. Find the value of χ^2 . Is there good association between the two ?

<i>Hair colour</i>					
		<i>Fair</i>	<i>Brown</i>	<i>Black</i>	<i>Total</i>
<i>Eye Colour</i>	<i>Blue</i>	15	5	20	40
	<i>Grey</i>	20	10	20	50
	<i>Brown</i>	25	15	20	60
	<i>Total</i>	60	30	60	150

Solution :

Null Hypothesis H_0 : The two attributes, hair and eye colour are independent.

Table of expected frequencies :

$\frac{60 \times 40}{150} = 16$	$\frac{30 \times 40}{150} = 8$	$\frac{60 \times 40}{150} = 16$	40
$\frac{60 \times 50}{150} = 20$	$\frac{30 \times 50}{150} = 10$	$\frac{60 \times 50}{150} = 20$	50
$\frac{60 \times 60}{150} = 24$	$\frac{30 \times 60}{150} = 12$	$\frac{60 \times 60}{150} = 24$	60
60	30	60	150

Calculation of χ^2 :

Observed Frequency (O_i)	Expected Frequency (E_i)	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
15	16	1	0.0625
5	8	9	1.125
20	16	16	1
20	20	0	0
10	10	0	0
20	20	0	0
25	24	1	0.042
15	12	9	0.75
20	24	16	0.666
Total			3.6458

$$\therefore \chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = 3.6458$$

i.e, Calculated $\chi^2 = 3.6458$

Tabulated χ^2 for $(3 - 1)(3 - 1) = 4$ d.f. at 5% level of significance is 9.488

Since calculated $\chi^2 <$ tabulated χ^2 , we accept the null hypothesis H_0 .

i.e., The hair colour and eye colour are independent

i.e., The hair colour and eye colour are not associated.

Example 4 : Two researchers adopted different sampling techniques while investigating some group of students to find the number of students falling into different intelligence level. The results are as follows :

Researchers	Below Average	Average	Above Average	Genius	Total
X	86	60	44	10	200
Y	40	33	25	2	100
Total	126	93	69	12	300

Would you say that the sampling techniques adopted by the two researchers are significantly different. [Given 5% value of χ^2 for 2 degrees and 3 degrees of freedoms are 5.991 and 7.82 respectively].

Solution :

Null Hypothesis H_0 : There is no significant difference between the two researchers.

Table of expected frequencies :

$\frac{126 \times 200}{300} = 84$	$\frac{93 \times 200}{300} = 62$	$\frac{69 \times 200}{300} = 46$	$\frac{12 \times 200}{300} = 8$	200
$\frac{126 \times 100}{300} = 42$	$\frac{93 \times 100}{300} = 31$	$\frac{69 \times 100}{300} = 23$	$\frac{12 \times 100}{300} = 4$	100
126	93	69	12	300

Calculation of χ^2 :

Observed Frequency (O_i)	Expected Frequency (E_i)	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
86	84	4	0.0476
60	62	4	0.0645
44	46	4	0.0869
10	8	4	0.5
40	42	4	0.0952
33	31	4	0.129
25	23	4	0.1739
2	4	4	1
			2.0971

$$\text{Now } \chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = 2.0971$$

i.e., Calculated $\chi^2 = 2.0971$

Tabulated χ^2 for $(2 - 1) (4 - 1) = 3$ d.f. at 5% level of significance is 7.82.

Since calculated $\chi^2 < \text{tabulated } \chi^2$, we accept the null hypothesis H_0 .

i.e., There is no significant difference between the two researchers.

Example 4 : Four methods are under development for making discs of a super conducting material. Fifty discs are made by each method and they are checked for super conductivity when cooled with liquid.

	<i>1st Method</i>	<i>2nd Method</i>	<i>3rd Method</i>	<i>4th Method</i>
<i>Super Conductors</i>	31	42	22	25
<i>Failures</i>	19	8	28	25

Test the significant difference between the proportions of super conductors at 0.05 level. [JNTU 2006S (Set No. 2)]

Solution : 1. Null Hypothesis H_0 :

$$p_1 = p_2 = p_3 = p_4$$

2. Alternative Hypothesis H_1 :

$$p_1 \neq p_2 \neq p_3 \neq p_4$$

3. Level of significance, $\alpha = 0.05$

4. Computations.

	1st Method	2nd Method	3rd Method	4th Method	Total
Super Conductors	31	42	22	25	120
Failures	19	8	28	25	80
Total	50	50	50	50	200

Table of expected frequencies

$\frac{50 \times 120}{200} = 30$	$\frac{50 \times 120}{200} = 30$	$\frac{50 \times 120}{200} = 30$	$\frac{50 \times 120}{200} = 30$	120
$\frac{50 \times 80}{200} = 20$	$\frac{50 \times 80}{200} = 20$	$\frac{50 \times 80}{200} = 20$	$\frac{50 \times 80}{200} = 20$	80
50	50	50	50	200

Calculation for χ^2

Observed Frequency (O_i)	Expected Frequency (E_i)	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
31	30	1	1/30
42	30	144	144/30
22	30	64	64/30
25	30	25	25/30
19	20	1	1/20
8	20	144	144/20
28	20	64	64/20
25	20	25	25/20
			19.5

$\therefore$ Calculated $\chi^2 = 19.5$

Tabulated χ^2 for $(4 - 1) (4 - 1) = 9$ d.f. at 5% level of significance is 7.815.

Since calculated $\chi^2 >$ tabulated χ^2 , we reject the null hypothesis H_0 .

i.e., There is a significant difference between the proportions.

Example 2 : From the following data, find whether there is any significant liking in the habit of taking soft drinks among the categories of employees. Use Chi-square distribution test with level of significance 0.05.

Employees

<i>Soft Drinks</i>	<i>Clerks</i>	<i>Teachers</i>	<i>Officers</i>
<i>Pepsi</i>	<i>10</i>	<i>25</i>	<i>65</i>
<i>Thumsup</i>	<i>15</i>	<i>30</i>	<i>65</i>
<i>Fanta</i>	<i>50</i>	<i>60</i>	<i>30</i>

[JNTU 2006S (Set No. 2), (A) Dec. 2015, June 2016, Dec. 2017]

Solution :

- Null Hypothesis H_0 :** There is a significant liking in the habit of taking soft drinks.
- Alternative Hypothesis H_1 :** There is no significant liking.
- Level of significance :** $\alpha = 0.05$ (Given)

4. Computations.

<i>Soft Drinks</i>	<i>Clerks</i>	<i>Teachers</i>	<i>Officers</i>	<i>Total</i>
<i>Pepsi</i>	10	25	65	100
<i>Thumsup</i>	15	30	65	110
<i>Fanta</i>	50	60	30	140
<i>Total</i>	75	115	160	350

Table of Expected Frequencies:

$\frac{75 \times 100}{350} = 21.4$	$\frac{115 \times 100}{350} = 32.9$	$\frac{160 \times 100}{350} = 45.71$
$\frac{75 \times 110}{350} = 23.6$	$\frac{115 \times 110}{350} = 36.1$	$\frac{160 \times 110}{350} = 50.3$
$\frac{75 \times 140}{350} = 30$	$\frac{115 \times 140}{350} = 46$	$\frac{160 \times 140}{350} = 64$

Calculations for χ^2

Observed Frequency (O_i)	Expected Frequency (E_i)	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
10	21.4	129.96	6.073
25	32.9	62.41	1.897
65	45.7	372.49	8.151
15	23.6	73.96	3.134
30	36.1	37.21	1.031
65	50.3	216.09	4.3
50	30	400	13.333
60	46	196	4.261
30	64	1156	18.062
			60.2425

$\therefore$ Calculated $\chi^2 = 60.2425$

Tabulated χ^2 for $(3 - 1)(3 - 1) = 4$ d.f at 5% level of significance is 9.488.

Since calculated $\chi^2 >$ tabulated χ^2 , we reject the null hypothesis H_0 .

Thus we conclude that the habit of taking soft drink depends on categories of employees.

Example 2 : In an investigation on the machine performance, the following results are obtained :

	No. of units inspected	No. of defectives
Machine 1	375	17
Machine 2	450	22

Test whether there is any significant performance of two machines at $\alpha = 0.05$.

(or Can we conclude that both the machines are equally efficient at 0.05 level of significance).

[JNTU 2004S, 2005S, (K) May 2013 (Set No. 4)]

Solution :

1. Null Hypothesis H_0 : There is a significant performance of two machines.

2. Alternative Hypothesis H_1 : There is no significant performance.

3. Level of significance, $\alpha = 0.05$

4. Computations.

	<i>No. of units inspected</i>	<i>No. of defectives</i>	<i>Total</i>
<i>Machine 1</i>	375	17	392
<i>Machine 2</i>	450	22	472
<i>Total</i>	825	39	864 = N

Table of expected frequencies :

$\frac{825 \times 392}{864} = 374.3$	$\frac{39 \times 392}{864} = 17.69$
$\frac{825 \times 472}{864} = 450.69$	$\frac{39 \times 472}{864} = 21.3$

Calculation of χ^2 :

Observed Frequency (O_i)	Expected Frequency (E_i)	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
375	374.3	0.49	0.00131
17	17.69	0.4761	0.0269
450	450.69	0.4761	0.0011
22	21.3	0.49	0.023
			0.05231

$$\text{Now } \chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = 0.05231$$

i.e., calculated $\chi^2 = 0.05231$

Tabulated χ^2 for $(2 - 1) (2 - 1) = 1$ d.f at 5% level of significance is 3.84.

Since calculated $\chi^2 <$ tabulated χ^2 , we accept the null hypothesis H_0 .

i.e., There is a significant performance of two machines.

Example 8 : 1000 Students at college level were graded according to their I.Q and the economic conditions of their home. Use χ^2 test to find out whether there is any association between condition at home and I.Q ($\alpha = 0.05$).

I.Q.

Economic Condition	High	Low	Total
Rich	460	140	600
Poor	240	160	400
Total	700	300	1000

[JNTU April 2009 (Set No. 3)]

Solution : Set up the null hypothesis H_0 :

There is association between condition at Home and IQ.

Table of Expected Frequencies

$700 \times 600/1000 = 420$	$300 \times 600/1000 = 180$	600
$700 \times 400/1000 = 280$	$300 \times 400/1000 = 120$	400
700	300	1000

Calculations for χ^2

Observed frequencies (O_i)	Estimated frequencies (E_i)	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
460	420	1600	$1600/420 = 3.809$
140	180	1600	$1600/180 = 8.880$
240	280	1600	$1600/280 = 5.714$
160	120	1600	$1600/120 = 13.33$

Degrees of freedom = $(2 - 1)(2 - 1) = 1$

Calculated $\chi^2 = \sum \frac{(O - E)^2}{E} = 31.733$

Tabulated χ^2 for $\alpha = 0.05$ for 1 degree of freedom = 3.84

Calculated $\chi^2 >$ Tabulated χ^2

Hence we reject the null hypothesis.

There is no association between condition at home and IQ.

10.19 CHI-SQUARE TEST FOR POPULATION VARIANCE

Suppose that a random sample x_i ($i = 1, 2, \dots, n$) is drawn from a normal population with mean μ and variance σ^2 . To test the hypothesis that the population variance σ^2 has a specified value σ_0^2 .

Let the Null Hypothesis be $H_0 : \sigma^2 = \sigma_0^2$ (or $\sigma = \sigma_0$)

Then the Alternative Hypothesis is $H_1 : \sigma^2 \neq \sigma_0^2$ (or $\sigma^2 > \sigma_0^2$)

The test statistic is $\chi^2 = \sum \frac{(x_i - \bar{x})^2}{\sigma_0^2} = \frac{ns^2}{\sigma_0^2}$

where $s^2 = \text{sample variance} = \frac{\sum (x_i - \bar{x})^2}{n}$ and $ns^2 = (n-1)S^2$

Assuming that H_0 is true, the test statistic χ^2 follows Chi-square distribution with $(n-1)$ d.f.

Conclusion : If the calculated value of $\chi^2 >$ the tabulated value, we reject the Null Hypothesis H_0 . Otherwise, we accept H_0 .

SOLVED EXAMPLES

Example 2 : A random sample of size 20 from a normal population gives a mean of 42 and a variance of 25. Test the hypothesis that the population standard deviation is 8 at 5% level of significance.

Solution : Given $n = \text{sample size} = 20$, $\bar{x} = \text{sample mean} = 42$ and

$s^2 = \text{sample variance} = 25$. $\therefore s = 5$

Let the Null Hypothesis be $H_0 : \sigma = \sigma_0 = 8$

Then the Alternative Hypothesis is $H_1 : \sigma \neq 8$

The test statistic is $\chi^2 = \frac{ns^2}{\sigma_0^2} = \frac{20(25)}{(8)^2} = 7.8125$

$\therefore$ Calculated $\chi^2 = 7.8125$

d.f. $= n - 1 = 20 - 1 = 19$

Tabulated value of χ^2 at 5% level with 19 d.f. is 30.14

Since calculated value of $\chi^2 <$ the tabulated value of χ^2 , we accept the Null Hypothesis H_0 and conclude that the population standard deviation is 8 at 5% level for 19 d.f.

Example 1 : A firm manufacturing rivets wants to limit variations in their length as much as possible. The lengths (in cms) of 10 rivets manufactured by a new process are

2.15	1.99	2.05	2.12	2.17
2.01	1.98	2.03	2.25	1.93

Examine whether the new process can be considered superior to the old if the old population has standard deviation 0.145 cm ?

[JNTU 2004, 2005, 2006S, (H) Nov. 2010 (Set No. 4)]

Solution: We have

$$n = 10, \bar{x} = \frac{\sum x_i}{n} = \frac{20.68}{10} = 2.068$$

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n} = \frac{0.09096}{10} = 0.0091$$

$$\text{and } \sigma_0 = 0.145$$

1. **Null Hypothesis** $H_0 : \sigma^2 = \sigma_0^2$
2. **Alternative Hypothesis** $H_1 : \sigma^2 > \sigma_0^2$
3. Level of significance, $\alpha = 0.05$
4. Assuming that H_0 is true, the test statistic is

$$\chi^2 = \frac{ns^2}{\sigma_0^2} = \frac{0.09096}{(0.145)^2} = 4.326 \quad [\because \sigma_0 = 0.145 \text{ (given)}]$$

$$\therefore \text{Calculated } \chi^2 = 4.8$$

$$\text{d.f.} = n - 1 = 10 - 1 = 9$$

Tabulated χ^2 for 9 d.f. at 5% level of significance is 16.919.

Since calculated $\chi^2 < \text{tabulated } \chi^2$, we accept the null hypothesis H_0 . i.e., The new process cannot be considered superior to the old process.

REVIEW QUESTIONS

1. Explain briefly the Student's 't' - test
2. What is test statistic, 't' to test the difference of means for small samples. [JNTU (A) June 2016]
3. Explain briefly the Variance Ratio test (or F - test) [JNTU (K) Dec. 2013 (Set No. 1)]
4. When we apply F-test of significance. [JNTU (A) Dec. 2017]
5. (i) Explain briefly the χ^2 (Chi-Square) test.
(ii) Explain χ^2 - test as a test of Goodness of fit.

EXERCISE 10(B)

1. A machine which produces mica insulating washers for use in electric devices is set to turn out washers having a thickness of 10 mm. A sample of 10 washers has an average thickness 9.52 mm with a S.D of 0.6 mm. Calculate student's t.
2. Ten cartons are taken at random from an automatic filling machine. The mean net height of the 10 cartons is 11.802 and S.D is 0.15 OZ. Does the sample mean differ significantly from the intended weight of 12OZ ? Given $t_{0.05}$ for 9 d.f = 2.26.

3. A soap manufacturing company was distributing a particular brand of soap through a large number of retail shops. Before a heavy advertisement campaign, the mean sales per week per shop was 140 dozens. After the campaign, a sample of 26 shops was taken and the mean sales was found to be 147 dozens with S.D 16. Can you consider the advertisement effective.
4. A random sample of 16 values from a normal population showed a mean of 41.5 inches and the sum of squares of deviations from this mean equal to 135 square inches. Show that the assumption of a mean of 43.5 inches for the population is not acceptable.
5. Prices of shares of a company on the different days in a month were found to be 66, 65, 69, 70, 69, 71, 70, 63, 63, 64 and 68. Discuss whether the mean price of the shares in the month is 65. [JNTU (A) Nov. 2011 (Set No. 1)]
6. Experience shows that a fixed dose of a certain drug causes an average increase of pulse rate of 10 beats per minute with a S.D of 4. A group of 9 patients given the same dose showed the following increase : 13, 15, 14, 10, 8, 12, 16, 9, 20. Test at 5% level of significance whether this group is different in response to the drug.
7. Ten specimens of copper wires drawn from a larger lot have the following breaking strength (in kg unit) 578, 572, 570, 568, 572, 578, 570, 572, 569, 548. Test whether the mean breaking strengths of the lot may be taken to be 578 kg weight.
8. An I.Q test was administered to 5 persons before and after they were trained. The results are given below.

I.Q before training	110	120	123	132	125
I.Q. after training	120	118	125	136	121

Test whether there is any change in I.Q. after the training programme.

9. Two salesmen *A* and *B* are working in a certain district. From a sample survey conducted by the Head office, the following results were obtained. State whether there is any significant difference in the average sales between two salesmen.

No. of sales	20	18
Average sales (in Rs.)	170	205
Standard deviation (in Rs.)	20	25

10. Memory capacity of 10 students were tested before and after training. State whether the training was effective or not from the following scores.

Before training	12	14	11	8	7	10	3	0	5	6
After training	15	16	10	7	5	12	10	2	3	8

[JNTU (H) Nov. 2010 (Set No. 2), (A) Dec. 2018]

11. In one sample of 10 observations, the sum of the squares of the deviations of the sample values from sample mean was 120 and in the other sample of 12 observations, it was 314. Test whether the difference in variances is significant at 5% in variances level. [JNTU (H) Nov. 2010 (Set No. 2)]

12. The random samples were drawn from two normal populations and the following results were obtained.

Sample I	16	17	18	19	20	21	22	24	26	27	–	–
Sample II	19	22	23	25	26	28	29	30	31	32	35	36

Obtain estimates of the variances of populations and test whether the two populations have the same variance.

13. Two random samples drawn from normal populations are :

I	20	16	26	27	23	22	18	24	25	19	–	–
II	27	33	42	35	32	34	38	28	41	43	39	37

Obtain the estimates of the variances of populations. Test whether the two populations have the same variance.

14. A die is thrown 120 times and the frequencies of various faces are as follows :

Face No.	1	2	3	4	5	6
Frequency	10	15	25	25	18	27

Test whether the die was fair.

15. 12 dice were thrown 4096 times and a throw of 6 was reckoned as success. The observed frequencies were as given below.

Success	0	1	2	3	4	5	6	7 and above
Frequencies	447	1145	118	796	380	115	24	8

Find the value of χ^2 on the hypothesis that the dice were unbiased and hence show that the data are consistent with this hypothesis so far as the χ^2 test is concerned. [JNTU (K) Nov. 2011 (Set No. 3)]

16. 1000 students at college level were graded according to their I.Q and the economic conditions of their homes. Use χ^2 test to find out whether there is any association between economic conditions at home and I.Q.

	I.Q		
Economic conditions	High	Low	Total
Rich	460	140	600
Poor	240	160	400
Total	700	300	1000

17. From the following information, state whether the two attributes i.e., condition of house and condition of child are independent.

	Condition of House	
Condition of child	Clean	Dirty
Clean	69	51
Fairly clean	81	20
Dirty	35	44

18. Measurements of the fat content of two kinds of ice creams, Brand *A* and Brand *B*, yielded the following sample data :

Brand <i>A</i> (%)	13.5	14	13.6	12.9	13
Brand <i>B</i> (%)	12.9	13	12.4	13.5	12.7

Test the null hypothesis $\mu_1 = \mu_2$ (where μ_1 and μ_2 are the respective true average fat contents of the two kinds of ice cream) against the alternative hypothesis $\mu_1 \neq \mu_2$ at the level of significance $\alpha = 0.05$ (or) Test the significant between the means at 0.05 level. [JNTU 2003, 2003S, 2005S (Set No. 3)]

19. The following are the samples of skills. Test the significant difference between the means at 0.05 level.

Sample - I	74.1	77.7	74.4	74	73.8	—
Sample - II	70.8	74.9	74.2	70.4	69.2	72.2

[JNTU 2007, 2008S (Set No. 2)]

20. The two random samples reveal the following data :

SampleNo.	Size	Mean	Variance
I	16	440	40
II	25	460	42

[JNTU (K) Nov. 2012 (Set No. 3)]

Test whether the samples come from the same normal population.

21. (i) Fit a Poisson distribution to the following data and test the goodness of fit.

<i>x</i>	0	1	2	3	4
<i>f</i>	214	92	20	3	1

(ii) Fit a Poisson distribution to the following data Test the goodness of fit at a 0.5 level of significance (LOS).

x	0	1	2	3	4	5	6	7
$f(x)$	305	366	210	80	28	9	2	1

[JNTU (A) June 2019]

22. Fit a Binomial distribution to the following data and test for the goodness of fit at 0.05 level of significance.

No. of Heads (x)	0	1	2	3	4	5
No. of Tosses (f)	38	144	342	287	164	25

23. A certain medicine is given to each of the 9 patients resulted in the following increase of blood pressure. Can it be concluded that the medicine will in general be accompanied by an increase in blood pressure using the data : 7, 3, -1, 4, -3, 5, 6, -4, -1 .

$$\left[\text{Hint : } t = \frac{\bar{x} - 0}{S / \sqrt{x-1}} (\because H_0 : \mu_1 = \mu_2) \right]$$

ANSWERS

1. $t = -2.528$.
2. Calculated $t = 4.0$. The sample mean differs significantly from the mean $\mu = 12$ OZ.
3. Calculated $t = 2.19$. The advertising is effective in increasing the sales
5. Calculated $t = 2.3247$; The mean price of the shares in the month is not 65.
6. $H_0 : \mu = 10$, calculated $t = 2.38$, significant
7. Calculated $t = 1.54$, $\mu = 5.78$
8. $t = 0.816$; no change in I.Q
9. $H_0 : \mu_1 = \mu_2$, $t = -4.65$, significant
10. $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 \neq \mu_2$, $t = -1.3646$, not significant
11. Calculated $F = 2.14$, not significant
12. $\sigma_1^2 = \sigma_2^2$, $F = 1.935$, not significant
13. $F = 2.14$ 14. $\chi^2 = 11.40$, significant
16. $\chi^2 = 31.74$ 17. $\chi^2 = 25.64$
19. $|t| = 4.06$
22. $\chi^2 = 8.92$. Binomial distribution gives a good fit at 5% level.

QUIZ

1. The t - test is applicable to samples for which n is _____
2. The main use of χ^2 - test is _____
3. The main use of F - test is _____
4. $t_{1-\alpha} =$ _____ []
 - (a) t_α
 - (b) $-t_\alpha$
 - (c) $t_{\alpha-1}$
 - (d) None
5. $F_{1-\alpha}(v_1, v_2) =$ _____ []
 - (a) $F_\alpha(v_2, v_1)$
 - (b) $\frac{1}{F_\alpha(v_1, v_2)}$
 - (c) $F_\alpha(v_1, v_2)$
 - (d) None
6. The shape of t -distribution is similar to that of []
 - (a) Chi-square distribution
 - (b) F -distribution
 - (c) Normal distribution
 - (d) None
7. If $\bar{x} = 47.5$, $\mu = 42.1$, $S = 8.4$ and $n = 25$ then t is []
 - (a) 3.21
 - (b) 4.5
 - (c) 3.12
 - (d) None
8. Chi-square distribution curve varies from []
 - (a) $-\infty$ to ∞
 - (b) 0 to ∞
 - (c) $-\infty$ to 0
 - (d) None
9. Which distribution is used to test the equality of population means []
 - (a) Chi-square distribution
 - (b) F -distribution
 - (c) t -distribution
 - (d) None
10. If two independent random samples of size $n_1 = 13$ and $n_2 = 7$, are [] taken from a normal population. The variance of the first sample will be atleast four times as that of a second sample then F is
 - (a) $\frac{1}{4}$
 - (b) 4
 - (c) 16
 - (d) None
11. A random variable X has F - distribution with (m, n) degrees of [] freedom, then $1/X$ has the same distribution with ____ degrees of freedom.
 - (a) (m, n)
 - (b) (n, m)
 - (c) $\left(\frac{1}{m}, \frac{1}{n}\right)$
 - (d) $\left(\frac{1}{n}, \frac{1}{m}\right)$

12. Range of F - distribution is []
 (a) $(-\infty, \infty)$ (b) $(-\infty, 0)$
 (c) $(0, \infty)$ (d) None
13. Range of t - distribution is []
 (a) $(-\infty, \infty)$ (b) $(-\infty, 0)$
 (c) $(0, \infty)$ (d) None
14. In a t - distribution of sample size n , the degrees of freedom are []
 (a) n (b) $n-1$
 (c) $n-2$ (d) $n+1$
15. Chi-square distribution is
 (a) continuous (b) symmetrical
 (c) multimodal (d) None
16. If χ_1^2 and χ_2^2 are two independent distributions with m and n degrees of freedom respectively then $\chi_1^2 + \chi_2^2$ will be chi-squared distribution with _____ degrees of freedom. []
 (a) mn (b) m/n
 (c) $m-n$ (d) $m+n$
17. The mean and variance of a χ^2 distribution with 8 degrees of freedom are _____ and _____ respectively. []
 (a) 7, 49 (b) 8, 64
 (c) 8, 16 (d) 7, 14
18. If the standard deviation of a Chi-square distribution is 10, then its degrees of freedom is []
 (a) 25 (b) 49
 (c) 50 (d) None

ANSWERS

1. $n < 30$
2. To test whether the deviations of the observed frequencies from the expected frequencies are significant are not.
3. To test the equality of population means by comparing sample variances.
4. b 5. b 6. c 7. a
8. b 9. b 10. b 11. b
12. c 13. a 14. b 15. a
16. d 17. c 18. c

STATISTICAL TABLES



Table – 1

n	r	p									
		0.10	0.20	0.25	0.30	0.40	0.50	0.60	0.70	0.80	0.90
1	0	0.9000	0.8000	0.7500	0.7000	0.6000	0.5000	0.4000	0.3000	0.2000	0.1000
	1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
2	0	0.8100	0.6400	0.5625	0.4900	0.3600	0.2500	0.1600	0.0900	0.0400	0.0100
	1	0.9900	0.9600	0.9375	0.9100	0.8400	0.7500	0.6400	0.5100	0.3600	0.1900
	2	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
3	0	0.7290	0.5120	0.4219	0.3430	0.2160	0.1250	0.0640	0.0270	0.0080	0.0010
	1	0.9720	0.8960	0.8438	0.7840	0.6480	0.5000	0.3520	0.2160	0.1040	0.0280
	2	0.9990	0.9920	0.9844	0.9730	0.9360	0.8750	0.7840	0.6570	0.4880	0.2710
	3	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
4	0	0.6561	0.4096	0.3164	0.2401	0.1296	0.0625	0.0256	0.0081	0.0016	0.0001
	1	0.9477	0.8192	0.7383	0.6517	0.4752	0.3125	0.1792	0.0837	0.0272	0.0037
	2	0.9963	0.9728	0.9492	0.9163	0.8208	0.6875	0.5248	0.3483	0.1808	0.0523
	3	0.9999	0.9984	0.9961	0.9919	0.9744	0.9375	0.8704	0.7599	0.5904	0.3439
	4	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
5	0	0.5905	0.3277	0.2373	0.1681	0.0778	0.0312	0.0102	0.0024	0.0003	0.0000
	1	0.9185	0.7373	0.6328	0.5282	0.3370	0.1875	0.0870	0.0308	0.0067	0.0005
	2	0.9914	0.9421	0.8965	0.8369	0.6826	0.5000	0.3174	0.1631	0.0579	0.0086
	3	0.9995	0.9933	0.9844	0.9692	0.9130	0.8125	0.6630	0.4718	0.2627	0.0815
	4	1.0000	0.9997	0.9990	0.9976	0.9898	0.9688	0.9222	0.8319	0.6723	0.4095
	5		1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
6	0	0.5314	0.2621	0.1780	0.1176	0.0467	0.0156	0.0041	0.0007	0.0001	0.0000
	1	0.8857	0.6554	0.5339	0.4202	0.2333	0.1094	0.0410	0.0109	0.0016	0.0001
	2	0.9841	0.9011	0.8306	0.7443	0.5443	0.3438	0.1792	0.0705	0.0170	0.0013
	3	0.9987	0.9830	0.9624	0.9295	0.8208	0.6563	0.4557	0.2557	0.0989	0.0158
	4	0.9999	0.9984	0.9954	0.9891	0.9590	0.8906	0.7667	0.5798	0.3447	0.1143
	5	1.0000	0.9999	0.9998	0.9993	0.9959	0.9844	0.9533	0.8824	0.7379	0.4686
	6		1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
7	0	0.4783	0.2097	0.1335	0.0824	0.0280	0.0078	0.0016	0.0002	0.0000	
	1	0.8503	0.5767	0.4449	0.3294	0.1586	0.0625	0.0188	0.0038	0.0004	0.0000
	2	0.9743	0.8520	0.7564	0.6471	0.4199	0.2266	0.0963	0.0288	0.0047	

(continued) Binomial Distribution Function $\sum_{r=0} P(X=r)$

n	r	p									
		0.10	0.20	0.25	0.30	0.40	0.50	0.60	0.70	0.80	0.90
8	0	0.4305	0.1678	0.1001	0.0576	0.0168	0.0039	0.0007	0.0001	0.0000	
	1	0.8131	0.5033	0.3671	0.2553	0.1064	0.0352	0.0085	0.0013	0.0001	
	2	0.9619	0.7969	0.6785	0.5518	0.3154	0.1445	0.0498	0.0113	0.0012	0.0000
	3	0.9950	0.9437	0.8862	0.8059	0.5941	0.3633	0.1737	0.0580	0.0104	0.0004
	4	0.9996	0.9896	0.9727	0.9420	0.8263	0.6367	0.4059	0.1941	0.0563	0.0050
	5	1.0000	0.9988	0.9958	0.9887	0.9502	0.8555	0.6846	0.4482	0.2031	0.0381
	6		0.9991	0.9996	0.9987	0.9915	0.9648	0.8936	0.7447	0.4967	0.1869
	7		1.0000	1.0000	0.9999	0.9993	0.9961	0.9832	0.9424	0.8322	0.5695
8				1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	
9	0	0.3874	0.1342	0.0751	0.0404	0.0101	0.0020	0.0003	0.0000		
	1	0.7748	0.4362	0.3003	0.1960	0.0705	0.0195	0.0038	0.0004	0.0000	
	2	0.9470	0.7382	0.6007	0.4628	0.2318	0.0898	0.0250	0.0043	0.0003	0.0000
	3	0.9917	0.9144	0.8343	0.7297	0.4826	0.2539	0.0994	0.0253	0.0031	0.0001
	4	0.9991	0.9804	0.9511	0.9012	0.7334	0.5000	0.2666	0.0988	0.0196	0.0009
	5	0.9999	0.9969	0.9900	0.9747	0.9006	0.7461	0.5174	0.2703	0.0856	0.0083
	6	1.0000	0.9997	0.9987	0.9957	0.9750	0.9102	0.7682	0.5372	0.2618	0.0530
	7		1.0000	0.9999	0.9996	0.9962	0.9805	0.9295	0.8040	0.5638	0.2252
	8			1.0000	1.0000	0.9997	0.9980	0.9899	0.9596	0.8658	0.6126
9					1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	
10	0	0.3487	0.1074	0.0563	0.0282	0.0060	0.0010	0.0001	0.0000		
	1	0.7361	0.3758	0.2440	0.1493	0.0464	0.0107	0.0017	0.0001	0.0000	
	2	0.9298	0.6778	0.5256	0.3828	0.1673	0.0547	0.0123	0.0016	0.0001	
	3	0.9872	0.8791	0.7759	0.6496	0.3823	0.1719	0.0548	0.0106	0.0009	0.0000
	4	0.9984	0.9672	0.9219	0.8497	0.6331	0.3770	0.1662	0.0474	0.0064	0.0002
	5	0.9999	0.9936	0.9803	0.9527	0.8338	0.6230	0.3669	0.1503	0.0328	0.0016
	6	1.0000	0.9999	0.9965	0.9894	0.9452	0.8281	0.6177	0.3504	0.1209	0.0128
	7		0.9999	0.9996	0.9984	0.9877	0.9453	0.8327	0.6172	0.3222	0.0702
	8		1.0000	1.0000	0.9999	0.9983	0.9893	0.9536	0.8507	0.6242	0.2639
	9				1.0000	0.9999	0.9990	0.9940	0.9718	0.8926	0.6513
	10					1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	0	0.3138	0.0859	0.0422	0.0198	0.0036	0.0005	0.0000			
11	1	0.6974	0.3221	0.1971	0.1130	0.0302	0.0059	0.0007	0.0000		
	2	0.9104	0.6174	0.4552	0.3127	0.1189	0.0327	0.0059	0.0006	0.0000	
	3	0.9815	0.8369	0.7133	0.5696	0.2963	0.1133	0.0293	0.0043	0.0002	
	4	0.9972	0.9496	0.8854	0.7897	0.5328	0.2744	0.0994	0.0216	0.0020	0.0000
	5	0.9997	0.9883	0.9657	0.9218	0.7535	0.5000	0.2465	0.0782	0.0117	0.0003
	6	1.0000	0.9980	0.9924	0.9784	0.9006	0.7256	0.4672	0.2103	0.0504	0.0028
	7		0.9998	0.9988	0.9957	0.9707	0.8867	0.7037	0.4304	0.1611	0.0185
	8		1.0000	0.9999	0.9994	0.9941	0.9673	0.8811	0.6873	0.3826	0.0896
	9			1.0000	1.0000	0.9993	0.9941	0.9698	0.8870	0.6779	0.3026
	10					1.0000	0.9995	0.9964	0.9802	0.9141	0.6862
	11						1.0000	1.0000	1.0000	1.0000	1.0000

(continued) Binomial Distribution Function $\sum_{r=0}^n P(X=r)$

		P									
n	r	0.10	0.20	0.25	0.30	0.40	0.50	0.60	0.70	0.80	0.90
12	0	0.2824	0.0687	0.0317	0.0138	0.0022	0.0002	0.0000			
	1	0.6590	0.2749	0.1584	0.0850	0.0196	0.0032	0.0003	0.0000		
	2	0.8891	0.5583	0.3907	0.2528	0.0834	0.0193	0.0028	0.0002	0.0000	
	3	0.9744	0.7946	0.6488	0.4925	0.2253	0.0730	0.0153	0.0017	0.0001	
	4	0.9957	0.9274	0.8424	0.7237	0.4382	0.1938	0.0573	0.0095	0.0006	0.0000
	5	0.9995	0.9806	0.9456	0.8821	0.6652	0.3872	0.1582	0.0386	0.0039	0.0001
	6	0.9999	0.9961	0.9857	0.9614	0.8418	0.6128	0.3348	0.1178	0.0194	0.0005
	7	1.0000	0.9994	0.9972	0.9905	0.9427	0.8062	0.5618	0.2763	0.0726	0.0043
	8		0.9999	0.9996	0.9983	0.9847	0.9270	0.7747	0.5075	0.2054	0.0256
	9		1.0000	1.0000	0.9998	0.9972	0.9807	0.9166	0.7472	0.4417	0.1109
	10				1.0000	0.9997	0.9968	0.9804	0.9150	0.7251	0.3410
	11					1.0000	0.9998	0.9978	0.9862	0.9313	0.7176
12						1.0000	1.0000	1.0000	1.0000	1.0000	
13	0	0.2542	0.0550	0.0238	0.0097	0.0013	0.0001	0.0000			
	1	0.6213	0.2336	0.1267	0.0637	0.0126	0.0017	0.0001	0.0000		
	2	0.8661	0.5017	0.3326	0.2025	0.0579	0.0112	0.0013	0.0001		
	3	0.9658	0.7473	0.5843	0.4206	0.1686	0.0461	0.0078	0.0007	0.0000	
	4	0.9935	0.9009	0.7940	0.6543	0.3530	0.1334	0.0321	0.0040	0.0002	
	5	0.9991	0.9700	0.9198	0.8346	0.5744	0.2905	0.0977	0.0182	0.0012	0.0000
	6	0.9999	0.9930	0.9757	0.9376	0.7712	0.5000	0.2288	0.0624	0.0070	0.0001
	7	1.0000	0.9980	0.9944	0.9818	0.9023	0.7095	0.4256	0.1654	0.0300	0.0009
	8		0.9998	0.9990	0.9960	0.9679	0.8666	0.6470	0.3457	0.0991	0.0065
	9		1.0000	0.9999	0.9993	0.9922	0.9539	0.8314	0.5794	0.2527	0.0342
	10			1.0000	0.9999	0.9987	0.9888	0.9421	0.7975	0.4983	0.1339
	11				1.0000	0.9999	0.9983	0.9874	0.9363	0.7664	0.3787
	12					1.0000	0.9999	0.9987	0.9903	0.9450	0.7458
13						1.0000	1.0000	1.0000	1.0000	1.0000	
14	0	0.2288	0.0440	0.0178	0.0068	0.0008	0.0001	0.0000			
	1	0.5846	0.1979	0.1010	0.0475	0.0081	0.0009	0.0001			
	2	0.8416	0.4481	0.2811	0.1608	0.0398	0.0065	0.0006	0.0000		
	3	0.9559	0.6982	0.5213	0.3552	0.1243	0.0287	0.0039	0.0002		
	4	0.9908	0.8702	0.7415	0.5842	0.2793	0.0898	0.0175	0.0017	0.0000	
	5	0.9985	0.9561	0.8883	0.7805	0.4859	0.2120	0.0583	0.0083	0.0004	
	6	0.9998	0.9984	0.9617	0.9067	0.6925	0.3953	0.1501	0.0315	0.0024	0.0000
	7	1.0000	0.9976	0.9897	0.9685	0.8499	0.6047	0.3075	0.0933	0.0116	0.0002
	8		0.9996	0.9978	0.9917	0.9417	0.7880	0.5141	0.2195	0.0439	0.0015
	9		1.0000	0.9997	0.9983	0.9825	0.9102	0.7207	0.4158	0.1298	0.0092
	10			1.0000	0.9998	0.9961	0.9713	0.8757	0.6448	0.3018	0.0441
	11				1.0000	0.9994	0.9935	0.9602	0.8392	0.5519	0.1584
	12					0.9999	0.9991	0.9919	0.9525	0.8021	0.4154
	13					1.0000	0.9999	0.9992	0.9932	0.9560	0.7712
14						1.0000	1.0000	1.0000	1.0000	1.0000	

(continued) Binomial Distribution Function $\sum_{r=0}^n P(X=r)$

n	r	p									
		0.10	0.20	0.25	0.30	0.40	0.50	0.60	0.70	0.80	0.90
15	0	0.2059	0.0352	0.0134	0.0047	0.0005	0.0000				
	1	0.5490	0.1671	0.0802	0.0353	0.0052	0.0005	0.0000			
	2	0.8159	0.3980	0.2361	0.1268	0.0271	0.0037	0.0003	0.0000		
	3	0.9444	0.6482	0.4613	0.2969	0.0905	0.0176	0.0019	0.0001		
	4	0.9873	0.8358	0.6865	0.5155	0.2173	0.0592	0.0094	0.0007	0.0000	
	5	0.9978	0.9389	0.8516	0.7216	0.4032	0.1509	0.0338	0.0037	0.0001	
	6	0.9997	0.9819	0.9434	0.8689	0.6098	0.3036	0.0951	0.0152	0.0008	
	7	1.0000	0.9958	0.9827	0.9500	0.7869	0.5000	0.2131	0.0500	0.0042	0.0000
	8		0.9992	0.9958	0.9848	0.9050	0.6964	0.3902	0.1311	0.0181	0.0003
	9		0.9999	0.9992	0.9963	0.9662	0.8491	0.5968	0.2784	0.0611	0.0023
	10		1.0000	0.9999	0.9993	0.9907	0.9408	0.7827	0.4845	0.1642	0.0127
	11			1.0000	0.9999	0.9981	0.9824	0.9095	0.7031	0.3518	0.0556
	12				1.0000	0.9997	0.9963	0.9729	0.8732	0.6020	0.1841
	13					1.0000	0.9995	0.9948	0.9647	0.8329	0.4510
	14						1.0000	0.9995	0.9953	0.9648	0.7941
	15							1.0000	1.0000	1.0000	1.0000
16	0	0.1853	0.0281	0.0100	0.0053	0.0003	0.0000				
	1	0.5147	0.1407	0.0635	0.0261	0.0033	0.0003	0.0000			
	2	0.7892	0.3518	0.1971	0.0994	0.0183	0.0021	0.0001			
	3	0.9316	0.5981	0.4050	0.2459	0.0651	0.0106	0.0009	0.0000		
	4	0.9830	0.7982	0.6302	0.4499	0.1666	0.0384	0.0049	0.0003		
	5	0.9967	0.9183	0.8103	0.6598	0.3288	0.1051	0.0191	0.0016	0.0000	
	6	0.9995	0.9733	0.9204	0.8247	0.5272	0.2272	0.0583	0.0071	0.0002	
	7	0.9999	0.9930	0.9729	0.9256	0.7161	0.4018	0.1423	0.0257	0.0015	0.0000
	8	1.0000	0.9985	0.9925	0.9743	0.8577	0.5982	0.2839	0.0744	0.0070	0.0001
	9		0.9998	0.9984	0.9929	0.9417	0.7728	0.4728	0.1753	0.0267	0.0005
	10		1.0000	0.9997	0.9984	0.9809	0.8949	0.6712	0.3402	0.0817	0.0033
	11			1.0000	0.9997	0.9951	0.9616	0.8334	0.5501	0.2018	0.0170
	12				1.0000	0.9991	0.9894	0.9349	0.7541	0.4019	0.0684
	13					0.9999	0.9979	0.9817	0.9006	0.6482	0.2108
	14					1.0000	0.9997	0.9967	0.9739	0.8593	0.4853
	15						1.0000	0.9997	0.9967	0.9719	0.8147
16							1.0000	1.0000	1.0000	1.0000	

(continued) Binomial Distribution Function $\sum_{r=0}^n P(X=r)$

n	r	p									
		0.10	0.20	0.25	0.30	0.40	0.50	0.60	0.70	0.80	0.90
17	0	0.1668	0.0225	0.0075	0.0023	0.0002	0.0000				
	1	0.4818	0.1182	0.0501	0.0193	0.0021	0.0001	0.0000			
	2	0.7618	0.3096	0.1637	0.0774	0.0123	0.0012	0.0001			
	3	0.9174	0.5489	0.3530	0.2019	0.0464	0.0064	0.0005	0.0000		
	4	0.9779	0.7582	0.5739	0.3887	0.1260	0.0245	0.0025	0.0001		
	5	0.9953	0.8943	0.7653	0.5968	0.2639	0.0717	0.0106	0.0007	0.0000	
	6	0.9992	0.9623	0.8929	0.7752	0.4478	0.1662	0.0348	0.0032	0.0001	
	7	0.9999	0.9891	0.9598	0.8954	0.6405	0.3145	0.0919	0.0127	0.0005	
	8	1.0000	0.9974	0.9876	0.9597	0.8011	0.5000	0.1989	0.0403	0.0026	0.0000
	9		0.9995	0.9969	0.9873	0.9081	0.6855	0.3595	0.1046	0.0109	0.0001
	10		0.9999	0.9994	0.9968	0.9652	0.8338	0.5522	0.2248	0.0377	0.0008
	11		1.0000	0.9999	0.9993	0.9894	0.9283	0.7361	0.4032	0.1057	0.0047
	12			1.0000	0.9999	0.9975	0.9755	0.8740	0.6113	0.2418	0.0221
	13				1.0000	0.9995	0.9936	0.9536	0.7981	0.4511	0.0826
	14					0.9999	0.9988	0.9877	0.9226	0.6904	0.2382
	15					1.0000	0.9999	0.9979	0.9807	0.8818	0.5182
	16						1.0000	0.9998	0.9977	0.9775	0.8332
17						1.0000	1.0000	1.0000	1.0000	1.0000	
18	0	0.1501	0.0180	0.0056	0.0016	0.0001	0.0000				
	1	0.4503	0.0991	0.0395	0.0142	0.0013	0.0001				
	2	0.7338	0.2713	0.1353	0.0600	0.0082	0.0007	0.0000			
	3	0.9018	0.5010	0.3057	0.1646	0.0328	0.0038	0.0002			
	4	0.9718	0.7164	0.5787	0.3327	0.0942	0.0154	0.0013	0.0000		
	5	0.9936	0.8671	0.7175	0.5344	0.2088	0.0481	0.0058	0.0003		
	6	0.9988	0.9487	0.8610	0.7217	0.3743	0.1189	0.0203	0.0014	0.0000	
	7	0.9998	0.9837	0.9431	0.8593	0.5634	0.2403	0.0576	0.0061	0.0002	
	8	1.0000	0.9957	0.9807	0.9404	0.7368	0.4073	0.1347	0.0210	0.0009	
	9		0.9991	0.9946	0.9790	0.8653	0.5927	0.2632	0.0596	0.0043	0.0000
	10		0.9998	0.9988	0.9939	0.9424	0.7597	0.4366	0.1407	0.0163	0.0002
	11		1.0000	0.9998	0.9986	0.9797	0.8811	0.6257	0.2783	0.0513	0.0012
	12			1.0000	0.9997	0.9942	0.9519	0.7912	0.4656	0.1329	0.0064
	13				1.0000	0.9987	0.9846	0.9058	0.6673	0.2836	0.0282
	14					0.9998	0.9962	0.9672	0.8354	0.4990	0.0982
	15					1.0000	0.9993	0.9918	0.9400	0.7287	0.2662
	16						0.9999	0.9987	0.9858	0.9009	0.5497
	17						1.0000	0.9999	0.9984	0.9820	0.8499
18							1.0000	1.0000	1.0000	1.0000	

(continued) Binomial Distribution Function $\sum_{r=0}^n P(X=r)$

n	r	p									
		0.10	0.20	0.25	0.30	0.40	0.50	0.60	0.70	0.80	0.90
19	0	0.1351	0.0144	0.0042	0.0011	0.0001					
	1	0.4203	0.0829	0.0310	0.0104	0.0008	0.0000				
	2	0.7054	0.2369	0.1113	0.0462	0.0055	0.0004	0.0000			
	3	0.8850	0.4551	0.2631	0.1332	0.0230	0.0022	0.0001			
	4	0.9648	0.6733	0.4654	0.2822	0.0696	0.0096	0.0006	0.0000		
	5	0.9914	0.8369	0.6678	0.4739	0.1629	0.0318	0.0031	0.0001		
	6	0.9983	0.9324	0.8251	0.6655	0.3081	0.0835	0.0116	0.0006		
	7	0.9997	0.9767	0.9225	0.8180	0.4878	0.1796	0.0352	0.0028	0.0000	
	8	1.0000	0.9933	0.9713	0.9161	0.6675	0.3238	0.0885	0.0105	0.0003	
	9		0.9984	0.9911	0.9674	0.8139	0.5000	0.1861	0.0326	0.0016	
	10		0.9997	0.9977	0.9895	0.9115	0.6762	0.3325	0.0839	0.0067	0.0000
	11		0.9999	0.9995	0.9972	0.9648	0.8204	0.5122	0.1820	0.0233	0.0003
	12		1.0000	0.9999	0.9994	0.9884	0.9165	0.6919	0.3345	0.0676	0.0017
	13			1.0000	0.9999	0.9969	0.9682	0.8371	0.5261	0.1631	0.0086
	14				1.0000	0.9994	0.9904	0.9304	0.7178	0.3267	0.0352
	15					0.9999	0.9978	0.9770	0.8668	0.5449	0.1150
	16					1.0000	0.9996	0.9945	0.9538	0.7631	0.2946
	17						1.0000	0.9992	0.9896	0.9171	0.5797
	18							0.9999	0.9989	0.9856	0.8649
	19							1.0000	1.0000	1.0000	1.0000
20	0	0.1216	0.0115	0.0032	0.0008	0.0000					
	1	0.3917	0.0692	0.0243	0.0076	0.0005	0.0000				
	2	0.6769	0.2061	0.0913	0.0355	0.0036	0.0002	0.0000			
	3	0.8670	0.4114	0.2252	0.1071	0.0160	0.0013	0.0001			
	4	0.9568	0.6296	0.4148	0.2375	0.0510	0.0059	0.0003			
	5	0.9887	0.8042	0.6172	0.4164	0.1256	0.0207	0.0016	0.0000		
	6	0.9976	0.9133	0.7858	0.6080	0.2500	0.0577	0.0065	0.0003		
	7	0.9996	0.9679	0.8982	0.7723	0.4159	0.1316	0.0210	0.0013	0.0000	
	8	0.9999	0.9900	0.9591	0.8867	0.5956	0.2517	0.0565	0.0051	0.0001	
	9	1.0000	0.9974	0.9861	0.9520	0.7553	0.4119	0.1275	0.0171	0.0006	
	10		0.9994	0.9961	0.9829	0.8725	0.5881	0.2447	0.0480	0.0026	0.0000
	11		0.9999	0.9991	0.9949	0.9435	0.7483	0.4044	0.1133	0.0100	0.0001
	12		1.0000	0.9998	0.9987	0.9790	0.8684	0.5841	0.2277	0.0321	0.0004
	13			1.0000	0.9997	0.9935	0.9423	0.7500	0.3920	0.0867	0.0024
	14				1.0000	0.9984	0.9793	0.8744	0.5836	0.1958	0.0113
	15					0.9997	0.9941	0.9490	0.7625	0.3704	0.0432
	16					1.0000	0.9987	0.9840	0.8929	0.5886	0.1330
	17						0.9998	0.9964	0.9645	0.7939	0.3231
	18						1.0000	0.9995	0.9924	0.9308	0.6083
	19							1.0000	0.9992	0.9885	0.8784
	20								1.0000	1.0000	1.0000

Table – 2

Poisson Distribution Function : $p(x, \lambda) = \sum_{r=0}^x e^{-\lambda} \frac{\lambda^r}{r!}$

$\lambda \backslash x$	0	1	2	3	4	5	6	7	8	9
0.001	0.999	0.000								
0.002	0.998	0.0019								
0.003	0.997	0.0002								
0.004	0.996	0.003								
0.005	0.995	0.004								
0.006	0.994	0.005								
0.007	0.993	0.006								
0.008	0.992	0.007								
0.009	0.991	0.008								
0.01	0.99	0.009								
0.02	0.980	1.000								
0.04	0.961	0.999	1.000							
0.06	0.942	0.998	1.000							
0.08	0.923	0.997	1.000							
0.10	0.905	0.995	1.000							
0.15	0.861	0.990	0.999	1.000						
0.20	0.819	0.982	0.999	1.000						
0.25	0.779	0.974	0.998	1.000						
0.30	0.741	0.963	0.996	1.000						
0.35	0.705	0.951	0.994	1.000						
0.40	0.670	0.938	0.992	0.999	1.000					
0.45	0.638	0.925	0.989	0.999	1.000					
0.50	0.607	0.910	0.986	0.998	1.000					
0.55	0.577	0.894	0.982	0.998	1.000					
0.60	0.549	0.878	0.977	0.997	1.000					
0.65	0.522	0.861	0.972	0.996	0.999	1.000				
0.70	0.497	0.844	0.966	0.994	0.999	1.000				
0.75	0.472	0.827	0.959	0.993	0.999	1.000				
0.80	0.449	0.809	0.953	0.991	0.999	1.000				
0.85	0.427	0.791	0.945	0.989	0.998	1.000				
0.90	0.407	0.772	0.937	0.987	0.998	1.000				
0.95	0.387	0.754	0.929	0.984	0.997	1.000				
1.00	0.368	0.736	0.920	0.981	0.996	0.999	1.000			
1.1	0.333	0.699	0.900	0.974	0.995	0.999	1.000			
1.2	0.301	0.663	0.879	0.966	0.992	0.998	1.000			
1.3	0.273	0.627	0.857	0.957	0.989	0.998	1.000			
1.4	0.247	0.592	0.833	0.946	0.986	0.997	0.999	1.000		
1.5	0.223	0.558	0.809	0.934	0.981	0.996	0.999	1.000		
1.6	0.202	0.525	0.783	0.921	0.976	0.994	0.999	1.000		
1.7	0.183	0.493	0.757	0.907	0.970	0.992	0.998	1.000		
1.8	0.165	0.463	0.731	0.891	0.964	0.990	0.997	0.999	1.000	
1.9	0.150	0.434	0.704	0.875	0.956	0.987	0.997	0.999	1.000	
2.0	0.135	0.406	0.677	0.857	0.947	0.983	0.995	0.999	1.000	

(Continued) Poisson Distribution Function : $p(x, \lambda) = \sum_{r=0}^x e^{-\lambda} \frac{\lambda^r}{r!}$

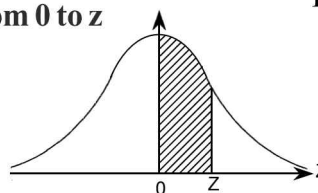
$\lambda \backslash x$	0	1	2	3	4	5	6	7	8	9
2.2	0.111	0.355	0.623	0.819	0.928	0.975	0.993	0.998	1.000	
2.4	0.091	0.308	0.570	0.779	0.904	0.964	0.988	0.997	0.999	1.000
2.6	0.074	0.267	0.518	0.736	0.877	0.951	0.983	0.995	0.999	1.000
2.8	0.061	0.231	0.469	0.692	0.848	0.935	0.976	0.992	0.998	0.999
3.0	0.050	0.199	0.423	0.647	0.815	0.916	0.966	0.988	0.996	0.999
3.2	0.041	0.171	0.380	0.603	0.781	0.895	0.955	0.983	0.994	0.998
3.4	0.033	0.147	0.340	0.558	0.744	0.871	0.942	0.977	0.992	0.997
3.6	0.027	0.126	0.303	0.515	0.706	0.844	0.927	0.969	0.988	0.996
3.8	0.022	0.107	0.269	0.473	0.668	0.816	0.909	0.960	0.984	0.994
4.0	0.018	0.092	0.238	0.433	0.629	0.785	0.889	0.949	0.979	0.992
4.2	0.015	0.078	0.210	0.395	0.590	0.753	0.867	0.936	0.972	0.989
4.4	0.012	0.066	0.185	0.359	0.551	0.720	0.844	0.921	0.964	0.985
4.6	0.010	0.056	0.163	0.326	0.513	0.686	0.818	0.905	0.955	0.980
4.8	0.008	0.048	0.143	0.294	0.476	0.651	0.791	0.887	0.944	0.975
5.0	0.007	0.040	0.125	0.265	0.440	0.616	0.762	0.867	0.932	0.968
5.2	0.006	0.034	0.109	0.238	0.406	0.581	0.732	0.845	0.918	0.960
5.4	0.005	0.029	0.095	0.213	0.373	0.546	0.702	0.822	0.903	0.951
5.6	0.004	0.024	0.082	0.191	0.342	0.512	0.670	0.797	0.886	0.941
5.8	0.003	0.021	0.072	0.170	0.313	0.478	0.638	0.771	0.867	0.929
6.0	0.002	0.017	0.062	0.151	0.285	0.446	0.606	0.744	0.847	0.916
	10	11	12	13	14	15	16			
2.8	1.000									
3.0	1.000									
3.2	1.000									
3.4	0.999	1.000								
3.6	0.999	1.000								
3.8	0.998	0.999	1.000							
4.0	0.997	0.999	1.000							
4.2	0.996	0.999	1.000							
4.4	0.994	0.998	0.999	1.000						
4.6	0.992	0.997	0.999	1.000						
4.8	0.990	0.996	0.999	1.000						
5.0	0.986	0.995	0.998	0.999	1.000					
5.2	0.982	0.993	0.997	0.999	1.000					
5.4	0.977	0.990	0.996	0.999	1.000					
5.6	0.972	0.988	0.995	0.998	0.999	1.000				
5.8	0.965	0.984	0.993	0.997	0.999	1.000				
6.0	0.957	0.980	0.991	0.996	0.999	0.999	1.000			

(Continued) Poisson Distribution Function : $p(x, \lambda) = \sum_{r=0}^x e^{-\lambda} \frac{\lambda^r}{r!}$

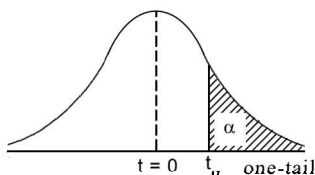
$\lambda \backslash x$	0	1	2	3	4	5	6	7	8	9
10.5	0.000	0.000	0.002	0.007	0.021	0.050	0.102	0.179	0.279	0.397
11.0	0.000	0.000	0.001	0.005	0.015	0.038	0.079	0.143	0.232	0.341
11.5	0.000	0.000	0.001	0.003	0.011	0.028	0.060	0.114	0.191	0.289
12.0	0.000	0.000	0.001	0.002	0.008	0.020	0.046	0.090	0.155	0.242
12.5	0.000	0.000	0.000	0.002	0.005	0.015	0.035	0.070	0.125	0.201
13.0	0.000	0.000	0.000	0.001	0.004	0.011	0.026	0.054	0.100	0.166
13.5	0.000	0.000	0.000	0.001	0.003	0.008	0.019	0.041	0.079	0.135
14.0	0.000	0.000	0.000	0.000	0.002	0.006	0.014	0.032	0.062	0.109
14.5	0.000	0.000	0.000	0.000	0.001	0.004	0.010	0.024	0.048	0.088
15.0	0.000	0.000	0.000	0.000	0.110	0.003	0.008	0.018	0.037	0.070
	10	11	12	13	14	15	16	17	18	19
10.5	0.521	0.639	0.742	0.825	0.888	0.932	0.960	0.978	0.988	0.994
11.0	0.460	0.579	0.689	0.781	0.854	0.907	0.944	0.968	0.982	0.991
11.5	0.402	0.520	0.633	0.733	0.815	0.878	0.924	0.954	0.974	0.986
12.0	0.347	0.462	0.576	0.682	0.772	0.844	0.899	0.937	0.963	0.979
12.5	0.297	0.406	0.519	0.628	0.725	0.806	0.869	0.916	0.948	0.969
13.0	0.252	0.353	0.463	0.573	0.675	0.764	0.835	0.890	0.930	0.957
13.5	0.211	0.304	0.409	0.518	0.623	0.718	0.798	0.861	0.908	0.942
14.0	0.176	0.260	0.358	0.464	0.570	0.669	0.756	0.827	0.883	0.923
14.5	0.145	0.220	0.311	0.413	0.518	0.619	0.711	0.790	0.853	0.901
15.0	0.118	0.185	0.268	0.363	0.466	0.568	0.664	0.749	0.819	0.875
	20	21	22	23	24	25	26	27	28	29
10.5	0.997	0.999	0.999	1.000						
11.0	0.995	0.998	0.999	1.000						
11.5	0.992	0.996	0.998	0.999	1.000					
12.0	0.988	0.994	0.997	0.999	0.999	1.000				
12.5	0.983	0.991	0.995	0.998	0.999	0.999	1.000			
13.0	0.975	0.986	0.992	0.996	0.998	0.999	1.000			
13.5	0.965	0.980	0.989	0.994	0.997	0.998	0.999	1.000		
14.0	0.952	0.971	0.983	0.991	0.995	0.997	0.999	0.999	1.000	
14.5	0.936	0.960	0.976	0.986	0.992	0.996	0.998	0.999	0.999	1.000
15.0	0.917	0.947	0.967	0.981	0.989	0.994	0.997	0.998	0.999	1.000

Areas under the Standard Normal Curve from 0 to z

$$Z = \frac{x - \mu}{\sigma}$$



z	0	1	2	3	4	5	6	7	8	9
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0754
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1256	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1916	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2258	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2518	.2649
0.7	.2580	.2612	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2996	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4654	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4979	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990
3.1	.4990	.4991	.4991	.4991	.4992	.4992	.4992	.4992	.4993	.4993
3.2	.4993	.4993	.4994	.4994	.4994	.4994	.4994	.4995	.4995	.4995
3.3	.4995	.4995	.4995	.4996	.4996	.4996	.4996	.4996	.4996	.4997
3.4	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4998
3.5	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998
3.6	.4998	.4998	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999
3.7	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999
3.8	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999
3.9	.5000	.5000	.5000	.5000	.5000	.5000	.5000	.5000	.5000	.5000

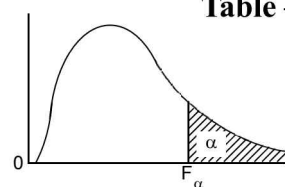
t_{α} - Critical Values of the t-Distribution with ν Degrees of Freedom Table – 4

ν	α								
	0.40	0.30	0.20	0.15	0.10	0.05	0.025	0.01	0.005
1	0.325	0.727	1.376	1.963	3.078	6.314	12.706	31.821	63.657
2	0.289	0.617	1.061	1.386	1.886	2.920	4.303	6.965	9.925
3	0.277	0.584	0.978	1.250	1.638	2.353	3.182	4.541	5.841
4	0.271	0.569	0.941	1.190	1.533	2.132	2.776	3.747	4.604
5	0.267	0.559	0.920	1.156	1.476	2.015	2.571	3.365	4.032
6	0.265	0.553	0.906	1.134	1.440	1.943	2.447	3.143	3.707
7	0.263	0.549	0.896	1.119	1.415	1.895	2.365	2.998	3.499
8	0.262	0.546	0.889	1.108	1.397	1.860	2.306	2.896	3.355
9	0.261	0.543	0.883	1.100	1.383	1.833	2.262	2.821	3.250
10	0.260	0.542	0.879	1.093	1.372	1.812	2.228	2.764	3.169
11	0.260	0.540	0.876	1.088	1.363	1.796	2.201	2.718	3.106
12	0.259	0.539	0.873	1.083	1.356	1.782	2.179	2.681	3.055
13	0.259	0.537	0.870	1.079	1.350	1.771	2.160	2.650	3.012
14	0.258	0.537	0.868	1.076	1.345	1.761	2.145	2.624	2.977
15	0.258	0.536	0.866	1.074	1.341	1.753	2.131	2.602	2.947
16	0.258	0.535	0.865	1.071	1.337	1.746	2.120	2.583	2.921
17	0.257	0.534	0.863	1.069	1.333	1.740	2.110	2.567	2.898
18	0.257	0.534	0.862	1.067	1.330	1.734	2.101	2.552	2.878
19	0.257	0.533	0.861	1.066	1.328	1.729	2.093	2.539	2.861
20	0.257	0.533	0.860	1.064	1.325	1.725	2.086	2.528	2.845
21	0.257	0.532	0.859	1.063	1.323	1.721	2.080	2.518	2.831
22	0.256	0.532	0.858	1.061	1.321	1.717	2.074	2.508	2.819
23	0.256	0.532	0.858	1.060	1.319	1.714	2.069	2.500	2.807
24	0.256	0.531	0.857	1.059	1.318	1.711	2.064	2.492	2.797
25	0.256	0.531	0.856	1.058	1.316	1.708	2.060	2.485	2.787
26	0.256	0.531	0.856	1.058	1.315	1.706	2.056	2.479	2.779
27	0.256	0.531	0.855	1.057	1.314	1.703	2.052	2.473	2.771
28	0.256	0.530	0.855	1.056	1.313	1.701	2.048	2.467	2.763
29	0.256	0.530	0.854	1.055	1.311	1.699	2.045	2.462	2.756
30	0.256	0.530	0.854	1.055	1.310	1.697	2.042	2.457	2.750
40	0.255	0.529	0.851	1.050	1.303	1.684	2.021	2.423	2.704
60	0.254	0.527	0.848	1.045	1.296	1.671	2.000	2.390	2.660
120	0.254	0.526	0.845	1.041	1.289	1.658	1.980	2.358	2.617
∞	0.253	0.524	0.842	1.036	1.282	1.645	1.960	2.326	2.576

Note : The above table gives the values of t for one-tail test (either left-tail or right-tail test). If we have to find the value of t for a two-tail test at a level, we take the value of $\alpha / 2$ for α . For example, the value of t at 5% level with 9 d.f. is $t_{0.025} = 2.262$ and the value of t at 1% level with 11 d.f. is $t_{0.005} = 3.106$

Critical Values of the F-Distribution

Table – 5



Values of $F_{0.05}(v_1, v_2)$									
v_1									
v_2	1	2	3	4	5	6	7	8	9
1	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04
120	3.92	3.07	2.68	2.45	2.29	2.17	2.09	2.02	1.96
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88

(Continued) Critical Values of the F-Distribution

Values of $F_{0.05}(v_1, v_2)$										
	v_1									
v_2	10	12	15	20	24	30	40	60	120	∞
1	241.9	243.9	245.9	248.0	249.1	250.1	251.1	252.2	253.3	254.3
2	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
3	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.52
4	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.62
5	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.36
6	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17	2.45	2.38	2.31	2.23	2.19	2.17	2.10	2.06	2.01	1.96
18	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
21	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
23	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.94	1.79	1.73
25	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
26	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
27	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
28	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
29	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
30	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.75	1.68	1.62
40	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
120	1.91	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35	1.25
∞	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

(Continued) Critical Values of the F-Distribution

Values of $F_{0.01}(v_1, v_2)$									
v_1									
v_2	1	2	3	4	5	6	7	8	9
1	4052	4999.5	5403	5625	5764	5859	5928	5981	6022
2	98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39
3	34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.35
4	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66
5	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16
6	13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98
7	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72
8	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91
9	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35
10	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94
11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63
12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39
13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19
14	8.86	6.51	5.56	5.04	4.69	4.46	4.28	4.14	4.03
15	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00	3.89
16	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78
17	8.40	6.11	5.18	4.67	4.34	4.10	3.93	3.79	3.68
18	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60
19	8.18	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52
20	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46
21	8.02	5.78	4.87	4.37	4.04	3.81	3.64	3.51	3.40
22	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35
23	7.88	5.66	4.76	4.26	3.94	3.71	3.54	3.41	3.30
24	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.26
25	7.77	5.57	4.68	4.18	3.85	3.63	3.46	3.32	3.22
26	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.29	3.18
27	7.68	5.49	4.60	4.11	3.78	3.56	3.39	3.26	3.15
28	7.64	5.45	4.57	4.07	3.75	3.53	3.36	3.23	3.12
29	7.60	5.42	4.54	4.04	3.73	3.50	3.33	3.20	3.09
30	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.07
40	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89
60	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72
120	6.85	4.79	3.95	3.48	3.17	2.96	2.79	2.66	2.56
∞	6.63	4.61	3.78	3.32	3.02	2.80	2.64	2.51	2.41

(Continued) Critical Values of the F-Distribution

Values of $F_{0.01}(v_1, v_2)$										
v_2	v_1									
	10	12	15	20	24	30	40	60	120	∞
1	6056	6106	6157	6209	6235	6261	6287	6313	6339	6366
2	99.40	99.42	99.43	99.45	99.46	99.47	99.47	99.48	99.49	99.50
3	27.23	27.05	26.87	26.69	26.60	26.50	26.41	26.32	26.22	26.13
4	14.55	14.37	14.20	14.02	13.93	13.84	13.75	13.65	13.56	13.46
5	10.05	9.89	9.72	9.55	9.47	9.38	9.29	9.20	9.11	9.02
6	7.87	7.72	7.56	7.40	7.31	7.23	7.14	7.06	6.97	6.88
7	6.62	6.47	6.31	6.16	6.07	5.99	5.91	5.82	5.74	5.65
8	5.81	5.67	5.52	5.36	5.28	5.20	5.12	5.03	4.95	4.86
9	5.26	5.11	4.96	4.81	4.73	4.65	4.57	4.48	4.40	4.31
10	4.85	4.71	4.56	4.41	4.33	4.25	4.17	4.08	4.00	3.91
11	4.54	4.40	4.25	4.10	4.02	3.94	3.86	3.78	3.69	3.60
12	4.30	4.16	4.01	3.86	3.78	3.70	3.62	3.54	3.45	3.36
13	4.10	3.96	3.82	3.66	3.59	3.51	3.43	3.34	3.25	3.17
14	3.94	3.80	3.66	3.51	3.43	3.35	3.27	3.18	3.09	3.00
15	3.80	3.67	3.52	3.37	3.29	3.21	3.13	3.05	2.96	2.87
16	3.69	3.55	3.41	3.26	3.18	3.10	3.02	2.93	2.84	2.75
17	3.59	3.46	3.31	3.16	3.08	3.00	2.92	2.83	2.75	2.65
18	3.51	3.37	3.23	3.08	3.00	2.92	2.84	2.75	2.66	2.57
19	3.43	3.30	3.15	3.00	2.92	2.84	2.76	2.67	2.58	2.49
20	3.37	3.23	3.09	2.94	2.86	2.78	2.69	2.61	2.52	2.42
21	3.31	3.17	3.03	2.88	2.80	2.72	2.64	2.55	2.46	2.36
22	3.26	3.12	2.98	2.83	2.75	2.67	2.58	2.50	2.40	2.31
23	3.21	3.07	2.93	2.78	2.70	2.62	2.54	2.45	2.35	2.26
24	3.17	3.03	2.89	2.74	2.66	2.58	2.49	2.40	2.31	2.21
25	3.13	2.99	2.85	2.70	2.62	2.54	2.45	2.36	2.27	2.17
26	3.09	2.96	2.81	2.66	2.58	2.50	2.42	2.33	2.23	2.13
27	3.06	2.93	2.78	2.63	2.55	2.47	2.38	2.29	2.20	2.10
28	3.03	2.90	2.75	2.60	2.52	2.44	2.35	2.26	2.17	2.06
29	3.00	2.87	2.73	2.57	2.49	2.41	2.33	2.23	2.14	2.03
30	2.98	2.84	2.70	2.55	2.47	2.39	2.30	2.21	2.11	2.01
40	2.80	2.66	2.52	2.37	2.29	2.20	2.11	2.02	1.92	1.80
60	2.63	2.50	2.35	2.20	2.12	2.03	1.94	1.84	1.73	1.60
120	2.47	2.34	2.19	2.03	1.95	1.86	1.76	1.66	1.53	1.38
∞	2.32	2.18	2.04	1.88	1.79	1.70	1.59	1.47	1.32	1.00

Table – 6

 χ^2_{α} - Critical Values of the Chi-squared Distribution with ν Degrees of Freedom

ν	0.30	0.25	0.20	0.10	0.05	0.025	0.02	0.01	0.005	0.001
1	1.074	1.323	1.642	2.706	3.841	5.024	5.412	6.635	7.879	10.827
2	2.408	2.773	3.219	4.605	5.991	7.378	7.824	9.210	10.597	13.815
3	3.665	4.108	4.642	6.251	7.815	9.348	9.837	11.345	12.838	16.268
4	4.878	5.385	5.989	7.779	9.488	11.143	11.668	13.277	14.860	18.465
5	6.064	6.626	7.289	9.236	11.070	12.832	13.388	15.086	16.750	20.517
6	7.231	7.841	8.558	10.645	12.592	14.449	15.033	16.812	18.548	22.457
7	8.383	9.037	9.803	12.017	14.067	16.013	16.622	18.475	20.278	24.322
8	9.524	10.219	11.030	13.362	15.507	17.535	18.168	20.090	21.955	26.125
9	10.656	11.389	12.242	14.684	16.919	19.023	19.679	21.666	23.589	27.877
10	11.781	12.549	13.442	15.987	18.307	20.483	21.161	23.209	25.188	29.588
11	12.899	13.701	14.631	17.275	19.675	21.920	22.618	24.725	26.757	31.264
12	14.011	14.845	15.812	18.549	21.026	23.337	24.054	26.217	28.300	32.909
13	15.119	15.984	16.985	19.812	22.362	24.736	25.472	27.688	29.819	34.528
14	16.222	17.117	18.151	21.064	23.685	26.119	26.873	29.141	31.319	36.123
15	17.322	18.245	19.311	22.307	24.996	27.488	28.259	30.578	32.801	37.697
16	18.418	19.369	20.465	23.542	26.296	28.845	29.633	32.000	34.267	39.252
17	19.511	20.489	21.615	24.769	27.587	30.191	30.995	33.409	35.718	40.790
18	20.601	21.605	22.760	25.989	28.869	31.526	32.346	34.805	37.156	42.312
19	21.689	22.718	23.900	27.204	30.144	32.852	33.687	36.191	38.582	43.820
20	22.775	23.828	25.038	28.412	31.410	34.170	35.020	37.566	39.997	45.315
21	23.858	24.935	26.171	29.615	32.671	35.479	36.343	38.932	41.401	46.797
22	24.939	26.039	27.301	30.813	33.924	36.781	37.659	40.289	42.796	48.268
23	26.018	27.141	28.429	32.007	35.172	38.076	38.968	41.638	44.181	49.728
24	27.096	28.241	29.553	33.196	36.415	39.364	40.270	42.980	45.558	51.179
25	28.172	29.339	30.675	34.382	37.652	40.646	41.566	44.314	46.928	52.620
26	29.246	30.434	31.795	35.563	38.885	41.923	42.856	45.642	48.290	54.052
27	30.319	31.528	32.912	36.741	40.113	43.194	44.140	46.963	49.645	55.476
28	31.391	32.620	34.027	37.916	41.337	44.461	45.419	48.278	50.993	56.893
29	32.461	33.711	35.139	39.087	42.557	45.772	46.693	49.588	52.336	58.302
30	33.530	34.800	36.250	40.256	43.773	46.979	47.962	50.892	53.672	59.703

